

Prepladder Version X Qbank

Subject: Forensic_Medicine

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Postmortem autopsy and torture

1. Which of the following statements about medicolegal autopsy is wrong?

- A. Always complete autopsy
- B. Consent of kin is required
- C. Requisition from police or magistrate is required
- D. Objective reporting to the police or magistrate is required

2. Which of the following statements regarding clinical autopsy is false?

- A. Consent from kin is required
- B. Requested by a clinician
- C. Always complete
- D. Tests are done in a medical laboratory

3. Given below are a few statements on various kinds of autopsy. Which of them is true? A psychological autopsy is done after suicide. Virtopsy combines 2D surface scanning/CAD photogrammetry with radiologic imaging like MRI. Psychological autopsy involves data collection. The ability to identify the infection status of antemortem from postmortem wounds is limited in Virtopsy. Easy to identify postmortem artifacts using Virtopsy

- A. All are true
- B. All except 2, 4 and 5
- C. All except 2 and 4
- D. All except 2 and 5

4. 2-5% of autopsies cannot determine the cause of death even after extensive lab investigations. What are such autopsies called?

- A. Obscure autopsy
 - B. Negative autopsy
 - C. Virtual autopsy
 - D. Clinical autopsy
-

5. A 60-year-old male patient was found deceased under unknown circumstances, and a detailed autopsy was requested to investigate possible causes of death. During the procedure, the police surgeon decided to use an evisceration technique that involves dissecting and examining each organ system individually rather than removing the organs as a whole. Which of the following techniques is he most likely using?

- A. Letulle
 - B. Rokitansky
 - C. Gohn
 - D. Virchow
-

6. A forensic pathologist is conducting an autopsy on a middle-aged female who was found dead under suspicious circumstances. The case requires a thorough examination of the thoracic and abdominal organs while maintaining an aesthetic appearance for potential display. Which incision technique is most appropriate to ensure optimal exposure of internal organs and accommodate forensic display requirements?

- A. I-shaped Incision
 - B. Y-shaped Incision
 - C. Modified Y-shaped Incision
 - D. T-shaped Incision
-

7. A 35-year-old male is found dead in his apartment under suspicious circumstances. Upon initial external examination, petechial hemorrhages are noted around the eyes. The scene suggests a possible asphyxial death, as there are no overt signs of trauma. During the autopsy by the police surgeon, which among the following has to be opened first to identify definitive signs of asphyxia?

- A. Thoracic cavity
 - B. Abdominal cavity
 - C. Cranial cavity
 - D. Pelvic cavity
-

8. A 28-year-old man is found dead at home and poisoning is suspected. During the autopsy, the forensic pathologist decides to dissect the stomach to examine its contents. Which of the following is the most appropriate procedure for stomach dissection to preserve evidence of poisoning and ensure accurate forensic analysis?

- A. Open the stomach along the greater curvature and discard the contents
 - B. Open the stomach along the lesser curvature and collect the contents for toxicology
 - C. Open the stomach along the greater curvature and collect the contents for toxicology
 - D. Incise the stomach along the pylorus and drain the contents into the small intestine
-

9. A 45-year-old man was found dead under suspicious circumstances, and his body was buried without a post-mortem examination. Months later, new evidence suggests foul play, and the family now requests further investigation. The local police inspector wishes to exhume the body to re-examine the cause of death. Which of the following authorities is legally empowered to order an exhumation in this case? A) Police Inspector B) First Class Magistrate C) Sub-Inspector of Police D) Medical Officer E) Sub-Divisional Magistrate

- A. All except D
- B. All except C and D
- C. Only B and E
- D. None of the above

10. A 45-year-old man who had been incarcerated for political dissent is reported to have died under suspicious circumstances. The autopsy reveals multiple injuries consistent with physical torture. In considering international guidelines for the medical investigation of such cases, which key document outlines principles and standards for identifying and investigating torture for a Police Surgeon?

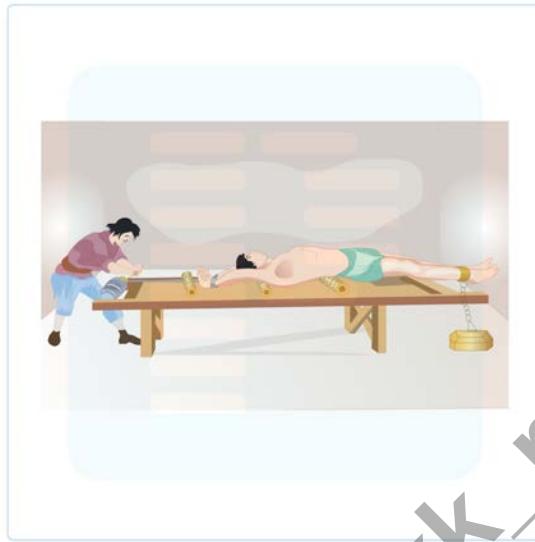
- A. Declaration of Tokyo
- B. Istanbul Protocol
- C. Declaration of Helsinki
- D. Nuremberg Code

11. What is the type of torture mentioned in the image given below?



- A. Scaphism
- B. Dry Submarine
- C. Picana
- D. El Plantón

12. What is the type of torture mentioned in the image given below?



- A. Chepuwa
- B. Belana
- C. Sawhorse
- D. Operating table

13. What is the type of torture mentioned in the image given below?



- A. Telefono
- B. Falanga
- C. Passing the board
- D. Nail torture

14. What is the type of torture mentioned in the image given below?



- A. La Bandera
- B. Parrot Perch
- C. Crucifixion
- D. Jack Knife Position

15. What is the type of torture mentioned in the image given below?



- A. Wet Submarine
- B. Black Slave
- C. Iron Chair
- D. Judas cradle

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	4
Question 4	2
Question 5	3
Question 6	3
Question 7	1
Question 8	3
Question 9	3
Question 10	2
Question 11	4
Question 12	2
Question 13	1
Question 14	2
Question 15	1

Solution for Question 1:

Correct Answer: B) Consent of kin is required

Explanation:

Consent of kin is not required for medicolegal autopsy but is required for pathological autopsy.

Medicolegal autopsy/necropsy/postmortem examination - specialized scientific examination of a deceased body conducted primarily for legal purposes.

A complete autopsy is a detailed examination of the entire body, both internally and externally, including the brain, chest and abdominal organs, and musculoskeletal systems.

Objectives	Identification and determination of cause and time of death For newborn infants, determine whether the child was born alive and viable For mutilated, fragmented, or skeletal remains, determine if they are human, the probable cause of death, and the time since death Restore the body to the best possible cosmetic condition before releasing it to relatives
Essential Requirements	Performed by a registered medical practitioner with specialized training or experience in forensic medicine (forensic pathology) Meticulous examination Complete autopsy (Option A ruled out) Materials sent to forensic science laboratory Preservation of records and evidence Objective reporting to the police or magistrate is required. (Option D ruled out) Consent of kin is not required (Option B) Requisition from police or magistrate (Option C ruled out)

Solution for Question 2:

Correct Answer: C) Always complete

Explanation:

Clinical autopsy can be partial or complete

Clinical/Pathological Autopsy:

- Conducted to determine the cause of death
- Requires consent from the next of kin or relatives (Option A ruled out)
- Requested by a clinician (Option B ruled out)
- The autopsy report is provided to the clinician

- Necessary tests are done in the concerned medical laboratory (Option D ruled out)
- Complete or partial autopsy, depending on the consent given (Option C)

Solution for Question 3:

Correct Answer: D) All except 2 and 5

Explanation:

Virtual autopsy/Virtopsy	
Virtopsy is a non-invasive alternative to traditional autopsy using imaging instead of physical dissection.	
Technology	Utilizes 3D surface scanning, MSCT, MRI, and CAD photogrammetry to create a digital, objective record. (Option D) This can be shared online for secondary opinions and presented as evidence in court if required.
Features	Provides high-resolution, colored 3D images in various sections Allows visualization of body surfaces, deeper tissues, organs, coronary vessels, pulmonary emboli, soft tissue trauma, and bone injuries
Limitations	High cost of equipment Radiologic imaging limitations, interference from metal Limited ability to distinguish internal organ colors Incomplete database for diagnosis Challenges in identifying infection status , antemortem vs. postmortem wounds (Options B and C ruled out) Difficulty in identifying postmortem artifacts and subtle tissue injuries, and differentiating certain pathological conditions (Option D)

Psychological autopsy - Meticulous collection of data after someone's suicide to understand the circumstances of their death.

Solution for Question 4:

Correct Answer: B) Negative autopsy

Explanation:

In a negative autopsy, the cause of death remains unknown

Obscure autopsy (Option A ruled out)	Negative autopsy (Option B)
The cause of death may not be clear at the time of dissection but can often be determined after detailed clinical, and laboratory investigations, and interviews Examples: Epilepsy, reflex vagal inhibition, uremia, and anesthetic overdose	The cause of death remains unknown even after extensive laboratory investigations Occurs in 2-5% of all postmortem cases

Solution for Question 5:

Correct Answer: C) Ghon

Explanation:

Evisceration techniques in autopsy:

En Masse (Letulle) (Option A ruled out)	Removing most internal organs in one full swoop Rapid technique Dissection afterward is lengthy Leaves all attachments intact
Virchow's Method (Option D ruled out)	Organs are removed individually Useful if the pathological focus is on a single organ
In Situ Dissection (Rokitansky) (Option B ruled out)	Organs are dissected within the body with minimal evisceration Rarely performed, but useful in cases involving highly transmissible diseases
En block (Ghon) (Option C)	Separate block dissection and organ separation Relatively quick Preserves most important inter-organ relationships

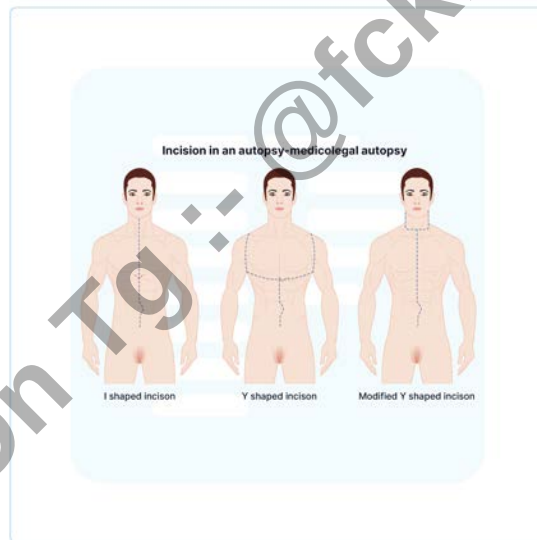
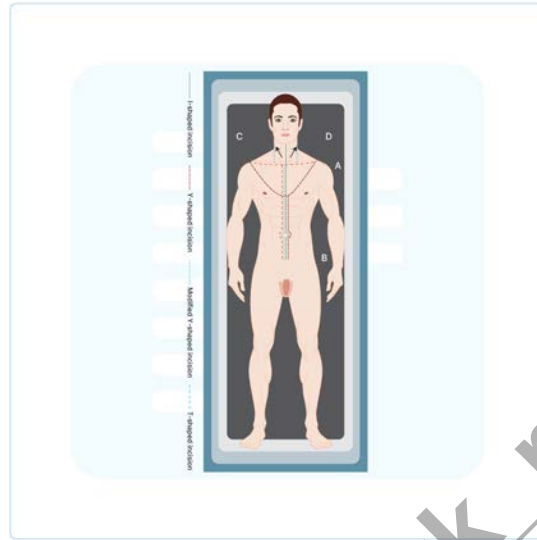
Solution for Question 6:

Correct Answer: C) Modified Y-shaped Incision

Explanation

Modified Y-shaped Incision is ideal for female cases, as it provides excellent exposure of the thoracic and abdominal organs while preserving the aesthetic appearance of the body for potential display

Type of Incision	Description	Indications
I-shaped Incision (Option A ruled out)	Common and straightforward. Extends from above the thyroid cartilage to the pubic symphysis, avoiding the umbilicus and injury sites.	Favored for its simplicity .
Y-shaped Incision (Option B ruled out)	Two cuts from each side of the neck starting from the mastoid, converging at the manubrium sterni, then continuing to the pubic symphysis.	Ideal for examining neck organs in asphyxial deaths .
Modified Y-shaped Incision (Option C)	Two incisions from the anterior axillary fold, curving beneath the breasts to meet at the xiphisternum, then a vertical incision to the pubic symphysis.	Preferred for female cases, especially for display.
X-shaped incision	Generally not done The incision of the back and limbs. To find and evaluate hidden subcutaneous hemorrhages	Preferred in custodial deaths



T-shaped Incision (Option D): Although this incision provides access to the abdominal cavity, it is rarely used in standard autopsy practice, and is not associated with improved aesthetic outcomes.

Solution for Question 7:

Correct Answer: A) Thoracic cavity

Explanation:

- In cases of suspected asphyxial death, the thoracic cavity is typically opened first.
- The order of opening is the trunk, cranium, and the neck. (To get a clear field of study for neck structures after draining of blood from neck vessels via head and trunk)
- Asphyxial deaths frequently involve abnormalities in the lungs and airway, such as congestion, petechiae, froth in the airways, or other indicators of respiratory failure.
- This helps confirm if the cause of death is indeed related to impaired oxygen exchange or suffocation.
- Examining the heart may also reveal signs of right-sided heart failure due to hypoxia.
- Thorax opened first in pulmonary embolism
- Abdomen is first to be opened in stillbirth

Order of Opening Cavities	
Head Injury Cases	The skull opened before the thorax and abdomen.
Asphyxial Deaths	The trunk opened first, then the skull.

Abdominal cavity (Option B): While the abdominal cavity could be opened to assess for trauma or distension, these findings are less likely to provide direct clues in cases of asphyxia unless abdominal trauma is suspected, which is not indicated here. Thus, it is typically examined after the thoracic cavity.

Cranial cavity (Option C): The cranial cavity is usually opened after the thorax unless there is clear evidence of head trauma or a need to rule out a central nervous system cause of death. In asphyxial deaths, intracranial findings are secondary, such as cerebral edema due to prolonged hypoxia.

Pelvic cavity (Option D): Trauma to the pelvic organs or fractures is more relevant in cases involving physical injury or sexual assault. These findings are not directly associated with asphyxial deaths, so this cavity is not examined first.

Solution for Question 8:

Correct Answer: C) Open the stomach along the greater curvature and collect the contents for toxicology

Explanation :

In forensic autopsies, the stomach is usually opened along the greater curvature to minimize disruption of stomach anatomy and allows the pathologist to collect and examine the stomach contents effectively, which is critical in cases of suspected poisoning.

Procedure for Stomach Dissection:

Solution for Question 9:

Correct Answer: C) Only B and E

Explanation :

In India, only specific magistrates can order exhumation:

- First Class, District, Sub-Divisional, Executive Magistrates.
- Law enforcement cannot order exhumation independently.

Exhumation	
Legal disinterment of a previously buried body. Conducted when post-burial examination is necessary.	
Common Reasons	Criminal Cases: Re-examine suspected homicides disguised as suicides/accidents. Investigate deaths from criminal abortion or negligence. Retrieve evidence (e.g., bullets). Civil Cases: Insurance/worker's compensation claims. Professional negligence cases. Inheritance disputes or identity verification. Correct burial errors (If the wrong body was buried)
Procedure Requirements	Timing & Personnel: Must be conducted during daylight (start in the morning). A medical officer, magistrate, and police officer are required. Documentation: Verify grave location and markers. Collect soil samples (above, below, and beside the body/coffin). Document body position and appearance. Create a detailed grave diagram. Photograph grave, coffin, and body.

Handling Guidelines	Avoid altering the body (no use of disinfectants or pesticides). Use plank or plastic sheet for removal if decomposition is minimal. Transport the body to the mortuary for post-mortem examination.
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Solution for Question 10:

Correct Answer: B) Istanbul Protocol

Explanation :

- The Istanbul Protocol, formally known as the UN Manual on the Effective Investigation and Documentation of Torture and Other Cruel, Inhuman, or Degrading Treatment or Punishment, is a key international document that provides guidelines for the medical and legal investigation of torture.
- It outlines detailed procedures for documenting physical and psychological evidence of torture, essential in cases where torture victims seek justice.
- This is different from the Declaration of Istanbul. The Declaration of Istanbul is based on Organ Trafficking and Transplant Tourism. This document provides ethical guidance for organ donation and transplantation practices. It aims to prevent organ trafficking and illegal travel for transplants.

Declaration	About
Declaration of Tokyo (Option A ruled out)	Guidelines for doctors concerning torture in relation to detention and imprisonment.
Declaration of Helsinki (Option C ruled out)	Ethical principles for medical research involving human subjects.
Nuremberg Code (Option D ruled out)	Set of research ethics principles for human experimentation.
Geneva Convention	Treatment of combatants and civilians during war and conflict.
Malta	The Role of Doctors in Hunger Strikes

Solution for Question 11:

Correct Answer: D) El Plantón

Explanation:

The given image depicts El Plantón

Torture: Defined as deliberate, systemic, or wanton infliction of physical or mental suffering by one or more persons acting alone or on the orders of any authority to force another person to yield information or for any other reason.

Physical torture subtypes

Types of Torture	Description
El Plantón (Option D)	Victim is forced to stand for extended periods in the 'Army Corps' posture or body folding or on one leg. Dependent edema and petechiae.
Scaphism (Option A)	An ancient form of execution involving being trapped between two boats and subjected to insects and exposure.
Dry Submarine (Option B)	Form of suffocation where the victim's head is wrapped in a plastic bag, causing breathing difficulty. Lung petechiae are seen.
Picana/Electric cattle prod (Option C)	An electric torture method that applies electric shocks to the genital region. Reddish brown lesions at points of contact with burning and scarring.

Solution for Question 12:

Correct Answer: B) Belana

Explanation:

The given image depicts Belana

Torture: Defined as deliberate, systemic or wanton infliction of physical or mental suffering by one or more persons acting alone or on the orders of any authority to force another person to yield information or for any other reason.

Physical torture subtypes

Types of torture	Description
Belana or The roller (Option B)	The pole is placed on the back of the victim and is pressed down with great weight and then rolled up over the legs and the body. Crushing of soft tissue and damage to muscles of the legs and the body.
Chepuwa (Option A)	The thighs of the victim are placed under two wooden rods and pressed to induce severe pain. {{caption_text}}
Sawhorse or 'el cabellete' (Option C)	Forced straddling on a sharp object like a saw. Perineal bruising was seen.
Operating table or 'el quirofano' (Option D)	Beating on the abdomen while lying on a table with the upper half of the body unsupported. Bruise, ruptured abdominal viscera.

Solution for Question 13:

Correct Answer: A) Telefono

Explanation:

The given image depicts Telefono

Torture: Defined as deliberate, systemic or wanton infliction of physical or mental suffering by one or more persons acting alone or on the orders of any authority to force another person to yield information or for any other reason.

Physical torture subtypes

Types of torture	Description
Telefono/Ear torture (Option A)	Repeated and simultaneous slapping of both the ears by open hands of the torturer. Ruptured tympanic membrane.
Falanga or Bastinado (Option B)	Most common form of torture. Victim is suspended upside down and then beaten on the soles by blunt instruments such as sticks, clubs or rifle butts. Haematoma of the soles.

Passing the board or Beating buttocks (Option C)	Victim is forced to the ground and beaten viciously with the board club or baton on the buttock.
Nail torture (Option D)	Victims' nails (either fingernails or toenails) are subjected to various forms of abuse, such as being pulled out, crushed, or having sharp objects driven under them.

Solution for Question 14:

Correct Answer: B) Parrot Perch

Explanation:

The given image depicts Parrot Perch

Torture: Defined as deliberate, systemic or wanton infliction of physical or mental suffering by one or more persons acting alone or on the orders of any authority to force another person to yield information or for any other reason.

Physical torture subtypes

Types of torture	Description
La Bandera (Option A)	Involves hanging the victim by their wrists, often with weights attached to the feet, causing intense pain in the arms and shoulders. Nerve and muscle damage with joint injuries seen.
Parrot Perch/Jack or 'Pau de arara' (Option B)	Victim is suspended head down from a horizontal pole placed under the knees with the wrist bound to the ankles.
Crucifixion/Palestine hanging (Option C)	The victim is nailed or tied to a cross, leaving them to hang in an agonizing position, often leading to death by asphyxiation or exhaustion.
Jack Knife Position (Option D)	The victim is bent forward, usually with their legs straight and their upper body folded down toward the knees, causing muscle cramps and restricted breathing.

Solution for Question 15:

Correct Answer: A) Wet Submarine

Explanation:

The given image depicts a Wet Submarine

Torture: Defined as deliberate, systemic or wanton infliction of physical or mental suffering by one or more persons acting alone or on the orders of any authority to force another person to yield information or for any other reason.

Physical torture subtypes

Types of torture	Description
Wet Submarine (Option A)	Involves submerging the victim's head in water or a similar liquid to create a sensation of drowning. Fecal matter and other debris in the airways.
Black Slave (Option B)	Hot metal skewer is inserted up the anus of the victim. Perianal and rectal burns were seen.
Iron Chair/Confession chair /Interrogation chair (Option C)	The victim is tied to an iron chair equipped with either wooden or iron spikes, which could be heated from below.
Judas cradle (Option D)	The victim is first stripped and then suspended over a pointed pyramid. The torture using ropes and pulleys, used to raise and lower the victim so that the point would penetrate the anus, vagina or scrotum.

Previous Year Questions

1. A married female died in unnatural condition within 3 years of marriage. Her father filed a report to police and inquiry will be done under which section?

- A. 194 BNSS/174 CrPC
- B. 196 BNSS/176 CrPC
- C. 116 BNS & 194 BNSS
- D. 106 BNS & 194 BNSS

2. A villager was assaulted by 4-5 people & brought to hospital in serious condition. His dying declaration was being taken in hospital. What is correct regarding this process?

- A. Magistrate can record the statement
- B. Doctor can record the statement
- C. Leading questions can be asked
- D. Oath is compulsory before recording the statement

3. A forensic expert received a summons from the district court to record his evidence in a criminal case, but two days prior, he already received a summons for a civil case in the same district court for the same date. What should the doctor do?

- A. Inform both courts and wait.
- B. Attend the court for which the summons came first.
- C. Attend the court for the criminal case.
- D. Do not attend either court.

4. Who order the autopsy in case of RTA?

- A. Forensic expert
 - B. Police
 - C. Lawyer
 - D. forensic doctor
-

5. What is the responsibility of the doctor in recording the dying declaration of a victim affected by a fire?

- A. To assess compos mentis
 - B. To record the statement even in the presence of magistrate
 - C. Cross-examine the person
 - D. Put the person under oath before declaration
-

6. What is the appropriate course of action following the observation of cigarette burn marks on the forearm of a 14-year-old girl who presents to a doctor with a fractured forearm, claiming to have fallen and hit her hand?

- A. To inform higher authorities
 - B. To do a complete physical examination
 - C. To tell or discuss with colleagues that she is a case of abuse
 - D. To call local social worker for help
-

7. Death registration in India done within?

- A. 21 days
 - B. 28 days
 - C. 30 days
 - D. 40 days
-

8. Who will conduct the investigation in the case where a woman, who had been married for 4 years, died and left a suicide note stating that she was subjected to dowry-related torture?

- A. Police
 - B. Village headman
 - C. Judge
 - D. Magistrate
-

9. An 11 year old girl was found to be quiet and shy in her classroom with intermittent crying, for the past few days. It was discovered that her uncle was touching her genitals inappropriately. The duty of the principal is to report this to?

- A. Magistrate
 - B. Police
 - C. Child welfare committee
 - D. Parents
-

10. According to the amendment in the medical termination of pregnancy (MTP) Bill of 2020, until what gestational age is it permissible to induce MTP in mentally ill rape victims?

- A. 20 weeks
 - B. 22 weeks
 - C. 24 weeks
 - D. 28 weeks
-

11. As an internal medicine consultant at a private clinic, you are handling the case of a patient who has chronic renal disease and needs hemodialysis. You provide dietary and fluid management recommendations and closely monitor the patient. Unfortunately, the patient decides to take ayurvedic supplements and occasionally disregards your instructions without informing you. Consequently, the patient's condition deteriorates, leading to respiratory distress. Subsequently, the patient's attendant files a legal complaint, accusing the doctor of negligence. Which of the following can serve as a defense for the doctor in court?

- A. Vicarious liability
 - B. Res Ipsa loquitur
 - C. Therapeutic misadventure
 - D. Contributory negligence
-

12. In a civil negligence case against a doctor, who is responsible for providing evidence?

- A. Patient
 - B. Police not under the rank of sub-inspector
 - C. Magistrate
 - D. Doctor
-

13. What is the term used to describe the unethical practice wherein a doctor receives payment in exchange for referring a patient to a radiologist for a CT scan?

- A. Dichotomy
 - B. Medical maloccurrence
 - C. Criminal negligence
 - D. Commission
-

14. A surgeon performs a hysterectomy for uterine fibroids after obtaining relevant informed consent from the patient. Despite adequate precaution, the ureter was injured intraoperatively. Under which of the following principles will the doctor not be held responsible for this act?

- A. Novus actus interveniens
 - B. Medical maloccurrence
 - C. Res ipsa loquitur
 - D. Physician error
-

15. A 16-year-old girl was rescued from a village in Bihar where she had been sold off by her father for Rs. 25000 to a 45-year-old man. On examination, she was found to be 12 weeks pregnant. She was granted permission to abort the pregnancy, taking into consideration the age of the girl and the circumstances of her pregnancy. A doctor who will be eligible to perform the abortion for this girl should have performed at least how many MTP prior to this?

- A. 10
 - B. 15
 - C. 25
 - D. 35
-

16. Which section of the Indian Penal Code (IPC) pertains to cases involving medical negligence?

- A. Sec 301 IPC
 - B. Sec 304 B IPC
 - C. Sec 304 A IPC
 - D. Sec 302 IPC
-

17. Which of the following courts has the authority to commute a death sentence?

- A. Session court
 - B. Additional session court
 - C. High court
 - D. Session court and additional session court
-

18. To perform MTP for a mentally challenged female, consent is obtained from?

- A. Husband/partner
 - B. Legal guardian
 - C. Doctor
 - D. Magistrate
-

19. Which of the following is punishable under the MTP Act?

- A. MTP done up to 24 weeks

- B. Done by government-authorized center
 - C. Done by experienced RMP
 - D. Non-entry in MTP-register, not sending report to DDHS
-

20. Medical termination of pregnancy can be done upto 24 weeks in case of?

- A. Congenital malformation
 - B. Minors
 - C. Contraception failure
 - D. Rape victim
-

21. A doctor was summoned as a public witness to appear and record his evidence in court. What is correct about this witness?

- A. Oral evidence
 - B. Common witness
 - C. Expert witness
 - D. Hostile witness
-

22. Which section of the Indian Penal Code (IPC) is associated with the penalization of perjury?

- A. 191
- B. 192
- C. 193
- D. 197

23. According to the law, nothing is an offense if a person is of an unsound mind and is not able to understand that the nature and consequences of his/her act are wrong. Which rule of insanity applies in this case?

- A. Durham's rule
 - B. Mc Naughton's rule
 - C. Curren's rule
 - D. Irresistible impulse rule
-

24. The court summons a member of the public who witnessed the incident to provide testimony and document their account. What kind of evidence does the witness offer to the court?

- A. Oral evidence
 - B. Hearsay evidence
 - C. Subpoena
 - D. Circumstantial evidence
-

25. While recording evidence in the court of law, the opposing lawyer asked the witness, "Did A kill B in your presence?" The witness said, "Yes". Which procedural route would this type of question be permitted in the court of law?

- A. Examination-in-chief
- B. Direct examination
- C. Cross-examination
- D. Re-direct examination

26. The authority who investigates the death of a 14-year-old girl who was abducted by a man and subsequently died while in police custody will be?

- A. Police
 - B. Coroner
 - C. Medical examiner
 - D. Judicial magistrate
-

27. Under which doctrine is the doctor not responsible when a patient with life-threatening injuries arrives at the casualty without time for consent and the doctor initiates life-saving procedures with the utmost care, but unfortunately, the patient still succumbs to death?

- A. Res ipsa loquitur
 - B. Doctrine of anticipation
 - C. Doctrine of extended consent
 - D. Doctrine of conjugated consent
-

28. What is the penalty for causing the death of a patient due to the use of counterfeit medication/ spurious drug?

- A. 5 years
 - B. 7 years
 - C. Life imprisonment
 - D. 1 year
-

29. A surgeon returns home from a party after many pegs of alcohol and is called to perform an emergency operation. During the operation, the assisting staff noticed the surgeon's handshaking and the instruments falling. He eventually nicks an artery, and the patient collapses. Under which of the following terms will this incident be tried?

- A. Criminal negligence
 - B. Civil negligence not amounting to criminal negligence
 - C. Therapeutic misadventure
 - D. Dichotomy
-

30. In relation to the punishment of perjury, which specific section of the IPC addresses this matter?

- A. 191
 - B. 192
 - C. 193
 - D. 197
-

31. What should the doctor do when faced with an unconscious 19-year-old female patient who presented to the emergency department, exhibiting signs of hydrocephalus on a non-contrast CT scan, and requiring immediate neurosurgical intervention, while also lacking any accompanying individual or means of identification?

- A. Do not operate without consent
- B. Wait for arrival of relatives
- C. Operate without consent

D. Take consent by calling the relatives

32. Who is responsible for handling the investigation in the case of a custodial death?

- A. Police inspector
 - B. Jail superintendent
 - C. Superintendent of Police
 - D. Judicial magistrate
-

33. Which article in the constitution does not pertain to children?

- A. Article 21A
 - B. Article 24
 - C. Article 45
 - D. Article 42
-

34. Please arrange the following clauses of IPC 320 in sequential order. Destruction or permanent impairing of the power of any member or joint Permanent disfiguration of the head or face Emasculation Fracture or dislocation of a bone or tooth

- A. 1, 2, 3, 4
 - B. 3, 2, 1, 4
 - C. 3, 1, 2, 4
 - D. 1, 4, 2, 3
-

35. What type of consent is obtained when a 45-year-old female patient is informed about the advantages and potential complications of a hysterectomy, and she gives her agreement for the procedure?

- A. Informed consent
 - B. Implied consent
 - C. Opt-out
 - D. Passive consent
-

36. In the court proceedings, the defense lawyer presents a leading question, to which the prosecutor objects. The appeal made by the prosecutor is accepted by the judge. Leading questions are not prohibited in which of the following scenarios?

- A. Re-examination
 - B. Cross-examination
 - C. Examination in chief
 - D. Dying declaration
-

37. Which of the following statements does not apply during cross-examination in a court of law?

- A. Witness is examined by the defense lawyer
 - B. Leading questions are not allowed
 - C. Objective of cross-examination is to weaken case of opponent
 - D. There is no time limit to cross examination
-

38. In the scenario where an individual has perpetrated a criminal act and is apprehended by the police, they are subsequently presented before a criminal court. Upon establishment of the accused's age as being less than a certain threshold, the court instructs the police to transfer the case to the juvenile courts. What is the age of the accused?

- A. 21
 - B. 17
 - C. 20
 - D. 24
-

39. The first-hand knowledge rule is applicable to which of the following?

- A. Common witness
 - B. Hand writing expert
 - C. Doctor
 - D. Hostile witness
-

40. When an individual commences employment with a second employer, the new employer assumes responsibility for the individual's actions, under which doctrine?

- A. Vicarious liability
 - B. Doctrine of negligent choice
 - C. Contributory negligence
 - D. Borrowed servant doctrine
-

41. If a healthcare professional is found guilty of professional misconduct, they must provide the patient's records promptly upon request, within:

- A. 72 hours
 - B. 48 hours
 - C. 24 hours
 - D. 7 days
-

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	3
Question 4	2
Question 5	1
Question 6	2
Question 7	1
Question 8	4
Question 9	2
Question 10	3
Question 11	4
Question 12	1
Question 13	1
Question 14	2
Question 15	3
Question 16	3
Question 17	3
Question 18	2

Question 19	4
Question 20	4
Question 21	3
Question 22	3
Question 23	2
Question 24	1
Question 25	3
Question 26	4
Question 27	3
Question 28	3
Question 29	1
Question 30	3
Question 31	3
Question 32	4
Question 33	4
Question 34	3
Question 35	1
Question 36	2
Question 37	2
Question 38	2
Question 39	1

Question 40	4
Question 41	1

Solution for Question 1:

In cases of a married woman's unnatural death within 7 years of marriage, a Magistrate-led inquiry is mandatory under Section 176 CrPC (or Section 196 BNSS) . This ensures a thorough and impartial investigation, especially when foul play, suicide, or dowry-related violence is suspected.

In India, there are two types of inquests: police inquest and magistrate inquest.

1. Police Inquest (Section 194, BNSS)

- Most common type of inquest in India.
- Conducted by a police officer , generally not below the rank of Head Constable .

The police officer, upon receiving information about the unnatural death, visits the site.

↓

Involves two or more witnesses from the local area, called panchas , to assist in the investigation. The report is referred to as a panchnama .

If the cause of death appears to be accidental or non-suspicious, the body is handed over to the family for funeral proceedings.

In case of suspicious circumstances , the police officer forwards the body for a postmortem to a certified medical officer and sends the report to the Magistrate .

2. Magistrate Inquest (Section 196, BNSS)

- Conducted by an Executive Magistrate , or Judicial Magistrate .
- It's mandatory in specific circumstances, including: Death, the disappearance of a person, or alleged rape in police custody or any custody authorized by the court Deaths due to police firing Deaths in jail Deaths of women within 7 years of marriage if, Commits suicide In any circumstances raising a reasonable suspicion that some other person committed an offence Any relative of the woman has made a request on this behalf There is any doubt regarding the cause of death Deaths where there's suspicion of rape before death Deaths with suspicion of foul play or where a relative of the deceased has made a request

- Death, the disappearance of a person, or alleged rape in police custody or any custody authorized by the court
- Deaths due to police firing
- Deaths in jail
- Deaths of women within 7 years of marriage if, Commits suicide In any circumstances raising a reasonable suspicion that some other person committed an offence Any relative of the woman has made a request on this behalf There is any doubt regarding the cause of death
- Commits suicide
- In any circumstances raising a reasonable suspicion that some other person committed an offence
- Any relative of the woman has made a request on this behalf
- There is any doubt regarding the cause of death
- Deaths where there's suspicion of rape before death
- Deaths with suspicion of foul play or where a relative of the deceased has made a request
- Also conducted if the case is referred by police due to suspicion arising during their investigation
- An inquiry by a Judicial Magistrate or Metropolitan Magistrate is required in instances of custodial death, disappearance, or custodial rape where foul play is suspected or there are allegations of an offense. In all other situations, the inquiry can be conducted by an Executive Magistrate.
- The magistrate is responsible for forwarding the body for postmortem examination to a Civil Surgeon or another authorized doctor within 24 hours of the death.
- Death, the disappearance of a person, or alleged rape in police custody or any custody authorized by the court
- Deaths due to police firing
- Deaths in jail
- Deaths of women within 7 years of marriage if, Commits suicide In any circumstances raising a reasonable suspicion that some other person committed an offence Any relative of the woman has made a request on this behalf There is any doubt regarding the cause of death
- Commits suicide
- In any circumstances raising a reasonable suspicion that some other person committed an offence
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- There is any doubt regarding the cause of death

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- Deaths with suspicion of foul play or where a relative of the deceased has made a request
- Commits suicide
- In any circumstances raising a reasonable suspicion that some other person committed an offence
- Any relative of the woman has made a request on this behalf
- There is any doubt regarding the cause of death

Differences Between Police and Magistrate Inquests:

Feature	Police Inquest	Magistrate Inquest
Conducted by	Police officer	Magistrate (DM, SDM, Judicial, or Executive Magistrate)
Involvement of witnesses	Requires signatures of panchas	Does not require witness signatures
Witness statements	Admissible in court	Not admissible in court
Can order arrest	No, but can arrest in cognizable offenses	Can issue arrest warrants
Exhumation	Cannot order exhumation	Can order exhumation of a body
Autopsy orders	Sends bodies for autopsy indiscriminately	Orders autopsy only when necessary
Forensic analysis	Cannot order chemical analysis of viscera	Can order chemical analysis

Solution for Question 2:

- Ideally, a Magistrate should record the dying declaration to ensure authenticity and legality.
- In exceptional cases, when a Magistrate is unavailable and death is imminent, a doctor or even a police officer may record the statement

Dying Declaration vs Dying Deposition

Feature	Dying Declaration	Dying Deposition
---------	-------------------	------------------

Recorder	Anyone—Magistrate/doctor/village headman/police/any member of the public	Always recorded by a Magistrate
Oath (Option D)	Not required	Must be taken
Accused or his counsel	Not present	Always present
Cross-examination	Not done	Done
Legal value	Comparatively less	Much more
Admissibility if declarant survives	Not admitted, but has corroborative value	Fully admitted
Nature	Merely recording of statement	Complete court procedure
Type of evidence	Documentary	Oral
Status in India	Followed	Not followed
Leading Qs (Option C)	Not Permitted	Permitted

Doctor can record the statement (Option B): Not ideal.

- A doctor can record the dying declaration only if a Magistrate is unavailable and death is imminent.
- However, the Magistrate is the preferred authority for legal validity.

Solution for Question 3:

- Criminal cases take precedence over civil cases due to their public importance and urgency.
- The forensic expert must attend the criminal case and inform the civil court about the conflict, requesting a rescheduled date.
- Ignoring a court summons can lead to legal consequences, so prioritizing the criminal case is mandatory.

Summons/Subpoena:

- A subpoena is a legal order mandating a witness to appear before a court on a specific date, time, and location to provide testimony or evidence, under penalty.

- A subpoena is issued by the court in writing, signed by the presiding officer, and carries the court's official seal (Sec. 63 BNSS) .
- Delivered to the witness by a police officer, court official, or public servant.
- The witness must sign and return a duplicate copy of the subpoena to acknowledge receipt (Sec. 64 BNSS).

Types of Subpoenas

- The witness is required to bring specific documents or items to the court (e.g., medico-legal reports or postmortem findings).
- For expert witness.
- The witness is required to appear in court to provide oral testimony.
- For common witnesses.

Punishment for Failing to Comply with a Subpoena/Summons:

1. Civil Cases:

- The witness may be ordered to pay damages for failing to attend court. This involves compensation for the inconvenience caused to the parties involved in the case.

2. Criminal Cases:

If a witness fails to appear in court without a justified reason, the following actions can be taken:

- Under 63 – 71 BNSS , the court may issue a notice to the witness to explain their absence.
- The court may issue notice under Section 389, BNSS and after hearing the witness, if it finds that the witness neglected to attend the court without a justified reason, may punish him under Section 208 and 209, BNS , and sentence him/her to: A fine (Rs.1000/-), OR Imprisonment (6 months), OR Both fine and imprisonment.
- A fine (Rs.1000/-), OR
- Imprisonment (6 months), OR
- Both fine and imprisonment.
- Also, the Court may issue bailable or non-bailable warrants to secure the presence of the witness (S.206 to 209, BNS and S.90, BNSS).
- A fine (Rs.1000/-), OR
- Imprisonment (6 months), OR
- Both fine and imprisonment.

Prioritization of Summons:

- Criminal vs. Civil Courts : If a witness is summoned by both criminal and civil courts for the same day, they must attend the criminal court , which takes precedence.

- Higher vs. Lower Courts : When summoned by two courts of the same jurisdiction (e.g., civil or criminal), the witness must attend the higher court first.
- Multiple Summons : If the witness receives multiple summonses from the same type of court, they should attend the court that issued the first summons and notify the other.

Inform both courts and wait (Option A): While informing both courts is a good step, simply waiting without attending either court is not advisable, as it could lead to legal consequences.

Attend the court for which the summons came first (Option B): The order in which the summons were received does not determine priority. The nature of the case (criminal vs. civil) takes precedence.

Do not attend either court (Option D): Ignoring court summons can result in contempt of court or other legal penalties.

Solution for Question 4:

Correct Option B - Police:

In cases of Road Traffic Accidents (RTA), the police are responsible for ordering an autopsy to determine the cause of death. This is part of their duty to investigate the circumstances surrounding the death and to ensure proper legal procedures are followed. The police will request an autopsy if there is a need to understand the cause of death more clearly or if it is required for legal reasons.

Incorrect Options:

Option A - Forensic expert:

A forensic expert conducts the autopsy but does not order it. The role of the forensic expert is to perform the examination and provide a report based on their findings.

Option C - Lawyer:

A lawyer may request an autopsy as part of legal proceedings, but they do not have the authority to order an autopsy in cases of RTA. Their role is generally more related to the legal representation of the parties involved.

Option D - Forensic doctor:

Similar to the forensic expert, the forensic doctor performs the autopsy but does not order it. The forensic doctor provides medical expertise in the examination and interpretation of findings.

Solution for Question 5:

Correct Option A - To assess compos mentis:

- When a dying declaration is being recorded, the role of the doctor is to assess the mental capacity of the person making the statement, also known as "compos mentis."
- The doctor evaluates whether the person is of sound mind and capable of providing a coherent and reliable statement.
- This assessment helps determine the admissibility and credibility of the dying declaration as evidence in legal proceedings.

Incorrect Options:

Option B - To record the statement even in the presence of a magistrate : It is not the role of the doctor to record the statement in the presence of a magistrate

Option C - Cross-examine the person : Cross-examination is a legal procedure conducted by legal professionals, such as lawyers or prosecutors. It is not the role of the doctor to cross-examine the person giving the dying declaration.

Option D - Put the person under oath before declaration: The doctor's role in recording a dying declaration does not involve putting the person under oath.

Solution for Question 6:

Correct Option B - To do a complete physical examination:

- When a 14-year-old girl presents with a fractured forearm and reports that she fell and hit her hand, it is essential for the doctor to conduct a thorough physical examination to assess the extent of the injury and evaluate any associated findings.

Incorrect Options :

Option A - To inform higher authorities : While it is important to address suspicions of abuse Before involving higher authorities, it is crucial to gather more information through a comprehensive examination.

Option C - To tell or discuss with colleagues that she is a case of abuse : Discussing the case with colleagues can provide additional insights, but it is necessary to gather more information through a comprehensive examination before making any definitive conclusions.

Option D - To call a local social worker for help : To call a local social worker for help: In cases where abuse is suspected, it is crucial to gather more information through a thorough physical examination and, if needed, reporting to higher authorities before contacting a social worker

Solution for Question 7:

Correct Option A - 21 days:

- In India, death registration is required to be done within 21 days (Option A) of the occurrence of the death. The Registrar of Births and Deaths, usually located at the local municipality or panchayat, is responsible for recording deaths and issuing death certificates.

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Incorrect Options:

- Option B, C & D: These options do not correspond to the prescribed time frame for death registration in India, which is 21 days.

Solution for Question 8:

Correct Option D - Magistrate:

- Usually, the death of a married lady within 7 years of marriage is presumed to be a case of dowry death.
- In this case, even though it is not mentioned that she was tortured for dowry forcing her to suicide, it is still assumed a dowry death.
- Death/suicide from torture/ requesting or pressures of dowry is investigated (inquest) by an executive magistrate .

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In this case, even though it is not mentioned that she was tortured for dowry forcing her to suicide, it is still assumed a dowry death.

Death/suicide from torture/ requesting or pressures of dowry is investigated (inquest) by an executive magistrate .

Incorrect Options:

Option A - Police: Police would have done the inquest if death occurred from unnatural causes like suicide, murder, homicide, etc.

Option B - Village headman: A village headman does not carry out magistrate inquests.

Option C - Judge: A judge does not carry out an inquest in the above scenario

Solution for Question 9:

Correct Option B - Police:

- Reporting the incident to the police is crucial because they have the authority and responsibility to investigate cases of child abuse, including inappropriate touching or sexual abuse.

Incorrect Options:

Option A - Magistrate : The police are the appropriate authority to handle the initial investigation and take immediate action to protect the child.

Option C - Child welfare committee : Child welfare committees play a crucial role in child protection, but the immediate action in cases of abuse is to report to the police.

Option D - Parents : While it is important to inform the parents or guardians of the child about the incident, the primary responsibility of the principal in such cases is to report to the police.

Solution for Question 10:

Correct Option C - 24 weeks:

- As per the Medical Termination of Pregnancy (Amendment) Bill, 2020, the limit for termination of pregnancy in a mentally ill woman or a woman who is a survivor of rape has been increased from 20 weeks to 24 weeks of gestation.

As per the Medical Termination of Pregnancy (Amendment) Bill, 2020, the limit for termination of pregnancy in a mentally ill woman or a woman who is a survivor of rape has been increased from 20 weeks to 24 weeks of gestation.

Incorrect Options:

Option A - 20 weeks: The Medical Termination of Pregnancy (Amendment) Bill, 2020, has increased the gestational limit for termination of pregnancy in certain cases from 20 weeks to 24 weeks.

Option B - 22 weeks: As per the Medical Termination of Pregnancy (Amendment) Bill, 2020, the gestational age limit for termination of pregnancy in cases of rape, incest, and vulnerable women, including mentally ill women, has been increased from 20 weeks to 24 weeks.

Option D - 28 weeks: Beyond 24 weeks, termination of pregnancy is only allowed in cases of substantial fetal abnormalities or if the continuation of the pregnancy poses a risk to the life of the mother.

Solution for Question 11:

Correct Option D - Contributory negligence:

- Contributory negligence can be used as a defense in this situation, as the patient failed to follow the doctor's instructions and chose to take alternative therapies without disclosing it. This contributed to his worsening condition and difficulty in breathing. Contributory negligence refers to the patient's own actions or negligence that contributed to their injuries, and it can reduce the liability of the doctor or healthcare provider in a legal claim.

Contributory negligence can be used as a defense in this situation, as the patient failed to follow the doctor's instructions and chose to take alternative therapies without disclosing it. This contributed to his worsening condition and difficulty in breathing. Contributory negligence refers to the patient's own actions or negligence that contributed to their injuries, and it can reduce the liability of the doctor or healthcare provider in a legal claim.

Incorrect Options:

Option A - Vicarious liability: Vicarious liability refers to the legal responsibility of a person in employee-employer contract.

Option B - Res Ipsa loquitur: Res Ipsa loquitur is a legal doctrine that allows a plaintiff to establish a presumption of negligence on the part of the defendant based on the facts of the case

Option C - Therapeutic misadventure: Therapeutic misadventure is a term used to describe an unintended outcome of a medical treatment or procedure that results in harm to the patient.

Solution for Question 12:

Correct Option A - Patient:

- In a civil negligence case against a doctor, the burden of proof lies with the patient. The patient is the one making the claim against the doctor and alleging negligence. Therefore, it is the patient's responsibility to provide evidence and prove that the doctor's actions or omissions fell below the standard of care expected, resulting in harm or injury.

In a civil negligence case against a doctor, the burden of proof lies with the patient. The patient is the one making the claim against the doctor and alleging negligence. Therefore, it is the patient's responsibility to provide evidence and prove that the doctor's actions or omissions fell below the standard of care expected, resulting in harm or injury.

Incorrect Options:

Option B - Police not under the rank of sub-inspector: The police are not directly involved in civil negligence cases against doctors.

Option C - Magistrate: In civil negligence cases against doctors, the burden of proof does not lie with the magistrate. Their role is to oversee the proceedings and apply the relevant laws.

Option D - Doctor: The doctor is the defendant in a civil negligence case, meaning they are the party being accused of negligence. It is not the doctor's burden to prove their innocence

Solution for Question 13:

Correct Option A - Dichotomy:

- The dichotomous method, in which one doctor refers a patient to another for a fee.

The dichotomous method, in which one doctor refers a patient to another for a fee.

Incorrect Options:

Option B - Medical Maloccurrence: The phrase "medical maloccurrence" refers to any instances of a bad or undesirable outcome that is unrelated to the quality of care provided.

Option C - Criminal negligence: Criminal negligence occurs when someone acts in a way that clearly endangers the security of human life. It refers to performing something that is improper or failing to carry out an obligation.

Option D - Commission: A commission is a fee a broker or other financial professional charges for their services in facilitating the purchase or sale of a financial asset.

Solution for Question 14:

Correct Option B - Medical maloccurrence:

- Medical maloccurrence refers to an unfortunate event or outcome that occurs during medical treatment despite the exercise of due care and skill. In this case, the ureter injury during a hysterectomy, despite adequate precaution, can be considered a medical

maloccurrence.

Medical maloccurrence refers to an unfortunate event or outcome that occurs during medical treatment despite the exercise of due care and skill. In this case, the ureter injury during a hysterectomy, despite adequate precaution, can be considered a medical maloccurrence.

Incorrect Option:

Option A - Novus actus interveniens : Even if the defendant has acted negligently, a subsequent intervening action breaks the chain of causation with the loss or damage sustained and so the defendant is not liable. Therefore, novus actus interveniens is not applicable in this context.

Option C - Res ipsa loquitur : Res ipsa loquitur is a legal doctrine that translates to "the thing speaks for itself." It applies when an injury occurs under circumstances that indicate negligence without direct evidence of it.

Option D - Physician error : Physician error refers to mistakes or failures in the actions or decisions of the physician that result in harm to the patient.

Solution for Question 15:

Correct Option C - 25:

- According to the Medical Termination of Pregnancy Act in India, a doctor must have performed at least 25 MTPs (Option C) to be eligible to perform the abortion for this girl. This criterion ensures that the doctor has sufficient experience and expertise in performing safe and legal abortions.

Incorrect Options:

- Option A, B & D - 10, 15 & 35: These options do not meet the requirement of having performed at least 25 MTPs as per the law.

Solution for Question 16:

Correct Option C - Sec 304 A IPC:

- 304A IPC defines punishment of criminal medical negligence, as rash, careless or gross negligence. Punishment includes up to 2 years imprisonment with a fine.

304A IPC defines punishment of criminal medical negligence, as rash, careless or gross negligence. Punishment includes up to 2 years imprisonment with a fine.

Incorrect Options:

Option A - Sec 301 IPC: 301 IPC defines punishment for culpable homicide causing the death of a person, other than the person whose death was intended.

Option B - Sec 304B IPC: 304B IPC defines punishment for dowry death (death of a female by suicide, accident, burns - within 7 years of marriage). Punishment is not less than 7 years and up to life imprisonment.

Option D - Sec 302 IPC: 302 IPC defines punishment for murder. Imprisonment for life, death penalty along with a fine.

Solution for Question 17:

Correct Option D -High court

- High courts have the authority to review and hear appeals against death sentences imposed by lower courts. During the appeal process, a high court can confirm, reverse, or modify the sentence based on the merits of the case under article 226 and 227.
- If a high court finds that the death sentence was unjustly imposed, it has the power to commute the sentence to life imprisonment or another lesser sentence.

Incorrect Options:

Options A, B, and D can not commute a death sentence.

Solution for Question 18:

Correct Option B - Legal guardian:

- In cases where a mentally challenged female is unable to provide consent for a medical procedure such as Medical Termination of Pregnancy (MTP), the consent of her legal guardian is typically required.

Incorrect Options:

Options A, C and D are incorrect.

Solution for Question 19:

Correct Option D - Non-entry in MTP-register, not sending report to DDHS:

- Failure to maintain proper records or report abortions to the appropriate authorities can lead to legal consequences.
- Specifically, non-entry in the MTP register and failure to send a report to the District Deputy Health and Family Welfare Officer (DDHS) are considered punishable offenses under the MTP Act.

Incorrect Options:

Option A - MTP done up to 24 weeks: The Medical Termination of Pregnancy (MTP) Act allows for abortions to be performed up to 24 weeks of gestation under certain conditions, such as when there is a risk to the life or physical or mental health of the woman, or if there is a substantial risk that the child would be born with physical or mental abnormalities.

Option B - Done by government-authorized center: The MTP Act permits abortions to be performed by registered medical practitioners (RMPs) in government-approved facilities, as well as by private practitioners in registered private clinics.

Option C - Done by experienced RMP: Under the MTP Act, abortions can be performed by registered medical practitioners (RMPs) who have undergone training and certification in the provision of abortion services.

Solution for Question 20:

Correct Option D -Rape victim:

- Medical termination of pregnancy can be done upto 24 weeks in case of rape victims

Incorrect Options:

Option A - Fetus malformation : Medical termination of pregnancy can be done beyond 24 weeks if the fetus is found to have severe malformations or abnormalities that are incompatible with life or would result in significant physical or mental challenges for the child.

Option B - Minors : Minors generally require parental or guardian consent for medical procedures, including abortion. The laws regarding abortion for minors can vary by country and jurisdiction, but typically there are provisions for involving parents or guardians in the decision-making process.

Option C - Contraception failure: Medical termination of pregnancy is not typically allowed solely due to contraception failure. However, if the failure of contraception leads to unwanted pregnancy and the woman's health or life is at risk, abortion might be considered based on medical assessment.

Solution for Question 21:

Correct Option C - Expert witness

- This is the correct answer. An expert witness is someone who possesses specialized knowledge, skills, or expertise in a particular field that is relevant to the case. In the given scenario, a doctor is summoned as a witness. If the doctor's testimony is related to their medical expertise and is meant to provide insight and explanation to the court on medical matters, then they would be considered an expert witness

Incorrect Options:

Option A - Oral evidence: Oral evidence refers to the testimony or statements provided by witnesses verbally in court. This option does not accurately describe the scenario in which a doctor is summoned as a witness. The scenario mentioned doesn't specifically emphasize the mode of evidence (oral or written) but focuses on the role of the witness.

Option B - Common witness: There is no standard term called "common witness." Witnesses can be categorized based on various factors, but this terminology isn't commonly used. The

scenario doesn't provide enough information to suggest that the doctor in question is a "common witness."

Option D - Hostile witness: A hostile witness is a term used to describe a witness who is uncooperative or openly antagonistic towards the party that called them to testify. This option is not applicable to the scenario provided, where a doctor is summoned as a public witness to record evidence in court.

Solution for Question 22:

Correct Option C - Section 193 of the Indian Penal Code (IPC):

- Section 193 of the IPC deals with the offense of giving false evidence in a judicial proceeding. It specifies that whoever intentionally gives false evidence in any stage of a judicial proceeding, or fabricates false evidence intending it to be used in a judicial proceeding, shall be punished with imprisonment and/or a fine.
- This section specifically addresses the punishment of perjury, which involves knowingly making false statements while under oath during a court proceeding.

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This section specifically addresses the punishment of perjury, which involves knowingly making false statements while under oath during a court proceeding.

Incorrect Options:

Option A - Section 191: Section 191 of the IPC deals with the offense of giving false evidence

Option B - Section 192: Section 192 of the IPC relates to the offense of fabricating false evidence.

Option D - Section 197: Under Section 197, if a person in an official capacity knowingly issues or signs a false certificate, they can be punished with imprisonment and or a fine.

Solution for Question 23:

Correct Option B - Mc Naughton's rule:

- McNaughton's rules state that "Nothing is an offense done by a person who is incapable of knowing that he is doing is wrong due to mental illness".

McNaughton's rules state that "Nothing is an offense done by a person who is incapable of knowing that he is doing is wrong due to mental illness".

Incorrect Options:

Option A - Durham's rule: Durham's Rule states that a criminal cannot be convicted of a crime if the act was a result of a mental disease or defect the criminal had at the time of the incident .

Option C - Curren's Rule: Curren's Rule "an accused person will not be criminally responsible if, at the time of committing the act, he did not have the capacity to regulate his conduct to the requirements of law, as a result of mental disease or defect."

Option D - Irresistible impulse rule: According to the Irresistible Impulse rule in jurisdictions defendants typically must present sufficient evidence to prove the existence of mental illness and that the mental illness caused the inability to control one's actions or conform one's conduct to the law.

Solution for Question 24:

Correct Option A - Oral evidence:

- "All statements which the court permits or requires to be made before it by a witness, in relation to matters of fact under inquiry, such statements are called as oral evidence" Hence anything which is spoken or expressed by the mouth of a witness who is called for trial in the Court is referred to as oral evidence.

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Incorrect Options:

Option B - Hearsay evidence: Hearsay evidence refers to a testimony that is provided by a witness and is not based on any personal communication.

Option C - Subpoena: Subpoena is a formal written order that requires a person to appear before a court, or other legal proceedings and testify, or produce documentation.

Option D - Circumstantial evidence: Circumstantial evidence is evidence of facts that the court can draw conclusions from .

Solution for Question 25:

Correct Option C - Cross-examination:

- Cross-examination is the correct procedural route in the court of law for this type of question. During cross-examination, the opposing lawyer has the opportunity to question the witness to test their credibility, challenge their statements, and uncover any inconsistencies or biases, by asking leading questions.. Cross-examination plays a critical role in the adversarial system, allowing both sides to present their case and test the evidence presented by the opposing party.

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Incorrect Options:

Option A - Examination-in-chief: Examination-in-chief refers to the questioning of a witness by the lawyer who called them to testify. This phase is typically used to elicit direct evidence and establish the witness's version of events.

Option B - Direct examination: Direct examination involves the lawyer questioning their own witness to present evidence and establish their case. Like examination-in-chief, it aims to elicit direct evidence and the witness's version of events.

Option D - Re-direct examination: Re-direct examination occurs after cross-examination and allows the lawyer to clarify or rebut issues that arose during cross-examination.

Solution for Question 26:

Correct Option D - Judicial magistrate:

- The inquest in the given scenario would typically be conducted by a Judicial magistrate.

The inquest in the given scenario would typically be conducted by a Judicial magistrate.

Incorrect Options:

Option A - Police : The police are responsible for investigating crimes, apprehending suspects, and maintaining law and order.

Option B - Coroner : They determine the cause and circumstances of death and may hold an inquest to gather evidence and information related to the death.

Option C - Medical examiner : A Medical Examiner is a forensic pathologist or medical doctor. They may provide valuable information and evidence during an inquest, but they are not typically responsible for conducting the inquest itself.

Solution for Question 27:

Correct Option C - Doctrine of extended consent:

- This doctrine holds that in emergency situations where a patient is unable to give consent, and there is no time to obtain consent from a legal surrogate, a healthcare provider is permitted to perform necessary life-saving procedures.
- The underlying principle is that it is presumed the patient would consent to urgent medical intervention if they were able.
- This doctrine applies directly to the described scenario.

Incorrect Options:

Option A - Res ipsa loquitur:

- This is a legal doctrine that means 'the thing speaks for itself'.

- It is used in tort law when the cause of injury is obvious, and it is presumed that negligence must have occurred because the event typically does not happen without negligence.
- This doctrine is not applicable in this case as it is related to proving negligence rather than consent.

Option B - Doctrine of anticipation:

- This term is not commonly recognized in medical or legal practice as it relates to patient consent or emergency medical treatment.
- It does not apply to the scenario described.

Option D - Doctrine of conjugated consent:

- This term is not standard in medical or legal contexts related to patient consent.
- It does not apply to emergency situations and consent procedures.

Solution for Question 28:

Correct Option C - Life imprisonment:

- The punishment for the death of a patient by spurious drug
- The death of a patient caused by the administration of a spurious drug is a serious offense. In many jurisdictions, the punishment for such an act is life imprisonment. use of counterfeit or adulterated drugs, can have severe consequences for the health and lives of individuals.
- It serves as a deterrent and emphasizes the importance of ensuring the safety and quality of drugs in healthcare systems.

The punishment for the death of a patient by spurious drug

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It serves as a deterrent and emphasizes the importance of ensuring the safety and quality of drugs in healthcare systems.

Incorrect Options:

- Option A, B and D are incorrect. Refer to the explanation for the correct answer.

Option A, B and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 29:

Correct Option A - Criminal negligence:

- The incident depicted within the question is an example of criminal negligence , which refers to a disregard for the safety of others that constitutes a criminal act.
- Criminal negligence includes a breach of the duty of care owed to the patient and can result in criminal charges being brought against the responsible party.
- In this case, the surgeon was under the impact of liquor, which impaired his capacity to perform the operation securely, and he eventually caused hurt to the patient.

The incident depicted within the question is an example of criminal negligence , which refers to a disregard for the safety of others that constitutes a criminal act.

Criminal negligence includes a breach of the duty of care owed to the patient and can result in criminal charges being brought against the responsible party.

In this case, the surgeon was under the impact of liquor, which impaired his capacity to perform the operation securely, and he eventually caused hurt to the patient.

Incorrect Options:

Option B - Civil negligence not amounting to criminal negligence:

- Civil negligence not producing criminal negligence: Civil negligence refers to a breach of the duty of care owed to a patient or other party that comes about in harm or damage. E.g.: doctor has given the wrong prescription.
- E.g.: doctor has given the wrong prescription.

Civil negligence not producing criminal negligence: Civil negligence refers to a breach of the duty of care owed to a patient or other party that comes about in harm or damage.

- E.g.: doctor has given the wrong prescription.

E.g.: doctor has given the wrong prescription.

Option C - Therapeutic misadventure:

- This term refers to an antagonistic outcome or harm during medical treatment if the treatment was given with due care and consideration.

This term refers to an antagonistic outcome or harm during medical treatment if the treatment was given with due care and consideration.

Option D - Dichotomy:

- This term refers to a division or contrast between two things that are or are spoken to as being restricted or completely distinctive.

This term refers to a division or contrast between two things that are or are spoken to as being restricted or completely distinctive.

Solution for Question 30:

Correct Option C - 193:

- Punishment for False Evidence and Perjury : This section deals with the punishment for giving false evidence (perjury) in any stage of a judicial proceeding or fabricating false evidence for the purpose of being used in any stage of a judicial proceeding. The punishment for this offense is imprisonment of either description for a term which may extend to seven years, and a fine.

Punishment for False Evidence and Perjury : This section deals with the punishment for giving false evidence (perjury) in any stage of a judicial proceeding or fabricating false evidence for the purpose of being used in any stage of a judicial proceeding. The punishment for this offense is imprisonment of either description for a term which may extend to seven years, and a fine.

Incorrect Options:

Option A - Section 191

- Definition of Giving False Evidence : This section defines what constitutes giving false evidence. It states that a person is said to give false evidence if they, being legally bound by an oath or by law to state the truth, make a false statement which they know or believe to be false.

Option B - Section 192

- Fabricating False Evidence : This section defines the act of fabricating false evidence. It explains that whoever causes any circumstance to exist or makes any false entry in any book or record, intending that such circumstance, false entry, or false statement may appear in evidence in a judicial proceeding and cause any person who is to decide on such

proceeding to form an erroneous opinion on a point material to the result of such proceeding, is said to fabricate false evidence.

Option D - Section 197

- This section deals with the punishment for giving false certificates or false information.

Solution for Question 31:

Correct Option C - Operate without consent:

- Since, the patient requires urgent medical intervention and she is unconscious. Also, she has no relatives accompanying her, so their consents couldn't be taken in this case scenario. She might die if the doctor waits for her relatives to come and give consent.
- In such a scenario (emergency), section 92 IPC would be taken into consideration.
- 92 IPC states that: Nothing is an offense which is done in good faith and for the benefit of person (or) the guardian in a situation when taking the consent is not possible (like in emergencies)

Since, the patient requires urgent medical intervention and she is unconscious. Also, she has no relatives accompanying her, so their consents couldn't be taken in this case scenario. She might die if the doctor waits for her relatives to come and give consent.

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92 IPC states that: Nothing is an offense which is done in good faith and for the benefit of person (or) the guardian in a situation when taking the consent is not possible (like in emergencies)

Incorrect Options:

Option A - Do not operate without consent: Patient was presented with an emergency requiring urgent neurological intervention. the patient may acquire some irreversible damages and may also lead to the death of the patient, if we wait for consent. Hence, the correct step would be to operate without consent.

Option B - Wait for arrival of relatives: If the doctor waits for the relatives to come to give consent, some irreversible damages to the patient might occur and the condition might become life-threatening.

Option D - Take consent by calling the relatives: Taking consent by calling the relatives might be a good step, but since the patient is in an emergency situation, section 92 IPC permits the doctor

to operate without consent.

Solution for Question 32:

Correct Option D - Judicial magistrate:

- Section 176 (I) of the Code of Criminal Procedure (CrPC) states that if a person in custody dies or disappears or a woman is raped in custody, the Judicial Magistrate (Executive Magistrate) has the power to order an inquiry.
- Conditions for magistrate inquest: A – Death in Asylum B – Death in hostels C – Custodial death/rape D – Dowry death E – Exhumation
- A – Death in Asylum
- B – Death in hostels
- C – Custodial death/rape
- D – Dowry death
- E – Exhumation

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Conditions for magistrate inquest:

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- E – Exhumation

A – Death in Asylum

B – Death in hostels

C – Custodial death/rape

D – Dowry death

E – Exhumation

Incorrect Options:

- Option A, B and C are incorrect. Refer to the explanation for the correct answer.

Option A, B and C are incorrect. Refer to the explanation for the correct answer.

Solution for Question 33:

Correct Option D - Article 42:

- 42," is the correct answer as it is not directly related to children.
- Article 42 of the Indian Constitution focuses on the provision of just and humane conditions of work and maternity relief.

42," is the correct answer as it is not directly related to children.

Article 42 of the Indian Constitution focuses on the provision of just and humane conditions of work and maternity relief.

Incorrect Options:

Option A - Article 21A: Article 21A of the Indian Constitution relates to the Right to Education . It states that the State shall provide free and compulsory education to all children in the age group of 6 to 14 years.

Option B - Article 24: Article 24 of the Indian Constitution deals with the prohibition of child labor.

Option C - Article 45: Article 45 of the Indian Constitution pertains to the provision of early childhood care and education for children below the age of six years .

Solution for Question 34:

Correct Option C - 3, 1, 2, 4:

Grievous hurt: It comes under 320 IPC.

Incorrect Options: Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 35:

Correct Option A - Informed consent:

- Informed consent refers to an intentional assertion made by a patient to experience a medical or surgical treatment or procedure after being provided with data about the proposed treatment's benefits, dangers, and options.
- Informed consent requires that the person is competent to form the choice, has been given adequate data to form the choice, and has unreservedly given their consent without constraint or undue influence.

Informed consent refers to an intentional assertion made by a patient to experience a medical or surgical treatment or procedure after being provided with data about the proposed treatment's benefits, dangers, and options.

Informed consent requires that the person is competent to form the choice, has been given adequate data to form the choice, and has unreservedly given their consent without constraint or undue influence.

Incorrect Options:

Option B - Implied consent:

- Consent is given by means of a gesture/ an act .
- It is when a patient's activities recommend that they have consented to a treatment or method if they have not given express verbal or written consent.
- This type of consent is typically utilized in emergencies

Consent is given by means of a gesture/ an act .

It is when a patient's activities recommend that they have consented to a treatment or method if they have not given express verbal or written consent.

This type of consent is typically utilized in emergencies

Option C - Opt-out:

- Opt-out is a consent that assumes a patient has agreed to a treatment or strategy unless they expressly decay it.
- This consent is regularly utilized for scheduled medical strategies such as blood tests or immunizations.

Opt-out is a consent that assumes a patient has agreed to a treatment or strategy unless they expressly decay it.

This consent is regularly utilized for scheduled medical strategies such as blood tests or immunizations.

Option D - Passive consent:

- It is when a patient does not effectively give consent, but their silence or need of complaint is taken as implied consent.
- This consent is ordinarily utilized for routine therapeutic procedures such as school vaccinations.

It is when a patient does not effectively give consent, but their silence or need of complaint is taken as implied consent.

This consent is ordinarily utilized for routine therapeutic procedures such as school vaccinations.

Solution for Question 36:

Correct Option B - Cross-examination:

- Leading question is a question where the answer is yes or no.
- Leading questions are asked only in cross examination

Leading question is a question where the answer is yes or no.

Leading questions are asked only in cross examination

Incorrect Options:

Option A - Re-examination: Leading questions are generally not permitted amid re-examination, so this alternative isn't correct.

Option C - Examination in chief: Leading questions are by and large not permitted during examination in chief, so this option isn't correct.

Option D - Dying declaration: Leading questions not permitted amid dying declaration, so this alternative isn't correct.

Solution for Question 37:

Correct Option B - Leading questions are not allowed:

- Leading questions are not allowed is an incorrect statement regarding cross-examination in a court of law. In fact, leading questions are often used during cross-examination to elicit specific information or to challenge the witness's testimony, under section 143 IEA. Therefore, the statement "Leading questions are not allowed" does not apply during cross-examination in a court of law.

Incorrect Options:

Option A - Witness is examined by the defense lawyer: That statement applies during cross-examination in a court of law. The defense lawyer has the right to cross-examine the witness in order to test their credibility and the reliability of their testimony.

Option C - Objective of cross-examination is to weaken case of opponent: The objective of cross-examination is to challenge the credibility, accuracy, and completeness of the witness's testimony, and to weaken the case presented by the opposing party. The cross-examining lawyer aims to extract information that may undermine the witness's testimony or support their own case.

Option D - There is no time limit to cross examination: The specific time limits may vary depending on the jurisdiction and the specific rules of the court, but they are generally in place to ensure that the trial proceeds in an efficient and timely manner.

Solution for Question 38:

Correct Option B - 17:

- To transfer the case to juvenile court, the person has to be a juvenile. Any person, male or female, who is below 18 years of age is a juvenile (Age: <18 years).
- Any crime or an offense done by a juvenile would be tried in the juvenile justice board .

Incorrect Options:

- Option A, C and D: All the choices except option B is more than juvenile age. Hence, they are incorrect.

Solution for Question 39:

Correct Option A - Common witness:

- The first-hand knowledge rule, also known as the firsthand or personal knowledge rule, is applicable to a common witness. A common witness is an individual who has observed or experienced an event directly and can provide testimony based on their own personal knowledge of the event.
- The first-hand knowledge rule requires that a witness's testimony be based on facts or information that they themselves have perceived through their own senses

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The first-hand knowledge rule requires that a witness's testimony be based on facts or information that they themselves have perceived through their own senses

Incorrect Options:

Option B - Hand writing expert : They rely on their specialized knowledge and training rather than personal observation or direct experience of the events related to the handwriting.

Option C - Doctor: A doctor may provide expert testimony based on their professional knowledge and experience rather than personal observation or direct experience of a specific event.

Option D - Hostile witness : A hostile witness is a witness who is unwilling or uncooperative in providing testimony. the first-hand knowledge rule is not specifically applicable to hostile witnesses.

Solution for Question 40:

Correct Option D - Borrowed servant doctrine:

- When a person starts working for a second employer, the new employer may assume responsibility for their actions under the Borrowed Servant Doctrine.
- This doctrine applies when an employee is temporarily loaned or "borrowed" by another employer. In such cases, the borrowing employer assumes control over the borrowed employee and becomes responsible for their actions within the scope of their employment during the loaned period.

Incorrect Options :

Option A - Vicarious liability : Vicarious liability refers to a situation where one person or entity is held responsible for the actions of another person or entity.

Option B - Doctrine of negligent choice : The Doctrine of negligent choice refers to a legal principle that holds a party liable for choosing a negligent individual or entity to perform a task.

Option C - Contributory negligence : Contributory negligence assigns partial blame to a plaintiff for their own injuries or damages due to their failure to exercise reasonable care.

Solution for Question 41:

Correct Option A - 72 hours:

- In cases of professional misconduct, when a patient requests their medical records, the healthcare provider is obligated to provide the records within a specific timeframe. The correct timeframe for providing patient records on demand in the case of professional misconduct is typically within 72 hours.

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typically within 72 hours.

Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation for the correct answer.

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BNS Sections

1. Match the BNS sections with the offense BNS Section Offense 1.274 a. Adulteration of food or drink intended for sale 2. 275 b. Sale of adulterated drugs 3. 276 c. Sale of noxious food or drink 4. 277 d. Adulteration of drugs

BNS Section	Offense
1.274	a. Adulteration of food or drink intended for sale
2. 275	b. Sale of adulterated drugs
3. 276	c. Sale of noxious food or drink
4. 277	d. Adulteration of drugs

- A. 1-a, 2-c, 3-d, 4-b
- B. 1-c, 2-d, 3-b, 4-a
- C. 1-a, 2-b, 3-c, 4-d
- D. 1-b, 2-a, 3-d, 4-c

2. A 28 year old woman undergoes an illegal abortion and is rushed to the emergency department after collapsing at home. On arrival, she is unresponsive and later dies. Upon forensic examination, the immediate cause of death is determined. Which of the following is NOT typically considered an immediate cause of death in cases of criminal abortion?

- A. Fat embolism
- B. Septicemia
- C. Amniotic fluid embolism
- D. Vagal inhibition

3. Bestiality is a term given to which of the following unnatural sexual offenses?

- A. Anal intercourse
 - B. Oral intercourse
 - C. Female homosexuality
 - D. Sexual intercourse with an animal
-

4. Which of the following acts cannot be considered as rape?

- A. Inserting an article into the urethra
 - B. Manipulating any part of a woman's body to cause penetration
 - C. Penetrating wife without consent
 - D. Applying mouth-to-mouth
-

5. Match the BNS sections with the offences

BNS Section	Offense
1. 97	a. Importing a girl under 21 years into India from a foreign country for illicit intercourse
2. 137 (1) (b)	b. Abducting a minor from lawful guardianship
3. 96	c. Abducting a child under 10 years to take any moveable property
4. 141	d. Procuring a girl under 18 years for illicit intercourse

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1. 97	a. Importing a girl under 21 years into India from a foreign country for illicit intercourse
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3. 96	c. Abducting a child under 10 years to take any moveable property
4. 141	d. Procuring a girl under 18 years for illicit intercourse

- A. 1-a, 2-b, 3-c, 4-d
 - B. 1-c, 2-b, 3-d, 4-a
 - C. 1-d, 2-c, 3-b, 4-a
 - D. 1-b, 2-c, 3-a, 4-d
-

6. A man attacks a woman in public and tries to disrobe her. What is the punishment he will get for this?

- A. 3-7 years of imprisonment and fine
 - B. 1-3 years of imprisonment
 - C. 7-10 years of imprisonment
 - D. 4-7 years of imprisonment
-

7. Acid attack comes under which of the following BNS sections?

- A. BNS 124
 - B. BNS 119 (1)
 - C. BNS 123
 - D. BNS 119 (2)
-

8. A 32 year old man gets into an argument and intentionally causes severe injuries to another person, resulting in a broken arm and significant physical pain. The police decided to charge him under the appropriate section of the BNS. Which section of the BNS is appropriate in this context?

- A. He is charged under the section BNS 115 (1)
- B. He is charged under the section BNS 115 (2)

- C. He is charged under the section BNS 117 (2)
 - D. He is charged under the section BNS 124 (2)
-

9. A 35 year old man was assaulted, leading to several injuries and is hospitalized. You are tasked with determining whether the case fits the legal classification of grievous hurt under BNS 116 for some of the expected events. Which of the following scenarios would NOT be classified as grievous hurt?

- A. The person dies due to an injury that, under normal circumstances, was not intended to cause death.
 - B. The person suffers an abdominal injury causing the intestines to protrude and is hospitalized for 22 days.
 - C. The person sustains a deep injury without any fracture or permanent disfigurement.
 - D. The person is in a coma for 20 days following a severe head injury.
-

10. A 40 year old man was involved in an altercation and sustained multiple injuries. You are evaluating whether these injuries qualify as grievous hurt under BNS 116. Which of the following scenarios does NOT meet the criteria for classifying the injury as grievous hurt?

- A. He is rendered impotent after sustaining a serious genital injury.
 - B. He loses a single tooth due to a punch on his face
 - C. He attains a cut injury on his face that temporarily alters his appearance.
 - D. He got his wrist fractured due to a heavy fall
-

11. Which of the following correctly defines the term "injury" as per legal terminology?

- A. Harm caused to a person's reputation, body, or property, irrespective of legality.
- B. Any harm legally or illegally caused to a person's mind, body, or property.
- C. Any harm caused to a person's physical body.
- D. Any harm illegally caused to a person, in body, mind, reputation, or property.

12. A 11-year old child accused of murder was let go free after the trial. What could be the likely explanation in this case?

- A. BNS 20 - depending on the mental maturity
- B. BNS 20 - never held liable
- C. BNS 21 - depending on the mental maturity
- D. BNS 21 - never held liable

13. Match the BNS sections with the offense BNS Section Offense 1. 104 a. Attempt to commit suicide. 2. 107 b. Punishment for murder by life convict. 3. 109 c. Abetment of suicide of child or insane person. 4. 226 d. Attempt to murder.

BNS Section	Offense
1. 104	a. Attempt to commit suicide.
2. 107	b. Punishment for murder by life convict.
3. 109	c. Abetment of suicide of child or insane person.
4. 226	d. Attempt to murder.

- A. 1-b, 2-c, 3-d, 4-a
- B. 1-c, 2-d, 3-b, 4-a
- C. 1-a, 2-b, 3-c, 4-d
- D. 1-b, 2-a, 3-d, 4-c

14. A 60 year old man gets into a heated argument with his neighbor during a community event. In a fit of rage, he pushes the neighbour hard, causing him to fall and hit his head on a concrete surface. The neighbor sustains a severe head injury and dies a few hours later in the hospital. He did not intend to kill but was reckless during the confrontation. Which section of the BNS would he be most likely be charged under?

- A. BNS 301
- B. BNS 104
- C. BNS 105
- D. BNS 102

15. A surgeon deliberately administers a lethal dose of anesthesia to a patient during a routine surgery because of a personal grudge. The patient dies on the operating table. Investigations reveal that the surgeon was fully aware that the dose would cause death. Under which section of the BNS would the surgeon be charged?

- A. BNS 301
- B. BNS 302
- C. BNS 101
- D. BNS 102

16. A 32 year old man, unaware that he is HIV positive, has been in a relationship with a woman for the past year. Neither he nor his partner has undergone recent medical check-ups, and they have been engaging in unprotected intercourse. Unfortunately, she tests positive for HIV as well. This act comes under which BNS?

- A. BNS Section 271
 - B. BNS Section 272
 - C. BNS Section 106A
 - D. BNS Section 109
-

17. Match the following 1) BNS 227 a) Giving false evidence under oath 2) BNS 229 b) Punishment for providing false evidence under oath 3) BNS 234 c) Punishment for issuing a false certificate 4) BNS 238 d) Punishment for disappearance of vital evidence

1) BNS 227	a) Giving false evidence under oath
2) BNS 229	b) Punishment for providing false evidence under oath
3) BNS 234	c) Punishment for issuing a false certificate
4) BNS 238	d) Punishment for disappearance of vital evidence

- A. 1-b, 2-a, 3-c, 4-d
 - B. 1-b, 2-a, 3-d, 4-c
 - C. 1-a, 2-b, 3-c, 4-d
 - D. 1-a, 2-b, 3-d, 4-c
-

18. Person presented to ED with complaints of severe chest pain for 1 hour. CT chest done revealed the presence of Aortic dissection. Patient was taken up for an emergency endovascular surgery without obtaining consent as there were no caregivers present at the time of presentation. Patient succumbed to hypotensive shock during the procedure. The surgeon will not be held liable under

- A. BNS 26
- B. BNS 27

C. BNS 28

D. BNS 30

19. BNS 27 deals with

- A. Act done in good faith for the benefit of a child (<12 years) with guardian's consent
 - B. Act done in good faith for a person's benefit after obtaining lawful consent
 - C. Act not intended to cause death/grievous hurt done by a person >18 years of age
 - D. Consent is not valid when it is given by fear/misconception
-

20. Husband physically assaults his wife following a verbal altercation between them. On enquiry, it is known that the Husband was under the influence of alcohol during this act and his wife reveals that he is a chronic alcoholic, drinks on a daily basis. Which section deals with the liable act done by her husband?

- A. BNS 20
 - B. BNS 22
 - C. BNS 23
 - D. BNS 24
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	4
Question 4	3
Question 5	2
Question 6	1
Question 7	1
Question 8	3
Question 9	1
Question 10	3
Question 11	4
Question 12	3
Question 13	1
Question 14	3
Question 15	3
Question 16	1
Question 17	3
Question 18	4

Question 19	1
Question 20	4

Solution for Question 1:

Correct Answer: A) 1-a, 2-c, 3-d, 4-b

Explanation:

BNS Section	Offense
274	Adulteration of food or drink intended for sale. Punishment: Imprisonment upto 6 months or fine. Bailable
275	Sale of noxious food or drinks. Punishment: Imprisonment upto 6 months or fine. Bailable
276	Adulteration of drugs. Punishment: Imprisonment upto 6 months or fine. Non bailable
277	Sale of adulterated drugs. Punishment: Imprisonment upto 6 months or fine. Bailable

- BNS section 286: Negligent conduct with respect to poisonous substances.

Solution for Question 2:

Correct Answer: B) Septicemia

Explanation:

While septicemia is a common complication in criminal abortions, it usually takes time to develop and is considered a delayed cause of death rather than an immediate cause of death.

Immediate cause	Delayed cause (onset within 48 hrs to 72 hrs)	Remote cause
-----------------	---	--------------

1) Vagal inhibition 2) Air embolism 3) Hemorrhage 4) Fat embolism 5) Amniotic fluid embolism 6) Rarely poisoning.	1) Septicemia 2) Pyaemia 3) Confined local infection and toxemia 4) General peritonitis 5) Tetanus.	1) Jaundice and renal failure 2) Bacterial endocarditis 3) Pulmonary embolism.
---	---	--

- Fat embolism (Option A): Fat embolism can occur during illegal abortions, especially if bones are fractured or fatty tissues are disturbed, leading to embolic events.
- Amniotic fluid embolism (Option C): Although rare, amniotic fluid embolism can be a cause of sudden death in both criminal and legal abortions, particularly if the uterine wall is breached.
- Vagal inhibition (Option D): Vagal inhibition, due to sudden pain or manipulation, can lead to reflex cardiac arrest, an immediate cause of death in such cases.

Solution for Question 3:

Correct Answer: D) Sexual intercourse with an animal

Explanation:

Sexual intercourse by a human with a lower animal is called bestiality

- Sodomy -anal intercourse (Option A)
- Buccal coitus -intercourse through the mouth (Option B)
- Tribadism -female homosexuality (Option C)
- Bestiality -sexual intercourse by a human with a lower animal

Solution for Question 4:

Correct Answer: C) Penetrating wife without consent

Explanation:

In India, sexual intercourse by a man with his wife if she is not under 18 years of age (old definition not under 15 years) is not considered as rape.

Rape:

BNS	BNS 63 (Old: Section 375 IPC)
Definition	Unlawful sexual intercourse by a man with a woman through force or under certain conditions
Acts considered	Penetrates his penis into the vagina, mouth, urethra, or anus Inserts any object or body part (not his penis) or applies his mouth to the vagina, mouth, urethra, or anus (Option A ruled out) (Option D ruled out) Manipulates any part of the woman's body to cause penetration into the vagina, urethra, or anus (Option B ruled out)
Circumstances	Against her will Without her consent With her consent obtained through fear of death or hurt With her consent, when she believes he is her husband due to impersonation When she cannot understand the nature of consent due to unsoundness of mind, intoxication, or being drugged Statutory rape: With or without her consent, if she is under 18 years of age When she is unable to communicate consent
Exception	Medical interventions or procedures Sexual intercourse by a man with his wife if she is not under 18 years of age (Option C)

Solution for Question 5:

Correct Answer: B) 1-c, 2-b, 3-d, 4-a

Explanation:

Kidnapping: Carrying away a person from lawful guardianship by illegal means.

New	Old	About
BNS 97	Section 369 IPC	Kidnapping or abducting a child under 10 years with the intent to take dishonestly any moveable property from the child
BNS 137 (1)(b)	Section 361 IPC	Kidnapping or abducting a minor from lawful guardianship: Boy under 16 years Girl under 18 years

BNS 96	Section 366A IPC	Procuring a girl under 18 years old for illicit intercourse or for prostitution
BNS 141	Section 366B IPC	Importing a girl under 21 years into India from a foreign country for illicit intercourse

Solution for Question 6:

Correct Answer: A) 3-7 years of imprisonment and fine

Explanation:

Public disrobing is considered as an indecent assault and is punishable under Section BNS 76 for 3-7 years imprisonment and fine

Assault:

- An offer, threat, or attempt to apply force to another person's body in a hostile manner BNS 130 (Old: IPC Section 351)

Indecent Assault:

Definition	Unwanted sexual behavior or touching of a female without her consent, intending to outrage her modesty
Legal Aspect	BNS 74 Old: Section 354 IPC 1-5 years imprisonment and/or fine The offense is cognizable, non-bailable, and non-compoundable Tried by any Magistrate
Whoever assaults or uses criminal force on any woman or abets such act with the intention of disrobing or compelling her to be naked	BNS 76 Old: Section 354-B IPC 3-7 years imprisonment and fine (Option A)

Solution for Question 7:

Correct Answer: A) BNS 124

Explanation:

Acid attack is under section Section BNS 124 (1) and Section BNS 124 (2).

Acid attack:

Definition	Throwing or attempting to throw acid or any corrosive substance that can cause bodily injury, disfigurement, scars, temporary or permanent disability, or grievous hurt
BNS sections	Section BNS 124 (1) and Section BNS 124 (2) (Option A)
Punishment	Imprisonment from 5 to 7 or 10 years to life and fine

Solution for Question 8:

Correct Answer: C) He is charged under the section BNS 117 (2)

Explanation :

- BNS 117 (2): Specifies the punishment for voluntarily causing grievous hurt, which can result in imprisonment for up to seven years and a fine. (Option C) Old: Section 325 IPC
- Specifies the punishment for voluntarily causing grievous hurt, which can result in imprisonment for up to seven years and a fine. (Option C)
- Old: Section 325 IPC
- BNS 115 (1): Defines voluntarily causing hurt. (Option A ruled out) Old: Section 321 IPC
- Defines voluntarily causing hurt. (Option A ruled out)
- Old: Section 321 IPC
- BNS 115 (2): Deals with the punishment for voluntarily causing hurt, which carries a lesser penalty than grievous hurt, with imprisonment up to one year or a fine of up to ₹1,0000. (Option B ruled out) Old: Section 323 IPC, where fine was up to ₹1000 .
- Deals with the punishment for voluntarily causing hurt, which carries a lesser penalty than grievous hurt, with imprisonment up to one year or a fine of up to ₹1,0000. (Option B ruled out)
- Old: Section 323 IPC, where fine was up to ₹1000 .
- BNS 124 (2): Covers voluntarily throwing or attempting to throw acid, with a minimum imprisonment of 5 years and a fine. (Option D ruled out) Old: Section 326B IPC

- Covers voluntarily throwing or attempting to throw acid, with a minimum imprisonment of 5 years and a fine. (Option D ruled out)
- Old: Section 326B IPC
- Specifies the punishment for voluntarily causing grievous hurt, which can result in imprisonment for up to seven years and a fine. (Option C)
- Old: Section 325 IPC
- Defines voluntarily causing hurt. (Option A ruled out)
- Old: Section 321 IPC
- Deals with the punishment for voluntarily causing hurt, which carries a lesser penalty than grievous hurt, with imprisonment up to one year or a fine of up to ₹1,0000. (Option B ruled out)
- Old: Section 323 IPC, where fine was up to ₹1000 .
- Covers voluntarily throwing or attempting to throw acid, with a minimum imprisonment of 5 years and a fine. (Option D ruled out)
- Old: Section 326B IPC

Solution for Question 9:

Correct Answer: A) The person dies due to an injury that, under normal circumstances, was not intended to cause death.

Explanation:

This situation involves an unintended fatal outcome, which falls under the category of culpable homicide or accidental death, not grievous hurt. Grievous hurt pertains to severe injuries, not necessarily causing death.

BNS 116 (Old Section 320 IPC) defines following injuries as grievous:

- Emasculation (cutting off the penis, castration, or causing loss of power of erection due to spinal injury).
- Permanent privation of the sight of either eye.
- Permanent privation of the hearing of either ear.
- Privation of any member (part, organ, limb) or joints
- Destruction or permanent impairing of powers of any member or joint.

- Permanent disfiguration of head or face.
- Fracture or dislocation of a bone or a tooth.
- Any hurt which endangers life , or which causes the sufferer to be, during the space of 20 days , in severe bodily pain, or unable to follow his daily routine.

The person suffers an abdominal injury causing the intestines to protrude and is hospitalized for 22 days (Option B): This is classified as grievous hurt because it causes serious injury that can endanger life and requires significant medical intervention.

The person sustains a deep injury without any fracture or permanent disfigurement (Option C): A deep injury may cause considerable pain or disability, but without a fracture or permanent disfigurement, it does not meet the legal criteria for grievous hurt.

The person is in a coma for 20 days following a severe head injury (Option D): Prolonged unconsciousness or coma due to head injury is considered grievous hurt, as it severely affects the victim's capacity to follow ordinary pursuits.

Solution for Question 10:

Correct Answer: C) He attains a cut injury on his face that temporarily alters his appearance.

Explanation:

A temporary cut or injury that does not result in permanent disfigurement does not meet the criteria for grievous hurt.

BNS 116 (Old Section 320 IPC) defines following injuries as grievous:

Solution for Question 11:

Correct Answer: D) Any harm illegally caused to a person, in body, mind, reputation, or property.

Explanation :

IPC	Term	BNS	Definition
44	Injury	Sec 2(14)	The word "injury" denotes any harm illegally caused to any person, in body, mind, reputation, or property. (Option D)
45	Life	Sec 2(17)	The word "life" denotes the life of a human being, unless the contrary appears from the context.
46	Death	Sec 2(6)	The word "death" denotes the death of a human being, unless the contrary appears from the context.

Harm caused to a person's reputation, body, or property, irrespective of legality (Option A): The term "injury" specifically denotes illegal harm, not general harm regardless of legality.

Any harm legally or illegally caused to a person's mind, body, or property (Option B): Injury refers to illegal harm, not legal or illegal.

Any harm caused to a person's physical body (Option C): Injury encompasses more than just physical harm; it includes harm to mind, reputation, and property.

Solution for Question 12:

Correct answer: C) BNS 21 - depending on the mental maturity

Explanation:

BNS 21 deals with children between 7-12 years cannot be held liable if they lack sufficient maturity/understanding.

BNS 20 (Option B):

- Child below 7 years of age will not be held criminally responsible for his/her act.
- Minimum age of criminal responsibility: 7 years (BNS) 5 years (Indian Railways Act)
- 7 years (BNS)
- 5 years (Indian Railways Act)

- Minimum age of full criminal responsibility: 18 years
- 7 years (BNS)
- 5 years (Indian Railways Act)

BNS 21 (Option C):

- A child between 7-12 years of age can be held liable for any crime depending on their mental maturity.
- Mental maturity is determined by the - Court of Law .

Solution for Question 13:

Correct Answer: A) 1-b, 2-c, 3-d, 4-a

Explanation:

BNS Section	Offense
104	Punishment for murder by life convict. Punishment: Death penalty
105	Punishment for culpable homicide not amounting to murder. Punishment : Part I: Life imprisonment or up to 10 years and fine. Part II: Up to 10 years imprisonment or fine.
107	Abetment of suicide of child or insane person. Punishment : Death penalty or life imprisonment or up to 10 years and fine.
108	Abetment of suicide. Punishment: Up to 10 years imprisonment and fine.
109	Attempt to murder. Punishment: Up to life imprisonment or up to 10 years and fine. If injury occurs, may extend to life imprisonment.
110	Attempt to commit culpable homicide. Punishment : Up to 3 years imprisonment or fine, or both. If injury occurs, may extend to 7 years.
226	Attempt to commit suicide. Punishment: Up to 1 year imprisonment or fine.

Solution for Question 14:

Correct Answer: C) BNS 105

Explanation:

BNS Section 105 applies because the act caused death, but without the intention to kill. The act was reckless and dangerous, leading to death, but it does not meet the criteria for murder under Section 103.

100 BNS: Culpable homicide

Culpable homicide not amounting to murder:

- Defined under Section 105 of the BHARATIYA NYAYA SANHITA(BNS)
- It refers to a situation where a person causes the death of another, but the act does not fulfill all the conditions necessary for it to be classified as murder under Section 101.
- Death caused due to grave and sudden provocation.
- Death caused without premeditation in the heat of the moment during a sudden fight.
- Death caused in good faith while exercising the right to self-defense.

Punishment under Section 105 BNS:

The section is divided into two parts:

Part I:

- If the act is done with the intention of causing death or causing bodily injury likely to cause death, the punishment can be life imprisonment or imprisonment for up to 10 years and a fine.

Part II:

- If the act is done with the knowledge that it is likely to cause death but without any intention to cause death or such injury, the punishment can be imprisonment for up to 10 years or a fine, or both.

BNS 301 (Option A): Trespassing on burial places. Imprisonment for 1 year or fine and non-bailable.

BNS 104 (Option B): Punishment for murder by life convict. Punishment is death and non-bailable.

BNS 102 (Option D): Culpable homicide by causing death of person other than person whose death was intended.

Solution for Question 15:

Correct Answer: C) BNS 101

Explanation:

BNS Section 101 applies as the surgeon deliberately administered a lethal dose, knowing it would result in death. The act was intentional and premeditated, which qualifies as murder under this section.

BNS Section 100: Culpable homicide

Culpable homicide amounting to murder:

- Defined under Section 101 of the BHARATIYA NYAYA SANHITA (BNS)
- Intention to cause death: The person committing the act has the clear intention of causing the death of the victim.
- Intention to cause bodily injury likely to cause death: The act is done with the intention of causing such bodily injury as the offender knows is likely to cause death.
- Knowledge of high probability of death: The act is done with the knowledge that it is so imminently dangerous that it must, in all probability, cause death or such bodily injury as is likely to cause death.
- The punishment for murder under Section 103 of the BNS is death or life imprisonment and the person may also be liable to pay a fine.

BNS 301 (Option A): Trespassing on burial places. Imprisonment for 1 year or fine and non-bailable.

BNS 302 (Option B): Uttering words with deliberate, intent to wound religious feelings. Imprisonment for 1 year or fine and non-bailable.

BNS 102 (Option D): Culpable homicide by causing death of person other than person whose death was intended.

Solution for Question 16:

Correct Answer: A) BNS Section 271

Explanation :

BNS Section 269 applies to anyone who negligently or recklessly performs an act that could spread an infectious disease dangerous to life, such as unintentional transmission of HIV.

BNS Section	Offense
BNS Section 269 (Option A)	This section pertains to negligent acts that are likely to spread infection of a disease dangerous to life. The punishment can be imprisonment for up to six months or a fine or both.
BNS Section 272 (Option B)	This section deals with malignant acts likely to spread infection of disease dangerous to life. The punishment can be imprisonment for up to two years or fine.
BNS Section 106 (Option C)	This section deals with causing death by negligence. It applies when death is caused by a rash or negligent act that does not amount to culpable homicide. The punishment for such an offense is imprisonment for up to two years, or a fine, or both.
BNS Section 109 (Option D)	This section pertains to attempt to murder. The punishment can extend to imprisonment for life, or imprisonment of up to 10 years and a fine.

Solution for Question 17:

Correct answer: C) 1-a, 2-b, 3-c, 4-d

Explanation:

Correct match:

1) BNS 227	a) Giving false evidence under oath
2) BNS 229	b) Punishment for providing false evidence under oath
3) BNS 234	c) Punishment for issuing a false certificate
4) BNS 238	d) Punishment for disappearance of vital evidence

BNS related to false evidence/false certificate/disappearance of evidence:

BNS 227	Giving false evidence/statement under oath by a witness (Perjury)
BNS 228	Fabrication of evidence
BNS 229	Punishment for providing false evidence under oath (Punishment for perjury) (imprisonment of upto 7 years + fine)
BNS 234	Punishment for issuing a false certificate (imprisonment of upto 7 years + fine)
BNS 238	Punishment for disappearance of vital evidence (imprisonment of upto 7 years + fine)
BNS 239	Intentional omission of vital information by person legally bound to inform

Solution for Question 18:

Correct Answer: D) BNS 30

Explanation:

Aortic dissection is a surgical emergency without which the patient would die, and does not warrant an obtained consent under emergency circumstances and so the Surgeon will not be held liable for any unforeseen consequences during/after the procedure under BNS 30 . This is known as the Doctrine of Anticipation.

BNS related to consent:

BNS 25	Act not intended to cause death/grievous hurt done by a person >18 years of age after rightful consent.
--------	---

BNS 26 (Option A)	Act done in good faith for a person's benefit after obtaining lawful consent.
BNS 27 (Option B)	Act done in good faith for the benefit of a child (<12 years) or a person with unsound mind with guardian's consent
BNS 28 (Option C)	Consent given by a person under fear or threat/under misconception of fact/child under 12 years of age/person of unsound mind/under intoxicated state is considered invalid.
BNS 30 (Option D)	An act done in good faith for the benefit of a person without their consent is not an offense. This is true if the circumstances make it impossible for the person to give consent. Known as Doctrine of Anticipation.

Solution for Question 19:

Correct answer: A) Act done in good faith for the benefit of a child (<12 years) with guardian's consent

Explanation:

BNS 27 deals with the Act done in good faith for the benefit of a child (<12 years) or a person of unsound mind.

BNS related to consent:

BNS 25 (Option C)	Any act which causes harm except the act intended to cause death or grievous hurt is not an offense if it is done with the consent of a person who is above 18 years of age. Consent can be either Implied (or) Expressed. E.g., two boxers fighting each other in an olympic match.
BNS 26 (Option B)	If a person commits an act done in good faith and with the consent for the benefit of that person, it will not be considered an offense. E.g., a surgeon performing surgery after getting consent.

BNS 27 (Option A)	Any act done in good faith for the benefit of a child (<12 years) or a person of unsound mind after obtaining guardian's (caregiver's) consent. Neither the Doer nor the Caregiver would be held responsible for any unforeseen consequences. E.g., a doctor doing necessary medical examination on a child (<12years) after obtaining guardian's consent.
BNS 28 (Option D)	Consent is not valid if it is given by a person under fear or threat /under misconception of fact/child under 12 years of age /person of unsound mind /under intoxicated state . E.g., a female giving consent for sex to her boss under fear of losing her job is considered invalid.

Solution for Question 20:

Correct answer: D) BNS 24

Explanation:

Her husband would be held liable under BNS 24 - deals with voluntary intoxication (as there is no history of the husband being forced to drink).

BNS dealing with Criminal liability:

BNS 20 (Option A)	Act done by a child below 7 years of age - NOT held liable
BNS 21	Act done by a child between 7-12 years of age can/cannot be held liable depending on the mental maturity (assessed by The court of law)
BNS 22 (Option B)	Act of a person of unsound mind/insane - NOT held liable (McNaughten's rule)
BNS 23 (Option C)	Act of a person done under involuntary intoxication - NOT held liable
BNS 24 (Option D)	Act of a person done under voluntary intoxication - Held liable

Traces of evidence & Sexual offences

1. What is the primary purpose of applying toluidine blue to the anogenital area during the examination of rape victim?

- A. To enhance the visibility of foreign objects
- B. To identify areas of micro-injury
- C. To check for signs of infection
- D. To prepare for DNA collection

2. Which of the following is NOT true regarding the POCSO Act?

- A. It is designed to protect female children under the age of 18 from sexual abuse
- B. It also includes protection against digital sexual crimes and exploitation against children.
- C. The Act was amended in 2019, making punishments stricter.
- D. Special courts have been established to ensure the swift disposal of POCSO cases

3. A 28-year-old male presents to the emergency department following an alleged sexual assault by a man. The victim recalls that the assault occurred approximately 4 hours ago. Which of the following examination findings would least likely support evidence of the current sodomy assault?

- A. Linear abrasions or contusions in the perianal region
- B. Dried semen around the anus
- C. Paradoxical dilatation of anus during bimanual traction of buttocks
- D. Prolapse of rectal mucosa

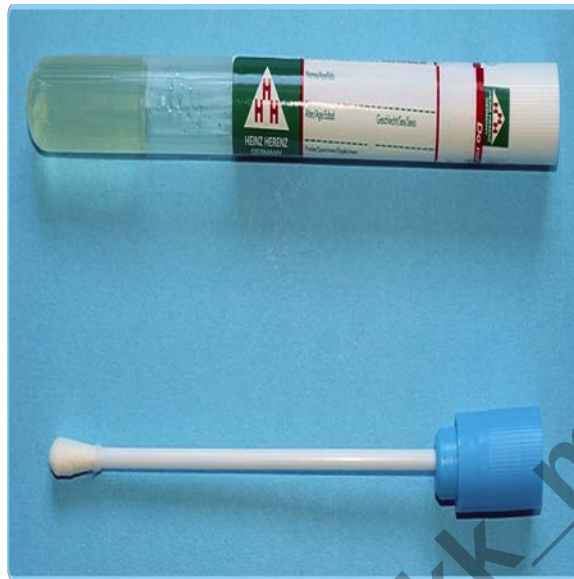
4. A 40-year-old male is arrested after being caught secretly watching women undress in a public locker room. He admits that he derives sexual pleasure from watching others undress and sometimes engaging in sexual acts. What is the likely paraphilic disorder?

- A. Frotteurism
 - B. Voyeurism
 - C. Fetishism
 - D. Eonism
-

5. A 32-year-old male presents to the psychiatrist with complaints of unusual sexual desires. He reports feeling intense sexual gratification when inflicting pain on his partner during consensual intimate encounters. He denies any interest in other forms of sexual activity and mentions that causing pain is the primary source of his sexual arousal. What is the most likely paraphilic disorder?

- A. Algolagnia
 - B. Uranism
 - C. Mooning
 - D. Cunnilingus
-

6. Which of the following is a part of the SAFE kit? 1. 2. 3. 4. 5.







- A. 1, 2, 3
- B. 2, 3, 4, 5
- C. 1, 2, 3, 4, 5
- D. 1, 2, 3, 5

7. A 20-year-old male was brought for medicolegal examination after being accused of raping a girl 2 hours back. The examining physician noted the cheesy secretion around the corona glandis and conducted a thorough assessment of the genital organs, including a penile swab. Which of the following is incorrect?

- A. Cheesy material is inconsistent with his recent sexual intercourse
- B. Lugol's iodine test turns brown if vaginal cells are present
- C. His consent for examination is needed
- D. Lugol's iodine can be done till 4 days after rape

8. A 28-year-old woman testified in court that she did not give consent during an alleged sexual assault. The case involves an offense under Section 63 of the BNS. During cross-examination, the defense lawyer attempts to question her about her past sexual relationships. The court intervenes and prevents this line of questioning. Which of the following provisions are relevant to this case?

- A. BSA Section 48
- B. BNS Section 48
- C. BNSS Section 48
- D. None of the above

9. During a news report, a journalist accidentally reveals the name of a woman who has accused her neighbor of rape under BNS Section 63. The report goes viral, causing distress to the victim. If the journalist is charged under BNS section 72, what is the maximum punishment they could face?

- A. 1 year of imprisonment plus fine
- B. 2 years of imprisonment plus a fine
- C. 5 years of imprisonment without fine
- D. A warning from the court with fine

10. A 26-year-old man is proven guilty of committing an offense of rape that resulted in the victim being in a persistent vegetative state. He is sentenced to 20 years of imprisonment. Under which BNS section is he most likely convicted?

- A. BNS Section 65
- B. BNS Section 66
- C. BNS Section 67
- D. BNS Section 68

11. During a forensic examination, a 25-year-old woman reports that she was raped by a colleague at a workplace event. The medical examination reveals no physical injuries but confirms non-consensual sexual intercourse. According to the Bharatiya Nyaya Sanhita, what is the minimum punishment the accused would face?

- A. 5 years of rigorous imprisonment
 - B. 7 years of rigorous imprisonment
 - C. 10 years of rigorous imprisonment
 - D. Life imprisonment
-

12. Under Section 63 of the Bharatiya Nyaya Sanhita (BNS), which of the following scenarios does NOT constitute rape?

- A. A man engages in sexual intercourse with a woman by falsely promising marriage, knowing he does not intend to marry her.
 - B. A husband forces his wife, who is above 18 years of age, into sexual intercourse without her consent.
 - C. A man engages in consensual sexual intercourse with a woman who is 17 years old.
 - D. A man has consensual sexual intercourse with a woman who is mentally incapacitated and unable to provide informed consent.
-

13. Which of the following best represents an application of Locard's Exchange Principle in forensic investigations?

- A. The use of DNA analysis to determine genetic relationships
- B. The transfer of sweat and residue forming latent fingerprints on surfaces
- C. The identification of drug metabolites in urine samples

D. The analysis of blood pressure to assess cause of death

14. Which of the following statements regarding IPC 377 is incorrect?

- A. Pederasty involves an adult engaging in sexual activity with a child
 - B. Butch is typically the active agent
 - C. Minimum punishment for offenses is 10 years of imprisonment
 - D. Sapphism involves sexual relationships with a catamite
-

15. A 30-year-old man approaches a lawyer seeking advice after discovering that his wife has been having an extramarital affair. He expresses anger and frustration, wanting to understand his legal options regarding this situation. Which of the following statements correctly reflects the legal status of adultery in India?

- A. It's a criminal offense punishable by imprisonment under IPC
 - B. The man can file a criminal case against his wife's lover
 - C. The man can pursue a civil case for divorce
 - D. A serious offense that can lead to both criminal and civil liabilities
-

16. Forensic analysts find hair samples at a crime scene and need to determine the origin of hair. They note the following characteristics: the medulla is thick, and the scale pattern shows coronal scales. What type of hair are they most likely examining?

- A. Human hair
- B. Animal hair
- C. Synthetic hair
- D. Insect hair

17. During an autopsy, forensic experts find hair strands embedded in the victim's scalp with a swollen bulb and ruptured sheath. What does this evidence suggest about the circumstances of the victim's death?

- A. The hair fell out naturally.
 - B. The victim was forcibly restrained.
 - C. The hair was dyed before death.
 - D. The victim had a medical condition.
-

18. Which of the following statements regarding semen analysis tests is incorrect?

- A. The fluorescence test results are bluish-white.
 - B. The Acid phosphatase test is positive in aspermia
 - C. The CPK test detects fresh seminal stains
 - D. Light microscopy is confirmatory
-

19. A 28-year-old woman presents with allegations of sexual assault. A P/V swab was taken, and a test as shown below was performed. Which of the following tests was likely done?



- A. Teichmann test
- B. Barberio test
- C. Takayama test
- D. Florence test

20. A forensic laboratory conducts a Florence Test on a suspected semen stain. The test yields dark brown rhombic crystals that resemble hemin crystals, indicating a positive result. Which of the following statements is true regarding the Florence Test?

- A. It is a confirmatory test for the presence of semen.
- B. The formation of rhombic crystals is solely due to the presence of spermatozoa.
- C. False positives can occur if other substances containing choline are present.
- D. The test does not require follow-up confirmatory tests.

21. A forensic scientist is analyzing a suspected bloodstain and decides to perform the Takayama test for confirmation. Which of the following statements about the Takayama test is correct?

- A. It can only be performed on fresh blood samples.
 - B. The presence of pink, feathery crystals indicates a positive result for blood.
 - C. It provides information about the species of origin of the blood sample.
 - D. It is less sensitive than the Teichmann test.
-

22. A forensic laboratory is using the Teichmann test to confirm the presence of blood in a suspected bloodstain. Which of the following statements regarding this test is not true?

- A. The test requires the addition of sodium chloride and glacial acetic acid to the blood sample.
 - B. The presence of brownish rhombic-shaped crystals indicates a positive result for blood.
 - C. The test can reliably identify the species of origin of the blood sample.
 - D. False negatives may occur due to the age of the bloodstain or excessive heating.
-

23. Which of the following tests is used to confirm the presence of blood by detecting red, feathery crystals of pyridine ferroprotoporphyrin?

- A. Teichmann Test
 - B. HemaTrace Test
 - C. Takayama Test
 - D. Precipitin Test
-

24. A forensic investigator is analyzing a crime scene for potential blood stains. They decide to use a presumptive test that can detect trace amounts of blood but may also give false positives with other substances like bleach or rust. Which of the following is the most appropriate test for this scenario?

- A. Kastle Meyer Test
 - B. Hemastix
 - C. Luminol
 - D. Forensic Bluestar®
-

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	3
Question 4	2
Question 5	1
Question 6	3
Question 7	3
Question 8	1
Question 9	2
Question 10	2
Question 11	3
Question 12	2
Question 13	2
Question 14	4
Question 15	3
Question 16	2
Question 17	2
Question 18	3

Question 19	2
Question 20	3
Question 21	2
Question 22	3
Question 23	3
Question 24	3

Solution for Question 1:

Answer: B) To identify areas of micro-injury

Explanation:

Toluidine blue is a contrast agent used to highlight micro-injuries in the anogenital area, aiding in the forensic examination.

BNSS Section 184 - Medical examination of Victim of Rape

Step	Details
Request for Examination	Any police officer can request it within 24 hours.
Consent	Must obtain consent from the woman Examination supervised by a female RMP.
Initial Examination	Ask the victim to remove clothing. Examine and collect evidence from clothes. Look for foreign objects (hairs, seminal stains) (Option A) Search for physical injuries.
Detection of Micro Injuries	Apply toluidine-blue contrast to the anogenital area.
Swabs Collection	4 swabs: Smear: Prepare a smear for microscopic examination. Sperm: Use saline for motility testing. DNA Profiling (Option D) Chemical Tests: Acid phosphatase, creatinine phosphokinase, lactate dehydrogenase.
Examination Position	Examine in lithotomy position under good lighting.

Hymen Examination	Use Glaister-Keene rods or hymenoscope Look for tears (usually posterior at 5-7 o'clock position).
Vaginal Examination	Use a speculum for vaginal examination.
Cervical Examination	Use a colposcope for cervical examination.
Post-Examination Care	Treat STDs, injuries, or potential pregnancy. Provide psychological support.

To check for signs of infection (Option C): Infection assessment typically involves clinical evaluation

Solution for Question 2:

Answer: A) It is designed to protect female children under the age of 18 from sexual abuse.

Explanation: The POCSO Act is gender-neutral, and it protects all children, not just female children, under the age of 18 from sexual abuse.

Protection of Children from Sexual Offences (POCSO) Act

Enacted in 2012, it is law in India designed to protect children under the age of 18 from sexual abuse, sexual harassment, and pornography.

Key Features of the POCSO Act:

- Gender-neutral law (Option A)
- It covers penetrative and non-penetrative assault, sexual harassment, and use of children for pornography
- Burden of proof is on the accused, unlike many other criminal laws where the burden is on the prosecution
- Mandatory reporting by adults and failure to do so can be punishable
- Special courts for the trial of offenses to ensure a child-friendly environment
- Identity of the victim is to be protected, and the proceedings are to be conducted in camera to prevent exposure

Recent Changes and Developments in the POCSO Act:

- 2019 Amendment (Option C ruled out) Stricter Punishments: Death penalty for aggravated sexual assault against children The minimum punishment for penetrative sexual assault increased to 10 years, and for aggravated penetrative sexual assault increased to 20 years, with the possibility of life imprisonment or death in extreme cases.
- Stricter Punishments: Death penalty for aggravated sexual assault against children
- The minimum punishment for penetrative sexual assault increased to 10 years, and for aggravated penetrative sexual assault increased to 20 years, with the possibility of life imprisonment or death in extreme cases.
- Fast-track Courts to ensure swift disposal of POCSO cases, given the backlog of pending cases. (Option D ruled out)
- Clarifications on 'Touch' and Sexual Intent: The intent behind physical contact is crucial in determining whether it qualifies as an offense under POCSO.
- The intent behind physical contact is crucial in determining whether it qualifies as an offense under POCSO.
- Digital Crimes Against Children: (Option B ruled out) Provisions related to the use of children for pornographic and online exploitation purposes have been emphasized with stricter penalties.
- Provisions related to the use of children for pornographic and online exploitation purposes have been emphasized with stricter penalties.
- Stricter Punishments: Death penalty for aggravated sexual assault against children
- The minimum punishment for penetrative sexual assault increased to 10 years, and for aggravated penetrative sexual assault increased to 20 years, with the possibility of life imprisonment or death in extreme cases.
- The intent behind physical contact is crucial in determining whether it qualifies as an offense under POCSO.
- Provisions related to the use of children for pornographic and online exploitation purposes have been emphasized with stricter penalties.

Solution for Question 3:

Answer: C) Paradoxical dilatation of anus during bimanual traction of buttocks

Explanation:

- Lateral buttock traction test: Paradoxical dilatation of the anus, observed during the bimanual traction of the buttocks, is typically seen in habitual passive agents who have a history of chronic anal intercourse.
- It suggests a loss of muscle tone over time and would not be a reliable sign of a recent assault within 4 hours.

Medical Examination in Sodomy

- Active agent: Person who penetrates
- Passive agent: Person who receives it (Anal canal/ Rectal canal/ Perianal area)

Non-Habitual Passive Agent	Habitual Passive Agent
Pain/ tenderness with edema is present Fresh or dried semen in the anus or perianal area (depends on duration) (Option B ruled out) Perianal injuries Linear abrasions (Option A ruled out) Contusions Lacerations Fresh bleeding or dried blood stains may be present Fecal matter around the anus, on the perineum Prolapse of rectal mucosa (Option D ruled out) Loss of skin fold in the perianal region due to hematoma Anal muscle- dilated, irritable, and tender	No pain/ tenderness Perianal hair is shaved usually Absorption of perianal fat with funnel-shaped depression of anus Perianal skin is thickened and keratinized Dilation of the anus with loss of mucosal rugosity Prolapse of rectal mucosal rim Loss of muscle tone and does not contract readily when the skin around it is pinched Hemorrhoids Lateral buttock traction test: Paradoxical dilatation of anus during bimanual traction of buttocks (Option C)

Solution for Question 4:

Answer: B) Voyeurism

Explanation:

Voyeurism involves deriving sexual pleasure from watching others, typically while they are undressing or engaging in sexual acts.

Category	Description
Transvestism/Eonism	Wearing clothes of the opposite sex (Common in males) (Option D)
Fetishism	Sexual satisfaction from inanimate objects like bras or underpants (Common in males) (Option C)
Voyeurism/Scotophilia/ Peeping Tom	Watching someone undress or bathe (Specifically females) Mixoscopia: Sexual gratification from watching others engage in sex.

Frotteurism	Rubbing private parts against a female's buttocks or thighs (Option A)
Masturbation/Onanism/ Ipsation	Self-stimulation of genitalia.
Coprolalia	Obscene language.
Scatalogia	Obscene telephone calls.
Partialism	Sexual interest focused on a specific body part other than the genitalia.
Klismaphilia	Sexual gratification from enemas.
Ecouterism	Sexual arousal from hearing sounds associated with sexual activity.
Nymphomania	Excessive sexual desire in females.
Satyriasis	Excessive sexual desire in males.
Dippoldism	Teacher beating a student for sexual satisfaction (Named after German schoolmaster Dippold).
Gerontophilia	Sexual attraction to elderly individuals.
Pedophilia	Sexual attraction to children.
Narratophilia	Sexual arousal by reading erotic material.
Somnophilia	Sexual arousal from a stranger who is asleep.
Pyromania	Sexual satisfaction from setting fires.
Playing the Bagpipes	Intercourse under the armpits.
Troilism	Sexual activity involving three people.

Solution for Question 5:

Answer: A) Algolagnia

Explanation:

Algolagnia is a paraphilic disorder characterized by deriving sexual pleasure from inflicting pain on a partner

Category	Description
Sadism/Algolagnia	Getting satisfaction by giving pain to a partner.
Masochism/ Passive Algolagnia	Getting satisfaction by receiving pain from a partner.
Bondage	Combination of Sadism and Masochism.
Lust Murder	Murder acts as a stimulus for sexual arousal.
Urolagnia/Undinism/ Urophilia	Sexual arousal from urine smell, touch, or sight.
Uranism	Sexual gratification by fingering, fondling, licking, or sucking the genitalia of the opposite sex (Option B)
Coprophilia	Sexual arousal from fecal smell and touch.
Sexual Oralism	Fellatio: Oral stimulation of male genitalia. Cunnilingus: Oral stimulation of female genitalia (Option D)
Exhibitionism	Showing private parts. Streaking: Running naked in public. Mooning: Showing bare buttocks (Mnemonic: Moon resembles buttocks) (Option C) Flashing: Momentarily exposing body parts by quickly moving clothing (Mnemonic: like a flashlight)
Necrophilia	Sexual activity with a dead body.
Necrophagia	Eating parts of a dead body.

Solution for Question 6:

Correct Answer: C) 1, 2, 3, 4, 5

Explanation:

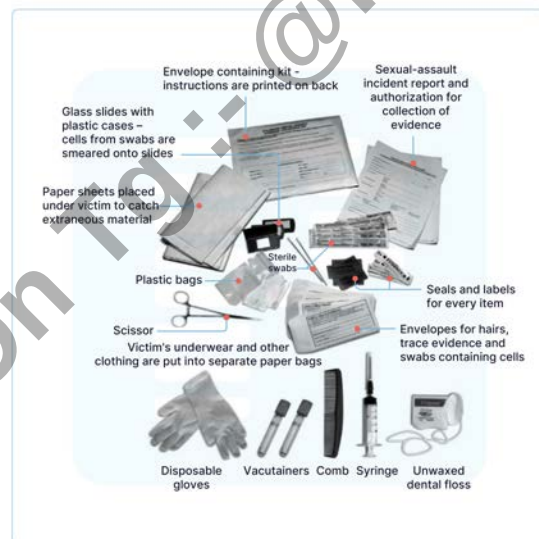
All the given images are a part of the SAFE kit

- The Safe Kit, also known as the Sexual Assault Evidence Collection Kit, Sexual Assault Forensic Evidence (SAFE) Kit, or Physical Evidence Recovery Kit (PERK)

- It is a crucial tool used by medical personnel to gather and preserve physical evidence following an allegation of sexual assault.
- The kit was developed by Louis Vitullo and is often referred to as the Vitullo Kit.

Contents of the Safe Kit

- Lips
- Cheeks
- Thighs
- Vagina
- Anus
- Buttocks
- Victim's clothes
- Head hair
- Pubic hair
- Blood samples



Solution for Question 7:

Answer: C) His consent for examination is needed

Explanation:

In cases of alleged rape, consent from the accused is not required for a medicolegal examination under Section 51 of the BNSS.

Examination of Accused

Legal Basis	Examination under S.51 BNSS Conducted with reasonable force No consent needed Request by SI
Burden of Proof	In cases of rape, the burden of proof is on males.
General Examination	Assess behavior, mental status, clothes for stains/tears, Presence of female hair.
Genital Examination	Check for semen/sperm Smegma (cheesy-like secretions) at corona glandis Presence of smegma on an accused can be a useful clue and help refute a false rape charge. if present, it is inconsistent with recent sexual intercourse. (Option A) However, the absence of smegma does not prove that rape did not occur (Can be rubbed off and show -ve results) The smegma is rubbed off during intercourse which takes about 24 hours for re-deposition.
Vaginal Cell Examination	Examine the penis and glans for vaginal cells using Papanicolaou stain and Lugol's iodine.
Penile Swab Procedure	Obtain a swab, allow it to dry, and expose it to Lugol's iodine vapors Brown indicates vaginal cells (glycogen present) (Option B) Lugol's iodine can be done till 4 days after rape (Option D)

Solution for Question 8:

Answer: A) BSA section 48

Explanation:

Bharatiya Sakshya Adhiniyam (BSA) Section 48 states that in a prosecution for an offence under section 64 to 78 of the BNS or for attempt to commit any such offence, where the question of consent is in issue, evidence of the character of the victim or of such person's previous sexual experience with any person shall not be relevant on the issue of such consent or the quality of consent.

Bharatiya Nyaya Sanhita (BNS) Section 48 (Option B) is related to abetment outside India for offence in India. Hence, it is irrelevant in this scenario.

Bharatiya Nagarik Suraksha Sanhita (BNSS) Section 48 (Option C) covers the obligation of a person making an arrest to inform about the arrest to a relative or friend, making this option irrelevant.

Solution for Question 9:

Answer: B) 2 years of imprisonment plus a fine

Explanation: Under BNS Section 72, if a journalist publishes or prints the name or any matter that may reveal the identity of a victim of rape, they can face a punishment of imprisonment that may extend up to 2 years, along with a fine.

BNS 72:

Whoever prints or publishes the name or any matter which may make known the identity of any person (victim) against whom an offence under section 64 to section 71 is alleged or found to have been committed shall be punished with imprisonment of either description for a term which may extend to two years and shall also be liable to fine. (Option B)

Exceptions

- If it is for investigation purposes
- Victim is authorizing
- Printing or publication of the judgment of any High Court or Supreme Court

Solution for Question 10:

Correct Answer: B) BNS Section 66

Explanation:

Section 66, BNS specifically addresses the punishment for rape causing the victim to be in a persistent vegetative state or resulting in death , with rigorous imprisonment for a term which shall not be less than twenty years, but which may extend to imprisonment for life, which shall mean imprisonment for the remainder of that person's natural life, or with death.

Section 65, BNS - Punishment for causing death or resulting in persistent vegetative state of victim (Option A):

Section 67, BNS - Sexual intercourse by husband upon his wife during separation (Option C):

Whoever has sexual intercourse with his own wife, who is living separately, whether under a decree of separation or otherwise, without her consent, shall be punished with imprisonment of either description for a term which shall not be less than two years but which may extend to seven years, and shall also be liable to fine.

Section 68, BNS - Sexual intercourse by a person in authority (Option D):

Whoever,

Solution for Question 11:

Correct Answer: C) 10 years of rigorous imprisonment

Explanation:

The minimum duration of punishment in a case of rape is 10 years according to Bharatiya Nyaya Sanhita (BNS), section 64.

Section 64 of BNS, punishment of rape, states that:

- 1) Whoever, except in the cases provided for in sub-section (2) commits rape, shall be punished with rigorous imprisonment of either description for a term which shall not be less than ten years , but which may extend to imprisonment for life, and shall also be liable to fine.
- 2) Whoever,

Solution for Question 12:

Correct Answer: B) A husband forces his wife, who is above 18 years of age, into sexual intercourse without her consent.

Explanation:

Section 63 of the Bharatiya Nyaya Sanhita (BNS) states that sexual intercourse or sexual acts by a man with his own wife, the wife not being under eighteen years of age, is not rape.

Rape definition and conditions

Section 63, BNS defines rape:

Exceptions:

- A medical procedure or intervention shall not constitute rape.
- Sexual intercourse or sexual acts by a man with his wife, the wife not being under eighteen years of age, is not rape. (Option D)

Note: Even if a girl gives consent below 18 years, it is known as statutory rape.

Solution for Question 13:

Answer: B) The transfer of sweat and residue forming latent fingerprints on surfaces

Explanation:

Locard's Exchange Principle involves the transfer of material when two objects come into contact, as seen with latent fingerprints formed by sweat and residue.

Locard's Exchange Principle

- States that "when two objects come into contact, material is always transferred between them."
- Crucial in forensic investigations, especially in sexual assault cases.

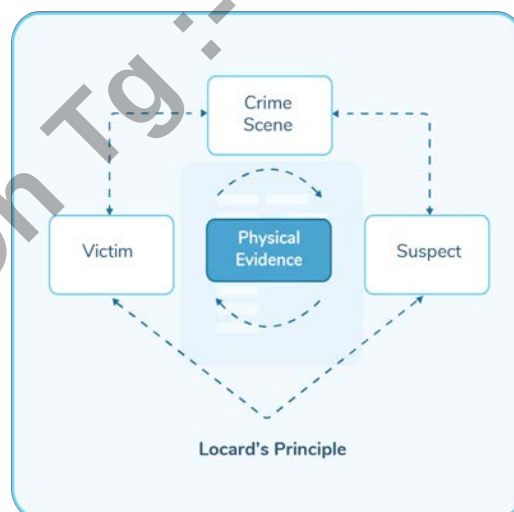
Applications:

- Fingerprint Development: Latent fingerprints form from sweat and residue transfer onto surfaces. Various techniques are used to make these prints visible.
- Latent fingerprints form from sweat and residue transfer onto surfaces.
- Various techniques are used to make these prints visible.

- Trace Evidence in Sexual Assault: Key types of evidence: fibers, hair, biological fluids. Helps establish links between victim and assailant, supporting case details.
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- Helps establish links between victim and assailant, supporting case details.
- Latent fingerprints form from sweat and residue transfer onto surfaces.
- Various techniques are used to make these prints visible.
- Key types of evidence: fibers, hair, biological fluids.
- Helps establish links between victim and assailant, supporting case details.

Timeliness of Evidence Collection:

- Recovery of trace evidence diminishes over time.
- Sperm detection: Motile sperm: up to 12 hours in the vagina. Non-motile sperm: up to 72 hours.
- Motile sperm: up to 12 hours in the vagina.
- Non-motile sperm: up to 72 hours.
- Motile sperm: up to 12 hours in the vagina.
- Non-motile sperm: up to 72 hours.



The use of DNA analysis to determine genetic relationships (Option A) refers to DNA analysis, which is not directly related to material transfer between objects.

The identification of drug metabolites in urine samples (Option C) deals with drug testing, which does not involve physical contact between objects.

The analysis of blood pressure to assess cause of death (Option D) concerns physiological assessment and is unrelated to material exchange.

Solution for Question 14:

Answer: D) Sapphism involves sexual relationships with a catamite

Explanation:

This statement is incorrect, as Sapphism specifically refers to relationships between females, not catamite.

Non-natural sexual offenses: IPC 377

Non-natural sexual offenses refer to acts of a sexual nature that deviate from traditional or consensual sexual activities as defined by law. These offenses often involve coercion, violence, or acts that violate societal norms and laws regarding sexual behavior.

Type of Offense	Description	Active Agent	Passive Agent
Anal Offenses (Anal/ Greek love/ Sodomy/ Buggery)	Pederasty: Adult-child sexual relationship. (Option A)	Pedarast (Adult)	Catamite (child)
Gerontophilia : Attraction to the elderly.	Gerontophile (Adult)	Old person	
Oral Sex (Coitus per os, Sin of Gomorrah)	Fellatio: Oral stimulation of male genitalia.	Varies	Varies
Cunnilingus: Oral stimulation of female genitalia.			
Bestiality	Sexual acts involving animals.	Humans	Lower animals
Lesbianism/Sapphism/Tribadism	Sexual relationships between females.	Butch/Dyke (Option B)	Femme

- Minimum Punishment: 10 years imprisonment (Option C)
- Consent: Punishment applies when there is no consent.
- Minors: Mandatory punishment if minors (<18 years) are involved.

Solution for Question 15:

Answer: C) The man can pursue a civil case for divorce

Explanation:

In India, adultery was decriminalized by a Supreme Court ruling in 2018, meaning it is no longer punishable as a criminal offense under the Indian Penal Code (IPC)

Natural sexual offences: Peno-vaginal route

Type of Offense	Description	Legal Status in India
Rape	Forceful sexual intercourse	Criminal offense Defined under IPC Sec 375
Incest	Sexual relationship between blood relations Complexes: Electra complex (Daughter-Father) Oedipus complex (Son-Mother) Pharaon (Brother-Sister)	Not a crime (no IPC provision)
Adultery	Extramartial affairs	Decriminalized Civil case; grounds for divorce (Option C)

Solution for Question 16:

Answer: B) Animal hair

Explanation:

This conclusion is based on two key characteristics observed in the animal hair: a thick medulla and the presence of coronal scales.

Characteristic	Animal Hair	Human Hair (Option A)
Medulla	Thick	Narrow

Cortex	Thin	Thick
Scale Pattern	Coronal scales (C for Cattle)	Imbricate scales (I for Insaan)
Texture	Coarse and thick	Fine and thin
Medullary Structure	Continuous medulla	Discontinuous or absent medulla
Pigmentation	Pigment is at the center	Pigment is at the periphery
Medullary Index (MI)	More than 0.5	0.33 or less



Solution for Question 17:

Answer: B) The victim was forcibly restrained.

Explanation:

- In this case, hair strands with a swollen bulb and ruptured sheath indicate that the hair was forcibly removed rather than falling out naturally.
- This suggests a struggle or violent interaction, where the victim may have been restrained or attacked, leading to the forcible pulling of hair.

Study of Hair: Trichology

- Significance of Hair in the detection of: Growth rate: Approximately 0.44 mm/day. DNA analysis: Can provide genetic information. Barr bodies: Indicate sex (more prevalent in females). Race identification: Certain characteristics can suggest racial background.
- Growth rate: Approximately 0.44 mm/day.
- DNA analysis: Can provide genetic information.
- Barr bodies: Indicate sex (more prevalent in females).
- Race identification: Certain characteristics can suggest racial background.
- Three Parts of Hair: Root: Anchors the hair in the follicle, and contains the bulb where growth occurs. Shaft: The visible part of the hair, made up of keratin. Tip: The end of the hair, which can show wear and damage patterns.
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- Root: Anchors the hair in the follicle, and contains the bulb where growth occurs.
- Shaft: The visible part of the hair, made up of keratin.
- Tip: The end of the hair, which can show wear and damage patterns.

Medicolegal importance

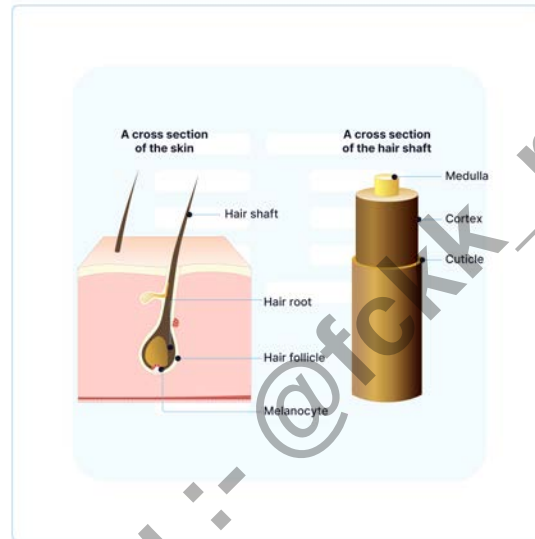
- Blunt trauma: Flattening/splitting of hair shaft.
- Sharp trauma: Clean, uniform cut surface.
- Burned cases: Swollen, fragile, curled, twisted hair with peculiar odor (Singeing of hair)
- Naturally fallen Hair : Distorted bulb, absent sheath (Option A)
- Forcibly plucked Hair : Ruptured sheath, swollen bulb, irregular shape.
- Diseases : Lighter color in kwashiorkor, malnutrition, vitamin deficiencies; fungal tunneling can produce transverse lines (Option D)
- Dyeing : Color in the cuticle, the microscopical examination can confirm (Option C)

Cross-Section of Hair

- Outermost layer: Cuticle: Composed of imbricate (overlapping) scales that protect the inner layers.
- Cuticle: Composed of imbricate (overlapping) scales that protect the inner layers.
- Middle layer: Cortex: Contains the bulk of the hair, providing strength, color, and texture.

- Cortex: Contains the bulk of the hair, providing strength, color, and texture.
- Innermost layer:
- Cuticle: Composed of imbricate (overlapping) scales that protect the inner layers.
- Cortex: Contains the bulk of the hair, providing strength, color, and texture.

Medulla: The central core of the hair, which may vary in thickness and presence, contributing to individual characteristics.



Solution for Question 18:

Answer: C) The CPK test detects fresh seminal stains

Explanation:

The CPK test detects old seminal stains (up to 6 months old), not fresh stains.

Fluorescence Test (Option A)

- Substance Detected : Seminal stain
- Result : Bluish white color under UV light

Acid Phosphatase Test (Option B)

- Substance Detected : Acid Phosphatase

- Result : Positive even in aspermia (absence of sperm)
- Normal Range : 320-360 Bodansky units

CPK (Creatinine phosphokinase)

- Substance Detected : Old seminal stains
- Mnemonic : CPK - Old (Pakahua means old in Hindi)
- Normal Range : 660 IU/ml
- Notes : Detects stains up to 6 months old.

Ammonium Molybdate Test

- Substance Detected : Phosphorus content of seminal fluid

Ag tests:

- Glycoprotein P30 test: Prostate Specific Ag (PSA)
- MHS-5: Seminal vesicle-specific Ag
- MAB-4eb: Sperm specific Ag

LDH Isoenzyme Detection

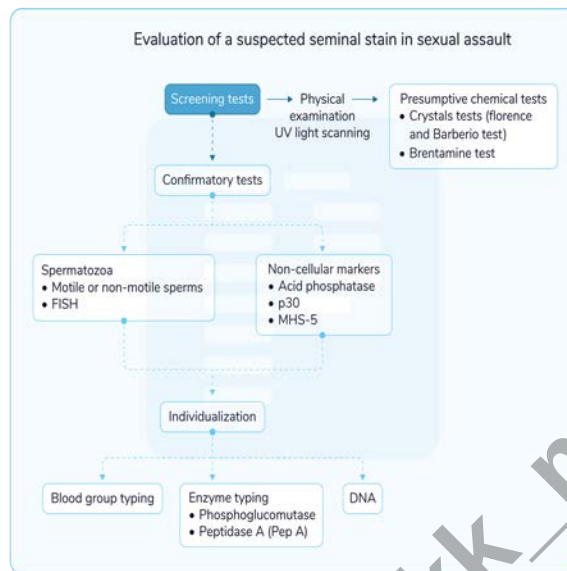
- Substance Detected : Sperm
- Appearance/Result : Absolute proof = One unbroken sperm
- Notes : Most specific test for semen.

Phadebas Test

- Substance Detected : Saliva (L-amylase)
- Notes : Detects saliva in cases of oral sex.

Confirmatory Tests (Option D)

- Gold Standard : The presence of spermatozoa under light microscopy.
- Light microscopy allows for the direct observation of sperm cells in a sample.
- The detection of spermatozoa confirms the presence of seminal fluid and is a reliable indicator.



Solution for Question 19:

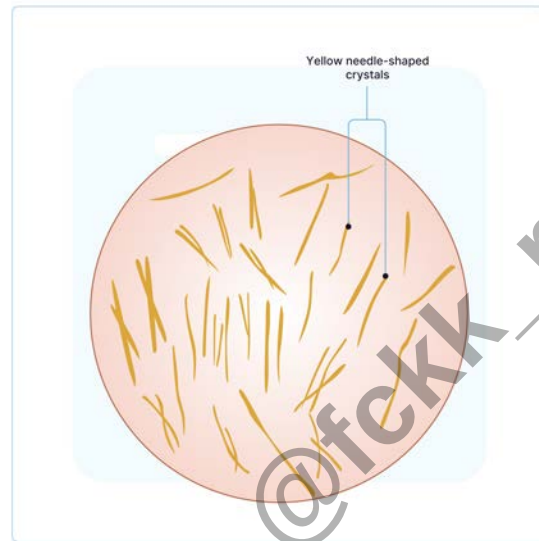
Answer: B) Barberio test

Explanation:

- The given image shows the Barberio test, indicated by the presence of yellow needle-shaped crystals.
- The Barberio test specifically detects Spermine, a compound derived from the prostate, and is used to identify seminal fluid.

Test Name	Substance Detected	Result	Reagent Used	Mnemonic	Notes
Barberio Test	Spermine	Yellow needle-shaped crystals of Spermine Picrate	Picric acid	Barbar picks your hair with a needle	Detects Spermine from the prostate; confirms the presence of seminal fluid.

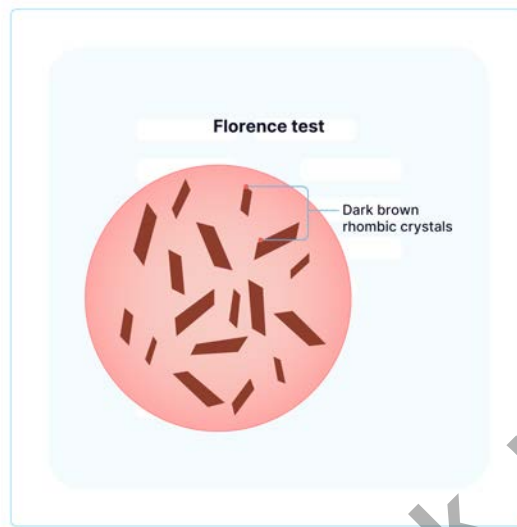
Florence Test	Choline	Dark brown rhombic crystals of Choline iodide	KI (Potassium iodide)	Florence has 'C', C for Choline	It detects choline from the seminal vesicle and is used to identify seminal fluid.
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Teichmann test (Option A): This test detects hemoglobin or blood, not seminal fluid.

Takayama test (Option C) : Similar to the Teichmann test, the Takayama test is used for hemoglobin detection.

Florence Test (Option D) : This test detects choline and produces different crystal formations (dark brown rhombic crystals).



Solution for Question 20:

Answer: C) False positives can occur if other substances containing choline are present.

Explanation:

Florence Test is a presumptive screening tool that can yield false positives if other substances containing choline, such as certain plant or animal materials, are present.

The Florence Test: Presumptive Test for Semen

The Florence Test is a preliminary screening method used to detect the presence of semen stains. It is not definitive and should be followed by confirmatory tests. (Option A ruled out)

Procedure

Step	Description
Extraction	Extract the suspected semen stain.
Drying	Dry the extract on a glass slide.
Reagent Application	Place a coverslip over the dried extract and add Florence solution (8% iodine in water with 5% potassium iodide).

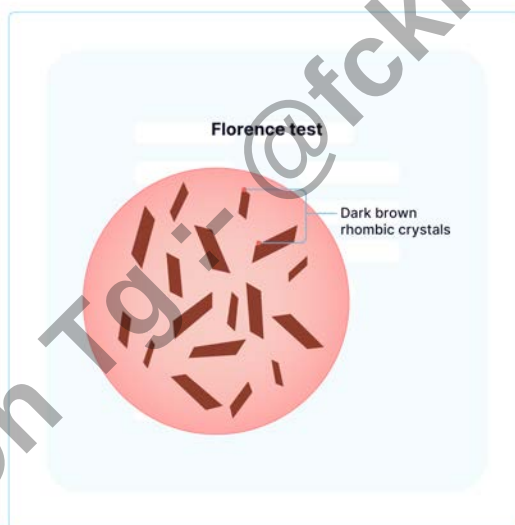
Results

Result Type	Observation
Positive Result	Formation of dark brown rhombic crystals (choline periodide) in clusters or rosettes, indicating the presence of semen. (Option B ruled out)
Negative Result	Absence of characteristic crystals, suggesting no semen is present.

Limitations

- False Positives: Other choline-containing substances (e.g., certain plants or animal matter) may yield false positive results.

False Negatives: Low choline content or decomposed stains may result in false negatives.



Importance of Confirmatory Tests (Option D ruled out)

- Requirement: A positive Florence Test necessitates follow-up with confirmatory tests, including: Microscopic examination for spermatozoa Quantitative acid phosphatase tests Detection of protein p30
- Microscopic examination for spermatozoa
- Quantitative acid phosphatase tests
- Detection of protein p30
- Microscopic examination for spermatozoa
- Quantitative acid phosphatase tests
- Detection of protein p30

Solution for Question 21:

Answer: B) The presence of pink, feathery crystals indicates a positive result for blood.

Explanation:

The presence of pink, feathery crystals formed during the Takayama test indicates a positive result for blood. This test is effective on both fresh and older bloodstains but does not provide information about the species of origin, which requires further testing.

Takayama Test

- The Takayama test (Hemochromogen crystal test) is a confirmatory blood test.
- More reliable than the Teichmann test, especially for older bloodstains. (Option D ruled out)

Test Procedure

Step	Description
Sample Preparation	Place a small bloodstain sample on a glass slide and cover with a coverslip.
Reagent Introduction	Introduce Takayama reagent (sodium hydroxide, pyridine, glucose) under the coverslip.
Heating	Briefly heat the slide.
Crystal Formation	Observe for pink, feathery crystals of reduced alkaline hematin (hemochromogen) under a microscope.

Advantages of Takayama Test

- Sensitivity: Effective on smaller blood quantities.
- Aged Stains: Performs well on older, dried bloodstains. (Option A ruled out)
- Reliability: Generally more dependable than the Teichmann test.

Limitations

- The test does not determine the species of origin (human or animal); additional tests are necessary. (Option C ruled out)

Solution for Question 22:

Answer: C) The test can reliably identify the species of origin of the blood sample.

Explanation:

The Teichmann test does not provide information about the species of origin of the blood; additional tests are needed for that.

The Teichmann Test

- Confirmatory test for blood (Hemin crystal test).
- Verifies presence of blood after a presumptive positive result.

Procedure

Step	Description
Sample Preparation	Small sample of suspected bloodstain on a glass slide.
Reagents Added	Add a few crystals of sodium chloride and drops of glacial acetic acid. (Option A ruled out)
Heating	Gently heat the slide.
Crystal Formation	Formation of brownish rhombic-shaped crystals (hemin or hematin chloride) if blood is present. (Option B ruled out)

Crystal Characteristics

- Hemin crystals may appear singly or in clusters.
- Visual representation can assist in identification (see Figure 30.4).

Limitations (Causes of False Negatives)

Condition	Effect on Test
Age of the Stain	Reliability decreases with age. (Option D ruled out)
Washing or Chemical Treatment	May prevent positive results.
Excess Sodium Chloride	Interferes with crystal formation.
Moisture in Glacial Acetic Acid	Can lead to negative results.
Overheating	Degrades blood components. (Option D ruled out)

Solution for Question 23:

Answer: C) Takayama Test

Explanation:

The Takayama test confirms the presence of blood by detecting red, feathery crystals of pyridine ferroprotoporphyrin formed by the reaction of hemoglobin with pyridine.

Confirmatory Tests for Blood Stains in Forensic Medicine

- Confirmatory tests are critical for identifying blood presence and origin in forensic investigations.
- Methods include chemical, immunological, and mass spectrometry techniques for reliable results.

Chemical Confirmatory Tests

Test	Description
Takayama Test	Detects hemoglobin by forming red, feathery crystals of pyridine ferroprotoporphyrin.
Teichmann Test (Option A ruled out)	Converts hemoglobin to hemin, creating brownish-yellow rhomboid crystals, confirming blood presence.

Immunological Tests

Test	Description
Precipitin Test (Option D ruled out)	Identifies species origin through antibody-antigen reactions, using antisera specific to human blood.
HemaTrace Test (Option B ruled out)	Uses monoclonal antibodies to detect human hemoglobin, showing a visible pink band for confirmation.

Mass Spectrometry Techniques

- Thread Spray Mass Spectrometry : Analyzes whole blood directly on textiles, differentiating species based on lipid composition and hemoglobin chains. It offers high sensitivity and specificity.

Solution for Question 24:

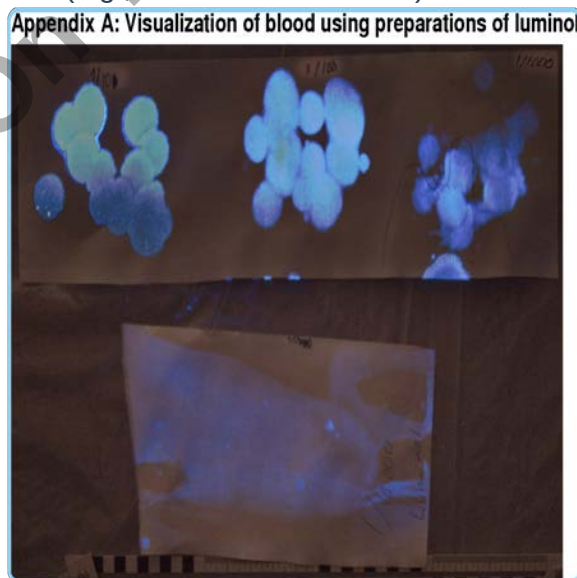
Explanation:

Luminol is widely used for detecting trace amounts of blood. It reacts with hemoglobin and produces a chemiluminescent glow in the dark. However, it can yield false positives with substances like bleach or metals.

Blood Stain Screening Tests in Forensic Medicine

Presumptive Tests

- Luminol : Detects trace blood stains by reacting with hemoglobin. Produces a chemiluminescent glow in the dark. Prone to false positives (e.g., with metals or bleach).
- Detects trace blood stains by reacting with hemoglobin.
- Produces a chemiluminescent glow in the dark.
- Prone to false positives (e.g., with metals or bleach).
- Detects trace blood stains by reacting with hemoglobin.
- Produces a chemiluminescent glow in the dark.
- Prone to false positives (e.g., with metals or bleach).



- Forensic Bluestar® : (Option D ruled out) Offers rapid results and can be used for DNA analysis afterward. Similar to Luminol but also has a risk of false positives.

- Offers rapid results and can be used for DNA analysis afterward.
- Similar to Luminol but also has a risk of false positives.
- Kastle Meyer Test : (Option A ruled out) High specificity but lower sensitivity. Detects blood at 1:16,384 dilution.
- High specificity but lower sensitivity.
- Detects blood at 1:16,384 dilution.
- Hemastix : (Option B ruled out) More sensitive than Kastle Meyer (detects blood at 1:64 dilution). Less specific, with false positives from substances like tomatoes or rust.
- More sensitive than Kastle Meyer (detects blood at 1:64 dilution).
- Less specific, with false positives from substances like tomatoes or rust.
- Offers rapid results and can be used for DNA analysis afterward.
- Similar to Luminol but also has a risk of false positives.
- High specificity but lower sensitivity.
- Detects blood at 1:16,384 dilution.
- More sensitive than Kastle Meyer (detects blood at 1:64 dilution).
- Less specific, with false positives from substances like tomatoes or rust.

Principles of Blood Detection Tests:

Test	Working Principle
Kastle-Meyer	Hemoglobin catalyzes the oxidation of phenolphthalein, resulting in a pink color.
Hemastix	Reagent reacts with hemoglobin, causing a color change based on the concentration of hemoglobin.
Luminol	Chemiluminescent reaction occurs when luminol reacts with hemoglobin under alkaline conditions, producing a blue glow.
Forensic Bluestar®	Similar to luminol, but with enhanced sensitivity; it reacts with hemoglobin to produce a bright blue luminescence.

Sexual Jurisprudence

1. A 30-year-old woman comes to the clinic with her newborn baby. She explains that her husband passed away six months ago, shortly after she became pregnant. The baby was born healthy, and she is excited to introduce him to the world. What type of child is this?

- A. Atavism
- B. Illegitimate child
- C. Supposititious child
- D. Posthumous child

2. A 32 y/o woman and her partner are having difficulty conceiving due to poor sperm motility. The doctor recommends artificial insemination. Which of the following methods would likely offer the highest success rate for this couple?

- A. Intracervical insemination (ICI)
- B. Intrauterine insemination (IUI)
- C. Intratubal insemination
- D. Intrauterine tuboperitoneal insemination (IUTPI)

3. According to the Hindu Marriage Act, of 1955, which of the following is considered a ground for a void marriage?

- A. Impotence
- B. Bigamy
- C. Unsoundness of mind
- D. Consent obtained by force

4. Which of the following is not considered a ground for divorce under the Hindu Marriage Act, of 1955?

- A. Change of religion
 - B. Cruelty
 - C. Financial disputes
 - D. Virulent leprosy
-

5. A 32-year-old woman arrives at the hospital in labour. During delivery, it is noted that one of her twins is white, while the other is black whereas both the parents are of Caucasian descent. The mother recalls having had intercourse with two different partners within a short timeframe during her ovulation. What phenomenon best explains this situation?

- A. Superfetation
 - B. Superfecundation
 - C. Lithopedion
 - D. Fetus Papyraceus
-

6. A couple is considering artificial insemination with donor sperm (AID). What legal issue is most likely to arise if the procedure is performed without the husband's consent?

- A. Genetic defects in the child
- B. Legitimacy and paternity disputes
- C. Transmission of infectious diseases
- D. Increased risk of psychological issues in the child

7. Which term refers to the inability to initiate or maintain the sexual arousal pattern in females or an incapacity to achieve orgasm?

- A. Impotence
 - B. Erectile Dysfunction
 - C. Quod
 - D. Frigidity
-

8. Which of the following features distinguishes a fimbriated hymen from a torn hymen due to trauma?

- A. Asymmetrical notches
 - B. Posterior location of notches
 - C. Mucous membrane covering intact notches
 - D. Extension of notches to the vaginal wall
-

9. A 24 y/o woman is anxious about societal perceptions of her virginity after her first sexual experience. Which of the following statements is true regarding virginity and defloration?

- A. Virginity can be medically assessed by examining the hymen.
 - B. Defloration is solely marked by the hymen's rupture during first intercourse.
 - C. Virginity is a social construct that varies across cultures.
 - D. False virgins always present with small and firm breasts.
-

10. A medical student is studying the anatomical variations of the hymen. Which of the following statements regarding hymen types and their implications is true?

- A. A single central opening characterises a septate hymen.
 - B. The cribriform hymen features multiple small openings in the membrane.
 - C. An infantile hymen typically has multiple small openings.
 - D. The annular hymen typically have two side-by-side openings.
-

11. Which of the following is not a feature of the Indian Council of Medical Research (ICMR) guidelines on surrogacy?

- A. The surrogate mother must be under 45 years of age.
 - B. The ART clinic is allowed to advertise for surrogate mothers.
 - C. Surrogacy should only be considered for those physically or medically unable to carry a child.
 - D. A woman can be a surrogate mother up to three times in her lifetime.
-

12. A woman presents with amenorrhea, breast tenderness, and morning sickness. Which category of pregnancy signs do these symptoms belong to?

- A. Presumptive
 - B. Probable
 - C. Positive
 - D. Diagnostic
-

13. Which of the following is classified as an induced abortion performed to protect the mother's life?

- A. Criminal Abortion
 - B. Legal Abortion
 - C. Spontaneous Abortion
 - D. Septic Abortion
-

14. Which of the following is an indication for pregnancy termination under the eugenic grounds of the MTP Act?

- A. Cardiac disease
 - B. Intractable hyperemesis gravidarum
 - C. Exposure to teratogenic drugs
 - D. Rape
-

15. Which section of the BNS involves life imprisonment for causing a miscarriage without the woman's consent?

- A. 88
 - B. 89
 - C. 90
 - D. 91
-

16. Which of the following drugs is known for being an emmenagogue used in criminal abortions?

- A. Lead

- B. Quinine
 - C. Savin
 - D. Cantharides
-

17. Which of the following methods of violence is most likely to directly injure the uterus and cause membrane rupture?

- A. Inserting foreign objects like knitting needles
 - B. Strenuous activities like horse riding
 - C. Using pastes with toxic ingredients
 - D. Injecting fluids into the uterus with a syringe
-

18. A 10-year-old boy exhibits extreme jealousy toward his mother's close relationship with her new partner, showing signs of distress when she pays attention to anyone else. What psychological complex might this behaviour suggest?

- A. Pharaoh Complex
 - B. Electra Complex
 - C. Oedipus Complex
 - D. Narcissistic Complex
-

19. A 28-year-old female presents to a clinic with a sore throat and discomfort following a recent sexual encounter. During the examination, the clinician notes abrasions on her mouth and lips. Which sexual act is most likely associated with the injuries observed in this patient?

- A. Sodomy

- B. Sin of Gomorrah
 - C. Tribadism
 - D. Bestiality
-

20. Which of the following statements best describes the role of the Sexual Assault Forensic Examination (rape kit) concerning victim care?

- A. The examination focuses solely on collecting evidence and does not provide medical treatment.
 - B. The examination can help victims access immediate medical care and provide resources for their health needs.
 - C. The examination requires all victims to file a police report and then provides resources for their health needs.
 - D. The examination is primarily for legal purposes and does not address the psychological well-being of the victim.
-

21. What is the expected result of a Lateral Traction Test in an individual who is not accustomed to anal penetration?

- A. Relaxation and dilation of the anal sphincter up to 4-5 cm
 - B. Reflex constriction of the anal sphincter
 - C. No response in the anal sphincter
 - D. Complete prolapse of the rectum
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	2
Question 4	3
Question 5	2
Question 6	2
Question 7	4
Question 8	3
Question 9	3
Question 10	2
Question 11	2
Question 12	1
Question 13	2
Question 14	3
Question 15	2
Question 16	3
Question 17	1
Question 18	3

Question 19	2
Question 20	2
Question 21	2

Solution for Question 1:

Correct Answer: D) Posthumous child

Explanation:

A posthumous child refers to a child born after the father has died. The baby in this scenario fits this definition perfectly.

Atavism (Option A) : This term denotes the reappearance of a characteristic in an individual after several generations of absence, often due to genetic recombination. A child may resemble their grandparents rather than their parents, highlighting the complex nature of heredity. It does not relate to the status of the child's birth, so this option is incorrect.

Illegitimate child (Option B) :

- Also known as a bastard this describes a child born out of lawful wedlock or not within a competent time after the dissolution of marriage.
- In this scenario, the child is born to a woman who was married, making this option incorrect.
- Certain conditions can render a child illegitimate, including if the alleged father is under the age of puberty, physically incapable of fathering children, not sexually accessible to the mother at the time of conception, or has blood group incompatibility.
- Legitimate Child : A child born during a lawful marriage or within 280 days after the marriage has been dissolved by divorce or death of the husband, as long as the mother remains unmarried.

Supposititious child (Option C) : This term describes a fictitious child where a woman may pretend to be pregnant and later present a living child as her own. This can involve substituting one child for another or claiming a child after an abortion, typically for financial gain or to claim property. In this case, there is no indication of deception or substitution, so this option does not apply.

Solution for Question 2:

Correct Answer: B) Intrauterine insemination (IUI)

Explanation:

This method involves injecting sperm directly into the uterus, which bypasses the cervix and can improve the chances of conception, particularly in cases of poor sperm motility.

Artificial Insemination:

Artificial insemination is a fertility treatment designed to help individuals and couples who have difficulty conceiving naturally.

Sperm Count Cutoff for Fertility Treatments:

- IUI (Intrauterine Insemination): Recommended when the sperm count is between 5-10 million per mL .
- ICSI (Intracytoplasmic Sperm Injection): Recommended when sperm count is between 1-5 million per mL .

Artificial Insemination Methods			
Type	Method	Advantages	Limitations
IUI (Intrauterine Insemination) (Option B)	Sperm is injected into the uterus.	High success, minimal pain	-
ICI (Intracervical Insemination) (Option A ruled out)	Semen is placed in the cervix, which then travels to the uterus and fallopian tubes on its own.	Low cost, suitable for various couples	Lower success rate Not available option for women with thick cervical mucous.
IUTPI (Intrauterine Tuboperitoneal Insemination) (Option D ruled out)	Sperm injected into Fallopian tubes	Bypasses infertility issues	-
Intratubal Insemination (Option C ruled out)	In this procedure, a catheter goes through the cervix and uterus and sperm is injected only into the Fallopian tubes	Effective for cervical mucus issues	More invasive

Solution for Question 3:

Correct Answer: B) Bigamy

Explanation:

Bigamy (entering into another marriage without dissolving the first) is classified as a void marriage, null from the time of inception.

The Hindu Marriage Act, 1955, outlines the legal framework for Hindu marriages in India, addressing the

- Grounds for void and voidable marriages
- Grounds for divorce

Grounds for Void and Voidable Marriages:

Void Marriage	Voidable Marriage
Null from the time of inception.	Remains valid until annulled by the court
Bigamy (entering into another marriage without dissolution of the first).	Impotence (Option A ruled out)
Prohibited degree of relationship (blood relations), unless permitted by custom.	Unsoundness of mind of either party at the time of marriage (Option C ruled out)
Sapinda relationship (relationship extending to 3rd generation in the line of ascent through the mother and 5th generation through the father)	Consent obtained by force, fraud, or misconception of facts (Option D ruled out)
	Pregnancy of the woman by another person, unknown to the husband at the time of marriage

Solution for Question 4:

Correct Answer: C) Financial disputes

Explanation:

Financial disputes are not listed as grounds for divorce under the Hindu Marriage Act of 1955.

Grounds for Divorce (Hindu Marriage Act, 1955)

- Adultery : Voluntary sexual intercourse with any person other than his/her spouse.
- Cruelty : Willful and unjustifiable conduct to cause danger to life, limb, or health of another (including mental health). (Option B ruled out)
- Desertion : Abandonment of one spouse without reasonable cause and without consent or against the wish of the other.
- Apostasy : Change of religion. (Option A ruled out)
- Unsoundness of mind .
- Virulent leprosy and sexually transmitted diseases including AIDS. (Option D ruled out)
- Renouncing the world .
- Additional grounds for women : Husband convicted of rape, sodomy, or bestiality.

Solution for Question 5:

Correct Answer: B) Superfecundation

Explanation:

Superfecundation describes the fertilization of two ova released during the same ovulation cycle, resulting from separate acts of intercourse with different partners. This can lead to fraternal twins who are half-siblings, which explains the observed differences in their appearances at birth.

Superfecundation	Superfetation (Option A ruled out)
This refers to the fertilization of two ova released from the ovary during the same ovulation cycle , occurring through two separate acts of intercourse at short intervals. The term also describes cases where two different males father fraternal twins, known as heteropaternal superfecundation . This can result in twins who are half-siblings, exemplified by one baby being white and the other black. Medico-legal Aspect: Significant differences in the appearance of the two babies may lead to suspicions of adultery or infidelity.	This occurs when two ova are fertilized at different times during the ovulation cycle , specifically involving the fertilization of a second ovum in a pregnant woman. In this case, one fetus tends to be more developed than the other. They may be born simultaneously, showing different levels of maturation, or at different times, which can range from 1 to 3 months apart. The likelihood of superfetation is higher in women with a septate or double uterus.

Lithopedion (Option C) : Known as a "stone baby," this condition occurs when a fetus dies during an ectopic abdominal pregnancy and becomes calcified. It is not relevant to the situation of twin

pregnancies.

Abdominal Pregnancy with Five Years of Evolution (Lithopedion): A Case Report. Open Journal of Obstetrics and Gynecology. 06. 515-519. 10.4236/ojog.2016.68065. " data-hash="" data-license="CC BY 4.0" data-source="https://www.researchgate.net/figure/Calcified-abdominal-pregnancy-lithopedion-after-dissection-of-uncalcified-ovular-membrane_fig3_305713563" data-tags="" height="1701" src="https://image.prepladder.com/content/ynHsCxT8gKmHpEW2P01h1736159345.png" width="2396" />

Fetus Papyraceus (Option D) : This term refers to a condition in twin pregnancies where one fetus may die and become flattened and mummified due to pressure from the other. It does not apply to the case described, as it does not involve two distinct fetuses from different fathers.



Intraoperative photograph showing papyraceous fetus and atrophic placenta.

Solution for Question 6:

Correct Answer: B) Legitimacy and paternity disputes

Explanation:

Performing artificial insemination with donor sperm (AID) without the husband's consent could result in legal problems regarding the child's paternity, raising questions about the child's

legitimacy and the father's legal rights.

Precautions and Legal Problems in Artificial Insemination:

- **Informed Consent** : Informed consent is essential for medical procedures. In artificial insemination (AI), obtaining consent from both the woman and her partner (if applicable) is crucial. This should include details about the sperm source, associated risks, and legal implications.
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- **Donor Screening** : Sperm donors should undergo thorough screening for genetic and infectious diseases to reduce the risk of transmitting health issues to the child and to prevent potential legal disputes over genetic defects.
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- **Confidentiality** : Maintaining confidentiality is vital in AI, especially for donor insemination (AID), where the donor's identity and the details of the procedure should be protected to avoid legal and ethical conflicts.
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- Informed consent is essential for medical procedures. In artificial insemination (AI), obtaining consent from both the woman and her partner (if applicable) is crucial. This should include details about the sperm source, associated risks, and legal implications.
- Sperm donors should undergo thorough screening for genetic and infectious diseases to reduce the risk of transmitting health issues to the child and to prevent potential legal disputes over genetic defects.
- Maintaining confidentiality is vital in AI, especially for donor insemination (AID), where the donor's identity and the details of the procedure should be protected to avoid legal and ethical conflicts.

Medico-legal Considerations in AI:

- **Litigation Risk** : Doctors may face lawsuits if a defective child is born. Screening the donor for genetic defects can help prevent this.
- **Marriage Nullity & Divorce** : AI for sterility is not grounds for divorce, but AI due to impotence is. If AID (Artificial Insemination by Donor) is done without the husband's consent, it is considered cruelty and can lead to divorce and legal action.

- Legitimacy : A child born via AIH (Artificial Insemination by Husband) is legitimate. But via AID is considered illegitimate and cannot inherit property.
- Adultery : No adultery is committed since there is no physical coitus involved. Indian law also states that women cannot be punished for adultery.
- Incest Risk : There is a potential risk of incest between the offspring of AI and the donor's children.
- Unmarried Women/Widows : Legally, there is no bar for unmarried or widowed women to undergo AID. The child would be legitimate. If a deceased husband's sperm is used, the child is still legitimate.

Genetic defects in the child (Option A): While a potential concern, this is not a primary legal issue stemming from the lack of consent.

Transmission of infectious diseases (Option C): This is related to the screening process of donor sperm, not a legal issue related to consent.

Increased risk of psychological issues in the child (Option D) : This could be a concern but is not a primary legal issue arising from a lack of spousal consent.

Solution for Question 7:

Correct Answer: D) Frigidity

Explanation:

Frigidity describes the inability to initiate or maintain the sexual arousal pattern in females, characterised by a lack of desire for sexual intercourse or the inability to achieve orgasm.

Causes of Frigidity:

- Vaginismus- Spasmodic contraction of the vagina due to increased sensitivity in the region. It is due to a definite "cramp-like spasm" of the adductor muscles instead of a rhythmic contraction.
- Neurological disorders
- Hypothyroidism
- Use of sedatives or depressants
- Menopause

Impotence (Option A) : This refers to the inability of an individual to engage in sexual intercourse and attain satisfaction, meaning they cannot copulate.

Causes of Male Impotence:

- Psychological: Most common cause of impotence. Anxiety, depression, severe stress, fear and tension
- Anxiety, depression, severe stress, fear and tension
- Vasculogenic: The most common organic cause Cause: Hypertension (decreased blood flow).
- Cause: Hypertension (decreased blood flow).
- Honeymoon Impotence: It is temporary and is caused due to anxiety.
- Other causes of Impotence Neurogenic Congenital Acquired causes like hydrocele.
- Neurogenic
- Congenital
- Acquired causes like hydrocele.
- Anxiety, depression, severe stress, fear and tension
- Cause: Hypertension (decreased blood flow).
- Neurogenic
- Congenital
- Acquired causes like hydrocele.

Erectile dysfunction (Option B) : This is the inability to develop and sustain an erection that is adequate for satisfactory sexual intercourse, occurring without the presence of an ejaculatory issue like premature ejaculation. It is usually a term used for males.

Quod (Option C) also known as impotence quode hanc is when a male might experience impotence with one specific female but not with another.

Solution for Question 8:

Correct answer: C) Mucous membrane covering intact notches

Explanation:

Hymen Tear	Fimbriated Hymen
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Often, it results from sexual intercourse or trauma. It can be associated with injuries to surrounding areas, such as the perineum and labia. Usually occurs at the lateral edges of the labia and extends into the vaginal wall. Traumatic tears may not have mucous membrane covering and can be located posteriorly. (Option B ruled out)

Characterised by a wavy or irregular border with symmetrical notches. (Option A ruled out) Notches are typically found anteriorly and do not penetrate the vaginal wall. (Option D ruled out) The mucous membrane remains intact around natural notches. It requires careful examination to differentiate it from traumatic tears.

Hymen: The hymen is a thin mucous membrane at the vaginal opening, which changes in size during childhood and puberty.

Hymen Rupture Causes:

Most commonly occurs due to sexual intercourse.

Other causes include:

- Accidental injuries (e.g., falls).
- Masturbation with large objects.
- Medical examinations and procedures.
- Insertion of foreign objects.
- Infections (e.g., diphtheria, fungal infections).
- Poor hygiene leads to scratching.
- Use of sanitary tampons.

Solution for Question 9:

Correct Answer: C) Virginity is a social construct that varies across cultures.

Explanation:

Virginity is primarily a social construct influenced by cultural values, rather than a medical concept with objective criteria. While misconceptions about the hymen persist, the notion of virginity varies significantly across cultures.

Definitions:

- Virginity : A social, cultural, and religious construct. (Option C) Refers to the state of never having engaged in sexual intercourse. Lacks a medical or scientific definition, highlighting its subjective nature across different societies.
- A social, cultural, and religious construct. (Option C)

- Refers to the state of never having engaged in sexual intercourse.
- Lacks a medical or scientific definition, highlighting its subjective nature across different societies.
- Defloration : Refers specifically to the rupture of the hymen during the first vaginal intercourse. It is increasingly recognized that hymen condition does not reliably indicate sexual activity. (Option B ruled out)
- Refers specifically to the rupture of the hymen during the first vaginal intercourse.
- It is increasingly recognized that hymen condition does not reliably indicate sexual activity. (Option B ruled out)
- A social, cultural, and religious construct. (Option C)
- Refers to the state of never having engaged in sexual intercourse.
- Lacks a medical or scientific definition, highlighting its subjective nature across different societies.
- Refers specifically to the rupture of the hymen during the first vaginal intercourse.
- It is increasingly recognized that hymen condition does not reliably indicate sexual activity. (Option B ruled out)

Hymen and Virginity:

- Intact hymens are inaccurately linked to virginity.
- Factors other than intercourse can rupture the hymen, including- accidents, medical exams or insertion of objects into the vagina.
- Hymen's appearance varies. It is usually thin. It could also be tough, fleshy and cartilaginous.
- Variability and potential for healed injuries make hymen examination an unreliable indicator of sexual history. (Option A ruled out)

Category	True Virgin	False Virgin
Genital Signs	Intact, rigid, inelastic hymen; admits only a fingertip, causing pain. Thick, fleshy labia majora that close the vaginal orifice; small, pink labia minora covered by labia majora. Narrow vagina with prominent rugae; full finger cannot be admitted. Present fossa navicularis, intact fourchette, narrow vestibule, small clitoris, intact posterior commissure.	Intact hymen that may be loose, elastic, or thick; admits two fingers easily. Less fleshy labia majora; vaginal orifice may be visible; enlarged, pigmented labia minora. Roomy vagina with less prominent rugae; full finger can be admitted. Disappearance of fossa navicularis; torn and healed fourchette; wider vestibule; enlarged clitoris; possibly torn posterior commissure. (Option D ruled out)

Extra-Genital Signs	Small, hemispherical, firm breasts; pink areola.	Larger, pendulous, flabby breasts; pigmented areola.
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Solution for Question 10:

Correct Answer: B) The cribriform hymen features multiple small openings in the membrane.

Explanation:

The cribriform hymen is characterised by multiple openings, which is an accurate description.

Hymen Types:

The hymen, a mucous membrane fold at the vaginal opening, can have several types:

Type	Description
Annular	Centrally located opening (Option D ruled out)
Semilunar	Anteriorly located opening
Infantile	Small, linear central opening (Option C ruled out)
Septate	Two side-by-side openings (Option A ruled out)
Cribriform	Multiple openings (Option B)
Vertical	Vertical orientation of the opening
Imperforate	No opening present

Causes of Hymen Rupture:

Cause	Description
Sexual Intercourse	The most common cause of hymen rupture.
Masturbation	The use of large objects can cause a rupture.
Accidents	Falls onto sharp objects may lead to a rupture.
Medical Procedures	Gynaecological exams or surgeries can cause a rupture.
Foreign Body Insertion	Possible in cases of abuse or forced sexual activity

Solution for Question 11:

Correct Answer: B) The ART clinic is allowed to advertise for surrogate mothers.

Explanation:

ICMR guidelines prohibit ART clinics from advertising for surrogate mothers or engaging in commercial dealings for surrogacy. The couple is responsible for finding the surrogate.

Surrogate Mother:

A surrogate mother is a woman who carries a child for another couple or individual, intending to give up the baby after birth. Surrogacy can be:

- Traditional surrogacy : The surrogate is the biological mother.
- Gestational surrogacy : The surrogate carries an embryo that is not biologically hers, typically through in vitro fertilisation (IVF).
- Commercial Surrogacy: Previously common in India, commercial surrogacy involved contractual agreements with financial compensation for surrogates beyond medical and pregnancy costs but is now prohibited under the 2018 Surrogacy (Regulation) Bill.
- Altruistic Surrogacy: Supported by the 2018 bill, altruistic surrogacy allows surrogates to receive coverage only for medical and pregnancy-related expenses, emphasizing a non-monetary, ethical approach without profit.

ICMR Guidelines on Surrogacy:

- Surrogates can be known, unknown, or a relative of the couple. If a relative, she should be from the same generation as the intended mother.
- Considered only for those who are physically or medically unable to carry a pregnancy to term. (Option C ruled out)
- Genetic parents must legally adopt the child born via surrogacy.
- Payment to the surrogate mother must cover all pregnancy-related expenses, documented in an agreement.
- ART clinics cannot advertise for surrogate mothers or engage in commercial surrogacy; the couple is responsible for finding a surrogate. (Option B)
- The surrogate mother must be under 45 years of age. (Option A ruled out)

- A woman can be a surrogate no more than three times in her lifetime. (Option D ruled out)
- Clinics must guide couples through surrogacy, ensuring surrogates meet health criteria for full-term pregnancy.

Solution for Question 12:

Correct answer: A) Presumptive

Explanation:

Pregnancy Signs		
Presumptive Signs (Option A)	Probable Signs (Option B ruled out)	Positive Signs (Option C ruled out)
Amenorrhea Breast Changes: Tenderness, enlargement, darkened nipples, colostrum secretion. Morning Sickness Quickening: First fetal movements felt around 16-18 weeks. Pigmentation Changes Chadwick's Sign: Bluish colour of the vaginal mucosa by the 8th week. Increased urination Fatigue Mood/Body Changes	Abdominal Enlargement Hegar's Sign: Softening of the uterine isthmus (6-10 weeks). Goodell's Sign: Softened cervix by 6th week. Palmer's Sign: Uterine contractions felt early (4-8 weeks). Piskacek's Sign: Asymmetrical uterine enlargement. Ballotement: Fetal rebound during an exam by 4-5 months. Immunological Pregnancy Tests: Detect hCG in urine/serum.	Fetal Heart Sounds: Heard via stethoscope by 18-20 weeks. Fetal Movements: Palpable by 20-22 weeks. Ultrasound: Fetal visualisation from 4-5 weeks. Radiographic Imaging: Fetal skeleton visible from 16 weeks.

Diagnostic (Option D): This term is not typically used in pregnancy sign classification.

Solution for Question 13:

Correct answer: B) Legal Abortion

Explanation:

Abortion		
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Medical Definition	Abortion is the expulsion of an embryo or fetus weighing 500g or less, which is not capable of surviving independently. This typically corresponds to 22 weeks of gestation.	
Legal Definition	Abortion is the expulsion of the products of conception from the uterus at any period before full term.	
Types of Abortion		
Natural/ Spontaneous Abortion	Occurs without external induction and often coincides with menstrual flow. (Option C ruled out) It is most common in the first trimester of pregnancy.	
Artificial/Induced Abortion: Willful termination of pregnancy before fetal viability.	Legal/ Justifiable Abortion	Performed to protect the life or health of the mother and follows the MTP Act. (Option B) Complications increase with gestational age, with higher rates in mid-trimester abortions.
Criminal/ Illegal Abortion	Induced unlawfully, often before the third month, with a high risk of complications such as infection and uterine injury. (Option A ruled out) Complications include hemorrhagic shock, septicemia, and secondary infertility.	

Septic Abortion (Option D): It refers to a complication of abortion involving infection of the uterus and products of conception, not an intentionally induced abortion.

Solution for Question 14:

Correct answer: C) Exposure to teratogenic drugs

Explanation:

Indications for MTP	
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Therapeutic (Options A and B ruled out)	Pregnancy that endangers the woman's life or may cause severe physical/mental injury (e.g., cardiac disease, hypertension, epilepsy, intractable hyperemesis gravidarum, breast/ cervical cancer).
Eugenic (Option C)	Risk of serious physical or mental abnormalities in the fetus (e.g., exposure to teratogenic drugs, infections like rubella, structural abnormalities like anencephaly).
Socio-economic	Unplanned pregnancies in low socio-economic conditions or failure of contraception.
Humanitarian (Option D ruled out)	Pregnancy resulting from rape.

Timing for MTP:

- Up to 12 weeks: One doctor's opinion is required.
- 12 to 20 weeks: Consent of two doctors is needed.
- Beyond 20 weeks: Only allowed on therapeutic grounds to save the woman's life (one doctor's decision).
- Emergency cases: Any registered doctor can terminate if necessary to save the woman's life.

Solution for Question 15:

Correct answer: B) 89

Explanation:

Criminal Abortion - Legal Aspects		
BNS (Bharatiya Nyaya Sanhita, 2023)	IPC Section (Indian Penal Code, 1860)	Definition
88 (Option A ruled out)	312	Voluntary criminal abortion by anyone, including the pregnant woman, can lead to up to 3 years imprisonment and/or fine. If the woman is "quick with child" (fetal movement felt), the punishment can extend up to 7 years and a fine.

89 (Option B)	313	Causing a miscarriage without the woman's consent is punishable by life imprisonment or up to 10 years in prison, with a fine.
90 (Option C ruled out)	314	If a miscarriage attempt leads to the woman's death, the responsible person faces up to 10 years in prison and a fine. If done without her consent, the punishment can be life imprisonment or up to 10 years.
91 (Option D ruled out)	315	Actions to prevent a child from being born alive or causing death after birth can result in up to 10 years imprisonment, fine, or both.
92	316	Causing the death of a "quick unborn child" (after fetal movement) is considered culpable homicide, punishable by up to 10 years imprisonment and a fine.

Criminal Abortion Methods

- Upto end of 1st month: Violent exercises
- Upto end of 2nd month: Abortifacient drugs
- 3rd or 4th month: Mechanical interference

Solution for Question 16:

Correct answer: C) Savin

Explanation:

Abortifacient Drugs	
Description	Abortifacient drugs are substances used in illegal abortions, posing severe health risks to women and potentially leading to death. Dosages and administration are often handled by unqualified individuals, increasing complications.
Mechanisms of Action	

Ecbolics	Stimulate uterine contractions to expel the fetus. Examples: Ergot, Hydrastis canadensis, quinine, lead, pituitary extract - oxytocin, synthetic estrogens. (Option A and B ruled out)
Emmenagogues (Option B)	Increase menstrual flow , potentially causing abortion in high doses. Examples: Savin , borax, apiol, rue.
Irritants	Cause reflex uterine contractions through gastrointestinal or genitourinary irritation. GIT Irritants: Strong laxatives like calomel, castor oil and saline cathartics. Genitourinary Irritants: Oils like pennyroyal, tansy and potassium permanganate, which can cause severe damage.
Systemic Poisoning	Induces abortion through toxic effects on the body. Examples: Lead, copper, iron, and certain organic irritants like cantharides, and unripe papaya. (Option D ruled out)

Solution for Question 17:

Correct answer: A) Inserting foreign objects like knitting needles

Explanation:

Methods of Violence in Criminal Abortion	
General Violence	Local Violence

Abdominal Pressure : Applying force to the abdomen (kicking, kneeling, tight bandages, etc.) aiming to cause injury. Strenuous Activities: Engaging in extreme physical activities like horse riding, jumping, or falling to induce trauma. (Option B ruled out) Cupping : Using heated cups on the abdomen to create suction, possibly separating the placenta. Temperature Extremes : Alternating between hot and cold baths, aiming to shock the system.	Syringing : Injecting harmful fluids (like soap or chemicals) into the uterus, risking infection and air embolism. (Option D ruled out) Syringe Aspiration : Inserting a syringe into the cervix to aspirate the gestational sac, leading to aspiration and expulsion of contents. Membrane Rupture : Using sharp objects (knitting needles, pencils) to puncture the amniotic sac, risking haemorrhage and infection. (Option A) Abortion Stick : Inserting sticks soaked in irritants (like calotropis) to cause uterine contractions. Cervical Dilation : Forcing open the cervix with objects to cause contractions. Air Insufflation : Pumping air into the uterus to detach the foetus, risking air embolism. Electric Current : Using electrical current on the cervix to induce contractions content expulsion. Toxic Pastes : Injecting harmful pastes into the uterus, leads to tissue damage like necrosis and infection. (Option C ruled out)
Causes of Death in Criminal Abortion	
Immediate	Delayed
Vagal Inhibition Air Embolism Fat Embolism Haemorrhage Amniotic Fluid Embolism	Septicaemia Generalised Peritonitis Tetanus Local infection Toxemia Pyaemia

Solution for Question 18:

Correct answer: C) Oedipus Complex

Explanation:

The boy's jealousy towards his mother's attention to others indicates an unconscious desire for exclusivity with her, characteristic of the Oedipus complex.

Oedipus Complex	Electra Complex	Pharaoh Complex
Refers to a son's sexual desire for his mother.	Refers to a daughter's sexual desire for her father. (Option B ruled out)	Refers to a brother's sexual desire for his sister. (Option A ruled out)

- These complexes are associated with psychological issues and may be rooted in early-life trauma or unresolved conflicts.
- Individuals exhibiting these complexes may have psychopathic tendencies.

Narcissistic Complex (Option D) : While narcissism involves self-centeredness, it does not specifically relate to the jealousy exhibited in the context of family dynamics and maternal attachment. It is usually characterized by self-love which may or may not be associated with genital excitation.

Solution for Question 19:

Correct answer: B) Sin of Gomorrah

Explanation:

Sin of Gomorrah describes oral sexual acts and could lead to abrasions in the mouth and throat, making it the most likely cause of the patient's symptoms.

Unnatural sexual Acts	
Sodomy (Option A ruled out)	Definition : Anal intercourse, historically associated with the town of Sodom. Parties Involved : This can occur between males or between a male and female. Legal Aspects : Consensual sodomy is not punishable; non-consensual acts lead to punishment for the active agent. Contexts : Common in confined environments (e.g., prisons, barracks) where individuals are isolated for long periods. Evidence : Physical examination may show signs like abrasions or relaxed anal muscles in habitual cases.
Sin of Gomorrah (Buccal Coitus) (Option B)	Definition : Oral sexual act involving a male organ in another's mouth. Historical Context : Referred to as a "sin" in the town of Gomorrah. Legal Aspects : Punishable when performed without consent. Physical Evidence : Possible abrasions or teeth marks on the penis; semen in the victim's mouth may be detected up to nine hours after the act. Risks : Potential for accidental death if the penis blocks the airway.

Bestiality (Option D ruled out)	Definition : Sexual acts between humans and animals, typically involving smaller animals or pets. Common Animals : Includes livestock (e.g., calves, sheep) and household pets (e.g., dogs, cats). Physical Evidence : Animal hair or secretions on the person, and human sperm or injuries in the animal's genital area. Associated Beliefs : Sometimes linked to superstitions, like the idea of curing certain diseases through intercourse with animals.
Tribadism (Female Homosexuality) (Option C ruled out)	Definition : Sexual activity between women for mutual gratification, also known as lesbianism. Common Acts : Includes kissing, body contact, genital apposition, and use of objects for stimulation. Historical Context : Linked to the Isle of Lesbos and sometimes called "Sapphism." Social Dynamics : This can lead to intense emotional connections, and historically noted cases include jealousy and possessiveness among partners.

Solution for Question 20:

Correct answer: B) The examination can help victims access immediate medical care and provide resources for their health needs.

Explanation:

The Sexual Assault Forensic Examination is designed not only to collect evidence but also to ensure that victims receive necessary medical treatment, addressing both physical health and immediate care needs.

Sexual Assault Forensic Examination Kit (also known as a rape kit)

- Purpose : Preserves DNA evidence from the body, clothes, or personal items after a sexual assault and provides important medical care.
- No Requirement to Report : Victims can undergo an exam without reporting the crime, allowing evidence to be stored in case of later reporting. (Option C ruled out)
- Contents : The kit includes items like swabs, bags, documentation forms, and instructions for evidence collection (contents may vary by location).
- Exam Procedure : Typically includes immediate medical care, health history review, head-to-toe examination, evidence collection (e.g., DNA, blood, hair samples), and documentation of injuries.

- Follow-Up Care : Includes treatment for STIs, emergency contraception, and resources for further medical support.
- Examiners : These are conducted by trained professionals like Sexual Assault Nurse Examiners (SANEs) or Sexual Assault Forensic Examiners (SAFEs).
- Benefits : Helps in potential prosecution, connects DNA to databases for future investigations, and addresses physical health needs after an assault.

The examination focuses solely on collecting evidence and does not provide medical treatment (Option A) : This option is incorrect. While evidence collection is a key component, the Sexual Assault Forensic Examination also includes important medical care, such as treating injuries and providing preventive measures for STIs and emergency contraception.

The examination is primarily for legal purposes and does not address the psychological well-being of the victim (Option D) : Sexual Assault Forensic Examination does take into account the psychological well-being of the victim, often providing access to mental health resources and support during and after the examination.

Solution for Question 21:

Correct answer: B) Reflex constriction of the anal sphincter

Explanation:

Individuals unaccustomed to anal penetration usually exhibit reflex constriction of the anal sphincter.

Lateral Traction Test

Purpose : Used to assess the anal sphincter's response, potentially indicating habitual anal penetration.

Procedure : A thumb is placed on each side of the anus, and lateral traction is applied to observe sphincter reaction.

Expected Results :

- Reflex Constriction : Typically observed in individuals who are not accustomed to anal penetration. (Option C & D ruled out)
- Relaxation and Dilation : In habitual anal penetration, the sphincter may relax, with the anal opening dilating up to 4–5 cm, sometimes revealing the rectum (Option A ruled out)

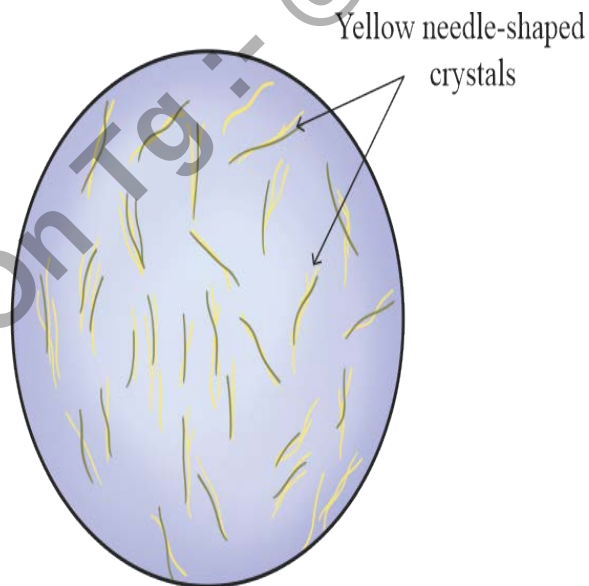
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Previous Year Questions

1. 27 weeks of gestation with congenital anomalies. whose presence is not required for authorisation of MTP?

- A. Obstetrician
- B. Lawyer
- C. Pediatrician
- D. Sonologist

2. Examination of a suspicious stain shows the following finding. The test is?



- A. Barberios test
- B. Florence test
- C. Teichman test
- D. Takayama test

3. A child was born after 8 months of father's death. Grandparents filed a case that the baby is not their son's. DNA is positive. The child is called ?

- A. Suppositious child
 - B. Fabricated child
 - C. Posthumous child
 - D. Illegitimate child
-

4. Recording of evidence of a female in sexual Assault in court is done in?

- A. Open court proceedings
 - B. Closed court proceedings
 - C. Hearing on different location
 - D. Hearing on camera
-

5. A man wears a female dress and high heels and gets pleasure by doing so. He was not attracted to the same gender. This condition is?

- A. Transvestic fetishism
 - B. Voyeurism
 - C. Sexual dysphoria
 - D. Gay orientation
-

6. Declaration of Oslo by World Medical Association in 1970 deals with?

- A. Therapeutic Abortion
 - B. Torture, degradation or cruel treatment of prisoners
 - C. Ethical aspects for Medical Researches
 - D. Hunger
-

7. Match the following sexual paraphilias:- A. Eonism B. Necrophilia C. Frotteurism D. Exhibitionism Sexual gratification by rubbing private parts against another person Exposure of one's genitals to an unsuspecting stranger Sexual gratification by intercourse with dead body Sexual pleasure by wearing clothes of the opposite sex

A. Eonism B. Necrophilia C. Frotteurism D. Exhibitionism

- A. A4, B3, C1, D2
 - B. A2, B3, C4, D1
 - C. A4, B1, C3, D2
 - D. A1, B4, C2, D3
-

8. What is the appropriate IPC section for the act of inducing abortion without the consent of the woman?

- A. 312
 - B. 313
 - C. 314
 - D. 315
-

9. All are grounds for divorce except :

- A. Adultery
 - B. Incurable Leprosy
 - C. Poverty
 - D. Insanity
-

10. What is the term for a child born when a woman pretends to be pregnant and later uses the child to extort a man?

- A. Supposititious child
 - B. Superfecundation
 - C. Posthumous child
 - D. Superfetation
-

11. Which of the following defines rape?

- A. 375 IPC
 - B. 376 IPC
 - C. 377 IPC
 - D. 498 (A) IPC
-

12. Which test is employed to distinguish human blood from blood of other species among the options provided?

- A. Teichman test
- B. Precipitin test

- C. Barberio test
 - D. Takayama test
-

13. Please describe the test depicted in the image below:



- A. Barberio test
 - B. Florence test
 - C. Takayama test
 - D. Teichmann test
-

14. Edmond Locard is most famous for?

- A. Study of fingerprints
- B. Theory of exchange
- C. Stature estimation
- D. Forensic ballistics

15. Which confirmatory test is utilized by forensic experts to establish the presence of a bloodstain at a crime scene?

- A. Spectroscopic test
 - B. Kastle meyer test
 - C. Benzidine test
 - D. Orthotoluidine test
-

16. What is the substance identified by Barberio's test, which was conducted on the stains from a sexual assault incident involving an 18-year-old female and a 20-year-old male?

- A. Spermine
 - B. Acid phosphatase
 - C. Choline
 - D. Fructose
-

17. During the forensic examination of semen from a rape victim, the identification of distinctive yellow needle-like crystals raises attention. What is the specific test employed to ascertain the presence of sperm?

- A. Barberio test
 - B. Acid phosphatase test
 - C. Papanicolaou smear
 - D. Florence test
-

18. What is the most appropriate term to describe the behavior of a man who consistently derives sexual pleasure from causing physical harm to his partner, often resulting in cuts and burns on their arms?

- A. Voyeurism
 - B. Fetishism
 - C. Sadism
 - D. Masochism
-

19. Which organization is related to the Protection of Children from Sexual Offences (POCSO) Act?

- A. Child and welfare committees
 - B. Women and child safety department
 - C. Children rights protection committee
 - D. All of the above
-

20. A registered medical practitioner is requested by a survivor of sexual violence to examine her. Within what time must the practitioner take a vaginal swab to look for the presence of spermatozoa?

- A. Within 72 hours
 - B. Within 36 hours
 - C. Within 5 days
 - D. Within 48 hours
-

21. During the visit to the outpatient department, a female patient aged 18 years was examined. The genital examination disclosed the following observations. These observations suggest: Labia majora - separated Labia minora - flabby Fourchette tear present Roomy vagina seen with intact hymen

- A. True virgin
- B. False virgin
- C. Premenstrual stage
- D. Molestation

22. As a medical officer, what actions should you take when a 14-year-old girl accompanied by her mother presents to the outpatient department (OPD) with a history of penovaginal penetration by a neighbor, with consent given by the girl? The girl expresses her refusal to undergo an examination. Document informed refusal Do not inform the police Counsel the mother and daughter Examine with necessary force Inform the police

- A. 1, 2, 3 are true
- B. 2, 3, 4 are true
- C. 1, 3, 5 are true
- D. 3, 4, 5 are true

23. Frotteurism is characterized as _____:

- A. Obtaining sexual pleasure by wearing clothes of opposite sex
- B. Desire to seek surgery to become member of opposite sex
- C. Sexual gratification by rubbing his private parts against another person
- D. Exposure of one's genitals to an unsuspecting stranger

24. A man continues to call females. Achieves sexual gratification by talking obscenity & sharing obscene pictures. The condition is?

- A. Frotteurism
 - B. Voyeurism
 - C. Scatologia
 - D. Coprophilia
-

25. What does Barberio's test detect when yellow needle-shaped rhombic crystals are observed under a microscope?

- A. Semen
 - B. Blood
 - C. Sputum
 - D. CSF
-

26. An 18 year old female was sexually assaulted by a 20 year old male. There was presence of stains on the clothes of the victim. Barberio's test was performed on the stains. Which of the following is detected by this test ?

- A. Spermine
 - B. Acid phosphatase
 - C. Choline
 - D. Fructose
-

27. According to the MTP act of 1971, what is the maximum gestational age at which medical termination of pregnancy is permitted?

- A. 28 weeks
 - B. 22 weeks
 - C. 24 weeks
 - D. 16 weeks
-

28. Which of the following techniques is considered abnormal when used for criminal abortion?

- A. Dilatation and curettage
 - B. Mifepristone
 - C. Abortion stick
 - D. Vacuum aspiration
-

29. Typically, what is the condition of the hymen in cases of sexual assault involving a young child?

- A. Ruptured since it is located superficially
 - B. Ruptured since it is very thin
 - C. Unruptured since it is deeply situated
 - D. Unruptured since it is highly elastic
-

30. What is the appropriate term to describe a person who sits naked on their balcony, deriving sexual pleasure from observing others looking at them, while overlooking a park?

- A. Voyeurism
 - B. Fetishism
 - C. Exhibitionism
 - D. Masochism
-

31. A person of eonism derives pleasure from?

- A. Wearing clothes of opposite sex
 - B. Kissing and licking of anus by a sexual partner
 - C. Underclothing of female
 - D. Seeing a female undressing
-

32. Section 314 IPC deals with?

- A. Causing miscarriage with the consent of the mother
 - B. Causing miscarriage without the consent of the mother
 - C. Death of the mother by act done with intent to cause miscarriage
 - D. Causing the death of the quick unborn child by an act amounting to culpable homicide
-

33. What test is performed when a 10-year-old boy presents to the emergency department with difficulty walking and pain around the anus, and yellow needle-shaped rhombic crystals are observed in a specimen taken from the perianal region when tested with picric acid?

- A. Barberio test
- B. Florence test
- C. Teichmann test

D. Acid phosphatase test

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	3
Question 4	2
Question 5	1
Question 6	1
Question 7	1
Question 8	2
Question 9	3
Question 10	1
Question 11	1
Question 12	2
Question 13	4
Question 14	2
Question 15	1
Question 16	1
Question 17	1
Question 18	3

Question 19	1
Question 20	1
Question 21	2
Question 22	3
Question 23	3
Question 24	3
Question 25	1
Question 26	1
Question 27	3
Question 28	3
Question 29	3
Question 30	3
Question 31	1
Question 32	3
Question 33	1

Solution for Question 1:

Correct Option B - Lawyer:

A lawyer's presence is not required for the authorization of a Medical Termination of Pregnancy (MTP) for congenital anomalies. The primary requirement is the involvement of a medical team with specific specialists. The lawyer's role typically comes into play in legal proceedings or when there are legal challenges, but not in the medical decision-making process for MTP.

Incorrect Options:

Option A - Obstetrician:

Required: An obstetrician is essential for evaluating and managing the medical aspects of the pregnancy, including assessing the viability and risks involved with the continuation or termination of the pregnancy.

Option C - Pediatrician:

Required: A pediatrician's input is crucial when congenital anomalies are involved, as they provide insight into the nature and severity of the anomalies, which helps in making informed decisions about the MTP.

Option D - Sonologist:

Required: A sonologist, who performs and interprets ultrasound scans, plays a critical role in diagnosing congenital anomalies and assessing the gestational age of the fetus. This information is vital for decision-making regarding MTP.

Solution for Question 2:

Correct Option A - Barberios test:

The Barberios test is used for the detection of semen and is characterized by the formation of yellow, needle-shaped crystals when a stain is treated with certain reagents. This test is specific for identifying seminal fluid.

Incorrect Options:

Option B - Florence test: The Florence test detects the presence of blood by forming yellow, needle-shaped crystals. However, in practice, it's often referred to as the Florence Reaction, which is not commonly used to identify semen.

Option C - Teichman test: The Teichman test is also used for blood detection but is characterized by the formation of rhomboid-shaped crystals, not needle-shaped ones.

Option D - Takayama test: The Takayama test detects blood and forms pink or reddish-brown crystals, not yellow needle-shaped crystals.

Solution for Question 3:

Correct Option C - Posthumous child:

A posthumous child is one who is born after the death of one or both parents. In this case, since the child was born after the father's death but is confirmed by DNA to be the biological child of the deceased father, the correct term is "posthumous child."

Incorrect Options:

Option A - Suppositious child: A suppositious child refers to a child who is falsely claimed to be the legitimate child of a deceased person, but this term is not commonly used in modern legal or medical contexts. It does not apply to the situation described.

Option B - Fabricated child: A fabricated child is not a standard term used in legal or medical contexts. It suggests falsification but is not relevant in the context of a child's legitimacy or paternity.

Option D - Illegitimate child: An illegitimate child is one born out of wedlock. However, since the issue in this case is related to the timing of birth relative to the father's death rather than marital status, "illegitimate child" is not the most accurate term here.

Solution for Question 4:

Correct Option B - Closed court proceedings:

In cases involving sexual assault, closed court proceedings are used to ensure the privacy and protection of the victim. These proceedings are not open to the public, which helps maintain the victim's confidentiality and reduce the emotional impact of the court process. Closed court sessions help protect the victim from public scrutiny and potential re-traumatization.

Incorrect Options:

Option A - Open court proceedings: Open court proceedings are accessible to the public and can expose the victim to unnecessary stress and scrutiny. Such settings do not provide the privacy needed for sensitive cases like sexual assault.

Option C - Hearing on different location: While a hearing might be conducted from a different location for logistical reasons (e.g., via video link), this is not specifically designed to protect the privacy of the victim in the same way that closed court proceedings do.

Option D - Hearing on camera: Although on-camera recordings may be used to record testimony, this does not replace the need for closed court proceedings. On-camera recordings are part of the procedure to ensure that evidence is preserved and presented appropriately, but they are generally conducted within the framework of closed court proceedings to ensure the privacy of the victim.

Solution for Question 5:

Correct Option A - Transvestic fetishism:

Transvestic fetishism involves wearing clothing traditionally associated with the opposite sex for sexual arousal. It is characterized by deriving sexual pleasure from cross-dressing, without necessarily having an attraction to the same gender. The person in this scenario finds pleasure in wearing female clothing and high heels but is not attracted to the same sex.

Incorrect Options:

Option B - Voyeurism: Voyeurism involves gaining sexual pleasure from watching others who are unaware that they are being watched, particularly in intimate or private situations. It is not related to cross-dressing or deriving pleasure from wearing the clothing of the opposite sex.

Option C - Sexual dysphoria: Sexual dysphoria (more commonly referred to as gender dysphoria) is the distress or discomfort that may occur when an individual's gender identity does not align with their assigned sex at birth. This condition involves a disconnect between one's experienced gender and the gender assigned at birth, not the act of cross-dressing for pleasure.

Option D - Gay orientation: Gay orientation refers to an attraction to individuals of the same gender. The scenario described does not involve sexual attraction to the same sex but rather involves deriving pleasure from wearing clothing typically associated with the opposite sex.

Solution for Question 6:

Correct Option A - Therapeutic Abortion:

- The Declaration of Oslo discusses the role of doctors in therapeutic abortion.
- It was adopted by the 24th World Medical Assembly Oslo, Norway, August 1970 and amended by the 35th World Medical Assembly Venice, Italy, October 1983

Incorrect Options:

Option B - Torture, degradation or cruel treatment of prisoners:

- Torture, degradation or cruel treatment of prisoners was discussed in the declaration of Tokyo.

Option C - Ethical aspects for Medical Researches:

- Ethical aspects for Medical Researches was discussed in the declaration of Helsinki.

Option D - Hunger:

- Role of doctors in the Hunger strike was discussed in the declaration of Malta.

Solution for Question 7:

Correct Option A - A4, B3, C1, D2:

- Eonism- Sexual pleasure by wearing clothes of the opposite sex
- Necrophilia- Sexual gratification by intercourse with dead body
- Frotteurism- Sexual gratification by rubbing private parts against another person
- Exhibitionism- Exposure of one's genitals to an unsuspecting stranger

Incorrect Options:

- Options B, C, and D: These are incorrect options

Solution for Question 8:

Correct Option B - IPC Section 313:

- In the Indian Penal Code (IPC), Section 313 deals with the act of causing miscarriage without a woman's consent.
- This section makes it a criminal offense to cause a woman to miscarry against her will. It states that anyone who causes a woman, who is with child, to miscarry without her consent can be punished with imprisonment that may extend up to ten years, along with a fine.

In the Indian Penal Code (IPC), Section 313 deals with the act of causing miscarriage without a woman's consent.

This section makes it a criminal offense to cause a woman to miscarry against her will. It states that anyone who causes a woman, who is with child, to miscarry without her consent can be punished with imprisonment that may extend up to ten years, along with a fine.

Incorrect Options:

Option A - IPC Section 312: It deals with voluntarily causing a miscarriage and carries a lesser punishment compared to Section 313.

Option C - IPC Section 314 : Whoever, with intent to cause the miscarriage of a woman with child, does any act which causes the death of such woman.

Option D - IPC Section 315 : Act done with intent to prevent child being born alive or to cause it to die after birth.

Solution for Question 9:

Correct Option C - Poverty:

- Poverty is not a ground for divorce.

Incorrect Options:

Option A - Adultery : Adultery is ground for divorce.

Option B - Incurable Leprosy : Incurable Leprosy is not a grounds for divorce.

Option D - Insanity : Insanity is a grounds for divorce.

Solution for Question 10:

Correct Option A - Supposititious child:

- Supposititious child / fictitious child/ substituted child is a fraudulent child that is said to conveniently appear before the people/ court to blackmail with intentions of property/money etc.

Incorrect Options :

Option B - Superfecundation: It is when 2 ova are fertilized in 1 menstrual cycle through 2 different acts of sexual intercourse.

Option C - Posthumous child : It is the child born after the death of the father.

Option D - Superfetation : It is when 2 ova are fertilized in 2 menstrual cycles through 2 different acts of sexual intercourse (2 Ts)

Solution for Question 11:

Correct Option A - 375 IPC:

- Gives definition of rape

Incorrect Options:

Option B - 376 IPC: Gives punishment for rape

Option C - 377 IPC: Gives punishment for unnatural sexual offenses Option D - 498 (A) IPC: Gives Punishment for causing physical/ mental cruelty of a woman by husband / relatives

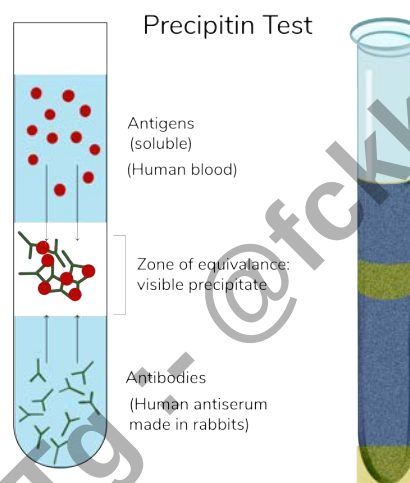
Solution for Question 12:

Correct Option B - Precipitin test:

- The precipitin test is used to differentiate human blood from the blood of other species.
- This test is based on the principle that antibodies in serum will react with and precipitate their corresponding antigens.

The precipitin test is used to differentiate human blood from the blood of other species.

This test is based on the principle that antibodies in serum will react with and precipitate their corresponding antigens.



Incorrect Options:

Option A - Teichman test: This test is used to detect the presence of hemoglobin in a blood stain.

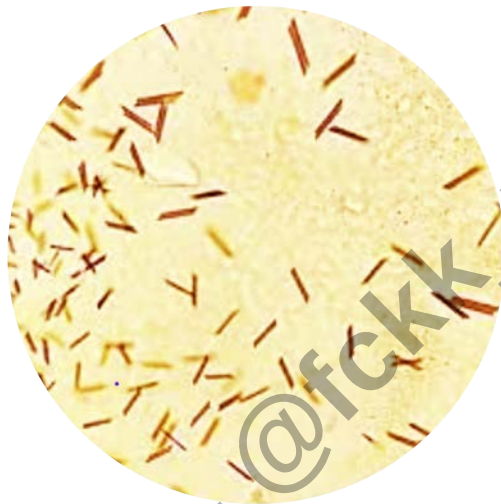
Option C - Barberio test: This test is used to detect seminal stains .

Option D - Takayama test: This test is used to detect the presence of blood in a forensic sample,

Solution for Question 13:

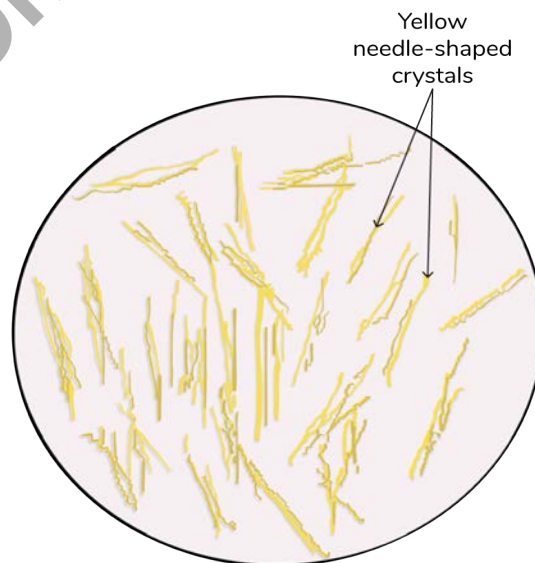
Correct Option D - Teichmann test:

- Teichmann test is one of the microchemical confirmatory test used to detect the presence of trace evidence of blood.
- In this test, the sample is made to react with NaCl and glacial acetic acid and then seen under the microscope.
- Presence of brown rhombic crystals indicate the presence of blood stains in the sample. These type of crystals are found due to the formation of hematin chloride .



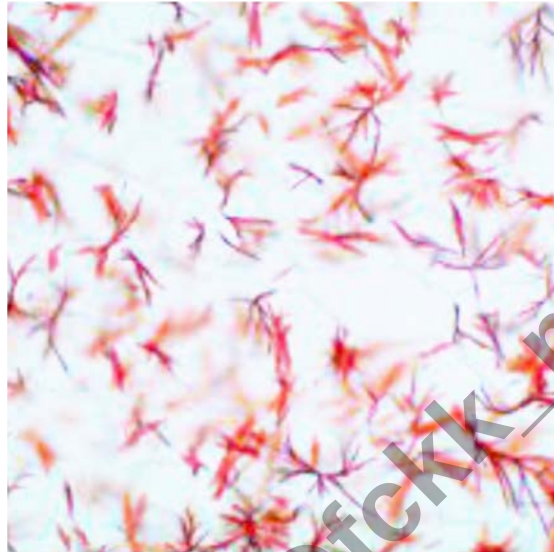
Incorrect Options:

Option A - Barberio test: It includes confirming yellow crystals that develop when semen is subjected to an aqueous solution of picric acid under a microscope.



Option B - Florence test: This test is done to detect the presence of choline.

Option C - Takayama test: A confirmatory test used to find blood stains is the Takayama test.



Solution for Question 14:

Correct Option B - Theory of exchange:

- Locard is most famous for his theory of exchange, also known as Locard's Exchange Principle.
- This principle states that whenever two objects come into contact, there will be an exchange of materials between them.

Incorrect Options:

Option A - Study of fingerprints: While Edmond Locard did contribute to the field of forensic science, he is not primarily known for the study of fingerprints.

Option C - Stature estimation: Stature estimation involves determining a person's height based on skeletal remains.

Option D - Forensic ballistics: Forensic ballistics involves the analysis of firearms, ammunition, and the examination of ballistic evidence.

Solution for Question 15:

Correct Option A - Spectroscopic test:

- When it comes to identifying a bloodstain, the confirmatory test used is a spectroscopic test (Option A). Spectroscopic tests, such as infrared spectroscopy or UV-visible spectroscopy, analyze the absorption or emission of light by the sample to identify specific compounds or substances present in the bloodstain

When it comes to identifying a bloodstain, the confirmatory test used is a spectroscopic test (Option A). Spectroscopic tests, such as infrared spectroscopy or UV-visible spectroscopy, analyze the absorption or emission of light by the sample to identify specific compounds or substances present in the bloodstain

Incorrect Options:

Option B - Kastle Meyer test: The Kastle Meyer test is a presumptive test It is not a confirmatory test.

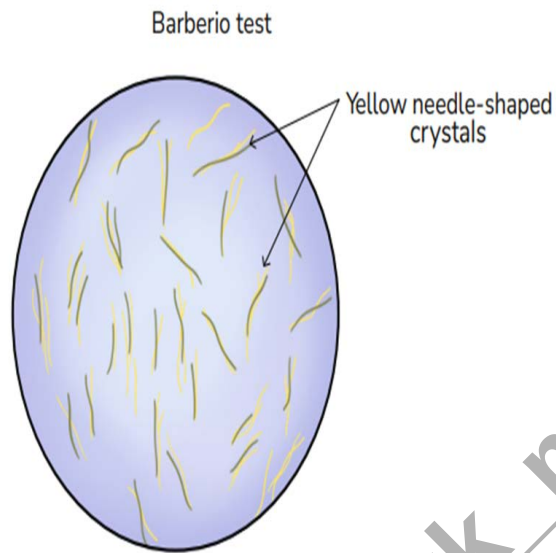
Option C - Benzidine test : The benzidine test is another presumptive test for detecting blood It is not a confirmatory test.

Option D - Orthotoluidine test : The orthotoluidine test is a colorimetric test used to detect the presence of blood. It is also a presumptive test and not a confirmatory test.

Solution for Question 16:

Correct Option A - Spermine:

- Barberio's Test is used to detect spermine crystals , using picric acid. Spermine is an important component of prostatic secretion and is found in semen. On adding picric acid to the sample, yellow needle shaped (rhombic) crystals are seen (also called spermine picrate crystals).



Incorrect Options:

Option B - Acid phosphatase: Acid phosphatase test is done in cases where old semen is suspected to be present in the crime scene. Positive even in the cases of Aspermia (absence of sperms)

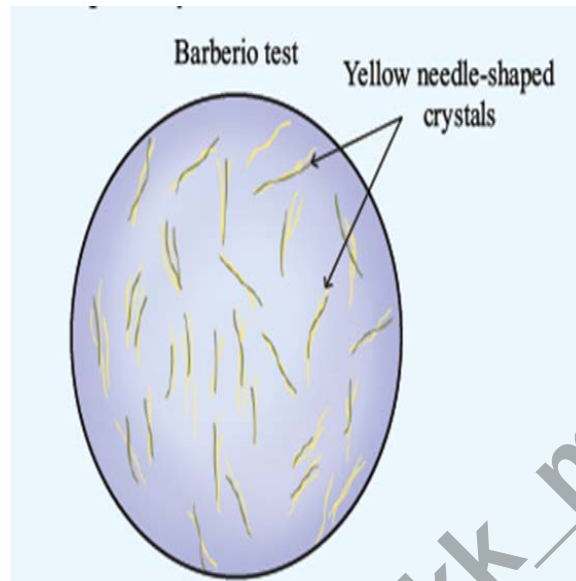
Option C - Choline: Choline crystals are detected by florence test.

Option D - Fructose: Although fructose forms an important part of semen, it is not used often in semen analysis in sexual offenses.

Solution for Question 17:

Correct Option A - Barberio test:

- Barberio test: The reagent used here is picric acid.
- Spermine from the prostate reacts with picric acid to form spermine picrate crystals, which are yellow and needle-shaped.



Incorrect Options:

Option B - Acid phosphatase test:

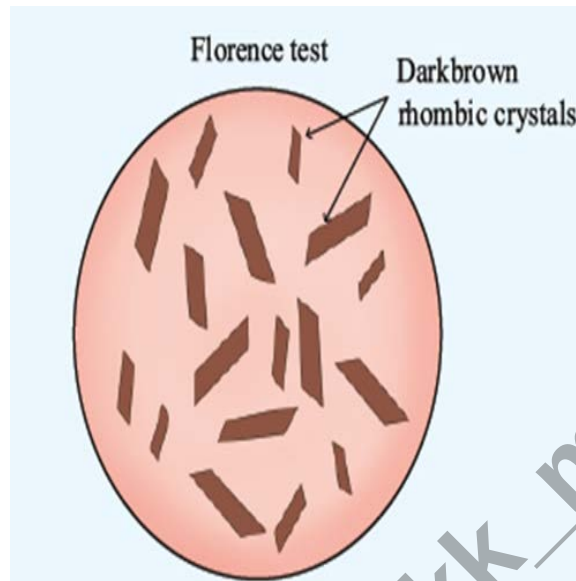
- This test can be used to detect seminal fluid, as acid phosphatase is present in high concentrations in semen. The test does not involve the formation of yellow needle-like crystals.

Option C - Papanicolaou smear:

- A Papanicolaou smear, commonly known as a Pap smear, is a screening test used to detect cervical cancer and abnormalities in cervical cells.

Option D - Florence test:

- Detects choline, which is derived from seminal vesicles.
- Reagent used: Potassium iodide (KI).
- Type of crystal formed: Choline iodide crystals, which are dark brown and rhombic in shape.



Solution for Question 18:

Correct Option C - Sadism:

- The behavior described in the scenario is the core characteristic of sadism, which is a sexual disorder that involves gaining pleasure from the experience of inflicting pain, humiliation, or suffering on others.

The behavior described in the scenario is the core characteristic of sadism, which is a sexual disorder that involves gaining pleasure from the experience of inflicting pain, humiliation, or suffering on others.

Incorrect Option:

Option A - Voyeurism: Voyeurism is the act of observing individuals who are naked or engaged in sexual activity without their knowledge or consent, for the purpose of sexual gratification.

Option B - Fetishism: Fetishism is a type of sexual disorder in which an individual is sexually aroused by non-human objects or specific body parts that are not typically considered sexually arousing, such as shoes, leather, or feet.

Option D - Masochism: Masochism is a sexual disorder in which an individual derives sexual pleasure from receiving pain or humiliation.

Solution for Question 19:

Correct Option A - Child and welfare committees:

- Child and welfare committees play an important role under POCSO act, cases registered under this act need to be reported to CWC within 24 hours of recording of the complaint.

Incorrect Options:

- Option B, C & D are incorrect. Refer to the explanation for the correct answer.
-

Solution for Question 20:

Correct Option A - Within 72 hours:

- When a survivor of sexual violence requests a medical examination, it is crucial to collect evidence promptly to increase the chances of detecting and preserving relevant forensic evidence, such as the presence of spermatozoa.
- The recommended timeframe for collecting a vaginal swab to look for the presence of spermatozoa is within 72 hours of the assault.

Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation for the correct answer.
-

Solution for Question 21:

Correct Option B - False virgin:

- The given findings, including separated labia majora, flabby labia minora, a tear in the fourchette, and a roomy vagina with an intact hymen, are indicative of a false virgin.
- A false virgin is a term used to describe an individual who has experienced sexual intercourse or other activities that have caused changes to the external genitalia, despite having an intact hymen.

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A false virgin is a term used to describe an individual who has experienced sexual intercourse or other activities that have caused changes to the external genitalia, despite having an intact hymen.

Incorrect Options:

Option A - True virgin : The findings described in the question indicate changes in the external genitalia that are consistent with previous sexual activity. Therefore, the term "true virgin" does not apply in this case.

Option C - Premenstrual stage : The premenstrual stage refers to the phase of the menstrual cycle that occurs before menstruation, and it does not involve the specific physical changes described in the question.

Option D - Molestation: The findings in the question do not directly indicate molestation. The given findings are more suggestive of previous consensual sexual activity rather than molestation.

Solution for Question 22:

Correct Option C - 1, 3, 5 are true:

- As per the given scenario, Consent for the examination of the victim should be taken as she is 14- year old.
- If the victim is less than 12 years of age or has an unsound mind, The consent needs to be taken from the guardian.
- Counselling is done if she refuses for examination.
- If after the counseling she refuses to get examined then informal refusal has to be documented.

As per the given scenario, Consent for the examination of the victim should be taken as she is 14- year old.

If the victim is less than 12 years of age or has an unsound mind, The consent needs to be taken from the guardian.

Counseling is done if she refuses for examination.

If after the counseling she refuses to get examined then informal refusal has to be documented.

Incorrect Options:

- Option A, B and D are incorrect. Refer to the explanation for the correct answer.

Option A, B and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 23:

Correct Option C - Sexual gratification by rubbing his private parts against another person:

- Frotteurism refers to the sexual gratification derived from rubbing one's genitals against another person without their consent.
- It typically occurs in crowded places or situations where the person can engage in this behavior discreetly
- Frotteurism is considered a paraphilic disorder and is associated with non-consensual sexual behavior.

Incorrect Options:

Option A - Obtaining sexual pleasure by wearing clothes of the opposite sex: This option describes cross-dressing or transvestism.

Option B - Desire to seek surgery to become a member of the opposite sex: This option refers to gender dysphoria.

Option D - Exposure of one's genitals to an unsuspecting stranger: This option describes describes exhibitionism

Solution for Question 24:

Correct Option C - Scatologia:

- Based on the provided information, the patient most likely has scatologia.
- Scatologia refers to a paraphilic disorder in which a person achieves sexual arousal and gratification by engaging in obscene or sexually explicit talk or behavior . It involves the use of obscene language, sharing obscene pictures, or engaging in explicit conversations with others, often without their consent.

Based on the provided information, the patient most likely has scatologia.

Scatologia refers to a paraphilic disorder in which a person achieves sexual arousal and gratification by engaging in obscene or sexually explicit talk or behavior . It involves the use of obscene language, sharing obscene pictures, or engaging in explicit conversations with others, often without their consent.

Incorrect Options:

Option A - Frotteurism: This is a paraphilic disorder in which a person derives sexual pleasure from rubbing against or touching non-consenting individuals in crowded places. It typically involves non-consensual physical contact.

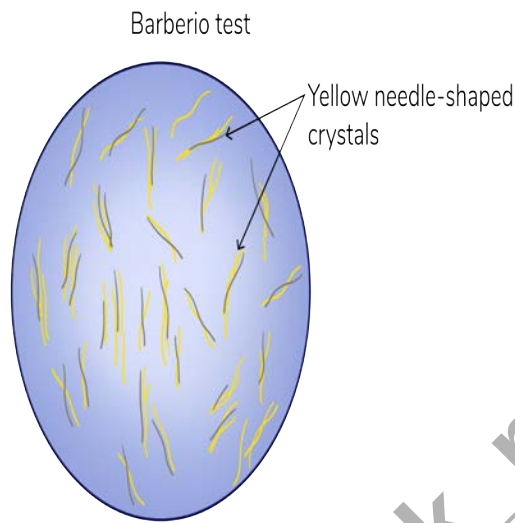
Option B - Voyeurism: Voyeurism intense sexual interest in observing others while they are undressing, naked, or engaged in sexual activities, without their knowledge or consent.

Option D - Coprophilia: Coprophilia is a paraphilic disorder in which a person derives sexual pleasure from feces or associated activities

Solution for Question 25:

Correct Option A - Semen:

- Test reaction depends on the presence of Spermine which comes from the prostate.
- Reaction gives yellow, needle shaped, rhombic crystals of Spermine picrate.
- Reagent - Picric acid.



Incorrect Options:

Option B - Blood: Takayama is one of the confirmatory tests for the blood staining. It gives pink-feathery crystals

Option C - Sputum: Sputum analysis is generally done in case of mycobacterial infections such as Tuberculosis.

Option D - CSF: CSF analysis is done to help diagnose the brain and spinal diseases

Solution for Question 26:

Correct Option A - Spermine:

- Barberio's test is a chemical test used in forensic analysis to detect the presence of spermine.
- Spermine is a naturally occurring polyamine found in human seminal fluid, which is released during ejaculation.
- Therefore, if Barberio's test is positive for spermine, it indicates the presence of semen and can be an important forensic evidence in cases of sexual assault.

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Therefore, if Barberio's test is positive for spermine, it indicates the presence of semen and can be an important forensic evidence in cases of sexual assault.

Incorrect Options:

Option B - Acid phosphatase : While acid phosphatase can also be used as a marker for semen detection, Barberio's test specifically targets the detection of spermine, not acid phosphatase.

Option C - Choline : Choline is not specifically associated with the detection of semen or sexual assault. Barberio's test is not designed to detect choline.

Option D - Fructose : Barberio's test is not specifically designed to detect fructose.

Solution for Question 27:

Correct Option C - 24 weeks:

- According to MTP act, Abortion can be done till 20-24 weeks with the upper limit of 24 weeks.
- According to this act, consent of women is of utmost priority.

Incorrect Options:

- Option A, B and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 28:

Correct Option C - Abortion stick:

- Seaweed called laminaria is used as an abortion stick and can be sterilised, rolled, and dried to form a thin stick or "tent" that can be put into the cervix to stimulate dilation.
- Criminal abortion methods include: Up to the end of first month: Violent exercise Up to the end of second month: Abortifacient drugs 3 rd or 4 th month: Mechanical interference

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- Up to the end of second month: Abortifacient drugs
- 3 rd or 4 th month: Mechanical interference

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- Up to the end of first month: Violent exercise
- Up to the end of second month: Abortifacient drugs
- 3 rd or 4 th month: Mechanical interference

Up to the end of first month: Violent exercise

Up to the end of second month: Abortifacient drugs

3 rd or 4 th month: Mechanical interference

Incorrect Options:

Option A - Dilatation and curettage: The uterine tissue is removed by the doctor using a surgical tool called a curette, which can be either a suction device or a cutting instrument.

Option B - Mifepristone: Mifepristone is a medication that inhibits progesterone, a hormone required for a pregnancy to progress.

Option D - Vacuum aspiration: In the first trimester, curettage by vacuum aspiration is the most common method of abortion.

Solution for Question 29:

Correct Option C - Unruptured since it is deeply situated:

- In cases of rape or sexual abuse of a young child, the hymen is more likely to be unruptured rather than ruptured.
- This is because the hymen is situated deeper inside the vaginal canal in children and is not easily accessible or visible externally.

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This is because the hymen is situated deeper inside the vaginal canal in children and is not easily accessible or visible externally.

Incorrect Options:

Option A - Ruptured since it is superficially: The hymen is not superficially located in children. It is positioned deeper inside the vagina.

Option B - Ruptured since it is very thin: While the hymen is thin, its thinness does not necessarily imply that it will be ruptured in cases of rape or sexual abuse.

Option D - Unruptured since it is highly elastic: The elasticity of the hymen enables it to stretch, but it does not guarantee that it will remain unruptured in cases of sexual abuse

Solution for Question 30:

Correct Option C - Exhibitionism:

- The behavior described in the scenario, where a guy derives sexual gratification by sitting nude on his balcony and observing people who look at him, is best described as exhibitionism. Exhibitionism is a paraphilic disorder characterized by the act of exposing one's genitals to unsuspecting individuals in order to achieve sexual arousal or gratification.

The behavior described in the scenario, where a guy derives sexual gratification by sitting nude on his balcony and observing people who look at him, is best described as exhibitionism. Exhibitionism is a paraphilic disorder characterized by the act of exposing one's genitals to unsuspecting individuals in order to achieve sexual arousal or gratification.

Incorrect Options:

Option A - Voyeurism: Voyeurism is the act of gaining sexual pleasure or gratification from observing others who are naked or engaged in sexual activities without their consent.

Option B - Fetishism: Fetishism is a paraphilic disorder in which individuals derive sexual arousal or gratification from specific objects, materials, or body parts. It typically involves a fixation on nonliving objects or non-genital body parts.

Option D - Masochism: Masochism refers to the sexual arousal or gratification derived from experiencing pain or humiliation.

Solution for Question 31:

Correct Option A - Wearing clothes of opposite sex:

- Eonism is a practice where individuals derive pleasure or satisfaction from wearing the clothes of the opposite sex. It's not about voyeurism or fetishizing specific items of clothing or sexual practices but about the act of adopting the appearance of the opposite sex

Eonism is a practice where individuals derive pleasure or satisfaction from wearing the clothes of the opposite sex. It's not about voyeurism or fetishizing specific items of clothing or sexual practices but about the act of adopting the appearance of the opposite sex

Incorrect Options:

Option B - Kissing and licking of the anus by a sexual partner: This sexual activity, often referred to as anilingus, doesn't have any specific connection to eonism.

Option C - Underclothing of a female: Underclothing of a female eonism isn't about this particular item of clothing. It's about the act of dressing in clothing of the opposite sex as a whole.

Option D - Seeing a female undressing: This option describes voyeurism

Solution for Question 32:

Correct Option C - Death of the mother by act done with intent to cause miscarriage:

- Section 314 of the Indian Penal Code (IPC) . It deals with the death of the mother by an act done with the intent to cause a miscarriage. It is punishable by imprisonment of either kind for a time that may be as long as 10 years , as well as by a fine if the act was committed without the woman's permission.

Section 314 of the Indian Penal Code (IPC) . It deals with the death of the mother by an act done with the intent to cause a miscarriage. It is punishable by imprisonment of either kind for a time that may be as long as 10 years , as well as by a fine if the act was committed without the woman's permission.

Incorrect Options:

Option A - Causing miscarriage with the consent of the mother: This is covered by Section 312 of the IPC.

Option B - Causing miscarriage without the consent of the mother : This is addressed under Section 313 of the IPC.

Option D - Causing the death of the quick unborn child by an act amounting to culpable homicide: This is addressed under IPC section 316. Anyone who commits any act that, if it resulted in death, would constitute culpable homicide

Solution for Question 33:

Correct Option A - Barberio test:

- This test is used to detect spermine.
- It includes the microscopic confirmation of spermine picrate crystals, which form when semen is subjected to an aqueous solution of picric acid, in the form of needle-shaped crystals.

Incorrect Options:

Option B - Florence test : In this test, when the Florence reagent (Potassium Iodide + Iodine + Water) is put to the slide, choline periodide crystals with a rhomboidal shape start to form.

Option C - Teichmann test : The Teichmann test is a forensic test that is used to confirm the presence of blood

Option D - Acid phosphatase test : The acid phosphatase test is also used in forensics, particularly to detect semen. It relies on a color change reaction when acid phosphatase reacts with a specific substrate.

Medical Jurisprudence

1. Which of the following does not fall under an exception to informed consent?

- A. A medical examination ordered by a police officer for an individual arrested on suspicion of drug trafficking
- B. Emergency surgical treatment for a patient after a road traffic accident
- C. A psychiatric evaluation ordered by a court for a prisoner showing signs of severe mental illness
- D. A patient diagnosed with tuberculosis refusing treatment, and the physician administering medication under the guise of placebo treatment to manage the condition for public health reasons

2. A 50-year-old woman undergoes a surgical procedure and experiences severe postoperative complications due to the surgeon's failure to follow standard protocols. As a result, she files a lawsuit seeking compensation for her injuries. In this scenario, which of the following statements accurately describes the nature of the lawsuit?

- A. The lawsuit is a criminal negligence case because it involves severe harm and a violation of medical standards.
- B. The lawsuit is a civil negligence case because it seeks compensation for harm caused by the surgeon's actions.
- C. The lawsuit cannot proceed because the patient consented to the surgery.
- D. The lawsuit is a tort case that does not involve any financial compensation.

3. A 68-year-old woman with severe coronary artery disease undergoes coronary bypass surgery. Prior to the operation, the surgeon discusses the potential

complications, including a 2-5% risk of mortality due to inherent surgical risks. Despite the surgery being performed with appropriate care, the patient unfortunately dies during the procedure. Which legal principle best explains the surgeon's liability in this case?

- A. Novus Actus Interveniens
 - B. Calculated Risk Doctrine
 - C. Vicarious Liability
 - D. Respondeat Superior
-

4. Which of the following is NOT considered a type of infamous conduct or professional misconduct according to the guidelines for medical practitioners?

- A. Performing an illegal abortion operation
 - B. Issuing false or misleading certificates for administrative purposes
 - C. Refusing emergency treatment to a patient
 - D. Advertising in the yellow pages of the telephone directory
-

5. A 78-year-old patient with terminal cancer is experiencing unbearable pain and requests to end their life peacefully. The patient's family consults with the medical team about options. Which of the following accurately describes the type of euthanasia they are considering?

- A. Involuntary euthanasia
 - B. Passive euthanasia
 - C. Voluntary euthanasia
 - D. Active euthanasia
-

6. A 65-year-old patient in Mumbai wants to ensure their end-of-life wishes are respected if they become incapacitated. Which of the following statements about Living wills in India is correct?

- A. Living wills are not legally recognized in India
 - B. A Living will must be verified by a Chief Judicial Magistrate
 - C. Can designate a person to make medical decisions on their behalf
 - D. If two living wills are made, the first one will be valid.
-

7. A 35-year-old bus driver is diagnosed with epilepsy during a routine medical check-up. The patient requests the doctor not to disclose this information to anyone. In this scenario, which of the following actions by the doctor would be most appropriate?

- A. Maintain absolute confidentiality as requested by the patient
 - B. Immediately report the condition to the patient's employer
 - C. Explain the potential risks to the patient and persuade them to allow disclosure
 - D. Disclose the information without the patient's consent
-

8. A 65-year-old male with terminal cancer is being treated in a palliative care unit. The physician is aware that disclosing the full extent of his poor prognosis may cause severe distress and worsen his psychological state. However, his family is requesting detailed information about his condition to be disclosed to the patient. Which of the following ethical principles is most appropriate for the physician to apply in this situation?

- A. The physician should use full disclosure and inform the patient of all risks, benefits, and prognosis, allowing the patient to make an informed decision about further care.
- B. The physician should apply therapeutic privilege, withholding certain information if it is believed that disclosing the full prognosis could harm the patient's emotional well-being.

- C. The physician should rely on therapeutic privilege but must always seek approval from the patient's family before withholding any information.
- D. The physician should provide the information to the family only, as full disclosure primarily concerns the patient's relatives in situations involving terminal illness.
-

9. Which BNS section denotes that consent given by an insane person is invalid?

- A. 25 BNS / IPC 87
- B. 26 BNS / IPC 88
- C. 28 BNS / IPC 90
- D. 30 BNS / IPC 9
-

10. A 48-year-old patient with ischemic heart disease is discharged following a cardiac procedure. Upon discharge, the doctor prescribes medications without reviewing the patient's current medication list, resulting in reduced effectiveness of the new drug. The patient also fails to adhere to the regimen, taking the medication only when he "felt like it." This leads to a severe cardiac event, prompting the patient to file a lawsuit, alleging negligence on the doctor's part. Which legal principle would most likely serve as the doctor's defense in this case?

- A. Composite negligence
- B. Vicarious liability
- C. Contributory negligence
- D. Res ipsa loquitur
-

11. A patient is undergoing an elective hysterectomy. During the surgery, the surgeon inadvertently nicks a blood vessel, leading to excessive bleeding. The surgeon immediately identifies the issue and successfully controls the bleeding,

preventing further complications. Despite the patient's stable recovery, the patient files a negligence claim against the surgeon. Which defense is most likely to be successful in this case?

- A. Contributory negligence
 - B. Therapeutic misadventure
 - C. Error of judgment
 - D. Medical Maloccurrence
-

12. A patient undergoes surgery for a broken leg. After the operation, the patient develops a severe infection due to the use of contaminated surgical instruments. The patient had no involvement in the handling of the instruments and did not contribute to the negligence. Which legal doctrine best applies to this situation regarding the surgeon's liability?

- A. Doctrine of Res Ipsa Loquitur
 - B. Doctrine of Common Knowledge
 - C. Novus Actus Interveniens
 - D. Borrowed Servant Doctrine
-

13. A 65-year-old female patient undergoes elective thyroidectomy due to hyperthyroidism. During the procedure, the surgeon inadvertently damages the recurrent laryngeal nerve, resulting in vocal cord paralysis. The patient had been fully informed of the risks involved and consented to the surgery. Which of the following best describes the outcome of this incident?

- A. Medical Maloccurrence
- B. Borrowed Servant Doctrine
- C. Products Liability
- D. Composite Negligence

14. How long should the following types of medical records be retained in India: Outpatient Records (OPD), Inpatient Records (IPD), and Medico-legal Records?

- A. 5 years, 15 years, 15 years
 - B. 10 years, 23 years, 25 years
 - C. 5 years, 15 years, 10 years
 - D. 3 years, 10 years, 10 years
-

15. Which of the following statements is false regarding the Bolam Test and the Bolitho Test?

- A. The Bolam Test allows a doctor to avoid negligence if their actions align with a responsible body of medical opinion.
 - B. The Bolitho Test requires that the accepted medical opinion is logically defensible.
 - C. The Bolam Test involves active scrutiny from the court regarding the logic of medical opinions.
 - D. The Bolitho Test allows a finding of negligence if an expert's opinion is indefensible, regardless of peer support.
-

16. Match the following International Declarations with their guidelines. Geneva a. Terminal Illness Venice b. Therapeutic abortion Ottawa c. Hippocratic Oath Oslo d. Child health.

Geneva	a. Terminal Illness
Venice	b. Therapeutic abortion
Ottawa	c. Hippocratic Oath

Oslo	d. Child health.
------	------------------

- A. 1-d, 2-c, 3-b, 4-a
 - B. 1-c, 2-a, 3-d, 4-b
 - C. 1-a, 2-c, 3-b, 4-d
 - D. 1-b, 2-c, 3-d, 4-a
-

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Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	2
Question 4	3
Question 5	3
Question 6	3
Question 7	3
Question 8	2
Question 9	3
Question 10	3
Question 11	2
Question 12	1
Question 13	1
Question 14	4
Question 15	3
Question 16	2

Solution for Question 1:

Answer: D) A patient diagnosed with tuberculosis refusing treatment, and the physician administering medication under the guise of a placebo treatment to manage the condition for public health reasons

Explanation:

- Informed consent is provided when a patient understands the nature of their condition, the proposed treatment, the outcomes of the suggested treatment, alternative treatment options, and the associated risks and benefits. The patient has the right to either accept or decline the procedure.
- Consent refers to a voluntary agreement, permission, or compliance, which may be either expressed or implied.
- It is exempt in certain circumstances.
- While patients with notifiable diseases (e.g., tuberculosis) may be treated to protect public health, using a placebo without the patient's knowledge violates ethical guidelines. Even in cases of public health concerns, deceptive treatment is not an acceptable practice and does not fall under exceptions to informed consent.

Exceptions to Informed Consent :

- Emergencies (Option B ruled out)
- Medical examination ordered by a police officer for an arrested individual [Sec 51 & 52 BNSS] (Option A ruled out)
- Therapeutic privilege
- Therapeutic waiver
- Medico-legal postmortems
- Psychiatric examination or treatment mandated by a court order (Option C ruled out)
- Use of placebos
- Prisoners
- A person suffering from disease under the 'notified' category (to notify the authorities only)
- Treatment of notifiable diseases for greater community interest.

Implied consent:

- Occurs when consent is not explicitly stated or written but is inferred through actions.
- For example, when a patient visits a hospital for an illness, it is understood that they are consenting to a physical examination to determine the cause of their illness.
- This form of consent is legally valid.

Substituted consent:

- When a person is unable to provide informed consent for medical treatment, proxy consent must be obtained from a close relative.
- The typical order of priority is the spouse, followed by an adult child, then a parent, and finally a sibling.

Presumed consent/opt-out: Refers to the patient's refusal of the procedure, test, or treatment.

Solution for Question 2:

Answer: B) The lawsuit is a civil negligence case because it seeks compensation for harm caused by the surgeon's actions.

Explanation:

The patient in this scenario has experienced severe complications following a surgical procedure due to the surgeon's negligence in adhering to standard protocols. This clearly indicates a potential breach of the duty of care expected from medical professionals.

Feature	Civil Negligence	Criminal Negligence
Offense	No specific and clear violation of law	Must have specifically violated a particular criminal law
Negligence	Simple absence of care and skill	Gross negligence, inattention, or lack of competence
Conduct of physician	Compared to a generally accepted simple standard of conduct	Not compared to a single test
Consent for act	Good defense, cannot recover damages	Not a defense, can be prosecuted
Trial by	Civil court	Criminal court
Evidence	Strong evidence is sufficient	Guilt must be proved beyond reasonable doubt
Punishment	Liable to pay damages	Imprisonment, fine, or both
Contributory negligence	Defense for doctor	Not a defense

Double jeopardy	Cannot be tried twice for the same offense	Can be tried twice for the same crime
Damage	Repairable damage or harm to patient	Irreparable damage to the patient
Dispute	Between two parties in their individual capacities	Between the State and the offending doctor
Complainant	Sufferer party is the complainant	Public prosecutor on behalf of the State is the complainant

The lawsuit is a criminal negligence case because it involves severe harm and a violation of medical standards (Option A):

- Criminal negligence involves actions that result in severe harm or death, leading to criminal charges.
- However, medical negligence is typically addressed in civil court unless the actions are grossly negligent or intentional (e.g., fraud, assault).
- In this case, the woman is seeking compensation for her injuries rather than pursuing criminal charges against the surgeon. Thus, it does not meet the threshold for criminal negligence.

The lawsuit cannot proceed because the patient consented to the surgery (Option C):

- Consent to surgery does not relieve a surgeon from liability for negligence.
- While informed consent is crucial, it pertains to the patient's understanding of the risks and benefits of the procedure.
- If the surgeon fails to adhere to standard protocols, causing harm, the patient can still file a negligence lawsuit.

The lawsuit is a tort case that does not involve any financial compensation (Option D):

- A tort is a civil case involving an offense that inflicts harm on another individual by breaching a legally protected right, except for contractual disputes. Tort cases may involve compensation claims
- While the lawsuit is indeed a tort case (as it involves a civil wrong and the pursuit of damages), it specifically seeks financial compensation for injuries sustained due to negligence.
- Hence, this statement is inaccurate because it contradicts the fundamental purpose of the lawsuit, which is to seek damages for the alleged harm caused by the surgeon's failure to follow protocols.

Solution for Question 3:

Correct Answer: B) Calculated Risk Doctrine

Explanation:

The calculated risk doctrine applies here because the risk of mortality was clearly communicated and is recognized as an inherent risk of the procedure, indicating that the surgeon acted within accepted standards of care.

- Res ipsa loquitur does not apply when injuries may occur despite reasonable care.
- Defense for Doctors : Doctors can present evidence that accepted treatments have unavoidable risks.
- Example: A patient dying during a coronary bypass surgery is seen as an inherent risk (2-5%).
- This refers to an event that breaks the chain of causation from the original negligent act.
- Negligence Element : Requires proof of negligence to establish a break in causation.
- Often applied in incidents like leaving surgical instruments inside a patient.
- Pleas are rarely accepted by courts.
- Employers are liable for the negligent acts of their employees under certain conditions. Also called the Master-Servant rule.
- Conditions : Employer-employee relationship. Conduct occurs within the scope of employment. Incident occurs while on the job.
- Employer-employee relationship.
- Conduct occurs within the scope of employment.
- Incident occurs while on the job.
- Principal doctors can be held responsible for the negligence of their assistants.
- Temporary Relationships : Surgeons assisting each other may create an employer-employee relationship thus being liable for each other's actions.
- Borrowed Servant Doctrine : Employees (e.g., nurses) can serve multiple employers.
- Hospital Liability : Hospitals are liable for the acts of employees under their supervision, but not for senior staff if due care in hiring is proven.
- Direct Supervision : Physicians are responsible for the acts of interns and residents they supervise.
- Defense Against Liability : Employers can avoid liability by proving the employee was not negligent or was acting outside the scope of employment.

Solution for Question 4:

Answer: C) Refusing emergency treatment to a patient

Explanation: Refusing emergency treatment to a patient is not explicitly listed as a type of infamous conduct or professional misconduct. It will be considered under medical negligence.

Professional Misconduct / Infamous Conduct

Definition	Any conduct of the doctor that might reasonably be regarded as disgraceful or dishonorable or judged by professional men of good repute and competence. It involves abuse of professional position.
Examples	1. Performing illegal abortions (Option A) 2. Issuing false or misleading certificates (Option B) 3. Not maintaining medical records for 3 years 4. Not displaying registration number 5. Absence from duty in rural postings 6. Fee splitting (dichotomy) 7. Advertising in certain forms (telephone directory, etc.) (Option D) 8. Performing artificial insemination without informed consent

Solution for Question 5:

Answer: C) Voluntary euthanasia

Explanation: In this case, the patient is requesting to end their life peacefully, which aligns with voluntary euthanasia. The patient is capable of making this decision and providing consent for the same.

Types of Euthanasia

Type	Definition	Key Features
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Voluntary Euthanasia (Option C)	Wherein the individual requests euthanasia, either during illness or before, if complete incapacitation is expected.	- Patient's consent is obtained - Patient is capable of making the decision
Non-voluntary Euthanasia	Where an individual is incapable of perception and feeling, and hence cannot decide or distinguish between life and death.	- Patient cannot give informed consent - Often involves coma patients or severely defective infants
Involuntary Euthanasia (Option A)	Where an individual may distinguish between life and death, and any medical killing is involuntary, i.e. against the will of the person.	- Considered murder - Ethically, morally, and legally wrong
Active Euthanasia (Option D)	Positive merciful act, to end useless suffering or a meaningless existence	- Act of commission - Administration of lethal doses of drugs - Using measures that would hasten death
Passive Euthanasia (Option B)	Discontinuing or not using extraordinary life-sustaining measures to prolong life	- Act of omission - Allowing death by not resuscitating - Not using measures that would delay death

- In India, passive euthanasia is legal under controlled and well-defined conditions involving the withdrawal or withholding of life-sustaining treatment in patients with terminal or irreversible conditions, such as a persistent vegetative state.
- The other types of euthanasia are not legal in India.

Solution for Question 6:

Answer: C) Can designate a person to make medical decisions on their behalf

Explanation: Living will allow individuals to express their wishes for end-of-life care and can designate a person to make medical decisions on their behalf.

Living Will or Advance Medical Directive

Legal Status	Legally recognized since 2018 Supreme Court judgment (Option A)
Purpose	Express wishes for end-of-life care and designate decision-maker

Eligibility	Not limited to terminally ill patients; can be executed by any competent adult
Authentication	Must be attested by a Judicial Magistrate First Class (JMFC) (Option B)
Authorities involved	Primary Medical Board and Secondary Medical Board
Validity	If two living wills are made, the last one will be considered valid (Option D)

Solution for Question 7:

Answer: C) Explain the potential risks to the patient and persuade them to allow disclosure

Explanation:

- This scenario falls under the concept of privileged communication, which involves balancing patient confidentiality with public safety. The doctor should explain to the patient the potential risks their condition poses to public safety, given their occupation as a bus driver.
- The doctor should then attempt to persuade the patient to allow disclosure to the appropriate authorities.
- If the patient still refuses after this explanation, the doctor may then consider disclosure without consent due to the overriding public safety concern.

Privileged communication:

Definition	A statement made bonafide upon any subject matter by a doctor to the concerned authority having corresponding interest due to his legal, social, or moral duty to protect the interests of the community
Nature	An exchange of information between two individuals in a confidential relationship, and an exception to professional secrecy
To be privileged	Must be made to the person who has a duty towards it. If made to more than one person or to a person who has not got a direct interest in it, the plea of privilege fails

Doctor's responsibility	Should first persuade the patient to obtain his consent before revealing the proper authority; however, disclosure can be done without consent if consent is not forthcoming
Examples of privileged communication	Potential threat of grave harm to safety or health of the patient and the public Conditions affecting public safety (e.g., epilepsy in bus driver, pilot) Notifiable diseases Suspected crime or assault Patient's own interest (e.g., suicidal tendencies)

Solution for Question 8:

Answer: B) The physician should apply therapeutic privilege, withholding certain information if it is believed that disclosing the full prognosis could harm the patient's emotional well-being.

Explanation:

Full Disclosure :

- Refers to the physician's ethical responsibility to share all relevant information about the patient's diagnosis, prognosis, and treatment options.
- This allows the patient to make informed decisions, emphasizing autonomy and respect for their right to know the full extent of their condition.
- In this scenario, full disclosure would mean the physician informs the patient of all the details regarding their prognosis, even if it could cause psychological distress.

Therapeutic Privilege :

- Therapeutic privilege allows a physician to withhold certain information if revealing it could harm the patient's emotional or mental health.
- This is used only in cases where full disclosure may cause more harm than benefit.
- In the given scenario, the physician may choose not to disclose the full severity of the prognosis if it is deemed that this information would negatively impact the patient's mental state.

(Option A): Full disclosure may not always be appropriate, especially in sensitive situations where revealing all information could cause psychological harm.

(Option C):

- Therapeutic privilege does not require the physician to seek approval from the family before withholding information.
- The decision to withhold information is primarily based on the physician's judgment about the potential harm to the patient, though family involvement may be helpful in some cases.

(Option D):

- Full disclosure primarily concerns the patient, not the family.
- While family members can be involved in discussions, the ethical duty to disclose information focuses on the patient's right to know about their own condition.

Solution for Question 9:

Answer: C) 28 BNS / IPC 90

Explanation:

The consent given by an insane or intoxicated person, who is unable to understand the nature and consequences of that to which he gives his consent is invalid (28 BNS / IPC 90)

Provisions under the Act	BNS Section	Corresponding IPC Section
Any act done without intent to cause death or serious harm, but done with consent, is not considered an offense.	25 (Option A)	87
An act done in good faith with the consent of the person, for their benefit, is not an offense.	26 (Option B)	88
An act done in good faith for the benefit of a child under 12 years old or an insane person, with the consent of a guardian, is not an offense.	27	89
Consent is invalid if: Given under fear or misconception Given by an insane person	28 (Option C)	90

The exceptions under sections 25, 26, and 27 do not apply to acts that independently constitute offenses, or are intended or likely to cause harm to the person giving consent or on whose behalf consent is given.	29	91
Acts done in good faith without consent are permissible in the following situations: During an emergency When the person is incapable of giving consent When there is no guardian to give consent	30 (Option D)	92
Acts causing slight harm are not considered offenses.	33	95

Solution for Question 10:

Answer: C) Contributory negligence

Explanation:

- In this scenario, Contributory Negligence is the appropriate legal principle.
- In this case, the patient failed to follow the prescribed medication regimen which significantly contributed to the worsening of their condition. The doctor also failed to review the medications that the patient was already using. Thus, while the doctor may have some degree of liability, the patient's actions play a crucial role in reducing the doctor's responsibility.

Contributory Negligence:

- Contributory negligence refers to any unreasonable behavior or lack of care on the part of the patient that contributes to the injury, along with the doctor's negligence. Without the patient's behavior, the injury would not have occurred.
- Contributory negligence serves as a good defense for doctors in civil cases .
- The burden of proof lies on the doctor to show that the patient was negligent and that the patient's actions or omissions contributed significantly to the harm.
- However, doctors are still expected to foresee potential patient harm and provide adequate warnings. For example, a doctor must warn a patient about the consequences of not

following medication instructions or lifestyle advice.

- If contributory negligence is proven, the damages awarded by the court to the patient may be reduced, as the patient's actions contributed to the harm.

Examples of Contributory Negligence :

- The patient fails to give an accurate medical history, leading to incorrect treatment.
- The patient does not follow the doctor's instructions regarding medications, tests, or diet.
- The patient discharges themselves against medical advice.

Composite negligence (Option A): Involves a situation where the patient did not contribute to their injury, but two or more other parties are responsible for causing harm. In composite negligence, all parties are jointly and severally liable for the damages.

Vicarious liability (Option B): This applies when an employer (such as a hospital or a medical institution) is held responsible for the actions of an employee (e.g., a doctor or nurse). This is not relevant in this case.

Res ipsa loquitur (Option D): A legal doctrine that suggests negligence is so obvious that it speaks for itself. This doctrine does not apply here as the harm was not solely due to the doctor's actions but was influenced by the patient's choices.

Solution for Question 11:

Answer: B) Therapeutic misadventure

Explanation:

In this case, Therapeutic Misadventure is the most appropriate defense.

- The surgeon unintentionally caused a complication (nicking a blood vessel) during the surgery, but it was quickly managed without any long-term harm to the patient.
- Since the complication was accidental and not due to negligence, this defense can be used effectively.
- The surgeon must prove that the event was an unforeseen complication of the surgery and not due to lack of skill or care.

Defenses Against Negligence:

Limitation : A case must be filed within a specified time (usually 2 years) from the date of the alleged negligence.

Contributory negligence (Option A) : In this scenario, the patient did not contribute to the complication during surgery. Therefore, this defense does not apply as the patient did not act negligently in any way.

Error of judgment (Option C) : This situation is not about a poor decision but an accidental complication (nicking a blood vessel).

Medical maloccurrence (Option D) : It refers to adverse outcomes that result from unavoidable risks or unpredictable complications, unrelated to the quality of care provided and not implying negligence. In contrast, therapeutic misadventure involves unintended injuries or harm directly caused by errors or mishaps during medical management.

Solution for Question 12:

Correct Answer: A) Doctrine of Res Ipsa Loquitur

Explanation:

Doctrine of Res Ipsa Loquitur applies because the injury (infection) would not have occurred without negligence, the surgeon had exclusive control over the instruments, and the patient was not contributory negligent. It allows the patient to establish negligence without needing expert testimony.

Doctrine of Res Ipsa Loquitur

- Means "the thing speaks for itself"; Used when negligence is obvious.
- Expert Evidence : Patients don't need to prove negligence through expert testimony; the burden shifts to the doctor to prove innocence.
- Conditions : Injury would not have occurred without negligence. Doctor had exclusive control over the treatment or instrument causing injury. Patient was not contributory negligent.
- Injury would not have occurred without negligence.
- Doctor had exclusive control over the treatment or instrument causing injury.
- Patient was not contributory negligent.
- Examples : Blood transfusion errors (infected blood, mismatches). Failure to administer necessary vaccines. Overprescribing medication. Wrong-site surgeries. Leaving surgical

instruments inside the patient. Failing to remove swabs, leading to complications.

Prolonged splinting causing loss of use of a limb. Doctrine of Common Knowledge (Option B)

Applies when negligence is evident from common knowledge, not requiring specialized

medical expertise. Expert Evidence : Experts are not needed for matters of common

knowledge. Patient's Burden: The patient must show the act of negligence (commission or omission) but doesn't need to prove the standard of care.

- Blood transfusion errors (infected blood, mismatches).
- Failure to administer necessary vaccines.
- Overprescribing medication.
- Wrong-site surgeries.
- Leaving surgical instruments inside the patient.
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- Injury would not have occurred without negligence.
- Doctor had exclusive control over the treatment or instrument causing injury.
- Patient was not contributory negligent.
- Blood transfusion errors (infected blood, mismatches).
- Failure to administer necessary vaccines.
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- Patient's Burden: The patient must show the act of negligence (commission or omission) but doesn't need to prove the standard of care.

Doctrine of Common Knowledge (Option B)

Novus Actus Interveniens (Option C): This doctrine refers to an unrelated event that breaks the chain of causation. In this scenario, there is no unrelated event that caused the injury; it directly resulted from the use of contaminated instruments.

Borrowed Servant Doctrine (Option D): This doctrine applies to situations where an employee (like a nurse) is temporarily under the control of another employer (like a surgeon). It does not directly relate to the negligence of the surgeon in this case, as it does not address the issue of instrument contamination.

Solution for Question 13:

Correct Answer: A) Medical Maloccurrence

Explanation:

This incident represents a medical maloccurrence because the complication arose from an unavoidable risk associated with the surgery, despite the surgeon providing quality care and obtaining informed consent.

Medical Maloccurrence

- A less than ideal medical outcome not linked to the quality of care.
- Types: Anticipated medical/surgical complications (unavoidable risks). Unpredictable complications (unavoidable). Complications from informed patient-doctor decisions that seem inappropriate in hindsight.
- Anticipated medical/surgical complications (unavoidable risks).
- Unpredictable complications (unavoidable).
- Complications from informed patient-doctor decisions that seem inappropriate in hindsight.
- Characteristics: Often unrelated to the quality of care. May occur despite good medical attention (e.g., idiosyncratic drug responses, nerve damage during surgery).
- Often unrelated to the quality of care.
- May occur despite good medical attention (e.g., idiosyncratic drug responses, nerve damage during surgery).
- Anticipated medical/surgical complications (unavoidable risks).
- Unpredictable complications (unavoidable).

- Complications from informed patient-doctor decisions that seem inappropriate in hindsight.
- Often unrelated to the quality of care.
- May occur despite good medical attention (e.g., idiosyncratic drug responses, nerve damage during surgery).

Borrowed Servant Doctrine (Option B)

- An employee can serve multiple employers (e.g., a nurse assisting a surgeon is a "borrowed servant" during surgery). Negligence occurring due to borrowing of temporary staff from another department.

Products Liability (Option C)

- Responsibility for injury/death caused by a defective product.
- Plaintiff's Burden: Prove manufacturer deviated from due care (design, manufacture, testing, warnings). Prove defect caused injury/death.
- Prove manufacturer deviated from due care (design, manufacture, testing, warnings).
- Prove defect caused injury/death.
- Manufacturer's Responsibility: Must prove safety and effectiveness of products.
- Prove manufacturer deviated from due care (design, manufacture, testing, warnings).
- Prove defect caused injury/death.

Composite Negligence (Option D)

- Injury occurs without the injured person's negligence, due to the combined negligence of multiple parties.
- All wrongdoers are jointly liable for damages, and the injured person can sue any or all of them.

Solution for Question 14:

Correct Answer: D) 3 years, 10 years, 10 years

Explanation:

Outpatient Records (OPD) should be retained for 3 years, Inpatient Records (IPD) for 10 years, and Medico-legal Records for 10 years.

Retention Timeline for Medical Records in India

- Outpatient Records (OPD) in hospital: 3 years
- Inpatient Records (IPD) in hospital: 10 years (Option A, B, & D ruled out)
- Medico-legal Records: 10 years
- Records during pending court cases: Indefinitely (beyond stipulated time)
- Records after court decision: At least 2 years
- In-patient records (individual doctors): 3 years from the start of treatment
- Access to documents for patients: Within 72 hours
- Consumer Protection Act (filing suit): 2 to 3 years from date of treatment (may be relaxed by the court)

Solution for Question 15:

Correct Answer: C) The Bolam Test involves active scrutiny from the court regarding the logic of medical opinions.

Explanation:

The Bolam Test does not involve active scrutiny from the court regarding the logic of medical opinions; it primarily relies on the consensus of medical professionals. In contrast, the Bolitho Test includes this scrutiny.

Feature	Bolam Test	Bolitho Test
Core Principle	A doctor is not negligent if their actions align with those of a responsible body of medical opinion. (Option A ruled out) No negligence if actions align with peer practice	The court must ensure that the medical opinion is not only widely accepted but also logically defensible, considering risks and benefit. (Option B ruled out) If an expert's opinion fails to meet this logical standard, it may be deemed indefensible, leading to a finding of negligence regardless of peer support. (Option D ruled out) Requires logical defensibility of peer opinions
Judicial Role	Minimal scrutiny; relies on expert consensus	Active scrutiny; court evaluates logic behind opinions

Criticism	Rigid; can uphold substandard care	Adds complexity; requires robust expert justification
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Solution for Question 16:

Correct Answer: B) 1-c, 2-a, 3-d, 4-b

Explanation:

International Declarations	Description
Geneva	Hippocratic Oath
Tokyo	Torture
Sydney	Definition of death
Oslo	Therapeutic Abortion
Helsinki	Human experimentation
Venice	Terminal illness
Lisbon	Rights of patients
Malta	Hunger strike
Washington	Biological warfare
Ottawa	Child Health
Hong Kong	Old abuse

Infanticide

1. Regarding infanticide, feticide, and filicide in India, which of the following statement is incorrect?

- A. Infanticide is punishable by death or imprisonment for life
- B. Feticide is the killing of a fetus at any time prior to birth
- C. The killing of a living child in the mother's womb who has not yet breathed cannot be considered infanticide
- D. Female feticide is the act of aborting a fetus because it is female

2. A stillborn fetus is discovered during a routine prenatal checkup. The fetus shows skin slippage on the face and back, soft and flaccid tissues, and a sweetish disagreeable smell, but no gas formation. The skull shows mild cranial compression but no overlapping of cranial sutures. Based on these findings, approximately how long ago did fetal death likely occur?

- A. Less than 12 hours ago
- B. Between 12-24 hours ago
- C. Between 36-48 hours ago
- D. More than 96 hours ago

3. A 35-year-old woman presents with suspected intrauterine fetal death at 36 weeks gestation. Which of the following findings would be LEAST useful in confirming recent fetal demise?

- A. Spalding's sign on ultrasonography
- B. Presence of skin slippage on the fetus's face and back

- C. Robert's sign on ultrasonography
 - D. Absence of fetal heart sounds on electronic fetal monitoring
-

4. Match the following tests with their correct descriptions: 1. Wreden's test a. Compares the weight of lungs to body weight ratio before and after respiration 2. Ploucquet's test b. Checks for air bubbles in the stomach and intestines 3. Fodere's test (Static test) c. Examines the presence of air in the middle ear 4. Breslau's Second Life test d. Observes the weight change of lungs after respiration

1. Wreden's test	a. Compares the weight of lungs to body weight ratio before and after respiration
2. Ploucquet's test	b. Checks for air bubbles in the stomach and intestines
3. Fodere's test (Static test)	c. Examines the presence of air in the middle ear
4. Breslau's Second Life test	d. Observes the weight change of lungs after respiration

- A. 1-b, 2-a, 3-d, 4-c
 - B. 1-b, 2-d, 3-a, 4-c
 - C. 1-c, 2-d, 3-a, 4-b
 - D. 1-c, 2-a, 3-d, 4-b
-

5. A forensic pathologist is examining the lungs of a newborn during an autopsy. The lungs appear mottled in color and occupy most of the thoracic cavity. However, when subjected to the hydrostatic test, portions of the lung tissue sink while others float. Microscopically, some areas show expanded alveoli lined with flat squamous cells, while other areas show closed alveoli lined with cuboidal cells. What is the most likely explanation for these conflicting findings?

- A. The infant was stillborn, and the floating lung portions are due to putrefaction
- B. The infant was live-born but suffered from severe respiratory distress syndrome
- C. The infant was live-born and partially breathed before death

D. The findings are inconclusive and could be due to artificial respiration attempts

6. A forensic examiner is evaluating two fetuses found under different circumstances. The examiner performs Breslau's First Life Test on Baby A and Breslau's Second Life Test on Baby B. The findings are as follows: For Baby A, after performing the hydrostatic test, the lungs sink when compressed and placed back in water. For Baby B, air bubbles are observed escaping from the stomach and intestines after an incision is made underwater. What do these findings suggest about the live birth status of both babies?

- A. Baby A was stillborn, while Baby B was born alive and breathed.
 - B. Baby A was born alive and breathed, while Baby B was stillborn.
 - C. Both Baby A and Baby B were born alive and breathed.
 - D. Both tests are inconclusive due to potential contamination.
-

7. A newborn is examined shortly after delivery. The physician observes a localized swelling on the baby's head that appears several hours after birth. The swelling is confined to one side of the skull, is firm to the touch, and does not cross the suture lines. What is the most likely diagnosis for this condition?

- A. Caput Succedaneum
 - B. Cephalhematoma
 - C. Hydrocephalus
 - D. Intracranial hemorrhage
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8. A 3-month-old infant is found unresponsive in their crib by a caregiver early in the morning. The infant had been healthy and fed well the previous evening. The caregiver reports the baby typically sleeps on their stomach. An autopsy is

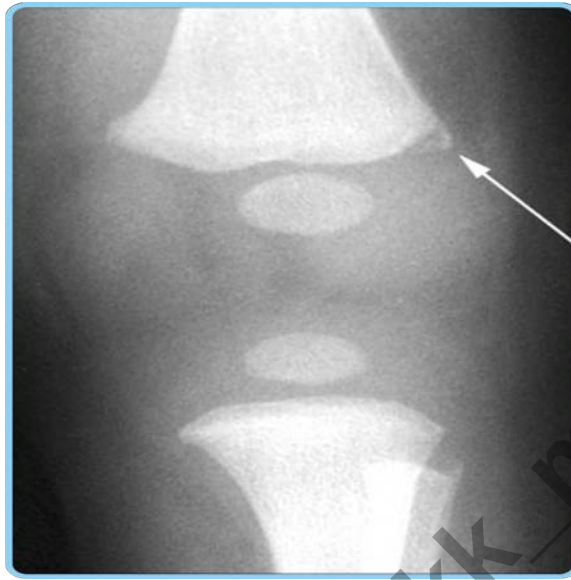
performed, revealing no conclusive cause of death. Which of the following statements is most accurate based on the scenario?

- A. Sleeping on the stomach significantly reduces the risk of death.
 - B. The autopsy typically reveals specific pathological findings.
 - C. Low birth weight or premature infants have an increased risk of sudden death.
 - D. Maternal smoking has no post-natal association with infant mortality.
-

9. A 6-year-old child is frequently admitted with gastrointestinal distress, including vomiting and abdominal pain. The mother provides elaborate details about the symptoms and insists on multiple tests despite negative results. Symptoms appear only when the mother is present, and the child seems healthy when separated. What is the most likely diagnosis for the child's condition?

- A. Gastroenteritis
 - B. Munchausen Syndrome by Proxy
 - C. Functional abdominal pain
 - D. Anxiety disorder
-

10. A 2-year-old girl presents with persistent crying, refusal to eat, and multiple bruises. The parents report frequent falls. On exam, there are bruises on the abdomen and back, a healing burn on the thigh, and rib tenderness. X-ray done is shown below. The parents give vague explanations and claim the child "bruises easily." A CT scan reveals subdural hematoma without a clear trauma history. What is the most likely diagnosis?



- A. Osteogenesis imperfecta
- B. Caffey syndrome
- C. Munchausen syndrome by proxy
- D. Hemophilia

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	1
Question 4	4
Question 5	3
Question 6	1
Question 7	2
Question 8	3
Question 9	2
Question 10	2

Solution for Question 1:

Correct Answer: C) The killing of a living child in the mother's womb who has not yet breathed cannot be considered infanticide

Explanation: The causing of death of a living child in the mother's womb may amount to culpable homicide, even though the child may not have breathed or been completely born (punishable under Sec. 299 IPC) .

The various types of infanticides are listed in the table below:

Term	Definition	Legal Aspects
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Infanticide	Killing of an infant at any time from birth up to the age of 12 months	Charged under Section (Option A) Punishable by death or imprisonment for life and fine Applies even if the child is killed in the womb and may not have breathed or been completely born (Culpable homicide- Sec.92 BNS)
Feticide (Option B)	Killing of the fetus at any time prior to birth	Causing the death of a quick unborn child amounts to culpable homicide Punished with imprisonment up to 7 years and/or fine if with woman's consent. (Sec.88 BNS) If without the woman's consent up to 10 years with or without a fine. (Sec 89 BNS)
Female Feticide (Option D)	Act of aborting a fetus because it is female	Covered under laws against feticide and sex determination
Neonaticide	Killing of a child within 24 hrs	
Filicide	The deliberate act of killing a child by the parents	Treated as murder under Section 103 BNS

Solution for Question 2:

Correct Answer: C) Between 36-48 hours ago

Explanation: The baby shows signs of maceration such as skin slippage (occurs as early as 12 hours) & cranial compression (occurs after 36 hours) but no overlapping of cranial sutures (occurs >96 hours), which indicates fetal demise occurred after 36 hours but before 96 hours.

Maceration: It is a process of aseptic autolysis of dead fetus inside the womb. This occurs when the dead child remains in the uterus for about 3–4 days surrounded with liquor amnii and with exclusion of air.

Maceration changes in intrauterine death

Time Since Death	Maceration changes observed
12 hours	Earliest sign: Skin slippage of face, back, or abdomen
24 hours	Skin becomes brown or purplish in color

> 36 hours	Cranial compression becomes visible
72 hours (3 days)	Desquamation over 75% of body surface
> 96 hours (4 days)	Overlapping of cranial sutures
> 1 week	Mouth is widely open
Throughout process	Fetus is soft and flaccid Sweetish disagreeable smell No gas formation Internal organs show autolytic decomposition (except lungs and uterus)

Less than 12 hours ago (Option A) is incorrect because skin slippage is already present, which doesn't occur until at least 12 hours after death.

Between 12-24 hours ago (Option B) is incorrect because while skin slippage is present, the cranial compression indicates more time has passed.

More than 96 hours ago (Option D) is incorrect because while cranial compression is present, overlapping cranial sutures are absent, which would be expected after 96 hours.

Solution for Question 3:

Correct Answer: A) Spalding's sign on ultrasonography.

Explanation: Spalding's sign on ultrasonography is the least useful for confirming recent fetal demise. It typically appears about 7 days after fetal death, indicating prolonged intrauterine retention. By the time this sign is visible, the fetal death is no longer recent.

Radiological Signs of Dead-born fetus

Sign	Imaging Modality	Description	Timeline
Absence of Fetal Heart Motion (Option D)	Ultrasound	No detectable cardiac activity on real-time imaging	Immediate indicator of fetal death-
Absence of Fetal Movements	Ultrasound	No spontaneous or induced fetal movements during examination	Immediate indicator of fetal death

Robert's Sign (Option C)	Ultrasound	Gas bubbles in fetal heart chambers and great vessels	May appear by 12 hours, but can be difficult to interpret
Hyperflexion of Spine	X-ray, Ultrasound	Abnormal curvature of the fetal spine due to loss of muscle tone	Variable, typically seen in later stages
Spalding's Sign (Option A)	Ultrasound	Overlapping of fetal skull bones due to loss of brain tissue consistency	Appears about 7 days after fetal death
Loss of Rib Parallelism	X-ray	Ribs lose their normal parallel arrangement and overlap present	Later sign, associated with collapse of thoracic cavity
Deuel's Halo Sign	X-ray	Indicates elevation of the subcutaneous fat layer from the underlying bone caused by edema	Not reliable sign, as can be seen in both living and dead fetus

Presence of skin slippage on the fetus's face and back (Option B): Skin slippage or maceration is a sign of fetal death that occurs after the fetus has been dead for several days, indicating the decomposition process.

Solution for Question 4:

Correct Answer: D) 1-c, 2-a, 3-d, 4-b

Explanation:

Tests for Stillborn vs Liveborn

Test Name	Description	Key Points
Wreden's test	Examines the presence of air in the middle ear	- Absence of gelatinous embryonic connective tissue - Presence of air in the middle ear after live birth
Ploucquet's test	Compares the weight of lungs to body weight ratio before and after respiration	- Ratio before respiration: 1/70 of body weight - Ratio after respiration: 1/35 of body weight - Demonstrates establishment of respiration

Fodere's test (Static test)	Observes the weight change of lungs after respiration	- Weight becomes approximately double after respiration due to increase in blood flow to lung beds after breathing - Not constant due to factors like edema, pneumonitis, or amniotic fluid inhalation
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Solution for Question 5:

Correct Answer: C) The infant was live-born and partially breathed before death

Explanation: The mixed findings suggest that the infant was live-born and partially breathed before death. This is explained by:

- The mottled appearance and partial thoracic cavity occupation (some aeration)
- The mixed hydrostatic test results (some areas floated, others sank)
- The microscopic findings of both expanded alveoli with flat cells (respired areas) and closed alveoli with cuboidal cells (unrespired areas)

Unrespired vs Respired lung

Feature	Unrespired Lung (Stillborn)	Respired Lung (Live-born)
Color	Bluish red	Mottled
Volume	Small	Large, covers heart
Thoracic cavity	Not full	Occupies fully
Pleura	Loose, wrinkled	Taut, stretched
Margins	Sharp	Rounded
Surface	Smooth	Uneven
Consistency	Dense, firm, non-crepitant (liver-like)	Soft, spongy, elastic, crepitant
Ploucquet's test (Lung weight: body weight ratio)	1:70	1:35
Fodere's static test	30-40 g	60-70 g
Diaphragm	4th-5th rib level	6th-7th rib level

Blood oozing	Little frothless blood	Abundant frothy blood
Flotation (Hydrostatic test)	Whole and parts sink in water	Floats in water
Alveoli	Not expanded	Expanded
Microscopically	Alveolar sacs closed, lined with cuboidal/columnar cells	Sacs dilated, lined with flat squamous cells with prominent vascularity

The infant was stillborn, and the floating lung portions are due to putrefaction (Option A): While putrefaction can cause gas formation that makes tissues float, the microscopic findings of expanded alveoli with flat cells in some areas indicate genuine respiration occurred.

The infant was live-born but suffered from severe respiratory distress syndrome (Option B): Severe respiratory distress syndrome typically results in collapsed alveoli throughout the lungs, not a mix of expanded and closed alveoli as observed here.

The findings are inconclusive and could be due to artificial respiration attempts (Option D): While artificial respiration could cause partial aeration, it typically results in a more uniform pattern. The mixed microscopic findings suggest natural, partial respiration.

Solution for Question 6:

Correct Answer: A) Baby A was stillborn, while Baby B was born alive and breathed.

Explanation:

The hydrostatic test showed that Baby A's lungs sank, indicating they were unrespired and thus confirming stillbirth. In contrast, Baby B showed air bubbles escaping from the stomach and intestines, suggesting that respiration occurred, confirming live birth.

Hydrostatic Test/ Raygat Test/ Breslau's First Life Test

- Principle: Based on the specific gravity of the lungs; before respiration, it is 1040–1050, and after respiration, it is 940–950, allowing the lungs to float.
- Procedure: Dissect out the fetal lungs and tie them at the hilar region. Place both lungs in a trough of water and observe.
- Dissect out the fetal lungs and tie them at the hilar region.
- Place both lungs in a trough of water and observe.

- Inference: If they sink: Unrespired lung. If they float: Remove from water, cut into small pieces, and compress between sponges, then re-test in water. If they sink: Unrespired lung. If they float: Respired lung. Floating indicates residual air in the lungs from breathing after birth.
- If they sink: Unrespired lung.
- If they float: Remove from water, cut into small pieces, and compress between sponges, then re-test in water. If they sink: Unrespired lung. If they float: Respired lung.
- If they sink: Unrespired lung.
- If they float: Respired lung.
- Floating indicates residual air in the lungs from breathing after birth.
- False positive: Caused by putrefying gasses or artificial inflation.
- False negative: Caused by conditions like atelectasis, edema, feeble respiration, pneumonia, or congenital syphilis.
- A hydrostatic test is not necessary if: The fetus shows congenital anomalies (e.g., anencephaly) The fetus is macerated or mummified The umbilical cord has separated and scarred The stomach contains milk Bruises on the lungs indicate artificial respiration efforts The fetus is born before 180 days of gestation
- The fetus shows congenital anomalies (e.g., anencephaly)
- The fetus is macerated or mummified
- The umbilical cord has separated and scarred
- The stomach contains milk
- Bruises on the lungs indicate artificial respiration efforts
- The fetus is born before 180 days of gestation
- Dissect out the fetal lungs and tie them at the hilar region.
- Place both lungs in a trough of water and observe.
- If they sink: Unrespired lung.
- If they float: Remove from water, cut into small pieces, and compress between sponges, then re-test in water. If they sink: Unrespired lung. If they float: Respired lung.
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- If they float: Respired lung.
- Floating indicates residual air in the lungs from breathing after birth.
- If they sink: Unrespired lung.
- If they float: Respired lung.
- The fetus shows congenital anomalies (e.g., anencephaly)
- The fetus is macerated or mummified

- The umbilical cord has separated and scarred
- The stomach contains milk
- Bruises on the lungs indicate artificial respiration efforts
- The fetus is born before 180 days of gestation

Breslau's Second Life Test

- Steps: Remove the stomach and intestines, tying double ligatures at each end. Submerge them in water. Make an incision between the ligatures.
- Remove the stomach and intestines, tying double ligatures at each end.
- Submerge them in water.
- Make an incision between the ligatures.
- Interpretation: If air bubbles emerge, it indicates that respiration has occurred, confirming live birth. The presence of milk in the stomach also serves as positive evidence of live birth.
- If air bubbles emerge, it indicates that respiration has occurred, confirming live birth.
- The presence of milk in the stomach also serves as positive evidence of live birth.
- Remove the stomach and intestines, tying double ligatures at each end.
- Submerge them in water.
- Make an incision between the ligatures.
- If air bubbles emerge, it indicates that respiration has occurred, confirming live birth.
- The presence of milk in the stomach also serves as positive evidence of live birth.

Baby A was born alive and breathed, while Baby B was stillborn. (Option B): If Baby A had been born alive, the lungs would have floated during the hydrostatic test, indicating respiration. The sinking of the lungs confirms stillbirth. Baby B's air bubbles indicate that it was born alive.

Both Baby A and Baby B were born alive and breathed. (Option C): Baby A's lungs did not float, which indicates it was stillborn. Therefore, it cannot be concluded that both babies were born alive.

Both tests are inconclusive due to potential contamination. (Option D): Baby A's sinking lungs definitively indicate stillbirth and Baby B's air bubbles confirm live birth, making the tests conclusive.

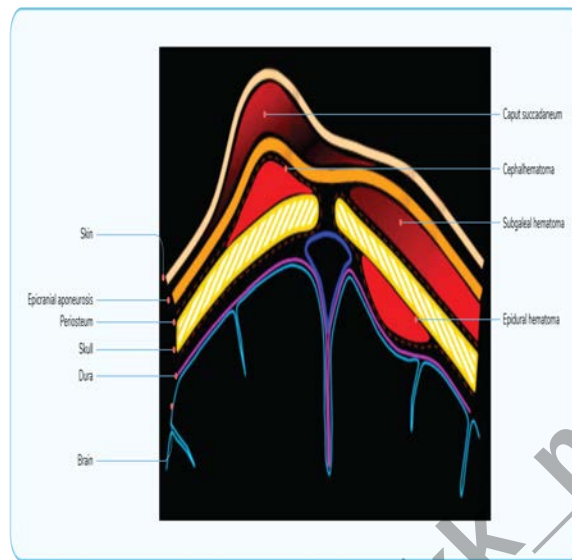
Solution for Question 7:

Correct Answer: B) Cephalohematoma

Explanation:

Cephalohematoma refers to a localized collection of blood between the skull and the periosteum, often appearing several hours after birth. It is typically unilateral, firm to the touch, and confined by the suture lines, matching the characteristics noted in the scenario.

Feature	Cephalohematoma	Caput Succedaneum (Option A)
Definition	Blood collection between the periosteum and skull due to a ruptured emissary vein, often from trauma (e.g., forceps delivery).	Swelling from fluid accumulation between scalp layers, typically from prolonged pressure (e.g., during labor).
Situation	Generally unilateral, commonly over the parietal bone.	Can be bilateral, covering a broader area of the scalp.
Impulse on Crying (Seen in hydrocephalus)	No impulse seen.	No impulse seen.
Limitation by Suture Line	Confined by sutures.	No restriction by sutures.
Underlying Pathology	May involve a skull fracture.	Non-pathological; usually harmless.
Occurrence	Does not appear at birth, and usually develops later.	Present at birth.
Development and Disappearance	Appears 12–24 hours after birth, and resolves in 6–8 weeks. Maybe followed by jaundice.	Disappears on its own within 24 hours. Maybe followed by jaundice.
Medico-Legal Importance	Can assist in determining how long the infant survived post-birth based on regression.	Provides clear evidence of in-utero life, though, in prolonged labor, fetal death may occur before birth.



Hydrocephalus (Option C) involves fluid accumulation in the brain ventricles, leading to overall head enlargement rather than localized swelling.

Intracranial hemorrhage (Option D) involves bleeding within the skull, typically presenting with more severe symptoms and not as a confined scalp swelling.

Solution for Question 8:

Correct Answer: C) Low birth weight or premature infants have an increased risk of sudden death.

Explanation:

Infants who are low birth weight or born prematurely are at a higher risk for sudden death due to various factors, including underdeveloped respiratory systems and increased vulnerability.

Sudden Infant Death Syndrome (SIDS)

- The sudden and unexpected death of a seemingly healthy infant, with no explanation after a complete autopsy. It is an autopsy diagnosis, not a clinical one.
- Features Incidence: 0.2–0.4% of all live births. Most vulnerable between 2 weeks and 2 years, peaking at 2–4 months. The higher death rate in males (M: F ratio 3:2). More prevalent in low and middle-class families with poor housing conditions, larger families, and a

lack of health awareness. Typically found dead in the early morning or after the first feed. Increased risk among twins, particularly those who are premature or of low birth weight. Increased risk linked to maternal smoking (pre- or postnatal) and drug abuse. (Option D)

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- Typically found dead in the early morning or after the first feed.
- Increased risk among twins, particularly those who are premature or of low birth weight.
- Increased risk linked to maternal smoking (pre- or postnatal) and drug abuse. (Option D)
- Cause Prolonged sleep apnea, leads to hypoxia. Respiratory infections cause viremia and sleep depression in the respiratory center. Nasal edema and mucus narrowing airways. Possible hypersensitivity to cow's milk causing laryngeal spasm. Accidental suffocation from bedclothes or overlying by a caregiver. The increased risk is associated with prone or side sleeping positions. (Option A) Miscellaneous causes include metabolic deficiencies and environmental factors.
- Prolonged sleep apnea, leads to hypoxia.
- Respiratory infections cause viremia and sleep depression in the respiratory center.
- Nasal edema and mucus narrowing airways.
- Possible hypersensitivity to cow's milk causing laryngeal spasm.
- Accidental suffocation from bedclothes or overlying by a caregiver.
- The increased risk is associated with prone or side sleeping positions. (Option A)
- Miscellaneous causes include metabolic deficiencies and environmental factors.
- Postmortem Findings Generally negative or non-specific pathological findings . (Option B)
The trachea may contain milky vomit, sometimes blood-stained. Multiple petechial hemorrhages on the heart, lungs, and thymus (agonal). Common pulmonary edema. Possible milk or blood-stained froth at the mouth or bedding, with hands clenched around bedclothes.
- Generally negative or non-specific pathological findings . (Option B)
- The trachea may contain milky vomit, sometimes blood-stained.
- Multiple petechial hemorrhages on the heart, lungs, and thymus (agonal).
- Common pulmonary edema.
- Possible milk or blood-stained froth at the mouth or bedding, with hands clenched around bedclothes.

- Medico-Legal Aspects SIDS is considered a natural death; however, parents may be wrongfully suspected of criminal involvement or negligence. Some infanticide cases may be misrepresented as SIDS.
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- Possible hypersensitivity to cow's milk causing laryngeal spasm.
- Accidental suffocation from bedclothes or overlying by a caregiver.
- The increased risk is associated with prone or side sleeping positions. (Option A)
- Miscellaneous causes include metabolic deficiencies and environmental factors.
- Generally negative or non-specific pathological findings . (Option B)
- The trachea may contain milky vomit, sometimes blood-stained.
- Multiple petechial hemorrhages on the heart, lungs, and thymus (agonal).
- Common pulmonary edema.
- Possible milk or blood-stained froth at the mouth or bedding, with hands clenched around bedclothes.
- SIDS is considered a natural death; however, parents may be wrongfully suspected of criminal involvement or negligence.
- Some infanticide cases may be misrepresented as SIDS.

Solution for Question 9:

Correct Answer: B) Munchausen Syndrome by Proxy

Explanation:

Munchausen Syndrome by Proxy

- Munchausen Syndrome by Proxy is a form of abuse where a caregiver fabricates or induces symptoms in a child for attention.
- Children may present with vague complaints like vomiting, diarrhea, fever, or seizures, often inflicted intentionally by the parent (e.g., using anticoagulants, suffocation, ipecac syrup).
- The caregiver derives personal benefit, such as attention or sympathy, at the child's expense.
- Diagnosis The illness does not conform to expected patterns or courses. Symptoms are not supported by lab or imaging results. Injuries or wounds may fail to heal properly. The child's condition may worsen in the presence of the caregiver and improve when separated. Positive drug tests for substances not prescribed to the child. The child may have been admitted to various hospitals and seen by numerous physicians.
- The illness does not conform to expected patterns or courses.
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- Injuries or wounds may fail to heal properly.
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- The child may have been admitted to various hospitals and seen by numerous physicians.
- The illness does not conform to expected patterns or courses.
- Symptoms are not supported by lab or imaging results.
- Injuries or wounds may fail to heal properly.
- The child's condition may worsen in the presence of the caregiver and improve when separated.
- Positive drug tests for substances not prescribed to the child.
- The child may have been admitted to various hospitals and seen by numerous physicians.

Gastroenteritis (Option A): This condition typically presents with clear, objective symptoms such as diarrhea and fever, which would be consistent across different caregivers. The lack of objective findings in this scenario makes gastroenteritis unlikely.

Functional Abdominal Pain (Option C): While this condition can cause recurrent abdominal pain, it usually lacks the distinct pattern of worsening in the presence of a caregiver. The clear

connection to the mother's presence makes this diagnosis less likely.

Anxiety Disorder (Option D): Although anxiety can manifest in physical symptoms, it wouldn't typically cause the specific pattern observed here, where the child's symptoms are linked to the presence of the mother. This option does not explain the mother's insistence on testing and the lack of objective findings.

Solution for Question 10:

Correct Answer: B) Caffey syndrome

Explanation:

- Caffey syndrome or Battered baby syndrome is strongly suggested when a child presents with multiple injuries at different stages of healing, and the parents provide a vague or inconsistent explanation.

The rib fractures, bruises, and a healing burn—combined with intracranial hemorrhage without significant trauma—point to ongoing physical abuse.



Battered Baby Syndrome (Caffey Syndrome)

A battered child is one who has sustained repeated physical harm due to intentional violence inflicted by a parent or guardian. Other names include Caffey-Kempe syndrome, maltreatment

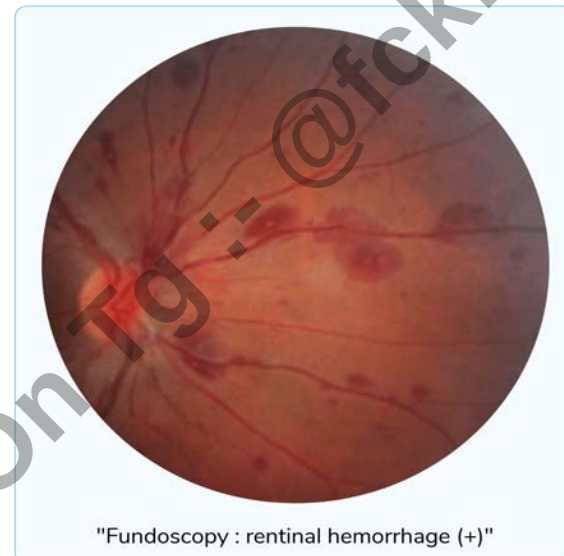
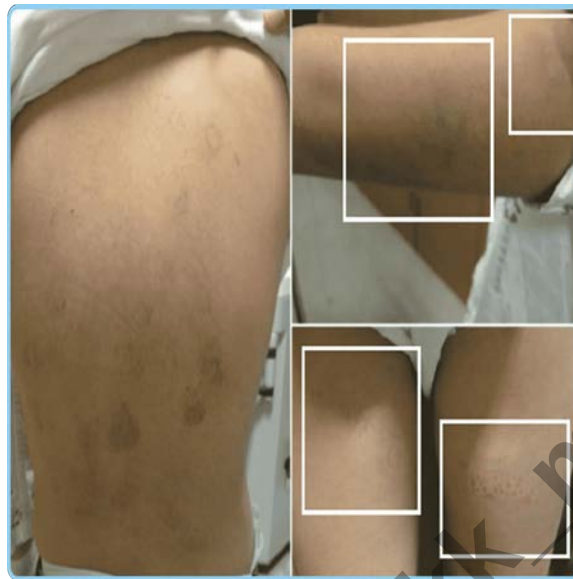
syndrome, or parent-infant traumatic stress syndrome .

Features Raising Suspicion of Abuse:

- **Inconsistent History:** Parents may provide vague explanations for injuries, such as a fall, which do not match the type or timing of the injuries.
- **Delayed Medical Care:** Parents often seek medical help only when the child's condition worsens.
- **Injuries in Various Stages:** Multiple injuries at different healing stages are often present.

Parental Admission: Parents may later confess to mild punishment but downplay the severity.





- Surface Injuries: Bruises, abrasions, and lacerations, including characteristic labial frenulum tears.
- CNS Injuries: Encephalopathy, retinal hemorrhages, and subdural hematoma (SDH), indicative of shaken baby syndrome.
- Eye Injuries: Retinal hemorrhages and possible lens dislocation.
- Visceral Injuries: Damage to the spleen, liver, or intestines.
- Burns: Cigarette burns or scalds from hot liquids are often found.
- Skeletal Injuries: Fractures (transverse, spiral, metaphyseal chip fractures), Rib fractures and the characteristic "string of beads" appearance on X-rays are common.
- Cardiovascular Injuries: Rib fractures, lung and heart contusions.

- Genitourinary Injuries: Consider sexual abuse.

Osteogenesis imperfecta (Option A) causes frequent fractures but does not account for bruising, burns, or intracranial hemorrhage.

Munchausen syndrome by proxy (Option C) involves a caregiver fabricating or inducing illness in a child, but the repeated physical injuries in this case make direct physical abuse more likely.

Hemophilia (Option D) could explain bruising but not the presence of fractures and burn marks .

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Asphyxial Death

1. A 30 y/o man is found dead after an incident where he fell into cold water. Witnesses report that he surfaced briefly, then collapsed and became unresponsive. No significant water is found in his lungs upon autopsy, but signs of cardiac arrest are evident. Which of the following is the most likely cause of death?

- A. Wet drowning
- B. Hydrocution
- C. Dry drowning
- D. Near drowning

2. A 30 y/o male is brought to the emergency department after being found hanging from a ceiling fan. On physical examination, which of the following findings is most indicative of antemortem hanging?

- A. Protrusion of the tongue
- B. Dribbling of saliva
- C. Subconjunctival hemorrhages
- D. Blue lips and oral mucous membrane

3. A 45 y/o man is found unconscious in an alley with marks around his neck suggesting strangulation. The police suspect the use of a ligature tightened rapidly with a lever. Which of the following types of strangulation best describes this scenario?

- A. Throttling
- B. Mugging

- C. Banded
 - D. Garroting
-

4. A 28 y/o construction worker is trapped under a heavy beam and found unresponsive. He has deep purple discoloration above the compression point, pale skin below, and petechiae on his face and neck. He is pronounced dead at the scene. What type of asphyxia is most likely?

- A. Traumatic Asphyxia
 - B. Positional Asphyxia
 - C. Drowning
 - D. Gagging
-

5. Which of the following statements regarding the given image is correct?



- A. It can cause death through asphyxia and vagal inhibition, requiring at least 2 minutes of pressure.

- B. Results in minimal visible injuries on the neck.
- C. Key postmortem findings include only superficial bruises and no internal injuries.
- D. Suicidal death is a possibility.

6. A forensic pathologist examines a drowning victim's dead body and conducts tests for diatoms and chloride levels in the heart. The victim was found in a lake, and a water sample from the scene was collected for analysis. Which of the following statements is false ?

- A. In freshwater drowning, chloride levels in the left heart are lower.
- B. In saltwater drowning, chloride levels in the left heart are higher.
- C. Finding diatoms in bone marrow indicates that the individual ingested water while alive.
- D. The absence of diatoms in tissues can definitively rule out drowning.

7. Match the following signs of drowning. 1. Sehrt's Sign a) Free liquid in the paranasal sinus. 2. Sveshnikov's sign b) Water in the middle ear due to violent respiratory efforts while drowning. 3. Sabinsky's Sign c) Micro rupture in gastric mucosa due to stretching. 4. Ueno's Sign d) Small and anaemic spleen due to vasoconstriction.

1. Sehrt's Sign	a) Free liquid in the paranasal sinus.
2. Sveshnikov's sign	b) Water in the middle ear due to violent respiratory efforts while drowning.
3. Sabinsky's Sign	c) Micro rupture in gastric mucosa due to stretching.
4. Ueno's Sign	d) Small and anaemic spleen due to vasoconstriction.

- A. 1-d, 2-b, 3-a, 4-c
- B. 1-c, 2-a, 3-d, 4-b
- C. 1-a, 2-c, 3-b, 4-d

D. 1-b, 2,c, 3-d, 4-a

8. A forensic pathologist is conducting an autopsy on a body recovered from a body of water. During the examination, several findings are noted. Based on the following statements regarding postmortem findings in drowning, which of the following is true? Frothy fluid, resulting from the churning of water, mucus, and air, is present at the mouth and nostrils, indicating drowning. Cutis anserina indicates prolonged immersion in warm water. Washerwoman's hand appears as wrinkled and sodden skin due to prolonged submersion. A cadaveric spasm is the immediate stiffening of muscles holding debris, indicating drowning. Paltauf's hemorrhage is a postmortem finding not associated with drowning. Water and bleeding in the middle ear confirm antemortem drowning. The heart shows equal amounts of blood on both sides in drowning victims.

- A. 2, 5, 6, 7
 - B. 1, 3, 4, 6
 - C. 1, 4, 5, 6
 - D. 2, 3, 5, 7
-

9. A 25 y/o man is found dead near a water body after being submerged for an unknown period. Autopsy reveals ballooned, pale pink lungs that retain their shape and do not collapse after removal. Blood tests show significant hyponatremia and hyperkalemia. Which of the following best describes his condition?

- A. Dry drowning
 - B. Seawater drowning
 - C. Freshwater drowning
 - D. Hydrocution
-

10. A 3-month-old infant is found unresponsive in bed next to a sleeping adult who has a strong odor of alcohol. The infant's nose appears flattened, and there are pale areas around the face. Which type of suffocation is most likely responsible for the infant's death?

- A. Burking
 - B. Overlaying
 - C. Traumatic Asphyxia
 - D. Positional Asphyxia
-

11. Which of the following methods of suffocation is most commonly associated with a reflex vagal inhibition that can lead to sudden cardiac arrest?

- A. Smothering
 - B. Gagging
 - C. Choking
 - D. Café Coronary
-

12. A 35 y/o man is found unresponsive in his apartment. The scene suggests possible asphyxia due to a hanging. Upon examination, the forensic team notes several physical findings. Which of the following findings would most likely not be observed in this case of asphyxial death?

- A. Tardieu spots
 - B. Bright red discolouration of the skin
 - C. Cyanosis
 - D. Congestion of viscera
-

13. Which of the following findings is most characteristic of ligature strangulation?

- A. Distinct bruising of subcutaneous tissues beneath the ligature
 - B. Presence of Tardieu's spots primarily on the chest and abdomen
 - C. Significant cervical vertebrae fracture
 - D. Absence of postmortem staining above the ligature
-

14. What is a common characteristic of sexual asphyxia that differentiates it from judicial hanging?

- A. Intent to cause death
 - B. Use of elaborate setups and self-release mechanisms
 - C. Resulting cervical spine fracture
 - D. Presence of distinct ligature marks
-

15. A forensic team is called to investigate a case of judicial hanging in an obese individual. The victim was found suspended from a significant height with the knot positioned at the back of the neck. During the autopsy, a transverse tear in the carotid artery is observed, consistent with a long drop hanging. Which of the following findings is most likely associated with this case?

- A. Dribbling of saliva.
 - B. Ligature marks.
 - C. Amussat's Sign
 - D. Simon's Sign
-

16. Which of the following is NOT a cause of early death in hanging?

- A. Asphyxia
 - B. Brain Infarction
 - C. Venous Congestion
 - D. Reflex Vagal Inhibition
-

17. Match the following neck structure compressed with their respective pressures needed to cause death in hanging. 1.Jugular Veins a) 10 kg 2.Carotid Arteries b) 30 kg 3.Trachea c) 2 kg 4.Vertebrae Arteries d) 15 kg

1.Jugular Veins	a) 10 kg
2.Carotid Arteries	b) 30 kg
3.Trachea	c) 2 kg
4.Vertebrae Arteries	d) 15 kg

- A. 1-d, 2-a, 3-c, 4-b
 - B. 1-c, 2-b, 3-a, 4-d
 - C. 1-b, 2-c, 3-d, 4-a
 - D. 1-c, 2-a, 3-d, 4-b
-

18. In which type of hanging is the knot positioned at the nape of the neck?

- A. Incomplete Hanging
 - B. Typical Hanging
 - C. Atypical Hanging
 - D. Near-Hanging
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	1
Question 5	1
Question 6	4
Question 7	2
Question 8	2
Question 9	3
Question 10	2
Question 11	4
Question 12	2
Question 13	1
Question 14	2
Question 15	3
Question 16	2
Question 17	4
Question 18	2

Solution for Question 1:

Correct Answer: B) Hydrocution

Explanation: Hydrocution (immersion syndrome) occurs when sudden exposure to cold water triggers vagal inhibition, leading to cardiac arrest without significant water aspiration.

- This fits the clinical scenario where the man collapsed shortly after cold water immersion.

Type of Drowning	Features
Dry Drowning (Option C ruled out)	Definition : Death from laryngeal spasm with minimal water entering the lungs. Mechanism : Asphyxia occurs without significant water inhalation. Diagnosis is made by inference and exclusion, as no anatomical evidence is present. Occurrence : May occur with relatively dry lungs if circulation continues briefly after water removal. Resuscitative efforts may lead to water absorption into hypertonic plasma, keeping lungs distended. Laryngeal spasm may prevent further water entry, allowing circulatory function to clear intra-alveolar fluid.
Hydrocution (Immersion Syndrome)	Definition : Sudden death due to cardiac arrest from vagal inhibition caused by cold water exposure. Mechanism : Cold water stimulates nerve endings on the skin. Water impact on the epigastrium can trigger the vagal reflex. Cold water in the ear, nasal passages, pharynx, and larynx stimulates mucosal nerve endings. Risk Factors : Inexperienced divers or those diving feet first. Alcohol consumption increases the risk due to vasodilation.
Near/Post Immersion Syndrome (Secondary Drowning) (Option D ruled out)	Definition : Refers to a submersion victim resuscitated and surviving for 24 hours. Complications : Hypoxemia leads to potential brain damage. Electrolyte disturbances and pulmonary oedema. Haemoglobinuria, sepsis-pneumonia, metabolic acidosis, and chemical pneumonitis. Cerebral oedema, cardiac arrhythmias, and myocardial anoxia.

Wet drowning (Option A) : In wet drowning, water would enter the lungs, but in this scenario, no significant amount of water was found in the lungs.

Solution for Question 2:

Correct Answer: B) Dribbling of saliva

Explanation:

Dribbling of saliva occurs due to pressure and friction from the ligature on the submandibular glands during life, indicating antemortem hanging.

Post-mortem Findings in Hanging:

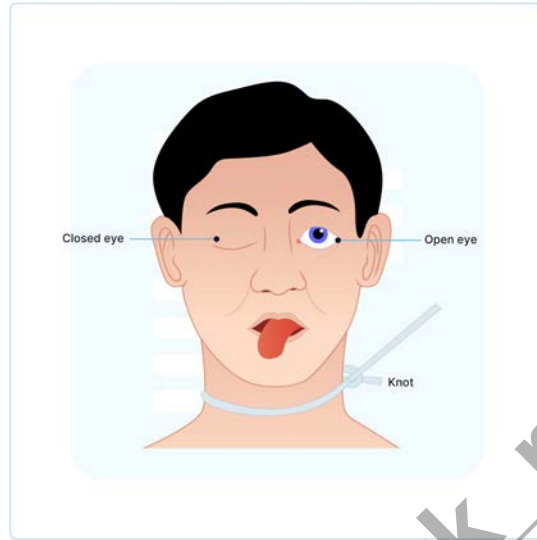
External Findings:

Finding	Description
Face	Swollen and cyanosed due to impaired venous return. Prominent eyeballs from increased pressure. Dilated pupils (indicative of antemortem hanging). Subconjunctival hemorrhages. Protrusion and swelling of the tongue may show bite marks and small hemorrhages. Bleeding from nose and/or ears. Blue lips and oral mucous membranes. Dribbling of saliva indicates excessive salivation from ligature pressure (indicative of antemortem hanging). (Option B)
Neck	Ligature mark ('furrow') typically above the hyoid bone; size/shape depends on ligature material. The skin at the ligature site is initially pale, later yellowish-brown, dry, and parchment-like. Postmortem blisters may appear on the skin near the furrow. Neck veins above the furrow may be distended. The neck becomes slender and elongated due to suspension. Flexed neck to the side opposite the knot.

Other Body Parts	Tardieu's spots on the forehead, eyelids, conjunctiva, and temples. Cyanosis of fingernails. Purple postmortem staining in lower limbs (glove and stocking pattern). Clenched hands. Penile turgidity and involuntary semen discharge in males. Involuntary fecal and urine discharge in both sexes. Signs of asphyxia may be absent in complete hanging.

La Facies Sympathique:

- Involves the sympathetic nervous system , which controls pupil dilation.
- Neck compression affects sympathetic pathways, leading to unilateral miosis (constriction of the pupil) on the side opposite the ligature.
- May also include ptosis (drooping of the eyelid) due to the same disruption in sympathetic innervation.



Protrusion of the tongue (Option A) occurs due to ligature pressure but is not specific to antemortem conditions.

Subconjunctival hemorrhages (Option C): This may occur post-mortem or due to other causes.

Blue lips and oral mucous membrane (Option D): This finding is related to hypoxia and does not specifically indicate antemortem conditions.

Solution for Question 3:

Correct Answer: D) Garroting

Explanation:

This method specifically involves using a ligature tightened rapidly by a lever, leading to a quick loss of consciousness.

Types of Strangulation:

Type	Description
------	-------------

Manual Strangulation (Throttling) (Option A ruled out)	Involves compression of the victim's neck using fingers, palms, or hands. About 50% of deaths result from vagal inhibition due to pressure on the carotid nerve plexus. Pressure needs to be applied for at least 2 minutes for death to occur.
Mugging (Option B ruled out)	Involves compressing the victim's neck using the bend of the elbow or knee. Typically a surprise attack from behind, often leaving minimal injury marks. Also known as a "chokehold," which is prohibited in wrestling due to its danger.
Bansdola (Option C ruled out)	A brutal method using a bamboo stick or similar object to crush the neck. Involves tying sticks across the neck or pressing a foot or knee against the throat while lying down. Leaves a distinct bruise across the trachea corresponding to the width of the object used.
Garroting (Option D)	Uses a ligature that is tightened by twisting with a lever (Spanish windlass). Causes rapid loss of consciousness and collapse. Historically, it has been used as a method of execution in countries like Spain, Portugal, and Turkey.

Solution for Question 4:

Correct Answer: A) Traumatic Asphyxia

Explanation:

The scenario describes respiratory arrest due to mechanical fixation from the heavy beam, leading to marked congestion and discoloration above the compression level (purple skin), along with petechiae, which are classic indicators of traumatic asphyxia.

Features	Traumatic Asphyxia	Positional Asphyxia (Option B ruled out)
Definition	Respiratory arrest due to mechanical fixation of the chest by unyielding substances or external pressure. (Perthe's syndrome)	Asphyxia results from the body being in a position that impairs breathing, often due to neck twisting or compression of the trachea.
Mechanism	Prevents normal chest wall movements, leading to respiratory failure. Often associated with significant compression of the chest and abdomen	Abdominal weight compresses the diaphragm, hindering thoracic organ function and respiratory movement. Impaired venous return to the heart can contribute to asphyxia.
Common Causes	Crowded situations leading to trampling Crushing accidents (e.g., falls of earth/stones in mines, building collapses). Heavy objects (e.g., vehicles or machinery) pressing on the body.	Physical restraint in a prone position (face down). Falling and wedging in a well or narrow space. Indirect compression (jack-knife position) where thighs and knees push against the chest.
Postmortem Findings	Intense congestion and petechial hemorrhages in the head, neck, and upper chest. Skin colour above the compression level: deep purple or purple-red; below this level: pale or mildly cyanosed.- Masque ecchymotique Swelling, congestion, and petechiae on the face, lips, and scalp.	Marked congestion and cyanosis in the face and neck. Presence of petechiae on the face.

Choking (Option C) : While choking involves airway obstruction, the details of the scenario do not suggest that the patient is choking on an object, as the cause of death is related to external pressure from the heavy beam.

Gagging (Option D) involves airway obstruction due to something blocking the throat or mouth, but in this case, the cause of death is due to external chest compression from the heavy beam.

Solution for Question 5:

Correct Answer: A) It can cause death through asphyxia and vagal inhibition, requiring at least 2 minutes of pressure.

Explanation:

The given image describes Throttling (manual strangulation) .

- Throttling results in death through mechanisms like asphyxia (obstruction of the airway) and vagal inhibition (pressure on the carotid sinus leading to a drop in heart rate and blood pressure).
- The requirement of applying pressure for a minimum of 2 minutes to cause death is consistent with the physiological effects of throttling.

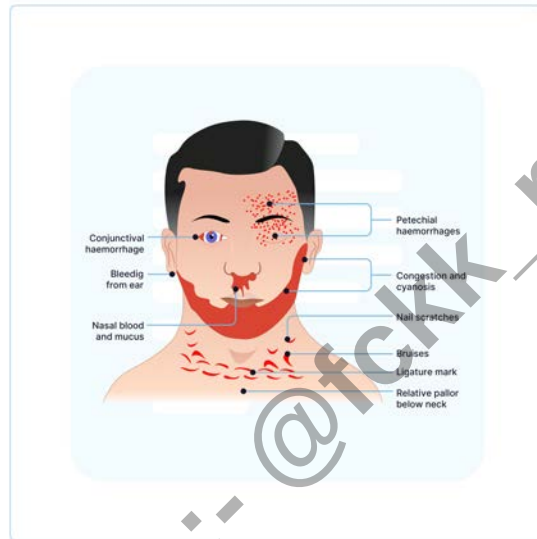
Throttling (Manual Strangulation):

- Definition : A method of strangulation where the assailant compresses the victim's neck with their hands.
- Mechanisms of Death : Asphyxia : Obstruction of the airway due to neck pressure. Cerebral Anoxia : Restricted blood flow to the brain due to pressure on blood vessels, leading to potential brain damage. Vagal Inhibition : Stimulation of the vagus nerve can cause a sudden drop in heart rate and blood pressure, potentially resulting in cardiac arrest.
- Asphyxia : Obstruction of the airway due to neck pressure.
- Cerebral Anoxia : Restricted blood flow to the brain due to pressure on blood vessels, leading to potential brain damage.
- Vagal Inhibition : Stimulation of the vagus nerve can cause a sudden drop in heart rate and blood pressure, potentially resulting in cardiac arrest.
- Fatality Requirement : Sustained pressure for at least 2 minutes is typically required for lethality.
- Asphyxia : Obstruction of the airway due to neck pressure.
- Cerebral Anoxia : Restricted blood flow to the brain due to pressure on blood vessels, leading to potential brain damage.
- Vagal Inhibition : Stimulation of the vagus nerve can cause a sudden drop in heart rate and blood pressure, potentially resulting in cardiac arrest.

Postmortem Findings:

Examination Type	Findings
External Examination (Option B ruled out)	Bruises and Abrasions: On the front and sides of the neck. Signs of Asphyxia: Cyanosis, facial swelling, Tardieu's spots, bleeding from nose and ears. Injuries on Other Body Parts: Additional signs of violence (abrasions, bruises, scratches).

Internal Examination (Option C ruled out)	Haemorrhages: In subcutaneous tissues under bruises. Muscle Damage: Tears or lacerations in neck muscles. Hyoid Bone Fracture: Significant diagnostic indicator, typically from inward compression. Laryngeal and Tracheal Damage: Fractures of thyroid and potentially cricoid cartilages. Haemorrhages in Mouth/Throat: Bruising and lacerations on the tongue, haemorrhages in mucous membranes.
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Medico-legal Questions:

- Determining Manner of Death : Suicidal Throttling : Considered impossible; hands would relax upon unconsciousness. (Option D ruled out) Homicidal Throttling : Presumed unless proven otherwise; common in homicides, especially involving vulnerable victims. Accidental Throttling : Rare, may occur during playful or affectionate contact.
- Suicidal Throttling : Considered impossible; hands would relax upon unconsciousness. (Option D ruled out)
- Homicidal Throttling : Presumed unless proven otherwise; common in homicides, especially involving vulnerable victims.
- Accidental Throttling : Rare, may occur during playful or affectionate contact.
- Suicidal Throttling : Considered impossible; hands would relax upon unconsciousness. (Option D ruled out)
- Homicidal Throttling : Presumed unless proven otherwise; common in homicides, especially involving vulnerable victims.
- Accidental Throttling : Rare, may occur during playful or affectionate contact.

Solution for Question 6:

Correct Answer: D) The absence of diatoms in tissues can definitively rule out drowning.

Explanation:

The absence of diatoms does not definitively exclude drowning; other factors may contribute to their presence in the body.

Aspect	Diatom Test	Gettler Test
Features	Diatoms are microscopic unicellular algae with silicon skeletons that are chemically inert and resistant to strong acids. In antemortem drowning, diatoms enter the bloodstream through ruptured alveolar walls, reaching organs like the lungs, heart, brain, liver, and bone marrow. Diatoms resist putrefaction, making the test useful even in decomposed bodies. A 50g tissue sample is treated with nitric acid and examined for diatoms. A water sample from the drowning site is collected for comparison of diatom species and proportions. Limitations : Possible contamination from food, air, or environment. The test cannot rule out contamination, but it is still the "gold standard" for diagnosing typical drowning.	Normal Chloride Levels: The chloride content in the right and left sides of the heart is typically similar, around 600 mg/100 ml. A 25 mg% or more difference in chloride levels between the two sides suggests antemortem drowning.
Interpretation	Presence: Finding diatoms in tissues (bone marrow , blood, etc.) strongly indicates antemortem drowning, suggesting the individual inhaled diatom-rich water. Detection in multiple tissues enhances reliability, indicating systemic absorption and confirming drowning. (Option C ruled out) Absence: While the absence of diatoms does not rule out drowning, it raises questions about the death's circumstances. Prior water ingestion or contact with diatom-rich environments could allow diatoms to enter circulation without drowning.	Freshwater Drowning: The chloride content in the left heart is lower than in the right heart due to dilution from the water. (Option A ruled out) Saltwater Drowning: In saltwater drowning, the chloride content in the left heart is higher than in the right heart due to hemoconcentration and mixing with saltwater. (Option B ruled out) No Change in Chloride Content: Observed in deaths from laryngeal spasm, vagal inhibition, putrefaction, patent foramen ovale, or when the saline content of the drowning medium is similar to that of blood.

Solution for Question 7:

Correct Answer: B) 1-c, 2-a, 3-d, 4-b

Explanation:

Signs in Drowning	
Sehrt's Sign	Micro rupture in gastric mucosa due to over-stretching on ingestion of fluid.
Sveshnikov's sign	Free liquid in the paranasal sinus.
Sabinsky's Sign	Small and anemic spleen due to vasoconstriction.
Ueno's Sign	Water in the middle ear due to violent respiratory efforts while drowning.
Wydler's Sign	Gastric content in a container with three layers (Foam-top, liquid- middle, solid- bottom).
Drowning Index	Weight of both lungs + weight of pleural effusion / Weight of spleen If > 14.1%, it suggests AM drowning.

Reference:

Solution for Question 8:

Correct Answer: B) 1, 3, 4, 6

Explanation:

PM Findings in Drowning	
External Findings	Internal Findings

Pale & cyanosed face from putrefaction. Eyes half-open/closed, suffused conjunctiva, dilated pupils, subconjunctival hemorrhages (Antemortem drowning). Froth : Fine, copious, tenacious, lathery froth at mouth and nostrils is characteristic of antemortem drowning, and reappears after wiping or chest pressure. Mechanism: The churning of water, mucus, and retained air in airways creates froth. (Statement 1) In other conditions like strangulation, and electric shock, the froth is not as fine or tenacious. Cutis anserina (goose flesh): The puckered appearance of skin due to the contraction of erector muscles in cold water, occurs immediately after antemortem submersion or by rigor mortis. (Statement 2) Washerwomen's hand : Wrinkled, sodden, and bleached appearance of hands and feet due to prolonged submersion, occurring postmortem in 3–4 hours and fully affecting the hands by 24 hours. (Statement 3) Cadaveric spasm : Immediate stiffening of muscles holding grass, sand, or debris in clenched hands— antemortem proof of drowning. (Statement 4) Rigor mortis appears early after a violent struggle (Antemortem drowning).

Lungs : Ballooning with frothy fluid, rib indentations, heavy and waterlogged. Paltau's hemorrhage : Red-gray mottling with areas of hemorrhage due to effusion of blood along interlobular septa. (Statement 5) Heart : Right side full of dark venous blood, left side empty (a sign of asphyxia). (Statement 7) Gettler test: Measures chloride content to differentiate antemortem drowning. Freshwater: Lower chloride in the left heart. Saltwater: Higher chloride in the left heart. Stomach : Water is present in 70% of cases, indicating antemortem drowning if liquid was not swallowed voluntarily. Brain : Congested with softened grey-white junction (Antemortem). Middle ear : The presence of water and bleeding in the middle ear confirms antemortem drowning. (Statement 6) Diatom test : The presence of diatoms in lungs and bone marrow suggests antemortem drowning.

Solution for Question 9:

Correct Answer: C) Freshwater drowning

Explanation: The scenario suggests Freshwater drowning which is characterized by pale, ballooned lungs that retain their shape and electrolyte disturbances like hyponatremia and hyperkalemia due to haemodilution.

- Wet Drowning : Rapid death from cardiac arrest due to water entering the lungs.
- Dry Drowning : This happens due to laryngospasm without water entering the lungs.
- Near/Delayed/Secondary Drowning : Symptoms occur after initial recovery.
- Hydrocution/Immersion Syndrome : Physiological responses due to sudden immersion in cold water.

The 2 types of Wet drowning are

Features	Freshwater Drowning	Seawater Drowning (Option B ruled out)
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Pathophysiology	Haemodilution : Inhalation increases blood volume by 50%, causing anemia, hemolysis, and myocardial hypoxia. Electrolyte Changes : Marked hyponatremia and hyperkalemia; calcium decrease leads to arrhythmias. Surfactant Dysfunction : Freshwater denatures the surfactant which causes alveolar collapse and pulmonary oedema. Cardiac Load : Increased pulmonary oedema and electrolyte imbalances decrease cardiac output and cause hypotension.	Haemoconcentration : High salinity ($\geq 3\%$ NaCl) draws water into the lungs, causing severe pulmonary oedema and elevated plasma sodium. Electrolyte Changes : Increased blood viscosity leads to bradycardia and myocardial failure; altered potassium and sodium levels(hyponatremia) Red Cell Changes : Crenation of red blood cells impairs oxygen transport. Myocardial Anoxia : Asphyxia from pulmonary oedema and cardiac stress results in hypoxia and potential arrest.
Lung characteristics	Size : Ballooned but light; pale pink colour; emphysematous consistency. Shape after Removal : Retained, no collapse; crepitus with little froth and no fluid.	Size : Ballooned and heavy; purplish or bluish colour; soft, jelly-like consistency. Shape after Removal : Flattened; no crepitus; copious fluid and froth.
Fatal period	Death typically occurs within 4-5 minutes	Death may take 8-10 minutes.

Dry drowning (Option A) : In dry drowning, water does not enter the lungs, so the autopsy finding of ballooned lungs filled with fluid does not match.

Hydrocution (Option D) : Hydrocution involves sudden cardiac arrest upon immersion, without the lung findings or electrolyte imbalances seen in this case.

Solution for Question 10:

Correct Answer: B) Overlaying

Explanation: Overlaying correctly describes the scenario where the infant is suffocated due to the weight of the adult sleeping beside them. The signs of flattened facial features and pale appearance further support this diagnosis.

Overlaying	Burking	Wedging
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Definition : Compression suffocation due to the weight of an adult (e.g., mother) on an infant during sleep. Mechanism : Limits thoracic movements, reducing or preventing respiratory exchange. Commonly occurs when the adult is under the influence of alcohol. Findings : May show flattening of the nose and face. The pale appearance of the affected areas; nostrils may be filled with frothy, possibly blood-stained fluid. Typical signs of asphyxia. Cause of Death : May result from neurogenic cardiac arrest	Definition : A method of homicidal smothering and traumatic asphyxia. Historical Context : Named after William Burke and William Hare, who killed 16 people in the late 1820s in Edinburgh for anatomical dissection. Method : Victims were lured, given alcohol, and then subdued by kneeling or sitting on their chests to close their mouths and noses. One accomplice would drag the victim's body, aiding in concealment. (Option A ruled out)	Definition : Suffocation caused by the body being compressed between two firm surfaces. Common Observations : Often seen in children. Mechanism : Prevents normal breathing due to the physical pressure from both sides.
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Traumatic Asphyxia (Option C) : This is generally associated with external pressure from heavy objects rather than the body weight of a person.

Positional Asphyxia (Option D) : This involves impaired breathing due to body position, but the scenario suggests external compression from an adult, making overlaying more likely.

Solution for Question 11:

Correct Answer: D) Café Coronary

Explanation:

Café Coronary is a condition characterized by sudden collapse and death while eating, often in intoxicated individuals.

- The high blood alcohol content anesthetizes the gag reflex, and reflex vagal inhibition can lead to a sudden cardiac arrest.
- Definition : Asphyxia where air entry to the lungs is prevented by means other than pressure on the neck or drowning.
- Caused by breathing in a vitiated atmosphere (deficient in oxygen).

Methods of Suffocation	Features
------------------------	----------

Smothering (Option A ruled out)	Definition : Closing the nose and mouth using hands or foreign substances (e.g., mud, cloth). Autopsy Findings : External : Pallor may be present around the mouth/nose, pressure marks, distinct nail marks, petechiae, contusion on the inner lip, and stains on the bedding. Internal : Potential internal organ haemorrhages.
Gagging (Option B ruled out)	Definition : Asphyxia caused by a cloth or material obstructing the mouth and nose. Mechanism : A gag can block the pharynx, initially allowing nasal airflow. Obstruction worsens due to saliva and mucus, leading to complete airway blockage. Symptoms : Facial congestion. Fine petechiae on the face, sclerae, and conjunctivae. Intent : Always homicidal.
Choking (Option C ruled out)	Definition : Airway obstruction due to an object lodged in the hypopharynx. Accidental Choking : Often occurs during meals, especially while laughing or crying; vomited matter may be inhaled under intoxication or unconsciousness. Immediate Symptoms : Acute respiratory distress; complications may develop later. Autopsy Findings : Thickened epiglottis and folds Mucus accumulation in the throat. Lungs exhibit interalveolar oedema, congestion, and haemorrhages.
Café Coronary	Definition : Sudden collapse and death while eating, often in intoxicated individuals. Symptoms : Cyanosis and violent coughing. Collapse without significant distress. Cause of Death : Reflex vagal inhibition leading to sudden cardiac arrest; high blood alcohol content anaesthetises the gag reflex. A poorly chewed food bolus may obstruct the larynx. Autopsy Findings : Obstruction may be found in the mouth, larynx, or trachea, with food particles in mucus in the trachea/bronchi, differentiating from cases of post-mortem oesophageal food forcing. Signs of asphyxia may be absent. Treatment: Heimlich Maneuver : Stand behind the patient, interlock fingers, and apply pressure to the stomach.

Solution for Question 12:

Correct Answer: B) Bright red discoloration of the skin

Explanation:

Bright red discolouration of the skin is not a typical feature of asphyxia; instead, asphyxia often leads to cyanosis (bluish discolouration).

Asphyxia:

Asphyxia is a condition resulting from insufficient oxygen reaching the tissues, leading to hypoxia and potentially death.

Signs of Asphyxial Death:

- Tardieu Spots (Tardieu's or Bayard's Ecchymoses): (Option A ruled out) Small, round, dark-red spots (pin-head sized). Typically found in areas with less support for capillaries: conjunctiva, face, epiglottis, subpleural lung surfaces, heart, meninges, and thymus. More visible in fair-skinned individuals and fresh bodies; fades with putrefaction. Not exclusive to asphyxia; can occur in other conditions such as electrocution, poisoning, coronary thrombosis, and bleeding disorders.
- Small, round, dark-red spots (pin-head sized).
- Typically found in areas with less support for capillaries: conjunctiva, face, epiglottis, subpleural lung surfaces, heart, meninges, and thymus.
- More visible in fair-skinned individuals and fresh bodies; fades with putrefaction.
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- More visible in fair-skinned individuals and fresh bodies; fades with putrefaction.
- Not exclusive to asphyxia; can occur in other conditions such as electrocution, poisoning, coronary thrombosis, and bleeding disorders.



- Cyanosis: (Option C ruled out) Characterized by a bluish discoloration due to decreased oxygen tension in blood and increased reduced haemoglobin (≥ 5 g/dL). Commonly observed in lips, nose, ear lobules, and internally in lungs, meninges, liver, spleen, and kidneys.
- Characterized by a bluish discoloration due to decreased oxygen tension in blood and increased reduced haemoglobin (≥ 5 g/dL).
- Commonly observed in lips, nose, ear lobules, and internally in lungs, meninges, liver, spleen, and kidneys.
- Congestion of viscera (Option D ruled out)
- Characterized by a bluish discoloration due to decreased oxygen tension in blood and increased reduced haemoglobin (≥ 5 g/dL).
- Commonly observed in lips, nose, ear lobules, and internally in lungs, meninges, liver, spleen, and kidneys.

Solution for Question 13:

Correct Answer: A) Distinct bruising of subcutaneous tissues beneath the ligature

Explanation:

In ligature strangulation, the ligature compresses the neck, leading to trauma and bruising of the subcutaneous tissues and muscles underneath the ligature.

- This bruising is a direct result of the force applied by the ligature, which can cause small blood vessels to rupture, resulting in bleeding into the surrounding tissues.

Causes of Death in Ligature Strangulation:

Postmortem Findings:

External Examination:

Finding	Description
Face	Congested, swollen, and cyanosed. Tardieu's spots (petechiae) on the forehead, temples, eyelids, and conjunctiva. (Option B ruled out) Prominent eyes with congested conjunctiva, dilated pupils, and possible subconjunctival haemorrhages. Cyanosis of lips, fingernails, and earlobes. Postmortem staining above the ligature. (Option D ruled out) Swollen, dark-coloured, protruding tongue, possibly bitten. Bloodstained frothy fluid and mucus in mouth and nostrils.
Neck	Ligature mark ('furrow'): well-defined groove, typically the same width as the ligature, often horizontal. More prominent on the front and sides of the neck. Red base with congested or ecchymosed margins; imprint pattern may be present. Can be indistinct if the ligature is soft; it may appear as a deep groove with a narrow cord ('cheese cutter method'). May show an oblique appearance if the victim was dragged post-strangulation.

Other Body Parts	Deep, prominent postmortem staining. Involuntary discharge of urine and faecal matter; less common seminal ejaculation. Clenched hands; possible genital turgidity. Possible abrasions, fingernail marks, and contusions on face, and arms, indicating struggle. Horizontal, irregular abrasions on the neck may indicate attempts to remove ligature.

Internal Examination:

- On dissection- Contusion and haemorrhage- of tissue, muscles of the neck, tonsils and glands.
- Fracture of the thyroid cartilage is commonly seen.
- Fracture of the hyoid bone is not very common as it is usually at or below the level of the thyroid.

Feature	Hanging	Strangulation
Face	Usually, pale, petechiae are less common	Congested, livid with petechiae
Tongue swelling and protrusion	Less marked	More marked
Bleeding from nose, ears, mouth	Less common	More common
Ligature mark ('furrow') Direction Continuity Level in the neck Base	Oblique Noncontinuous Above thyroid Pale, hard, parchment-like	Transverse Continuous Below thyroid Soft and reddish
Abrasions and ecchymoses about the edges of the mark	Not common	Common

Involuntary discharge of feces and urine	Less common	More common
Involuntary discharge of seminal fluid	More common	Less common
Bruising of neck muscles	Less common	More common
Subcutaneous tissue under the mark	White, hard, glistening	Ecchymosed
Hyoid bone fracture	May occur	Uncommon
Thyroid fracture	Unlikely	More common
Larynx and trachea fracture	Unlikely	May be found
Emphysematous bullae on lungs	Sometimes present	Very common
Carotid arteries	Damage may be seen in the intima	Rare
Design	History of previous unsuccessful attempts of suicide may be available	No such history is available
Signs of struggle	Absent	Always present, unless taken unaware

Significant cervical vertebrae fracture (Option C): This is more characteristic of manual strangulation or hanging, where greater force is applied. Ligature strangulation typically does not cause vertebral fractures.

Solution for Question 14:

Correct Answer: B) Use of elaborate setups and self-release mechanisms

Explanation:

This characteristic is typical of sexual asphyxia, where individuals may create complex arrangements to induce hypoxia while retaining a release method.

Sexual Asphyxia vs. Judicial Hanging:

Similarities:

Aspect	Description
--------	-------------

Asphyxia Mechanism	Both involve asphyxia caused by a ligature around the neck.
External Findings	May include a ligature mark, cyanosis, and petechiae.

Differences:

Aspect	Sexual Asphyxia (Autoerotic Asphyxia)	Judicial Hanging
Intent (Option A ruled out)	Achieve sexual gratification; death is accidental.	Capital punishment; is intended to cause death.
Execution	Involves elaborate setups, bondage, and self-release mechanisms. (Option B)	Highly controlled; specific knot placement (submental) and calculated drop height.
Injuries	Usually less severe; aims to induce hypoxia without fatal damage.	Designed to cause rapid death through forceful cervical spine fracture (hangman's fracture-spondylolisthesis of C2 over C3).
Ligature Mark (Option D ruled out)	May be less prominent due to padding; often not distinctive.	Typically distinct, located below the chin or ear.
Cervical Spine Fracture (Option C ruled out)	Rarely occurs.	A hallmark feature.
Scene Findings	Evidence of sexual activity is often present.	Takes place in a controlled environment; no signs of sexual activity.

Solution for Question 15:

Correct Answer: C) Amussat's Sign

Explanation:

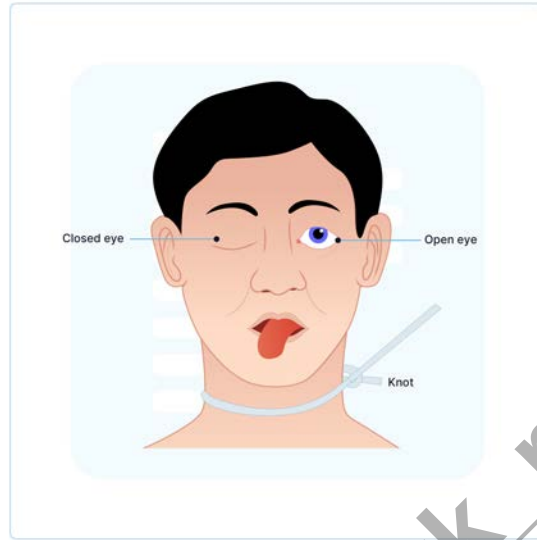
The scenario describes the typical presentation of Amussat's Sign, which is characterized by transverse carotid artery tears, particularly in cases of long drop hangings with posteriorly placed knots, especially in obese victims.



- While dribbling of saliva and ligature marks are significant, Amussat's Sign specifically indicates carotid artery injuries in hanging.
- Simon's Sign indicates hemorrhages in the lumbar spine but is not directly linked to the mechanics of hanging.

Dribbling of Saliva (Option A):

- Dribbling of saliva is the surest sign of antemortem hanging.
- Excessive salivation occurs while the person is alive due to pressure and friction from the ligature on the submandibular glands.
- Saliva dribbles from the angle of the mouth opposite to the side of the knot.
- If the knot is at the nape of the neck, saliva dribbles across the middle of the lower lip.
- While this is commonly seen in hangings, it does not indicate the specific injury (carotid artery tears) described in the scenario.
- Le facie sympathique is a vital sign in hanging characterized by unilateral miosis and possibly ptosis on the side opposite the ligature.



Ligature marks (Option B) are typically found above the hyoid bone and indicate constriction caused by the ligature.

- Although important, these marks primarily show the location and type of ligature used rather than the mechanics of suspension described in the scenario, such as long drops and carotid artery injury, which are more characteristic of Amussat's Sign.

Simon's Sign (Option D) refers to fractures of the anterior vertebral bodies seen in hanging cases where there is a significant drop or fall.

- While related to long-drop hangings, it refers to skeletal injury, not the vascular damage (carotid tears) described in this case.



Simon's bleeding

Bleeding on the anterior aspects of the intervertebral discs of the lumbar region - Simon's bleeding.

Solution for Question 16:

Correct Answer: B) Brain Infarction

Explanation:

Brain infarction is associated with delayed deaths rather than early deaths by hanging.

- Early death typically results from immediate causes such as asphyxia, venous congestion, or reflex vagal inhibition.
- Brain infarction is a later complication that may occur due to reduced blood flow or other factors following the hanging incident.

Death in Hanging

The usual period for death is 3–5 minutes and this can extend to 5–8 minutes of suspension leading to death.

- Occurs if cervical vertebrae are fractured.
- This can happen if there is inhibition of the heart.
- Usually results from asphyxia.
- If coma is the cause, death occurs more slowly.

Causes of Death from Hanging

Delayed Deaths

Rarely, delayed deaths may result from:

- Aspiration pneumonia
 - Edema of the lungs or larynx
 - Infections
 - Brain infarction (Option B)
 - Hypoxic encephalopathy
 - Brain abscess
-

Solution for Question 17:

Correct Answer: D) 1-c, 2-a, 3-d, 4-b

Explanation:

Neck structure compressed	Compression Pressure	Result
Jugular Veins	2 kg	Venous congestion leading to cessation of cerebral circulation
Carotid Arteries	2.5 - 10 kg	Cerebral anemia/ anoxia
Trachea	15 kg	Asphyxia
Vertebral Arteries	8 - 30 kg	Cerebral anemia/ anoxia

Solution for Question 18:

Correct Answer: B) Typical Hanging

Explanation:

Typical hanging refers to cases where the knot is centrally positioned over the occiput, specifically at the nape of the neck.

Hanging

- Hanging is a form of asphyxia resulting from the suspension of the body by a ligature around the neck, with the constricting force primarily being the body weight.
- Near-Hanging: Survivors of a hanging injury who reach the hospital are classified as near-hanging cases. (Option D ruled out)

Types of Hanging:

- Based on Knot Position: Typical Hanging: The knot is positioned centrally over the occiput, at the nape of the neck. Atypical Hanging: The knot is located anywhere other than the occiput. (Option C ruled out)
- Typical Hanging: The knot is positioned centrally over the occiput, at the nape of the neck.
- Atypical Hanging: The knot is located anywhere other than the occiput. (Option C ruled out)

- Based on the Degree of Suspension: Complete Hanging: The body is fully suspended, with no part touching the ground, and the constricting force is the weight of the body. Incomplete (Partial) Hanging: A part of the body (toes, feet) is touching the ground, or the individual is in a sitting, kneeling, lying, or prone position, with the weight of the head acting as the constricting force. (Option A ruled out)
- Complete Hanging: The body is fully suspended, with no part touching the ground, and the constricting force is the weight of the body.
- Incomplete (Partial) Hanging: A part of the body (toes, feet) is touching the ground, or the individual is in a sitting, kneeling, lying, or prone position, with the weight of the head acting as the constricting force. (Option A ruled out)
- Typical Hanging: The knot is positioned centrally over the occiput, at the nape of the neck.
- Atypical Hanging: The knot is located anywhere other than the occiput. (Option C ruled out)
- Complete Hanging: The body is fully suspended, with no part touching the ground, and the constricting force is the weight of the body.
- Incomplete (Partial) Hanging: A part of the body (toes, feet) is touching the ground, or the individual is in a sitting, kneeling, lying, or prone position, with the weight of the head acting as the constricting force. (Option A ruled out)



- Based on Intent: Suicidal Accidental Homicidal Autoerotic
- Suicidal
- Accidental
- Homicidal
- Autoerotic

- Types of Knots in Hanging Slip Knot (Running Noose): Created by attaching a rope to itself, forming a loop. The loop can be tightened to create a running noose. Binding Knot (Fixed Knot): The rope wraps around the neck at least once. Secured by tying the two ends of the rope together, forming a fixed noose.
 - Slip Knot (Running Noose): Created by attaching a rope to itself, forming a loop. The loop can be tightened to create a running noose.
 - Created by attaching a rope to itself, forming a loop.
 - The loop can be tightened to create a running noose.
 - Binding Knot (Fixed Knot): The rope wraps around the neck at least once. Secured by tying the two ends of the rope together, forming a fixed noose.
 - The rope wraps around the neck at least once.
 - Secured by tying the two ends of the rope together, forming a fixed noose.
 - Suicidal
 - Accidental
 - Homicidal
 - Autoerotic
 - Slip Knot (Running Noose): Created by attaching a rope to itself, forming a loop. The loop can be tightened to create a running noose.
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 - Secured by tying the two ends of the rope together, forming a fixed noose.
 - Created by attaching a rope to itself, forming a loop.
 - The loop can be tightened to create a running noose.
 - The rope wraps around the neck at least once.
 - Secured by tying the two ends of the rope together, forming a fixed noose.
-

Previous Year Questions

1. Identify the type of homicide:



- A. Mugging
- B. Garrotting
- C. Burking
- D. Banskola

2. A 24 year old man who was accused of murder was taken into police custody. As a part of the interrogation, the prisoner was kept in a prone position with both his wrists and ankles bound behind his back and secured with a rope. What is this method known as ?

- A. Choke/ Carotid hold
- B. Carotid steeper hold
- C. Hog tie

D. Bar arm

3. What is the term used to describe the act of a friend causing the death of another person by sitting on their chest and blocking their nose and mouth, following an argument over a shared girlfriend, while under the influence of alcohol?

- A. Burking
 - B. Smothering
 - C. Traumatic asphyxia
 - D. Overlaying
-

4. In a case where an autopsy is performed on a person who died by hanging, a ligature mark is observed on the lower one-third of the neck. The person's tongue appears to be protruding, and the head was found hanging to the left side with saliva dripping from the left corner of the mouth. Additionally, both pupils are dilated, and the right eye remains open. Which structure is most likely being compressed, leading to the right eye remaining open in this case?

- A. Right internal jugular vein
 - B. Cervical sympathetic chain
 - C. Left vagus nerve
 - D. Right internal carotid artery
-

5. What is the specific name of the test performed by the doctor during the autopsy where they tied the bronchus and submerged the lung in water to determine if it floats or sinks?

- A. Gettler's test

- B. Ploucquet's test
- C. Hydrostatic test
- D. Diatom's test

6. What is the cause of death in a case where a deceased individual has a ligature fully encircling the neck at a horizontal level below the thyroid, as observed during post-mortem examination, and no evidence of saliva dribbling?

- A. Throttling
- B. Ligature strangulation
- C. Gagging
- D. Hanging

7. What is this finding suggestive of:



- A. Drowning after Organophosphate poisoning

- B. Antemortem drowning
 - C. Drowning after Arsenic poisoning
 - D. None of the above
-

8. What type of strangulation method is the Spanish windlass?

- A. Bansdola
 - B. Garroting
 - C. Mugging
 - D. Throttling
-

9. A girl was found deceased in a lake and her body was retrieved. Which of the following is false regarding ante mortem drowning?

- A. Water in the stomach
 - B. Froth on nose and mouth
 - C. No Mud and vegetation in respiratory tract
 - D. Cadaveric spasm in hand muscle
-

10. In judicial hanging, the cause of death is due to:

- A. Vagal inhibition
 - B. Cervical vertebral fracture or dislocation
 - C. Cerebral hypoxia
 - D. Asphyxia
-

11. In which type of hanging does the weight of the head act as a constricting force?

- A. Complete
 - B. Partial
 - C. Typical
 - D. Atypical
-

12. A person was found dead in the bushes with his hands tied. On examination of the body, there was cyanosis petechial hemorrhages, abrasions, and bruises around the mouth and lips. There was associated facial congestion and edema. What is the least likely cause of death here?

- A. Asphyxia
 - B. Suicide
 - C. Smothering
 - D. Homicide
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	1
Question 4	2
Question 5	3
Question 6	2
Question 7	2
Question 8	2
Question 9	3
Question 10	2
Question 11	2
Question 12	2

Solution for Question 1:

Correct Option C - Burking:

- Burking is a type of homicide caused by smothering and traumatic asphyxia. In this method, the attacker kneels or sits on the chest of the victim and closes the victim's mouth and nose with his or her hands.

Incorrect Options:

Option A - Mugging: Mugging is a method of strangulation caused by holding the neck of the victim in the bend of the elbow. The attacker usually attacks from behind.

Option B - Garrotting: Garrotting is a method of strangulation caused by a ligature being thrown around the neck and quickly tightening it.

Option D - Bansdola: Bansdola is a method of strangulation that uses two bamboo sticks.

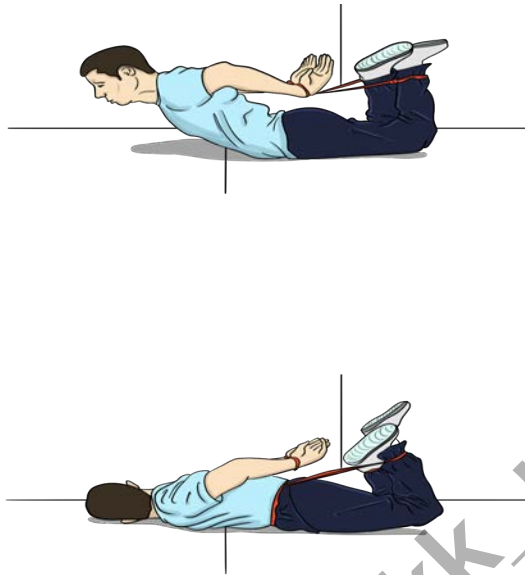
Solution for Question 2:

Correct Option C - Hog tie:

- Hog tie is a method of restraining an individual by binding their wrists and ankles together behind their back, typically using ropes or other restraints.
- This position renders the person immobilized and limits their ability to move or escape. It is commonly used in law enforcement or security situations to prevent individuals from fleeing or posing a threat.

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Incorrect Options:

Option A - Choke/ Carotid hold : A choke or carotid hold involves applying pressure to the neck, specifically targeting the carotid arteries and jugular veins, with the intention of causing temporary unconsciousness

Option B - Carotid steeper hold: The term "carotid steeper hold" does not refer to a recognized or commonly used technique.

Option D - Bar arm : "Bar arm" does not correspond to a specific method of restraint or positioning described in the scenario

Solution for Question 3:

Correct Option A - Burking:

- Burking accurately describes the act of sitting on the victim's chest and obstructing their airway, resulting in asphyxia and death.



Incorrect Options:

Option B - Smothering: Smothering refers to the act of covering or suffocating someone, usually by placing an object or substance over their nose and mouth to prevent them from breathing.

Option C - Traumatic asphyxia : Traumatic asphyxia is a condition that occurs when there is a sudden, severe compression of the chest or abdomen, leading to a disruption of normal breathing.

Option D - Overlaying: Overlaying, also known as accidental suffocation, occurs when an adult or object unintentionally covers an infant's airway during sleep, leading to suffocation.

Solution for Question 4:

Correct Option B - Cervical sympathetic chain:

- The sympathetic chain, also known as the sympathetic trunk or ganglia, is part of the autonomic nervous system.
- In this case, the compression of the cervical sympathetic chain is the probable reason for the open right eye. When the sympathetic chain is affected, it can lead to an interruption in the innervation of the smooth muscles in the eyelids, resulting in the inability of the right eyelid to close.

Incorrect Options:

Option A - Right internal jugular vein : The right internal jugular vein is a major blood vessel located in the neck. Its compression would not directly cause the opening of the right eye.

Option C - Left vagus nerve: The vagus nerve is a cranial nerve that provides parasympathetic innervation to many organs in the body. its compression would not directly cause the specific symptom mentioned in the question, which is the open right eye.

Option D - Right internal carotid artery : The right internal carotid artery is a major blood vessel that supplies oxygenated blood to the brain. Its compression is called ammusat's sign but would not directly cause the opening of the right eye.

Solution for Question 5:

Correct Option C - Hydrostatic test:

- The test described, where the bronchus is tied and the lung is immersed in water to check whether it floats or sinks, is known as the Hydrostatic test.
- In the context of the described test during autopsy, the correct answer is the Hydrostatic test. This test is performed to assess the buoyancy of the lung, as the presence of air within the lung tissue causes it to float in water. It is used as a supportive indication of drowning, as drowning often leads to the aspiration of water into the lungs, resulting in increased lung buoyancy.

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In the context of the described test during autopsy, the correct answer is the Hydrostatic test. This test is performed to assess the buoyancy of the lung, as the presence of air within the lung tissue causes it to float in water. It is used as a supportive indication of drowning, as drowning often leads to the aspiration of water into the lungs, resulting in increased lung buoyancy.

Incorrect Options:

Option A - Gettler's test :

- It is based on the chloride concentration in the blood samples of heart chambers [right side & left side]

- Normally, chloride concentration in both right side chambers & left side chambers should be equal

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Option B - Ploucquet's test:

- Ploucquet's test is a test used to detect ratio between
-

Ploucquet's test is a test used to detect ratio between

Weight of the lung

Weight of the baby

Option D - Diatom's test:

- Diatom's test is a forensic examination that involves analyzing the presence of diatoms, in various tissues and fluids to determine if a person was submerged in water.

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Solution for Question 6:

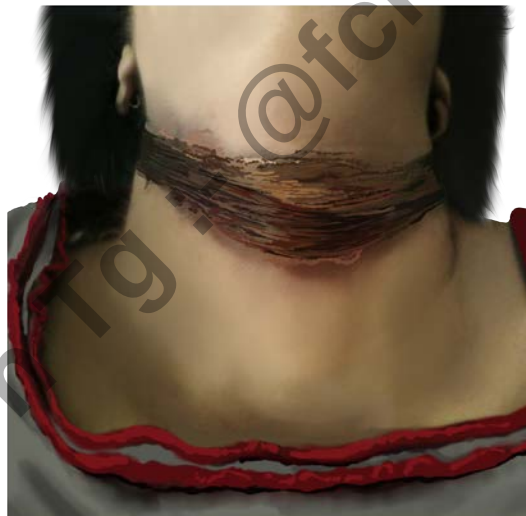
Correct Option B - Ligature strangulation:

- Ligature strangulation is the right cause of death in this situation.
- The presence of a ligature that encompassed the neck and was below the thyroid level indicates a ligature check, which may be a classic sign of ligature strangulation.
- The absence of saliva spilling is additionally consistent with this sort of asphyxia, as the airway is regularly compressed without hindrance to the oral cavity

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Incorrect Options:

Option A - Throttling:

- This will cause bruises or abrasions on the neck and encompassing areas, but it isn't steady with the level of ligature mark seen in this case.

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Option C - Gagging:



- Gagging includes airway obstacles with a foreign object, such as a cloth or piece of cloth, put within the mouth.
- This could cause saliva to spill from the mouth and isn't steady without saliva.

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Option D - Hanging:



- The ligature mark, in this case, is level and below the level of the thyroid, which isn't steady, with the ordinary vertical ligature mark seen hanging.

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Solution for Question 7:

Correct Option B - Antemortem drowning:

- Violent respiratory struggle causes churring effect, which mixes Mucus + surfactant + water + air & results in production of Froth.
- In post-mortem drowning as this violent respiratory struggle is missing, froth can't be seen.
- Thus, it is a sign of Antemortem drowning.

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Incorrect Options:

- Option A, C & D are incorrect. Refer to the explanation for the correct answer.

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Solution for Question 8:

Correct Option B - Garrotting:

- The term "Spanish windlass" refers to a specific type of garrotting, a method of strangulation.
- In this method, a cord or similar material is wrapped around a person's neck, and a stick or other similar object is twisted to tighten the cord, leading to strangulation. This particular type of garrotting is named after a technique historically used in Spain.

Incorrect Options:

Option A - Bansdola: Bansdola is a technique where a bamboo or wooden block is used to cause pressure on the neck.

Option C - Mugging: Mugging usually refers to an act of robbery or assault, often involving physical force.

Option D - Throttling: Throttling refers to strangulation with the hands or fingers.

Solution for Question 9:

Correct Option C - No Mud and vegetation in respiratory tract:

- This statement is false as in the case of antemortem drowning there is fine froth in the respiratory tract along with mud, sand and vegetation.
- There is no mud and vegetation in the case of post-mortem drowning.

Incorrect Options:

Option A - Water in the stomach : In cases of antemortem drowning, it is common to find water in the stomach due to the inhalation or ingestion of water during the drowning process.

Option B - Froth on nose and mouth : Froth on the nose and mouth can be observed in cases of antemortem drowning. It occurs due to the mixing of air and water during the struggle for breath.

Option D - Cadaveric spasm in hand muscle : Cadaveric spasm refers to the postmortem stiffening of muscles that occurs in the position they were in at the time of death.

Solution for Question 10:

Correct Option B - Cervical vertebral fracture or dislocation:

- The most common cause of death in judicial hanging is cervical vertebral fracture or dislocation. The most common cause of death in hanging is a combination of asphyxia and cerebral venous congestion.



- The knot in judicial hanging is commonly placed at the side of the neck (on the right side at the right angle of the mandible). But, the most effective/ ideal site is said to be below the chin/ submental.
- Cause of death: With proper judicial hanging, there will be rupture of brainstem between pons and medulla. Results in instant and irreversible loss of consciousness and irreversible apnea. There is fracture dislocation at C2– C3 level . Bilateral fracture of either of pedicles or laminae of C2 is also Hangman's fracture .

Incorrect Options:

Option A - Vagal inhibition: Vagal inhibition is one of the causes of death in case of ligature strangulation

Option C - Cerebral hypoxia: Cerebral hypoxia/ anoxia is not the cause of death in case of judicial hanging.

Option D - Asphyxia: Hanging is the most common asphyxial method of death. Asphyxia along with venous congestion is the most common cause of death in hanging.

Solution for Question 11:

Correct Option B - Partial:

- In an incomplete type of hanging also known as partial hanging, some part of the body is touching the ground. Constricting Force here will be the weight of the head (around 4-5 kg).

Incorrect Options:

Option A - Complete: In the complete type of hanging, the constricting force is the whole body.

Option C - Typical: In typical, position of the knot is the occiput. But the whole body is suspended, hence the whole body weight is the constricting force.

Option D - Atypical: In atypical type of hanging, the position of the knot is any place other than the occiput. But in this type also just like typical, the whole body is suspended and hence the constricting force will be the whole body weight

Solution for Question 12:

Correct Option B - Suicide:

- Although cyanosis may be present in some techniques of suicide, bruises around mouth and lips, along with petechial hemorrhages and abrasions points towards injury by force.
- Asphyxia, smothering and homicide can be possible in such situations.
- However, suicidal attempts are usually painless, and may not fit in this scenario.

Incorrect Options:

Option A - Asphyxia: Usual signs include Cyanosis, petechial hemorrhages and congestion.

Option C - Smothering: Done by pillow or hand, smothering has features like hematoma, contusions on the lips, mouth and oral orifices. facial congestion and edema in the description fit in this

Option D - Homicide: It may present various features depending on the mode of homicide. abrasions, bruises, petechial hemorrhages are possible in homicidal attempts.

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Mechanical injuries

1. A 45-year-old woman comes to the emergency department with a bruise on her thigh after bumping into a table. The bruise is noted to be blue. Upon further examination, she mentions that she noticed the bruise about a day ago. What is the most likely age of the bruise based on its current color?

- A. Day 4
- B. 5 to 6 days
- C. Few hours to 3 days
- D. 2 weeks

2. A 45-year-old woman presents for a follow-up appointment after undergoing surgery for an abdominal incision. The physician observes that the wound is beginning to heal, with granulation tissue forming in the wound bed. What is the most likely duration of healing indicated by the presence of vascularized granulation tissue?

- A. 7 days
- B. 12 hours
- C. 2–3 days
- D. 4–6 days

3. A 50-year-old woman presents to the emergency department after hitting her head on a metal beam. On examination, the wound has sharp edges but lacks undermining, and tissue bridges are visible. There is no significant underlying soft tissue present between the skin and the skull. What is the most likely type of injury?

- A. Incised wound
 - B. Split laceration
 - C. Shearing laceration
 - D. Chop wound
-

4. A victim of an attack displays several injuries on the palms, flexor sides of the fingers, and interdigital spaces and are irregular and ragged cuts. Which of the following best describes the nature of these injuries?

- A. Passive defense injuries sustained while raising arms for protection
 - B. Active defense injuries sustained while attempting to grasp the weapon
 - C. Lacerations from a blunt weapon
 - D. Injuries caused by falling onto a sharp object
-

5. A 30-year-old woman was attacked with an axe and suffered a chop wound from it. Which of the following is true about a chop wound?

- A. The wound is a laceration on the surface and does not involve the underlying bone
 - B. The dimensions of the wound do not correspond to the weapon size
 - C. The margins of the wound are always irregular
 - D. The heel of the axe leaves a deeper wound than the toe of the axe
-

6. An old method of suicide conducted by Japanese samurai warriors as a ritual is shown below. Identify the cause of death in such cases.



- A. Hypovolemic shock
- B. Asphyxia
- C. Sudden cardiac collapse
- D. Hypoxemia

7. Assertion: The appearance and shape of stab wounds are significantly influenced by the orientation of Langer's lines on the skin. Reason: Stab wounds that are parallel to Langer's lines tend to be narrow and slit-shaped, while those that cut across the lines may gape due to skin elasticity and the tension created along these lines. Analyze the above statements and answer accordingly.

- A. Both assertion and reason are correct and the reason is the right explanation
 - B. The assertion is wrong but the reason is correct
 - C. Both assertion and reason are wrong
 - D. The assertion is true but the reason is wrong
-

8. A forensic surgeon is examining multiple stab wounds on a homicide victim. The wounds are wedge-shaped as in the image: Which of the following weapons is likely the cause of the injury here?



- A. Scissors
- B. Single-edged knife
- C. Double-edged knife
- D. Broken glass bottleneck

9. A forensic surgeon is examining a stab wound on a victim. Which of the following statement/s is/are true about the dimensions of the wound? 1. Breadth of the wound is equal to the thickness of the weapon. 2. Depth is the greatest dimension of a stab wound. 3. The edges of the wound are clean-cut. 4. The length of the wound is usually slightly less than the width.

- A. 1, 2, and 3
- B. 2, 3, and 4
- C. 1 and 3
- D. All are correct

10. A homicidal case with a stab wound over the abdomen is being examined. The weapon has passed through and through with a larger entry wound and a smaller exit wound. Which type of wound is this classified as?

- A. Puncture wound
 - B. Perforating wound
 - C. Penetrating wound
 - D. Concealed wound
-

11. A forensic pathologist examines a body and notes the following findings. What do these findings suggest?



- A. The individual experienced a moment of hesitation before the fatal injury.
- B. The injuries were caused by a blunt object.
- C. The victim was attacked with a sharp instrument from a distance.

D. The wounds are consistent with self-inflicted injuries.

12. During an autopsy of a 50-year-old male, the forensic pathologist observes a wound shown in the figure. Which of the following is correct regarding this type of wound?



- A. Shallow at beginning, deeper towards end
 - B. Irregular margins with bruising
 - C. No significant hemorrhage
 - D. Clean, well-defined edges
-

13. A 32-year-old male was involved in a high-speed motor vehicle accident. Upon examination, the direction of the vehicle's movement can be inferred from the orientation of the skin flap, as shown in the image below. What is the most likely type of injury in this case?



- A. Chop wound
- B. Cut laceration
- C. Stretch laceration
- D. Tear laceration

14. Which of the following is not a cause of injury due to sharp force?

- A. Laceration
- B. Incision
- C. Stab
- D. Chop

15. During an autopsy, the forensic pathologist observes an injury on the thigh, which shows irregular edges, bruising, and embedded dirt. On further examination, the deeper tissues show uneven tearing, with visible nerve and blood vessel damage. What type of injury is most consistent with these findings?

- A. Incised wound caused by a sharp object.
 - B. Laceration caused by blunt force trauma.
 - C. Abrasion caused by friction.
 - D. Puncture wound from a sharp object.
-

16. Which of the following statements accurately describes the differences between a bruise and hypostasis?

- A. Bruises are caused by ruptured vessels, but hypostasis from distension of vessels.
 - B. Bruises typically have clearly defined margins, but hypostasis merges with the surrounding area.
 - C. Bruises are often seen in the most dependent parts of the body, but hypostasis is only present at the site of injury.
 - D. Bruises do not cause elevation of the affected area, while hypostasis may cause swelling.
-

17. A medicolegal case of assault is being examined at the emergency with the following contusions. What is the most likely explanation for the appearance of these bruises?



- A. Natural bruises due to physical trauma
- B. Artificial bruising to falsely accuse someone
- C. Post-mortem bruising from blood settling
- D. Bruising due to pre-existing medical condition

18. A patient with a history of a fall from a bike a day before, comes with the following findings.



- A. Spectacle hematoma
- B. Grey Turner's sign
- C. Cullen's sign
- D. Battle sign

19. A 30-year-old man presents with multiple bruises on his body following a fight. Upon examination, the physician notes a specific pattern of bruising on his back as shown: What is this type of bruise called?



- A. Battle sign
- B. Spectacle hematoma
- C. Railway line type
- D. Automobile tyre pattern

20. A 75-year-old woman presents to the emergency department after a fall. She gives a history of landing on her buttocks, where she has developed bruises. Which factor is NOT likely influencing the formation of her bruises?

- A. Age
- B. Gender
- C. Color of skin
- D. Gravity shifting of blood

21. Which of the following statements correctly differentiates between postmortem and antemortem abrasions?

- A. Antemortem abrasions are yellowish-white or pale, whereas postmortem abrasions are reddish-brown to dark brown in color.
 - B. Postmortem abrasions are typically found on bony prominences, while antemortem abrasions can occur on any part of the body.
 - C. Congestion is definitive of postmortem abrasions.
 - D. Both postmortem and antemortem abrasions exhibit bleeding with clot formation.
-

22. A 35-year-old male is being evaluated for injuries as part of a medico-legal case. During the examination, an abrasion wound is noted to be brown in color. Based on the appearance of the wound, what is the most likely time since the injury occurred?

- A. 12-24 hours
 - B. 2-3 days
 - C. 4-5 days
 - D. 6-7 days
-

23. Which of the following statements is correct regarding the forensic findings given in the following image?



- A. Pressure abrasion marks are always superficial and heal without scarring.
- B. Continuous vertical force can cause damage to both the epidermis and underlying structures.
- C. Ligature marks caused by hanging do not show epidermal crushing.
- D. Abrasions from strangulation are typically caused by horizontal shearing forces.

24. A 30-year-old man was involved in a road traffic accident and presented with abrasion marks visible on his body. The abrasion pattern appears consistent with a distinct type of injury mechanism. Based on the image provided, what type of abrasion is most likely in this scenario?



- A. Linear Abrasion
- B. Grazed Abrasion
- C. Imprint Abrasion
- D. Pressure Abrasion

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	2
Question 4	2
Question 5	4
Question 6	3
Question 7	1
Question 8	2
Question 9	2
Question 10	2
Question 11	1
Question 12	4
Question 13	3
Question 14	1
Question 15	2
Question 16	1
Question 17	2
Question 18	2

Question 19	3
Question 20	4
Question 21	2
Question 22	3
Question 23	2
Question 24	3

Solution for Question 1:

Correct Answer: C) Few hours to 3 days

Explanation:

The above clinical scenario shows a blue-colored bruise that can be considered to be around a few hours to 3 days of age .

Methods used to estimate the age of a bruise include:

- The healing of a bruise happens by destroying and removing the extravasated blood.
- The age of the bruise can be analyzed by its color.
- The color change rate is different from person to person and even bruise to bruise in the same person.

Color of bruise	Age of bruise
Red (oxygenated blood)	Fresh bruise/ 1-2 hours
Blue (deoxyhemoglobin)	A few hours to 3 days
Bluish-black to brown (hemosiderin)	4th day (Option A)
Greenish (haematoidin/biliverdin)	5 to 6 days (Option B)
Yellow (bilirubin)	7 to 12 days
Normal	2 weeks (Option D)

- Factors affecting the color of bruise: Depth of bleeding Amount of bleeding Environmental lighting Overlying skin color

- Depth of bleeding

- Amount of bleeding

- Environmental lighting

- Overlying skin color

- Exceptions: In subconjunctival ecchymoses , the color change is from bright red to yellow and then disappears due to hemoglobin being kept oxygenated by air Meningeal hemorrhages also have a similar color change as to subconjunctival hemorrhages due to the oxygen supply from CSF Healing of bruises is very slow in older people A bruise attained during carbon monoxide poisoning is bright red

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- Depth of bleeding

- Amount of bleeding

- Environmental lighting

- Overlying skin color

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- Healing of bruises is very slow in older people

- A bruise attained during carbon monoxide poisoning is bright red

Solution for Question 2:

Correct Answer: C) 2–3 days

Explanation:

The presence of vascularized granulation tissue is a key indicator of the wound healing process and typically occurs during the 2-3 days post-injury phase.

Age of Incised Wound

Duration	Gross Findings	Microscopic Findings
Fresh	Red with clotted blood.	Capillary dilation, neutrophil margination and emigration, reactive changes in histiocytes.
12 hours	Margins red, swollen, and adherent; blood and lymph present.	Reactive changes in fibroblasts, monocytes in exudate. (Option B)
24 hours	Continuous layer of endothelial cells covers surface; scab of the dried clot.	Endothelial growth at edges; formation of vascular buds.
2–3 days	-	Formation of vascularized granulation tissue (Fibroblast activity) (Option C)
4–6 days	-	Formation of new fibrils. (Option D)
7 days	Scar formation.	Scar formation. (Option A)

Medico-Legal Importance of Incised Wounds

- **Weapon Identification:** Suggests a sharp-edged weapon; the appearance can occasionally indicate weapon type, although this is less definitive than with stab wounds.
- **Force Direction:** Offers clues about the direction of the applied force.
- **Injury Age Assessment:** Aids in determining how long ago the injury occurred.
- **Injury Severity:** Typically not life-threatening unless located over major arteries (e.g., wrist or neck), where it could be fatal.
- **Wound Analysis:** The position and characteristics of the wound may reveal the manner of injury (e.g., suicide, accident, or homicide).

Feature	Suicidal cut-throat wounds	Homicidal cut-throat wounds
Situation	Left side of the neck, crossing the front of the throat High, above the thyroid cartilage	Both sides of the neck Low, at or below the thyroid cartilage
Direction	Oblique, above downwards, left to right for right-handed persons	Transverse or below upwards
Number of Wounds	Multiple (20-30), superficial, parallel, merging with main wound	Multiple, intersecting at a deep level

Edges	Ragged due to overlapping superficial incisions	Sharp and clean-cut; beveling may be observed
Hesitation Cuts	Present	Absent
Wounds in Other Parts	May include cuts on wrists, groin, thighs	Severe injuries may be present on the head; no wrist wounds
Defense Wounds	Absent; may have unintentional cuts	Present unless the victim was caught unaware
Hands	May grasp the weapon firmly due to cadaveric spasm	Fragments of clothing or hair from the assailant may be grasped
Weapon at Site	Usually present	Typically absent
Vessels	Carotid artery may escape injury	Jugular vein and carotid artery likely cut
Blood Stains	May show splashing downward on clothes if in front of a mirror	Found on both palms from attempts to cover the wound
Circumstantial Evidence	Usually in quiet places (e.g., bedroom) with possible suicide notes	Scene disturbance and external footprints may be present

- Accidental Wounds: Commonly found around the hands.
- Defense Wounds: Present on forearms and palms from attempts to block an attack.
- Self-Mutilation: Superficial, multiple injuries inflicted intentionally by individuals with mental health issues, often avoiding vital areas.

Solution for Question 3:

Correct Answer: B) Split laceration

Explanation: Split lacerations are commonly seen in areas where the skin is close to bone, such as the scalp, with minimal underlying soft tissue. Although the edges may appear sharp, the presence of tissue bridges confirms the diagnosis of a laceration rather than an incised wound.

Types of Laceration

- Split laceration
- Avulsion laceration

- Stretch laceration
- Tear laceration
- Cut laceration

Types of Laceration	
	Split Laceration (Option B)
These occur when skin is crushed between two objects , such as the scalp being compressed between the skull and a hard surface like the ground or a blunt object. Seen on areas where the skin is close to bone , with minimal underlying tissue E.g., scalp, eyebrows, cheekbones, jaw, iliac crest, perineum, shin. Lacerations without excessive skin crushing can have sharp edges , resembling incised wounds. Also known as incised looking laceration . These are not undermined and show tissue bridges.	
	Avulsion (Shearing Laceration) (Option C)
Caused by a shearing force at an acute angle, detaching skin or organs from their attachments. The shearing and grinding force from a heavy object, like a lorry wheel passing over a limb, can cause the skin to separate from the underlying tissues, a condition known as avulsion or "flaying." This often results in crushed muscles and fractured bones. The detached skin may have extensive abrasions due to the tyre's friction, but one portion is still in continuity with adjacent skin. Lacerations from shearing forces can leave the skin intact while separating the underlying soft tissue from the fascia, resulting in a blood-filled pocket. This type of injury is often observed on the back of the thighs in pedestrians hit by motor vehicles.	

Incised wounds (Option A) have clean, sharp edges from a cutting instrument. The presence of tissue bridges in this case indicates it is a split laceration rather than a clean cut.

Chop wounds (Option D) involve significant tissue damage from extreme pressure. The sharp-edged, localized laceration with minimal underlying tissue damage does not fit the profile of a chop wound.

Solution for Question 4:

Correct Answer: B) Active defense injuries sustained while attempting to grasp the weapon

Explanation:

The wound in the given scenario points to an active defense wound where the victim has tried to grasp the weapon to prevent the attack and sustained injuries to the palm and on the fingers which would have been the case with a double-edged weapon.

Defence wounds are caused by the immediate and instinctive reaction of the victim to save himself from an attack and are usually seen on the extremities.

They are divided into two types:

	Active defense wounds	Passive defense wounds
Mechanism of injury	The victim tries to grasp the weapon.	The victim raises the arms, legs, or hands.
Site	Depends on the type of weapon. A single-edged weapon causes a single cut on the palm or the bends of fingers or thumb. A double-edged weapon produces cuts on both the palm and fingers. A typical knife defense wound is seen between the base of the thumb and the index finger.	Extensor or ulnar surfaces of the forearms, wrists, back of the hands, knuckle. (vs Option A) Rarely on the shin and feet. Arms and posterior aspect of lower limbs and back.
Type of Injuries	Irregular and ragged cuts due to loosening of skin tension while gripping the knife.	Fractures of the carpal bones, metacarpals, and digits (mostly blunt trauma). (vs Option D)
Medicolegal importance	Defense wounds indicate homicide. They are absent if the victim is: Unconscious Taken by surprise Attacked from the back Under the influence of drugs or alcohol	



Lacerations from a blunt weapon (Option C) are usually split lacerations when a part of the body gets crushed against a hard surface and causes injury. They have sharp edges (incised-looking laceration) unless much skin is involved and are seen mostly over areas with underlying bony prominences like eyebrows, forehead, etc.

Injuries caused by falling onto sharp objects (Option D) can be puncture, penetrating, or perforating wounds, which have clean-cut margins and usually have the dimensions of the weapon causing it.

Solution for Question 5:

Correct Answer: D) The heel of the axe leaves a deeper wound than the toe of the axe

Explanation:

The lower end or heel of an axe leaves a deeper wound than the upper end or toe of the axe due to the direction of attack and the force of injury being greater at the heel than the toe.

Chop wounds/Slash wounds	
--------------------------	--

These are deep gaping wounds inflicted by a blow with the moderately sharp cutting edge of a heavy weapon when applied with significant force. It is a combination of blunt and sharp force injury.	
Weapons used	Hatchet Axe Sword Broad heavy knife Chopper Saber Meat cleaver
Features of the wound	The axe's lower end (heel) strikes first, leaving a deeper wound than the upper end (toe). It is an incised wound on skin with an underlying comminuted fracture or deep groove in the bone. (vs Option A) Dimensions of the wound correspond to the cross-section of the penetrating blade. (vs Option B) Margins are sharp and may have abrasions, bruising, and some lacerations with severe injury to underlying organs. (vs Option C)
Medicolegal importance	Most injuries are homicidal. Few are accidental by machinery (propeller injuries). Rarely suicidal. Wounds give details of the weapon used and the direction of attack.



Solution for Question 6:

Correct Answer: C) Sudden cardiac collapse

Explanation:

The given scenario points to Harakiri, which is a method of suicide done by the victim using a short sword to inflict a single large stab L-shaped wound in the abdomen leading to a sudden drop in intraabdominal pressure and thus causing cardiac collapse and death.

Hara-kiri/Seppukku	
Hara-kiri refers to an unusual method of suicide where the victim inflicts a single large stab wound on the abdomen. It is associated with Japanese samurai warriors.	
Characteristics of the wound	It is a single large stab wound made by the victim by moving the sword across the abdomen (usually left to right) and then turning it up to form an L-shape .
Mechanism of injury	The wound is made by a short sword while in a sitting position or By falling onto a ceremonial sword. A well-fixed knife can produce a wound like this, but only with enough momentum to do so.
Cause of death	The large wound leads to the intestines pulling out of the body, which causes a sudden drop in intrabdominal pressure and cardiac return . This leads to sudden cardiac collapse and death .





Hypovolemic shock (Option A): this is seen in victims who have lost large volumes of blood causing hypovolemic shock and death. This is not possible here since in Harakiri, the wound is so extensive that the death is sudden due to a drop in intraabdominal pressure and cardiac collapse.

Asphyxia (Option B): is seen as the cause of death in cases of hanging where there is a lack of cerebral blood flow and respiratory asphyxia leading to loss of oxygen supply to the brain.

Hypoxemia (Option D): this is seen in cases of drowning where the immersion or submersion in a liquid medium leads to hypoxia and acidosis of tissues.

Solution for Question 7:

Correct answer: A) Both assertion and reason are correct and the reason is the right explanation

Explanation:

Lines of Langer:

- The arrangement of the dense network of collagen and elastic fibers in the dermis is referred to as the cleavage direction, while its linear representation on the skin is known as Langer's lines.

Impact of skin tension:

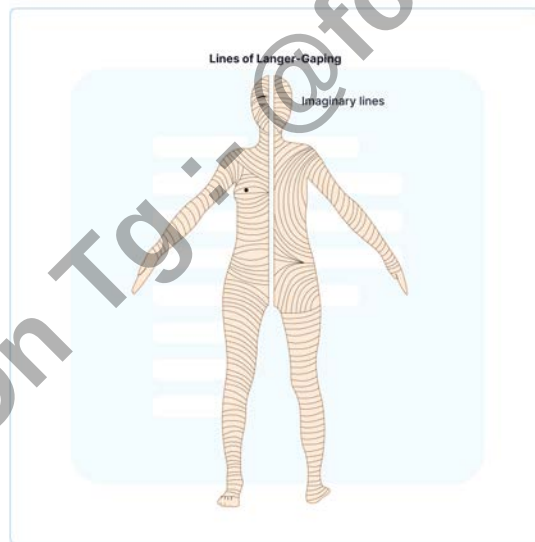
- Skin tension and Langer's lines can change round skin defects into elongated wounds that resemble stab wounds.
- They can also affect genuine stab wounds, making them appear artificially lengthened.
- The shape of the wounds also depends on the cleavage lines of Langer.

Assessing wound angles:

- It may be necessary to bring the wound margins together to determine if a wound angle is sharp or blunt.
- Holding the edges together aids in a clearer evaluation of the angles.

Variability in injury presentation:

- Identical weapons can cause different types of injuries based on their location and orientation on the body, influenced by skin elasticity and the alignment of Langer's lines.
- A stab wound parallel to the Langer's lines appears slit-shaped and narrow
- A stab wound that cuts through the Langer's lines transversely will gape.



Solution for Question 8:

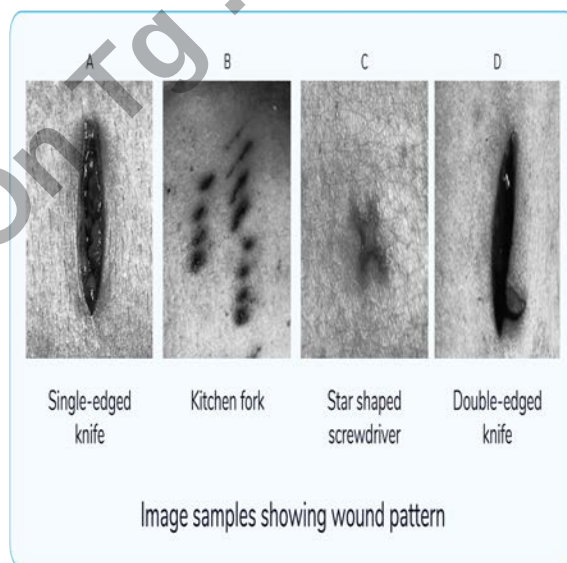
Correct Answer: B) Single-edged knife

Explanation:

The wound described in the scenario as well as in the image is a wedge-shaped stab wound which is caused by a single-edged weapon like a knife.

The shape of a stab wound depends on the weapon used:

Weapon used	Shape of wound
Single-edged weapon	Triangular or wedge-shaped surface wound One side is sharp, and the other is rounded, blunt, or squared off
Double-edged weapon (Option C)	Elliptical or slit-like wound Both sides are sharp or pointed
Round object	Circular wound
Pointed square weapon	Cross-shaped injury
Broken bottle (Option D)	Clusters of wounds of different shapes, sizes, depths, and irregular margins
Screwdriver	Slit-like stab wound with squared ends
Scissors (Option A)	Closed: flat "Z" shaped Open: like a knife
Ice picks	Small, round, or slit-like
Fork	Group of 3-4 wounds depending on the prongs of the fork



Solution for Question 9:

Correct answer: B) 2, 3, and 4

Explanation:

Wound description: when describing a stab wound, it's important to note the length, width, direction, and appearance of the wound's margins and angles.

Margins:

- The edges of the wound are clean-cut (statement 3) and typically show no abrasion or bruising.
- In cases of full blade penetration, a patterned abrasion or bruising may occur due to the hilt guard striking the skin.
- The margins are regular, sharp, and well-defined; however, wounds caused by pointed or conical instruments may have lacerated edges.

Length:

- The length of the wound is usually slightly less than the width (statement 4) of the weapon because of skin stretching.

Breadth:

- The breadth is generally greater than the thickness of the blade (statement 1) due to gaping; proper approximation of the edges is necessary for accurate measurement.

Depth:

- Depth is the greatest dimension of a stab wound (statement 2) and corresponds to the length of the weapon entering the body.
- If the entire length of the weapon penetrates without producing an exit wound, this depth is critical.

Safety note:

- It is unsafe to determine the depth of a stab wound by probing in the emergency room, as this can disturb any loose clot and lead to severe hemorrhage.
 - Probing may also create a false track by passing between fascial planes or within muscles.
 - Depth assessment should be conducted in the operating theater during wound repair.
-

Solution for Question 10:

Correct Answer: B) Perforating wound

Explanation:

The wound in the above scenario is a perforating wound as the weapon has passed through-and-through the body.

Stab wound/Punctured wound	
These are inflicted by penetration with long, narrow instruments having pointed (sometimes blunt) ends into the depth of the body, which is deeper than its length and width.	
Weapons used	Knife (most common) Pieces of glass (broken glass bottle) Scissors Screwdrivers Pens Ice picks Forks
Types of stab wounds	Puncture wound: (Option A) Soft tissues are involved Penetrating wound: (Option C) Enters a body cavity or a viscus Perforating wound: Through-and-through puncture wounds The weapon enters the body from one side and comes out the other The wound of entry is larger than the exit
Significance of the wound dimensions	Length is slightly less than the width of the weapon due to the stretching of the skin. Breadth is more than the thickness of weapon due to gaping. Depth, the greatest dimension of the wound, corresponds to the length of the weapon's blade, which has completely entered the body without perforation. Margins or edges of the wound are clean cut with usually no abrasion or bruising. Margins by a pointed or conical weapon have lacerated edges If the hilt guard of the weapon strikes the skin, a regular, sharp, and well-defined patterned abrasion or bruising is formed



Concealed wound (Option D) is a wound that is present in hidden areas of the body like the axilla, nostrils, etc., and is not a through-and-through wound.

Solution for Question 11:

Correct Answer: A) The individual experienced a moment of hesitation before the fatal injury.

Explanation:

The presence of small, superficial cuts around a deeper wound suggests a possible moment of doubt or fear before inflicting the fatal injury. These small cuts can be indicative of hesitation, often observed in scenarios of self-infliction or in cases where the victim may have struggled before the lethal wound.

Hesitation cuts:



- Hesitation cuts (also known as tentative cuts or trial wounds) are multiple, small, superficial cuts that typically involve only the skin.
- These cuts are often found at the beginning of a deeper incised wound.
- Psychological Aspect : They suggest that the individual may have hesitated while gathering the courage to make a decisive cut.
- Forensic Observation : In cases of homicidal sharp force injuries, multiple superficial, roughly parallel incised wounds may appear on the neck, often adjacent to a single deep, lethal incised wound, particularly in instances where torture is involved.

Hesitation cuts are associated with sharp objects, not blunt ones . The nature of the wounds indicates a sharp instrument was likely used. (Option B)

The close, parallel cuts suggest the attack occurred at close range , not from a distance. (Option C)

While hesitation marks can appear in self-inflicted wounds, this option is too specific without confirming intent . The wounds could also result from an attack by another individual. (Option D)

Solution for Question 12:

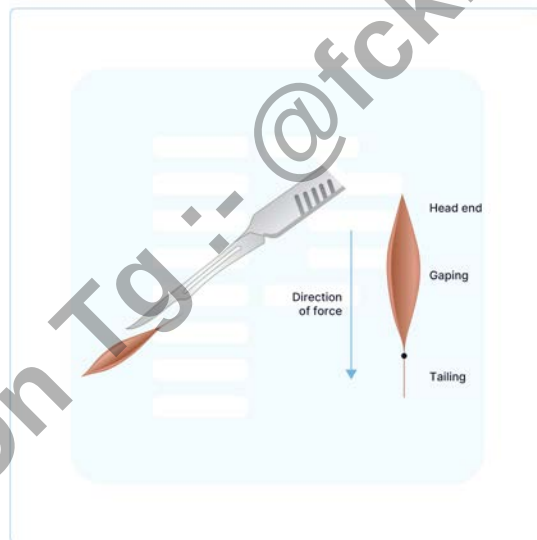
Correct Answer: D) Clean, well-defined edges

Explanation:

The given image shows an Incised wound with clean and well-defined edges .

Incised wounds

- An incised wound is a clean cut through the skin and subcutaneous tissues, typically longer than it is deep , caused by a sharp-edged instrument like a knife or razor.
- It results from pressure and friction against the tissue.



Wound Structure:

- Margins: Clean, well-defined edges (Option D) , often everted; may appear inverted with adhered muscle fibers. No contusions or abrasions (Option B) , but may show wrinkling with multiple incisions. No bridging tissue.
- Width: Greater than the weapon's cutting edge due to tissue retraction.
- Length: Exceeds width and depth, with no relation to the weapon's edge.
- Shape: Usually spindle-shaped, with more gaping in the center ; gaping is influenced by the direction of cuts across skin fibers.
- Hemorrhage: Significant bleeding occurs from cleanly cut vessels. (Option C)

Tailing	Bevelling
Deeper at the beginning due to greater knife pressure (the "head" of the wound). (Option A) Becomes shallower toward the exit. Indicating the cutting direction.	A wound created by an obliquely entering blade features a beveled margin on one side, and an undermined area on the other side. Characteristic of sharp weapons Associated with homicidal injuries. Suggest the positions of the assailant and victim .

Solution for Question 13:

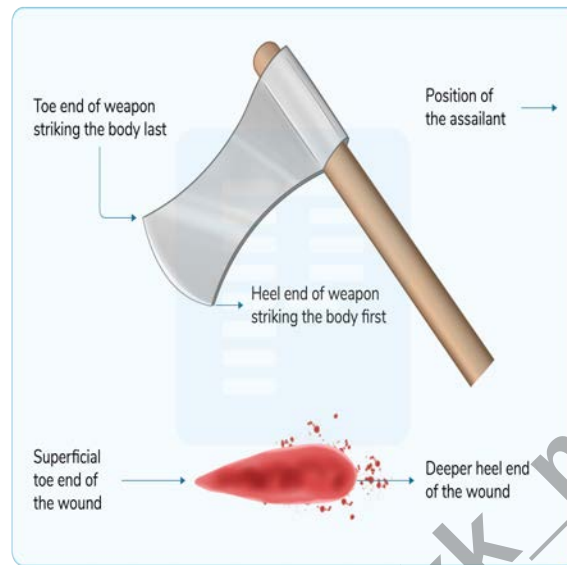
Correct Answer: C) Stretch laceration

Explanation:

Stretch laceration occurs from overstretching of skin, commonly seen in motor vehicle accidents. The presence of a skin flap, irregular tissue tearing, and embedded foreign materials are typical characteristics. Additionally, the skin flap's orientation helps infer the direction of the vehicle's movement, which is consistent with stretch lacerations.

Types of Laceration	
	Stretch laceration (Option C)
Caused by overstretching of fixed skin, leading to localized pressure and tearing, which creates a skin flap. Commonly seen in motor vehicle accidents, with the skin flap potentially indicating the vehicle's direction. Can occur from kicking or bone deformity after a fracture, leading to compound injuries. Results in irregular skin and tissue tearing, with a risk of foreign materials entering the wound.	
Tear lacerations (Option D)	Cut Lacerations (Option B)
Result from impact with irregular or semi-sharp objects (e.g., car door handles, broken glass). Characterized by a deeper starting point that tapers off towards the end, representing another form of overstretching.	Produced by heavy, sharp-edged instruments (e.g., axes, hatchets). Typically, they have abraded margins with bruising, although the term "cut laceration" is often considered less precise.

Chop/ Slash wound (Option A)



- Deep, gaping wounds caused by a blow from a sharp-cutting edge of a heavy weapon (e.g., hatchet, axe, sword, chopper).
- Wound size corresponds to the cross-section of the blade.
- Undermining occurs in the direction of chop.
- The heel (lower end) of the axe typically strikes first, leading to deeper wounds than the toe (upper end).
- The deeper end indicates the assailant's position.
- Clean slicing through bone at impact; rebound can create irregular fragments; may lead to complete or incomplete amputations.
- Mostly homicidal injuries to exposed body parts (head, face, neck, extremities)

Solution for Question 14:

Correct Answer: A) Laceration

Explanation:

Laceration results from blunt force trauma rather than sharp force. It leads to irregular, jagged wounds caused by tearing of the tissue.

Classification of Mechanical Injuries

Blunt Force Injuries	Sharp Force Injuries
Abrasion: Superficial damage to the skin (scraping, rubbing off the outer layer). Contusion: Bruise caused by blood vessels rupturing under the skin due to impact. Laceration: Irregular tear in the skin or tissue caused by crushing or blunt trauma. Fractures/Dislocations: Breaks or misalignment of bones due to strong blunt impact.	Incision: A clean-cut wound where the length is greater than the depth. Stab Wound: A deep wound where the depth is greater than the length. Chop Wound: A deep wound often associated with underlying fractures caused by heavy-bladed tools like an axe.

Solution for Question 15:

Correct Answer: B) Laceration caused by blunt force trauma.

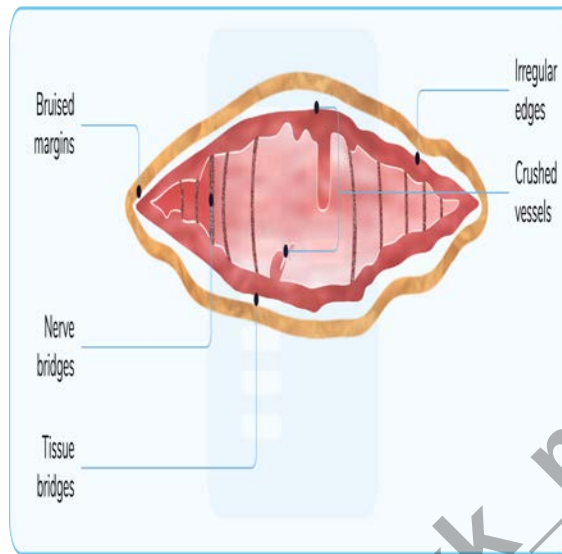
Explanation:

The irregular edges, bruising, and embedded dirt indicate a laceration resulting from blunt force . Such wounds often show uneven tearing and damage to underlying structures like nerves and blood vessels, consistent with the autopsy findings.

Laceration

- Laceration is the tearing or splitting of the skin, mucous membranes, muscles, or internal organs due to a shearing or crushing force from a blunt impact over a broad area.
- When blunt force causes severe bruising and damage to deeper tissues, it is referred to as a crush injury .

Characteristics



- **Irregular Margins** : Ragged, uneven edges with pointed or blunt ends, often showing small tears.
- **Swallow Tails** : Tearing at the ends of lacerations, diverging.
- **Bruising** : Seen around the wound, more pronounced in areas impacted at an angle or with downward force.
- **Tissue Bridges** : Deeper tissues are unevenly torn, leaving nerve, vessel, and connective tissue bridges.
- **Hair Damage** : Crushed hair bulbs; hair and skin tags may be embedded in the wound.
- **Hemorrhage** : Less severe bleeding as arteries are crushed and torn irregularly , retracting and allowing blood to clot .
- **Soiling** : Debris like mud, wood splinters, sand, glass, or hair can become embedded, carrying medico-legal significance.
- **Wound Gaping** : Impact from an angle causes tearing on the opposite side and abrasion on the impact side, leading to gaping.

Incised wounds (Option A) have clean, smooth edges and do not show the uneven tearing or bruising present in this case.

Abrasions (Option C) involve superficial skin damage and typically lack deep tissue involvement and significant bruising.

Puncture wounds (Option D) are small and deep, usually causing localized damage without the extensive tearing or irregular margins seen here.

Solution for Question 16:

Correct Answer: A) Bruises are caused by ruptured vessels, but hypostasis from distension of vessels

Explanation:

Bruises are caused by ruptured vessels like veins, venules, and arterioles, which may be superficial or deep, whereas hypostasis is due to the distension of vessels that occurs after death.

Aspect	Bruise	Hypostasis/ Postmortem staining
Cause (Option A)	Ruptured vessels that may be deep or superficial.	Distension of vessels with blood in the dermis.
Site (Option C)	At the site of and surrounding the injury. May appear anywhere in the body.	Over extensive areas of the most dependant parts.
Appearance (Option D)	Swollen.	No elevation of the affected area.
Epidermis	May be abraded.	Not abraded.
Margins (Option B)	Merge with the surrounding area.	Clearly defined.
Color	Different colors according to the age of the bruise.	Uniform, bluish-purple.
Incision	Extravasation of blood into surrounding tissues and are firmly clotted. Subcutaneous tissues are deep reddish-black.	Blood is seen in blood vessels. Subcutaneous tissues are pale.

Solution for Question 17:

Correct Answer: B) Artificial bruising to falsely accuse someone

Explanation:

The scenario here shows a bruise that is dark brown and irregularly shaped with well-defined margins and redness and inflammation of the surrounding tissues that point to artificial bruises.

Contusions can be divided as follows:

Artificial bruising	True bruising (Option A)
These are bruises created to make a false charge of assault . They are usually present on accessible parts of the body. They appear dark brown and irregular in shape. Redness and inflammation are seen in the surrounding tissues. Produced by the juice of a marking nut rubbed on the body. Well-defined margins with small vesicles. Itching is present. Other substances used: Plumbago Madar/calotropis Semicardium	The cause of true bruising is trauma. They can be located anywhere in the body. They undergo typical color changes. Redness and inflammation are seen on the site. They are usually round with well-defined, diffuse, and irregular margins with no vesicles. They have extravasated blood as content and are not associated with itching.

Postmortem bruising from blood settling (Option C):

- An injury produced before death may lead to a bruise later due to further escape of blood.
- A deep bruise may appear several hours or 1-2 days later, and deeper extravasation of blood may not appear at all.
- Caused by percolation and rapid hemolysis of stagnant blood, leading to the pigment being diffused locally.
- This produces a stain on the surface called a come-out bruise .
- Some bruises are difficult to detect, and hence, the examination of the whole body under UV light helps locate them.

Bruising due to pre-existing medical conditions (Option D) like arteriosclerosis, purpura, leukemia, hemophilia, scurvy, etc., causes easy bruising but appears different from an artificial bruise, and a clear history of these diseases is present.

Solution for Question 18:

Correct answer: B) Grey Turner's sign

Explanation:

The sign here is the Grey Turner's sign which is the ecchymosis seen over the flank or side of the abdomen due to extensive retroperitoneal hemorrhage which is a type of migratory or ectopic bruise.

Gravity shifting of blood (ectopic/migratory bruise):

- Bruises may appear at sites distant from the actual injury (e.g., black eyes).
- The blood tracks along fascial planes or between muscle layers, following the path of least resistance, leading to bruising in superficial tissue areas.
- Therefore, the location of a bruise does not always reflect the site of injury.

Grey Turner's sign:

- Ecchymosis observed over the flank or side of the abdomen which is indicative of extensive retroperitoneal hemorrhage.
- This sign typically develops within 24–48 hours.

Cullen's Sign : (Option C)

- Bluish-black discoloration around the periumbilical area resulting from extensive retroperitoneal or intra-abdominal hemorrhage.
- Common causes include ruptured ectopic pregnancy or acute pancreatitis.
- Other examples: Spectacle hematoma (Option A) around the eyes after a blunt trauma to the forehead Bruise behind the ear due to basal skull fracture known as Battle sign (Option D) Bruise in thigh due to fractured pelvis Bruise around the knee due to a kick on the calf
- Spectacle hematoma (Option A) around the eyes after a blunt trauma to the forehead
- Bruise behind the ear due to basal skull fracture known as Battle sign (Option D)
- Bruise in thigh due to fractured pelvis
- Bruise around the knee due to a kick on the calf
- Spectacle hematoma (Option A) around the eyes after a blunt trauma to the forehead
- Bruise behind the ear due to basal skull fracture known as Battle sign (Option D)
- Bruise in thigh due to fractured pelvis
- Bruise around the knee due to a kick on the calf

Solution for Question 19:

Correct Answer: C) Railway line type

Explanation:

The above clinical scenario shows patterned contusions that appear as two parallel, linear hemorrhages with normal skin in between known as tram track or railway line-type contusions.

Bruises can be divided into three types based on their location:

- Intradermal bruise: in the immediate subepidermal layer. Made by impact with a patterned object and hemorrhage is sharply defined
- Made by impact with a patterned object and hemorrhage is sharply defined
- Subcutaneous bruise: in the subcutaneous tissue. Located often in the fatty layer with blurred edges Caused commonly by a blunt object
- Located often in the fatty layer with blurred edges
- Caused commonly by a blunt object
- Deep bruise: deeper into the subcutaneous tissues.
- Made by impact with a patterned object and hemorrhage is sharply defined
- Located often in the fatty layer with blurred edges
- Caused commonly by a blunt object

Contusions/bruises can be classified into two types based on the way they appear:

Ectopic/percolated/migratory bruise	Patterned bruise
In deep bruises, blood tracks along the fascial planes and may cause the bruise at a different site. This is also called percolated or migratory contusion. Examples: Spectacle hematoma (Option B) around the eyes after a blunt trauma to the forehead Bruise behind the ear due to basal skull fracture known as Battle sign (Option A) Bruise in thigh due to fractured pelvis Bruise around knee due to a kick on calf Grey Turner's sign: Ecchymosis seen over the flank or side of the abdomen due to extensive retroperitoneal hemorrhage. Takes 24-48 hours to develop Cullen's sign: Bluish-black discoloration of the periumbilical skin due to extensive retroperitoneal or intraabdominal bleeding. Causes can be ruptured ectopic pregnancy or acute pancreatitis	The shape and size of the bruise resemble a part of the object that caused it. Shapes of bruises by different weapons: A solid body like a hammer or fist causes round bruises A doughnut bruise due to objects like a cricket ball A rod, stick, or whip leads to two parallel, linear hemorrhages with normal skin in between known as railway line or tram line bruises Blows from a whip are elongated and partially encircle a limb or the body Tire patterns from an automobile accident (Option D)

Solution for Question 20:

Correct Answer: D) Gravity shifting of blood

Explanation:

In the given case scenario, gravity shifting of blood is not a factor influencing the formation of bruises, as the bruises are at the site of trauma rather than at a different site.

Mechanism of injury of contusions	
A contusion or bruise is a leak of blood into the tissues due to the rupture of blood vessels (veins, venules, arterioles) caused due to injury by a blunt object. Some blunt objects that can cause contusions are fist, stones, sticks, bar, whip, hammer, axe, wooden handle, poker, foot with shoe or boot, etc.	
Factors influencing the formation of bruises	
Condition and type of tissues	Vascular and loose tissues may form large bruises even with a small degree of violence. Examples: face, vulva, and scrotum. Firm fibrous tissues, strongly supported tissues, and those covered by a thick dermis produce a comparatively small bruise for moderate trauma. Examples: abdomen, back, scalp, palms, and soles. Injury of tissues over bony surfaces shows prominent bruising.
Age (Option A)	Children bruise easily. Older people bruise easily.
Gender (Option B)	Women bruise more easily than men. Obese people bruise easily due to the high volume of subcutaneous tissue.
Colour of the skin (Option C)	Bruising is more easily visible in fair-skinned people than in dark-skinned. Contusions are better felt in dark-skinned people than seen.
Natural diseases	Prominent bruising is seen with minor trauma in patients with: Arteriosclerosis Whooping cough in children due to violent coughing can cause bruising Purpura hemorrhagica Leukemia Hemophilia Scurvy Vitamin K Prothrombin deficiency Phosphorus poisoning Older persons on systemic steroids
Gravity shifting of the blood	In deep bruises, blood tracks along the fascial planes and may cause the bruise at a different site. This is called ectopic bruising or percolated or migratory contusion. Examples: Spectacle hematoma or black eye in blunt trauma to the forehead. Bruise behind ear due to basal skull fracture (Battle's sign). Bruise in the neck from fracture of the jaw. Bruise around the ankle from a kick on the calf.

Solution for Question 21:

Correct Answer: B) Postmortem abrasions are typically found on bony prominences, while antemortem abrasions can occur on any part of the body.

Explanation:

Postmortem abrasions are often the result of handling or movement of the body after death, and they typically occur over bony prominences where the skin is easily damaged. In contrast, antemortem abrasions can occur anywhere due to trauma experienced while the person is still alive.

Postmortem Abrasion Vs Antemortem Abrasion

Feature	Postmortem Abrasion	Antemortem Abrasion
Image		
Intra-vital reaction or congestion (Option C)	Absent	Present
Site (Option B)	Typically found on bony prominences	Can occur on any part of the body
Color of abrasion (Option A)	Yellowish-white or pale	Reddish-brown to dark brown
Scab formation	Absent or minimal	Present and well-formed
Exudates	Less or absent	Increased (due to inflammatory reaction)
Elevation (Scab)	Less raised	More raised due to active healing
Bleeding/ Color changes (Option D)	No active bleeding	Active bleeding with clot formation
Healing process	No healing process	Healing starts with inflammation

Solution for Question 22:

Correct Answer: C) 4-5 days

Explanation:

By 4-5 days, the wound has undergone healing processes, and the scab typically appears brown, indicating desiccation and progress in wound healing.

Direction of Force in Abrasion and Color of Scab with Time

Direction of Force in Abrasion

The direction of force applied in an abrasion can be identified based on specific characteristics of the skin damage. At the end of an abrasion, the skin shows distinct features that help determine the direction of the force, particularly through epithelial tags or aggregated lines.

Characteristics of the end of an abrasion:

Multiple Lines	Indicate the endpoint of the abrasion. Reflect the direction of the applied force.
Uneven and Longitudinal Lines	Show irregularity at the end of the abrasion. Suggest pulling or scraping of the skin.
Heaped-up or Aggregated Lines	Also known as Epithelial tags (Terminal tags) Skin tissues dragged opposite to the force direction. Point toward the end of the abrasion, indicating the force direction E.g., tags heaped at the right end, if the force was left to right.

Color of Scab and Aging of Abrasion

A scab forms as blood or lymph exudate dries up on the surface of the abrasion, providing valuable information regarding the age of the injury. The color progression of the scab is indicative of how long the abrasion has been present.

Color changes of a scab over time:

Time Post-Injury	Scab Color	Description
Before 12 hours	No scab	Exudate is still fluid and not yet dried.
12-24 hours	Bright red	Fresh blood and exudate are present.
2-3 days	Reddish-brown	Scab darkens, indicating it is ageing.
4-5 days	Brown	Scab continues to darken, turning brown.
6-7 days	Black	Scab becomes black, marking the late stage of healing.

After 7 days	Falls off	Scab typically falls off after this period.
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Solution for Question 23:

Correct Answer: B) Continuous vertical force can cause damage to both the epidermis and underlying structures. Explanation :

The image shows pressure abrasion caused by ligature strangulation . It involves a continuous vertical force that can crush the epidermis and even extend to deeper tissues, potentially leading to necrosis. This differs from superficial abrasions, which do not cause such depth of injury.

Pressure Abrasion (Crushing Abrasion)



Mechanism: Caused by a continuous vertical force (Option D) applied to the skin, crushing the epidermis (Option C) .

Features: Deeper skin injuries (Option A) are often seen with prolonged pressure, potentially leading to necrosis.

Examples: Ligature marks in hanging or strangulation.

Solution for Question 24:

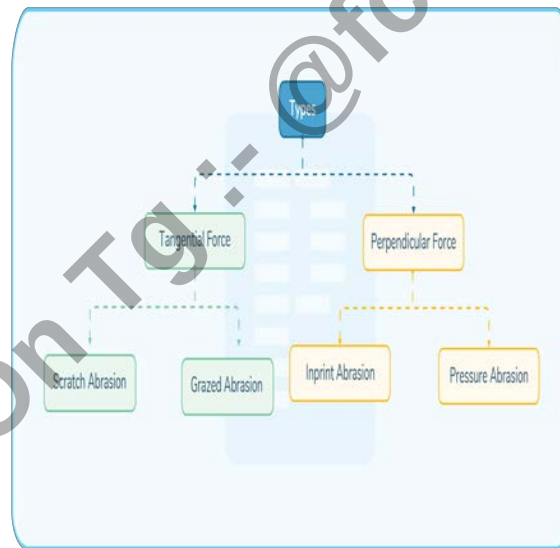
Correct Answer: C) Imprint Abrasion

Explanation:

Imprint abrasions occur when a patterned object, like a vehicle tyre or a textured surface, presses against the skin, leaving a distinct impression of the object. In this case, the patterned nature of the injury suggests an imprint abrasion.

Abrasion:

Abrasion is a superficial injury caused by the removal or damage of the skin's outer layer (epidermis) due to friction or pressure. It is classified into four main types:



Scratch Abrasion (Linear Abrasion)	Grazed Abrasion (Sliding Abrasion)
Mechanism: Caused by tangential (side-to-side) force, producing a long and narrow abrasion. Features: Linear in nature, with length being more significant than width. Examples: Fingernail abrasion, thorn abrasion, and pin abrasion.	Mechanism: Occurs when skin slides over a rough surface, resulting in superficial scrapes. Also Known As: Grinding abrasion, gravel rash, brush-burn. Features: Seen in road traffic accidents or falls, especially on exposed areas like knees or elbows.
Imprint Abrasion (Patterned Abrasion)	Pressure Abrasion (Crushing Abrasion)

Mechanism: Results from compressive force applied perpendicular to the skin, leaving a distinct pattern of the object. Also Known As Impact abrasion, contact abrasion, patterned abrasion. Examples: Tyre marks, bite marks.

Mechanism: Caused by a continuous vertical force applied to the skin, crushing the epidermis. Features: Deeper skin injuries are often seen with prolonged pressure, potentially leading to necrosis. Examples: Ligature marks in hanging or strangulation

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Regional injuries

1. What is the most commonly fractured bone in cases of basilar skull fractures?

- A. Temporal bone
- B. Occipital bone
- C. Sphenoid bone
- D. Frontal bone

2. In which of the following scenarios is a diastatic skull fracture most likely to occur?

- A. An adult hit by a broad surface object
- B. A child with increased intracranial pressure
- C. A teenager experiencing a sports-related injury
- D. An elderly person falling onto their buttocks

3. A 32 year old male is brought to the emergency after sustaining a gunshot wound to the head during a robbery. The bullet trajectory is reported as oblique. Which type of skull fracture is most commonly associated with this presentation?



- A. Depressed fracture
- B. Gutter fracture
- C. Comminuted fracture
- D. Elevated fracture

4. A "motorcyclist fracture," typically results from what type of impact?

- A. Side impact
- B. Direct blow to the chin
- C. Heavy blow to the vertex
- D. Blunt force to the occiput

5. A 25-year-old male presents with facial trauma after being struck in the eye during a sports accident. Which radiographic finding is most indicative of a blowout fracture of the orbit?

- A. Spider web-like pattern on X-ray
 - B. Ping-pong ball dent appearance
 - C. Teardrop sign
 - D. Circular crack around the skull base
-

6. A 45 year old woman falls and strikes the back of her head on the ground. Upon examination, she exhibits signs of a contrecoup injury affecting the frontal lobe. Which of the following statements about her injury is incorrect?

- A. The injury results from the brain striking the frontal skull after an initial impact on the occipital region.
 - B. Contrecoup injuries are commonly associated with RTAs.
 - C. Contrecoup injuries primarily affect the areas directly beneath the point of impact.
 - D. The rotational forces during the fall could contribute to the contrecoup mechanism.
-

7. In the context of intracranial haemorrhages, which of the following statements accurately describes the relationship between the layers of the cranial anatomy and the associated blood vessels?

- A. The epidural space contains bridging veins.
 - B. The subdural space is located between the pia mater and the arachnoid mater.
 - C. The subarachnoid space contains arteries of the circle of Willis.
 - D. The middle meningeal artery is located between the dura mater and the arachnoid mater.
-

8. A 25 year old man is brought to the ED after a fall from a height. He was unconscious upon arrival, and a CT scan was done which revealed the following.

What is the most likely underlying cause of this patient's condition?



- A. Rupture of bridging veins
- B. Rupture of the middle meningeal artery
- C. Venous thrombosis
- D. Subarachnoid hemorrhage

9. A forensic pathologist examines a body that has suffered extensive thermal injury. Which of the following characteristics would most likely indicate the presence of a heat hematoma as a postmortem finding?

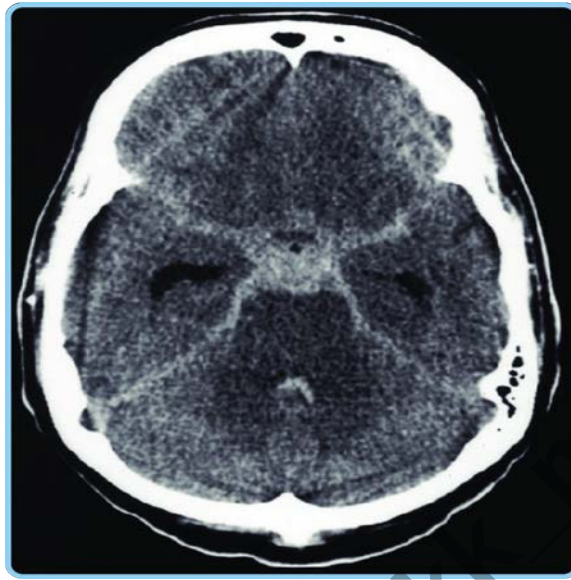
- A. Absence of carboxyhemoglobin
- B. Reddish brown discolouration of the hematoma
- C. Rupture of the meningeal artery
- D. Honeycomb appearance of a blood clot

10. A 65 year old woman presents to the emergency department with confusion and headache that developed gradually over the past two weeks after a minor fall. A CT scan reveals an isodense lesion along the cerebral convexities. What is the most likely diagnosis?



- A. Acute Subdural Hemorrhage
- B. Subacute Subdural Hemorrhage
- C. Chronic Subdural Hemorrhage
- D. Epidural Hematoma

11. A 50 y/o man is found unconscious in his home after complaining of severe headache. A CT of the patient is given below. His medical history is significant for hypertension and smoking. Which of the following is the most likely underlying cause in this patient?



- A. Ruptured berry aneurysm
- B. Trauma from a fall
- C. Hypertensive encephalopathy
- D. Acute subdural haematoma

12. A retired boxer presents with a gradual decline in his coordination, memory, and speech over the past few years. His wife reports personality changes, including paranoia and social withdrawal. O/E, he displays dysarthria and broad-based gait. Which of the following is the most likely diagnosis?

- A. Parkinson's disease
- B. Alzheimer's disease
- C. Dementia pugilistica
- D. Multiple sclerosis

13. A 65 year old man with a history of hypertension presents to the ED with a sudden onset of weakness in his right arm and leg. He also complains of a severe headache and nausea. On examination, he has a decreased level of consciousness and right-sided hemiparesis. A CT scan shows a hyperdense lesion in the left basal ganglia. Which of the following is the most likely cause of his condition?

- A. Epidural haematoma
 - B. Subarachnoid haemorrhage
 - C. Hypertensive intracerebral haemorrhage
 - D. Subdural haematoma
-

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Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	1
Question 5	3
Question 6	3
Question 7	3
Question 8	2
Question 9	4
Question 10	2
Question 11	1
Question 12	3
Question 13	3

Solution for Question 1:

Correct answer: A) Temporal bone

Explanation:

The temporal bone is the most commonly fractured in basilar skull fractures. These fractures can lead to serious complications, including damage to cranial nerves VII (facial nerve) and VIII (vestibulocochlear nerve) and the carotid artery, as well as potential cerebrospinal fluid (CSF) leaks.

Thickness of skull bones		
	Male	Female
Frontal	7.8 mm	8.6 mm
Parietal	9.6 mm	10.1 mm
Occipital	10.1 mm	10 mm
Temporal	6 mm	6 mm
Temporal bone fracture		
Longitudinal Temporal Bone Fracture		Transverse Temporal Bone Fracture
The most common type (70-90% of temporal bone fractures). Runs parallel to the long axis of the petrous temporal bone. Otic Capsule is spared. Conductive hearing loss due to ossicle involvement and tympanic membrane involvement. Facial nerve involvement occurs in about 25% of cases.		Less common (10-30% of temporal bone fractures). Extends perpendicular to the petrous temporal bone's long axis. Otic Capsule is involved, leading to sensorineural hearing loss. Ossicles and tympanic membranes are generally unaffected. Higher incidence of facial nerve involvement.

Occipital bone (Option B), Sphenoid bone (Option C), Frontal bone (Option D): While the occipital, sphenoid, and frontal bones can be involved in basilar skull fractures, they are not as commonly fractured as the temporal bone. Their occurrences are significantly less frequent compared to fractures of the temporal bone.

Solution for Question 2:

Correct answer: B) A child with increased intracranial pressure

Explanation:

Fissured Fracture (Linear Fracture)	Diastatic Fracture (Sutural Fracture)
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The most common skull fracture accounts for about 70% of cases. Linear cracks may involve the vertex or skull base. It can affect the full thickness of the bone or just the inner/outer layers. The fracture line is irregular and fine, sometimes hard to detect on X-rays. It runs parallel to the compressive force. No depression of bone fragments.	Involves the separation of cranial sutures. More common in children. Often occurs with other skull fractures, especially fissured fractures. Commonly seen at the sagittal suture.
Causes: Impact against broad surfaces (e.g., falls or road traffic accidents). (Option A ruled out) Falls on feet or buttocks, transmitting force upwards. (Option D ruled out)	Causes: Blunt trauma to the head. Increased intracranial pressure leads to suture separation. (Option B)
Narrow fissure skull fracture	

A teenager experiencing a sports-related injury (Option C): Sports-related injuries typically do not involve sutural separation, which is specific to diastatic fractures.

Solution for Question 3:

Correct answer: B) Gutter fracture

Explanation:

Depressed Fracture	Comminuted Fracture	Gutter Fracture
Occurs when a fragment of the skull is pushed inward towards the brain. Caused by blows from heavy objects with a small surface (e.g., hammers, bricks). (Option A ruled out) The shape of the depressed area may resemble the object that caused the injury. Variants: Elevated Fracture : One end of the fractured fragment is raised above the skull, while the other is pushed inward. Caused by sharp objects like an axe. (Option D ruled out) Pond Fracture : Seen in infants, this smooth depression occurs without a fracture line, resembling a dent in a ping-pong ball. Caused by obstetrics forceps blade.	Involves multiple intersecting fracture lines, breaking the bone into several fragments. Caused by high-impact trauma (e.g., vehicular accidents, crushing injuries). (Option C ruled out) It may occur alongside other fractures (e.g., fissured or depressed fractures). Spider web-like patterns on an X-ray are characteristic of comminuted fractures. In severe cases, bone fragments may penetrate the brain or become displaced.	A groove or channel is formed in the skull bone, often due to an oblique bullet wound. (Option B) Frequently associated with comminuted and depressed fractures of the inner skull table. Bone fragments can cause injury to the meninges or brain.
Depressed fracture	Comminuted fracture	Gutter fracture

Solution for Question 4:

Correct answer: A) Side impact

Explanation:

Ring Fracture	Motorcyclist Fracture (Hinge Fracture)
Encircles the foramen magnum at the skull base. Runs from the sella turcica to the petrous ridges and posterior fossa. It can extend to the middle ear and roof of the nose.	Transverse fracture at the skull base, splitting it and creating a "hinge" - nodding face sign . It extends from one petrous ridge across the sella turcica to the other petrous ridge.
Caused by: (Option B, C and D ruled out) Falls landing on feet/buttocks (force transmitted through the spine). Heavy blows to the top of the skull/ vertex. Twisting of the head on the spine. Blows to the occiput or chin.	Often caused by side impacts . (Option A)
Ring fracture	Hinge fracture

Solution for Question 5:

Correct answer: C) Teardrop sign

Explanation:

Blowout fracture is a fracture of the bony orbit (eye socket) resulting from blunt trauma to the eye. It causes fractures to the medial wall and floor of the orbit. The teardrop sign is a significant radiographic finding in blowout fractures , indicating the herniation of orbital soft tissues, such as fat and muscles, into the maxillary sinus. This often leads to pain, restricted eye movements and diplopia.



Blowout fracture of right orbit showing Teardrop sign

Spider web-like pattern on X-ray (Option A): The spider web-like pattern on an X-ray is characteristic of comminuted fractures, which involve multiple fracture lines creating a fragmented appearance.

Ping-pong ball dent appearance (Option B): The ping-pong ball dent appearance is associated with pond fractures, which occur in infants and young children. This type of fracture is characterised by a smooth, concave indentation in the skull without a distinct fracture line.

Circular crack around the skull base (Option D): A circular crack around the skull base is indicative of a ring fracture, which encircles the foramen magnum and is associated with severe trauma.

Solution for Question 6:

Explanation:

Coup injuries primarily affect the areas directly beneath the point of impact, while contrecoup injuries affect the side opposite the impact.

Coup vs. Contrecoup Injuries		
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Features	Coup	Contrecoup
Definition	Brain injury directly beneath the point of impact.	Brain injury on the side opposite the impact.
Mechanism	Caused by direct impact force.	Occurs when a moving head is suddenly decelerated by hitting a firm surface eg. striking the head on the ground after a fall during RTAs. (Option B ruled out)
Occurrence	More common when a moving object strikes the head.	Common in falls or blows to a stationary head.
Common scenarios	Common in moving head striking an object, e.g., hitting a car dashboard in an accident.	Seen in falls, where the head hits the ground or an unyielding surface, causing injury on the opposite side of the impact.
Organs involved	It can be seen in the brain, heart, spleen and lungs. (More commonly in the brain)	

Theories of Contrecoup Injuries:

- Shear Strain Theory (Holbourn): Brain particles are pulled apart due to rotational forces. (Option D ruled out)
- Vacuum Theory: A vacuum or cavity forms in the cranial cavity, exerting suction on the brain opposite the point of impact, causing damage.
- Deceleration Theory: Sudden arrest of the head causes the brain to strike the skull at the opposite pole of impact.

Common scenarios:

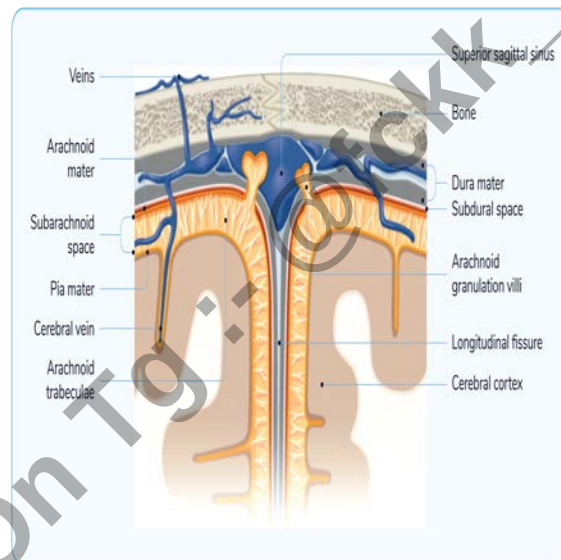
- Occipital impact: Produces frontal contrecoup injury due to brain displacement against the frontal lobes. (Option A ruled out)
- Temporal or parietal impact: Results in contralateral contrecoup injury on the opposite side of the brain.
- Top of the head impact: Can cause ventral surface contusion of the cerebral hemisphere.
- Frontal region impact: May not produce occipital contrecoup injury due to the smooth internal surface of the posterior cranial fossa

Solution for Question 7:

Correct Answer: C) The subarachnoid space contains arteries of the circle of Willis

Explanation: The subarachnoid space is situated between the arachnoid mater and pia mater and contains major arteries, including those of the circle of Willis.

- This space is usually filled with clear, colourless cerebrospinal fluid.
- Head trauma or rupture of an intracranial aneurysm causes a subarachnoid haemorrhage (bleeding in the subarachnoid space).
- The most common specific locations of intracranial aneurysms are at the middle cerebral artery bifurcation and along the anterior communicating artery.



Meningeal layers

Intracranial haemorrhages

- The first layer is the skull vault (cranium), followed by the dura mater, arachnoid mater, pia mater, and brain tissue (cerebrum).
- The space between the skull and dura contains the middle meningeal artery.
- The middle meningeal artery is most prominent at the pterion, the point where the frontal, parietal, sphenoid, and temporal bones meet.

Space	Location	Blood vessel involved
Epidural Space	Between the skull vault and dura mater. The collection of blood here leads to epidural hematoma.	Middle meningeal artery (Option D ruled out)

Subdural Space	Between the arachnoid and dura mater. The collection of blood here leads to subdural hematoma. (Option B ruled out)	Bridging veins (which drain into the dural venous sinus) (Option A ruled out)
Subarachnoid Space	Between the arachnoid and pia mater	Arteries of the circle of Willis (Middle cerebral artery, Anterior communicating artery -most commonly) (Option C)

Solution for Question 8:

Correct Answer: B) Rupture of the middle meningeal artery

Explanation: The CT scan of the patient showing a biconvex accumulation of blood between the skull and dura mater after a fall confirms the diagnosis of EDH. EDH is primarily caused by trauma leading to the rupture of the middle meningeal artery.

Anatomy : The dura mater is a strong, grey-bluish connective tissue membrane. Its outer layer adheres to the skull, and the inner layer merges with the arachnoid.

Causes :

- Primarily caused by trauma, resulting in the skull moving relative to the dura and creating an extradural space.
- Injury Sites : Commonly due to fractures of the squamous temporal bone, leading to injury of the middle meningeal artery (posterior branch).

Clinical Features :

- Initial Symptoms : History of head injury; may include temporary unconsciousness.
- Lucid Interval : A period (30-40% of cases) after initial unconsciousness where the patient regains awareness, lasting 2-4 hours to a week before neurological symptoms decline.
- Progression : Increased intracranial pressure can lead to confusion, lethargy, coma, and neurological signs such as contralateral weakness, dilated pupils (initially on the side of haemorrhage), and possible decerebrate rigidity.
- Fatality: Typically associated with a minimum haemorrhage volume of 100 ml, often due to respiratory failure from brainstem compression.

Imaging : Brain CT scans show a biconvex or lens-shaped mass typical of extradural haemorrhage .



Treatment :

- Craniotomy : Recommended for acute symptomatic extradural hematomas to facilitate complete hematoma evacuation.
- Trephination (Burr Hole Evacuation) : Employed when advanced surgical expertise is unavailable, serving as a crucial intervention to reduce mortality.

Rupture of bridging veins (Option A) : This type of injury is associated with subdural haemorrhage , not extradural haemorrhage, and would not produce a biconvex mass on CT.



Venous thrombosis (Option C) : While it can lead to increased intracranial pressure, it does not typically result in a localized biconvex mass, as seen in extradural haemorrhage.

Subarachnoid haemorrhage (Option D) : This condition presents with diffuse blood in the subarachnoid space rather than a well-defined mass

Solution for Question 9:

Correct Answer: D) Honeycomb appearance of a blood clot

Explanation:

The honeycomb appearance of a blood clot is indicative of a heat hematoma, which is a specific postmortem finding associated with thermal injuries.

A heat hematoma resembles an extradural haemorrhage in appearance and therefore needs to be differentiated from it.

Features	Extradural Haematoma	Heat Haematoma
Cause	Trauma to the skull.	Exposure of the head to intense heat leads to charring. (Option C ruled out)
Mechanism	Most commonly due to rupture of the meningeal artery	Shrinkage of meninges and brain, causing brain tissue to ooze out.
Location	Usually adjacent to Sylvian fissure and limited due to dural attachments at the suture line.	Anywhere, usually bilateral.
Distribution	Localised	Diffuse
Trauma signs	Signs of blunt force trauma are present.	Signs of blunt force trauma are absent.
Blood clot	A rubbery, reddish-purple blood clot is seen. Discoid shape.	A soft, friable clot of light chocolate colour is seen. (Option B ruled out) Honeycomb appearance, evenly distributed or sickle-shaped. (Option C)
Carboxyhemoglobin	Absent	Present (Option A ruled out)

Solution for Question 10:

Correct Answer: B) Subacute Subdural Hemorrhage

Explanation:

The gradual onset of symptoms and isodense appearance on CT are characteristic of subacute subdural haemorrhage, typically occurring days to weeks after injury.

- Anatomy : Subdural space is narrow, between the dura mater and arachnoid. Cerebral (bridging) veins cross this space to reach venous sinuses.
- Subdural space is narrow, between the dura mater and arachnoid.
- Cerebral (bridging) veins cross this space to reach venous sinuses.
- Causes : Rupture of parasagittal bridging veins or inferior cerebral veins.
- Occurrence : Common in children, the elderly, alcoholics, or those with brain atrophy. Occurs from minor trauma, often without fractures. Seen in falls, vehicle accidents, or assaults.
- Common in children, the elderly, alcoholics, or those with brain atrophy.
- Occurs from minor trauma, often without fractures.
- Seen in falls, vehicle accidents, or assaults.
- Subdural space is narrow, between the dura mater and arachnoid.
- Cerebral (bridging) veins cross this space to reach venous sinuses.
- Common in children, the elderly, alcoholics, or those with brain atrophy.
- Occurs from minor trauma, often without fractures.
- Seen in falls, vehicle accidents, or assaults.

Acute SDH	Subacute SDH	Chronic SDH
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Rapid onset after trauma Rupture of large bridging veins or cortical arteries. The most common symptoms are mental confusion, drowsiness, dilation of the pupil (ipsilateral) and contralateral involvement of hemiparesis. Death may occur if the haemorrhage volume is about 100 to 150 ml. Hyperdense on CT. (Option A ruled out)	Symptoms develop over days to weeks after injury. Usually from smaller bridging veins. Isodense on CT	Develops weeks or months later Often in the elderly; Slow bleeding leads to gradual symptoms. Symptoms arise weeks/months later; May lead to Subdural Hygroma : -Tear in the arachnoid allows CSF to collect in the subdural space, causing cerebral compression. Hypodense on CT (Option C ruled out)
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- CT Findings : Crescentic (crescent moon-shaped, concave, banana-shaped) appearance. More extensive than epidural haemorrhage (EDH), covering a larger area.
- Crescentic (crescent moon-shaped, concave, banana-shaped) appearance.
- More extensive than epidural haemorrhage (EDH), covering a larger area.
- Treatment: Decompression surgery
- Decompression surgery
- Crescentic (crescent moon-shaped, concave, banana-shaped) appearance.
- More extensive than epidural haemorrhage (EDH), covering a larger area.
- Decompression surgery

Epidural Hematoma (Option D) : This diagnosis is unlikely because epidural hematomas typically appear as biconvex (lens-shaped) lesions on CT, rather than the crescentic shape seen in subdural haemorrhages.



Solution for Question 11:

Correct Answer: A) Ruptured berry aneurysm

Explanation:

The CT scan shows evidence of subarachnoid haemorrhage, which is characterised by bleeding into the space between the brain and the meninges. It shows a classical star sign .

The most common cause of subarachnoid haemorrhage is a ruptured aneurysm , particularly a berry aneurysm. The patient's age, history of hypertension, and smoking are all risk factors for developing aneurysms.

Subarachnoid Haemorrhage (SAH):

Bleeding into the subarachnoid space (between the arachnoid and pia mater) which is normally filled with cerebrospinal fluid (CSF).

Causes of SAH:

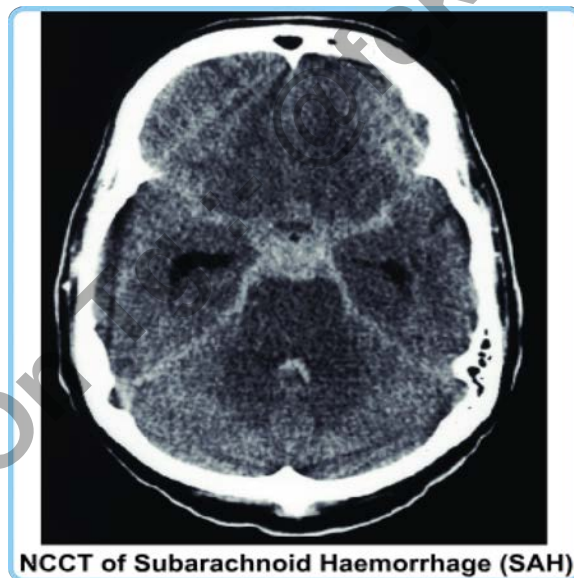
Traumatic Causes	Non-traumatic Causes
Head trauma (m/c) Cerebral contusions or lacerations Explosive blasts Asphyxia by strangulation Traumatic asphyxia Blows to the neck Accidents and falls Cervical manipulations (e.g., vertebral or basilar arteries) Rupture of traumatic intracerebral haemorrhage (ICH) Prolonged hyperextension of the head during bronchoscopy	Ruptured berry aneurysm (saccular aneurysm) (m/c) Arteriovenous malformations Atherosclerosis Leaking intracerebral haemorrhage Purpuric states Leukaemia

Signs & Symptoms:

- Sudden, severe "thunderclap headache" (reaches peak intensity within a minute).
- Nausea and vomiting.
- Neck stiffness, photophobia, drowsiness, or agitation.
- Depressed consciousness.
- Key findings on examination : Meningism and positive Kernig's sign.

Diagnosis:

- First line : Non-contrast CT scan (shows hyperdense blood collections along falx, basilar cisterns).
- Second line : Lumbar puncture (if CT is inconclusive) – shows blood-mixed CSF under increased pressure.



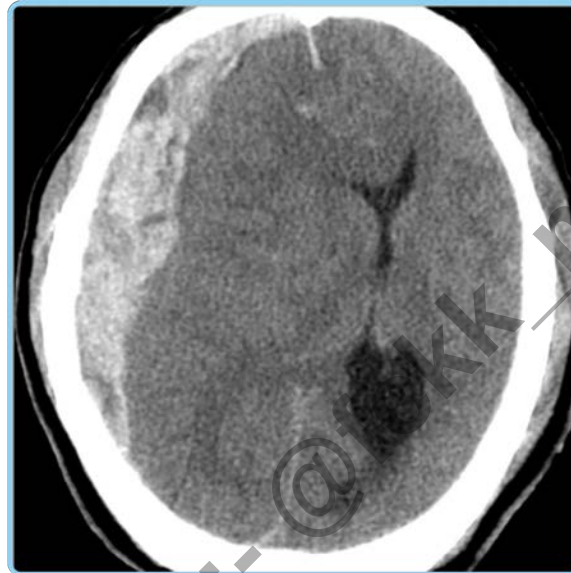
Medico-legal Aspects:

- Trauma-induced rupture : Can determine if trauma caused or contributed to berry aneurysm rupture (especially with acute neurological symptoms after injury).
- Vessel condition : Older individuals with atherosclerosis and hypertension have a higher rupture risk.
- Postmortem SAH : This can occur due to decomposition, requiring careful examination to distinguish from ante-mortem haemorrhage.

Trauma from a fall (Option B) : No trauma history is mentioned, making this less likely compared to an aneurysm rupture.

Hypertensive encephalopathy (Option C) : Hypertensive encephalopathy typically causes diffuse cerebral oedema and other changes, not isolated subarachnoid haemorrhage. The site is usually putamen which is the most common site of hypertensive bleed.

Acute subdural haematoma (Option D) : Subdural haematomas usually result from head trauma and appear differently on imaging compared to SAH.



Solution for Question 12:

Correct Answer: C) Dementia pugilistica

Explanation:

The patient's history of repeated head trauma (boxing), along with symptoms like incoordination, memory loss, personality changes, dysarthria and broad-based gait, is characteristic of punch-drunk syndrome (dementia pugilistica) .

Punch-Drunk Syndrome:

Also known as dementia pugilistica or boxer's encephalopathy , it results from repeated cerebral concussions, typically in boxers.

Signs and Symptoms:

- Deterioration of speed and reflexes

- Incoordination
- Personality changes (social instability, paranoia, delusions)
- Memory loss (progressing to dementia)
- Parkinsonian signs : Ataxia or intention tremors Shuffling, broad-based gait Dysarthria (difficulty speaking)
- Ataxia or intention tremors
- Shuffling, broad-based gait
- Dysarthria (difficulty speaking)
- Ataxia or intention tremors
- Shuffling, broad-based gait
- Dysarthria (difficulty speaking)

Autopsy findings:

- Chronic subdural haematoma (SDH)
- Attenuation of the corpus callosum
- Diffuse axonal injury (DAI)
- Cortical atrophy

Parkinson's disease (Option A) : Although Parkinson's disease shares some features like tremors and gait disturbances, this patient has a history of boxing and repeated head trauma, which makes punch-drunk syndrome more likely.

Alzheimer's disease (Option B) is primarily associated with memory loss and cognitive decline but the history of boxing and gait disturbances differentiate punch drunk syndrome from it.

Multiple sclerosis (Option D) can cause motor and cognitive symptoms but is less likely given the patient's history and specific symptoms linked to repeated head trauma.

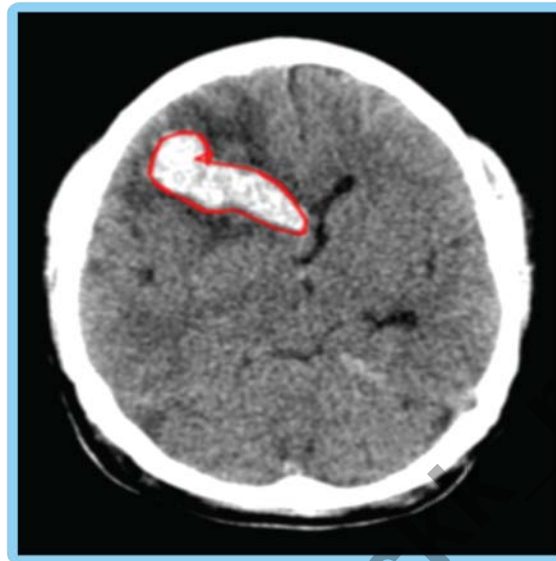
Solution for Question 13:

Correct Answer: C) Hypertensive intracerebral haemorrhage

Explanation:

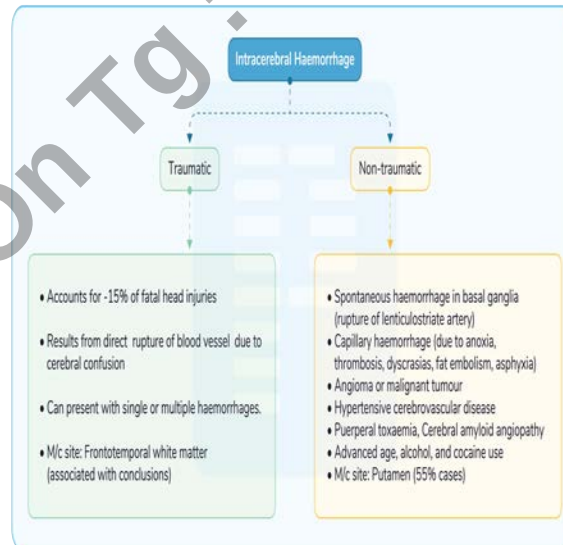
The patient has a history of hypertension and the CT findings (hyperdense lesion in the basal ganglia) are typical of hypertensive ICH, which commonly affects the basal ganglia. Putamen is

the most common site of hypertensive bleeding.



Intracerebral Haemorrhage (ICH): ICH refers to bleeding within the brain parenchyma, distinct from surface bleeding.

Causes of ICH: The most common cause is hypertension.



Clinical Features:

- Sudden onset of focal neurological deficit
- Decreased consciousness
- Signs of raised intracranial pressure (vomiting, headache)

- Contralateral hemiparesis
- Seizures are rare

Diagnosis:

- CT scan : ICH appears as a hyperdense lesion
- Comparison with other haemorrhages: EDH: Biconvex, lenticular (Option A ruled out) SDH: Crescentic (Option D ruled out) SAH: Hyperdense along the falx and basilar cisterns (Option B ruled out)
- EDH: Biconvex, lenticular (Option A ruled out)
- SDH: Crescentic (Option D ruled out)
- SAH: Hyperdense along the falx and basilar cisterns (Option B ruled out)
- EDH: Biconvex, lenticular (Option A ruled out)
- SDH: Crescentic (Option D ruled out)
- SAH: Hyperdense along the falx and basilar cisterns (Option B ruled out)

EDH	SDH	SAH
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Thermal injuries and RTA

1. A 25-year-old male sustains second-degree burns covering 40% of his total body surface area (TBSA) in an industrial accident. According to the Parkland formula, how much fluid should be administered within the first 24 hours?

- A. 4,000 mL
- B. 8,000 mL
- C. 12,000 mL
- D. 16,000 mL

2. Which of the following is a frequent type of injury seen in tertiary impact injuries?

- A. Contrecoup injury
- B. Tailgating injury
- C. Terrible triad injury
- D. Sparrow footmark injury

3. A forensic pathologist is tasked with determining whether the burns on a severely charred body occurred before or after death. During the examination, the pathologist observes a combination of findings, some of which are more suggestive of antemortem burns. Which of the following combinations of findings is least indicative of antemortem burns?

- A. Elevated carbon monoxide levels and soot in the trachea
- B. Pugilistic attitude and blisters filled with clear fluid
- C. Presence of an inflammatory reaction at the burn edges and cherry-red discoloration of the skin

D. Blisters filled with protein-rich fluid and soot in the upper respiratory tract

4. Which of the following is a classic pattern of injury seen in pedestrian children involved in road traffic accidents?

- A. Tailgating
 - B. Whiplash
 - C. Waddell's Triad
 - D. Terrible Triad
-

5. Which of the following is a pathognomonic feature of a lightning strike injury?

- A. Lichtenberg flowers
 - B. Joule burns
 - C. Crocodile skin lesions
 - D. Sparrow foot marks
-

6. Which factor is considered more dangerous in causing electrocution?

- A. AC current
 - B. DC current
 - C. Voltage = 50V
 - D. Dry skin
-

7. Which type of fracture is most commonly associated with seat belt injuries in motor vehicle accidents?

- A. Smith fracture
 - B. Chance fracture
 - C. Barton fracture
 - D. Colles fracture
-

8. Which among the following are usually associated with collisions over the dashboard during an RTA? 1) Posterior cruciate ligament injury 2) Posterior dislocation of hip 3) Patella fracture

- A. 1, 2
 - B. 2, 3
 - C. 1, 3
 - D. 1, 2, 3
-

9. What is the most likely structure to be affected in this illustration shown?



- A. DJ flexure of small intestine
- B. Mesentery
- C. Spleen
- D. Liver

10. Person sustained a blow to the front of the chest by the steering wheel in an RTA. Which site of the aorta is most likely to be affected?

- A. Ascending aorta - 1.5cms proximal to ligamentum arteriosum
- B. Ascending aorta - 1.5cms distal to ligamentum arteriosum
- C. Descending aorta - 1.5cms proximal to ligamentum arteriosum
- D. Descending aorta - 1.5cms distal to ligamentum arteriosum

11. What is the mechanism of injury and common level affected in fracture-dislocation seen with whiplash injury

- A. Flexion-hyperextension; C3-C4
 - B. Sideways movement of head; C3-C4
 - C. Flexion-hyperextension; C5-C6
 - D. Sideways movement of head; C5-C6
-

12. Windshield impact to front seat passenger may impact which among the following injuries

- A. Avulsion of forehead skin
 - B. Dicing injury
 - C. Sparrow foot marks
 - D. All of the above
-

13. What is a unique type of injury often associated with motorcyclists?

- A. Bone pearls
 - B. Lichtenberg flowers
 - C. Whiplash
 - D. Tailgating
-

14. Which of the following statements accurately describes 'Hide and die' syndrome?

- A. It occurs when hypothermic victims discard their clothing due to overheating sensations
- B. It is characterized by the victim pulling household items over themselves or hiding in unusual places, due to mental confusion from hypothermia

- C. It refers to a skin condition caused by prolonged exposure to cold, resulting in ulcerative lesions
 - D. It is the result of an intentional attempt by victims to escape from a dangerous situation
-

15. A police surgeon is tasked with determining whether the burns on a deceased individual occurred before or after death. The body was discovered in a residential fire, and toxicology revealed elevated carbon monoxide levels. Which among the following findings is more relatable to the elevated carbon monoxide levels?

- A. Absence of soot in the upper respiratory tract
 - B. Blisters filled with clear fluid and air
 - C. Presence of an inflammatory reaction at the burn edges
 - D. Both B and C
-

16. A 7 year old child is admitted in ER with burns to the entire posterior trunk and the left lower limb. Using the Lund and Browder chart, calculate the percentage of total body surface area (TBSA) affected?

- A. 25%
 - B. 35%
 - C. 36%
 - D. 40%
-

17. A 30-year-old male is brought to the emergency department with burns covering his anterior trunk and anterior right upper limb. Using the Wallace Rule of Nines, what percentage of total body surface area (TBSA) is affected, and how does this correlate with forensic medicine in terms of assessing the severity and cause of injury?

- A. 18%
- B. 22.5%
- C. 27%
- D. 36%

18. Destruction of superficial skin is classified under which category in Wilson's classification of burns?

- A. Epidermal
- B. Dermal
- C. Dermo-epidermal
- D. Deep

19. Match the following types of burns with its characteristics

Type of heat
Characteristic
1. Dry heat burn a. Vesicles over the burned area
2. Moist heat burn b. No vesicles present
3. Chemical burn c. Vesicles at the circumference of the burnt area

Type of heat	Characteristic
1. Dry heat burn	a. Vesicles over the burned area
2. Moist heat burn	b. No vesicles present
3. Chemical burn	c. Vesicles at the circumference of the burnt area

- A. 1-a, 2-b, 3-c
- B. 1-c, 2-b, 3-a
- C. 1-c, 2-a, 3-b
- D. 1-b, 2-c, 3-a

20. Which of the following statements regarding heat stroke is wrong?

- A. Sweating is present
 - B. Core body temperature of more than 40.5°C
 - C. Failure of thermoregulation
 - D. Sudden collapse
-

21. Which of the following statements regarding immersion syndrome and frostbite is false?

- A. Immersion syndrome affects the extremities
 - B. Pressure is applied in frostbite treatment
 - C. Dampness is required for immersion syndrome
 - D. Tetanus prophylaxis is given in frostbite cases with deep infection
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	2
Question 4	3
Question 5	1
Question 6	1
Question 7	2
Question 8	4
Question 9	2
Question 10	4
Question 11	3
Question 12	4
Question 13	4
Question 14	2
Question 15	3
Question 16	2
Question 17	2
Question 18	3

Question 19	3
Question 20	1
Question 21	2

Solution for Question 1:

Correct Answer: D) 16,000 mL

Explanation

- The Parkland formula is used to determine the amount of fluid (Ringer's lactate) to administer within the first 24 hours after a burn.
- For adults, the calculation is 4 ml/kg/% burn, while for children, it is 3 ml/kg/% burn.
- Half of the total volume is administered in the first 8 hours, and the remaining half is given over the next 16 hours.
- The formula can be expressed as:

$\text{TBSA (\%)} \times \text{Weight (kg)} \times 4 = \text{Volume (ml) to be administered}$

- For a patient weighing 70 kg with 40% TBSA burns, $4 \times 40 \times 70 = 11,200$ mL (over 24 hours)
- Half of the fluid (5600 mL) is given in the first 8 hours.
- The remaining (5600 mL) is given over the next 16 hours.
- The Parkland formula's role in forensic medicine is crucial in determining whether appropriate medical care was provided to burn victims.
- In cases of suspected neglect, assault, or improper care (such as withholding adequate resuscitation), the forensic evaluation of fluid management can provide critical evidence.
- Under-resuscitation or over-resuscitation may impact survival rates and can help forensic investigators determine whether negligence or foul play occurred during treatment.

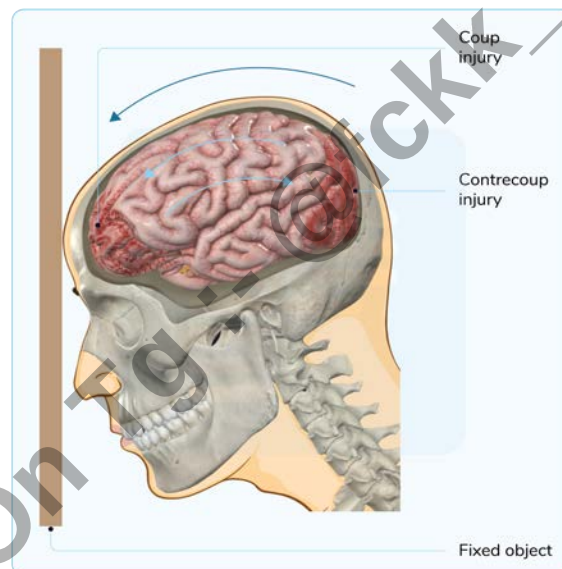
Medico-legal Aspect	
1st Degree burns	Simple
2nd Degree burns	Grievous
3rd Degree burns	Grievous

Solution for Question 2:

Correct Answer: A) Contrecoup injury

Explanation:

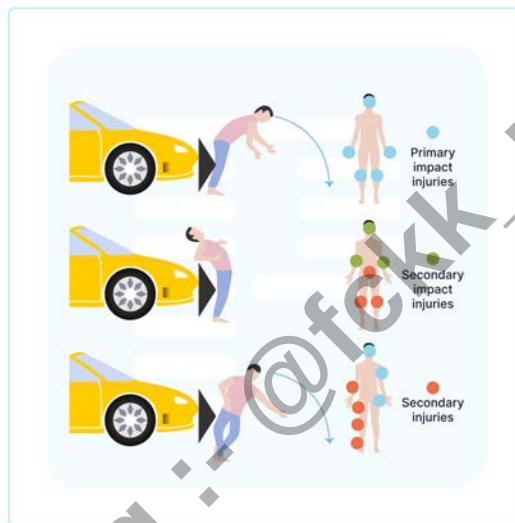
Contrecoup injury occurs when the brain strikes the opposite side of the skull from where the impact occurred, often seen in tertiary impact injuries.



Secondary injuries/Tertiary impact injuries

- Results from body parts, striking the ground, following primary and secondary impact and are more lethal.
- When the pedestrian is thrown to the ground, he/she sustains abrasions, bruises or lacerations over the bony prominences.
- Contrecoup injury: Brain damage is frequent without any associated skull fracture due to moving head of the victim, being suddenly stopped on impact, resulting in diffuse damage to the axons.
- Fracture of the skull and ribs due to direct contact and fracture of spine due to hyper flexion or extension.

- Sometimes, pedestrians are run over if knocked down by the vehicle. This will tend to occur if the pedestrian centre of gravity is lower than the impact site.
- Tyre tread marks on one surface of the body with graze like abrasions on the opposite side.
- Compression of the chest resulting in flail chest.
- Flaying injury: Avulsion injury when the wheel moves over a fleshy part causing degloving of skin and subcutaneous tissue by tearing it away from underlying tissue .
- Burning and singeing of skin and hair, resulting from discharge of hot exhaust.



Tailgating injury (Option B): Unique injury wherein the motorcyclist drives under the rear of the truck, causing head injuries and decapitation.

Terrible triad injury (Option C): This refers to a specific set of injuries to the elbow (dislocation, radial head fracture, coronoid fracture), not related to tertiary impact.

Sparrow foot marks (Option D): Peculiar facial lacerations due to contact with the shattered windscreen seen in transportation injuries.

Solution for Question 3:

Correct Answer: B) Pugilistic attitude and blisters filled with clear fluid

Explanation

While common in burn victims, pugilistic posture is a non-specific finding and can occur whether the individual was alive or dead at the time of the burn.

Non specific findings in burns cases				
Phenomenon	Description	Cause	Key Features	Medico-Legal Significance
Puppet Organs	Internal organs become shriveled, firm, and hardened by heat.	Prolonged exposure to high temperatures.	Organs resemble firm, "cooked" structures.	Shows prolonged heat exposure.
Heat Hematoma	Large artifact resembling extradural hematoma, containing 100–120 ml of blood.	Blood from longitudinal venous sinuses or diploic veins.	1.5 cm thick; the heat forces blood from calvarium through veins onto dura.	Resembles extradural hematoma.
Heat Ruptures	Skin splits resembling incised or lacerated wounds due to heat contraction of tissue.	Contraction of heated, coagulated tissue.	Fissures contain nerves, vessels, connective tissue; no clotted blood, irregular margins, no bruising.	Differentiates heat damage from violent injury.
Pugilistic Attitude	Flexed posture of arms and legs resembling a boxing stance.	Coagulation of muscle proteins and dehydration.	Flexor muscles contract more than extensors; seen in cremations.	No significance whether alive or dead pre-burn.
Heat Fractures	Fractures caused by heat, with long bones showing curved fractures.	Muscle shrinkage pulls on tendons and bones during exposure to high heat.	Heat fractures do not involve skull sutures; neurocranium remains as fragments up to 10 cm; hyoid can become fragile.	Identifies extreme heat exposure and helps distinguish heat fractures from trauma.
Heat Hematoma				
Heat Ruptures				
Pugilistic Attitude				
Heat Fractures				

Solution for Question 4:

Correct Answer: C) Waddell's Triad

Explanation:

Waddell's Triad is a classic pattern of injury seen in pediatric pedestrians struck by vehicles.

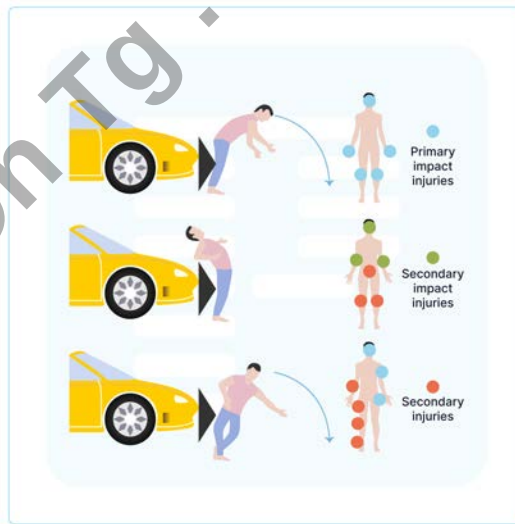
- Femoral fracture
- Intra-thoracic or abdominal injury
- Contralateral head injury.

Pedestrian Injuries

- Most common type of transportation injuries which are blunt force injuries, the result of travel on the ground, in the air and on water.

Three patterns of pedestrian injuries seen:

- primary impact injuries
- Secondary impact injuries
- Secondary injuries / Tertiary impact injuries



Primary Impact Injuries:

Injuries	Occur when the victim is struck by the vehicle, often following recognizable patterns. Common injuries include abrasions, lacerations, fractures of legs, thighs, or buttocks, and fractures of the fibula and tibia. When an adult is hit by a car, the impact is often at knee level, and injuries vary depending on the victim's position and vehicle orientation. Common cause of death is due to head injury and fracture dislocation of cervical spine, mainly at the atlanto-occipital joint.
Factors influencing injury behavior include	Position of the victim's feet (whether they were firmly placed or raised at impact). Speed of the vehicle (low speed results in being thrown onto the bonnet; at higher speeds, the victim is projected or scooped up). Road surface type (smooth, rough, gravel). Point of impact relative to the center of gravity. When the pedestrian is knocked down from behind with both feet fixed to the ground, there will be fracture of the bones of the lower limbs.
Waddell's Triad	A classic pattern in pedestrian children struck by vehicles. Fractured femoral shaft. Intrathoracic or intra-abdominal injuries. Contralateral head injury.
Injury Behavior Based on Position and Vehicle Impact	On Being struck from behind with feet not firmly on the ground: The victim's feet are thrown backwards. Possible head injuries from striking against the windscreen. Victims can be scooped up or fall to one side, sustaining further head injuries from striking the ground. Struck from the front: Injuries to chest, abdomen, with potential fractures of ribs or vertebrae. Pelvic or sacroiliac fractures, as well as fractures of the tibia or fibula from bumper impact. Bumper impact injuries: Soft tissue damage and fractures (typically wedge-shaped). Fractures may suggest the point of impact and movement direction of the vehicle. The location and measurement of the bumper fracture on the victim provide details about the vehicle's height and speed. Wedge type bumper fracture of the tibia seen. Marks may suggest if the victim was walking or lying on the road when struck. When the pedestrian walks into the side of a moving vehicle Glancing abrasions Crushing lacerations on the side or front of the face, chest and arms. Fracture of ribs with or without laceration. Sustain secondary injuries Fracture of the skull occurs due to direct impact of the vehicle on the head.

Secondary impact injuries:

- Often seen in cases of scooped up victims being thrown over the bonnet.
- The person sustains injuries by hitting his head against the windscreen, its rim or side pillars.
- Extensive abrasions, bruises, and lacerations may be seen.

Tailgating (Option A): Unique injury wherein the motorcyclist drives under the rear of the truck, causing head injuries and decapitation.

Whiplash (Option B): This injury typically occurs in motor vehicle accidents, where the neck is rapidly flexed and extended, usually due to rear-end collisions, but it does not relate to pedestrian injuries.

Terrible Triad (Option D): This refers to a specific injury pattern in the elbow involving dislocation, radial head fracture, and coronoid process fracture, not related to pedestrian injuries.

Solution for Question 5:

Correct Answer: A) Lichtenberg figures (Lichtenberg flowers)

Explanation:

Lichtenberg flowers: These are characteristic fern-like patterns that appear on the skin following a lightning strike.

Lightning stroke

- Lightning bolt(DC>1000 million V) is produced when the charged undersurface of a thundercloud sends its electrical charge to the ground.
- Death is caused by high voltage current due to cardiopulmonary arrest or electrothermal injuries.

Lichtenberg flowers/Arborescent markings:

- Pathognomic feature of lightning strike.
- Superficial, several inches long, thin, irregular, tortuous, dendritic red marks on the skin having resemblance to the branches of a tree.
- This Fern-like pattern of erythema in the skin is usually found over the shoulders or flanks.
- Appears within 1hr and disappears in 24-48 hrs if person survives.

Other features of lightning strike:

- Development of edema of skin at the point of entry wound.
- Linear burns and surface burns
- Fusing and magnetisation of metallic articles.
- Contusions, lacerations, rupture of tympanic membrane.

- Singed hair and patterned skin burns marks underneath metal articles of jewellery.



Joule burns (Option B): These are burns caused by the conversion of electrical energy into heat within the body and are not specific to lightning strikes. They can occur in any electrical injury.

Crocodile skin lesions (Option C): These are indicative of other types of electrical injuries, characterized by a specific pattern of skin damage but are not unique to lightning strikes.

Sparrow foot marks (Option D): Peculiar facial lacerations due to contact with the shattered windscreen seen in transportation injuries.

Solution for Question 6:

Correct Answer: A) AC current

Explanation:

Ac current is more dangerous because it causes continuous muscle contractions, making it harder to release the source of shock. It can also disrupt the heart rhythm, leading to fatal arrhythmias.

Electrocution(Electrical injuries)

- Refers to injury or death caused by electric shock.

- It occurs when an electric current passes through the body, disrupting normal bodily functions, particularly the heart, muscles, and nerves.

Electricity exerts two major effects on the body:

- Cellular depolarisation of nerves and muscles.
- Heat production

Factors influencing electrocution:

- Current Type: AC (alternating current) is more dangerous than DC (direct current) at similar voltage levels because AC causes continuous muscle contractions and is more likely to induce ventricular fibrillation.
- Voltage Level: Electrocution is rare at <100V and most deaths occur at >200V. Amperage is more important as it indicates the actual intensity of electricity, which passes through the body.
- Resistance of the Body: The human body's resistance varies by tissue type (blood, muscle, fat, and skin). Wet or broken skin decreases resistance, increasing the likelihood of severe injuries whereas dry skin offers higher resistance.
- Duration of Contact: Longer exposure leads to more severe injuries. Short contact may cause local burns, while prolonged contact can result in heart and respiratory failure.
- Pathway of Current: Death is more likely to occur if the brain stem or heart is in the direct path of the current.

Effects due to passage of electricity:

Low Tension Injuries (<1000V):

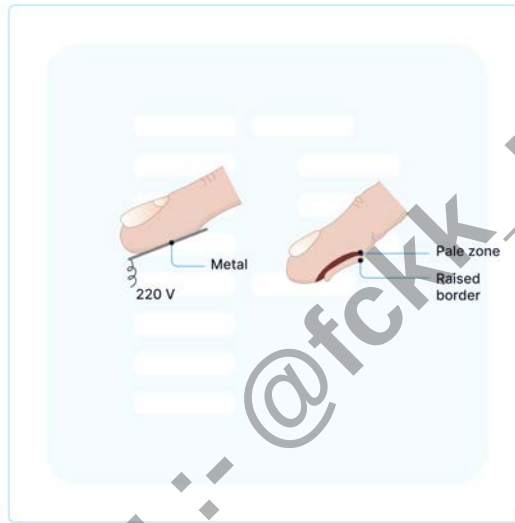
- Typically cause superficial burns at the points of entry and exit.
- Most common sites of low-voltage contact entry injury are hands(fingers) and that of exit injury is foot or opposite hand.
- Muscle contractions, tetany, and possible ventricular fibrillation may occur. Victims may suffer from arrhythmias or cardiac arrest if the current crosses the heart.

High Tension Injuries (>1000V):

- Injuries can be caused by three sources- flash, flame or the current itself.
- High voltage causes deeper burns, organ damage, and massive tissue destruction.
- It may lead to immediate cardiac or respiratory arrest. The current can cause thermal injuries, with electricity creating heat as it passes through the body.
- Crocodile skin lesions : Third degree burns or red brown punch out spark lesions.

Characteristics of Injuries:

- **Entry Wounds:** Typically small and well-defined, resembling a burn. In low tension injuries, the area might be red and blistered; in high tension injuries, there may be charring or significant tissue damage.
- **Exit Wounds:** Larger than the entry wound, often with greater tissue damage. High voltage can cause severe destruction, sometimes with a crater-like appearance.
- **Joule burns/Electrical burns:** specific and diagnostic of electrical burns characterised by no red line surrounding the burns with puckering of skin and surrounding area of pallor.



- **Flash or spark burns:** When the contact is less firm, so that an air gap exist between skin and conductor and causes the outer skin keratin to melt over a small area
- **Current pearls:** Small balls of molten metal derived from the metal of the contacting electrode.
- **Bone pearls/Wax drippings:** Heat generated by the current may melt the calcium phosphate, which is seen in x-ray of limbs as typical round dense foci.

Post-mortem Findings in Electrocution:

External Signs:

- Face is pale, eyes are congested and pupils are dilated.
- Presence of entry and exit wounds:
 - Burns at the points of contact.
 - Fractures or dislocation of bones

Internal Signs:

- Deep tissue burns
- Heart damage

- Lungs congested
- Focal necrosis with variable haemorrhage seen in the heart.
- Brain, meninges and parenchymatous organs are congested.
- Electrical petechiae due to combination of venous congestion due to cardiac arrest and sudden rise in blood pressure induced by muscle contractions.

Cause of death in electrocution:

- Ventricular fibrillation -M/C
- Paralysis of the respiratory muscle
- Inhibition of respiratory Centre
- Ventricular asystole

DC current (Option B): Less hazardous because it causes a single muscle contraction, which may allow the person to pull away from the source.

Voltage = 50V (Option C): Generally considered a low voltage and below the threshold usually considered dangerous for causing electrocution in dry conditions.

Dry skin (Option D): Dry skin provides higher resistance to electrical current, reducing the risk of electrocution compared to moist skin.

Solution for Question 7:

Correct Answer: B) Chance fracture of the lumbar spine

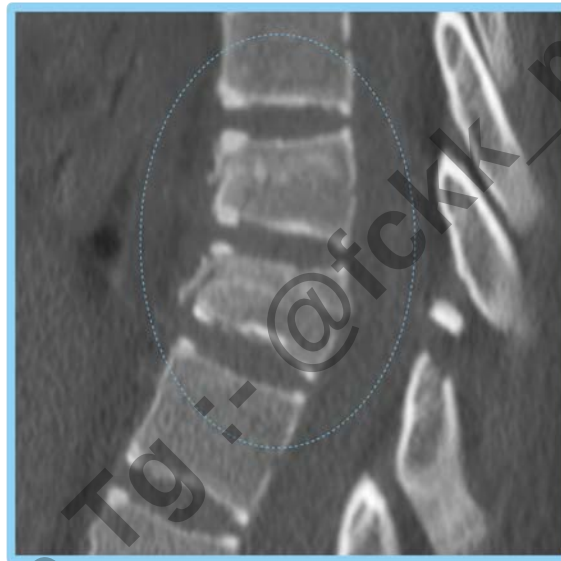
Explanation:

The Chance fracture/Seat belt fracture, a horizontal fracture through the vertebral body, pedicles and spinous process of the lumbar spine, is commonly associated with seat belt injuries.

Chance fracture:

- Horizontal or transverse fracture through the vertebral body, pedicles, and spinous process, commonly affecting the lumbar spine.
- Mechanism of Injury: Results from a flexion-distraction force (bending and pulling apart) on the spine, typically during sudden deceleration, such as in motor vehicle accidents with improper or no shoulder restraint.

- Flexion-distraction spinal injury with no rotation of the fracture fragments or lateral displacement.
- Often associated with seat belt injuries when only the lap belt is used, leading to significant force across the lumbar spine.
- Most commonly occurs in the thoracolumbar region (T10-L2).
- Clinical Presentation: Patients may report lower back pain and possibly abdominal pain due to associated organ injuries.
- Diagnosis : Diagnosed through X-ray or CT scan, showing a horizontal fracture line through the posterior elements of the spine.



- Treatment : immobilization and surgical stabilization. Usually reduced on Risser table by applying hyper extension to the injured thoracolumbar area.

Smith fracture (Option A): Distal radius fracture with palmar (volar) displacement, typically from a fall on a flexed wrist.

Barton fracture (Option C): Intra-articular distal radius fracture with dislocation of the carpal bones, usually from a fall on an outstretched hand.

Colles fracture (Option D): Distal radius fracture with dorsal displacement, commonly resulting from a fall on an extended wrist.

Solution for Question 8:

Correct answer: D) 1, 2, 3

Explanation:

All of the following - posterior cruciate ligament injury , posterior dislocation of hip and fracture-patella are usually seen in dashboard injuries.

Dashboard injury:

Affected	Front seat passenger (hip and knees flexed position)
3 Ps	Patella fracture Posterior cruciate ligament injury Posterior hip dislocation
Other injuries	Fracture - pelvis/femur Most common nerve injured: Sciatic nerve

Solution for Question 9:

Correct answer: B) Mesentery

Explanation:

Illustration shown is seat belt injury - Mesentery is most commonly affected due to seatbelt injuries.



Seat belts:

- Restrains the body against severe deceleration and hence preventing impact from windscreen, steering wheel and other obstruction. Though it slows down deceleration, it can jam at a certain point leading to injuries.
- Follows Inertia-reel mechanism.
- Primarily of two types: Shoulder diagonal (most commonly used in vehicles) Lap strap (aircrafts)
- Shoulder diagonal (most commonly used in vehicles)
- Lap strap (aircrafts)
- Shoulder diagonal (most commonly used in vehicles)
- Lap strap (aircrafts)

Seat Belt injuries:

Patterned contusion (known as Seat belt sign)	Drivers: starts from the right shoulder Passengers: starts from the left shoulder
Intra-abdominal (deceleration- compression)	Mesentery - most commonly affected (Option B) Small intestine: DJ flexure (primary), Ileocaecal junction (secondary) Liver, Spleen
Vertebral injuries (distraction- compression)	Distraction(separation of joint): seen on posterior side Compression: seen on anterior side Chance fracture: fracture involving all 3 columns of spine
Seat belt syndrome	Seat belt sign + intra-abdominal injuries + vertebral injuries

Solution for Question 10:

Correct answer: D) Descending aorta - 1.5cms distal to ligamentum arteriosum

Explanation:

Junction of Aortic arch with the descending aorta distal to origin of subclavian artery i.e 1.5cms distal to ligamentum arteriosum is the most likely site of tear in steering wheel injuries as the lower thoracic aorta at that point is bound to the anterior longitudinal ligament on the front of dorsal spine.

Steering wheel injury:

Sites affected	Sternum/Ribs (Flail chest) Chest (patterned bruises seen) Elbow/forearm (pressure transmission effect)
Aorta	Intimal tears in aorta (either partial or complete). Site: 1.5cms distal to ligamentum arteriosum (junction of aortic arch with descending aorta) (Option D) Multiple parallel intimal tears near the main transection is known as Ladder rung tears. Osseous punch: compression of aorta between sternum and thoracic vertebra.
Other organs involved	Heart: Hemopericardium -> Cardiac tamponade Lungs: Pincer contusions seen Liver > Spleen

Solution for Question 11:

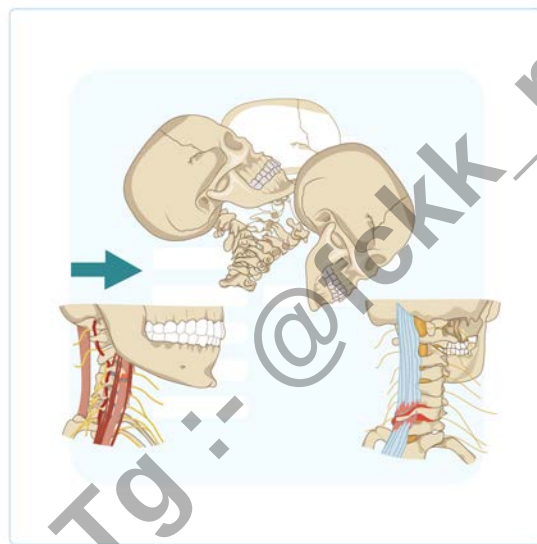
Correct answer: C) Flexion-hyperextension; C5-C6

Explanation:

Whiplash injury is caused due to violent acceleration-deceleration force leading to sudden flexion-hyperextension of head and can be associated with Fracture-dislocation of C5-C6 vertebra.

Whiplash injury: (soft tissue injury)

Mechanism	Acceleration: hyperextension followed by hyperflexion Deceleration: hyperflexion followed by hyperextension
Associated injuries	Atlanto-axial dislocation > C5-C6 fracture-dislocation Anterior longitudinal ligament and cervical ligaments are torn
Passenger	Most commonly front seat passenger is affected
Prevention	Use of head restraints (head rest) prevents hyperextension of neck



Solution for Question 12:

Correct answer: D) All of the above

Explanation:

Windshield impact to either driver/front seat passenger can cause sparrow foot marks , Dicing injury , avulsion of forehead skin , basilar skull fractures.

Windshield injury:

Sparrow foot marks	Multiple, small, punctate facial lacerations due to shattered glass pieces. Mechanism: Driver/passenger ramming onto the glass . Site: Predominantly over the face More specific for windshield injuries.
Dicing injury	More common than sparrow foot marks Multiple superficial cuts over the skin. Mechanism: Glass pieces falling towards the person . Site: All over the face and body Can also occur in rear seat passengers
Other injuries	Avulsion of forehead skin Basilar skull fractures Fracture-dislocation

Solution for Question 13:

Correct Answer: D) Tailgating

Explanation:

Tailgating/Under-running is a unique injury wherein the motorcyclist drives under the rear of the truck, causing head injuries and decapitation.

Motorcycle Injuries

- An accident that might result in minor injuries with an automobile, can result in death with a motorcycle.
- Motorcyclists experience death rate 35 times greater than the occupants of a car.
- Most common causes of motorcycle accidents are alcohol, drugs and environmental factors like potholes, bumps, reckless driving.
- Most common cause of motorcycle fatality is running off the road.
- Most injuries are due to ejection from the vehicle into the roads, due to high speed and instability of the vehicle.
- Primary injuries: Open fracture of tibia and fibula.
- Secondary injuries: Fractures of skull, ribs and cervical spine.
- Hinge fracture / Motorcyclist fracture: Transverse fracture of the base of the skull is common
- Temporo-parietal fractures are also common with ring fractures around foramen magnum.
- Passengers falling off the backs of the motorcycle will have lacerations of the back of the head, fractures of posterior cranial fossa.

- Blood samples should be analysed for the presence and amount of alcohol and drugs, if consumed since the question of contributory negligence may subsequently arise.

Bone pearls (Option A): Small, rounded calcifications seen in electrocution.

Lichtenberg flowers (Option B): Skin patterns seen after a lightning strike.

Whiplash (Option C): Neck injury caused by rapid back-and-forth movement, commonly seen in car accidents but not specific to motorcyclists.

Solution for Question 14:

Correct Answer: B) It is characterized by the victim pulling household items over themselves or hiding in unusual places, due to mental confusion from hypothermia

Explanation:

In the 'hide and die' syndrome, the body of a hypothermia victim is found in unusual locations resembling an attempt to "hide" or create a protective environment due to mental confusion and a possible hibernation reflex (confused with signs of homicide or robbery)

Hypothermia:

Definition	Core temperature below 36°C (96.8°F)
Signs	Disorientation Dulling of consciousness Loss of reflex Fall in respiration, heart rate, and blood pressure Red patches and pallor of the skin Edema of the face Stiffness of neck muscles
Autopsy findings	Patches of pink to brownish-pink discoloration Postmortem staining: Pink/bright red Cyanosis or pallor in extremities ("white deaths") Edema of feet and blistering of skin Stomach: Wischnowsky spots- brownish black colored gastric mucosal ulcers Viscera-congested, hemorrhagic

Phenomena seen in hypothermia victims	Paradoxical undressing - found undressed due to disorientation which increases heat loss (mistaken for evidence of sexual assault or homicide) (Option A ruled out) 'Hide and die' syndrome- found in unusual locations resembling an attempt to "hide" or create a protective environment due to mental confusion and a possible hibernation reflex (confused with signs of homicide or robbery) (Option B) Chilblain (Erythema Pernio)- red, itching skin lesions typically found on extremities due to cold exposure, often accompanied by edema and blistering.
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Solution for Question 15:

Correct Answer: C) Presence of an inflammatory reaction at the burn edges

Explanation:

Presence of an inflammatory reaction at the burn edges is more relatable to elevated carbon monoxide levels because both indicate that the individual was alive during the fire. Inflammatory reactions at burn edges occur only if the body is still capable of mounting a physiological response to the burn, which requires the individual to be alive.

S.No	Feature	Antemortem burns	Postmortem burns
1	Line of redness	Present	Absent
2	Vesicles (Option B and D ruled out)	Contain serous fluid, rich in albumin, chloride, and some polymorphs	Contain air; if fluid is present, it contains little albumin and no chloride
3	Base of vesicles	Red and inflamed	Dull, dry, hard and yellow
4	Soot in upper respiratory tract (Option A ruled out)	May be present	Absent
5	Inflammation and repair	Present along with pus and slough	Absent
6	Healing	Granulation tissue seen in old cases	Absent
7	Carboxyhemoglobin	Present	Absent

8	Enzyme reaction	Increase in enzymes in the periphery of burns	No such increase
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Solution for Question 16:

Correct Answer: B) 35%

Explanation

- Lund and Browder Chart adjusts the percentage of body surface area burned according to the patient's age, accounting for developmental changes in body proportions. For a 7-year-old: Posterior trunk: 18% of TBSA. Left lower limb: $(14 + 6 \times 0.5)\%$ of TBSA = 17% of TBSA. Total affected TBSA = 18% + 17% = 35%.
- Posterior trunk: 18% of TBSA.
- Left lower limb: $(14 + 6 \times 0.5)\%$ of TBSA = 17% of TBSA.
- Total affected TBSA = 18% + 17% = 35%.
- Posterior trunk: 18% of TBSA.
- Left lower limb: $(14 + 6 \times 0.5)\%$ of TBSA = 17% of TBSA.
- Total affected TBSA = 18% + 17% = 35%.
- The use of the Lund and Browder chart is vital in forensic cases to accurately assess injury severity in cases of burns in children.
- The estimation of Total Body Surface Area (TBSA) using Wallace's Rule of Nines is not precise for children due to their proportionately larger head surface area.

Body Surface Area Assessment in Children (Lund and Browder Chart)	
Baseline Percentages (Under 1 Year)	Head: 18% of Total Body Surface Area (TBSA) Each Leg: 14% of TBSA Trunk and Arms: Same as adult percentages
Age Adjustment Formula	For each year over age 1: Add 0.5% to each leg Subtract 1.0% from head Continue adjustments until adult values are reached.
Example Calculation	For a 3-year-old child: Start with baseline values Adjust for 2 years (age 1 to 3): Legs: $14\% + (2 \times 0.5\%) = 15\%$ each Head: $18\% - (2 \times 1.0\%) = 16\%$

Solution for Question 17:

Correct Answer: B) 22.5%

Explanation

- According to the Wallace Rule of Nines, the anterior trunk accounts for 18% of TBSA, and the anterior right upper limb contributes an additional 4.5%, making a total of 22.5% in this scenario.
- This rule is essential in estimating burn severity and planning fluid resuscitation.

Body Part	Estimated TBSA Adults	Labeled Image
Entire left arm	9%	
Entire right arm	9%	
Head & neck	9%	
Chest	9%	
Back of chest	9%	
Abdomen	9%	
Back of Abdomen	9%	
Perineum	1%	
Entire left leg	18%	
Entire right leg	18%	

Solution for Question 18:

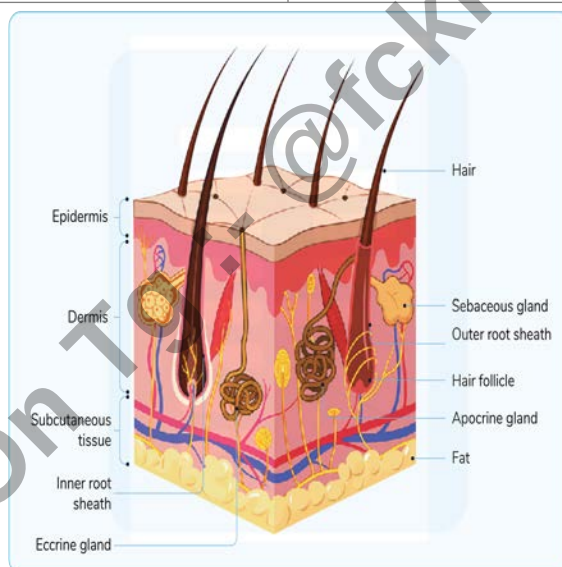
Correct Answer: C) Dermo-epidermal

Explanation:

Destruction of superficial skin or whole skin including the dermis is categorized under dermo-epidermal burns in Wilson's classification of burns

Wilson's classification of burns:

Category	Degree of damage
Epidermal (Option A ruled out)	Erythema Vesication with blister formation
Dermo-epidermal (Option C)	Destruction of superficial skin Destruction of whole skin including dermis
Deep (Option D ruled out)	Destruction of deep fascia, muscles Complete charring involving vessels, nerves, bones



Another classification of burns - Dupuytren's

Solution for Question 19:

Correct Answer: C) 1-c, 2-a, 3-b

Explanation:

Feature	Dry Heat Burn	Moist Heat Burn	Chemical Burn
Cause	Flame, heated body, or X-rays	Solid steam or liquid > 60°C	Corrosives
Site	At or above the site of contact	At and below the site of contact	At or below the site of contact
Splashing	Absent	Present	Present
Skin	Dry, wrinkled, and may be charred	Sodden, bleached	Corroded and devitalized
Vesicles	At the circumference of the burnt area	Over the burnt area	Usually not present
Redline	Present	Present	Absent
Color	Black	Bleached	Distinctive coloration
Charring	Present	Absent	Absent
Singeing	Present	Absent	Absent
Ulceration	Absent	Absent	Present
Scar	Thick, contracted	Thin, less contracted	Thick, contracted
Clothes	Burnt	Wet, not burnt	It may be burnt, with characteristic stains

Solution for Question 20:

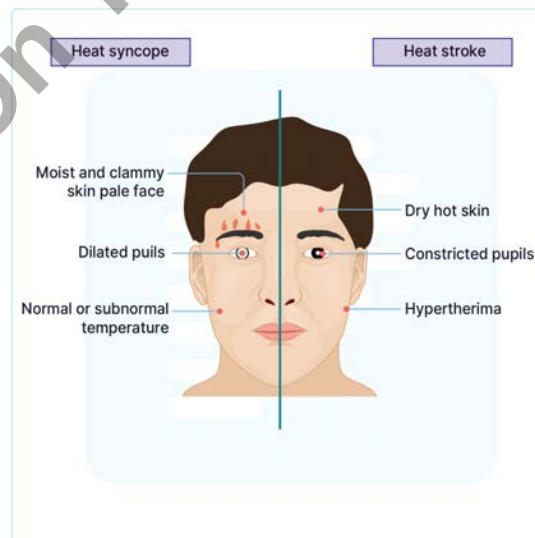
Correct Answer: A) Sweating is present

Explanation:

Sweating is absent in heat stroke but present in exertional heat stroke

Condition	Cause	Clinical Features	Postmortem Findings
-----------	-------	-------------------	---------------------

Heat Cramps (Miner's/Stoker's/Fireman's Cramps)	Fluid and electrolyte depletion (dilutional hyponatremia), following vigorous activity and profuse sweating.	Sudden severe muscle cramps (arms, legs, abdomen) Tender, hard, lumpy muscles Flushed face, dilated pupils, dizziness, vomiting Moist, cool skin, body temperature normal/slightly elevated	Non-specific
Heat Exhaustion	Prolonged strenuous activity with inadequate water/salt intake in a hot environment.	Nausea, vomiting, malaise, myalgia CNS symptoms: dizziness, fatigue, impaired judgment, psychosis (severe water depletion) Moist skin, increased pulse, rectal temperature $> 37.8^{\circ}\text{C}$	Non-specific
Heat Stroke	Failure of thermoregulation in extreme heat exposure (Option C ruled out)	Sudden collapse-loss of consciousness (Option D ruled out) Skin hot, initially wet, then dry Hyperventilation-tachycardia Core temp $> 40.5^{\circ}\text{C}$ (Option B ruled out) Pupils contracted Convulsions Unconsciousness Cerebral dysfunction High fever Absence of sweating (Option A)	Congested lungs, intrathoracic petechiae (children) Subendocardial hemorrhages Brain congestion, flattened convolutions, petechiae in the 3rd/4th ventricle walls



Solution for Question 21:

Correct Answer: B) Pressure is applied for frostbite

Explanation:

Pressure or friction is avoided in frostbite

Condition	Cause	Symptoms	Treatment
Immersion Syndrome (Trench/Immersion Foot/Hand)	Prolonged exposure to severe cold ($<10^{\circ}\text{C}$) and dampness (common in warfare, trenches, or sea immersion) (Option C ruled out)	Affects extremities (feet, hands) (Option A ruled out) Cold, numbness, burning, cyanosis	Air drying, protecting extremities from trauma and infection, gradual rewarming by air exposure at room temperature (avoid heat, massaging, or moistening the skin)
Frostbite	Exposure to extreme cold (-2.5°C), causing tissue freezing and blood supply obstruction	Blue-black discoloration of fingers, toes, nose, ears, face. Mild cases: numbness, prickling. Severe cases: paresthesia, stiffness, necrosis, gangrene, edema, blisters.	Rewarming: For superficial frostbite (frostnip): Firm steady pressure is applied with a warm hand (without rubbing), by placing fingers in the armpits and for the feet, by covering with dry socks For deep frostbite: Frozen extremity is immersed for several minutes in a moving water bath, heated to $40\text{--}42^{\circ}\text{C}$, until the distal tip of the part being thawed, flushes Pressure or friction is avoided (Option B) Anti-infective measures: Tetanus prophylaxis and antibiotics for deep infection (Option D ruled out)

Previous Year Questions

1. Leading questions can be asked in?

- A. Cross-examination
- B. Re-examination
- C. Anytime
- D. Examination in chief

2. Please match the following weapons with the potential injuries they may cause:

A. Axe B. RTA C. Blade D. Lathi 1. Incised wound 2. Train track bruise 3. Grazed abrasion 4. Chop wound

A. Axe B. RTA C. Blade D. Lathi

1. Incised wound 2. Train track bruise 3

- A. A-4, B-3, C-1, D-2
- B. A-1, B-3, C-4, D-2
- C. A-1, B-2, C-4, D-3
- D. A-4, B-1, C-3, D-2

3. Which injury is not seen in a person brought to hospital with a history of RTA ?

- A. Contusion
- B. Abrasions
- C. Laceration
- D. Chop wound

4. During a forensic autopsy, a gunshot victim is examined, and the doctor observes that the firearm injury has a stellate shape. What will be the range of gunshot?

- A. Close shot
 - B. Contact shot
 - C. Near shot
 - D. Distant shot
-

5. The dead body of a 30-year-old male drowning victim was brought in for autopsy. Upon examination, multiple cuts were observed on the victim's arm. What finding is likely to be seen in this scenario?



- A. Defense wound
- B. Self inflicted injury
- C. Accidental cut wound from falling

D. Washer-women hands

6. A 42 year old patient, a case of road traffic accident presented with EDH due to involvement of Middle Meningeal Artery. Where will the impact of collision be?

- A. Pterion
 - B. Angle of mandible
 - C. Lambdoid suture
 - D. Nasion
-

7. A 35-year-old female patient is admitted to the emergency department following an accidental burn incident. On examination, it is noted that she has sustained burn injuries to her right upper limb, the entire right lower limb, and the entire front and back of her abdomen. Based on the rule of nine, what is the estimated percentage of her total body surface area (TBSA) that is burned?

- A. 45%
 - B. 60%
 - C. 28%
 - D. 70%
-

8. Sequence of rigor mortis is?

- A. Center to periphery
- B. Head to foot
- C. Foot to head
- D. Simultaneously

9. Which of the following is the bleeding vessel in extradural hemorrhage?

- A. Middle meningeal artery
 - B. Bridging veins
 - C. Circle of Willis
 - D. Anterior cerebral artery
-

10. Identify the bullet range



- A. Distant range
 - B. Close contact
 - C. Close range
 - D. Can not be predicted
-

11. A person got stabbed in his anterior thigh and died. His body was brought for postmortem. What type of injury is this?

- A. Incised wound
 - B. Penetrating wound
 - C. Puncture wound
 - D. Blunt force injury
-

12. What type of fracture is caused by a hammer?

- A. Linear
 - B. Depressed
 - C. Hinge
 - D. Ring
-

13. A 3-year-old child presented with multiple burn injuries involving the entire head, neck & upper limb. What is the percentage of burn?

- A. 25-30%
 - B. 40-44%
 - C. 18-20%
 - D. 5-10%
-

14. In alleged history of gunshot injury, there is burning, blackening, tattooing around the wound. The injury is:

- A. Close shot entry wound
 - B. Close shot exit wound
 - C. Distant shot entry wound
 - D. Distant shot exit wound
-

15. Bullet fingerprinting involve:

- A. Primary markings
 - B. Secondary markings
 - C. Both
 - D. None
-

16. In a case where a gunshot victim was brought for autopsy, the entry wound displayed margins that were turned inward. Additionally, there was evidence of tattooing around the wound, but no signs of blackening or singeing of hair were observed. Based on this information, what is the likely firing range of the gunshot?

- A. Contact shot
 - B. Near shot
 - C. Close shot
 - D. Distant shot
-

17. What is bullet wipe ?

- A. Residual from barrel of gun
- B. Blackening
- C. Gutter fracture of skull

D. Tattooing

18. Which of the following is true about exhumation?

- A. The executive magistrate can order for exhumation
 - B. Police can order for exhumation
 - C. Postmortem can be done on exhumated body
 - D. CrPC 174 is related to exhumation
-

19. During measurement of a rifled barrel weapon, the caliber depends on?

- A. Distance between two opposite lands
 - B. Distance between two opposite grooves
 - C. Number of lead pellets
 - D. Mass and velocity
-

20. During a police investigation, some bones are uncovered from the crime scene. Which of the following test would aid in differentiating between human and canine bones, during the forensic analysis?

- A. Precipitin test
 - B. Acid-phosphatase
 - C. LDH
 - D. Teichmann test
-

21. Two handkerchiefs with blood stains are found at a crime scene. What is the most appropriate procedure for handling these items?

- A. Place them in same plastic bag and seal it for preservation
 - B. Rinse the bloodstains with water to prevent contamination
 - C. Preserve in an envelope/paper bag
 - D. Place them in different plastic bags and seal it for preservation
-

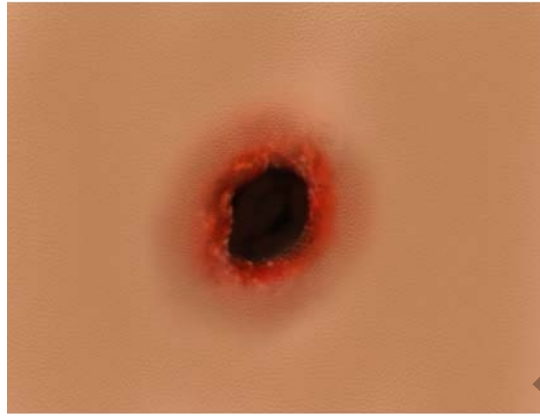
22. Most common organ involved in seat-belt injury is?

- A. Spleen
 - B. Liver
 - C. Pancreas
 - D. Hollow viscus
-

23. The most common organ injuries in air blast shock waves will be?

- A. Intestine
 - B. Lungs
 - C. Kidney
 - D. Heart
-

24. Identify the gun that caused the wound and the range of the shot.:



- A. Shotgun, intermediate range
- B. Shotgun, close range
- C. Pistol, near shot
- D. Pistol, close shot

25. A patient came to the casualty after a road traffic accident. A lacerated wound of 4x1 cm, which is bone – deep, is noted over the scalp of the forehead. Which of the following is the correct mechanism of this lacerated wound?

- A. Shearing force
- B. Impact of a semi sharp-edged object
- C. Overstretching of skin and subcutaneous tissue
- D. Tissue crushed between weapon and underlying bone

26. Which of the following is true about the cause of the given finding ?



- A. Lightning strike
- B. Electrocution
- C. Marbling
- D. Trickling down of acid

27. Which mechanism is responsible for the injuries sustained by a man brought to the emergency room with multiple small puncture wounds on the right side of his chest, below the nipple, and multiple small abrasions and bruising on the left lower abdomen following a blast incident?

- A. Flying missiles
 - B. Blast of air
 - C. Explosion burns
 - D. Building collapse
-

28. During the examination of a victim of assault, you notice a wound with ragged, irregular and uneven margins with tears diverging at an angle from the end of the main wound and typical swallowtails. Bleeding from this wound is not significant and shows crushed blood vessels and hair bulbs. You would document this wound as a/an.

- A. Incised wound
 - B. Chopped wound
 - C. Stab wound
 - D. Lacerated wound
-

29. Which of the following characteristics is not typically associated with a true bruise?

- A. Negative chemical tests
 - B. Progression of the color of bruise over time
 - C. Irregular margins
 - D. Presence of Itching
-

30. Identify the pattern of abrasion shown in the image below.



- A. Pressure abrasion
- B. Graze abrasion
- C. Ligature mark
- D. Imprint abrasion

31. The image given below is which type of wound?



- A. Contusion
- B. Abrasion
- C. Laceration
- D. Stab wound

32. Identify the following image?



- A. Hinderers line
- B. Langers lines
- C. Blaschko lines
- D. Dermatomes

33. Identify the lesion shown in the image below.:

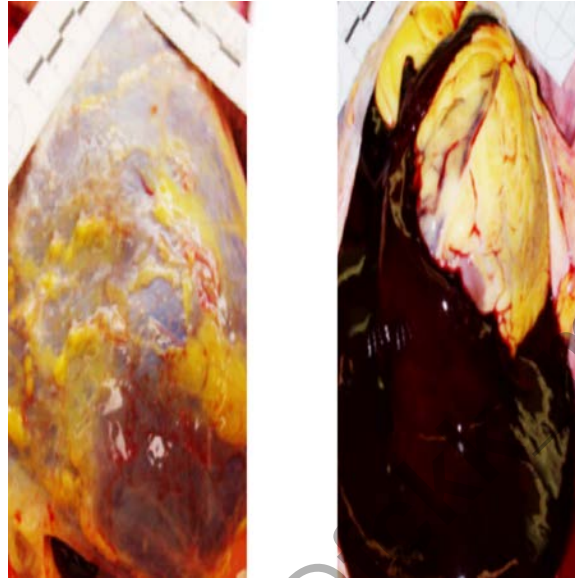


- A. Laceration
 - B. Incised wound
 - C. Laceration-looking incised wound
 - D. Incised looking laceration
-

34. During the examination of injury, hair bulbs are noted to be crushed. The probable type of wound is:

- A. Abrasion
 - B. Laceration
 - C. Stab injury
 - D. None
-

35. Identify the post-mortem finding in a patient with a stab wound on the left side of the chest.:



- A. Cardiac tamponade
- B. Fat necrosis
- C. Hemothorax
- D. Pulmonary embolism

36. Which of the following is classically associated with a lucid interval?

- A. Subdural haemorrhage
 - B. Parenchymal haemorrhage
 - C. Epidural hematoma
 - D. Subarachnoid haemorrhage
-

37. Please match the following weapons with their corresponding types of injuries.

A. Axe B. RTA C. Blade D. Lathi 1. Incised wound 2. Tram track bruise 3. Grazed abrasion 4. Chop wound

A. Axe B. RTA C. Blade D. Lathi

1. Incised wound 2. Tram track bruise 3

A. A-4, B-3, C-1, D-2

B. A-1, B-3, C-4, D-2

C. A-1, B-2, C-4, D-3

D. A-4, B-1, C-3, D-2

38. What is the probable etiology of the patient's burn injury who presents to the outpatient department (OPD) with a dampened, erythematous, blistered lesion on the right thigh, along with a demarcation line that extends distally?

A. Electric burn

B. Chemical burn

C. Dry heat

D. Moist heat

39. Identify the type of wound from the image.:



- A. Defense wounds
- B. Antemortem wound
- C. Postmortem wound
- D. Hesitation cuts

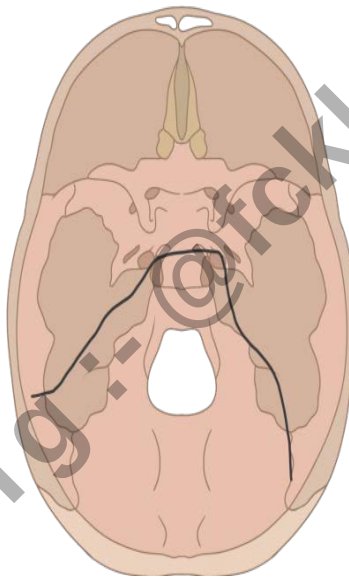
40. A dead body was found with a wound on the neck, with clean-cut edges, crushed tissues along with the disruption of the vertebral bones. Identify the type of wound.

- A. Incised wound
- B. Chop wound
- C. Lacerated wound
- D. Avulsion

41. Gutter fracture of skull bone is associated with impact due to:

- A. A sharp object
 - B. A blunt object
 - C. Fall from height
 - D. A bullet
-

42. Name the type of skull fracture.



- A. Depressed fracture
 - B. Hinge fracture
 - C. Pond fracture
 - D. Comminuted fracture
-

43. Identify the injury the image below :



- A. Abrasion
 - B. Laceration
 - C. Bruise
 - D. Imprint
-

44. Paradoxical undressing is seen in :

- A. Heart stress
 - B. Voyeurism
 - C. Hypothermia
 - D. LSD overdose
-

45. All of the following statements are true about the given condition except:



- A. Swabs should be taken immediately after moistening with sterile water
 - B. It is most commonly caused by incisors and canines
 - C. There are fewer chances of bacterial infection
 - D. Tetanus vaccination should be given
-

46. A middle aged man was brought to the ER after an assault with a sharp axe, identify the type of wound ?



- A. Incised wound
- B. Chop wound
- C. Lacerated wound
- D. Abrasion

47. A 12 year old boy presents with the following injury that he sustained on being tackled by an opponent during the course of a football match. As the casualty medical officer, you would enter the wound description in the case sheet as



- A. Incised wound
- B. Abrasion
- C. Laceration
- D. Bruise

48. Identify the image shown below:



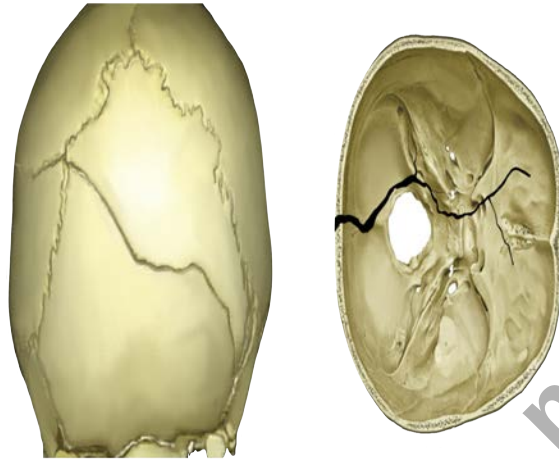
- A. Shot gun wound
- B. Tattooing
- C. Arsenic poisoning
- D. Graze wound

49. A 31-year-old man was found dead at home. He was brought to the mortuary by the police, and the following findings were noted during the examination: Which of the following is the likely cause of his death?



- A. Strangulation by ligature
- B. Hanging
- C. Throttling
- D. Drowning

50. Identify the type of skull fracture in the given image ?



- A. Ring
- B. Depressed
- C. Fissured
- D. Comminuted

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	4
Question 4	2
Question 5	1
Question 6	1
Question 7	1
Question 8	2
Question 9	1
Question 10	3
Question 11	2
Question 12	2
Question 13	1
Question 14	1
Question 15	3
Question 16	2
Question 17	1
Question 18	1

Question 19	1
Question 20	1
Question 21	3
Question 22	4
Question 23	2
Question 24	3
Question 25	4
Question 26	1
Question 27	1
Question 28	4
Question 29	4
Question 30	3
Question 31	1
Question 32	2
Question 33	4
Question 34	2
Question 35	1
Question 36	3
Question 37	1
Question 38	4
Question 39	1

Question 40	2
Question 41	4
Question 42	2
Question 43	1
Question 44	3
Question 45	3
Question 46	2
Question 47	3
Question 48	1
Question 49	1
Question 50	3

Solution for Question 1:

- A leading question suggests the answer within the question itself, often prompting a "yes" or "no" response.
- These questions are typically allowed during cross-examination but not during examination-in-chief or re-examination, except in special circumstances.

Examination of Witness :

The below sequence is based on the Sec. 138 IEA.

Oath

↓

Chief examination

Cross-examination

Re-examination

Aspect	Examination-in-Chief (Direct Examination)	Cross-Examination	Re-Examination
Conducted By	Party who calls the witness (e.g., public prosecutor)	Opposing party (typically defense)	Party who called the witness
Purpose	Present all relevant facts Provide expert interpretation if applicable	Challenge witness's credibility Elicit favorable facts Test accuracy	Clarify issues raised during cross-examination
Leading Questions	Not allowed unless the witness is declared hostile (Option D)	Allowed (Sec. 146 BSA)	Not allowed (Option B)
Nature of Questions	Open-ended questions to elicit narrative	Can include leading questions to control the narrative	Generally clarifying questions
Time Limit	Typically structured and may have time constraints	No set time limit; may last for hours or even days	Generally limited to clarification purposes
Judicial Oversight	Judge may intervene to ensure questions are relevant and not leading	Court can disallow indecent or scandalous questions (Sec. 154, BSA)	Judge monitors to ensure focus remains on clarification

Anytime (Option C): This is incorrect because leading questions are restricted during examination-in-chief and re-examination.

Solution for Question 2:

Correct Option A: A-4, B-3, C-1, D-2:

- (A) Axe: The correct match is option 4 - Chop wound. An axe is a cutting tool with a sharp blade on one side and a handle on the other. When an axe is used forcefully to strike a person, it can cause a deep, wide, and often irregular wound known as a chop wound. This type of wound is characterized by ragged edges and a combination of sharp and blunt force injury.
- An axe is a cutting tool with a sharp blade on one side and a handle on the other. When an axe is used forcefully to strike a person, it can cause a deep, wide, and often irregular wound known as a chop wound. This type of wound is characterized by ragged edges and a combination of sharp and blunt force injury.

- (B) RTA (Road Traffic Accident): The correct match is option 3 - Grazed abrasion. RTA refers to a Road Traffic Accident. In accidents involving vehicles, a person may slide or be dragged along the road surface, leading to abrasions. Grazed abrasions are shallow injuries where the outer layer of the skin is scraped off due to friction against the road surface. They usually appear as superficial scrapes or grazes.

- RTA refers to a Road Traffic Accident. In accidents involving vehicles, a person may slide or be dragged along the road surface, leading to abrasions. Grazed abrasions are shallow injuries where the outer layer of the skin is scraped off due to friction against the road surface. They usually appear as superficial scrapes or grazes.

- (C) Blade: The correct match is option 1 - Incised wound. A blade is a sharp-edged instrument, typically used for cutting. When a blade cuts through the skin, it creates an incised wound. Incised wounds have clean, smooth edges, and they are usually longer than they are deep.

- A blade is a sharp-edged instrument, typically used for cutting. When a blade cuts through the skin, it creates an incised wound. Incised wounds have clean, smooth edges, and they are usually longer than they are deep.

- (D) Lathi: The correct match is option 2 - Tram track bruise. A lathi is a long, heavy stick often used as a weapon or tool. When a lathi strikes a person, it can cause a patterned bruise known as a tram track bruise. Tram track bruises are characterized by parallel linear marks that resemble the tracks of a tram. They occur due to the impact of the cylindrical shape of the lathi.

- A lathi is a long, heavy stick often used as a weapon or tool. When a lathi strikes a person, it can cause a patterned bruise known as a tram track bruise. Tram track bruises are characterized by parallel linear marks that resemble the tracks of a tram. They occur due to the impact of the cylindrical shape of the lathi.

(A) Axe: The correct match is option 4 - Chop wound.

- An axe is a cutting tool with a sharp blade on one side and a handle on the other. When an axe is used forcefully to strike a person, it can cause a deep, wide, and often irregular wound known as a chop wound. This type of wound is characterized by ragged edges and a combination of sharp and blunt force injury.

An axe is a cutting tool with a sharp blade on one side and a handle on the other. When an axe is used forcefully to strike a person, it can cause a deep, wide, and often irregular wound known as a chop wound. This type of wound is characterized by ragged edges and a combination of sharp and blunt force injury.

(B) RTA (Road Traffic Accident): The correct match is option 3 - Grazed abrasion.

- RTA refers to a Road Traffic Accident. In accidents involving vehicles, a person may slide or be dragged along the road surface, leading to abrasions. Grazed abrasions are shallow

injuries where the outer layer of the skin is scraped off due to friction against the road surface. They usually appear as superficial scrapes or grazes.

RTA refers to a Road Traffic Accident. In accidents involving vehicles, a person may slide or be dragged along the road surface, leading to abrasions. Grazed abrasions are shallow injuries where the outer layer of the skin is scraped off due to friction against the road surface. They usually appear as superficial scrapes or grazes.

(C) Blade: The correct match is option 1 - Incised wound.

- A blade is a sharp-edged instrument, typically used for cutting. When a blade cuts through the skin, it creates an incised wound. Incised wounds have clean, smooth edges, and they are usually longer than they are deep.

A blade is a sharp-edged instrument, typically used for cutting. When a blade cuts through the skin, it creates an incised wound. Incised wounds have clean, smooth edges, and they are usually longer than they are deep.

(D) Lathi: The correct match is option 2 - Tram track bruise.

- A lathi is a long, heavy stick often used as a weapon or tool. When a lathi strikes a person, it can cause a patterned bruise known as a tram track bruise. Tram track bruises are characterized by parallel linear marks that resemble the tracks of a tram. They occur due to the impact of the cylindrical shape of the lathi.

A lathi is a long, heavy stick often used as a weapon or tool. When a lathi strikes a person, it can cause a patterned bruise known as a tram track bruise. Tram track bruises are characterized by parallel linear marks that resemble the tracks of a tram. They occur due to the impact of the cylindrical shape of the lathi.

Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 3:

Correct Option D - Chop wound:

- A chop wound typically involves a clean, sharp instrument like an axe or machete. In the context of a road traffic accident, especially involving vehicles, injuries caused by chopping weapons are uncommon. RTAs typically result in injuries associated with blunt force trauma,

sharp objects from the surroundings, or the vehicle itself.

Incorrect Options:

Option A - Contusion:

- Contusion, commonly referred to as a bruise, is characterized by injury to the dermis, where dermal blood vessels rupture, leading to extravasation of blood and the formation of a raised area or swelling. This occurs due to the force caused by blunt objects such as impacts during a road traffic accident. The margins of a contusion are irregular, contributing to its distinctive appearance.



Option B - Abrasions:

- Abrasions, or scrapes, occur when the skin rubs against a rough surface. They are frequently observed in RTAs, especially when individuals come into contact with the road or other abrasive surfaces.



Option C - Laceration:

- Lacerations are cuts or tears in the skin that can result from sharp objects or edges during an accident. They are commonly seen in RTAs, especially when there are broken glass or sharp metal components involved.



Solution for Question 4:

Correct Option B - Contact shot

- A contact shot occurs when the muzzle of the firearm is in direct contact with the victim's skin at the time of firing. This results in the deposition of soot, unburned powder particles, and even possible charring around the entrance wound. The gases and particles from the gunpowder can cause the surrounding skin to exhibit a stellate or star-shaped pattern due to the pressure and heat.

Incorrect Options:

Option A - Close shot: A close shot refers to a gunshot that is fired from a relatively short distance, usually within a few inches to a few feet. In a close shot, there might be stippling or soot deposition around the entrance wound, but the injury would not typically exhibit a stellate (star-shaped) appearance.

Option C - Near shot: A near shot, also known as a close intermediate shot, refers to a gunshot fired from a short distance but not in direct contact with the skin. The range is usually a few inches to a few feet. This might result in stippling or powder tattooing around the entrance wound due to the impact of unburned gunpowder particles on the skin. However, it is less likely to produce a stellate wound pattern as seen in contact shots.

Option D - Distant shot: A distant shot is fired from a considerable distance, and the effects on the body are determined by the range of fire. At a greater distance, the bullet's energy dissipates, and there may be no stippling or soot deposition around the entrance wound. The wound tends to be cleaner and smaller compared to closer shots. A stellate wound pattern is unlikely to be present in a distant shot.

Solution for Question 5:

Correct Option A - Defense wound

- When a person is attacked or struggling to defend themselves against an assailant, they might sustain what are known as defense wounds. These wounds typically occur on the flexor surface of the forearm, as individuals instinctively raise their arms to protect their face and body during an attack. The cuts on the victim's arm in this scenario are consistent with defense wounds, suggesting that the victim was trying to shield themselves from harm.

Incorrect Options:

Option B - Self-inflicted injury: Self-inflicted injuries are wounds that a person intentionally inflicts upon themselves. These injuries might result from self-harm or suicide attempts. However, in the given scenario, the presence of multiple cuts on the victim's arm does not necessarily suggest self-inflicted injury, as the location and pattern of the cuts seem more indicative of defensive actions rather than self-infliction.

Option C - Accidental cut wound from falling: An accidental cut wound from falling might result from an individual accidentally coming into contact with a sharp object while falling. However, this scenario is less likely in this case because the presence of multiple cuts on the flexor surface of the arm is not consistent with typical patterns of injuries sustained during a fall.

Option D - Washer-women hands: "Washer-women hands" is a term used to describe the appearance of the fingers and hands after prolonged exposure to water, like during laundry work. This option is not relevant to the scenario described, as it does not explain the presence of multiple cuts on the victim's arm or the context of the drowning incident.

Solution for Question 6:

Correct Option A - Pterion

- The impact of the collision resulting in an epidural hematoma (EDH) due to involvement of the Middle Meningeal Artery is likely to occur at the pterion. The pterion is a small, delicate area on the side of the skull where four bones come together: the frontal bone, the parietal bone, the temporal bone, and the sphenoid bone. This area is particularly vulnerable to traumatic injury due to its thin bone structure and the presence of the Middle Meningeal Artery, which runs just beneath the skull in this region. Trauma to the pterion can lead to a rupture of the Middle Meningeal Artery, causing bleeding between the dura mater and the skull, resulting in an epidural hematoma.

Incorrect Options:

Option B - Angle of mandible: The angle of the mandible is the outermost point of the lower jawbone. While trauma to this area can cause facial injuries, it is not directly associated with the development of an epidural hematoma due to involvement of the Middle Meningeal Artery.

Option C - Lambdoid suture: The lambdoid suture is the joint between the occipital bone (back of the skull) and the parietal bones. Trauma to this area can result in injuries to the posterior part of the skull, but it is not typically associated with epidural hematomas involving the Middle Meningeal Artery.

Option D - Nasion: The nasion is the intersection point between the frontal bone and the two nasal bones. Trauma to this area can lead to facial injuries, but it is not directly associated with the development of an epidural hematoma due to involvement of the Middle Meningeal Artery.

Solution for Question 7:

Correct Option A - 45%:

- For total body surface area, the Rule of 9, also known as the Wallace rule, is used.
- Calculation: Right upper limb: 9% Entire right lower limb: 18% Front of abdomen: 9% Back of abdomen: 9% Total = 9% + 18% + 9% + 9% = 45%
- Right upper limb: 9%
- Entire right lower limb: 18%
- Front of abdomen: 9%
- Back of abdomen: 9%
- Total = 9% + 18% + 9% + 9% = 45%
- Right upper limb: 9%
- Entire right lower limb: 18%
- Front of abdomen: 9%
- Back of abdomen: 9%
- Total = 9% + 18% + 9% + 9% = 45%

Incorrect Options:

- Options B, C, and D represent incorrect percentage of burned total body surface area.
-

Solution for Question 8:

Correct Option B - Head to foot:

Rigor mortis, or postmortem rigidity, typically starts in the smaller muscles and progresses to the larger muscle groups. The sequence generally begins with the muscles of the face and neck, followed by the upper limbs, and then extends to the trunk and lower limbs. Therefore, rigor mortis progresses from the head and upper parts of the body downwards towards the feet.

Incorrect Options:

Option A - Center to periphery: Rigor mortis does not progress from the center of the body to the periphery. It starts in the head and moves towards the feet, not the central parts to the outer extremities.

Option C - Foot to head: Rigor mortis does not start in the feet and move upwards to the head. It starts in the head and progresses downwards towards the feet.

Option D - Simultaneously: Rigor mortis does not occur simultaneously in all parts of the body. It progresses in a specific sequence, starting with the head and upper body, then moving downwards.

Solution for Question 9:

Correct Option A - Middle meningeal artery:

- The bleeding vessel responsible for extradural Hemorrhage (epidural Hemorrhage) is the middle meningeal artery.

Incorrect Options:

Option B - Bridging veins: Bridging veins are involved in subdural hemorrhage, not extradural hemorrhage.

Option C - Circle of Willis: Any rupture of arteries in the Circle of Willis can lead to subarachnoid hemorrhage, not extradural hemorrhage.

Option D - Anterior cerebral artery: The anterior cerebral artery is not involved in extradural hemorrhage.

Solution for Question 10:

Correct Option C - Close range:

When a gunshot wound shows blackening and tattooing around it, this indicates a close-range shot. Blackening is caused by the deposition of soot and gunpowder residue from the burning propellant, while tattooing is due to the impact of unburned powder particles. These signs are typically seen when the muzzle is relatively close to the target, usually within a few feet.

Incorrect Options:

Option A - Distant range:

At a distant range, the wound would not exhibit blackening or tattooing because the gunpowder residues would not have had a chance to settle on the skin. Distant range wounds usually present with a clean entry wound without additional residues.

Option B - Close contact:

Close contact wounds are typically characterized by a much more severe pattern, such as a star-shaped wound with burning and possibly charring of the skin. The presence of blackening and tattooing suggests a somewhat less extreme range, but still close.

Option D - Cannot be predicted:

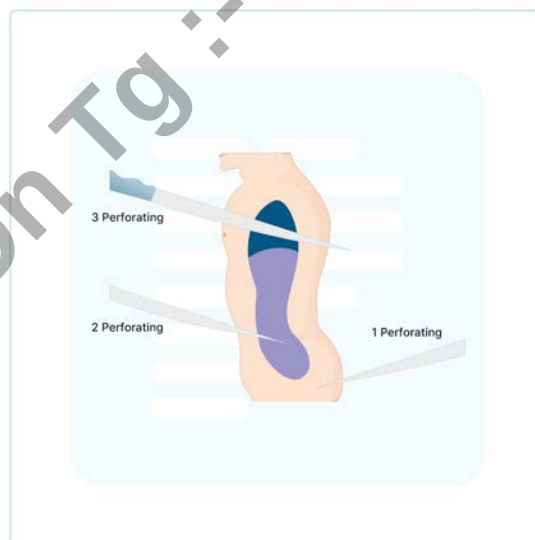
The presence of blackening and tattooing can indeed predict the bullet range with a reasonable degree of certainty. Therefore, this option is not accurate as the gunshot range can be estimated based on these features.

Solution for Question 11:

A penetrating wound is caused by a sharp object (e.g., a knife) that pierces the skin and enters deeper tissues or body cavities. In this case, the stab wound to the anterior thigh involves a sharp object penetrating the skin and underlying tissues, making it a penetrating wound.

Stab wound/Punctured wound	
----------------------------	--

These are inflicted by penetration with long, narrow instruments having pointed (sometimes blunt) ends into the depth of the body, which is deeper than its length and width.	
Weapons used	Knife (most common) Pieces of glass (broken glass bottle) Scissors Screwdrivers Pens Ice picks Forks
Types of stab wounds	Puncture wound: (Option C) Soft tissues are involved Penetrating wound: Enters a body cavity or a viscus Perforating wound: Through-and-through puncture wounds The weapon enters the body from one side and comes out the other The wound of entry is larger than the exit
Significance of the wound dimensions	Length is slightly less than the width of the weapon due to the stretching of the skin. Breadth is more than the thickness of weapon due to gaping. Depth, the greatest dimension of the wound, corresponds to the length of the weapon's blade, which has completely entered the body without perforation. Margins or edges of the wound are clean cut with usually no abrasion or bruising. Margins by a pointed or conical weapon have lacerated edges If the hilt/guard of the weapon strikes the skin, a regular, sharp, and well-defined patterned abrasion or bruising is formed



Incised wound (Option A): An incised wound is a clean cut caused by a sharp object (e.g., a razor or knife) and is typically longer than it is deep. A stab wound is deeper than it is long, so it does not qualify as an incised wound.

Blunt force injury (Option D): This involves trauma caused by a blunt object (e.g., a bat or hammer) and does not involve penetration of the skin or tissues. A stab wound is caused by a

sharp object, not blunt force.

Solution for Question 12:

A hammer is a heavy, blunt object that delivers a focused impact on the skull. This results in a depressed fracture, where the bone is pushed inward at the point of impact, creating a localized depression or indentation.

Depressed Fracture	Comminuted Fracture	Gutter Fracture
Occurs when a fragment of the skull is pushed inward towards the brain. Caused by blows from heavy objects with a small surface (e.g., hammers, bricks). The shape of the depressed area may resemble the object that caused the injury. Variants: Elevated Fracture : One end of the fractured fragment is raised above the skull, while the other is pushed inward. Caused by sharp objects like an axe. Pond Fracture : Seen in infants, this smooth depression occurs without a fracture line, resembling a dent in a ping-pong ball. Caused by obstetrics forceps blade.	Involves multiple intersecting fracture lines, breaking the bone into several fragments. Caused by high-impact trauma (e.g., vehicular accidents, crushing injuries). It may occur alongside other fractures (e.g., fissured or depressed fractures). Spider web-like patterns on an X-ray are characteristic of comminuted fractures. In severe cases, bone fragments may penetrate the brain or become displaced.	A groove or channel is formed in the skull bone, often due to an oblique bullet wound. Frequently associated with comminuted and depressed fractures of the inner skull table. Bone fragments can cause injury to the meninges or brain.
Depressed fracture	Comminuted fracture	Gutter fracture
Ring Fracture (Option D)	Motorcyclist Fracture (Hinge Fracture) (Option C)	
Encircles the foramen magnum at the skull base. Runs from the sella turcica to the petrous ridges and posterior fossa. It can extend to the middle ear and roof of the nose.	Transverse fracture at the skull base, splitting it and creating a "hinge" - nodding face sign . It extends from one petrous ridge across the sella turcica to the other petrous ridge.	
Caused by: Falls landing on feet/buttocks (force transmitted through the spine). Heavy blows to the top of the skull/ vertex. Twisting of the head on the spine. Blows to the occiput or chin.	Often caused by side impacts .	
Ring fracture	Hinge fracture	
Fissured Fracture (Linear Fracture) (Option A)	Diastatic Fracture (Sutural Fracture)	

The most common skull fracture accounts for about 70% of cases. Linear cracks may involve the vertex or skull base. It can affect the full thickness of the bone or just the inner/outer layers. The fracture line is irregular and fine, sometimes hard to detect on X-rays. It runs parallel to the compressive force. No depression of bone fragments.	Involves the separation of cranial sutures. More common in children. Often occurs with other skull fractures, especially fissured fractures. Commonly seen at the sagittal suture.
Causes: Impact against broad surfaces (e.g., falls or road traffic accidents) Falls on feet or buttocks, transmitting force upwards.	Causes: Blunt trauma to the head. Increased intracranial pressure leads to suture separation.
Narrow fissure skull fracture	Diastatic Fracture

Solution for Question 13:

Given Burn Areas:

Total Burn Percentage = $18\% + 9\% = 27\%$

The only option that fits within this range is A) 25-30%.

According to the Wallace rule of 9, this is how the area burned is calculated:

	Adults	Children	Infants
Head & neck	9%	18%	20%
Front of chest & abdominal wall	$9 \times 2 = 18\%$	18%	$10 \times 2 = 20\%$
Back of chest & abdominal wall	$9 \times 2 = 18\%$	18%	$10 \times 2 = 20\%$
Lower limb	$18 \times 2 = 36\%$	$13.5 \times 2 = 27\%$	$10 \times 2 = 20\%$
Upper limb	$9 \times 2 = 18\%$	18%	$10 \times 2 = 20\%$
Perineum	1%	1%	

Solution for Question 14:

Correct Option A - Close shot entry wound:

- When a gunshot injury occurs at close range, the muzzle of the firearm is in close proximity to the body. In such cases, the burning, blackening, and tattooing of the skin around the wound are commonly observed. These characteristics are due to the presence of unburned gunpowder particles and hot gasses discharged from the firearm.

Incorrect Options:

Option B - Close shot exit wound : In close shot exit wounds, the burning, blackening, and tattooing effects are usually not present. Close shot exit wounds are characterized by an irregular or stellate appearance, as the bullet exits the body and causes tissue disruption.

Option C - Distant shot entry wound: In distant shot entry wounds, there is a significant distance between the muzzle of the firearm and the body. The effects of burning, blackening, and tattooing are not typically observed in such cases. Distant shot entry wounds may appear as small, circular or oval-shaped holes without the characteristics associated with close-range shots.

Option D - Distant shot exit wound : Similar to distant shot entry wounds, distant shot exit wounds do not exhibit the effects of burning, blackening, and tattooing. These exit wounds are often characterized by a smaller size compared to close shot exit wounds.

Solution for Question 15:

Correct Option C - Both:

- Bullet fingerprinting is a technique used in forensic investigations to analyze the unique markings left on bullets or cartridge cases by the firearm used to discharge them.
- Primary markings include the lands and grooves on the inside of the barrel, which leave distinct striations on the bullet or cartridge case as it passes through the barrel. These primary markings are the key elements that make each firearm's tool marks unique.
- Secondary markings, may be present on a bullet or cartridge case due to factors such as the firing mechanism, extractor, or ejector of the firearm .
- When analyzing a bullet or cartridge case, forensic experts examine both the primary and secondary markings to compare them with test-fired bullets or cartridge cases from suspect

firearms

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Incorrect Options:

Option A - Primary markings: Primary markings alone do not encompass the complete analysis involved in bullet fingerprinting.

Option B - Secondary markings: Secondary markings alone are also not sufficient for bullet fingerprinting analysis.

Option D - None: None is incorrect because bullet fingerprinting does involve the examination of both primary and secondary markings.

Solution for Question 16:

Correct Option B - Near shot:

- Although an intermediate distance wound leaves blackening and tattooing but does not leave burring/singeing marks on the gunshot wound , it is not given in the option.
- The next best answer would be a near gunshot wound.
- This is because near gunshot wounds and intermediate gunshot wounds may or may not have blackening but definitely have tattooing present.

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This is because near gunshot wounds and intermediate gunshot wounds may or may not have blackening but definitely have tattooing present.

Incorrect Options:

Option A - Contact shot: Burning, blackening and tattooing are not found on the surface, but may be present within the wound.

Option C - Close Shot : Burning/ singeing, blackening and tattooing- all 3 are present.

Option D - Distant Shot : Burning/ singeing, blackening and tattooing- all 3 are absent .

Solution for Question 17:

Correct Option A - Residual from barrel of gun:

- Bullet wipe refers to the residue or material that is left behind on a target when a bullet passes through it. This residue is primarily composed of substances such as soot, lubricants, and metal particles that are present in the barrel of the gun.

Incorrect Options:

Option B - Blackening : Blackening typically refers to the darkening or discoloration of a surface due to various factors such as oxidation, charring, or exposure to heat.

Option C - Gutter fracture of the skull : Gutter fracture is a term used to describe a specific type of fracture in the skull, characterized by a groove or channel-like indentation.

Option D - Tattooing : Tattooing, in the context of firearms, refers to the deposition of gunpowder particles or other debris onto the skin when a firearm is discharged at close range.

Solution for Question 18:

Correct Answer: A - The executive magistrate can order for exhumation

- Exhumation - It is digging out of an already buried body legally from the grave

- exhumation is done only when there is written order from executive magistrate.
- The body can be exhumated by any government doctor.

Incorrect Option:

Option B Police can order for exhumation : This statement is incorrect

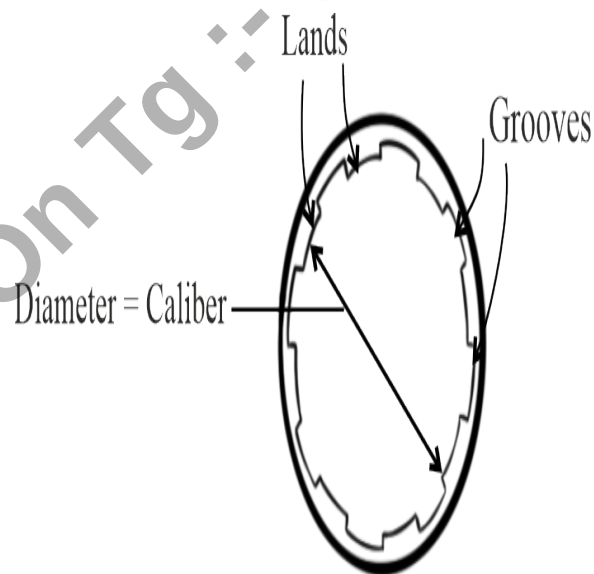
Option C Postmortem can be done on exhumated body : Autopsies can be performed on exhumated bodies

Option D CrPC 174 is related to exhumation : CrPC 176 is related to exhumation

Solution for Question 19:

Correct Option A - Distance between two opposite lands:

In a rifled weapon :



Caliber: Distance between two opposite lands.

Incorrect Options:

Option B - Distance between two opposite grooves:

- While the distance between two opposite grooves is indeed a measurement within the rifled barrel, it is not typically used to determine the caliber of the weapon.

Option C - Number of lead pellets:

Bore or Gauge : It is used for smooth bore weapons.

- The number of balls/pellets of equal size/weight made from 1 pound lead (454g)
- 12 bore = 12 balls

Option D - Mass and velocity:

- While the mass and velocity of a projectile fired from a rifled barrel weapon are important factors in its performance, they are not used directly to determine the caliber of the weapon.

Solution for Question 20:

Correct Option A - Precipitin test:

- The test which can give the species differentiation is the precipitin test.
- Precipitin tests can be done by extracting species-specific protein from bone into a solution and testing it against specific antisera of humans.
- The techniques such as electrophoresis or gel diffusion can now identify human tissue.

Incorrect Options:

Option B - Acid-phosphatase: This test can be used to detect seminal fluid.

Option C - LDH: LDH cannot be used for species differentiation.

Option D - Teichmann test: Teichmann crystal test, is a laboratory test used to detect the presence of blood. Specifically, it is used to identify hemoglobin, a protein found in red blood cells.

Solution for Question 21:

Correct Option C - Preserve in an envelope/paper bag:

- Paper bags or envelopes allow air to circulate, which helps prevent the growth of mold or bacteria on the blood stains.
- Items stored in envelopes or paper bags can be easily labeled and documented, ensuring proper chain of custody and traceability of the evidence.

Incorrect Options:

Option A - Place them in same plastic bag and seal it for preservation: Placing them in the same plastic bag risks cross-contamination between the two handkerchiefs, which can compromise the integrity of the evidence.

Option B - Rinse the bloodstains with water to prevent contamination: Rinsing the bloodstains with water can alter the evidence and potentially wash away important forensic information.

Option D - Place them in different plastic bags and seal them for preservation: Placing them in different plastic bags may not be necessary and can create additional waste. It's generally preferable to store related items together to maintain the context of the evidence.

Solution for Question 22:

Correct Option - D Hollow viscus:

- The most common organ involved in seat belt injury ideally is the mesentery.
- Since the given options do not contain mesentery the best option here is hollow viscus.

Incorrect Options:

Options A, B and C: Spleen and liver are the most common organs injured in blunt trauma to the abdomen.

Solution for Question 23:

Correct Option B - Lungs:

- The most common organ injured in an air blast shock wave is the tympanic membrane (TM); given that the options do not contain TM, the next best answer is the lung.
- Blast lung is a primary blast injury. The shock wave causes immediate lung injury, characterized by the rupture of alveolar capillaries.
- This leads to the influx of blood and extravasation of edema fluid into lung tissue, resulting in hemorrhagic foci that can be substantial depending on the level of blast loading.

Incorrect Options:

Options A, C, and D : Intestine, kidney, and heart are not the most common injuries in air blast.

Solution for Question 24:

Correct Option C - Pistol, near shot:

- When analyzing a gunshot wound, variables such as the size and shape of the wound, the depth of penetration, and the nearness of gunpowder buildup can give clues about the sort of gun utilized and the extent of the shot.
- In this case, the wound was caused by a gun, as shown by its size and shape, which are consistent with a small-caliber handgun.
- Furthermore, the wound was caused by a near shot, as shown by the presence of powder burns and the nonappearance of soot or stippling.
- Powder burns happen when gunpowder particles from the muzzle blast are stored on the skin, whereas sediment and stippling are caused by burning gunpowder and unburned powder particles, separately, which can, as it were, travel short distances.
- Hence, the need for soot or stippling proposes that the weapon was terminated at a near extent, inside a number of inches of the victim.

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Incorrect Options:

Option A - Shotgun, intermediate range:

- A shotgun wound regularly causes a large and irregularly molded wound, with a more extensive spread of pellets or shot.
- Besides, at middle ranges, the pattern of the shot can be observed on the skin, and there may be evidence of both powder burns and stippling.

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Option B - Shotgun, close range:

- At close range, a shotgun wound can cause extreme tissue harm and a widespread shot, which would be obvious on the skin as a design of different holes or craters.
- Besides, there would likely be evidence of powder burns, soot, and stippling.

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Option D - Pistol, close shot:

At a near range, a gun wound can cause a little entry wound and a larger exit wound, with proof of powder burns, soot, and stippling.

- In any case, in this case, there was no proof of sediment or stippling, showing that the weapon was fired from a marginally more distant distance.
- Gun fired inside the range of flame.
- Circular puncture wound is seen.

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Gun fired inside the range of flame.

Circular puncture wound is seen.

Solution for Question 25:

Correct Option D - Tissue crushed between weapon and underlying bone:

- Lacerated wounds are caused by blunt force trauma that results in a tear or cut in the skin and underlying tissues. In this particular case, the wound is described as being "bone-deep," which indicates that the injury extends to the underlying bone.
- Tissue crushed between weapon and underlying bone: This option correctly describes the mechanism of the lacerated wound. When tissue is crushed between a weapon or object and the underlying bone, it can result in a deep laceration that involves both the skin and the underlying tissues, reaching the bone

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Incorrect Options:

Option A - Shearing force : Shearing force refers to the application of a force that causes one layer of tissue to move parallel to another. This type of force can result in various types of injuries, such as abrasions and avulsions, but it is not typically associated with lacerated wounds. In a shearing force injury, the skin and tissues are more likely to be separated or torn apart rather than causing a deep laceration that extends to the bone.

Option B - Impact of a semi sharp-edged object : While an impact from a semi sharp-edged object could cause a laceration, it would not necessarily result in a wound that is bone-deep. Lacerations caused by sharp objects are typically superficial or involve only the soft tissues rather than reaching the underlying bone.

Option C - Overstretching of skin and subcutaneous tissue: Overstretching of the skin and subcutaneous tissue would typically result in a different type of injury, such as an abrasion or a stretch mark. It would not cause a deep laceration that extends to the bone.

Solution for Question 26:

Correct Option A - Lightning strike:

- The given finding is known as Lichtenberg's figures or filigree burns which are pathognomonic for lightning injury. It is a superficial, thin, tortuous, fern-like pattern of erythema in the skin. It indicates the path taken by the electrical discharge and tends to follow skin creases and long axis of the body.
- They are also known as arborescent burns or feathering as they have a resemblance to the branches of a tree. Lichtenberg's figures occur due to the rupture of small vessels. They are usually seen over the shoulder or flank within an hour of injury and disappear in 1-2 days.

Incorrect Options:

Option B - Electrocution :

- Electrocution or electrical burns can cause the following lesions: Joule burns : These are elevated blisters that form when the epidermis or the epidermal-dermal junction of the skin has been split by firm contact with an electrical wire. Spark lesion : The current jumps the gap as a spark when the contact is less firm, creating an air gap between the skin and conductor. As a result, a little portion of the keratin in the outer skin melts. The so-called spark lesion forms when the keratin cools down into a hard, brownish lump that is typically raised above the surrounding surface. Burns from high-tension currents are the cause of the crocodile skin effect. Sparking can happen over a significant distance of centimeters in high-voltage burns, such as those caused by high-tension grid transmission wires where the voltage is in the multi-kilovolt range. This may result in many spark lesions and a crocodile-skin effect.
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Option C - Marbling :

- Bacteria like clostridium multiply along the blood vessels & produces H₂S gas. H₂S combines with Hb & forms sulf-hemoglobin. This sulf-Hb stains vessel walls giving greenish, linear, branching pattern on the skin [Corresponding to vascular channels]. This is k/a Marbling. The appearance of marbling = 36-72 hrs after death. It helps to determine T.S.

Option D - Tricking down of acid :

- Trickling down of acid can cause redness, burns, blistering, and pain.

Solution for Question 27:

Correct Option A - Flying missiles:

- Flying missiles refer to objects propelled by the force of an explosion or blast.
- In the context of the question, the presence of multiple small puncture wounds on the right side of the chest and the pattern of injuries suggests that the man was likely struck by fragments or shrapnel propelled by the blast.
- These fragments can include metal shards, glass, or other debris that become airborne due to the force of the explosion. Therefore, considering the described injuries, flying missiles are the most likely mechanism responsible for these injuries.

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Incorrect Options:

Option B - Blast of air : A blast of air refers to the rapid displacement and movement of air caused by an explosion. While a blast of air can cause injuries, such as damage to the lungs, eardrums, or other internal organs, it typically does not cause the specific injuries described in the question, such as puncture wounds or abrasions. Therefore, this option is less likely to be the correct answer in this case.

Option C - Explosion burns : Explosion burns are injuries caused by exposure to the intense heat generated by an explosion. While burns can certainly occur during an explosion, the question does not mention any specific burn injuries. Instead, it describes puncture wounds, abrasions, and bruising. Therefore, this option is not the most likely mechanism responsible for the injuries described.

Option D - Building collapse : Building collapse refers to the structural failure of a building due to an explosion or other catastrophic events. While building collapse can certainly cause various types of injuries, including crush injuries, fractures, and internal injuries, it is not consistent with the specific pattern of injuries mentioned in the question (i.e., puncture wounds, abrasions, and bruising). Therefore, this option is unlikely to be the correct answer.

Solution for Question 28:

Correct Option D - Lacerated wound:

- The wound described in the scenario is most likely a lacerated wound due to its ragged, irregular, and uneven margins with tears diverging at an angle from the end of the main wound and typical 'swallowtails'.
- Bleeding from this wound is not significant and shows crushed blood vessels and hair bulbs.
- Lacerated wounds are typically caused by blunt force trauma and have an irregular shape with jagged margins. They can also be caused by tearing, such as in cases of dog bites or machinery accidents. Therefore, the wound should be documented as a lacerated wound.

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Incorrect Options:

Option A - Incised wound: An incised wound typically has clean and sharp margins and is caused by a sharp-edged instrument like a knife. The edges of an incised wound are not ragged, irregular, or uneven, and there are no tears diverging at an angle from the end of the main wound or "swallow tails."

Option B - Chopped wound : A chopped wound is a type of wound caused by a heavy, sharp-edged object, such as an axe or a machete. It typically has a clean, straight edge with minimal tissue damage.

Option C - Stab wound : A stab wound typically has a narrow and deep wound track with straight or slightly curved margins, and a clean and smooth wound surface. It is caused by a pointed object such as a knife, dagger, or ice pick penetrating the skin and underlying tissues.

Solution for Question 29:

Option D. Presence of vesicles :

- Itching is not typically associated with true bruises. This symptom is more characteristic of artificial bruises caused by substances that irritate the skin.

Option A. Negative chemical tests :

- True bruises usually test negative for the presence of certain chemicals that might be used to create artificial bruises. This is consistent with the definition of a true bruise, which results from the extravasation of blood due to blunt force trauma, not chemical irritation .

Option B. Progression of the color of bruise over time :

- One of the hallmark characteristics of true bruises is the progression of color changes over time due to the breakdown of hemoglobin in the tissue. This progression typically goes from red or purple to blue, green, yellow, and finally fading away .

.Option C. Irregular margins :

- True bruises often have irregular, diffuse margins due to the spread of blood within the tissues. This is in contrast to artificial bruises, which typically have well-defined, regular margins due to the application method of the irritant substance .

Solution for Question 30:

Correct Option C - Ligature mark:

- The given image shows a ligature mark.
- A ligature mark is an injury caused by the application of a ligature or a tightly bound material around the body, usually the neck.
- Ligature marks can be seen in cases of strangulation or hanging, where the ligature leaves a visible pattern on the skin.

Incorrect Options:

Option A - Pressure abrasion: A pressure abrasion is an injury caused by continuous or prolonged pressure on the skin. It can occur when a body part is pressed against a hard or rough surface for an extended period, resulting in skin damage. Pressure abrasions are commonly seen in cases of prolonged immobilization or compression.

Option B - Graze abrasion: A graze abrasion, also known as a superficial abrasion or a brush burn, is a shallow wound caused by friction or scraping of the skin against a rough surface. It typically appears as a linear or irregular-shaped scrape on the skin surface.

Option D - Imprint abrasion: An imprint abrasion occurs when an object or patterned surface leaves a distinct mark on the skin due to pressure or friction. The resulting abrasion reflects the shape, design, or texture of the object that caused it.

Solution for Question 31:

Correct Option A - Contusion:

- A contusion happens when trauma to the skin causes blood vessels to burst and leak blood into the surrounding area.
- This results in the typical discoloration we associate with bruises which can range from a bluish-purple to yellowish-green as it heals.
- While contusions are generally not serious and heal on their own they can sometimes be a sign of more severe internal damage.

Incorrect Options:

Option B - Abrasion: The given image does not indicate an abrasion. An abrasion is a partial thickness wound caused by damage to the skin and can be superficial involving only the epidermis to deep, involving the deep dermis. Abrasions usually involve minimal bleeding.

- Mild abrasions, also known as grazes or scrapes, do not scar or bleed because the dermis is left intact, but deep abrasions that disrupt the normal dermal structures may lead to the formation of scar tissue. A more traumatic abrasion that removes all layers of skin is called an avulsion.

Option C - Laceration: A laceration is a wound that results from soft bodily tissue being torn. This kind of wound frequently has uneven and jagged edges. The image is not indicative of laceration.

Option D - Stab wound: A stab wound is a specific form of penetrating trauma to the skin that results from a knife or a similar pointed object. While stab wounds are typically known to be caused by knives, they can also occur from a variety of implements, including broken bottles and ice picks. The image is not indicative of laceration.

Solution for Question 32:

Correct Option B - Langers lines:

- Langer's lines, also known as tension lines or cleavage lines, refer to the natural orientation of collagen fibers in the skin.
- Langer's lines correspond to the alignment of collagen fibers in the dermis, which is the second layer of the skin.
- They indicate the direction of the least skin tension and the alignment of the skin's elastic fibers as shown in the image in the question.
- Thus, making skin incisions parallel to the lines of Langer reduce the wound tension.

Incorrect Options:

Option A - Hinderers line:

- Hinderer's lines are 2 intersecting lines, one from the ala of the nose to the tragus and another from the lateral canthus of the eye. This is an incorrect option based on the image given in the question

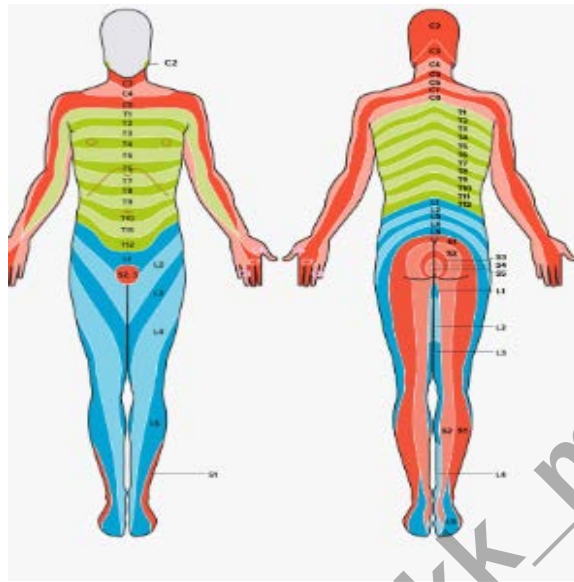
Option C - Blaschko lines:

- Blaschko's lines, also called the lines of Blaschko, are lines of normal cell development in the skin.
- These lines are only visible in those with a mosaic skin condition or in chimeras where different cell lines contain different genes.

Option D - Dermatomes:

- Dermatomes are commonly depicted in dermatome maps, which illustrate the specific regions of the body associated with each spinal nerve's dermatome. These maps are useful in clinical practice for diagnosing and localizing nerve-related conditions.

These maps are useful in clinical practice for diagnosing and localizing nerve-related conditions.



Solution for Question 33:

Correct Option D - Incised looking laceration:

- The most appropriate identification for the lesion shown in the image would be "Incised-looking laceration."
- Laceration produced without excessive skin crushing by blunt object on the areas where the skin is close to bone and subcutaneous tissues are scanty, may produce a wound which by linear splitting of tissues, may look like incised wound.

The most appropriate identification for the lesion shown in the image would be "Incised-looking laceration."

Laceration produced without excessive skin crushing by blunt object on the areas where the skin is close to bone and subcutaneous tissues are scanty, may produce a wound which by linear splitting of tissues, may look like incised wound.

Incorrect Options:

Option A - Laceration: A laceration is a type of wound that is typically irregular, with jagged or torn edges. It is caused by blunt force trauma or tearing of the skin and underlying tissues. In the given image, the wound does not exhibit the typical characteristics of a laceration, such as irregular edges, which makes this option incorrect.

Option B - Incised wound: An incised wound is a type of wound that is caused by a sharp object or instrument, resulting in clean, straight edges. In the given image, the wound does not have the characteristic clean and straight edges of an incise wound, making this option incorrect.

Option C - Laceration-looking incised wound: This option suggests that the wound appears like a laceration but actually has the characteristics of an incised wound. However, in the given image, the wound does not exhibit the typical features of an incised wound, such as clean and straight edges, making this option incorrect.

Solution for Question 34:

Correct Option B - Laceration:

- Laceration involves the tearing or cutting of tissues, which can crush the underlying hair bulbs.
- It often results from blunt force trauma, which can cause compression and crushing of the hair bulbs .
- Crushed hair bulbs indicate a significant force applied to the area, consistent with a laceration.

Laceration involves the tearing or cutting of tissues, which can crush the underlying hair bulbs.

It often results from blunt force trauma, which can cause compression and crushing of the hair bulbs .

Crushed hair bulbs indicate a significant force applied to the area, consistent with a laceration.

Characteristics of Lacerations:

- Trauma: Blunt trauma, may be heavy
- Damage: 3-Dimensional injury
- Margins: Irregular
- Floor: Vessels, nerve, and hair bulb are crushed.
- Structures: Epidermis +dermis + subcutaneous tissue.
- Swallow tail, tearing at the end, diverging.
- Tissue bridges and tissue fragments connect each other from the margin.
- Bleeding comparatively due to crushed blood vessels.

Trauma: Blunt trauma, may be heavy

Damage: 3-Dimensional injury

Margins: Irregular

Floor: Vessels, nerve, and hair bulb are crushed.

Structures: Epidermis + dermis + subcutaneous tissue.

Swallow tail, tearing at the end, diverging.

Tissue bridges and tissue fragments connect each other from the margin.

Bleeding comparatively due to crushed blood vessels.

Incorrect Options:

Option A - Abrasions: Abrasions typically involve superficial damage to the skin without significant underlying tissue injury. They do not usually cause crushing of hair bulbs.

Option C - Stab injuries: Stab injuries are caused by a pointed object piercing the skin. While they may cause damage to the underlying tissues, they do not typically result in crushed hair bulbs.

Option D - None: This option is incorrect because the description in the question indicates a specific type of wound characterized by crushed hair bulbs corresponding to a laceration.

Solution for Question 35:

Correct Option A - Cardiac tamponade:

- Cardiac tamponade caused due to stab injury.
- In this condition, blood or fluid collects in the sac surrounding the heart. This prevents the heart ventricles from expanding fully. The excess pressure from the fluid prevents the heart from working properly. As a result, the body does not get enough blood.
- When enough fluid builds up in the pericardial sac, squeezing the heart and causing a decrease in cardiac output and shock, cardiac tamponade, a medical or traumatic emergency, occurs.

Cardiac tamponade caused due to stab injury.

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When enough fluid builds up in the pericardial sac, squeezing the heart and causing a decrease in cardiac output and shock, cardiac tamponade, a medical or traumatic emergency, occurs.

Cardiac tamponade can occur due to:

- Dissecting aortic aneurysm (thoracic)
- End-stage lung cancer
- Heart attack (acute MI)
- Heart surgery
- Pericarditis caused by bacterial or viral infections
- Wounds to the heart

Dissecting aortic aneurysm (thoracic)

End-stage lung cancer

Heart attack (acute MI)

Heart surgery

Pericarditis caused by bacterial or viral infections

Wounds to the heart

Incorrect Options:

Option B - Fat necrosis: Epipericardial fat necrosis (EFN) is an inflammatory condition that affects the fat surrounding the heart's mediastinum.

Option C - Hemothorax : A hemothorax is a buildup of blood in the pleural space, which is the region between the visceral and parietal pleura. Tachypnea and respiratory discomfort are among the clinical indicators in such patients.

Option D - Pulmonary embolism: When a blood clot becomes lodged in a lung artery and prevents blood flow to a portion of the lung, it results in a pulmonary embolism. Most frequently, blood clots begin in the legs, move through the right side of the heart, and enter the lungs.

Solution for Question 36:

Correct Option C - Epidural hematoma:

- Epidural hematoma patients may experience a lucid interval of several hours prior to neurologic impairment because extradural blood buildup is frequently sluggish.
- Patient is unconscious and turns conscious and goes unconscious during a lucid interval.
- Treatment Aspiration Burr and Hole Craniotomy
- Aspiration
- Burr and Hole
- Craniotomy

Epidural hematoma patients may experience a lucid interval of several hours prior to neurologic impairment because extradural blood buildup is frequently sluggish.

Patient is unconscious and turns conscious and goes unconscious during a lucid interval.

Treatment

- Aspiration
- Burr and Hole
- Craniotomy

Aspiration

Burr and Hole

Craniotomy

Incorrect Options:

Option A - Subdural haemorrhage: This option is incorrect.

Option B - Parenchymal haemorrhage: Intraparenchymal haemorrhage is bleeding into the parenchyma of the brain, which is subsequently followed by a period of lucidity and neurologic decline.

Option D - Subarachnoid haemorrhage: Bleeding due to trauma, or rupture of an aneurysm (such as a saccular aneurysm) or arteriovenous malformation. Rapid time course. Patients complain of "worst headache of my life." Bloody or yellow (xanthochromic) lumbar puncture.

Solution for Question 37:

Correct Option A - A-4, B-3, C-1, D-2:

- (A) Axe: The correct match is option 4 - Chop wound . An axe is a cutting tool with a sharp blade on one side and a handle on the other. When an axe is used forcefully to strike a person, it can cause a deep, wide, and often irregular wound known as a chop wound. This type of wound is characterized by ragged edges and a combination of sharp and blunt force injury.
- An axe is a cutting tool with a sharp blade on one side and a handle on the other. When an axe is used forcefully to strike a person, it can cause a deep, wide, and often irregular wound known as a chop wound.
- This type of wound is characterized by ragged edges and a combination of sharp and blunt force injury.
- (B) RTA (Road Traffic Accident): The correct match is option 3 - Grazed abrasion. In accidents involving vehicles, a person may slide or be dragged along the road surface, leading to abrasions. Grazed abrasions are shallow injuries where the outer layer of the skin is scraped off due to friction against the road surface. They usually appear as superficial scrapes or grazes.
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- (C) Blade: The correct match is option 1 - Incised wound . A blade is a sharp-edged instrument, typically used for cutting. When a blade cuts through the skin, it creates an incised wound. Incised wounds have clean, smooth edges.
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- Incised wounds have clean, smooth edges.
- (D) Lathi: The correct match is option 2 - Tram track bruise. A lathi is a long, heavy stick often used as a weapon or tool. It can cause a patterned bruise known as a tram track bruise.
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Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation for the correct answer.

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Solution for Question 38:

Correct Option D - Moist heat:

- The most likely cause of the described burn injury, characterized by soddened injury, erythema, blisters, and a line of demarcation that extends down the limb, is "Moist heat."
- Moist heat burns typically occur when there is prolonged contact with hot liquids, steam, or moist objects. The moisture can lead to deep tissue damage and delayed healing, which matches the description provided.

The most likely cause of the described burn injury, characterized by soddened injury, erythema, blisters, and a line of demarcation that extends down the limb, is "Moist heat."

Moist heat burns typically occur when there is prolonged contact with hot liquids, steam, or moist objects. The moisture can lead to deep tissue damage and delayed healing, which matches the description provided.

Incorrect Options:

Option A - Electric burn : Electric burns are caused by contact with an electrical source, such as exposed wires or lightning. These burns often have characteristic entry and exit wounds and may not necessarily show soddened injury or a line of demarcation. While electric burns can cause erythema and blisters, they are not the most likely cause based on the information given.

Option B - Chemical burn : Chemical burns occur when the skin comes into contact with corrosive substances, such as acids or alkalis. They typically present with localized tissue damage and may show signs of erythema, blisters, and a line of demarcation. However, the description of a soddened injury is not consistent with a chemical burn, making it less likely to be the cause in this scenario.

Option C - Dry heat : Dry heat burns result from direct contact with hot objects, flames, or hot surfaces. While they can cause erythema, blisters, and a line of demarcation, they do not typically result in a soddened injury. Given the specific description dry heat is not the most likely cause.

Solution for Question 39:

Correct Option A - Defense wounds:

- Defense wounds are wounds that are dispensed by an individual who is attempting to defend themselves from an aggressor.
- If the weapon grasps while defending, the Palmar surface of 1st web space is the MC site affected. This is k/a Active defense injury
- If the person just tries to protect himself without grasping the weapon, the Medial/ Ulnar margin of the forearm is MC affected. It is k/a Passive defense injury
- Though Presence of defense injury is diagnostic to homicide , it is not mandatory to present in all cases.
- Defense injuries are absent when pt. is Attacked from behind Unconscious Asleep
- Attacked from behind
- Unconscious
- Asleep

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Though Presence of defense injury is diagnostic to homicide , it is not mandatory to present in all cases.

Defense injuries are absent when pt. is

- Attacked from behind
- Unconscious
- Asleep

Attacked from behind

Unconscious

Asleep

Incorrect Options :

Option B - Antemortem wound:

- Antemortem wounds happened before passing when the individual was still alive. The picture does not give sufficient data to decide whether the wound was made before or after

passing, so it cannot be classified as an antemortem wound based on the picture alone.

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Option C - Postmortem wound:

- Postmortem wounds are wounds that happen after death. The picture shows a wound that appears to have been made before death, with signs of drying and tissue harm around the edges. Hence, it cannot be classified as a postmortem wound.

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Option D - Hesitation cuts:

- Hesitation cuts are shallow, provisional cuts made by an individual who is reluctant to self-harm or commit suicide.
- Features: Multiple cuts, superficial, linear/parallel cuts. Seen in accessible parts of the body (forearm, thigh and abdomen) Suggestive of suicidal tendency.
- Multiple cuts, superficial, linear/parallel cuts.
- Seen in accessible parts of the body (forearm, thigh and abdomen)
- Suggestive of suicidal tendency.
- The wound within the image is more profound and serious than a hesitation cut.
- Moreover, there's no sign that the wound was self-inflicted, so it cannot be classified as a hesitation cut.

Hesitation cuts are shallow, provisional cuts made by an individual who is reluctant to self-harm or commit suicide.

Features:

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- Seen in accessible parts of the body (forearm, thigh and abdomen)
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Multiple cuts, superficial, linear/parallel cuts.

Seen in accessible parts of the body (forearm, thigh and abdomen)

Suggestive of suicidal tendency.

The wound within the image is more profound and serious than a hesitation cut.

Moreover, there's no sign that the wound was self-inflicted, so it cannot be classified as a hesitation cut.



Solution for Question 40:

Correct Option B - Chop wound:

- Based on the description provided, the type of wound is likely a "Chop wound."
- In this case, the presence of clean-cut edges, crushed tissues, and disruption of vertebral bones suggests a chop wound, indicating that a heavy and sharp-edged object was used with significant force to cause the injury.
- A chop wound occurs when a heavy and sharp-edged object, such as an axe or machete, is used with significant force. It typically results in a combination of a deep incised wound with associated crushing or fracturing of underlying structures, such as bones.

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Incorrect Options:

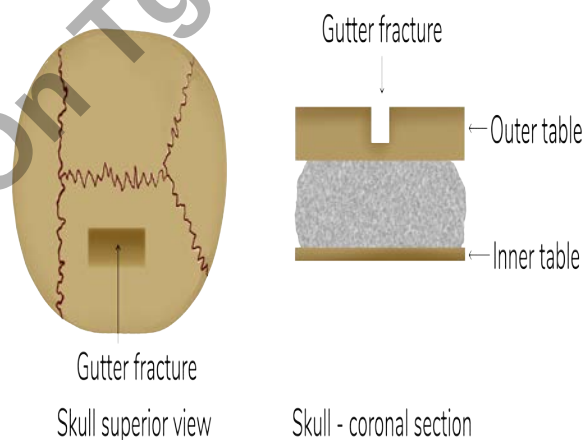
Option A - Incised wound : An incised wound is characterized by clean-cut edges, similar to a surgical or razor blade cut. It does not involve crushing or disruption of deeper tissues or bones.

Option C - Lacerated wound : Lacerated wounds are characterized by irregular, torn, or jagged edges. They are caused by blunt force trauma, often resulting in the tearing and stretching of tissues.

Option D - Avulsion : Avulsion refers to the tearing away of tissue or a body part from its normal position. It typically involves the separation of tissue or structure due to extreme force or trauma.

Solution for Question 41:

Correct Option D - A bullet:



- When a bullet just grazes/ glances/ taps/ slaps through the skull (there is no head-on collision), a part of the outer table is chipped out with an intact inner table, leaving behind a gutter-like groove. Hence, called the 'gutter fracture'.
- Gutter fracture is a long and narrow fracture. It is seen with a bullet glance/ graze/ slap of the skull. A part of the outer table will be chipped off/ grooved – 1st degree Gutter fracture.

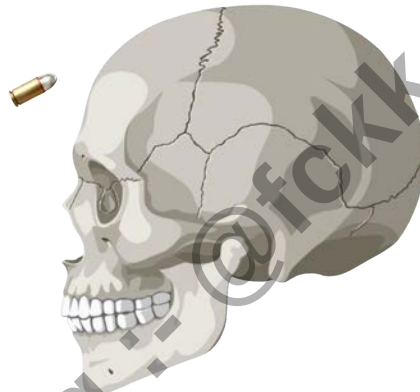
- 2 nd degree Gutter fracture: Fracture of the inner table (called the Depressed fracture).

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2 nd degree Gutter fracture: Fracture of the inner table (called the Depressed fracture).

GUTTER FRACTURE



Incorrect Options:

Option A - A sharp object: May lead to compound depressed fracture such as stiletto-heel fracture due to use of stiletto heel shoes (rare type).

Option B - A Blunt object: A heavy blunt blow leads to comminuted fracture. Multiple fragments are formed. There are intersecting lines and webs, known as spider web fracture. Fragments are displaced.

Option C - Fall from height: A person falling from a great height and landing on his feet results in ring fracture or foramen fracture . It is a fracture surrounding foramen magnum . Seen if someone falls from a height or falling on feet or falling on buttocks. Can be due to a blow on the chin or a blow on the vertex.

Solution for Question 42:

Correct Option B - Hinge fracture:

- In the given scenario where the image shows the base of the skull divided in half, the specific type of skull fracture is known as a "hinge fracture." A hinge fracture is a specific subtype of basilar skull fracture, which involves a separation or splitting of the base of the skull along a hinge-like mechanism.

In the given scenario where the image shows the base of the skull divided in half, the specific type of skull fracture is known as a "hinge fracture." A hinge fracture is a specific subtype of basilar skull fracture, which involves a separation or splitting of the base of the skull along a hinge-like mechanism.

Incorrect Options:

Option A - Depressed fracture: A depressed fracture is a type of skull fracture where the bone fragments are depressed or pushed inward. It does not correspond to the given image.

Option C - Pond fracture: Pond fracture is not a recognized term for a specific type of skull fracture.

Option D - Comminuted fracture: Comminuted fracture refers to a fracture in which the bone is fragmented into multiple pieces. It does not correspond to the given image.

Solution for Question 43:

Correct Option A - Abrasion:

- Based on the image, it is showing an abrasion on the palm. An abrasion is a superficial injury that occurs when the skin is scraped or rubbed against a rough surface, resulting in the removal of the superficial layers of skin. It typically appears as a shallow, irregular-shaped wound with a reddish or pinkish color.

Incorrect Options:

Option B - Laceration: A laceration is a type of injury characterized by a deep cut or tear in the skin and underlying tissues. It typically has clean or irregular edges.



Incised Looking Laceration

Option C - Bruise : A bruise is a discoloration of the skin caused by the rupture of blood vessels beneath the skin surface. It appears as a reddish, purple, or blue mark and is often the result of blunt force trauma ,it does not involve the removal of skin layers.



Option D - Imprint: An imprint refers to a pattern or mark left on the skin as a result of contact with an object or surface.

Solution for Question 44:

Correct Option C - Hypothermia:

- Paradoxical undressing is a phenomenon commonly observed in cases of severe hypothermia. Hypothermia is a condition characterized by a dangerously low body temperature, usually below 35 degrees Celsius (95 degrees Fahrenheit).
- This behavior is thought to result from the disorientation and confusion caused by hypothermia, as well as the dilation of blood vessels near the skin's surface, which can create a false sensation of warmth.

Incorrect Options:

Option A - Heart stress : It is not associated with paradoxical undressing.

Option B - Voyeurism : Voyeurism is a paraphilic disorder characterized by an individual's sexual interest and pleasure derived from observing others without their consent. It is unrelated to paradoxical undressing.

Option D - LSD overdose : Overdosing on LSD can cause a variety of physical and psychological effects, but paradoxical undressing is not a known symptom.

Solution for Question 45:

Correct Option C - There are fewer chances of bacterial infection:

- Human bites carry a high risk of polymicrobial infection by organisms such as *Staphylococcus* spp., *Streptococcus* spp., and *Eikenella corrodens* . Hence antibiotics are recommended.

Human bites carry a high risk of polymicrobial infection by organisms such as *Staphylococcus* spp., *Streptococcus* spp., and *Eikenella corrodens* . Hence antibiotics are recommended.

Incorrect Options:

Option A - Swabs should be taken immediately after moistening with sterile water: It is recommended to moisten swabs with sterile water before sampling. Moistening the swabs helps in better collection of potential biological material, such as saliva or cells.

Option B - It is most commonly caused by incisors and canines: These teeth are more likely to leave identifiable patterns and impressions in a bite mark.

Option D - Tetanus vaccination should be given: This statement is true.

Solution for Question 46:

Correct Option B - Chop wound :

- The given image caused by an axe is likely to be a chop wound. In the given image, the wound resembles an injury produced by a sharp axe. Such wounds may be inflicted with homicidal intent on the skull/scalp. , Chop wounds are usually deep gaping wounds caused by a blow with a sharp cutting edge of a fairly heavy weapon like an axe, sword, hatchet, etc.

The given image caused by an axe is likely to be a chop wound. In the given image, the wound resembles an injury produced by a sharp axe. Such wounds may be inflicted with homicidal intent on the skull/scalp. , Chop wounds are usually deep gaping wounds caused by a blow with a sharp cutting edge of a fairly heavy weapon like an axe, sword, hatchet, etc.

Incorrect Options:

Options A - Incised wound: It typically has well-defined, straight edges and minimal tissue damage.

Options C - Lacerated wound : A lacerated wound is a jagged, irregular wound caused by tearing or crushing of the skin and underlying tissues. It often has irregular edges and significant tissue damage.

Options D - Abrasion : An abrasion is a wound caused by friction, resulting in the removal of the superficial layers of the skin. It typically appears as a scrape or graze without deep penetration.

Solution for Question 47:

Correct Option C - Laceration:

- A laceration is a wound characterized by irregular, jagged edges caused by blunt force trauma or tearing of the skin and underlying tissues. In this case, the injury appears to have resulted from a forceful impact or tearing action, leading to a laceration.

A laceration is a wound characterized by irregular, jagged edges caused by blunt force trauma or tearing of the skin and underlying tissues. In this case, the injury appears to have resulted from a forceful impact or tearing action, leading to a laceration.

Incorrect Option:

Option A - Incised wound : An incised wound is a clean, smooth-edged wound typically caused by a sharp object.

Option B - Abrasion : An abrasion is a superficial wound caused by friction, often resulting in the removal of the superficial layers of the skin. It does not involve a deep tear or irregular edges.

Option D - Bruise : A bruise, also known as a contusion, is caused by blunt force trauma that damages blood vessels underneath the skin, leading to discoloration.

Solution for Question 48:

Correct Option A - Shot gun wound:

- The image shown depicts a typical pattern of injury caused by a shotgun. The appearance of multiple pellet or shot holes, irregular wound edges, and tissue destruction is characteristic of a shotgun wound.

The image shown depicts a typical pattern of injury caused by a shotgun. The appearance of multiple pellet or shot holes, irregular wound edges, and tissue destruction is characteristic of a shotgun wound.

Incorrect Options:

Option B - Tattooing: Tattooing refers to the deposition of tattoo ink or pigment into the skin. It is unrelated to the image shown, which depicts a penetrating injury.

Option C - Arsenic poisoning: Arsenic poisoning is a systemic condition, it does not cause wounds as shown in the image.

Option D - Graze wound: A graze wound is a superficial wound caused by a glancing impact, resulting in abrasion or scraping of the skin.

Solution for Question 49:

Correct Option A - Strangulation by ligature:

- Strangulation by ligature is correct because there is evidence of a ligature mark on the neck.
- The presence of a ligature mark makes strangulation by ligature or hanging as the likely cause.

Strangulation by ligature is correct because there is evidence of a ligature mark on the neck.

The presence of a ligature mark makes strangulation by ligature or hanging as the likely cause.

Incorrect Options:

Option B - Hanging: There is no indication of suspension or hanging-related findings.

Option C -Throttling: The absence of carotid intimal tear and muscle hematoma do not indicate forceful compression or pressure on the neck, which is consistent with throttling.

Option D - Drowning: Drowning is incorrect as there are no signs or information suggesting drowning in this case.

Solution for Question 50:

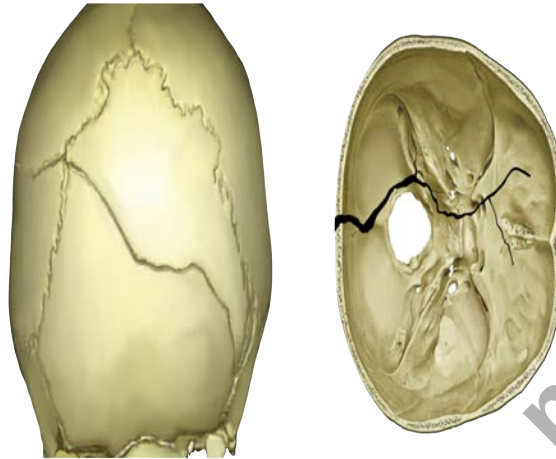
Correct Option C - Fissured:

The Fissured Fracture of skull image is given.

- Thin, Linear fracture line is seen on the skull vault.
- M/C type of skull fracture

Thin, Linear fracture line is seen on the skull vault.

M/C type of skull fracture



Incorrect Options:

Option A - Ring: Ring fracture is associated with the fall from height and landing on the foot. Fracture line is seen around the Foramen magnum like a ring.

Option B - Depressed: Due to the impact of the weapon, bone gets fractured and the fractured segment is depressed inside.

Depressed fracture



Option D - Comminuted fracture: Comminuted fracture is caused due to repeated blows with heavy force. Presents with multiple Fracture lines intersecting each other.

Displaced



Search On Tg :: @fckk_marrow

Forensic Ballistics

1. During a forensic analysis of a crime scene, investigators examined firearms and noted the distinctions between rifled and smooth-bore firearms. Which of the following is true regarding rifled and smooth-bore firearms

- A. Rifled firearms usually have multiple projectiles
- B. Smooth-bore firearms include handguns and revolvers
- C. Shotguns typically have a smooth barrel
- D. Smooth-bore firearms typically don't have chokes

2. A 45-year-old male was found dead after a suspected shooting incident involving a shotgun. During the investigation, experts examined the shotgun's characteristics to determine factors affecting its range. Which of the following does not enhance the range of shotguns?

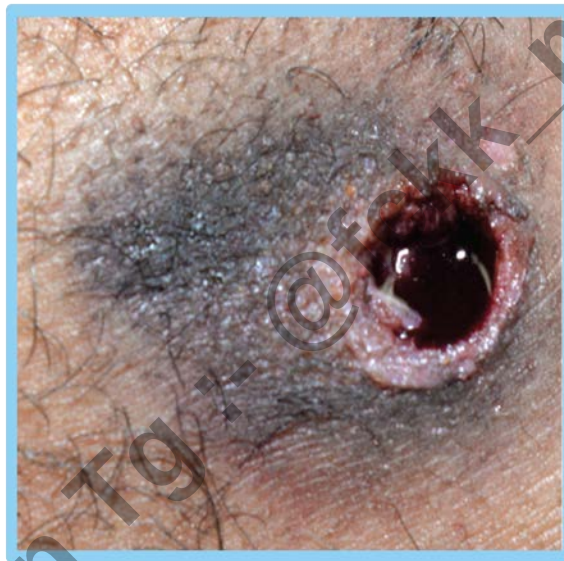
- A. Elongated and pointed bullet
- B. Barrel length
- C. Paradox gun
- D. Presence of a choke

3. A 38-year-old female is found deceased in an urban park with a single gunshot wound to the left side of her chest. Forensic analysis suggests the shot was fired from a distance, with no evidence of close-range effects like soot, muzzle imprint, or stippling. Which of the following is a feature of the entry wound?

- A. Fat and soft tissue expulsion
- B. Irregular, stellate-shaped wound

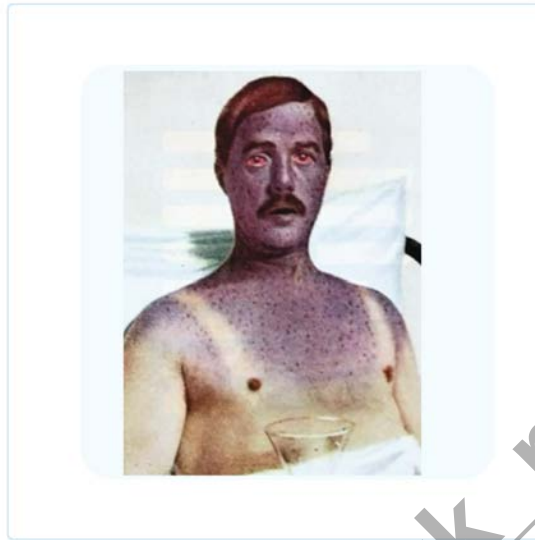
- C. Presence of everted margins
 - D. Presence of cherry red wound track
-

4. A 40-year-old male was found shot dead in a parking lot. Forensic examination revealed that the cause of death was a shotgun injury. Considering the type of injury, what is the probable range of the shot?



- A. <1 metre
 - B. 1-2 metres
 - C. 2-4 metres
 - D. >4 metres
-

5. A 40-year-old laborer was involved in an explosion at a chemical plant caused by a faulty pressure valve. Rescued from a confined space, he arrived at the emergency department with respiratory distress and injuries shown in the image. What type of injury describes this condition?



- A. Primary blast injury
- B. Secondary blast injury
- C. tertiary blast injury
- D. Quaternary blast injury

6. A 40-year-old male diver is brought to the emergency department after an underwater explosion and complains of severe abdominal pain, nausea, and difficulty breathing. Physical examination reveals tenderness in the abdomen. Which organ is most commonly injured in this patient due to the underwater blast?

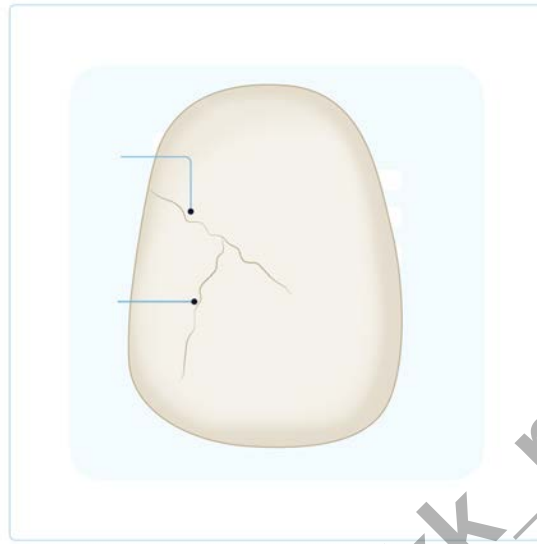
- A. Lungs
- B. Stomach
- C. Colon
- D. Small intestine

7. A 35-year-old male construction worker was involved in a workplace explosion. Positioned near the blast's epicenter, he was swiftly taken to the emergency department with multiple injuries, including punctate abrasions, contusions, and lacerations as depicted in the image. What type of blast injury is most likely in this case?



- A. Primary injury
- B. Secondary injury
- C. Tertiary injury
- D. Quaternary injury

8. In the forensic analysis of cranial fractures, Puppe's Rule is applied to determine the sequence of impacts. According to this rule, when two fracture lines intersect, as shown below, what can be inferred about their relationship?



- A. The line that ends at the intersection is the newer injury.
- B. The second fracture line crosses the first line.
- C. Both fracture lines are of equal age.
- D. Crossing fracture indicates a higher velocity impact.

9. A 28-year-old male was found dead in an abandoned warehouse due to a shotgun wound to the chest. Analysis indicated that the shotgun pellets caused an effect where multiple pellets, fired from a high range, impacted the same area, leading to significant damage and a larger wound channel. What effect is being described here?

- A. Billiard ball effect
- B. Balling effect
- C. Rayalseena phenomenon
- D. Kennedy phenomenon

10. A 40-year-old male was shot during a robbery while walking home at night. The weapon used was identified as a firearm, and forensic investigators recovered a bullet from the scene, as shown in the image below. Identify it.



- A. Hollow point bullet
- B. Souvenir bullet
- C. Incendiary bullet
- D. Dum dum bullet

11. A 30-year-old male was participating in a target shooting session at a local outdoor range when he accidentally fired a bullet that deflected off a metal target and struck another participant, resulting in his death. Which of the following best describes the bullet?

- A. Yawning bullet
- B. Ricochet bullet
- C. Tumbling bullet
- D. Tandem bullet

12. A forensic investigation was launched after a 20-year-old boy was found dead with a bullet in his body, showing signs of burning, blackening, and tattooing. At what distance does a shotgun produce a flame effect?

- A. 7 cm
 - B. 15 cm
 - C. 30 cm
 - D. 45 cm
-

13. A 30-year-old male was shot outside a nightclub late at night. Witnesses reported hearing gunshots and saw a dark sedan fleeing the scene. Police found shell casings and a bloodstained area, collecting gunshot residue (GSR) samples from the victim's hands and clothing. Which of the following is the most specific gunshot residue test?

- A. Harrison and Gilroy test
 - B. Atomic absorption Spectrometry
 - C. Neutron activation analysis.
 - D. SEM-EDXA
-

14. A 35-year-old male was found dead in an abandoned building with no weapon at the scene. Forensic examination revealed a rifle was used. Which of the following wound characteristics is true regarding distant gunshot injury?

- A. Presence of blackening and tattooing around the wound
- B. Presence of abrasion collar around the entry wound
- C. Clean, circular entry wound with everted edges
- D. Presence of multiple small pellet wounds

15. A forensic investigator is analysing the injuries caused by a bullet at a crime scene. Which of the following best describes the study of the interaction between a projectile and its target?

- A. Proximal Ballistics
 - B. Intermediate Ballistics
 - C. External Ballistics
 - D. Terminal Ballistics
-

16. A 50-year-old female is found deceased in her living room with a shotgun wound to the head. Considering the characteristics of the injury, what type of wound is this?



- A. Close shot wound
- B. Near shot wound
- C. Intermediate shot wound

D. Contact shot wound

17. In a forensic investigation of a 45-year-old male found dead from a gunshot wound, the scene indicates forced entry, with the front door ajar and a bullet and shell casing collected. Which of the following are not considered primary markings on the bullet?

- A. Diameter and width of lands and grooves
 - B. Number of lands and grooves
 - C. Degree of rifling twist
 - D. Surface irregularities from barrel imperfections
-

18. A 30-year-old male is found lying in the driveway with multiple gunshot wounds and is pronounced dead at the scene. Which component is responsible for the effects depicted in the wound image?



- A. Soot deposition

- B. Flame
 - C. Unburnt gunpowder
 - D. Weapon end
-

19. Forensic analysis of the cartridges recovered from the crime scene of a gunshot victim reveals the presence of primers. Which of the following is not used as a component of a primer?

- A. Tetrazine
 - B. Antimony sulphide
 - C. Barium Nitrate
 - D. Mercury fulminate
-

20. A 28-year-old male is found deceased in his apartment with a gunshot wound, and police find a firearm at the scene, along with traces of smokeless gunpowder on the victim's hands. What is the composition of single-base smokeless gunpowder?

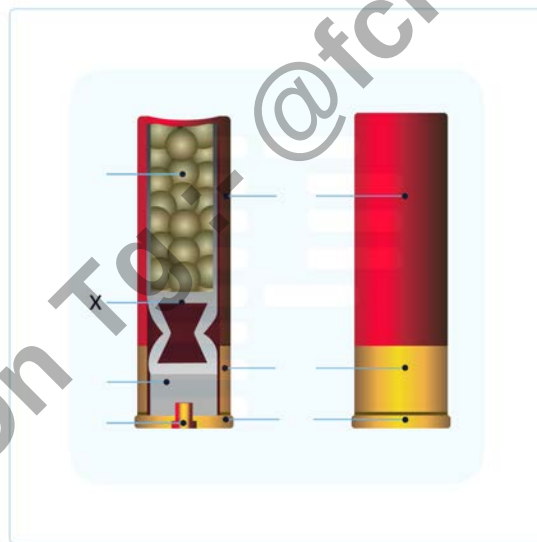
- A. Nitroguanidine
 - B. Nitroglycerine
 - C. Potassium nitrate
 - D. Nitrocellulose
-

21. A 30-year-old man presents to the emergency with severe burns and injuries after an accidental explosion while handling fireworks. Which of the following statements are true regarding the type of gunpowder commonly used in fireworks?
Produces less smoke compared to newer gun powders Produces more heat and

flame while burning 1gm of this powder produces 3000cc of gas It contains only Sulphur and Charcoal It is used in the composition of Semi smokeless gunpowder

- A. 1, 2, 4
- B. 1, 4, 5
- C. 2, 4, 5
- D. 2, 3, 5

22. Which of the following is not an advantage of the structure labeled “X” in the image?



- A. Effective sealing of bore
- B. Lubrication
- C. Allows gas to escape from the breech end
- D. Facilitates optimal pressure

23. During a forensic examination of a shooting incident, a medical examiner discovers both shotgun pellets and rifle bullets at the crime scene. The examiner notes the differences in their design and function. Which statement about these ammunition types is not correct?

- A. Shotgun cartridges contain multiple projectiles that spread upon firing.
 - B. Wads are used in all types of cartridges
 - C. Rifle bullets are generally elongated with a pointed tip
 - D. Rifle bullets and shotgun pellets are made of lead
-

24. During a forensic investigation, experts discussed the benefits of rifled firearms and how rifling impacts evidence collection. All the following are advantages of rifling except?

- A. Increases accuracy by stabilizing the bullet
 - B. Improves range by imparting spin to the projectile
 - C. Limits spread of pellets
 - D. Reduces wobbling of the bullet in flight
-

25. At a forensic lab, a team was examining evidence from a recent shooting. One forensic analyst pointed out that identifying the caliber of the gun can help in linking it to a specific weapon. What does the term "caliber" refer to in forensic ballistics?

- A. Distance between two opposite lands
 - B. Distance between two opposite grooves
 - C. Muzzle design influencing the dispersion of shot
 - D. Measurement of the diameter of the shotgun barrel
-

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Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	4
Question 4	1
Question 5	4
Question 6	3
Question 7	2
Question 8	1
Question 9	2
Question 10	4
Question 11	2
Question 12	2
Question 13	4
Question 14	2
Question 15	4
Question 16	4
Question 17	4
Question 18	3

Question 19	4
Question 20	4
Question 21	4
Question 22	3
Question 23	2
Question 24	3
Question 25	1

Solution for Question 1:

Correct Answer: C) Shotguns typically have a smooth barrel

Explanation:

Shotguns belong to the smooth-bore firearms, which usually have uniformly smooth barrels.

Difference between Rifled & Smooth Bore Firearms:

Feature	Rifled Firearms	Smooth Bore Firearms
Barrel Interior	Contains spiral grooves (Rifling)	Uniformly smooth barrel without grooves
Measurement	Caliber: Diameter between opposite lands, measured before rifling grooves are cut.	Bore: Defined by the number of lead balls of a specific size (diameter) that can be made from one pound of lead (454 grams).
Bullet Type	Typically fires a single projectile (bullet) (Option A)	Fires multiple pellets (shot) or a single slug
Accuracy	Higher due to rifling	Lower accuracy, designed for spread over a wide area.
Functionality	Imparts spin to the bullet for greater stability and precision	Fires multiple projectiles simultaneously for short-range effectiveness

Choke	Usually not applicable	May have chokes to control the spread of the shot (Option D)
Examples	Rifles, Pistols, Revolvers (Option B)	Shotguns



Solution for Question 2:

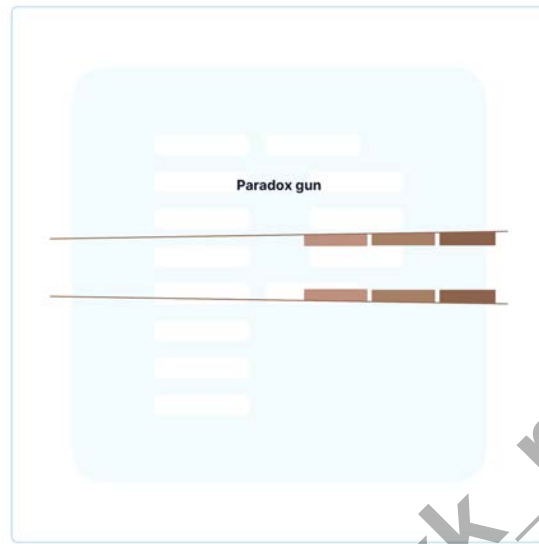
Correct Answer: A) Elongated and pointed bullet

Explanation:

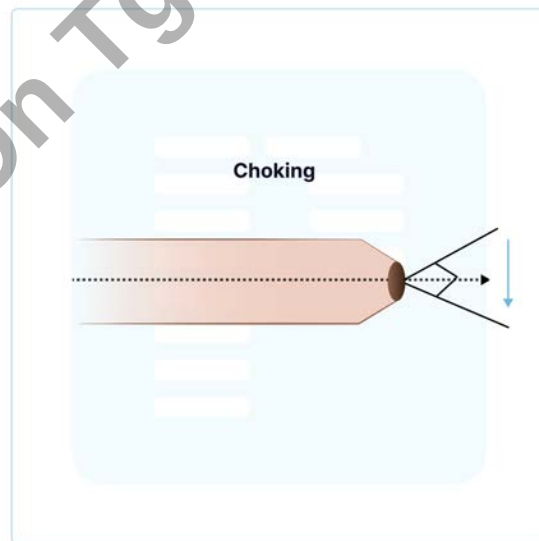
An elongated and pointed bullet does not enhance the effective range of shotguns.

Factors Enhancing the Range of Firearms

Paradox gun: A mixed gun incorporating characteristics of both a rifle and a shotgun, featuring a shotgun design with a rifled section in the barrel towards the muzzle end.



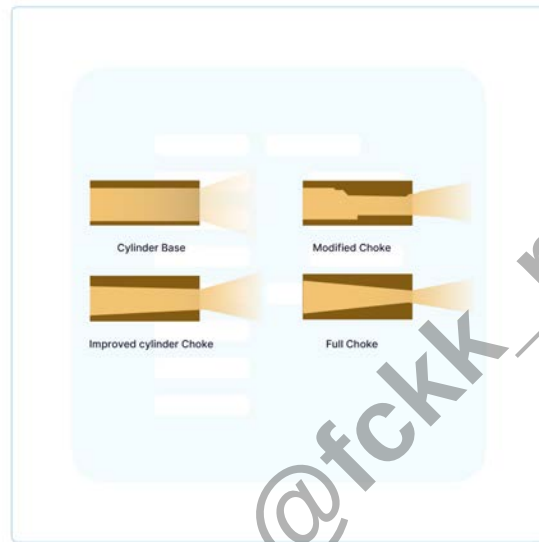
- Choking: Constriction at the muzzle end of a shotgun in order to: Reduce the dispersion of lead shot Enhance the velocity of the pellets
- Reduce the dispersion of lead shot
- Enhance the velocity of the pellets
- Reduce the dispersion of lead shot
- Enhance the velocity of the pellets



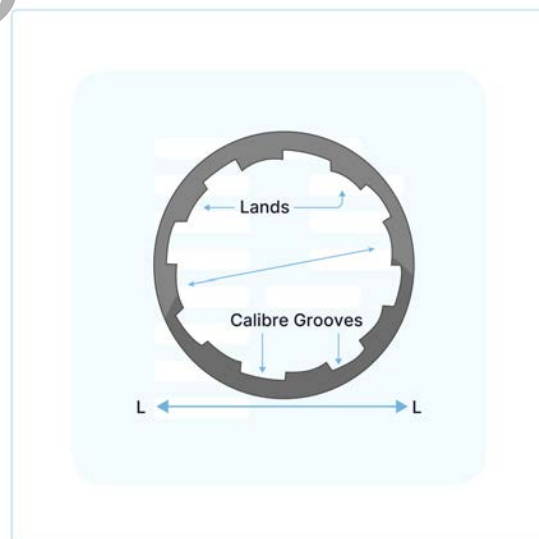
• Types of choking:

Type	Range
Cylindrical base (unchoked)	25 yards

Improved cylindrical choke	30 yads
Modified cylindrical choke	35 yards
Full choke	40 yards



- Elongated and pointed bullet (Option A) : Aerodynamic shape reduces air resistance, allowing the bullet to maintain velocity and travel farther.
- Lands and Grooves (Rifling) (Option B) : Imparts spin to the bullet, stabilising it in flight and improving accuracy and range.
- Calibre (Distance between two opposite lands) (Option C) : Larger calibres maintain stability and energy over longer distances, contributing to extended range.



- **Barrel length:** A longer barrel increases bullet velocity by allowing more time for propellant gases to act, enhancing range.
- **Ammunition Type:** High-performance bullets reduce drag, improving aerodynamic efficiency and range.
- **Muzzle Velocity:** Higher initial bullet speed reduces the effects of gravity and air resistance, extending range.

Solution for Question 3:

Correct Answer: D) Presence of cherry red wound track

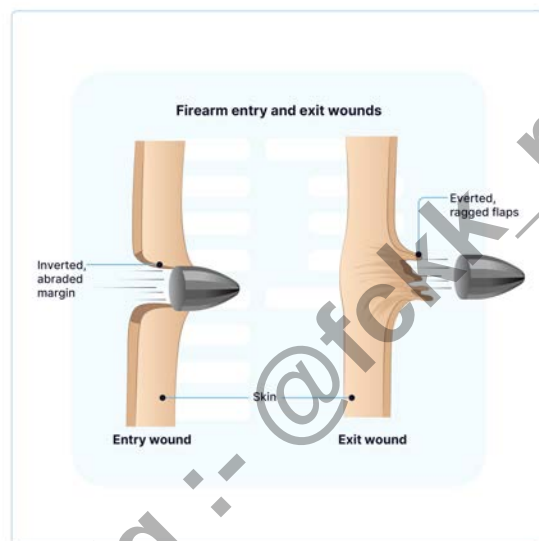
Explanation:

A cherry-red colour in the wound track is due to the reaction between carbon monoxide (CO) and haemoglobin in the blood at the entry wound site.

Features of the Entry Wound and Exit wound

	Entry wound	Exit wound
Size and shape (compared with a bullet)	Small, clean, circular wound	Large (except in contact shot), Irregular, stellate-shaped wound. (Option B)
Margins	Inverted	Everted (Option C)
Abrasion & Grease collar	Present	Absent (except in shored exit) *Shored exit: When the skin at an exit wound is supported by firm objects, tight clothing, or a body leaning against a hard surface, the wound appears as a circular defect with an abrasion margin similar to an entry wound. The pattern of the overlying material may also be imprinted on the wound's edges.
Burning, blackening & tattooing	Present	Absent
Skull	Inner table: Beveled Outer table: Clean cut	Inner table: Clean cut Outer table: Beveled
Cherry-red wound track	Present (might be) (Option D)	Absent

Bleeding	Less	More
Fat and soft tissue expulsion	Absent	Present (Option A)
Lead ring	May be present	Absent
Cloth fibres	Turned in	Turned out
Spectrograph	More metal found	Less metal found



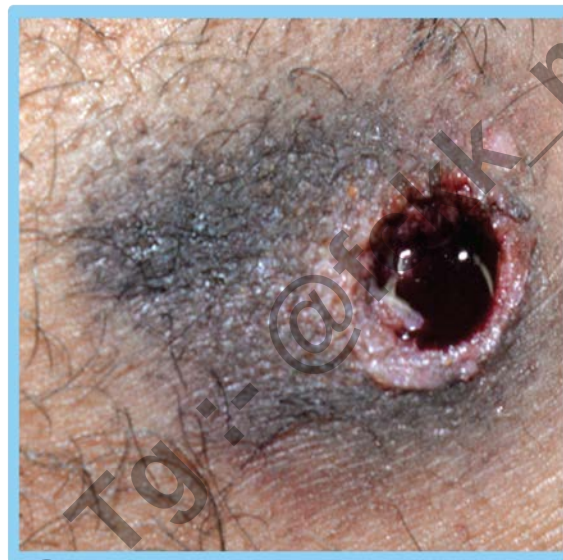
Solution for Question 4:

Correct Answer: A) < 1 metre

Explanation:

- A close-shot injury (< 1 metre) has a circular or elliptical entry wound with blackening, burning, tattooing, and singeing of hair.

The wound track and surrounding tissues appear cherry-red due to carbon monoxide absorption.



Gunshot wound characteristics based on Range of shooting:

Shooting Range	Range	Wound Description
Contact shot	Muzzle in direct contact	Circular entry wound with muzzle imprint, severe tissue damage, blackening, and burning of the wound edges. Cherry-red tissue due to carbon monoxide absorption.
Close-shot	< 1 meter	Circular or elliptical entry wound with blackening, burning, tattooing, singeing of hair, and cherry-red tissue discolouration due to carbon monoxide absorption.
Short-range	1-2 meters	Circular 4-5 cm wound (Rat hole) with irregular borders, no blackening, burning, or tattooing.

Intermediate shot	2-4 meters	Large, irregular wound surrounded by multiple satellite wounds.
Distant shot	> 4 meters	The entry wound fades, leaving multiple small pellet wounds.



Solution for Question 5:

Correct Answer: D) Quaternary blast injury

Explanation

The provided scenario and accompanying image of traumatic asphyxia, which is a characteristic of quaternary blast injury.

• Types of Blast injuries

Type of injury	Caused by	Features
Primary blast injury (Option A)	Blast waves	Commonly internal with no external signs. Ears: Tympanic membrane rupture Lungs: Alveolar, capillary, and membrane hemorrhage leading to blast lung (most common causes of death among survivors) GIT: Vomiting, melena, hematemesis, perforation.
Secondary blast injury (Option B)	Shrapnel, debris & flying objects (propelled by explosion)	Penetrating trauma Marshall triad: Includes bruises, abrasions, and punctate lacerations. Thoracic injuries
Tertiary blast injury (Option C)	Victims are thrown against solid objects by blast winds	Blunt or penetrating trauma. Bone fractures and coup-counter coup injury.
Quaternary blast injury (Option D)	Environmental contaminants, chemicals, radiations (nuclear bomb), virus and bacteria	Flash burns, crush injuries Fall resulting from the explosion Traumatic asphyxia and respiratory injuries Psychiatric injury Post-traumatic stress disorder (PTSD)

Solution for Question 6:

Correct Answer: C) Colon

Explanation:

In Underwater explosions, injuries to gas-filled organs are more frequent, with the Colon being the most commonly affected organ .

Blast injuries are complex physical traumas resulting from direct or indirect exposure to an explosion that causes sudden changes in atmospheric pressure.

Types of Blast Exposure:

- Air Blast: Causes less severe injuries due to energy dissipation.
- Immersion Blast: Force transmitted through water, leading to more severe injuries due to dense molecular packing. In enclosed spaces, shock waves can amplify, increasing injury severity.
- Solid Blast: Force transmitted through solids, affecting areas in contact with vibrating structures, leading to primary and secondary injuries.

Underwater Explosion:

- Injury to gas-filled organs is more frequent, with the Colon being the most commonly affected organ.
- Other potential injuries: Perforation of the stomach Small intestine Cecum Lung Injury (Less Common)
- Perforation of the stomach
- Small intestine
- Cecum
- Lung Injury (Less Common)
- Perforation of the stomach
- Small intestine
- Cecum
- Lung Injury (Less Common)

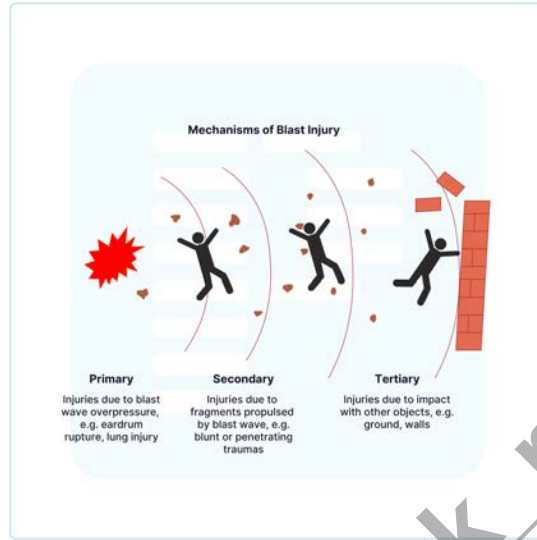
Solution for Question 7:

Correct Answer: B) Secondary injury

Explanation:

The provided case scenario, along with the accompanying image, illustrates Marshall's triad (contusion, abrasion and punctate laceration), which indicates a secondary blast injury.

Blast Injury: Blast injuries are complex physical traumas resulting from direct or indirect exposure to an explosion, leading to sudden changes in atmospheric pressure.



Classification of Blast Injuries:

Type	Mechanism	Features
Primary Injury	It results from blast waves and is often internal with no external signs.	Ears: Tympanic membrane rupture or inner ear disruption. Lungs: Alveolar-capillary membrane haemorrhage (blast lung), most common cause of death in survivors. GIT: Injuries to gas-filled organs are more common in underwater blasts.
Secondary Injury	It is resulting from shrapnel, debris & flying objects (propelled by explosion).	Penetrating trauma. The characteristic Marshall triad, which includes bruises, abrasions, and punctate lacerations, is evident. Commonly cause severe thoracic injuries, including lacerations of the heart and great vessels.
Tertiary Injury	Victims are thrown against solid objects by blast winds.	Blunt/Penetrating trauma. Includes bone fractures and coup-contrecoup injuries.
Quaternary Injury	Encompass all other injuries not classified above	Flash burns from extreme heat. Crush injuries and falls due to explosions. Respiratory injuries from toxic dust, gases, or radiation exposure. Psychiatric issues, such as post-traumatic stress disorder (PTSD), may affect individuals even if they are physically unharmed.

Solution for Question 8:

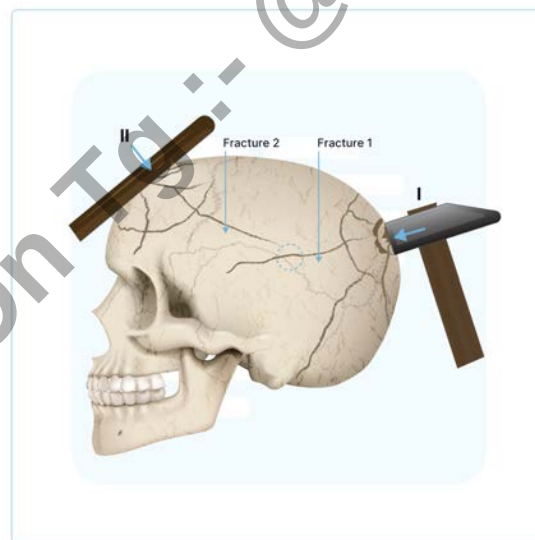
Correct Answer: A) The Line that ends at the intersection is the newer injury.

Explanation:

According to Puppe's rule, the line that ends at the intersection indicates a more recent injury after forming the initial line.

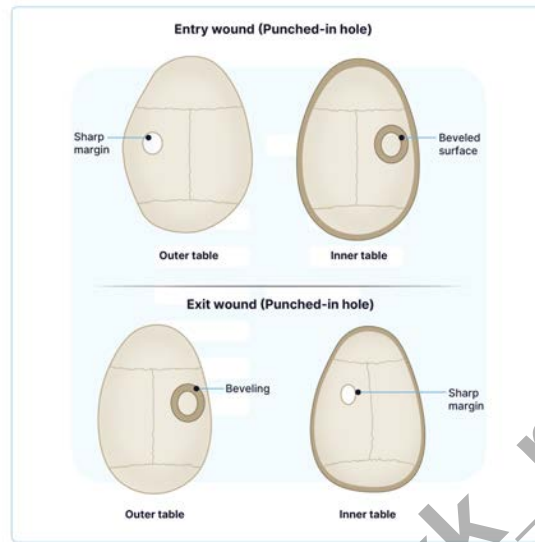
Puppe's Rule:

It states that the second fracture line does not cross the first when two fracture lines intersect. This principle helps determine the sequence of shots when multiple bullets have impacted the cranium, and it also applies to cases of multiple blunt force impacts on the skull.



Beveling of Skull:

- Entry wound: The entry wound presents as a punched-out hole with a clean cut on the outer table and a bevelled appearance on the inner table, where the bone remains unsupported, leading to chipping.
- Exit wound: The exit wound on the inner table shows a clean-cut hole with beveling on the outer table. It is typically larger than the entry wound due to the bullet's deformation and tumbling upon entering the skull.



Kronlein Shot: Rare injury caused by a high-velocity bullet, resulting in bursting of the skull, laceration of the dura, and evisceration of the brain. An autopsy of the skull could not determine the appearance of primary gunshot wounds and entry or exit wounds.

Solution for Question 9:

Correct Answer: B) Balling effect

Explanation:

The effect described above, where multiple pellets fired from a high range impact the same area and create a larger wound channel, is known as the balling effect.

Billiard ball effect vs Balling or welding

Billiard ball effect (Billiard Ball Ricochet Effect) (Option A)	Balling or welding (Option B)
When shotgun pellets enter tissue, the first pellets slow down and are struck from behind by subsequent pellets, causing them to scatter through the tissue like a cluster of billiard balls hit by the cue ball.	In distant shots, the pellets can merge into a compact mass, causing the lead shots to stick together and create a larger entry wound while exhibiting some dispersion. It occurs due to: Faulty manufacture (too much powder, poor pressure of wads, paraffin into cartridge case) Old ammunition

Very high dispersion (due to resistance)	Less dispersion
Low range (near shot resembles distant range)	High range (wound mimics intermediate shot)

Rayalseena phenomenon (Option C) : To disguise a murder as a firearm injury, a bullet is placed in an individual who was initially killed by a stab wound, misleading the investigating agency.

Kennedy phenomenon (Option D) : Identifying the wound becomes challenging due to surgical alterations.

Solution for Question 10:

Correct Answer: D) Dum dum bullet

Explanation:

The image depicts a Dum Dum bullet, which is a type of semi-jacketed bullet.

Types of Bullets

Type	Description	Images
Dum-dum bullet (Option D)	A jacketed bullet that leaves an area near the nose uncovered to expose the core.	
Hollow point bullet (Option A)	The modern Dum Dum bullet features a pit at the nose. Upon impact, pressure in the pit causes the lead ring to expand into a mushroom shape, resulting in increased soft tissue damage and a higher incapacitation index.	
Soft point bullet	A jacketed bullet with an open tip that exposes some lead inside expands upon impact but more slowly than a hollow-point bullet.	

Tandem bullet (Piggyback bullet)	Bullets eject one after another when the first bullet fails to exit the barrel and is pushed out by a subsequently fired bullet (faulty ammunition or a firearm that hasn't been used for years).	
Duplex bullet	It contains two projectiles by design (used in military rifles) and enters a target at different points.	
Frangible bullet	Designed to fragment or disintegrate upon impact, made of copper, powdered tungsten, lead, or iron. They do not ricochet.	
Souvenir or sleeping Bullet (Option B)	A bullet remains in the body for a long time without fresh bleeding, surrounded by a dense fibrous tissue capsule. It can result in lead poisoning.	
Armour-piercing bullet	A jacketed bullet (penetrates armour) with an inner core of hard, high-density metal like tungsten, tungsten carbide, depleted uranium, or steel, featuring a truncated cone.	
Incendiary bullet (Option C)	A type of military bullet designed to ignite a fire in the target.	
Tracer bullet	It leaves a visible mark or "trace" during flight, allowing the shooter to see where the shot impacts.	
Nontoxic shot or Green bullet	It replaces the copper-jacketed lead core with a copper-jacketed tungsten, steel, bismuth, tin, or nylon core. It does not release toxic lead (lead-free) into the environment (green bullet).	
Exploding bullet	A bullet designed to detonate or break apart uses an explosive or deflagrant, causing extensive damage to the victim. These bullets pose a risk to medical personnel, as they may explode during procedures if they fail to detonate in the body.	
Boat tail bullet	The cartridge's rear end is tapered rather than squared off to stabilise the projectile in flight.	

Poisoned bullets	Contains curare, ricin or aflatoxin.	
Wadcutter	Specifically designed for shooting paper targets at close range and high velocities.	

Solution for Question 11:

Correct Answer: B) Ricochet bullet

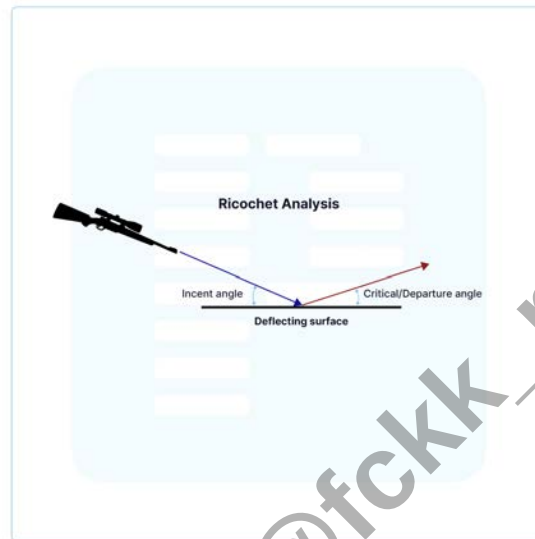
Explanation:

The scenario presented, in which a bullet deflects off a surface before striking the target, is characterised as a Ricochet Bullet .

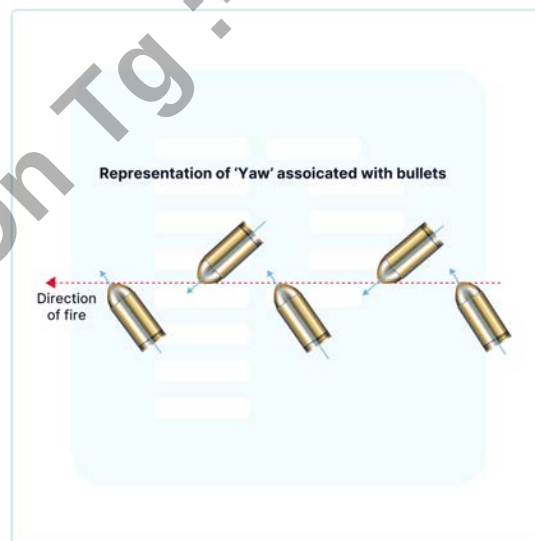
Ricochet Bullet:

- A ricochet bullet is a projectile that rebounds or deflects from its original path after striking an intermediate surface.
- The critical angle for ricochet is approximately 30 degrees .
- Wound Characteristics: No Abrasion Collar : Due to the loss of spin and rotation. Absence of Burning, Blackening, and Tattooing Atypical Wound Patterns : Resulting from the bullet's secondary movements after ricocheting.
- No Abrasion Collar : Due to the loss of spin and rotation.
- Absence of Burning, Blackening, and Tattooing
- Atypical Wound Patterns : Resulting from the bullet's secondary movements after ricocheting.
- Common Bullet Types Associated with Ricochets Round-nose bullets Full metal jacketed bullets Low-velocity bullets
- Round-nose bullets
- Full metal jacketed bullets
- Low-velocity bullets
- No Abrasion Collar : Due to the loss of spin and rotation.
- Absence of Burning, Blackening, and Tattooing
- Atypical Wound Patterns : Resulting from the bullet's secondary movements after ricocheting.

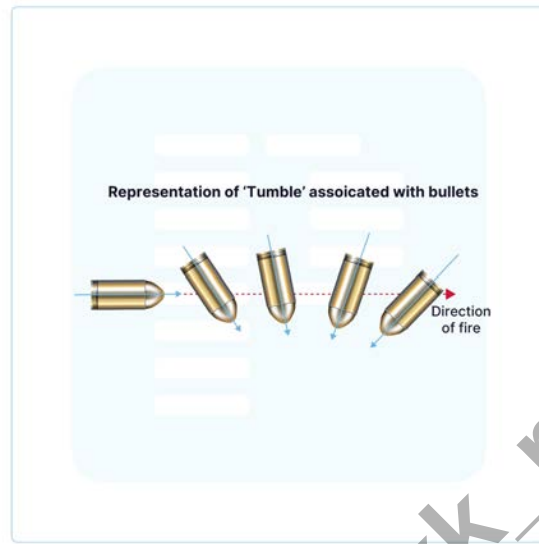
- Round-nose bullets
- Full metal jacketed bullets
- Low-velocity bullets



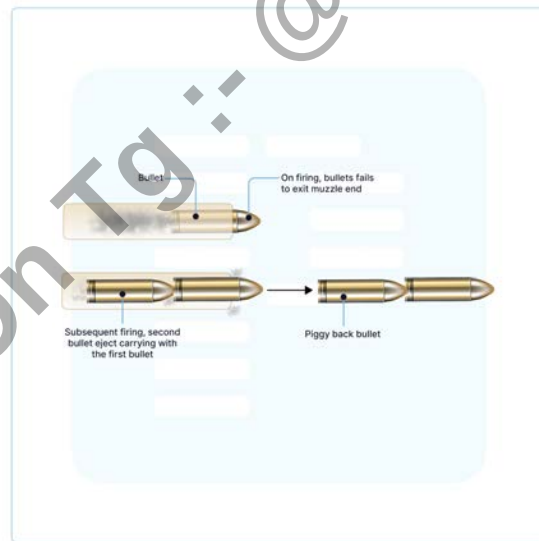
Yawing bullet (Option A) travels irregularly instead of travelling nose-on.



Tumbling bullet (Option C) rotates end to end during its motion.



Tandem bullet (Option D) refers to the simultaneous exit of two bullets (due to previously stuck bullet), also known as a piggyback bullet .



Solution for Question 12:

Correct Answer: B) 15 cm

Explanation:

The flame effect occurs at a distance of 15 cm with a shotgun.

Range of Effects of the Firearms:

Effects	Rifle	Shotgun
Flame (burning)	7 cm (Option A)	15 cm (Option B)
Smoke or soot (blackening)	30 cm (Option C)	45 cm (Option D)
Gunpowder (tattooing)	60-90 cm	60-90 cm



Solution for Question 13:

Correct Answer: D) SEM-EDXA'

Explanation:

The most specific Gunshot Residue (GSR) Test is SEM-EDXA (Scanning Electron Microscopy with Energy Dispersive X-ray Analysis).

Gunshot Residue (GSR) Tests :

Test	Description
------	-------------

Harrison and Gilroy test (Option A)	A qualitative calorimetric chemical test detects antimony, barium, and lead but is not specific to firearm discharge residues.
Atomic absorption Spectrometry (AAS) & Flameless atomic absorption spectrophotometry (FAAS) (Option B)	Uses high temperatures to vaporise metallic elements in primer residues for detection and quantification. Measuring antimony, barium, and lead from the primer, along with copper from the cartridge case or bullet jacketing.
Neutron activation analysis (Option C)	Identifies minute traces of elements in hair, nails, soil, glass, paint, gunshot residue (GSR), and drugs by detecting radioisotopes formed through irradiation in a nuclear reactor. Antimony and copper residues from the primer can be found on the suspect's hand.
SEM-EDXA (Scanning electron microscopy energy dispersive X-ray analysis) (Option D)	It is the most advanced tool for detecting minute traces of gunshot residue (GSR) on a suspect's body, providing qualitative rather than quantitative analysis. It is the most specific test.
Dermal Nitrate Test (Paraffin Test)	An obsolete and non-specific method for detecting gunshot residue (GSR) by identifying nitrates and nitrites on a suspect's hand

Solution for Question 14:

Correct Answer: B) Presence of abrasion collar around the entry wound

Explanation:

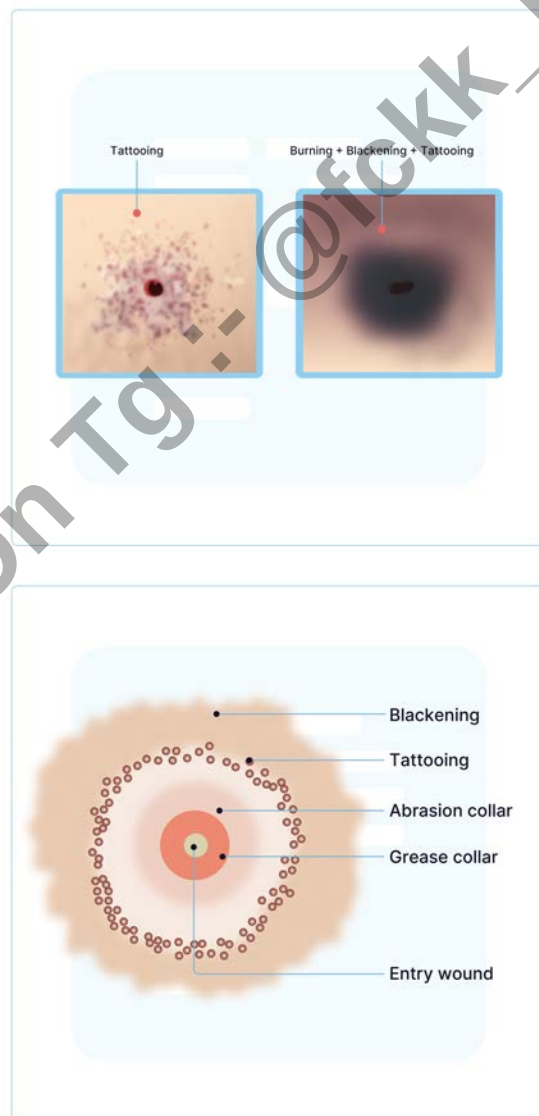
The entry wound of a distant shot from a rifled firearm will be circular in shape, featuring a grease collar and an abrasion collar, but will not exhibit blackening or tattooing.

Wound characteristics of a Rifled Firearm:

	Contact shot	Close shot or Flame Range (within 2–3 inches or 5–8 cm)	Near shot (Medium-Range or Intermediate Range, within 24 inches or 60 cm)	Distant shot (> 2 feet)
Muzzle impression	Present	Absent	Absent	Absent
Shape	Stellate or cruciate	Circular	Circular	Circular

Grease collar and Abrasion collar	Present	Present	Present	Present
Burning	Present	Present	Absent	Absent
Blackening	Present	Present	Absent	Absent
Tattooing	Present	Present	Present	Absent

- A grease collar is a dark ring around the entry wound caused by residues from the bullet, indicating skin contact.
- An abrasion collar is a ring of scraped skin around the entry wound, caused by the bullet's friction, revealing the angle of entry.



Presence of blackening and tattooing around the wound (Option A) : These are characteristics of close-range gunshot wounds, where gunpowder particles are deposited around the entry wound. In distant shots, there is typically no gunpowder residue, leading to a clean wound appearance without blackening or tattooing.

Clean, circular entry wound with everted edges (Option C) : Distant gunshot wounds usually present as clean, circular entry wounds with smooth edges, not everted edges.

Presence of multiple small pellet wounds (Option D) : This characteristic is indicative of shotgun injuries or wounds caused by projectiles that disperse multiple small pellets, not rifle wounds, which typically fire a single projectile.

Solution for Question 15:

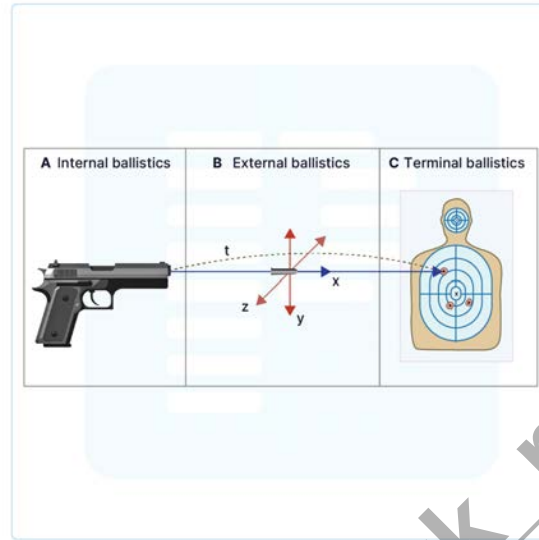
Correct Answer: D) Terminal Ballistics

Explanation:

Terminal Ballistics refers to the study of how a projectile behaves when it strikes and interacts with its target, including the damage inflicted upon tissues or materials.

Classification of Ballistics:

- Internal Ballistics (Proximal Ballistics): Study of the firearm and the projectile's behaviour from the moment the trigger is pulled until it exits the barrel.
- External Ballistics (Intermediate Ballistics): Analysis of a projectile's motion after it leaves the barrel and until it reaches the target.
- Terminal Ballistics: Study of the interaction between a projectile and its target upon impact.
- Wound Ballistics: Study of motion and effects of projectile in the tissue.



Solution for Question 16:

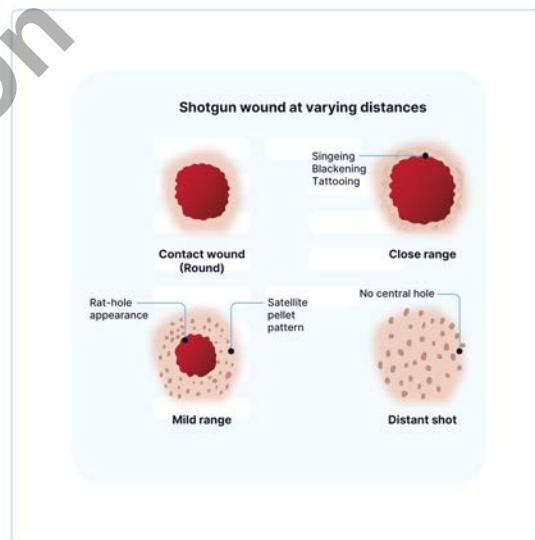
Correct Answer: D) Contact shot wound

Explanation:

- A contact shot wound shows a cruciate/ stellate entry wound, especially over thick bony areas like the skull, along with a muzzle impression and burning, blackening and tattooing in the surroundings.
- Blowback phenomenon: It occurs in contact gunshot wounds, particularly over thick, bony areas such as the skull. It results from the high energy and expanding gases generated when a firearm is discharged at close range. As the bullet penetrates the skin, the force of the gases creates a pressure wave that causes significant tissue displacement and injury.
- It occurs in contact gunshot wounds, particularly over thick, bony areas such as the skull.
- It results from the high energy and expanding gases generated when a firearm is discharged at close range.
- As the bullet penetrates the skin, the force of the gases creates a pressure wave that causes significant tissue displacement and injury.
- It occurs in contact gunshot wounds, particularly over thick, bony areas such as the skull.

- It results from the high energy and expanding gases generated when a firearm is discharged at close range.
- As the bullet penetrates the skin, the force of the gases creates a pressure wave that causes significant tissue displacement and injury.

Type of shotgun wounds				
Type of wound	Contact	Close (Option A)	Short/ Intermediate-range (Option C, D)	Distant
Entry wound	Cruciate/Stellate (Skull) or Circular	Circular	Rat-hole like	Multiple small pellet wounds with the mostly dispersed entry wound
Border of wound	Contused, Irregular and muzzled appearance	Clean-cut, irregular or lacerated	Abraded and scalloped borders	Irregular
Blackening	Present around the margins and even inside the wound.	Present	Absent	Absent
Burning	Absent			
Tattooing	Present			
Images				



Solution for Question 17:

Correct Answer: D) Surface irregularities from barrel imperfections

Explanation:

- Primary markings on bullets include the number, diameter, and width of lands and grooves (striations) and the degree of rifling twist .
- Surface irregularities serve as secondary markings .
- These markings allow forensic experts to connect bullets from a crime scene to a specific firearm, aiding in criminal investigations.

Primary Markings (Class characteristics)	Secondary Markings (Individual or accidental characteristics)
Determined by manufacturing specifications, design, and dimensions are most useful for identifying the make and model of the firearm involved. It consists of: Number, diameter and width of lands and grooves (Option A, B) Depth of grooves Direction and degree of rifling twist (Option C)	Surface irregularities (Option D) on the bullet result from imperfections in the barrel and the adhesion of bullet particles to the bore during firing (Metallic Fouling) .
It may be identical for bullets fired by two different weapons.	These are different for bullets fired by two different weapons, reflecting each firearm's distinct barrel characteristics.
It's more pronounced for jacketed bullets.	It is more pronounced when the grooves deeply mark the lead bullet.

The bullet found in the body, called the crime bullet , is compared under a Comparison Microscope with a test bullet fired from a suspected weapon (using the same brand and type of ammunition).

Solution for Question 18:

Correct Answer: C) Unburnt gunpowder

Explanation

The scenario presented involves a gunshot injury, and the highlighted structure in the image is tattooing, caused by unburnt gunpowder.

Discharge of the Gun Effects

Effects	Components causing the effect
Blackening (soot or smoke soiling or smudging)	Powder soot (carbon) deposition (Option A)
Tattooing or peppering or stippling	Unburnt or partially burnt gunpowder granules (Option C)
Burning or singeing	Flame (Option B)

Weapon end (Option D) creates a patterned abrasion or intradermal contusion caused by the muzzle end during close contact, with the shape depending on the type of firearm, ammunition, and the anatomical condition.



Solution for Question 19:

Correct Answer: D) Mercury fulminate

Explanation:

- The primer in a firearm is a small explosive device located at the base of the cartridge.

- When the trigger is pulled, the firing pin strikes the primer, igniting it; this produces a flame that ignites the gunpowder, propelling the bullet out of the cartridge due to increased pressure.
- In modern usage, lead azide and mercury fulminate are replaced by other components, such as lead peroxide, antimony, sulphide, and barium .

Composition of Primer (Pneumonic- BLAST)
B arium nitrate (Option C) L ead peroxide A ntimony sulphide (Option B) S typhnate T etrazine (Option A)

Solution for Question 20:

Correct Answer: D) Nitrocellulose

Explanation:

- Nitrocellulose or Gun cotton is the main component of single-base smokeless gunpowder.
- They produce less heat, smoke, and flame than traditional black gunpowder.
- They produce around 12000 to 13000 cc of gas.
- Smokeless gunpowder is of 3 types: Single base: Nitrocellulose Double base: Nitrocellulose + Nitroglycerine (Option B) Triple base: Nitrocellulose + Nitroglycerine + Nitroguanidine (Option C)
- Single base: Nitrocellulose
- Double base: Nitrocellulose + Nitroglycerine (Option B)
- Triple base: Nitrocellulose + Nitroglycerine + Nitroguanidine (Option C)
- They come in various G like flakes, discs, globules, squares, rectangles and cylinders.
- Single base: Nitrocellulose
- Double base: Nitrocellulose + Nitroglycerine (Option B)
- Triple base: Nitrocellulose + Nitroglycerine + Nitroguanidine (Option C)



Potassium nitrate (Option C), combined with sulfur and charcoal, forms the composition of Black Gunpowder.

Solution for Question 21:

Correct Answer: D) 2, 3, 5

Explanation:

- Most fireworks are made of black gunpowder, which mainly consists of Potassium nitrate (oxygen supplier or oxidizer) (75%), Charcoal (fuel) (15%), and Sulphur (increases density) (10%).
- This powder produces more heat, flame, and smoke while burning, and 1 gm of it produces up to 3000 cc of gas.
- Gunpowder is graded by the letter F, like F, FF, and FFF, where the number of Fs is directly proportional to the powder's fineness and burning speed.

These days, this is mainly replaced by smokeless powder and is only used in a few areas, such as fireworks, mining, and military use.



Types of Gunpowder:

Types of gunpowder	Black powder	Smokeless powder	Semi smokeless
Composition	Potassium nitrate Charcoal Sulphur	Consists of Nitrocellulose, Nitroglycerine, and Nitroguanidine	80% Black powder + 20% Smokeless powder
Gas produced per gram of powder	3000-4500 cc	12000-13000 cc	
Smoke produced	+++	+	
Power	+	+++	
Flame Temperature	Higher	Lower (due to Nitroguanidine)	
Burning of powder	Incomplete	Complete	
Wound produced	More blackening and tattooing	Less blackening and tattooing	

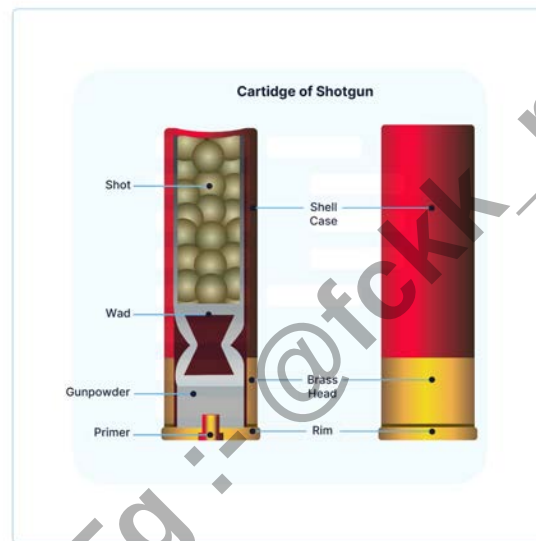
Solution for Question 22:

Correct Answer: C) Allows gas to escape from the breech end

Explanation:

The labeled structure in the image is the Wad , a part of the shotgun that prevents gas from escaping at the breech end.

- Positioned between the powder and the shot or placed on top of the shot, while the cardboard disc behind the shot charge stops the pellets from becoming trapped in the felt wad.



- Advantages of Wad Develops optimum pressure Seals the bore effectively Aids in lubrication Prevents gas from escaping at the breech end Separates propellant from the projectiles
- Develops optimum pressure
- Seals the bore effectively
- Aids in lubrication
- Prevents gas from escaping at the breech end
- Separates propellant from the projectiles
- Develops optimum pressure
- Seals the bore effectively
- Aids in lubrication
- Prevents gas from escaping at the breech end
- Separates propellant from the projectiles

Solution for Question 23:

Correct Answer: B) Wads are used in all types of cartridges

Explanation:

Wads are specifically used only in smooth-bore shotgun cartridges , not rifle cartridges.

Cartridge:

- A cartridge is a complete unit of ammunition consisting of the following components:
Cartridge case : Contains the percussion cap with primer. Propellant charge : Typically gunpowder. Projectile : This can be bullets or pellets.
- Cartridge case : Contains the percussion cap with primer.
- Propellant charge : Typically gunpowder.
- Projectile : This can be bullets or pellets.
- Cartridge case : Contains the percussion cap with primer.
- Propellant charge : Typically gunpowder.
- Projectile : This can be bullets or pellets.

Cartridge of Rifle	Cartridge of Shotgun
Contains a primer on the base and a cartridge case filled with gunpowder. Features a single projectile (bullet). Does not include a wad (Option B)	Contains lead shots (Option D) / Pellets (Multiple Projectiles) (Option A) Wads : Used only in smoothbore weapons (Shotguns). Advantages of Wads: Facilitates optimal pressure. Prevents gas escape from the bore. Aids in lubrication. Stops gas from escaping at the breech end. Maintains separation between the propellant and projectiles.

Bullet:

- A projectile that exits the muzzle of a rifled firearm upon discharge.
- Material : Typically made from soft metal, primarily lead (Option D) , often with added antimony and/or tin for increased hardness.
- Shape: Round nose soft bullet : Commonly used in revolvers and pistols. Rifle bullets : Generally elongated with a pointed tip. (Option C)
- Round nose soft bullet : Commonly used in revolvers and pistols.
- Rifle bullets : Generally elongated with a pointed tip. (Option C)
- Round nose soft bullet : Commonly used in revolvers and pistols.

- Rifle bullets : Generally elongated with a pointed tip. (Option C)
- Types: Lead bullets : Made entirely of lead, offering expansion upon impact.
Metal-Jacketed Bullets : A lead core encased in a metal jacket (usually copper) enhances accuracy and reduces barrel fouling. They also prevent deformation from dirt and impacts and reduce fragmentation to protect the firearm from damage.
- Lead bullets : Made entirely of lead, offering expansion upon impact.
- Metal-Jacketed Bullets : A lead core encased in a metal jacket (usually copper) enhances accuracy and reduces barrel fouling. They also prevent deformation from dirt and impacts and reduce fragmentation to protect the firearm from damage.
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- Metal-Jacketed Bullets : A lead core encased in a metal jacket (usually copper) enhances accuracy and reduces barrel fouling. They also prevent deformation from dirt and impacts and reduce fragmentation to protect the firearm from damage.

Solution for Question 24:

Correct Answer: C) Limits spread of pellets

Explanation:

Limiting the spread of pellets is a characteristic more associated with smooth-bore shotguns (not rifled firearms), achieved through a process known as Choking.

Choking	Rifling (Rifled Firearms)
Definition: Choking refers to the constriction at the muzzle end of a shotgun barrel. Types: Cylinder Bore: The barrel has a uniform diameter throughout. Choke Bore: The muzzle is slightly constricted. Degrees of Choke : Full, modified, improved cylinder, and cylinder. Advantages of Choking: Reduces shot spread. Increases explosive force. Enhances velocity and range. Dispersion: A fully choked barrel produces shot dispersion that is half that of a cylinder bore.	As the bullet travels through the bore, it engages with the rifling, imparting a spin that enhances accuracy and stability. Each weapon leaves a unique marking on the bullet, aiding forensic identification. Advantages of Rifling: Gyroscopic Stability Increased Accuracy (Option A) : Increased Range (stability and spin) (Option B) Reduced Wobbling (Option D) (tumbling end-over-end during flight) Greater Penetration Power



Solution for Question 25:

Correct Answer: A) Distance between two opposite lands

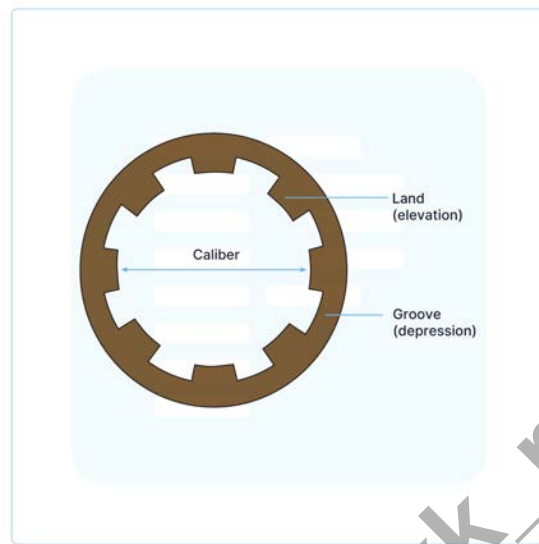
Explanation:

In forensic ballistics, "caliber" refers to the diameter of a rifled firearm's barrel, specifically measured between opposing lands. This measurement, typically expressed in inches or millimetres, is crucial for linking a bullet to a specific firearm.

Rifled Firearms:

- Rifled firearms have a bore characterised by a series of grooves that spiral along the barrel's length. These grooves (Option B), known as "rifling," alternate with raised areas called "lands." The caliber is determined by measuring the distance between these lands.
- The direction of rifling can be either right (clockwise) or left (counter-clockwise), with most handguns featuring a right-hand twist.

As the bullet travels through the barrel, it engages with the lands and grooves, imparting a spin that stabilises its flight and leaves distinctive markings on the bullet, which serve as identifiers for the firearm.



Muzzle constriction controlling shot spread (Option C) : This refers to the Choke , which narrows the barrel's end to control the dispersion of shot pellets as they exit, influencing the effectiveness of the shotgun at different ranges.

Measurement of shotgun bore diameter (Option D) : This describes the Gauge , indicating the size of the shotgun bore, which is determined by the number of lead balls corresponding to the diameter of shotgun bore that can be made from one pound of lead.



Forensic Identification

1. A museum curator is organizing an exhibit on the history of forensic science. They come across a set of tools and records from the late 19th century that include calipers, measuring tapes, and detailed cards with various bodily measurements and descriptions. The curator notices that these records consistently include 11 specific measurements, along with descriptions of physical features and photographs. What is the MOST likely purpose of this historical forensic method?

- A. To determine the age and cause of death in archaeological remains
- B. To create a standardized system for criminal identification
- C. To classify individuals based on racial characteristics for eugenics research
- D. To establish an observational method for normal human growth and development

2. A 6-year-old child is brought to court as a witness in a criminal case. Which of the following statements best describes the legal considerations regarding this child's testimony?

- A. The child cannot give evidence due to being under 7 years old.
- B. The child's testimony is automatically considered truthful.
- C. The child can give evidence if the court is satisfied with their understanding.
- D. The child's testimony will be undertaken by the guardian.

3. Which of the following criteria is included in Gustafson's method but excluded from Dalitz's method of age estimation from teeth?

- A. Root transparency
- B. Attrition

- C. Root resorption
 - D. Periodontal recession
-

4. Which of the following statements BEST describes the value and application of frontal sinus prints?

- A. Frontal sinus prints are useful for identification, but they change significantly after infections or injuries to the skull.
 - B. Frontal sinus prints are unique to each individual and remain stable after age 15, making them valuable for postmortem identification.
 - C. Frontal sinus prints are only useful for identifying individuals under 15 years of age, as they continue to alter in adulthood.
 - D. Frontal sinus prints provide a general indication of a person's age and sex, but are not unique enough for individual identification.
-

5. A forensic pathologist is examining a body found in a river. The deceased has multiple tattoos, including a faded amateur tattoo on the arm and a professional tattoo on the chest. The pathologist also notices a partially erased tattoo on the back. Which of the following statements is MOST likely to be true based on the information provided about tattoos?

- A. The faded amateur tattoo on the arm was likely created using red dye and is about 3 weeks old.
 - B. The professional tattoo on the chest probably used black or blue dye and could be identified in lymph nodes.
 - C. The partially erased tattoo on the back was most likely removed using chemical peeling with trichloroacetic acid.
 - D. All tattoos on the body can be accurately dated to determine the exact time they were created.
-

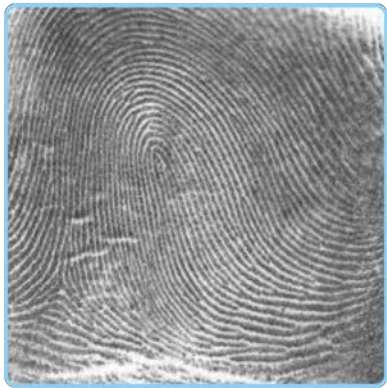
6. A forensic investigator is examining evidence from a crime scene in a restaurant. The following items have been collected: A wine glass with a partial fingerprint A napkin with a lipstick mark A fork with what appears to be saliva residue The investigator has a suspect in custody but lacks sufficient ridge detail from the partial fingerprint for a conclusive match. Which of the following approaches would be MOST appropriate for the investigator to pursue next?

- A. Focus solely on analyzing the lipstick mark on the napkin using chelioscopy techniques
 - B. Employ poroscopy to examine the sweat pores on the partial fingerprint from the wine glass
 - C. Disregard the evidence collected and seek alternative means of identification
 - D. Attempt to extract DNA from the saliva on the fork and compare it to the suspect's DNA
-

7. A forensic expert is called to testify in a court case involving disputed fingerprint evidence. During the trial, various aspects of fingerprint analysis are discussed, including abnormalities and development techniques. Which of the following statements about fingerprint abnormalities is INCORRECT?

- A. Ridge atrophy with alteration of ridge pattern can be seen in conditions such as celiac disease, dermatitis, and prolonged exposure to certain drugs.
 - B. People with certain genetic disorders like Naegeli-Franceschetti-Jadassohn syndrome may be born without fingerprints.
 - C. Latent prints require additional processing to be rendered visible and are often made from grease, dirt, oil, or blood.
 - D. Development techniques such as VMD, powdering, SPR, and iodine fuming can be used to visualize altered fingerprints
-

8. Match the following dactylography patterns with their correct type and frequency (%)
1. Loop a. X. 30-35% 2. Whorl b. Y. 5-10% 3. Arch c. Z. 60-70%

	a.		X. 30-35%
	b.		Y. 5-10%
	c.		Z. 60-70%

- A. 1-a-X, 2-c-Z, 3-b-Y
 B. 1-c-Z, 2-a-X, 3-b-Y
 C. 1-a-Z, 2-c-Y, 3-b-X
 D. 1-c-Y, 2-a-X, 3-b-Z

9. Match the following ages with their corresponding medicolegal significance: 1. 14 years a) Age of majority and can cast vote 2. 18 years b) Minimum age for appointment as President or Vice-President of India 3. 21 years c) Cannot be employed in factory jobs 4. 25 years d) Age for contesting membership of Parliament 5. 35 years e) Age of marriage for males

1. 14 years	a) Age of majority and can cast vote
2. 18 years	b) Minimum age for appointment as President or Vice-President of India
3. 21 years	c) Cannot be employed in factory jobs
4. 25 years	d) Age for contesting membership of Parliament
5. 35 years	e) Age of marriage for males

- A. 1-a, 2-c, 3-e, 4-b, 5-d
- B. 1-c, 2-a, 3-e, 4-b, 5-d
- C. 1-c, 2-a, 3-e, 4-d, 5-b
- D. 1-c, 2-e, 3-a, 4-d, 5-b

10. A team of forensic anthropologists is excavating a site where multiple sets of human remains have been discovered. They need to take precise measurements of various bones to aid in identification and analysis. As they set up their equipment, a junior member of the team asks about the best tool for taking linear measurements of long bones like femurs and tibias. Which instrument should the senior anthropologist recommend for this task?

- A. Sliding caliper
- B. Spreading caliper
- C. Hepburn-type osteometric board
- D. Digital radiography system

11. A forensic anthropologist is tasked with estimating the stature of an unidentified male skeleton. Only the femur is available for measurement. Which of the following multiplication factors should be used to calculate the stature based on the femur length?

- A. 3.3
- B. 3.6-3.8
- C. 4.48
- D. 5.30

12. A forensic odontologist receives dental records with the following notations: "#8 crown" (Universal System) "46 missing" (FDI System) "64 carious" (Modified FDI System for primary teeth) Which of the following correctly represents all these findings using only the FDI system for permanent dentition?

- A. 11 crown, 46 missing, 24 carious
- B. 21 crown, 46 missing, 14 carious
- C. 11 crown, 46 missing, 14 carious
- D. 21 crown, 46 missing, 24 carious

13. A 32-year-old patient presents with dental pain in the upper right first molar. During charting, the dentist notes that the tooth is marked as "6." The dentist then refers to a dental notation system where each tooth is numbered according to quadrants and the location of permanent teeth. Which of the following notation methods is being used?

- A. Zsigmondy-Palmer notation

- B. Haderup notation
 - C. Universal numbering system
 - D. FDI notation
-

14. A partially decomposed body was discovered at a crime scene, and authorities suspected foul play. On examination of dead body, a bullet was retrieved from the body and weapon marks were present on the clothing. To establish that a crime has taken place in this suspected homicide, which of the following is the most crucial factor?

- A. The presence of motive and opportunity for the crime
 - B. Identification of the dead body and proof of death by criminal act
 - C. Eyewitness testimony linking a suspect to the crime
 - D. Detailed reconstruction of the crime scene and sequence of events
-

15. A forensic pathologist is asked to estimate the age of a deceased infant based on the weight of its teeth, which is found to be 530 mg. Which of the following methods is the most appropriate for age estimation in this case?

- A. Stack's method
 - B. Boyde's method
 - C. Lamendin's method
 - D. Gustafson's method
-

16. A forensic odontologist is examining the remains of an unidentified individual found in a remote area. The examination reveals the following dental pattern: 28 teeth present All third molars are absent Second premolars are partially erupted

Canines are fully erupted Some primary (deciduous) molars are still present Based on this dental evidence, what is the most likely age range of the individual?

- A. 6-7 years
- B. 9-10 years
- C. 11-12 years
- D. 14-15 years

17. A 10-year-old girl visits the clinic for a routine examination. During the assessment, the physician notes the development of her upper and lower limb bones. Which of the following statements is FALSE?

- A. The head of the humerus fuses with the shaft at 17–18 years.
- B. The lower end of the radius fuses by 17–19 years.
- C. The greater trochanter of the femur fuses at 14–15 years.
- D. The upper end of the fibula fuses at 10–12 years.

18. Match the following skull sutures with their age of closure 1. Posterior fontanelle a) 2–4 years 2. Anterior fontanelle b) 45–50 years 3. Lambdoid suture c) At birth to 6 months 4. Metopic suture d) 1½–2 years

1. Posterior fontanelle	a) 2–4 years
2. Anterior fontanelle	b) 45–50 years
3. Lambdoid suture	c) At birth to 6 months
4. Metopic suture	d) 1½–2 years

- A. 1-a, 2-c, 3-d, 4-b
- B. 1-c, 2-d, 3-b, 4-a
- C. 1-d, 2-b, 3-c, 3-a

D. 1-d, 2-a, 3-b, 4-c

19. A 12-year-old boy is brought to the clinic for a routine check-up. During the examination, the physician notes the development of certain bones, particularly focusing on the clavicle and scapula. Based on the provided information, which statement is TRUE regarding the boy's skeletal development?

- A. The medial end of the clavicle fuses by 22 years.
 - B. The coracoid base of the scapula fuses at 10 years.
 - C. The xiphisternum fuses at 3 years postnatal.
 - D. The first sternebrae appears between 14–25 years.
-

20. A pregnant woman has an ultrasound that measures her fetus's Crown Heel Length (CHL) as 40 cm. What is the estimated age of the fetus in months?

- A. 7 months
 - B. 8 months
 - C. 9 months
 - D. 10 months
-

21. A forensic anthropologist is analyzing skeletal remains to determine the sex of an individual. Based on the following rules, which statement is TRUE?

- A. According to Ashley's Rule, females have a sternum length greater than 149 mm.
- B. Hyrtl's Law states that males have a sternum body length more than twice the length of the manubrium.
- C. Krogman's Table indicates that long bones alone provide 100% accuracy in sexing remains.

D. The skull has a higher accuracy rate than the pelvis.

22. A forensic anthropologist is tasked with determining the biological sex of skeletal remains found at an excavation site. The skeleton is relatively well-preserved, and specific morphological features are noted. The pelvis shows a wide subpubic angle, a broad sciatic notch, and a wide, circular pelvic inlet. Additionally, the skull has a smaller mastoid process and rounded chin. Based on these skeletal characteristics, which of the following is the most likely biological sex of the individual?

- A. Male
 - B. Female
 - C. Cannot be determined without DNA analysis
 - D. Hermaphroditic
-

23. A 30-year-old individual with ambiguous genitalia presents for forensic sex determination. Both sex chromatin (Barr body) study and gonadal biopsy are performed. After a thorough examination, it was noted that no Barr bodies were identified, and the gonadal tissue revealed immature seminiferous tubules. Based on this information, which of the following is the most appropriate interpretation?

- A. The absence of Barr bodies indicates Turner syndrome.
 - B. The individual is chromosomally male but may have androgen insensitivity syndrome (AIS).
 - C. The presence of seminiferous tubules confirms chromosomal female sex.
 - D. The absence of Barr bodies automatically excludes mosaicism.
-

24. An unidentified male body is brought to the mortuary for forensic examination. Upon evaluation, it is noted that the orbits of the skull are rounded, the palate is broad, and the teeth exhibit shovel-shaped incisors. The hair recovered from the body is straight and coarse. Based on the forensic findings, what is the most likely racial group of this individual?

- A. African descent
 - B. Caucasian descent
 - C. Asian descent
 - D. Hispanic descent
-

25. A young male is brought to the emergency department unconscious after a car accident. Despite resuscitation efforts, he is declared dead. The medical examiner is called to conduct a post-mortem examination. The patient has no identification, and his body was found far from any populated area. During the autopsy, the cephalic index (CI) of the skull is calculated to be 83. Based on this information, which population ancestry does this person most likely belong to?

- A. Aboriginal
 - B. Central European
 - C. Mongolian
 - D. Caucasian
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	2
Question 5	2
Question 6	2
Question 7	4
Question 8	2
Question 9	3
Question 10	3
Question 11	2
Question 12	1
Question 13	1
Question 14	2
Question 15	1
Question 16	3
Question 17	4
Question 18	2

Question 19	1
Question 20	2
Question 21	2
Question 22	2
Question 23	2
Question 24	3
Question 25	3

Solution for Question 1:

Correct Answer: B) To create a standardized system for criminal identification

Explanation: The description in the question matches the Bertillon system of anthropometry, which was developed as a standardized system for criminal identification .

Bertillon system of anthropometry

Developer	Alphonse Bertillon
Primary Purpose	To create a standardized system for criminal identification
Principle	• Measurements of various parts of the human body that do not alter after adult age (21 years) • Observation and recording of descriptive data, photograph and body marks
Measurements Included	• 11 specific measurements including: Height, AP diameter of head and trunk Span of outstretched arms Length of middle finger, left little finger, left forearm and left foot Length and breadth of right ear • Descriptive data: Color of hair, eyes, complexion, shape of nose, ears and chin • Body marks: Moles, scars and tattoo marks • Photographs: Front view and profile view of head
Tools Used	Calipers, measuring tapes, photographic equipment
Historical Significance	First scientific method of criminal identification, paving the way for modern forensic techniques

While anthropometry can be used in archaeological contexts, the Bertillon system was specifically developed for criminal identification of living individuals. Although anthropometry was sometimes misused for racial classification, this was not the purpose of Bertillon's system, which was focused on individual identification for law enforcement. While the system does involve human measurements, its primary purpose was not to establish growth and development baselines, but rather to identify criminals.

Solution for Question 2:

Correct Answer: C) The child can give evidence if the court is satisfied with their understanding.

Explanation: According to Sec. 118 IEA , for children below 12 years of age, competency for giving evidence depends upon understanding, and not on age. A child of any age can give evidence if the court is satisfied that the child is truthful.

The medicolegal importance for ages below 12 years is listed in the table below:

Age	Medicolegal Importance
5th lunar month (12 weeks of pregnancy)	• Till this duration, the decision for termination of pregnancy can be taken by one doctor.
5th-6th lunar month (20-24 weeks of pregnancy)	• Maximum period of gestation for use of the indication of MTP Act. 2 doctors required to take the decision of termination of pregnancy. • Above this age, termination is only on therapeutic grounds.
7 months (28 weeks of pregnancy)	• Fetus is considered to be viable. • Fetus born after this period, if it does not show any sign of life, is termed stillborn fetus (medically).
10th lunar month	• Fetus at this stage is a full term fetus.
5 years	• Custody of a minor who is below 5 years is with the mother.
7 years	• Below this age, child is not responsible for his criminal act, as he does not understand the nature and consequences of his act.

7-12 years	• Criminal responsibility: A child may or may not be held responsible for his act by the court, depending upon whether the child has attained sufficient maturity to understand the nature and consequence of the act .
10 years	• If a child below this age is removed from his lawful guardian for the purpose of robbing movable property from his/her possession, it will amount to kidnapping .
12 years	• A child under 12 years cannot give valid consent to suffer any harm which may occur from any act done in good faith and for his benefit. • A child above this age can give valid consent for examination (including medico-legal) and treatment (operative procedures and operations).

Option A is incorrect because there is no specific age threshold mentioned for giving evidence.

Option B is incorrect because the court must be satisfied about the child's truthfulness; it's not automatically assumed.

Option D is incorrect because child's testimony cannot be undertaken by the guardian.

Solution for Question 3:

Correct Answer: C) Root resorption

Explanation:

Dalitz's method for age estimation from teeth simplified Gustafson's system and excluded two of the six criteria: root resorption and cementum apposition.

Gustafson's Method:

Developed by Gosta Gustafson, this method estimates age by evaluating six specific regressive changes in teeth:

Each change is scored from 0 to 3, and the total score (Y) is used in the formula for age estimation .

- 0—No change
- 1—Beginning of change
- 2—Obvious change

- 3—Maximum change

$A + P + S + C + R + T = \text{total score (Y)}$

An increase in age shows a linear increase in the total score. The age is estimated using the following formula:

$\text{Age} = 11.43 + 4.56 \times Y \text{ (total score)}$

- The method is most accurate for individuals between 25–60 years old.
- It requires extracting and microscopically examining a longitudinal section of the tooth, limiting its use to deceased individuals or skeletal remains.

Dalitz's Method:

Dalitz refined Gustafson's method to improve accuracy by discarding less reliable factors like root resorption and secondary cementum apposition. He focused on scoring the following changes in the 12 anterior teeth on a 5-point scoring system from 0 to 4.

While more accurate, the Dalitz method's main drawback is its limitation to anterior teeth, excluding bicuspid (premolars) and molars from analysis.

Solution for Question 4:

Correct Answer: B) Frontal sinus prints are unique to each individual and remain stable after age 15, making them valuable for postmortem identification.

Explanation: The key features of frontal sinus prints include:

- They are unique to each individual
- Remain stable after age 15
- Valuable for postmortem identification

Frontal sinus print

Uniqueness	Unique to each individual
Stability	Permanent and fixed after 15 years of age (Option C)
Resilience	Rarely alter following infection or injury (Option A)

Comparison Method	Antemortem X-ray of skull (occipito-mental plane) compared with postmortem X-rays
Application	Useful for postmortem identification, especially when other methods are unavailable or inconclusive (Option D)

Solution for Question 5:

Correct Answer: B) The professional tattoo on the chest probably used black or blue dye and could be identified in lymph nodes.

Explanation: The statement is correct as black and blue dyes are commonly used in tattoos and pigmentations of the regional lymph nodes could be identified during autopsy.

Tattoos in Identification

Age of tattoo	Firm union, reddish/bluish scar: 5-6 days Pale, soft and sensitive (tender): 2 weeks-2 months Tough, brownish, glistening, wrinkled and little tender: 2-6 months Tough, white, glistening and non-tender: > 6 months
Dyes Used	Common dyes: Indigo, cobalt, carbon, vermilion, cadmium, selenium, Prussian blue and Indian ink Black, blue and red dyes are most commonly employed Depth of penetration: 50-3000 times a minute Pigment can migrate to nearby lymph nodes
Erasure Methods	Surgical methods: Dermabrasion (e.g., tannic acid, silver nitrate, trichloroacetic acid) Complete excision and skin grafting Non-surgical methods: Production of burns by means of red hot iron Scarification Using carbon dioxide snow Electrolysis Caustic or corrosive substances Chronic eczema may cause tattoo designs to disappear

According to age determination of tattoo, color of tattoo does not have any relation to age, and 3 weeks would fall into the pale, soft and sensitive category, not "faded". (Option A)

While trichloroacetic acid is used as an erasure method, the information is not enough in determining which method was most likely used. (Option C)

The age determination of tattoos can only provide ranges, not exact times, and only for relatively new tattoos (up to 6 months). (Option D)

Solution for Question 6:

Correct Answer: B) Employ poroscopy to examine the sweat pores on the partial fingerprint from the wine glass

Explanation: The most appropriate next step for the investigator would be to employ poroscopy to examine the sweat pores on the partial fingerprint from the wine glass because:

- It can provide identification when available ridges do not provide sufficient characteristics.
- Poroscopy examines sweat pores on fingerprint ridges, which are permanent, immutable, and individual.
- This method directly builds upon the existing fingerprint evidence.

Poroscopy and Cheiloscopy

Poroscopy	• Study of pore structure on papillary ridges of fingers as a means of identification • Discovered and developed by Edmond Locard in 1912 • Ridge characteristics and pores are permanent, immutable and individual • Useful to establish identity when available ridges do not provide sufficient characteristics • Ridges contain 9-18 pores • About 550-950 sweat pores per square centimeter in finger ridges, less (400) in the palms and soles
Cheiloscopy (Lip Prints)	• Study of lip prints for identification purposes • Lip prints are revealed at points of direct physical contact at crime scenes • Useful for personal identification • Credibility not firmly established in forensic science

Cheiloscopy (Option A) is less suitable because:

- The use of lip prints in evidence is limited, as their credibility has not been firmly established in forensic science.
- While it could provide additional information, it's not as reliable as poroscopy for identification purposes.

Disregarding the evidence (Option C) would be inappropriate, as there are still viable methods of analysis available.

DNA analysis (Option D), while potentially conclusive, would typically be a more time-consuming and expensive process. It would be a good follow-up if poroscopy doesn't yield conclusive results, but it's not the next best step. However, DNA Analysis is the gold standard for

identification.

Solution for Question 7:

Correct Answer: D) Development techniques such as VMD, powdering, SPR, and iodine fuming can be used to visualize altered fingerprints

Explanation: The development techniques mentioned (VMD, powdering, SPR, and iodine fuming) are used for visualizing latent prints, not altered fingerprints.

Types of fingerprint abnormalities

Altered Fingerprints	• Ridge atrophy with alteration of ridge pattern seen in celiac disease, dermatitis, eczema, psoriasis, acanthosis nigricans, scleroderma, and dry and atrophic skin (Option A) • Prolonged exposure to certain drugs (e.g., capecitabine) may cause loss of fingerprints • Elderly persons may have fingerprints that are difficult to obtain due to flattening of ridges
Absence of Fingerprints	• People with certain genetic disorders like Naegeli-Franceschetti-Jadassohn syndrome, dermatopathia pigmentosa reticularis, and adermatoglyphia are born without fingerprints (Option B) • Fingerprints can be erased permanently by criminals through burns, acids, plastic surgery, etc.
Latent Prints	• Require additional processing to be rendered visible • Often made from grease, dirt, oil, or blood (Option C) • Can be detected by various methods including powdering, chemical reagents, and alternate light sources • Fluorescence examination can sometimes detect latent prints • Development techniques include VMD, powdering, SPR, and iodine fuming

Solution for Question 8:

Correct Answer: B) 1-c-Z, 2-a-X, 3-b-Y

Explanation: Key points about dactylography and fingerprints:

Permanent impairment of fingerprint patterns	Leprosy, electric injury, and after exposure to radiation.
Ridge atrophy with alteration of ridge pattern	Celiac disease, dermatitis, eczema, psoriasis, acanthosis nigricans, scleroderma, and dry and atrophic skin.
Loss of fingerprints	Anti-cancer drug- capecitabine
Elderly persons	Fingerprints that are difficult to capture, since the elasticity of skin decreases with age
No fingerprints	Genetic disorders like Baird syndrome (congenital milia), Zinsser-Cole-Engman syndrome (dyskeratosis congenita), Naegeli-Franceschetti-Jadassohn syndrome, dermatopathia pigmentosa reticularis, and adermatoglyphia
Pattern Type	Frequency (%)
Loop (Ulnar/radial)	60-70
Whorl	30-35
Arch	5-10
Composite	1-5

Solution for Question 9:

Correct Answer: C. 1-c, 2-a, 3-e, 4-d, 5-b

Explanation: The medicolegal importance of various ages of 14 and above are listed in the table below:

Age	Medicolegal Importance
14 years	• Employment: Cannot be employed in factory jobs • Child under the age of 14 can work in non-hazardous industries including some agricultural work as per Child Labour (Prohibition and Regulation) Act

14-15 years	• A person can be engaged in non-hazardous factory jobs for a limited period during the day
15 years	• Sexual intercourse even with wife below this age amounts to rape • A person above 15 years of age can be employed in a factory like an adult, if he has a fitness certificate from a doctor
16 years	• Taking away a male under this age without consent of guardian amounts to kidnapping
18 years	• Statutory rape: Intercourse with a girl below this age, irrespective of whether with or without her consent amounts to rape • Judicial punishment: Below this age, an offender is juvenile and is tried in juvenile court and if convicted, sent to reformatory school (no imprisonment or death sentence) • Age of majority except when the individual is under guardianship of the court • Age of marriage for females • Can cast vote
21 years	• Age of marriage for males • If a girl below this age is 'imported' to India from foreign country for the purpose of illicit intercourse, the act amounts to kidnapping (Sec. 366 B IPC) • Person under the guardianship of the Court of Wards attains majority
25 years	• Age for contesting membership of Parliament and other legislative bodies • Age limit for entering in some government services
35 years	• According to Punjab Excise Act, a person below this age cannot buy and consume liquor • Minimum age for appointment as President, Vice-President and Governor of State in India • Prenatal diagnostic tests/procedures cannot be conducted if the pregnant female is below this age as per the PCPNDT Act
55-65 years	• Age of retirement from services under the government, statutory bodies, autonomous bodies/institutions or from judiciary services

Solution for Question 10:

Correct Answer: C) Hepburn-type osteometric board

Explanation: The lengths of bones are measured using a Hepburn-type osteometric board .

- The Hepburn-type osteometric board is a specialized tool used in forensic anthropology and bioarchaeology to take accurate linear measurements of bones.
- It typically consists of a flat board with a fixed end block and a sliding block , allowing for precise measurements of bone length.



Sliding callipers (Option A) are used in forensic anthropology but for smaller, more precise measurements up to 300mm. They are ideal for bone diameters or cranial dimensions, not for measuring full lengths of long bones like femurs or tibias.

Spreading callipers (Option B) are used in forensic anthropology to measure curved surfaces or rounded structures' diameters. They are less precise for linear measurements of long bones compared to osteometric boards.

Digital radiography system (Option D): While useful in forensic anthropology for non-invasive examination, age estimation, and identifying pathologies, digital radiography is not the primary tool for taking linear measurements of bones.

Solution for Question 11:

Correct Answer: B) 3.6-3.8

Explanation: The multiplication factor for calculating stature from femur length is 3.6-3.8.

For eg. If the femur length is measured to be 45 cm, the height of the individual would be 162.0 cm - 171.0 cm ($45 \times 3.6 = 162.0$ cm, $45 \times 3.8 = 171.0$ cm).

Multiplication factor for stature

Bone	Multiplication factor
Femur (Best bone for stature)	3.6-3.8
Tibia/Fibula	4.5
Humerus	5.3
Ulna	6.1
Radius	6.5

Solution for Question 12:

Correct Answer: A) 11 crown, 46 missing, 24 carious

Explanation:

Universal system:

For Permanent Dentition: Each of the 32 permanent teeth is numbered from 1 to 32. The numbering starts from the upper right third molar (tooth 1), moves across to the upper left third molar (16), then continues to the lower left third molar (17), and ends at the lower right third molar (32).

Right Left

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17

For Primary Dentition: The 20 primary teeth are labeled using uppercase letters A to T.

A	B	C	D	E	F	G	H	I	J
T	S	R	Q	P	O	N	M	L	K

FDI Notation:

- Permanent Dentition: The FDI system uses a two-digit number for each tooth. The first digit indicates the quadrant, and the second digit indicates the tooth's position within that quadrant. Quadrants: 1: Upper right 2: Upper left 3: Lower left 4: Lower right Teeth positions are numbered from 1 to 8, starting from the midline (central incisors) to the molars. Example: Tooth 11 is the upper right central incisor and tooth 48 is the lower right third molar.
- Quadrants: 1: Upper right 2: Upper left 3: Lower left 4: Lower right
- 1: Upper right
- 2: Upper left
- 3: Lower left
- 4: Lower right
- Teeth positions are numbered from 1 to 8, starting from the midline (central incisors) to the molars.
- Example: Tooth 11 is the upper right central incisor and tooth 48 is the lower right third molar.
- Quadrants: 1: Upper right 2: Upper left 3: Lower left 4: Lower right
- 1: Upper right
- 2: Upper left
- 3: Lower left
- 4: Lower right
- Teeth positions are numbered from 1 to 8, starting from the midline (central incisors) to the molars.
- Example: Tooth 11 is the upper right central incisor and tooth 48 is the lower right third molar.
- 1: Upper right
- 2: Upper left
- 3: Lower left
- 4: Lower right

FDI System (Permanent Teeth):

Quadrant 1 (Right Upper)	Quadrant 2 (Left Upper)														
18	17	16	15	14	13	12	11	21	22	23	24	25	26	27	28

Quadrant 4 (Right Lower)	Quadrant 3 (Left Lower)														
48	47	46	45	44	43	42	41	31	32	33	34	35	36	37	38

- Primary Dentition: The same two-digit system is used, but the quadrants for primary teeth are numbered 5 to 8. Example: Tooth 51 is the upper right central incisor and tooth 85 is the lower right second molar.
- Example: Tooth 51 is the upper right central incisor and tooth 85 is the lower right second molar.
- Example: Tooth 51 is the upper right central incisor and tooth 85 is the lower right second molar.

Quadrant 5 (Right Upper)	Quadrant 6 (Left Upper)								
55	54	53	52	51	61	62	63	64	65
Quadrant 8 (Right Lower)	Quadrant 7 (Left Lower)								
85	84	83	82	81	71	72	73	74	75

Modified FDI Notation:

The Modified FDI System is a variation of the FDI Notation and involves assigning a two-digit number to each tooth based on its quadrant and position, and remains intact. However, in the Modified FDI System, the numbers in lower quadrants start with a new line.

This means the lower quadrant for the lower right jaw will be #3 (in FDI it is #4).

- Quadrants in Modified FDI: 1: Upper right 2: Upper left 3: Lower right 4: Lower left
- 1: Upper right
- 2: Upper left
- 3: Lower right
- 4: Lower left
- 1: Upper right
- 2: Upper left
- 3: Lower right
- 4: Lower left
- In the Universal System, #8 represents the upper right central incisor.

- In the FDI system, this translates to 11 (1 for upper right quadrant, 1 for central incisor).
- This is already in the FDI system, representing the lower right first molar.
- It remains 46 in the FDI system for permanent teeth.
- "64 carious" refers to the maxillary left first primary molar in the FDI system for primary teeth.
- Since we need to convert this into the FDI system for permanent dentition, this would correspond to the permanent maxillary left first premolar (FDI 24).

Solution for Question 13:

Correct Answer: A) Zsigmondy-Palmer notation

Explanation:

Zsigmondy Palmer Notation for the numbering of teeth is as given below:

Permanent Teeth:

Right Upper Quadrant	Left Upper Quadrant														
8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8
Right Lower Quadrant	Left Lower Quadrant														
8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8

Deciduous/Primary Teeth:

Right Upper Quadrant	Left Upper Quadrant								
E	D	C	B	A	A	B	C	D	E
Right Lower Quadrant	Left Lower Quadrant								

E	D	C	B	A	A	B	C	D	E
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Haderup System:

In this system, values for upper jaw teeth are indicated with a '+' sign (and a '-' sign for lower jaw teeth). These signs are placed to the left (for left-side teeth) or right (for right-side teeth), followed by numbers 1 through 8 for permanent teeth.

Right Upper Quadrant	Left Upper Quadrant															
8+	7+	6+	5+	4+	3+	2+	1+	+1	+2	+3	+4	+5	+6	+7	+8	
Right Lower Quadrant	Left Lower Quadrant															
8-	7-	6-	5-	4-	3-	2-	1-	-1	-2	-3	-4	-5	-6	-7	-8	
Right Upper Quadrant	Left Upper Quadrant															
5+	4+	3+		2+		1+		+1		+2		+3		+4		+5
Right Lower Quadrant	Left Lower Quadrant															
5-	4-	3-		2-		1-		-1		-2		-3		-4		-5

The Haderup notation (Option B) uses a plus (+) sign for maxillary teeth and a minus (-) sign for mandibular teeth, combined with the tooth number (e.g., +6 for the upper right first molar and -6 for the lower right first molar).

In the Universal system (Option C), teeth are numbered from 1 to 32, starting from the upper right third molar (1) to the lower right third molar (32). The notation "6" would refer to the upper right canine in this system, and the Universal system does not utilize quadrant-based division, making it different from the description in the question.

In the FDI notation (Option D), teeth are numbered by a two-digit code: the first digit indicates the quadrant (1 for upper right, 2 for upper left, 3 for lower left, and 4 for lower right), and the second digit indicates the tooth within the quadrant (1 to 8 for permanent teeth). The single "6" in the question does not conform to this system.

Solution for Question 14:

Correct Answer: B) Identification of the dead body and proof of death by criminal act

Explanation:

To establish that a crime has occurred in a suspected homicide, it is essential to demonstrate Corpus delicti.

Corpus delicti

- the body of offence - It must be demonstrated that a crime has indeed taken place before an individual can be convicted of committing it.
- In homicide, it involves,

1) positive identification of the dead body , and

2) proof that the death occurred due to a criminal act (such as gunshot wounds or other inflicted injuries) - due to violence that was inflicted in a specific manner, at a particular time and place, by the individual or individuals accused of the crime, and no one else.

In the scenario described, the discovery of a bullet in the body and weapon marks on the clothing are critical in proving that the death was not accidental, but the result of a criminal act.

The presence of motive and opportunity for the crime (Option A) : is relevant for determining the involvement of a suspect but does not establish that a crime has actually occurred. Proof of criminal activity, such as injuries caused by weapons, must be shown first.

Eyewitness testimony linking a suspect to the crime (Option C) : Eyewitness testimony may help identify a perpetrator, but it does not prove that a crime has taken place. The primary task is to establish the fact that the death was due to unlawful actions.

Detailed reconstruction of the crime scene and sequence of events (Option D) : helps in understanding the circumstances surrounding the death, but it is not the first step in proving that a crime occurred. The focus should be on identifying the body and confirming that the death resulted from a criminal act.

Solution for Question 15:

Correct Answer: A) Stack's method

Explanation:

Stack's Method:

Stack's method is designed to estimate the age of infants and fetuses by weighing their erupting teeth.

- This method is applicable from 5 months in utero to 7 months postnatal.
- In the scenario provided, the weight of the teeth is 530 mg, which corresponds to 2 weeks postnatal according to Stack's reference table.

Prenatal age	Weight of teeth (mg)
28 weeks	60 mg
40 weeks	460 mg
Postnatal age	Weight of teeth (mg)
2 weeks	530 mg
30 weeks	1840 mg

Given the teeth weight of 530 mg, Stack's method is the most appropriate for estimating the infant's age.

Boyde's Method:

- Estimates age by counting cross striations (incremental lines) in the enamel, starting from the neonatal line.
- These lines represent daily enamel growth.
- It is suitable for estimating the age of infants by calculating the number of days since the formation of the neonatal line but does not rely on tooth weight.

Since the scenario involves tooth weight, Boyde's method is not the correct approach here. (Option B ruled out)

Lamendin's Method:

- Lamendin's method is used for age estimation in adults.
- It relies on measuring the periodontosis height and root transparency, both as a percentage of root height, which is then applied to a regression formula.
- This method is not applicable to infants or fetuses. (Option C ruled out)

Gustafson's Method:

- Gustafson's method is used for estimating age in adults by observing six dental changes, such as: Attrition Paradentosis Secondary dentine. Cementum apposition Root resorption. Transparency of root. Total score = A+ P+ S+ C+ R+ T
- Attrition
- Paradentosis
- Secondary dentine.
- Cementum apposition
- Root resorption.
- Transparency of root.
- Total score = A+ P+ S+ C+ R+ T
- Like Lamendin's method, it is not appropriate for infants or fetal age determination. (Option D ruled out)
- Attrition
- Paradentosis
- Secondary dentine.
- Cementum apposition
- Root resorption.
- Transparency of root.
- Total score = A+ P+ S+ C+ R+ T

Solution for Question 16:

Correct Answer: C) 11-12 years

Explanation:

Dental examination is important in determining the age of an individual. The eruption sequence and timing of teeth provide valuable information for age estimation.

In the above question:

Temporary (Deciduous or Primary) Dentition:

- Temporary or deciduous teeth are the first set of teeth to develop in humans. These teeth begin to erupt in infancy and are eventually replaced by permanent teeth.

- Total Number: 20 teeth (10 in each jaw: 2 incisors, 1 canine, and 2 molars in each quadrant). By the age of 2 to 3 years, all deciduous teeth typically erupt.

Permanent Dentition:

- Permanent teeth replace the deciduous teeth and are present in an individual throughout adulthood.
- Total Number: 32 teeth (16 in each jaw: 2 incisors, 1 canine, 2 premolars, and 3 molars in each quadrant).

Teeth Eruption Sequence:

Teeth	Ages of Eruption (Temporary)	Ages of Eruption (Permanent)	Mixed Dentition Period
Lower central incisors	6–8 months	6–8 years	6–8 years
Upper central incisors	7–9 months	6–8 years	6–8 years
Upper lateral incisors	7–9 months	7–9 years	7–9 years
Lower lateral incisors	10–12 months	7–9 years	7–9 years
Canines	17–18 months	11–12 years	11–12 years
First premolars	—	9–11 years	9–11 years
Second premolars	—	10–12 years	10–12 years
First molars	12–14 months	6–7 years	6–7 years
Second molars	20–30 months	12–14 years	12–14 years
Third molars	—	17–25 years	No mixed dentition (Permanent only)

Permanent dentition order of appearance mnemonic:

Mom (M1) Is (I1) IN (I2) PAIN (PM1) PAPA(PM2) CAN(CANINES) MAKE (M2)
MEDICINE (M3)

Start from 6 yrs (M1) ...Then ADD 1 yr for next permanent teeth according to mnemonic till M2 which is at 12 years AND then M3 will be at 17-25 yrs

Mixed Dentition:

Mixed dentition is a transitional stage where both primary and permanent teeth are present at the same time. This phase typically occurs between the ages of 6 and 14 years.

- The child will have a combination of both deciduous and permanent teeth. The first permanent molars erupt around 6 years, while the shedding of deciduous teeth continues until about 14 years.

6-7 years(Option A) : At this age, only the first permanent molars and some incisors would have erupted. Canines would not be fully erupted.

9-10 years(Option B) : While canines might start erupting, second premolars would not typically be visible yet.

14-15 years(Option D) : By this age, all permanent teeth except third molars should be fully erupted, and primary molars would typically be absent.

Solution for Question 17:

Correct Answer: D) The upper end of the fibula fuses at 10–12 years.

Explanation:

The upper end of the fibula fuses by 17–18 years, not 10–12 years.

Upper Limb	Centers of Ossification	Age of Appearance	Age of Fusion
Humerus	Head	1 year	At 5–6 years, the three fuse together (conjoint epiphysis), and at 17–18 years, fuse with the shaft. (Option A ruled out)
Greater tubercle	3 years		
Lesser tubercle	5 years		
Capitulum	1 year	At 14–15 years, all three fuses with the shaft	
Trochlea	9–10 years		
Lateral epicondyle	10–11 years		
Medial epicondyle	5–6 years	16 years	
Radius	Upper end	5–6 years	15–17 years

Lower end	1–2 years	17–19 years (Option B ruled out)	
Ulna	Upper end	8–9 years	15–17 years
Lower end	5–6 years	17–19 years	
Carpals	Pisiform	9–12 years	—
Other Carpals	The capitate and hamate bones ossify during infancy, with the capitate appearing first at 36th week(3-4 months age). Between ages 2 and 6, the number of visible carpal bones on an X-ray roughly corresponds to the child's age (e.g., 3 carpal bones indicate approximately 3 years old).		
Lower limb bone	Centers of Ossification	Age of Appearance	Age of Fusion
Hip Bone	Ischiopubic rami	—	7 years
Triradiate cartilage	—	12–14 years	
Iliac crest	15–16 years	19–21 years	
Ischial tuberosity	16–17 years	20–22 years	
Femur	Head	1 year	17–18 years
Greater trochanter	4 years	14–15 years (Option C ruled out)	
Lesser trochanter	14 years	16–17 years	
Lower end	9 months IUL (at birth)	17–18 years	
Tibia	Upper end	At birth	17–18 years
Lower end	1 year	17–18 years	
Fibula	Upper end	4 years	17–18 years
Lower end	2 years	16–17 years	
Tarsals	Calcaneum	5th month IUL	17–18 years

	Talus	7th month IUL	
	Cuboid	9th month IUL	

Solution for Question 18:

Correct Answer: B) 1-c, 2-d, 3-b, 4-a

Explanation:

Skull sutures with their age of closure:

Suture Closure	Age
Posterior fontanelle (occipital)	At birth to 6 months
Anterior fontanelle (bregma)	1½–2 years
Two halves of mandible	1–2 years
Metopic suture (between frontal bones)	2–4 years, may remain unfused
Basiocciput and basisphenoid	18–20 years (females) 20–22 years (males)
Lambdoid suture	45–50 years
Parieto-temporal	60–70 years

- Closure of skull sutures begins endocranially (inner side) 5–10 years earlier than ectocranially (outer side).
- Ectocranial closure is variable and may not occur, especially in the sagittal suture.
- The best age estimate comes from the sagittal suture, followed by the lambdoid and coronal sutures.
- Inner closure starts around 25 years of age.
- Outer closure timelines: Posterior one-third of the sagittal suture: 30–40 years Anterior one-third of the sagittal and lower half of the coronal: 40–50 years Middle of the sagittal and upper half of the coronal: 50–60 years
- Posterior one-third of the sagittal suture: 30–40 years
- Anterior one-third of the sagittal and lower half of the coronal: 40–50 years

- Middle of the sagittal and upper half of the coronal: 50–60 years
- Posterior one-third of the sagittal suture: 30–40 years
- Anterior one-third of the sagittal and lower half of the coronal: 40–50 years
- Middle of the sagittal and upper half of the coronal: 50–60 years

Solution for Question 19:

Correct Answer: A) The medial end of the clavicle fuses by 22 years.

Explanation:

The medial end of the clavicle fuses by 20–22 years.

Bone	Centers of Ossification	Age of Appearance	Age of Fusion
Sternum	Manubrium	5th-month IUL	60–70 years (manubrio-sternal joint)
1st sternebrae	5th-month IUL (Option D ruled out)	14–25 years (from below upwards; 3rd and 4th–15 years; 2nd and 3rd–20 years; 1st and 2nd–25 years)	
2nd and 3rd sternebrae	7th-month IUL		
4th sternebrae	10th-month IUL		
Xiphisternum	3rd year postnatal	40–45 years (with body) (Option C ruled out)	
Clavicle	Medial end	5th–7th week IUL (1st Ossification centre overall to appear in fetus)	20-22 years
Scapula	Coracoid base	10–11 years	14–17 years (Option B ruled out)
Acromion process	14–16 years	17–18 years	

Solution for Question 20:

Correct Answer: B) 8 months

Explanation:

Rule of Hasse:

- Rough method of calculating the age of fetus
- The length of fetus is measured from crown to heel in centimeters
- During first 5 months of pregnancy- square root of crown to heel length in centimeters gives approximate age of fetus in months
- During the last 5 months- crown to heel length in centimeters 5 gives age in months

Options A, C & D are numerically incorrect.

Solution for Question 21:

Correct Answer: B) Hyrtl's Law states that males have a body length more than twice the length of the manubrium.

Explanation:

Hyrtl's Law states that males have a sternum body length more than twice the length of the manubrium.

Ashley's Rule

- Used for Sternum bone.
- Sternum length in males >149mm.
- Sternum length in females <149mm. (Option A ruled out)

Hyrtl's Law

- Body is longer and more than twice the length of Manubrium in males.
- Shorter and less than twice the length of manubrium in females.

Krogman's Table The degree of accuracy in sexing adult skeletal remains.	
--	--

Entire skeleton	100%
Pelvis alone	95%
Skull alone	90% (Option D ruled out)
Pelvis plus skull	98%
Long bones alone	80% (Option C ruled out)

Solution for Question 22:

Correct Answer: B) Female

Explanation: The skeletal features described, such as the wide subpubic angle, broad sciatic notch, wide circular pelvic inlet, smaller mastoid process, and rounded chin, are characteristic of a female skeleton.

- These differences are primarily due to the role of the female pelvis in childbirth, requiring a wider structure for accommodating fetal passage.
- Other sexually dimorphic traits, such as a smaller mastoid process and a more rounded chin, are also commonly associated with female skeletons.

Distinguishing features of male and female skull:

Feature	Male skull	Female skull
General Appearance	Larger, heavier, rugged, marked muscular ridges	Smaller, lighter, walls thinner, smoother
Forehead	Receding, irregular, rough, less rounded	Vertical, round, full, infantile, smooth
Cranial capacity	More capacious (1450–1550 cc)	Less capacious (1300–1350 cc)
Glabella	Prominent	Less prominent
Orbits	Square, rounded margins, small	Rounded, sharp margins, large
Frontal and parietal eminence	Less prominent	Prominent
Zygomatic arch	Prominent	Not prominent

Occipital area	Prominent	Not prominent
Mastoid process	Large, round, blunt	Small, smooth, pointed
Digastric groove	Deep	Shallow
Palate	Large, U-shaped, broad	Small, parabolic
Foramen magnum	Relatively large, long	Small, round
External auditory meatus	Bony ridge along upper border prominent	Often absent

Distinguishing features of male and female pelvis:

Feature	Male Pelvis	Female Pelvis
General Appearance	Massive, rougher, prominent muscular markings	Less massive, slender, smoother, muscular markings not prominent
Shape	Deep funnel	Flat bowl
Acetabulum	Large, 52 mm diameter	Small, 46 mm diameter
Obturator foramen	Large, oval, base upwards	Small, triangular, apex forwards
Greater sciatic notch	Narrow, deep, small	Broad, shallow, large
Ischial tuberosity	Inverted	Everted
Body of pubis	Narrow, triangular	Broad, square, pits on the posterior surface, if borne children
Subpubic angle	V-shaped, sharp angle, 70°–75°	U-shaped, rounded, broader angle, 90°–100°
Pelvic brim or inlet	Heart-shaped	Circular or elliptical-shaped
Pelvic cavity	Conical, funnel-shaped	Broad, round
Pelvic outlet	Smaller	Larger

Male skeletons (Option A) generally exhibit a narrower subpubic angle (less than 70 degrees), a narrow sciatic notch, and a heart-shaped pelvic inlet. Additionally, male skulls tend to have more prominent mastoid processes, a square chin, and more pronounced brow ridges.

While DNA analysis (Option C) is an advanced and definitive method for determining biological sex, in this case, the pelvic and skull features are distinctive enough to strongly suggest that the individual was female.

Hermaphroditism (Option D) , or intersex conditions, presents with skeletal features that are ambiguous or a blend of both male and female characteristics. Intersex conditions are diagnosed with more advanced techniques (e.g., genetic analysis or histology), and the skeletal features described in this scenario fit within the normal range for a biologically female individual.

Solution for Question 23:

Correct Answer: B) The individual is chromosomally male but may have androgen insensitivity syndrome (AIS).

Explanation:

Sex Chromatin (Barr Body) :

- Barr bodies represent the inactivated X chromosome in individuals with two X chromosomes (typically females, 46, XX).
- In males (46, XY), there is only one X chromosome, so Barr bodies are absent.
- The presence or absence of Barr bodies can help distinguish chromosomal sex, though it cannot detect mosaicism or some intersex conditions.

Gonadal Biopsy :

- A gonadal biopsy allows for histological examination of the gonads.
- In this case, the presence of seminiferous tubules suggests the presence of testicular tissue, which is typical of male gonads. However, disorders such as AIS can result in ambiguous genitalia despite the presence of testicular structures.

Androgen Insensitivity Syndrome (AIS) :

- In AIS, individuals with a 46, XY karyotype have testes and produce androgens, but their bodies are unable to respond to these hormones.
- This results in underdeveloped or ambiguous male genitalia or the appearance of female external genitalia despite having male gonads and a male chromosome pattern.

The absence of Barr bodies and the finding of seminiferous tubules support the interpretation of the gonadal male sex. However, the ambiguous genitalia suggests a disorder of sexual development (DSD), and AIS is a possible explanation.

The absence of Barr bodies indicates Turner syndrome (Option A):

- Turner syndrome (45, X) is characterized by a female phenotype with a single X chromosome.
- Individuals with Turner syndrome also lack Barr bodies, but they do not present with seminiferous tubules, which are structures specific to male gonads.
- Additionally, Turner syndrome typically presents with features like short stature, webbed neck, and primary amenorrhea, which were not described in this case.

The presence of seminiferous tubules confirms chromosomal female sex (Option C):

- The presence of seminiferous tubules indicates gonadal male sex, even if external genitalia or secondary sexual characteristics are ambiguous.
- Chromosomal female sex (46, XX) would not be associated with seminiferous tubules but with ovarian tissue.

The absence of Barr bodies automatically excludes mosaicism (Option D):

- Mosaicism refers to the presence of two or more distinct cell lines in an individual.
- It is possible for a mosaic individual to have a cell line that lacks Barr bodies (e.g., 46, XY) and another cell line that does (e.g., 46, XX).
- Therefore, the absence of Barr bodies in the examined sample does not automatically rule out mosaicism, and further genetic testing would be required to explore this possibility.

Solution for Question 24:

Correct Answer: C) Asian descent

Explanation:

The observed features of rounded orbits, broad palate, and shovel-shaped incisors point towards Mongoloids - East Asians.

Feature	Caucasian (White)	Mongolian (Yellow)	Negro (Black)
Skull	Rounded (Mesaticephalic)	Square (Brachycephalic)	Narrow and elongated (Dolicocephalic)
Face	Straight lower face - orthognathism	Large and flattened, malar bones prominent	Jaw projecting - prognathism, malar bones prominent

Hair			
Cross section	Round to oval	Cylindrical/ Round	Elliptical
Medulla	Fragmented	Non-fragmented	Fragmented
Cuticle	Medium	Thick	Absent
Palate	Triangular	Rounded	Rectangular
Teeth	Chisel-shaped Incisors Carabelli's Cusp -accessory cusp in maxillary first molar. Total of 5 cusps. Smaller Teeth Narrower Dental Arches	Shovel-shaped Incisors Peg-shaped Teeth Taurodontism-body of the tooth is enlarged at the expense of the roots Larger Teeth	Hypocone- a cusp on the upper molars Larger Teeth Less Frequent Shovel-shap ed Incisors
Limbs	Proportional to body	Smaller	Longer



Individuals of African descent (Option A) typically have rectangular orbits, a broader nasal aperture, and tightly coiled hair. Shovel-shaped incisors and straight hair are not characteristic features of African populations.

Caucasian individuals (Option B) usually have oval orbits, a narrower V-shaped palate, and wavy or straight hair that tends to be finer in texture. Shovel-shaped incisors are not a common finding in this group.

Hispanic individuals (Option D) can present with a combination of features from both Caucasian and Native American ancestries. While some may exhibit broad palates or rounded orbits, the combination of shovel-shaped incisors and coarse, straight hair is more characteristic of Asian descent than Hispanic populations.

Solution for Question 25:

Correct Answer: C)Mongolian

Explanation:

Anthropometric measurements, such as the cephalic index (CI) , are valuable in determining ancestry, which assists forensic teams in narrowing down the possible identity of the deceased.

- The cephalic index is calculated as:

$$CI = (\text{Maximum Skull Breadth} / \text{Maximum Skull Length}) * 100$$

- The skull breadth is measured between the two parietal eminences and the length is measured between the glabella and the external occipital protuberances with a calipers and not a measuring tape.

Skull Shape	Cephalic Index (CI)	Population
Dolichocephalic (Long-headed)	70-74.99	Pure Aryans, Aborigines, African (Negroid) (Option A)
Mesaticephalic (Medium-headed)	75-79.99	European (Caucasoid), Chinese, Indians. (Option D)
Brachycephalic (Short-headed)	80-84.99	East Asian (Mongoloid) (Option C)

- Hyperbrachycephalic (CI > 85): Extremely broad skulls, sometimes seen in Central European populations. (Option B)

In this scenario, a cephalic index of 83 falls into the brachycephalic range , which is typical of East Asian (Mongoloid) populations. This measurement, combined with other forensic evidence such as stature, skin color, and dental patterns, could guide the investigation toward identifying the deceased.

General Toxicology

1. A 30-year-old military officer suffered a burn to his back from the mishandling of hand grenades. He presents to the emergency department with severe pain, blistering, and a garlicky odor coming from the burn site as shown below. which of the following is inconsistent with it?



- A. Phosphorescent vomit
- B. Kidney failure
- C. Hypertrophy of the liver
- D. Full-thickness burn

2. A police officer is accused of sexually assaulting a child under the age of 12. The case is prosecuted under the POCSO Act. According to POCSO 2019, what is the potential punishment for the officer if convicted?

- A. 20 years to the death penalty

- B. 10 years to life imprisonment
 - C. 5 years to 10 years imprisonment
 - D. Life imprisonment without parole
-

3. When a patient is already dead upon arrival at the hospital, which of the following actions is incorrect?

- A. Only Death Report Form 2 is sent
 - B. A post-mortem is necessary in this case
 - C. The doctor informs the police
 - D. The doctor fills out MCCD Form 4A
-

4. Under the Registration of Birth and Death Act, 10(3), who is obligated to certify the cause of death when a patient passes away after being attended by a medical practitioner?

- A. Any licensed healthcare provider who saw the patient
 - B. RMP who attended the patient's last illness
 - C. A medical officer appointed by the government
 - D. A registered nurse who provided care
-

5. A 70-year-old man with poorly controlled diabetes mellitus develops a foot ulcer that becomes infected. Despite treatment, the infection worsens, leading to gangrene of the affected foot. He is later admitted to the hospital with septicemia and unfortunately succumbs to the illness. What is the antecedent cause of death in this scenario?

- A. Diabetes mellitus

- B. Gangrene
 - C. Septicemia
 - D. Foot ulcer
-

6. Which of the following statements regarding the Human Organ Transplantation Act (HOTA) is incorrect?

- A. Authorized by any donor above 18 years of sound mind
 - B. The most common donated organ is kidney
 - C. The most common donated tissue is skin
 - D. Commercial dealing in organs can result in imprisonment for up to 5 years.
-

7. In the case of children in conflict with the law, the Juvenile Justice Board consists of three members. Which of the following statements about the Board's composition is correct?

- A. All members must be magistrates
 - B. The Board must include a female social worker
 - C. The Board must include a judge from the Supreme Court
 - D. Only two members are necessary to form the Board
-

8. A sewer worker is brought to the emergency department after collapsing inside a sewage tank. He has respiratory failure, mydriasis, confusion, and a characteristic rotten egg odor. Which of the following is not useful in treating this patient?

- A. Sodium thiosulfate
- B. Sodium nitrite

- C. Amyl nitrite
 - D. High-flow oxygen
-

9. A 38-year-old man is brought to the emergency department after collapsing in a closed warehouse where he was working with industrial chemicals. Upon arrival, the patient is unconscious with a blood pressure of 80/40 mmHg, rapid breathing, and oxygen saturation of 95% on room air. His skin is noted to have a flushed, "brick-red" discoloration. ABG analysis shows severe metabolic acidosis with a pH of 7.05 and elevated lactate levels. The patient's coworkers reported a strong odor of burning plastic in the warehouse at the time of the incident. Lee-Jones test confirms the diagnosis. The medical team prepares to administer the antidote. Which of the following treatments is incorrect for this patient?

- A. Inhalation of Amyl nitrite
 - B. Intravenous Hydroxocobalamin
 - C. Hyperbaric oxygen
 - D. Intravenous Methylene blue
-

10. A 50-year-old woman is found dead in her apartment after a fire broke out in her house building. At autopsy, her skin and mucosal membranes are noted to have a cherry red discoloration, and there are cutaneous bullae on her wrists and calves. What is the most likely cause of these findings?

- A. Cyanide poisoning
 - B. Thermal burns from the fire
 - C. Carbon monoxide poisoning
 - D. Severe dehydration
-

11. A 9-year-old child is found dead after being intentionally neglected by caregivers, showing signs of severe malnutrition and wasting. Autopsy reveals a distended gallbladder and a complete absence of subcutaneous fat. What postmortem finding is least likely to be present in this case of starvation?

- A. Brown atrophy of the heart
- B. Ankle edema
- C. Enlarged liver
- D. Tissue-paper-like intestinal wall

12. A 55-year-old male worker in the battery industry presents with muscle cramps, Parkinson-like tremors, yellow teeth, and bronchitis. Blood tests show elevated levels of both cadmium and manganese. Urine analysis shows hematuria and proteinuria. Despite removal from exposure, his neurological symptoms worsen. Which treatment option would be particular for neurological symptoms?

- A. Dimercaprol (BAL)
- B. CaNa₂ EDTA
- C. DMSA
- D. Sodium para-aminosalicylic acid

13. Hatter's shake is seen in which of the following?

- A. Mercury poisoning
- B. Iron poisoning
- C. Lead poisoning
- D. Arsenic poisoning

14. A 43-year-old man presents with abdominal pain, diarrhea, and progressive numbness in his limbs after consuming food at a barbecue. Over the next few weeks, he develops hair loss and severe neurological deficits, including confusion and muscle weakness. He is non-hypertensive and diabetic. What is the most likely diagnosis?

- A. Acute gastroenteritis
- B. Guillain-Barré Syndrome
- C. Thallium poisoning
- D. Diabetic neuropathy

15. A 35-year-old male vineyard worker presents with abdominal pain, increased salivation, and bluish-green vomiting. His laboratory tests indicate hemolysis. Which of the following statements regarding the poisoning is incorrect?

- A. Shows ptyalism
- B. Interstitial pulmonary fibrosis may occur
- C. D-penicillamine is the chelating agent
- D. Never affects the eyes

16. Which of the following is not a characteristic of an ideal homicidal poison?

- A. Highly toxic
- B. Has an antidote
- C. Causes no noticeable postmortem changes
- D. Dissolves in food or drink without altering its appearance

17. A body is brought for autopsy. On postmortem examination, a garlic odor was detected in the stomach. Which of the following findings is inconsistent with this observation?

- A. Luminous in dark
 - B. Opaque appearance
 - C. Inflammable
 - D. Highly toxic
-

18. A 30-year-old male presents with respiratory distress, excessive salivation, and miosis after working on a farm. On arrival, his vital signs show a heart rate of 50 bpm and blood pressure of 100/60 mmHg. What is the next best step in management?

- A. Administer atropine
 - B. Start intravenous fluids
 - C. Perform decontamination
 - D. Intubate immediately
-

19. A 5-year-old child presents with abdominal pain, irritability, and developmental delays. On examination, the child shows a wrist drop. Which of the following chelating agents is the most appropriate first-line treatment for this condition?

- A. Dimercaprol (BAL)
 - B. Succimer (meso-DMSA)
 - C. D-penicillamine
 - D. Calcium disodium EDTA (Ca Na² EDTA)
-

20. A 35-year-old man presents to the emergency department after ingesting ethylene glycol. His symptoms include confusion, metabolic acidosis, and renal impairment. What is the most appropriate antidote for this patient?

- A. Sodium Thiosulfate
- B. Prussian Blue
- C. Fomepizole
- D. Naloxone

21. A 55-year-old male presents to the emergency department with altered mental status. The medical team is considering administering the "coma cocktail." Which of the following is not a component of the traditional coma cocktail?

- A. Dextrose
- B. Naloxone
- C. Flumazenil
- D. Midazolam

22. A 30-year-old male presents with suspected poisoning from an unknown substance. Which statement about the universal antidote is true?

- A. It consists of only activated charcoal.
- B. Tannic acid is safe for all types of poisoning.
- C. Its use has declined due to ineffectiveness and hepatotoxicity.
- D. Gastric lavage is recommended for all poisonings.

23. Which of the following statements about gastric lavage is false?

- A. Gastric lavage is most effective if performed within 1 hour of poison ingestion
 - B. It is contraindicated in patients with esophageal varices
 - C. Tap water should never be used for gastric lavage in adults
 - D. Lavage can cause complications such as aspiration
-

24. A 45-year-old man is found dead at home. On examination, his postmortem staining is as shown in the image. Which of the following is the most likely cause of death?



- A. Opium poisoning
 - B. Hydrogen sulfide poisoning
 - C. Carbon monoxide poisoning
 - D. Organochlorine poisoning
-

25. A forensic pathologist is examining a deceased individual and notes a garlic-like odor around the mouth. Which of the following poisons is most likely

associated with this characteristic smell?

- A. Hydrogen cyanide (HCN)
 - B. Zinc Phosphide
 - C. Phosphorus
 - D. Methyl salicylates
-

26. A 30-year-old male is brought to the emergency department with suspected poisoning after ingesting an unknown substance. A toxicological analysis is performed, and the following specimens are collected: blood, urine, gastric contents, and liver tissue. Which of the following specimens is most crucial for detecting heavy metal poisoning analysis?

- A. Blood
 - B. Urine
 - C. Hair
 - D. Liver
-

27. A 28-year-old man is brought to the emergency department for acute intoxication after using an illegal narcotic drug. Upon examination, he is found to be in possession of 300 mg of heroin. According to the Narcotic Drugs and Psychotropic Substances Act, 1985, what is the most appropriate legal consequence he would face for this offense?

- A. Rigorous imprisonment for up to 5 years and a fine of ₹1 lakh to ₹2 lakh
 - B. Rigorous imprisonment for 10 to 20 years and a fine of ₹1 lakh to ₹2 lakh
 - C. Rigorous imprisonment for 15 to 25 years and a fine of ₹1 lakh to ₹2 lakh
 - D. Rigorous imprisonment for 20 to 30 years and a fine of ₹1 lakh to ₹2 lakh
-

28. A 45-year-old woman is admitted to the emergency room after ingesting an unknown substance. The doctor provides immediate supportive care and stabilizes her condition. Upon suspicion of homicidal poisoning, what is the next legal duty of the doctor?

- A. Notify the patient's family before taking any further action
 - B. Inform the police after receiving forensic lab results
 - C. Preserve evidence such as food, vomitus, and stomach wash samples
 - D. Administer activated charcoal as the first step in treatment
-

29. Which of the following statements about an ideal suicidal poison is false?

- A. It should have no known antidote.
 - B. It should be easily accessible.
 - C. It should be highly palatable and sweet-tasting.
 - D. It should be highly toxic.
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	4
Question 4	2
Question 5	2
Question 6	3
Question 7	2
Question 8	1
Question 9	4
Question 10	3
Question 11	3
Question 12	4
Question 13	1
Question 14	3
Question 15	4
Question 16	2
Question 17	2
Question 18	3

Question 19	2
Question 20	3
Question 21	4
Question 22	3
Question 23	3
Question 24	3
Question 25	3
Question 26	3
Question 27	2
Question 28	3
Question 29	3

Solution for Question 1:

Correct Answer: C) Hypertrophy of the liver

Explanation:

- The given image shows a Phosphorus burn
- Phosphorus poisoning typically leads to fatty changes and yellow atrophy of the liver, not hypertrophy.

Stage	Symptoms
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First Stage	Burning pain in throat, stomach, and abdomen Nausea, vomiting, pain, diarrhea Onset: 1-2 days post-poisoning Fatty changes or liver failure Yellow atrophy of the liver Painful penetrating burns (2nd and 3rd degree) painful, necrotic, and yellowish with a characteristic smell of garlic Commonly full-thickness burns, result from chemical and thermal insults. (Option D) Garlicky odor in breath and excreta Luminous vomiting and feces (diagnostic) (Option A) Smoking stool syndrome (white faint fumes from stool).
Second Stage	Asymptomatic period Duration: 3-4 days.
Third Stage	Multi-organ failure Liver and RBC involvement Flapping tremor (Asterixis) Kidney failure (Option B) Jaundice Encephalopathy Systemic involvement.

Solution for Question 2:

Correct Answer: A) 20 years to the death penalty

Explanation:

- This is a case of aggravated penetrative sexual assault as the child is <12 years and the crime is committed by an authority figure
- Under POCSO 2019, aggravated penetrative sexual assault carries a punishment ranging from 20 years to the death penalty.

Penetrative Sexual Assault:

- Involves penetration of the penis or objects into the child's orifices (vagina, urethra, anus)
- Involves oral contact with the child's genitalia or anus

Aggravated Penetrative Sexual Assault:

- When penetrative sexual assault is committed under certain circumstances the severity increases and punishment becomes greater
- In the following cases, penetrative sexual assault becomes aggravated:

Category	Description
1. Number of Offenders	Two or more offenders
2. Relationship to Victim	Relatives of the child

3. Age of Child	Child aged < 12 years
4. Authority Figures	Armed forces personnel Insane/institutionalized children Trusted individuals or those in authority Police officers Public servants Jail/remand home/protection home staff Hospital staff The staff of educational or religious institutions Gang members
5. Use of Weapons or Violence	Using deadly weapons Causing grievous hurt
6. Impact on Victim	Making the child mentally ill, pregnant, or HIV positive Acts against disabled or pregnant children
7. Additional Circumstances	With intent to murder During communal violence Repeated acts by a guardian on the same child

Punishment in Aggravated penetrative sexual assault:

Legislation	Punishment	Additional Grounds for Aggravation
POCSO 2012	10 years to life imprisonment Plus a fine	N/A
POCSO 2019	20 years to the death penalty Plus a fine	Assault resulting in the death of a child During a natural calamity

Solution for Question 3:

Correct Answer: D) The doctor fills out MCCD Form 4A

Explanation:

When a patient is already dead upon arrival at the hospital, the MCCD Form 4A is not used. Instead, only the Death Report Form 2 is submitted.

Process	Details
Forms Involved	1. MCCD (Form 4A) 2. Death Report (Form 2)
Recipient	Registrar or Sub-Registrar of the state government
Timeframe for Issuance	A death certificate must be issued within 21 days

If the patient is already dead upon arrival at a hospital	Only Form 2 is sent (Option A) Form 4A is not used. The doctor informs the police (Option C) A post-mortem is conducted to determine the cause of death (Option B) This is classified as a medico-legal case.
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Solution for Question 4:

Correct Answer: B) RMP who attended the patient's last illness

Explanation:

RMP who attended the patient's last illness, is required to certify the cause of death as they have firsthand knowledge of the patient's condition. Other options do not fulfill the legal obligation specified in the Registration of Birth and Death Act, 10(3).

Aspect	RBD ACT-10(3)	Section 174(3) CrPC
Scope	Registration of Birth and Death Act	Police Inquest
Applicability	Obligatory for Registered Medical Practitioners (RMP) who attended the patient's last illness	Applicable when the cause of death is unclear
Fee for Cause of Death	No fee charged	N/A
Certification	Cause of death provided instantly by RMP	Police refer to a doctor for postmortem
Cause of Death	Certified by RMP	Identified through postmortem if needed

Options (A, C, and D): do not fulfill the legal obligation specified in the Registration of Birth and Death Act, 10(3).

Solution for Question 5:

Correct Answer: B) Gangrene

Explanation:

Gangrene serves as the antecedent cause that occurs between the primary cause (diabetes mellitus) and the immediate cause of death (septicemia).

Medical certification of cause of death (MCCD)

Category	Details/Examples
Doctor's Duty	To diagnose the person as dead, declare death, conclude and certify cause of death
Cause of Death	Underlying/Primary Cause: Disease or injury that initiated events (e.g., Diabetes mellitus)
Antecedent/Intermediate Cause: Occurs between primary and immediate causes (e.g., Gangrene)	
Immediate Cause: Final complication due to disease or injury (e.g., Septicemia)	
Example Pathway	Diabetes mellitus (Primary cause) → Gangrene (Intermediate cause) → Septicemia (Immediate cause)
Mortality Statistics	Underlying/Primary cause written in mortality statistics (e.g., Diabetes mellitus)
Contributory Conditions (Part II)	Conditions not related to the disease but contributed to death (e.g., Smoking, Chronic obstructive pulmonary disease (COPD), Diabetes mellitus)
Example Pathway (Part I)	Immediate Cause: Rupture of the myocardium → Antecedent Cause: Myocardial infarction → Underlying Cause: Coronary artery thrombosis → Atherosclerosis

Solution for Question 6:

Correct Answer: C)The most common donated tissue is skin

Explanation:

This statement is incorrect because the most commonly donated tissue is actually the cornea, not the skin.

Human Organ Transplantation Act (HOTA-1994)

Now, Transplantation of Human Organs and Tissues Act (THOTA-2014)

Category		Details	
Purpose		Removal, storage, and transplantation of human organs and tissues Prevention of commercial dealing.	
Definition of Human Organ		Any part of the human body that cannot be replicated if removed Requires declaration of brainstem death by authorized doctors.	
Declaration of Brain stem death by		Treating doctor, In-charge doctor, Neurologist or neurosurgeon, Intensivist, anesthetist, physician, Independent specialist authorized by the government.	
Types of Donation		Living Donation: Near relatives (parents, children, siblings, grandparents, grandchildren) Non-relatives (requires prior approval) Swap transplantation. Cadaveric Donation: Authorized by any donor above 18 years of sound mind (Option A) Will must be written before death in the presence of two witnesses If no will is written then the consent is given by the one who has the legal possession of the body. Unknown/Unclaimed dead body for more than 48 hours, consent is given by a doctor or in-charge doctor of the hospital. Medicolegal cases need autopsy surgeon approval.	
Most Commonly Donated Organs		Kidney > Liver > Heart (Option B) Other organs are the liver, thymus, intestine, lungs, and pancreas.	
Most Commonly Donated Tissues		Cornea > Bone > Tendon Other tissues are the cartilage, middle ear, veins, heart valves, skin, tendons, and ligaments.	
Solution name	Key components	Preservation duration	Primary uses
University of Wisconsin (UW) (Most common)	Raffinose, lactobionate	Up to 48 hours	Abdominal organs
Collins Solution	High-potassium or intracellular solution	Less than 6 hours	Kidneys, heart, liver, pulmonary patches
HTK Solution	Histidine hydrochloride, low potassium	12 to 18 hours	Hepatic grafts
Celsior Solution	Lactobionate, mannitol, reduced glutathione	4 to 5 hours	Clinical heart preservation

Punishment

- Human organ/tissue transplantation without authorization 10 years plus 20 lacs fine. Applicable for medical staff and paramedical staff.
- 10 years plus 20 lacs fine.
- Applicable for medical staff and paramedical staff.
- Additional punishment for the doctor The State Medical Council nullifies registration. 1 offense → 3 years. Subsequent offense → Penal erasure or professional death sentence.
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- 1 offense → 3 years.
- Subsequent offense → Penal erasure or professional death sentence.
- For commercial dealing 2-7 years plus a fine (Option D)
- 2-7 years plus a fine (Option D)
- 10 years plus 20 lacs fine.
- Applicable for medical staff and paramedical staff.
- The State Medical Council nullifies registration.
- 1 offense → 3 years.
- Subsequent offense → Penal erasure or professional death sentence.
- 2-7 years plus a fine (Option D)

Solution for Question 7:

Correct Answer: B) The Board must include a female social worker

Explanation:

The Juvenile Justice (Care and Protection of Children) Act, 2015:

- In India, the act focuses on the protection, treatment, and rehabilitation of children in conflict with the law and those in need of care and protection.
 - Applies to children under 18 years.
 - Children aged 16-18 involved in heinous crimes can be tried as adults.
 - Child Welfare Committees (CWC): Handle cases of children in need of care and protection.
 - Juvenile Justice Boards (JJB) : Handle cases involving children in conflict with the law.
- Three members of the juvenile justice board: One 1st class judicial magistrate and two social workers (one should be female). (Option B)

- Three members of the juvenile justice board: One 1st class judicial magistrate and two social workers (one should be female). (Option B)
- Rehabilitation and Social Reintegration: Focuses on rehabilitation through child care institutions, adoption, foster care, and sponsorship.
- Adoption Regulations: The act includes provisions for adoption, including inter-country adoption, with strict guidelines.
- Penalties for Offenses Against Children: Strict punishments for cruelty, exploitation, and abuse of children.
- Punishment of juveniles Normal offense: 7 years. Minors aged 16-18 accused of serious crimes like rape or murder will undergo psychiatric evaluation and face trial as adults. Other orders: Counselling, Fines, Special homes, Community service.
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- Minors aged 16-18 accused of serious crimes like rape or murder will undergo psychiatric evaluation and face trial as adults.
- Other orders: Counselling, Fines, Special homes, Community service.

Solution for Question 8:

Correct Answer: A) Sodium thiosulfate

Explanation: The sewer worker is probably having hydrogen disulfide poisoning in this scenario. Sodium thiosulfate is ineffective against hydrogen sulfide poisoning as the body spontaneously detoxifies sulfmethemoglobin with the administration of nitrites itself.

Hydrogen sulphide	
Physical Appearance	Colorless gas, "rotten egg" odor

Sources	Decay of organic sulfur-containing products such as fish, manure, sewage, septic tank contents Exposure while cleaning out sewers Industrial sources- paper mills, leather industry, petroleum distillation and refining Natural sources—volcanoes, caves, and sulfur springs
Mode of Action	Inhibits cytochrome oxidase- cellular hypoxia and anaerobic metabolism
Clinical Features	Respiratory arrest Mydriasis, urinary retention, seizures, tremor, hallucinations, confusion, and cardiac arrhythmias
Treatment	High-flow oxygen (Option D ruled out) Antidote: Amyl nitrite (Option C ruled out) Sodium nitrite (Option B ruled out) Sodium thiosulfate is not useful (Option A)
Autopsy Findings	Greenish discoloration of grey matter of the brain, viscera, and bronchial secretions Characteristic rotten egg smell

Nitrites mechanism of action: It induces methemoglobinemia formation, and H⁺S preferentially binds to methemoglobin over cytochrome oxidase, forming sulfmethemoglobin.

Solution for Question 9:

Correct Answer: D) Intravenous Methylene blue

Explanation: The patient is probably having cyanide poisoning in this scenario. Methylene blue is NOT an antidote for cyanide poisoning and should not be used.

Cyanide poisoning	
Physical Appearance	Gaseous state- Hydrogen cyanide (HCN) Liquid form- Hydrocyanic acid or Prussic acid Solid state- salt forms
Sources of poisoning	Electroplating , photographic processes, rubber, and plastic manufacturing industries Insecticide and rodenticide Drug toxicity: Sodium nitroprusside Military chemical warfare agents Burning of plastic furniture, silk, or wool Cigarette smoking Bitter almonds, peach, apricot, plum, apple, pear, cassava

Mode of Action	Rapid absorption across both skin and mucous membrane Cyanide salts combine with hydrochloric acid in the stomach to form HCN that is absorbed Inhibition of cytochrome oxidase causes histotoxic anoxia Neurotoxicity
Clinical Features	Inhalation: Dizziness, vertigo, and constriction of the throat Ingestion: Nausea, headache, loss of muscular power, hypotension, cardiovascular failure, and convulsion Skin: Brick-red color of skin and mucous membranes Anion gap metabolic acidosis and lactic acidosis
Diagnosis	Lee-Jones test Serum cyanide level
Treatment	Severe acidosis (pH < 7.1): Administer sodium bicarbonate Antidotal therapy (Eli Lilly cyanide kit approach) First step: Inhalation of Amyl nitrite (Option A ruled out) Second step: Sodium nitrite (3% solution) IV Third step: Sodium thiosulfate (25% solution) IV Other Antidotes 4-dimethylaminophenol (4-DMAP) Hydroxocobalamin (Vitamin B12): Combines with cyanide to form cyanocobalamin, which is excreted in the urine (Option B ruled out) Hyperbaric oxygen (Option C ruled out) Note: Methylene blue is NOT an antidote for cyanide and must NOT be used (Option D)
Autopsy Findings	Bitter almond's odour Brick-red color of skin and mucous membranes Cyanosis of extremities Hemorrhagic gastritis Disseminated petechiae in brain, meninges, pleura, lungs, and pericardium Pulmonary and cerebral edema

Note:

- Mechanism of action of nitrites: It converts hemoglobin to methemoglobinemia which causes the detachment of cyanide from the heme group of cytochrome oxidase.
- Mechanism of action of sodium thiosulfate: Forms non-toxic thiocyanate which is excreted in the urine.

Solution for Question 10:

Correct Answer: C) Carbon monoxide poisoning

Explanation: The cherry red discoloration of the skin and cutaneous bullae are classic findings in carbon monoxide poisoning, particularly in victims of house fires.

Carbon Monoxide	
Physical Appearance	Odourless, colorless, non-irritating gas, lighter than air
Sources of Poisoning	Incomplete combustion of almost of fuel (wood, charcoal, gas, kerosene) Automobile exhaust Fires: A common cause of accidental CO poisoning resulting in mass deaths - large building (hotel, theatre, block of flats, etc.) in flames Paint remover (especially methylene chloride) Tobacco smoke
Normal CO level in plasma	1 to 5 % 7 to 8 % in smokers
Mode of Action	230 to 270 times greater affinity for hemoglobin than oxygen Leftward shift of the oxyhemoglobin dissociation curve Reduced arterial oxygen content Inactivating mitochondrial cytochrome oxidase - cellular respiration is interfered CO-induced brain injuries are common in basal ganglia, cerebral white matter, hippocampus, and cerebellum.
Clinical Features	Depends on the concentration of CO in the blood CO exposure during pregnancy is teratogenic
Diagnosis	Carboxyhemoglobin level (COHb) in blood Arterial blood gases
Treatment	Oxygen (100%) through a tight-fitting mask or endotracheal tube, until COHb falls to 15 to 20% Hyperbaric oxygen: Inhalation of oxygen at a pressure greater than 1 atmosphere absolute
Autopsy Findings	Cherry red (pink) color of skin, inner aspects of lips, nail beds, tongue, palms and soles Cutaneous bullae (skin blisters) in the calves, buttocks, wrists, and knees Cherry pink color of blood and tissues Firmer white matter of the brain Necrosis and cavitation of basal ganglia
CO Concentration	Clinical manifestations
Mild (COHb < 30%)	Headache, nausea, vomiting, dizziness, exertional dyspnoea

Moderate (COHb 30 to 40%)	Chest pain, blurred vision, confusion, weakness, increasing dyspnoea, tachycardia, tachypnoea, ataxia, severe headache, syncope, flushing, cyanosis, perspiration, decreased vigilance, diminished manual dexterity, impaired sensorimotor task performance, prolonged reaction time, difficulty thinking, impaired judgment, loss of muscular control, tinnitus or roaring in the ears, drowsiness, hallucinations and cardiovascular toxicity
Severe (COHb > 40%)	Trismus, muscle spasms, convulsions, palpitations, disorientation, ventricular dysrhythmias, hypotension, myocardial ischemia, skin blisters, pulmonary edema, respiratory failure, involuntary evacuations, coma, collapse, death
COHb > 50 - 60 %	Potentially lethal

Cyanide poisoning (Option A): While it can occur in fires, especially when burning plastics, the cherry red discoloration of the skin and mucosal membranes is more characteristic of carbon monoxide poisoning. Cyanide poisoning presents with a brick-red color of skin and mucus membranes.

Thermal burns from the fire (Option B) : It is common in fires but would typically cause blistering and charring in areas exposed to direct flames. The presence of cutaneous bullae in non-exposed areas like the wrists and calves, along with the cherry red discoloration, points more toward carbon monoxide poisoning.

Severe dehydration (Option D): Prolonged exposure to heat or lack of water can cause severe dehydration, but it would not explain the cherry red discoloration or the blisters. Dehydration typically leads to dry mucous membranes, decreased skin turgor, and hypovolemic shock.

Solution for Question 11:

Correct Answer: C) Enlarged liver

Explanation: This is a case of homicidal starvation, which leads to general organ shrinkage due to emaciation, and the liver is typically reduced in size, not enlarged.

Starvation

It can be due to withholding of food or unsuitable food administration

Causes:

- Famines
- Being trapped in pits, caves, mines, or landslides
- Parental or guardian neglect (Such children are called baby-farmer)
- Wilful withholding of food
- Wilful refusal to take food

Cause of death:

- Exhaustion
- Circulatory failure
- Intercurrent infection
- MODS (Multiorgan dysfunction syndrome)

Fatal starvation:

Death typically occurs with

- 20% loss of total body protein
- Reaching 40% of the original weight
- Losing 70-90% of body fat

Fatal period:

- If completely withdrawn from both food and water: Death in 10-12 days
- If only food is withdrawn: Death in 6-8 weeks
- Newborns may survive for 7-10 days without food and water

Medicolegal aspect:

- Accident starvation
- Suicidal starvation
- Homicidal starvation Battered baby syndrome Elderly abuse
- Battered baby syndrome
- Elderly abuse
- Battered baby syndrome
- Elderly abuse

Postmortem findings:

External:

- Rigor mortis sets in and disappears early.

- Face is pale with inelastic and hyperpigmented skin.
- Ankle edema is present. (Option B ruled out)
- Hair is dry lustreless and brittle.

Internal:

- Extreme emaciation and the general reduction in size and weight of all organs, except the brain.
- Gall bladder is distended with bile.
- Necrosis and atrophy of the liver. (Option C)
- Brown atrophy of the heart. (Option A ruled out)
- Fat is completely absent in subcutaneous tissue sparing the female breast and orbit.
- Stress fracture due to demineralization and osteomalacia.
- Wall of the intestine appears like a tissue paper with atrophy of mucosa. (Option D ruled out)

Solution for Question 12:

Correct Answer: D) Sodium para-aminosalicylic acid

Explanation: Sodium para-aminosalicylic acid (PAS) is the preferred treatment for manganese-induced neurotoxicity, particularly for patients presenting with Parkinson-like symptoms due to chronic manganese exposure.

Cadmium Poisoning	
Source of poisoning	Battery, welding industries Contaminated soil, water, and food Smoking Most cases of poisoning result from occupational or industrial exposure
Mode of Action	Displacement or substitution of cadmium for zinc in critical metabolic processes Apoptosis Nephrotoxicity - renal cortex

Clinical Features	Nausea, metallic taste, salivation, vomiting, abdominal pain, diarrhea, and myalgia Sudden muscle cramps- cries out in pain (Itai-Itai disease, or Ouch-Ouch disease) Cadmium and Health Effect. Journal of environmental protection and ecology. 10. 84-93. " data-hash="" data-license="NA" data-source="https://www.researchgate.net/figure/tai-itai-disease-and-bone-structure_fig3_273123192" data-tags="" height="276" src="https://image.prepladder.com/content/QBMRYyvha90kUOj6yHna1735874280.png" width="500" /> Bone demineralization with osteomalacia and pathological fractures Yellow staining of teeth (yellow teeth line) Rhinitis and anosmia (due to damage to the olfactory nerve) Renal damage- microhaematuria, proteinuria
Treatment	Acute poisoning Chelation with CaNa2 EDTA or DMSA (Options B & C ruled out) BAL is contraindicated (Option A ruled out) Chronic poisoning No chelating agents Removal from exposure and supportive treatment
Manganese Poisoning	
Source of poisoning	Steel, batteries, and firework manufacturing industries Fertilisers and pesticides
Mode of Action	Neurotoxicity- selective disruption of dopamine neurons and production of the neurotoxins
Clinical Features	Dust or fumes exposure Bronchitis, pneumonia Parkinson-like syndrome (manganism) Manganese madness (Locura manganica) - insomnia, confusion, anxiety, hallucinations, and bizarre behavior Angurugu syndrome or Groote Eylandt syndrome- upper motor neuron, cerebellar, and oculomotor symptoms and signs Metal fume fever due to inhalation of manganese oxide fumes
Diagnosis	MRI shows manganese in the basal ganglia.
Treatment	To help with neurological symptoms- Sodium para-aminosalicylic acid is used (Option D) Chelation therapy with CaNa2 EDTA is useful in some patients Mostly supportive measures

Solution for Question 13:

Correct Answer: A) Mercury poisoning

Explanation:

- Hatter's shake is associated with mercury poisoning due to chronic exposure, particularly in industries like hat-making and glass-blowing.

- It features intention tremors that worsen during voluntary movements, starting in the hands and progressing to other body parts.

Mottled skin from mercury poisoning

mercury poisoning." Annals of internal medicine 139 1 (2003): 78-80 ." data-hash="" data-license="Annals of Internal Medicine " data-source="https://www.semanticscholar.org/paper/A-new-cutaneous-sign-of-mercury-poisoning.-Dantzig/251ad7b73182e58d5328a2601815210bf928b29f" data-tags="" height="356" src="https://image.prepladder.com/content/WUI8PvwuDOM4zZKnzsJQ1735874152.png" width="500" />

Acute Poisoning

Symptoms	Acrid metallic taste, the feeling of constriction in the throat, difficulty in breathing Corrosive lesions in mouth (grayish-white coating), glossitis, and ulcerative gingivitis (1-3 days later if survives) Renal tubular necrosis (PCT) Membranous colitis (diphtheric colitis)
Treatment	Gastric lavage with 5% Sodium Formaldehyde Sulfoxylate and Sodium bicarbonate Sodium formaldehyde Sulfoxylate is an antidote Chelating agent of choice: BAL

Chronic Poisoning (Hydra-Gyrim): Gingivostomatitis, Erethism, Tremors

Specific features	Details
Minamata Disease	Organic (Methyl) Mercury exposure from shellfish in Japan
Erethism	Shyness, depression, anxiety, loss of confidence, suicidal thoughts
Danberry Tremor	Coarse tremors in hands, lips, tongue; slurred speech; early symptom is worsening handwriting
Hatter's Shake	Affects mobility (walking, dressing)
Glass-Blower Shake	Similar to Hatter's Shake; it appears later
Concussio Mercurialis	Most severe form; bed-ridden with no activity
Mercurialis	Brown discoloration of anterior lens capsule; seen via slit lamp; Visual Acuity is Intact
Acrodynia (Pink Disease)	Puffy, painful hands/feet; pinkish color; pruritus, perspiration; skin peeling similar to Kawasaki disease

Solution for Question 14:

Correct Answer: C) Thallium poisoning

Explanation:

Thallium poisoning is consistent with the entire clinical picture, including gastrointestinal symptoms, progressive neurological deficits, hair loss, and a history of potential exposure through food.

Characteristic	Description
Symptoms	Alopecia (1-2 weeks) Painful polyneuropathy (GBS-like) Butterfly-pattern maculopapular rash Loss of lateral 2/3 of eyebrows (Sign de Sourcil) Mee's lines on nails (Other causes: Arsenic poisoning, selenium poisoning, renal failure and chemotherapy)
Antidote	Prussian Blue (Berlin Blue; potassium ferric hexacyanoferrate) which sequesters thallium ions in the intestine, preventing their absorption

Acute gastroenteritis (Option A): While the initial symptoms of abdominal pain and diarrhea could suggest this, the subsequent neurological deficits and hair loss are not typical for gastroenteritis.

Guillain-Barré Syndrome (Option B): is unlikely because it doesn't typically present hair loss.

Diabetic neuropathy (Option D): Although the patient has diabetes, the acute onset of these symptoms, especially following a specific event, points more towards poisoning rather than chronic diabetic complications.

Solution for Question 15:

Correct Answer: D) Never affects the eyes

Explanation:

- This is a case of copper poisoning.
- Copper poisoning can cause ocular manifestations, such as chalcosis oculi, where copper deposits in the eye occur.

Category	Details
Signs and Symptoms	Pain in throat and abdomen increased salivation (ptyalism) Bluish-green vomiting Hemolysis in RBC, liver, and kidney Copper depositions in tissues (chalcosis) Copper deposition in the eye (chalcosis oculi) Wilson disease (sunflower cataract, Kayser-Fleischer ring) Fig: Sunflower cataract Fig: KF ring Green lines in gums (Clapton lines) Hair turning green Interstitial pulmonary fibrosis and granuloma in vineyard sprayer's lung disease due to CuSO_4 exposure.
Toxic Compounds	Copper Sulfate (CuSO_4): Blue vitriol; fatal dose: 30g Copper Subacetate (Verdigris): Fatal dose: 15g Copper Arsenite (Scheele's Green) Copper Aceto-Arsenite (Paris Green)
Treatment	Gastric lavage with 1% Potassium ferrocyanide The chelating agent of choice is D-penicillamine.

Solution for Question 16:

Correct Answer: B) Has an antidote

Explanation:

An ideal homicidal poison should have no antidote .

Qualities of an Ideal Homicidal Poison

Quality	Description
Cost	Irrelevant
Availability	Easily accessible
Physical Characteristics	Colorless, odorless, tasteless
Toxicity	Highly toxic (Option A ruled out)
Antidote	No available antidote (Option B)
Solubility	Dissolves in food or drink without noticeable change (Option D ruled out)

Signs and Symptoms	Resemble natural disease or delayed onset to avoid suspicion
Metabolism	Rapidly destroyed or undetectable in bodily fluids
Detection	Undetectable by chemical or other tests
Postmortem Changes	Does not cause visible postmortem changes (Option C ruled out)
Examples	Arsenic, aconite, thallium, oleander, insulin

Legal Definition

Category	Intent
Medicine	Given with intent to save life
Poison	Given with intent to cause bodily harm

Symptoms Suggestive of Poisoning

Symptom
Sudden abdominal pain, nausea, vomiting, diarrhea, collapse
Sudden onset of coma with pupil constriction
Unexplained coma, especially in children

Solution for Question 17:

Correct Answer: B) Opaque appearance

Explanation:

- The presence of a garlic odor in the stomach during an autopsy suggests the possible ingestion of white phosphorus, which is known for its characteristic garlicky smell.
- Opaque appearance is the finding that is inconsistent with the observation of a garlic odor in the stomach, as this characteristic is more associated with red phosphorus, not white phosphorus.

Feature	White Phosphorus	Red Phosphorus
1. Color	White or yellow	Reddish-brown
2. Appearance	Crystalline, waxy, translucent	Amorphous or crystalline, opaque
3. Solubility	Organic solvents: Yes Water: Very low (~ 3 mg/l)	Organic solvents: No Water: Insoluble
4. Odor	Characteristic garlicky odor	Odorless
5. Taste	Garlicky	Tasteless
6. Chemiluminescence	Luminous in dark (Option A)	Non-luminous
7. Ignitability	Inflammable, spontaneous ignition in air at room temp (emits white fumes) and in chlorine (Option C)	Non-inflammable, ignites only at > 260°C; heat necessary for ignition in chlorine
8. Reaction with Aqueous Alkali	Produces phosphine	None
9. Toxicity	Highly toxic (Option D)	Low toxicity
10. Uses	Fertilizers, insecticides, rodenticides, incendiary bombs, smoke screens, fireworks	On the sides of the box

Solution for Question 18:

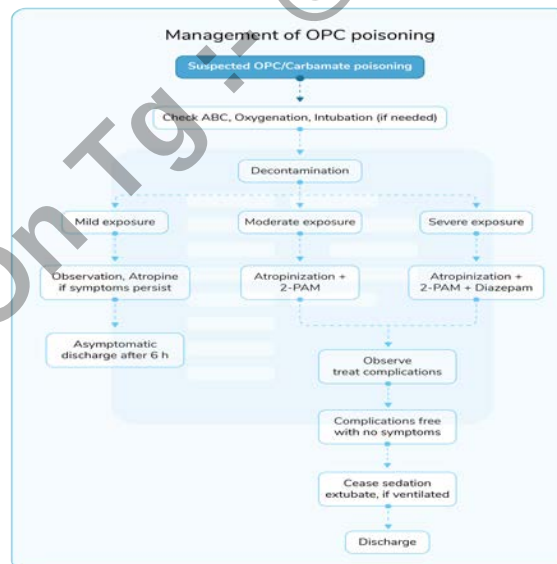
Correct Answer: C) Perform decontamination

Explanation:

- This case involves a 30-year-old male presenting with signs of organophosphate poisoning after farm exposure, characterized by respiratory distress, excessive salivation, and miosis.
- The next best step in management is decontamination, as it reduces the toxin burden and prevents further absorption, which is crucial before initiating other treatments.

Symptoms	Muscarinic: Lethargy, Salivation, Diarrhea, Lacrimation, Urination, Miosis, Muscle weakness, Bronchorrhea, Bronchospasm, Bradycardia, Emesis (Mnemonic: DUMB BELLS)
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Nicotinic: Muscle weakness, Fasciculation, Areflexia, Muscle paralysis	
CNS: Headache, Tremor, Confusion, Slurred speech, Coma, Convulsion	
Laboratory Diagnosis	RBC/True Cholinesterase (specific) Plasma/Pseudocholinesterase (Most common) Falls rapidly with organophosphorus poisoning Inexpensive, sensitive) Atropine test : If 2 mg of Atropine relieves symptoms, it confirms OP poisoning P-nitro phenol test for confirmation.
Management	Resuscitative measures (Airway-Breathing-Circulation) Decontamination Gastric lavage with 1:5000 KMnO ₄ Activated charcoal Atropine sulfate (2-4 mg IV every 10-15 minutes until atropinization) Signs of atropinization: Heart rate>80 beats/min, pupils no longer pinpoint, clear chest on auscultation with no wheeze, systolic BP>80mmHg, Dry axillae Oximes (Pralidoxime) to decrease muscarinic and CNS symptoms removes phosphate group reduces atropine requirement
Cause of Death	Respiratory Muscle Paralysis (RMP)



Solution for Question 19:

Correct Answer: B) Succimer (meso-DMSA)

Explanation:

Succimer (meso-DMSA) is the preferred oral chelating agent for lead poisoning in children due to its effectiveness and safety profile.

Overview of Chelating Agents

- Mechanism : Form stable complexes with metal ions to block interactions with biological systems. Promote elimination of metals via urine or feces.
- Form stable complexes with metal ions to block interactions with biological systems.
- Promote elimination of metals via urine or feces.
- Form stable complexes with metal ions to block interactions with biological systems.
- Promote elimination of metals via urine or feces.

Common Chelating Agents

Chelating Agent	Indication	Administration Route
Dimercaprol (BAL) (Option A ruled out)	Arsenic and mercury poisoning	Intramuscular (IM)
Succimer (meso-DMSA) (Option B)	Lead poisoning (especially in children)	Oral
Unithiol (DMPS)	Mercury and lead toxicity	Oral or Intravenous (IV)
D-penicillamine (Option C ruled out)	Copper and lead poisoning	Oral (but with side effects)
CaNa ²⁺ EDTA (Option D ruled out)	Lead poisoning	IV
Deferoxamine (DFO)	Iron overload (e.g., in thalassemia)	IV or IM
Prussian Blue	Thallium and cesium poisoning	Oral

Factors Affecting Chelation Efficacy

- Metal Type : Different metals require specific chelators due to varying affinities.
- Age : Age impacts both toxicity levels and chelation effectiveness, necessitating tailored treatments.
- Timing : Early use of chelators improves patient outcomes.

Solution for Question 20:

Correct Answer: C) Fomepizole

Explanation:

Fomepizole is a competitive inhibitor of alcohol dehydrogenase, it prevents the conversion of ethylene glycol into toxic metabolites. It is the preferred antidote over ethanol due to fewer side effects and easier administration.

Antidotes and their mechanisms:

Poisoning Type	Antidote	Mechanism
Arsenic	BAL (Dimercaprol)	IM, chelates arsenic for excretion.
Succimer (DMSA)	Oral chelation.	
DMPS	IV, transitioned to oral chelation.	
Copper	Potassium Ferrocyanide	Forms insoluble copper complexes, reducing absorption.
Penicillamine	Chelates copper.	
Cyanide	Sodium Nitrite	Forms methemoglobin to bind cyanide.
Sodium Thiosulfate (Option A ruled out)	Converts cyanide to thiocyanate for renal excretion.	
Hydroxocobalamin	Binds cyanide, forming cyanocobalamin.	
Dicobalt Edetate, Methylene Blue	Alternative antidotes.	
Ethylene Glycol	Ethanol/Fomepizole (Option C)	Inhibits alcohol dehydrogenase to prevent toxic metabolites.
Iron	Desferrioxamine	Chelates iron for excretion.
Magnesium Hydroxide/Sodium Bicarbonate Note: BAL should not be used in iron, cadmium, or selenium poisoning due to formation of toxic complexes.	Precipitate iron, reducing absorption.	
Lead	Sodium/Magnesium Sulfate	Forms insoluble lead sulfate.

Calcium Chloride/Gluconate	Promotes bone deposition of lead.	
CaNa ₂ EDTA + BAL	Chelation therapy for lead elimination.	
Mercury	Sodium Formaldehyde Sulfoxylate	Reduces toxic mercury ions.
BAL (Dimercaprol)	Traditional chelator; oral agents preferred.	
Methanol	Ethanol/Fomepizole	Compete for alcohol dehydrogenase, preventing toxic metabolites.
Opium	Naloxone (Option D ruled out)	Blocks opioid receptors, reversing opioid effects.
Organophosphate	Atropine	Blocks muscarinic receptors to manage cholinergic symptoms.
Pralidoxime (2-PAM)	Reactivates acetylcholinesterase.	
Phenol	Magnesium Sulfate	Absorbs phenol in the stomach.
Phosphorus	Potassium Permanganate	Oxidizes phosphorus into harmless phosphates.
Strychnine	Barbiturates	Depress CNS, preventing seizures.
Thallium	Prussian Blue (Option B ruled out)	Binds thallium in the intestine, preventing absorption.

Solution for Question 21:

Correct Answer: D) Midazolam

Explanation:

Midazolam is a benzodiazepine and is not part of the traditional coma cocktail; it is typically used to induce sedation rather than for treating altered mental status in emergency settings.

Coma Cocktail

The concept of a "coma cocktail" refers to a combination of treatments administered to patients with altered mental status, particularly in emergency settings

Components :

Treatment	Purpose
Dextrose (Option A ruled out)	Treats hypoglycemia
Naloxone (Option B ruled out)	For suspected opioid overdose
Thiamine	Prevents Wernicke's encephalopathy
Flumazenil (Option C ruled out)	Used for benzodiazepine overdose (not routine)

Current Perspectives

- Indiscriminate use can lead to misdiagnosis and harm patients.
- Flumazenil is risky in undiagnosed benzodiazepine dependence.
- Dextrose : Administer primarily for suspected hypoglycemia; can be given empirically if unsure.
- Naloxone : Appropriate for signs of opioid intoxication; avoid routine use without clinical signs.
- Thiamine : Reserve for individuals at risk (e.g., chronic alcohol users, malabsorption).
- Emphasis on accurate diagnosis rather than a standardized cocktail.
- Conduct thorough assessments and provide targeted therapies.
- Maintain cerebral perfusion pressure and supportive care

Solution for Question 22:

Correct Answer: C) Its use has declined due to ineffectiveness and hepatotoxicity.

Explanation:

Research indicates that activated charcoal is superior, and the presence of hepatotoxic components contributed to the decline of the universal antidote.

Universal Antidote:

- Definition: The term "universal antidote" implies a single solution for all poisonings, which is misleading.
- Decline in Use: No longer recommended since the mid-1980s and is currently unavailable.

Components of the Universal Antidote (Option A ruled out)

Component	Purpose
Powdered Charcoal	Adsorbs alkaloids
Magnesium Oxide	Neutralizes acids
Tannic Acid	Precipitates alkaloids, glycosides, and metals

Reasons for Discontinuation

- Activated Charcoal Superiority: Research shows activated charcoal is more effective than the universal antidote in reducing toxin absorption.
- Tannic Acid Issues: Interferes with activated charcoal's toxin absorption. Identified as hepatotoxic, posing risks to patients. (Option B ruled out)
- Interferes with activated charcoal's toxin absorption.
- Identified as hepatotoxic, posing risks to patients. (Option B ruled out)
- Interferes with activated charcoal's toxin absorption.
- Identified as hepatotoxic, posing risks to patients. (Option B ruled out)

Current Recommendations

- Avoid Universal Antidote: Emphasis on identifying the specific poison to administer the appropriate antidote.
- Activated Charcoal: Recommended as a first-line treatment if the poison is unknown.

Gastric Lavage Considerations

- Purpose: Stomach washing procedure to remove ingested poisons.
- It is contraindicated in corrosive poisoning, convulsant poisoning, comatose patients, volatile poisons and when there is a risk of hemorrhage or perforation. (Option D ruled out)

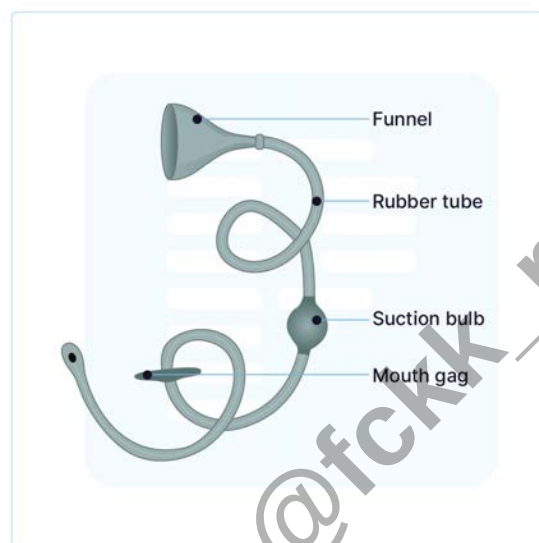
Solution for Question 23:

Correct Answer: C) Tap water should never be used for gastric lavage in adults

Explanation:

Tap water is generally acceptable for gastric lavage in adults unless specific solutions are required for certain poisons. However, saline is recommended for infants to avoid electrolyte imbalance.

Gastric Lavage in Poisoning Management



Gastric lavage is a procedure to remove ingested poison by introducing and withdrawing fluids through an orogastric tube. Its effectiveness depends on timing, proper technique, and patient-specific considerations.

Timing and Effectiveness

- Most effective within 1 hour of ingestion (Option A ruled out)
- Can be beneficial up to 4-6 hours in select cases
- Beyond 6 hours, poison likely passes into the intestines, reducing lavage efficacy

Procedure Steps

Step	Details
Patient Positioning	Trendelenburg: Head down to prevent aspiration Left lateral decubitus: Reduces chance of poison entering the duodenum
Tube Placement Confirmation	Inject air and listen for gastric bubbling sounds Watch for coughing or bubbles (indicates misplacement in trachea)
Fluid Administration	Fluid is administered and aspirated using a syringe attached to the tube

Fluid Selection	Tap water for most cases (Option C) Normal saline for infants Specific solutions (e.g., potassium permanganate) may be used for certain poisons
Residual Fluid	Leave a small amount of fluid to neutralize remaining poison

Contraindications

- Corrosive Poisoning : Avoid except for carbolic acid (phenol) to prevent perforation
- Convulsant Poisons : Risk of triggering seizures (e.g., strychnine)
- Comatose Patients : High aspiration risk
- Volatile Poisons/Hydrocarbons : May cause aspiration pneumonitis
- Risk of Hemorrhage/Perforation : Avoid if esophageal varices or recent surgery present (Option B ruled out)
- Additional Contraindications to Emetics Severe heart or lung diseases Advanced pregnancy CNS depression or seizures
- Severe heart or lung diseases
- Advanced pregnancy
- CNS depression or seizures
- Severe heart or lung diseases
- Advanced pregnancy
- CNS depression or seizures

Complications

Complication	Details
Aspiration (Option D ruled out)	Occurs in ~10% of cases. Risk of lung injury
Esophageal/Gastric Perforation	Can be life-threatening
Tube Misplacement	Misplaced into trachea, causing complications
Bleeding	Can occur with improper technique or pathology

Alternatives to Gastric Lavage

Alternative	Description
Emetics	Induce vomiting in conscious patients
Dilution	Large water intake to dilute corrosive substances

Endoscopic/Surgical Removal	Used for toxic foreign bodies or drug packets
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Solution for Question 24:

Correct Answer: C) Carbon monoxide poisoning

Explanation:

- The given image shows Cherry-red discoloration
- Cherry-red discoloration of the skin and tissues is characteristic of carbon monoxide poisoning due to the formation of carboxyhemoglobin (COHb), particularly when levels exceed 30%.

Postmortem Staining in Poisoning Cases

- Postmortem staining (Livor mortis) is the skin discoloration after death due to blood settling in dependent areas.
- The color of the staining can indicate certain types of poisoning:

Poison	Postmortem Staining Characteristics
Opium (Option A ruled out)	Purple or blackish staining
Carbon monoxide (Option C)	Cherry red discoloration (COHb > 30%)
Hydrogen sulfide (Option B ruled out)	Greenish discoloration in tissues
Datura	Asphyxial signs (no specific color)
Organochlorine/Kerosene (Option D ruled out)	Asphyxial signs (no specific color)

Solution for Question 25:

Correct Answer: C) Phosphorus

Explanation:

Phosphorus is known to produce a garlic-like odor, which is a key characteristic.

Smells Linked to Poisons

- Characteristic odors can aid in suspecting and diagnosing poisonings.
- Smells may be detected near the mouth and nose of individuals, living or deceased, especially during postmortem examinations.

Odor Characteristics of Various Poisons

Poison	Odor
Phosphorus, heavy metals (arsenic, selenium, thallium), parathion, malathion, alphos (Option C)	Garlic-like
Ethanol, methyl/propyl alcohol, chloroform, nitrites, acetone	Sweet and fruity
Paraldehyde, chloral hydrate	Acrid
Hydrogen sulfide (H ₂ S), mercaptans, disulfiram	Rotten eggs
Hydrogen cyanide (HCN) (Option A ruled out)	Bitter almond
Carbolic acid	Phenolic
Organophosphates	Kerosene-like
Zinc phosphide (Option B ruled out)	Fishy
Methyl salicylates (Option D ruled out)	Oil of wintergreen
Marijuana	Burnt rope, hemp

- The absence of a specific odor does not exclude poisoning; some poisons may lack distinct smells or have their odors masked by other substances.

Solution for Question 26:

Correct Answer: C) Hair

Explanation:

Hair accumulates heavy metals over time, making it valuable for detecting such poisoning.

Toxicological Analysis in Suspected Poisoning Cases

Toxicological analysis is essential for confirming or excluding poisoning as a cause of illness or death. It involves various specimens, tests, and result interpretations.

Specimens for Analysis

Specimen	Importance
Blood	Primary specimen; collect from femoral, iliac, or axillary veins to avoid contamination. (Option A ruled out) Preservative: Lithium heparin or EDTA tube, fluoride/oxalate, if alcohol is suspected
Urine	Useful for detecting parent compounds and metabolites of poisons. (Option B ruled out) Preservative: No preservative, sodium fluoride is added, if alcohol is suspected
Gastric Contents	Helps identify ingested poisons and their concentrations through vomitus and stomach contents analysis. Preservative: Saturated salt solution if there is no chance of acid poison. Rectified spirit if poisoned by acid.
Liver	Analyzes metabolism and concentration of drugs; critical when blood/urine concentrations are low. Bile can detect specific substances like chlorpromazine and morphine. (Option D ruled out) Preservative: Saturated salt solution if there is no chance of acid poison. Rectified spirit if poisoned by acid.
Hair and Nails	Detects heavy metal poisoning; metals accumulate in keratin. Neutron activation analysis identifies specific metals and exposure patterns. (Option C)
Scene Residues	Provides clues about the poison through materials found at the scene, such as tablets or empty containers.

Types of Toxicological Tests

Test Type	Description
Qualitative Tests	Determine presence/absence of specific drugs or toxins; positive indicates presence, negative suggests absence or low concentration.
Quantitative Tests	Measure specific concentration of drugs or toxins; crucial for assessing severity of toxicity for substances like acetaminophen, alcohols, barbiturates, and heavy metals.

Interpretation of Results

- **Negative Results** : A negative result does not exclude poisoning; it may indicate the substance was undetectable or below detection limits. Retesting may yield a positive result.
- **Quantitative Analysis** : Interpretation depends on ingestion timing, individual metabolism, and possible drug interactions.

Solution for Question 27:

Correct Answer: B) Rigorous imprisonment for 10 to 20 years and a fine of ■1 lakh to ■2 lakh

Explanation:

The NDPS Act stipulates rigorous imprisonment for 10 to 20 years and a fine of ■1 lakh to ■2 lakh for the possession of narcotic drugs, excluding small quantities.

NDPS Act, 1985

- The NDPS Act, 1985, is an Indian law regulating narcotic drugs and psychotropic substances.
- **Definitions** : Narcotic Drug : Includes coca leaves, cannabis (hemp), opium, poppy, and drugs derived from these. Psychotropic Substance : Any natural or synthetic substance listed in a Schedule of 76 drugs and their derivatives (e.g., amphetamine, pentobarbital, psilocybin, diazepam).
- **Narcotic Drug** : Includes coca leaves, cannabis (hemp), opium, poppy, and drugs derived from these.
- **Psychotropic Substance** : Any natural or synthetic substance listed in a Schedule of 76 drugs and their derivatives (e.g., amphetamine, pentobarbital, psilocybin, diazepam).
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- **Psychotropic Substance** : Any natural or synthetic substance listed in a Schedule of 76 drugs and their derivatives (e.g., amphetamine, pentobarbital, psilocybin, diazepam).

Punishments Under the NDPS Act

Offense	Punishment
Production, possession, transportation, import, sale, purchase, or use of narcotic drugs/psychotropic substances (excluding "ganga")	Rigorous imprisonment: 10-20 years (Options A, C, & D ruled out) ; Fine: ■1 lakh - ■2 lakh

Repeat offenses	Rigorous imprisonment: 15-30 years; Fine: ₹1.5 lakh - ₹3 lakh
Handling "ganga"	Imprisonment: 5 years; Fine: ₹50,000
Repeat offenses of handling "ganga"	Imprisonment: up to 10 years; Fine: ₹1 lakh
Possession of small quantities (e.g., 250 mg heroin, 5 g charas, 5 g opium, 125 mg cocaine)	Simple imprisonment: up to 1 year; Fine: unspecified amount
Possession of less than 500 g "ganga"	Imprisonment: up to 6 months
Trafficking (e.g., >1 kg pure heroin after previous convictions)	Maximum punishment: Death penalty

Related Legislation

- Prevention of Illicit Traffic in NDPS Act, 1988 : Provides for preventive detention and property seizure.

Solution for Question 28:

Correct Answer: C) Preserve evidence such as food, vomitus, and stomach wash samples

Explanation:

The doctor must preserve all relevant evidence (food, vomitus, excreta, stomach wash) in suspected poisoning cases.

Doctor's Duties in a Suspected Poisoning Case

Medical Duties:

- Stabilize the patient : Focus on airway, breathing, and circulation.
- Provide supportive care : Manage pain, oxygen therapy, seizures, and hyperthermia.

Legal Duties:

Duty	Details
Documentation	Record patient's details (name, age, sex, address, arrival time, etc.) and medical history.

Suspicion Confirmation	Collect samples (vomit, urine), observe symptoms, and consult with senior practitioners.
Hospitalization	Request hospital transfer after confirming homicidal poisoning, with patient consent if possible.
Preserve Evidence	Keep samples of food, drink, excreta, and stomach wash; failure to do so can lead to legal action.
Report to Authorities	Govt. Medical Officers must report all cases to police; private practitioners must report homicidal poisoning under Section 39 of the CrPC.
Dying Declaration	Ensure a dying declaration is recorded if the patient's condition is critical.
Death Certificate	Do not issue a death certificate in poisoning cases; inform the police.
Opinion on Poison	Provide an opinion only after receiving forensic science lab reports.
Information to Police	Provide all case details to the investigating officer when requested under Section 175 of CrPC.

Important Legal Points:

- Failure to preserve evidence can result in up to 7 years of imprisonment under Section 201 of IPC.
- Not reporting homicidal poisoning may lead to penalties under Sections 176 and 177 of IPC.
- Reporting suicidal or accidental poisoning is advised even if not legally mandatory, as other crimes (e.g., abetment) may be involved.

Notify the patient's family before taking any further action (Option A) : While notifying the family may be appropriate later, it should not delay immediate actions like preserving evidence and notifying authorities.

Inform the police after receiving forensic lab results (Option B) : The police must be informed immediately, not after waiting for forensic results.

Administer activated charcoal as the first step in treatment (Option D) : Administering activated charcoal may be part of the treatment, but preserving evidence and reporting to the police is the priority in cases of suspected poisoning.

Solution for Question 29:

Correct Answer: C) It should be highly palatable and sweet-tasting.

Explanation:

An ideal suicidal poison should be tasteless or have a pleasant taste, but not necessarily sweet-tasting, making this statement false.

Ideal Characteristics of a Suicidal Poison:

- Affordable : Should be inexpensive.
- Easily Available : Readily accessible. (Option B ruled out)
- Physical Traits : Tasteless or pleasant-tasting, with no unpleasant odor. (Option C)
- High Toxicity : Should be lethal. (Option D ruled out)
- No Antidote : Must lack an antidote. (Option A ruled out)
- Solubility : Easily dissolvable in food or drink.
- Effect : Should cause painless death.

Examples of Ideal Suicidal Poisons:

Poison	Category
Potassium Cyanide (KCN)	Cyanide compound
Hydrogen Cyanide (HCN)	Cyanide gas
Opium	Narcotic
Barbiturates	Sedative
Aluminum Phosphide (Alphos)	Phosphide compound
Organophosphates	Pesticides

Forensic Toxicology 1

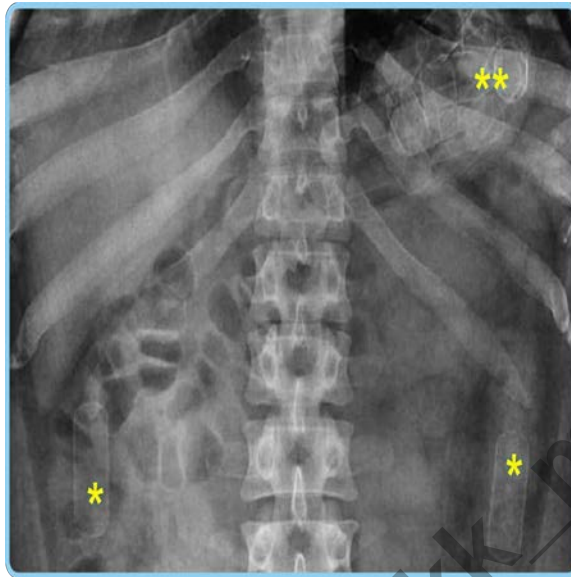
1. A 32-year-old man is found unconscious with a strong odor of antiseptic on his breath. He is rushed to the emergency department where he develops hypotension and renal failure. On examination, his angle of mouth is stained gray, and he has burns on his esophagus. The patient had a history of greenish discoloration of urine. What is the most likely cause of these findings?

- A. Oxalic acid poisoning
- B. Sulfuric acid poisoning
- C. Carbolic acid poisoning
- D. Nitric acid poisoning

2. Which of the following can cause priapism?

- A. Spanish fly
- B. Tumbu fly
- C. Botfly
- D. Screwworm fly

3. A police officer presents an X-ray of a suspect who is believed to be a body packer. Patient acts casual and his vitals are stable. The X-ray is as given. What is the most critical initial management step for this patient?



- A. Administer activated charcoal
- B. Perform a laparotomy
- C. Monitor and provide supportive care
- D. Do an urgent MRI

4. A 40-year-old man is brought to the emergency department after taking an overdose of morphine. He is unresponsive with a respiratory rate of 6 breaths per minute. Which laboratory finding would you expect in this patient?

- A. Elevated blood glucose levels
- B. Hypoxemia
- C. Metabolic alkalosis
- D. Hypokalemia

5. A 24 year old female was brought to the emergency department in comatose state and on examination, the vitals were BP:80/40, HR:42/min, RR:8/min,

Temperature 36 degree, Reflexes: sluggish, Miosis with conjunctival congestion, spO2:85%. Multiple skin scars were noted on her left forearm. The attenders gave a history of the frequent domestic strife and mood swings following which she would have a long unexplained absence from home. The duty doctor suspected drug abuse and started appropriate management. Which among the following chemical tests can be used for diagnosis in this case?

- A. Marquis test
- B. Marsh test
- C. McNally's test
- D. Gutzeit test

6. All among the following are true regarding the image except:



- A. Obtained from a plant called *Papaver somniferum*
- B. Extracted via vertical incision from unripe poppy
- C. Addictive potential is due to its action on Kappa (κ) receptors
- D. Pruritus is a common adverse effect

7. Patient was brought to ED after being bitten by a snake as shown below. Which of the following steps are part of management in this patient? 1) Immobilize the limb by applying crepe bandages 2) Cauterize the bite wound 3) No test dose to be given before starting ASV 4) Dose of ASV should be halved in children 5) ASV given until clotting time becomes normal



- A. 1, 3
- B. 1, 3, 5
- C. 2, 4, 5
- D. 3, 5

8. Which of the following snakes is known for its myotoxic venom?

- A. King Cobra
- B. Sea Snake
- C. Russell's Viper
- D. Banded Krait

9. Which of the following is a key difference between neurotoxic and vasculotoxic snake venoms?

- A. Neurotoxic venom local reaction is extensive and vasculotoxic venom local reaction is minimal
- B. Tongue paralysis is seen in vasculotoxic snakes.
- C. Neurotoxic venom leads to severe swelling at the bite site, while vasculotoxic venom causes respiratory paralysis.
- D. Neurotoxic venom causes wet gangrene , while vasculotoxic venom causes dry gangrene.

10. Which of the following is not a physical feature of a poisonous snake?

- A. Small belly scales
- B. Small head scales
- C. Long fang
- D. Compressed tail

11. What is the treatment of choice for the sting of the organism given below?



- A. Prazosin
- B. Atropine
- C. Calcium gluconate
- D. Scorpion antivenom

12. What is the mechanism of action of the poison found in these seeds (image)?



- A. Stimulation of anterior horn cells of spinal cord
- B. Inhibition of anterior horn cells of spinal cord
- C. Stimulation of posterior horn of spinal cord
- D. Inhibition of posterior horn of spinal cord

13. A forensic pathologist is conducting an autopsy on a 38-year-old male who died after an accidental exposure to hydrofluoric acid (HF) at an industrial plant. Which of the following postmortem findings is most characteristic of HF poisoning?

- A. Cherry-red discoloration of the blood
- B. Black tarry stool
- C. Severe hypocalcemia and calcium deposits in tissues
- D. Widespread gastric mucosal erosion

14. A forensic pathologist is performing autopsies on two individuals who died from chemical poisoning. One case involves nitric acid and the other oxalic acid.

Which of the following postmortem findings is most consistent with oxalic acid poisoning rather than nitric acid poisoning?

- A. Yellow discoloration of the skin with congestion of the larynx and trachea
 - B. Perforation of stomach
 - C. Soft, friable stomach wall with ulceration
 - D. Gelatinous brown stomach content with swollen, congested kidneys showing necrosis
-

15. A 40 year old man was found unconscious at home, with an open container of unknown acid nearby. He was rushed to the ER but died shortly after arrival. During autopsy, the police surgeon notes several injuries consistent with corrosive ingestion along with chalky white, brittle teeth. Which forensic finding would solidify the above finding in this case?

- A. Bright red lividity
 - B. Blackish staining and necrosis of the lips and mouth
 - C. Greenish discoloration of the skin
 - D. Yellowish staining and necrosis of the lips and mouth
-

16. In cases of poisoning by corrosive substances, gastric lavage can be performed in which of the following situations? A) Sulphuric acid poisoning B) Hydrochloric acid poisoning C) Carbolic acid (Phenol) poisoning D) Ammonia poisoning E) Oxalic acid poisoning

- A. All except D
 - B. All except A and B
 - C. Only C and E
 - D. Only E
-

17. What is the mechanism of action of the toxin present in this plant?



- A. Causes extrasystoles
- B. Paralyzes respiratory muscles
- C. Causes CNS depression
- D. CNS hyperexcitation

18. Given is the image of a poisonous plant. Which part of this plant is poisonous?



- A. Flower
 - B. All parts
 - C. Root
 - D. Seed
-

19. Which of the following statements on aconite is false?

- A. All parts of the plant are poisonous
 - B. Easily detected postmortem
 - C. Affects voltage-sensitive sodium channels
 - D. 1-2 g of root is lethal
-

20. Which of the following statements on strychnine poisoning is wrong?

- A. Rigor mortis sets in late
- B. Delayed putrefaction

- C. Strychnine can be detected in putrefied bodies and viscera
 - D. Brain and spinal cord should be preserved for analysis
-

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Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	3
Question 4	2
Question 5	1
Question 6	3
Question 7	4
Question 8	2
Question 9	4
Question 10	1
Question 11	1
Question 12	1
Question 13	3
Question 14	4
Question 15	2
Question 16	3
Question 17	1
Question 18	2

Question 19	2
Question 20	1

Solution for Question 1:

Correct Answer: C) Carbolic acid poisoning

Explanation :

Carbolic acid (phenol) poisoning presents with Gray/brown corrosions on mouth, esophageal burns, a strong antiseptic odor, CNS depression, hypotension and renal failure due to its toxic effects on the central nervous system, cardiovascular system and kidneys.

Phenol Poisoning (Carbolism)	
Physical Properties	Colorless, crystalline, turns pink on air exposure; slightly acidic, forms phenolate ion ($C_6H_5O^-$).
Uses	Antiseptic/disinfectant; Lysol (50% cresol), Dettol (chlorinated phenol with turpeneol).
Fatal Dose	1-2 g phenol or 25-50 ml of household phenol.
Fatal Period	3-4 hours.
Tissue Discoloration	Can lead to ochronosis (bluish-black discoloration in tissues like ears and sclera).
Signs and symptoms	Local Effects: Tingling → numbness → coagulation necrosis → painless white eschar, which turns brown and sloughs off. Gastrointestinal: Initial burning/tingling, later anesthesia; diarrhea and abdominal pain, vomiting rare. Respiratory: Phenol odor on breath, laryngeal/pulmonary edema, stertorous breathing, cyanosis. Metabolized by kidneys into hydroquinone/pyrocatechol; causes carboluria (dark green urine). Cardiovascular (CVS): Rapid, feeble, irregular pulse; cold, clammy skin; collapse, unconsciousness, coma. Muscular: Muscle spasms, convulsions CNS: Headache, giddiness, tinnitus, contracted pupils. Renal: Dark smoky green color of urine.

Postmortem Findings	External: Gray/brown corruptions on mouth, body, hands; phenolic odor, delayed putrefaction. Internal: Corroded GIT mucosa, leathery stomach, pulmonary edema, hemorrhagic nephritis.
Medico-legal Aspects	Used in suicides, accidental poisoning, rare for homicides (strong odor/taste), abortifacient.

Oxalic acid poisoning (Option A) : Oxalic acid causes renal damage, resulting in swollen, congested kidneys with tubular necrosis, along with gelatinous brown content in the stomach. It lacks the characteristic antiseptic odor associated with carbolic acid.

Sulfuric acid poisoning (Option B): Produces black burns, but there is no antiseptic odor, and sulfuric acid is more likely to cause severe external chemical burns rather than the systemic effects seen here.

Nitric acid poisoning (Option D): Causes yellow burns and severe systemic toxicity, but lacks the distinctive odor.

Solution for Question 2:

Correct Answer: A) Spanish fly

Explanation:

Blister bees/Spanish fly/*Cantharis vesicatoria*

- Active principle: Cantharidin
- Action: Locally irritant and nephrotoxic agent.
- Absorption: Readily absorbed from all surfaces including skin.



Signs and symptoms:

Externally:

- Redness
- Burning
- Formation of vesicles

On Ingestion:

- Burning sensation in the mouth, throat and abdomen.
- Difficulty in swallowing and speech.
- Dull pain felt in the loins, desire to micturate but urine is scanty and blood stained.
- Priapism in males and abortion in pregnant females.
- Patient becomes prostrated with convulsions and coma.

Fatal dose and period:

- 15-30mg of cantharidin or 1.5g of powder
- 24hrs

Treatment:

- Gastric lavage
- Demulcents(Not fat)
- Symptomatic treatment.

Post-mortem findings:

- Inflammation and vesicles seen in the mouth

Internally:

- Mucus membrane of the oesophagus and stomach is often swollen and engorged.
- Patches of ulceration and haemorrhages in GIT.
- Kidneys congested with haemorrhage.
- Lungs are edematous and congested.

Aconitine poisoning (Option A): Caused by consuming plants from the Aconitum genus (like monkshood), leading to severe neurological and cardiovascular symptoms.

Strychnine poisoning (Option C): Caused by ingesting strychnine, a potent neurotoxin leading to muscle spasms, seizures, and potentially death due to respiratory failure.

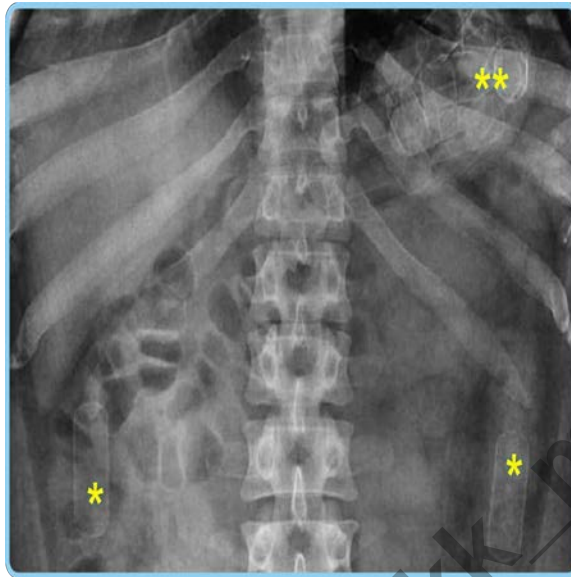
Oleander poisoning (Option D): Results from ingesting parts of the oleander plant, which contains cardiac glycosides, leading to nausea, vomiting, and potentially fatal heart issues.

Solution for Question 3:

Correct answer: C) Monitor and provide supportive care

Explanation:

X-ray is suggestive of an asymptomatic Heroin body packer . Most appropriate management would be to monitor and treat the patient symptomatically to excrete the packets.



Body packers:

Introduction	People who illegally carry narcotic drugs (Cocaine/Heroin), either by swallowing or insertion in the vagina/rectum as sealed pellets/packages.
Complications	GI obstruction/hemorrhage (caused by packets) Packet rupture - Narcotic intoxication
Imaging	Abdominal X-ray (first-line) Ultrasound or CT Scan (if unclear)
Management	Asymptomatic Patient: Whole Bowel Irrigation (caution: potential rupture) Alternatives: Allow packages to move into the colon Administer Enemas (Low-Volume Phospho Soda Enemas/ High-Volume Saline Enemas) Avoid Food Ingestion: Until all packages are passed. Encouraging Gastric Emptying :Metoclopramide 10 mg, 8-hourly
Symptomatic Patient: Continuous-infusion of Naloxone Whole bowel irrigation. Surgical intervention may be required.	

Solution for Question 4:

Correct answer: B) Hypoxemia

Explanation:

Morphine is an alkaloid found in opium

Morphine causes hypoxemia

Findings in Morphine poisoning:

- Hypoxia and respiratory acidosis is seen due to respiratory depression by acting on mu opiate receptors and in the medulla via mu and delta receptors.
- Hyperkalaemia occur following an overdose, especially in the presence of rhabdomyolysis and acute renal failure, not hypokalemia.
- Opioid Receptor Activation: Morphine binds to opioid receptors, particularly the mu-opioid receptors, which can influence various hormonal systems. This activation can lead to altered signaling pathways that may reduce gluconeogenic enzyme activity in the liver and can lead to hypoglycemia.

Solution for Question 5:

Correct answer: A) Marquis test

Explanation:

- From the above clinical scenario, the triad of coma, pinpoint pupils, and respiratory depression is said to be almost pathognomonic for opiate poisoning.
- Such cases may be due to therapeutic overdose, accidental overdose (in addicts), or deliberate overdose (suicidal).

The following tests are used for the diagnosis for opium poisoning (MDH)

- M - Marquis test
- D - Deninges test
- H - Husemann test

Stages of opium poisoning :

Stage	Symptoms/Characteristics
Stage of Excitement	Short duration, Euphoria Hallucinations, tachycardia Flushing of face, Conjunctival injection

Stage of Stupor	Headache,nausea,vomiting Weakness,heaviness in limbs Dizziness, miosis,cyanosis of face and lips
Stage of Narcosis/Coma	Deep coma (unarousable) Flaccid, relaxed muscles Pale face, Absent reflexes Congested conjunctiva Cold skin with profuse perspiration Pinpoint, non-reactive pupils Hypotension, Weak, feeble pulse, Hypothermia, Bradypnoea

Drugs used in Opioid Poisoning:

- Management of Acute Opioid poisoning: IV Naloxone
- DOC to prevent relapse: Naltrexone (oral)
- DOC for Opioid substitution therapy: Buprenorphine > Methadone

Marsh test (Option B): Used for the detection of Arsenic

McNally's test (Option C): Used for the detection of Salicylates

Gutzeit test (Option D): Used for the detection of Strychnine

Solution for Question 6:

Correct answer: C) Addictive potential is due to its action on Kappa (κ) receptors

Explanation:

Opium is extracted from the unripe poppy capsule (shown in the image).



Opium:

Introduction	Dried extract from the poppy plant (<i>Papaver somniferum</i>)
Extraction	Obtained by vertical incisions from unripe poppy capsule (toxic part)
Toxic principles	Phenanthrene group: Morphine, Codeine and Thebaine Benzylisoquinoline group: Papaverine and Noscapine
Mechanism of action (MOA)	Exhibits action via 3 receptors mainly - Mu(μ), Delta(δ) and Kappa(κ) receptors and release Endorphins. Addictive potential is due to its action on Mu(μ) and Delta(δ) receptors Action on Kappa(κ) receptor causes dysphoria - no addictive potential
Effects	Tolerability, Addiction, Dysphoria (κ -receptor) Pruritus is a common adverse effect. Other effects: Constipation, Respiratory depression, Urinary retention. Tolerability is developed for all actions except: 3 Cs Constipation Convulsions Constriction of pupils (Miosis)
Fatal dose	Opium: 2 g Morphine: 0.2 g Codeine: 80 mg Fatal period: 6-12 hours

Management	Support vitals : Use a respirator and other emergency procedures. Decontamination : Stomach wash with 1:5000 KMnO ₄ , leaving some solution in the stomach to oxidize any secreted alkaloid. Perform lavage even after IV/IM drug injection, as it is secreted in the stomach. Administer activated charcoal as the primary decontamination method. Enema with sodium sulfate twice daily. Whole-bowel irrigation for body packers. Antidote : Naloxone
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Solution for Question 7:

Correct answer: D) 3,5

Explanation:

Snake shown in the image is Russel's viper , which is hemotoxic.

Application of crepe bandage or cauterization of wound is not indicated for Viper bite.



Treatment of snake bite:

First Aid: ' CARRY NO RIGHT '	
-------------------------------	--

CARRY	Do not allow the victim to walk on their own, must be carried (in any form) to the hospital as soon as possible.
NO	No Suction No Cautery No Ligature No Tourniquet
RIGHT	R: Reassure the patient I: Immobilize the limb: by application of crepe bandages (Sutherland wraps) GH: Get to the Hospital immediately T: Tell history/symptoms
Pressure bandage	Used for immobilization of the limb by application of crepe bandage - known as Sutherland wraps . Contraindicated in viper bites. Ideal pressure: 50-70 mm Hg.
ASV	Mainstay of treatment. Used only when signs of systemic poisoning are present Polyvalent anti snake venom (PAV) used in India can neutralize (i) Cobra (ii) Krait (iii) Russel viper (iv) Saw-scaled viper bites. Dose: 1-2 ml/kg (!0 vials) irrespective of sex/age. Each vial: neutralizes 6-8 mg of snake venom Test dose is not required/advised to be given before initiating ASV. Ideally to be given within 4 hours of snake bite. To be given until clinical improvement (in cobra bites)/ Clotting time becomes normal (in viper bites).
Adjuvant treatment	Neostigmine 1.5mg: for neuromuscular symptoms (cobra bite) . Atropine 0.5mg: To antagonize muscarinic actions of Ach. Surgical debridement and FFP may be required in Viper bites.

Solution for Question 8:

Correct Answer: B) Sea snake

Explanation:

Sea snake venom is highly myotoxic, causing significant muscle damage, which can lead to rhabdomyolysis and acute kidney injury.



Clinical features of myotoxic snake venom:

- Common in Sea Snakes: Myotoxic venoms are characteristic of particularly sea snakes.
- Muscle necrosis (First finding): Myotoxic venoms cause direct damage to muscle fibers (skeletal muscle), leading to muscle cell death (necrosis).
- Rhabdomyolysis : Breakdown of muscle tissue results in the release of muscle contents, such as myoglobin, into the bloodstream.
- Elevated Creatine Kinase (CK): A marker of muscle damage, CK levels are significantly elevated in cases of myotoxic envenomation.
- Myoglobinuria : The release of myoglobin can overwhelm the kidneys, leading to myoglobin in urine, which can cause acute kidney injury.
- Muscle Weakness : Progressive muscle pain and weakness are common symptoms due to muscle breakdown.

Tests for neurotoxic and vasculotoxic snake bite

Neurotoxic Test	Vasculotoxic Test
Single Breath Count Test Count numbers in a single breath. Repeat the counting test every 15 minutes. If it decreases, it indicates ongoing respiratory failure.	Whole Blood Clotting Test (WBCT) Put whole blood in a clean and dry test tube. Wait for 20 minutes and analyse the blood in the test tube. After 20 minutes if blood clots: No snake bite or not vasculotoxic venom. After 20 minutes if blood does not clot: Suggests Viper snake poisoning.

King cobra (Option A): Primarily neurotoxic.

Russell's Viper (Option C): Primarily vasculotoxic.

Banded krait (Option D): Primarily neurotoxic.

Solution for Question 9:

Correct Answer: D) Neurotoxic venom causes wet gangrene , while vasculotoxic venom causes dry gangrene.

Explanation:

Ophitoxaemia characterises the clinical spectrum of snakebite envenomation.

Clinical features

Neurotoxic snakes (Elapidae)	Vasculotoxic (Viper)
Causes muscular, weakness of legs and paralysis of muscles of face, throat and respiration.	Causes enzymatic destruction of cell walls and coagulation disorders causing hemolysis.
Ptosis: B/L drooping of eyelid -M/C	DIC - Disseminated Intravascular Coagulopathy causing renal failure.
Local reaction: Minimal (Option A)	Local reaction: Extensive
Tongue paralysis seen (Option B)	Tongue not affected
Descending paralysis, Respiratory paralysis with dysphagia, dysphonia and dysarthria. (Option C)	Circulatory failure
Wet gangrene, early onset (Option D)	Dry gangrene, late onset
Pulse: Initially normal, then irregular.	Pulse: weak and irregular
Coagulation not affected	Coagulation with hemorrhagic manifestations
Normal pupils	Pupils dilated
Hypersalivation	Normal salivation
Staggering gait	Not affected specifically

Solution for Question 10:

Correct Answer: A) Small belly scales

Explanation:

Small belly scales are found in non-poisonous snakes

Poisonous and non-poisonous snakes

Poisonous snakes	Non-poisonous snakes
Head scales: Small, large scales seen with heat sensing pit antero-inferior to the eye.	Head scales: Large with certain exceptions.
Belly scales: Large and cover the entire breadth of belly.	Belly scales: Small like those on the back and do not cover the entire breadth.
Fangs: Long and canalized like a hypodermic needle.	Fangs: Short and solid.
Single row of scales.	Double rows of scales.
Tail compressed	Tail not markedly compressed.
Bite marks: Two fang marks, with or without small marks of other teeth.	Bite marks: Number of small teeth marks in row.
{{caption_text}}	

Features of common poisonous snakes (Identification and differences) and fatal doses

Snakes	Features and fatal doses
Common cobra	Head and neck: Hood present with spectacle marks Belly: Smooth scales Back: Spectacled white or yellow Colour: Brown/black/green Neurotoxic : Elapids Fatal dose(Dried form)= 15mg
King cobra	Head and neck: Hood present without spectacle marks Belly: Scales look shiny but dry to touch. Back: Yellow or black bands with broad chevron like markings. Colour: Yellow/black/green with white cross bands. Neurotoxic : Elapids Fatal dose(Dried form)= 12mg
Common krait	Head and neck: Covered with large shields. Belly: Creamy white Back: Single/double white bands with central row of hexagonal scales. Colour: Steel blue/black. Neurotoxic : Elapids Fatal dose(Dried form)= 2-5-6mg

Banded krait	Head and neck: Covered with large shields. Belly: Triangular in cross section. Back: Alternate black and yellowish bands. Colour: Steel blue/black. Neurotoxic : Elapids Fatal dose(Dried form)= 10 mg
Russell's viper	Head and neck: Flat triangular with distinct 'V' mark and small scales. Belly: White with broad plates. Back: Three rows of diamond shaped black spots. Colour: Brown/buff Vasculotoxic Fatal dose(Dried form)= 40 mg

Solution for Question 11:

Correct Answer: A) Prazosin

Explanation:

The image is of Indian red scorpion

The treatment of choice for Indian red scorpion sting (*Mesobuthus tamulus*) is prazosin.



Scorpion sting venom:

- Physical properties: Venom is clear, colorless, proteinous toxalbumin, having haemolytic and neurotoxic effects.

- Action: Venom is a potent autonomic stimulator, resulting in the release of massive amounts of catecholamines from the adrenals.

Signs and symptoms:

- Dysfunction of cranial nerve and hyperexcitability of skeletal muscles developed within hours.
- Local : Swelling, prominent, radiating pain, paraesthesia, and hyperesthesia which is accentuated by tapping on the affected area (Tap test)
- Systemic effects: Restlessness, blurred vision, abnormal eye movements, profuse sweating, slurred speech, muscular fasciculations, respiratory depression and pulmonary edema.
- Complications : Tachycardia, hypertension, arrhythmias, hyperthermia, rhabdomyolysis and acidosis.

Treatment:

- Limb is mobilised and pressure bandage is applied proximal to the site of sting.
- The site may be incised and washed with ammonia or borax.
- Prazosin therapy : Treatment of choice (1mg for adults, 500µg for children) is given orally and then after every 3 hours till extremities are warm, dry, and peripheral veins are visible.
- Scorpion antivenom (SAV) Specific antidote for scorpion venom.
- Calcium gluconate 10 ml of 10% solution slow IV given for pain, cramps and edema.
- Barbiturates given to sedate and control convulsion.
- Atropine to prevent pulmonary edema.

Post-mortem findings:

- Affected site is swollen
- Stinger area showing ecchymosis
- Pulmonary edema
- Myocardial infarction

Solution for Question 12:

Correct Answer: A) Stimulation of anterior horn cells of spinal cord

Explanation:

The given seeds are that of *Strychnos nux-vomica* and the poison is Strychnine which acts by stimulating all parts of the central nervous system, especially the anterior horn cells of the spinal cord

Strychnine:

Source	Seeds of <i>Strychnos nux-vomica</i> and other species of <i>Strychnos</i> plants found in India.	
Mechanism of action	Stimulates all parts of the central nervous system, especially the anterior horn cells. Greatly increases reflex excitability by removing normal inhibition of motor cell stimulation, causing widespread uncontrolled motor activity of the spinal cord (Option A) Slight stimuli like noise, light, or a breeze can cause violent generalized muscle spasms	
Differentiating Points	Strychnine Poisoning	Tetanus
1. History	History of poisoning	History of injury
2. Onset	Sudden onset	Gradual onset
3. Convulsions/Spasms	Generalized convulsions	Lockjaw (trismus)
4. Chest Movement	Chest fixed during convulsions	Not typically fixed
5. Relaxation Between Spasms	Complete relaxation between seizures	Relaxation between spasms is never complete
6. Duration Until Fatality	Ends fatally within a few hours	Death is rare within a few hours
7. Chemical Evidence	Chemical analysis reveals the poison	No specific chemical detection

Solution for Question 13:

Correct Answer: C) Severe hypocalcemia and calcium deposits in tissues

Explanation:

Hydrofluoric acid binds with calcium in the body, causing hypocalcemia and leading to the precipitation of calcium salts in tissues. This can be a characteristic finding in both clinical and postmortem evaluations of HF poisoning cases.

Hydrofluoric Acid (HF)	
Chemical Properties	Aqueous solution of hydrogen fluoride (HF) in concentrations from 0.5% (domestic) to 100% (industrial). Potent dehydrating agent; causes chemical burns.
Sources and Uses	Found in rust removers, detergents, cleaning agents. Used in glass etching, metal processing, fluorine compound production.
Mechanism of Toxicity	Corrosive: Hydrogen ions cause immediate tissue damage by coagulative necrosis. Fluoride ion toxicity: Fluoride ions penetrate tissues, leading to liquefactive necrosis and systemic toxicity.
Routes of Exposure	Dermal contact: Common, causes severe burns. Inhalation: Leads to respiratory distress. Ingestion: Causes gastrointestinal burns.
Symptoms	Dermal burns: Severe pain, redness, necrosis (may be delayed). Ocular injury: Conjunctivitis, corneal erosion, possible vision loss. Respiratory issues: Coughing, dyspnea, pulmonary edema (from inhalation). Gastrointestinal distress: Nausea, vomiting, abdominal pain (from ingestion). Neuromuscular: Confusion, seizures due to electrolyte imbalances (hypocalcemia, hyperkalemia).
Autopsy Findings	Severe burns in esophageal and gastric mucosa. Hyperemic lesions in the renal tubular system. Necrotic lesions in cardiac muscle.
Management	Promptly rinse the area with water or saline for at least 30 minutes. Apply 2.5% calcium gluconate gel every 10-15 minutes until pain subsides. Severe cases may require intravenous calcium gluconate, airway support, and monitoring of electrolytes in an ICU setting.

Cherry-red discoloration of the blood (Option A): This is typically associated with carbon monoxide poisoning, not HF poisoning.

Black tarry stool (melena) (Option B) : Melena is associated with upper gastrointestinal bleeding, which is unrelated to HF poisoning.

Widespread gastric mucosal erosion (Option D): This finding is more typical of ingestion of strong acids or corrosives but is not specific to HF poisoning.

Solution for Question 14:

Correct Answer: D) Gelatinous brown stomach content with swollen, congested kidneys showing necrosis

Explanation :

Oxalic acid causes renal damage, resulting in swollen, congested kidneys with tubular necrosis, along with gelatinous brown content in the stomach.

Gastric lavage with calcium lactate is indicated in oxalic acid poisoning.

Feature	HNO ₃ (Nitric Acid)	Oxalic Acid
Physical Properties	Colorless, pungent, choking, burning taste, fuming	Colorless, transparent, prismatic crystals, resembles MgSO ₄ and ZnSO ₄
Fatal Dose	10–15 ml	15–20 g
Fatal Period	12–24 hours	1–2 hours (may extend to 2 days–2 weeks with renal failure)
Signs and Symptoms	Oropharyngeal burns, pain in throat and epigastrium Dysphagia, dyspnea, dysphonia Eructation, vomiting Thirst	Fulminating: Immediate symptoms, coffee-ground vomiting, severe thirst, diarrhea, electrolyte imbalance, death within minutes Acute: Tingling, numbness, tetany, convulsions, renal failure Delayed: Nephritis, oxaluria, renal failure
Vomit	Acidic, contains altered blood and mucous	Coffee-ground vomitus
Teeth	Yellowish coating, not brittle	No specific change
Urination & Tenesmus	Present	Renal failure, scanty urine, oxaluria
Abdominal Findings	Abdominal distension due to gas Tenderness Stiffness (due to distension)	Burning pain, abdominal distension

Skin and Mucosa Erosion	Yellow discoloration over angles of mouth, lips, fingers	No external findings Mucosal bleaching, corrosion in mouth, esophagus, stomach
Perforation of Stomach	Less common	No perforation, gelatinous brown content in stomach (Option D) (Option B ruled out)
Postmortem Findings (Option A and D ruled out)	Yellow discoloration of skin Congestion of larynx, trachea Soft, friable stomach wall, ulcerated	Bleached mucosa (mouth, pharynx, esophagus) Red, velvety or blackish stomach wall Swollen, congested kidneys with necrosis
Medico-Legal Aspects	Accidental or suicidal , homicide rare	Usually accidental (mistaken for MgSO_4), rare in suicides or homicides, occasionally used as an abortifacient

Solution for Question 15:

Correct Answer: B) Blackish staining and necrosis of the lips and mouth

Explanation :

Sulfuric acid is highly corrosive and can cause characteristic blackish burns, particularly on the mucosal surfaces, such as the lips and mouth along with chalky white, brittle teeth. This is a key forensic finding in sulfuric acid ingestion cases.

Sulfuric Acid (H_2SO_4)	
Physical Properties: Colorless, odorless, with a burning taste; oily and non-fuming. Fatal Dose: 5–10 ml of concentrated acid. Fatal Period: 12–18 hours post-ingestion.	

Signs and Symptoms	Burning sensation and pain in the throat and epigastric region. Dysphagia, dyspnea, dysphonia Eructation and vomiting: Frequent belching and vomiting. Thirst: Present (worsened vomiting by drinking fluids) Vomitus: Strongly acidic, contains altered blood and mucous shreds. Teeth: Become chalky white and brittle. Constipation: Usually present. Urination: Suppressed. Tenesmus present Nature of stool: Presence of mucus and altered blood. Tenderness over the abdomen Abdominal stiffness due to peritonitis Abdominal distension: Uncommon. Skin and mucous membrane erosion: Especially on the mouth, lips, and fingers; blackening and excoriation occur. Stomach Perforation: Frequently seen.
Cause of Death	Shock, stomach perforation, peritonitis, laryngeal spasm, and malnutrition from esophageal stricture.
Postmortem Findings	Erosion of skin and mouth angles, black charring of the stomach, stomach perforation, and swelling in the liver and kidneys.
Perforation of stomach	Thinning of posterior wall of the stomach
Medico-Legal Aspects	Often occurs accidentally (mistaken for glycerin) or in suicide cases. Rarely associated with homicide. May be used as an abortifacient or for vitriolage (acid attacks).

- Bright red lividity (Option A): This is typically associated with carbon monoxide poisoning, not sulfuric acid poisoning.
- Greenish discoloration of the skin (Option C): This is more typical of certain copper compounds, not sulfuric acid poisoning..
- Yellowish staining and necrosis of the lips and mouth (Option D): This is typically associated with nitric acid poisoning.

Solution for Question 16:

Correct Answer: C) Only C and E

Explanation:

- Gastric lavage can be performed with caution for carbolic acid (Phenol) poisoning using lukewarm water, charcoal, olive oil, or magnesium sulfate. Care is taken to prevent further absorption of the corrosive substance.

- In oxalic acid poisoning gastric lavage with calcium lactate (2 tsp/lavage) is done

Type of Poison	Management
Mineral Acids (Sulphuric acid, Nitric acid, Hydrochloric acid)	Avoid gastric lavage/emetics. Dilute by giving milk or water. Administer 4 tablespoons of aluminum hydroxide gel. Provide demulcents (olive oil, milk, egg white, starch water, butter). Avoid bicarbonate or neutralizing agents. Prednisolone: 60 mg/day in divided doses. Treat shock with IV fluids, blood products (if bleeding), and antibiotics (if perforation is suspected). Tracheostomy for glottic edema. Nothing by mouth for a week, followed by IV nutrients, gradually reintroducing liquids, soft foods, and regular diet. Morphine for pain relief. Symptomatic treatment.
Organic Acids	Oxalic Acid Gastric lavage with calcium lactate (2 tsp/lavage). Antidotes: Limewater, calcium lactate, calcium gluconate/chloride (150 mg/kg orally) to form insoluble calcium oxalate for excretion. Calcium gluconate: 10 ml of 10% solution IV frequently. Parathyroid extracts: 100 units IM. Bowel management: Demulcent drinks, enemas, purgatives. Hemodialysis/exchange transfusion if needed. Symptomatic treatment.
Carbolic Acid (Phenol) Gastric lavage (done with caution) with lukewarm water, charcoal, olive oil, MgSO ₄ , or Na ₂ SO ₄ . Leave medicinal liquid paraffin or 30 g MgSO ₄ in the stomach after lavage. Provide demulcents. IV saline with 7 g NaHCO ₃ /l to manage circulatory depression, dilute carbolic acid and promote diuresis. If phenol contacts the skin: Remove contaminated clothing, wash with soap and water, and apply olive oil/methylated spirit. Emetics have a limited role.	
Alkalis	Gastric lavage/emetics contraindicated. Immediately dilute with water. Give demulcents (egg white, milk). Administer O ₂ inhalation in case of ammonia vapor poisoning. Maintain airway, tracheostomy if needed. Administer parenteral analgesics and antibiotics. Steroids contraindicated if esophageal perforation is present.

Note:

- Alkali will cause liquefactive necrosis-more toxic
- Acid will cause coagulative necrosis

Solution for Question 17:

Correct Answer: A) Causes extrasystoles

Explanation:

The given plant is Digitalis Purpurea, which In toxic doses, increases cardiac excitability and causes extrasystoles.

Digitalis Purpurea (Foxglove):

Source	Roots, leaves, and seeds contain several glycosides: digitoxin, digitalin, digitalein, and digitonin (most poisonous).
Mechanism of action	Glycosides act on the heart , improving heart function by prolonging the diastolic period. In toxic doses, they increase excitability and cause extrasystoles.
Treatment	ECG monitoring. Gastric lavage with tannic acid solution. Activated charcoal. Purgatives if needed. Atropine for bradycardia. Potassium chloride to reduce extrasystoles. Lignocaine, novocaine, or propranolol for cardiac arrhythmias. Trisodium EDTA may help lower serum calcium. Symptomatic treatment.

Solution for Question 18:

Correct Answer: B) All parts

Explanation:

Given is the image of Cerbera Thevetia (Yellow Oleander, Pila Kaner), all of its parts are poisonous.

Yellow Oleander:

Source	All parts of the plant are poisonous. (Option B)
Active Principles	Glycosides: Thevetin, thevotoxin, cerberin, peruvoside. Thevetin: A powerful cardiac poison. Thevotoxin: Less toxic than thevetin, similar to digitalis glycosides. Cerberin: Acts like strychnine.
Symptoms	Local: Sap may cause inflammation. Ingestion: Burning sensation in the mouth, tingling of the tongue, dry throat, vomiting, diarrhea, headache, dizziness. Later stages: Dilated pupils, drowsiness, loss of muscular power, rapid/weak/irregular pulse, falling blood pressure, heart block. Death: Due to peripheral circulatory failure.
Treatment	Gastric lavage and single-dose activated charcoal. IV sodium lactate molar solution and 5% glucose to combat acidosis. Atropine, Adrenaline, and Noradrenaline to counteract heart block and low blood pressure. Digoxin-specific antibody fragments to treat life-threatening cardiac arrhythmias. Symptomatic treatment.

Solution for Question 19:

Correct Answer: B) Easily detected postmortem

Explanation:

Aconite is difficult to detect postmortem

Aconite:





Species	Aconitum napellus, Aconitum ferox
Synonyms	Monk's hood , mitha zaher, bish, wolf's bane, women's bane, devil's helmet, or blue rocket
Source	All parts of the plant are poisonous (Option A ruled out) Root and root tubers are the most potent (often mistaken for horseradish root)
Active Principles	Aconitine, pseudo-aconitine, indaconitine, picroaconitine, aconine.
Mechanism of Action	Affects voltage-sensitive sodium channels, stimulates then paralyzes sensory nerves, secretory nerves, CNS, and myocardium. (Option C ruled out)
Symptoms (by system)	GIT: Nausea, vomiting, salivation, abdominal pain, diarrhea, burning/tingling of mouth, tongue, throat, followed by numbness. CVS: Hypotension, bradycardia , chest pain, ventricular tachycardia/fibrillation, slow/feeble pulse. CNS: Vertigo, restlessness, headache, giddiness. MS: Muscle weakness, twitching, spasms. RS: Slow, labored, shallow breathing. Ocular: Alternating pupil dilation/contraction (hippus) , diplopia, impaired vision. Others: Subnormal temperature, cold skin, yellow-green vision, tinnitus, paralysis of facial muscles. Death: Respiratory failure or ventricular fibrillation.

Fatal Dose	Root: 1-2 g. (Option D ruled out)
Treatment	No specific antidote. Gastric lavage with tannic acid/activated charcoal. Inotropic therapy for hypotension, atropine for bradycardia. Ventricular arrhythmia treated with amiodarone, flecainide; cardiopulmonary bypass for refractory cases.
Medico-legal Aspects	Ideal homicidal poison: cheap, easily available, small lethal dose, short fatal period, difficult to detect postmortem. (Option B) Accidental poisoning: mistaken for horseradish

Solution for Question 20:

Correct Answer: A) Rigor mortis sets in late

Explanation:

In the case of Strychnine poisoning, rigor mortis sets in early

Clinical Features	Opisthotonos - arched body with only the back of head and heels touching the ground: Emprosthotonos- forward curve Pleurothotonus- sideways curve Risus sardonicus- fixed grin:
Treatment	- Immediate Care: Place in a dark, quiet room. - Anesthesia: Chloroform or intravenous barbiturate if spasms allow. - Adsorption: Administer animal charcoal to absorb free strychnine, or tannic acid if charcoal is unavailable. - Antidotes: Intravenous barbiturates to stop convulsions. - Muscle relaxants: Mephenesin, intravenous diazepam for anticonvulsant effects.
Autopsy Findings	- Rigor mortis: Sets in almost immediately (Option A) and passes off early. - Stomach contents: Remains of seeds may be found. - Delayed putrefaction. (Option B ruled out) - Toxicological analysis: Strychnine can be detected in putrefied bodies and viscera (Option C ruled out) ; brain and spinal cord should be preserved for analysis. (Option D ruled out)

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Forensic Toxicology 2

1. A 32-year-old male weighing 100 kg was arrested for disorderly conduct at a market. Upon questioning, he confesses to consuming 100 grams of alcohol. Using the Widmark formula, calculate his Blood Alcohol Concentration (BAC). (The Widmark constant for men is 0.68)

- A. 167 mg%
- B. 127 mg%
- C. 137 mg%
- D. 147 mg%

2. A 24-year-old male with no history of violent behavior suddenly becomes agitated and aggressive after consuming a high dose of cannabis. He attacks several people without warning and later has no recollection of the incident. This condition, triggered by excessive cannabis use, is known as?

- A. Amotivational syndrome
- B. Hashish psychosis
- C. Run amok
- D. Korsakoff psychosis

3. A 30 year old male is brought to the emergency department by friends after acting erratically in public, hallucinating, and showing aggressive behaviour. He is agitated and confused and has dilated pupils and a rapid heart rate. While his vital signs are stable, he becomes more aggressive when approached. Which substance might be abused in this case?

- A. Hemlock

- B. Chloral hydrate
 - C. Cocaine
 - D. Phencyclidine
-

4. A 28 year old man is brought to the emergency department in an altered mental state after reportedly using a speedball for a more intense high. His friends mention he often combines a powerful stimulant with another drug to enhance his experience. What is the composition of a speedball that he is likely using?

- A. Heroin + LSD
 - B. Cocaine + Morphine
 - C. Cocaine + LSD
 - D. Heroin + Cocaine
-

5. A 35 year old man who works in a jewelry factory is brought to the emergency department drowsy, with sudden irritability, confusion, and vomiting over 24 hours. Examination shows a bluish line on his gums and pallor, and blood tests confirm acute heavy metal poisoning with cerebral oedema on the CT scan. Along with maintenance of airway, breathing, and circulation, what is the immediate management?

- A. Calcium gluconate
 - B. DMSA
 - C. D-penicillamine
 - D. Dimercaprol
-

6. A 6 year old girl was brought to the clinic by her parents, who reported symptoms including irritability, stomach pain, difficulty concentrating, decreased appetite, and fatigue. The examination showed signs of anaemia and facial circumoral pallor, leading the doctor to suspect heavy metal poisoning. What is the most common route of exposure to this poison in children?

- A. Inhalation of lead-contaminated air
- B. Ingestion of lead-contaminated water
- C. Ingestion of lead dust or paint chips
- D. Absorption through skin

7. A 30 year old construction manager presented with persistent abdominal pain, nausea, fatigue, and episodes of vomiting and diarrhoea over several months. His symptoms progressively worsened, and he developed skin changes, including darkening and hyperkeratosis on his palms and soles. Blood tests confirmed heavy metal poisoning. Which of the following is the preferred test and treatment for this condition?

- A. Neutron activation analysis and Naloxone
- B. Marsh test and BAL
- C. Atomic absorption spectroscopy and BAL
- D. Gutzeit test and Naloxone

8. A 39 year old man presents to the emergency department after consuming homemade alcohol, showing headache, dizziness, nausea, vomiting, blurred vision, confusion, and metabolic acidosis. Methanol poisoning is suspected, and treatment with fomepizole is initiated. Which mechanism explains how fomepizole mitigates methanol toxicity?

- A. It is a competitive inhibitor to Methanol
- B. Increases the excretion of Formic acid

- C. Used in treating Metabolic acidosis
 - D. It inhibits alcohol dehydrogenase
-

9. A 35 year old male presents with confusion, disorientation, nystagmus and impaired vision with a history of chronic alcoholism. On examination, he had ataxia and ophthalmoplegia. What is the likely condition that is causing these symptoms?

- A. Mallory-Weiss Syndrome
 - B. MarChiafava's Syndrome
 - C. Korsakoff Psychosis
 - D. Wernicke's Encephalopathy
-

10. A 45 year old man with a history of chronic alcohol use presents to the emergency department after abrupt cessation of drinking for three days. He exhibits agitation and tremors. On examination, he has diaphoresis. Which of the following symptoms may also occur in alcohol withdrawal 72 hours after cessation?

- A. Hypotension
 - B. Severe Bradycardia
 - C. Sedation
 - D. Disorientation
-

11. A 28 year old male was brought to the ED after multiple episodes of vomiting, confusion, and irritability following alcohol consumption at a party. He had a flushed face, sensory impairment, and a heart rate of 105. He was suspected to be in the 2nd phase of intoxication. What is the likely blood alcohol concentration (BAC)?

- A. 100 mg%
 - B. 300 mg%
 - C. 80 mg%
 - D. 150 mg%
-

12. A 25 year old male presents to the emergency department exhibiting severe agitation, hallucinations, and a heightened sense of paranoia after attending a rave. He reports feelings of overwhelming anxiety and visual distortions, claiming he sees colours and shapes that aren't there. Based on his symptoms, which substance is most likely responsible for his bad trip?

- A. MDMA
 - B. Amphetamine
 - C. LSD
 - D. Heroin
-

13. A 5-year-old child presents with developmental delay, irritability, and abdominal pain. Laboratory investigations for lead poisoning are ordered. Which of the following is the definitive indicator of lead poisoning?

- A. Microcytic-hypochromic anaemia with basophilic stippling
 - B. Increased levels of urinary coproporphyrin (>150 mcg/litre)
 - C. Blood lead level >40 micrograms/100 ml
 - D. Elevated Free Erythrocyte Protoporphyrin (FEP) levels
-

14. A 5-year-old girl is brought to the paediatrician due to irritability, decreased appetite, abdominal pain, and difficulty concentrating. She exhibits signs of

anaemia and pallor, as depicted in the image below. Blood tests confirm heavy metal poisoning. Which clinical manifestations are not typically associated with this condition?



- A. Burtonian lines
- B. Basophilic stippling
- C. Phossy jaw
- D. Calcium deposition in bone ends

15. A 4-year-old boy is brought to the clinic for irritability, poor appetite, and stomach pain. The family recently moved into an old house that is under renovation. Examination reveals pallor and a blue line on the gums. Blood smear findings are noted. What is the most likely finding on the blood smear in this child with suspected lead poisoning?

- A. Target cell
- B. Basophilic stippling
- C. Spherocytes
- D. Rouleaux formation

16. A 32-year-old male presents to the emergency department after ingesting the seeds given in the image. He has severe abdominal pain, bloody vomiting, dizziness, and a rapid, weak pulse. What is the mechanism of action of the toxic compound present in the seeds?



- A. Inhibits calcium ion channels, causing cell death.
- B. Blocks RNA polymerase, preventing protein synthesis
- C. Binds to ribosomal RNA, inhibiting protein synthesis.
- D. Disrupts mitochondrial function, leading to energy failure.

17. A 45-year-old construction worker presented with severe health issues over several weeks, including abdominal pain, persistent nausea, throat pain, fatigue, and intermittent diarrhoea. During hospitalisation, he developed confusion, slurred speech, and distinctive skin changes (shown in the image below) indicative of heavy metal poisoning. Which of the following clinical features is commonly associated with this condition?



- A.
- B.
- C.
- D.

18. A 40-year-old man was found dead in his home with no signs of forced entry. Family members reported he had been unwell for weeks, experiencing throat and abdominal pain, vomiting, and purging. Autopsy findings indicated emaciation, keratosis, and white streaks on his nails, suggesting heavy metal poisoning. Which of the following postmortem findings is not true in this case?

- A. Rigor mortis sets in quickly
 - B. Red velvety mucosa in the stomach
 - C. Subendocardial hemorrhage
 - D. Multiple haemorrhages in the stomach
-

19. A 30 year old male is brought to the emergency department after experiencing severe anxiety and chest pain following a night of partying. He reports using cocaine, and the medical team notes his elevated blood pressure and signs of agitation. Which of the following is the most likely complication of cocaine use?

- A. Respiratory depression
- B. Cardiac arrhythmia
- C. Urinary retention
- D. Optic neuritis and atrophy

20. Match the following drugs with their commonly known local names. Drugs Local names 1. Cocaine a. Smack 2. MDMA b. Weed 3. Heroin c. Molly 4. Cannabis d. White lady 5. LSD e. Acid

Drugs	Local names
1. Cocaine	a. Smack
2. MDMA	b. Weed
3. Heroin	c. Molly
4. Cannabis	d. White lady
5. LSD	e. Acid

- A. 1-c, 2-e, 3-a, 4-d, 5-b
- B. 1-d, 2-c, 3-a, 4-b, 5-e
- C. 1-c, 2-a, 3-b, 4-e, 5-d
- D. 1-b, 2-d, 3-e, 4-a, 5-c

21. Which of the following options correctly orders the potency of cannabis products from most potent to least potent?

- A. Charas > Hashish oil > Bhang > Ganja
 - B. Hashish oil > Bhang > Charas > Ganja
 - C. Hashish oil > Charas > Ganja > Bhang
 - D. Charas > Bhang > Ganja > Hashish oil
-

22. A 30-year-old man presents to the emergency department with symptoms of anticholinergic toxicity after ingesting an herbal preparation containing Datura seeds. Upon examination, he shows signs of confusion and dilated pupils. Which of the following statements about the seed morphology of Datura and Capsicum is incorrect?

- A. Datura seeds are large and thick, while capsicum seeds are small and thin
 - B. Datura seeds are kidney-shaped, while capsicum seeds are rounded
 - C. Datura seed has a smooth surface, while capsicum seeds have numerous depressions
 - D. Datura seeds are dark yellowish-brown, while capsicum seeds are pale yellow
-

23. A 24-year-old male presents to the emergency room after ingesting kidney-shaped seeds while camping. Upon examination, he exhibits dry mouth, dilated pupils, blurred vision, flushed skin, tachycardia, hallucinations, agitation, and disorientation. What is the most appropriate antidote for this poisoning?

- A. Naloxone
 - B. Physostigmine
 - C. Pyridostigmine
 - D. Fomepizole
-

24. A 40-year-old woman presents with blistering and severe itching on her arms after applying the juice from a black seed, as shown in the image. She also experiences nausea and abdominal pain. Which of the following statements is not true?



- A. Often used in malingering for artificial bruises.
- B. The toxic principle is Bhilawanol.
- C. Commonly used for Arrow and Cattle poisoning.
- D. Large oral doses can be potentially fatal

25. A 25-year-old female presents to the ER after ingesting seeds from a plant, as shown in the image. She reports severe abdominal pain, nausea, and bloody diarrhoea. Upon examination, she exhibits weakness, tachycardia, and dilated pupils. What is the primary active principle responsible for the toxic effects seen in this patient?



- A. Abrin
- B. Calotropin
- C. Crotin
- D. Ricin

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	4
Question 4	4
Question 5	4
Question 6	3
Question 7	3
Question 8	4
Question 9	4
Question 10	4
Question 11	4
Question 12	3
Question 13	3
Question 14	3
Question 15	2
Question 16	2
Question 17	3
Question 18	1

Question 19	2
Question 20	2
Question 21	3
Question 22	3
Question 23	2
Question 24	3
Question 25	1

Solution for Question 1:

Correct Answer: D) 147 mg%

Explanation:

Widmark Formula is $A = PCR$

A - Weight of alcohol P - Body weight C - Blood Alcohol Concentration R - is a constant (Fat- water ratio), Men - 0.68, Women - 0.55

- Alcohol Consumed = 100 g
- Body Weight = 100 kg
- r (Widmark factor for men) = 0.68
- Now, substitute the values into the formula: $C = A / PR = 100g / 1,00,000g \times 0.68 = 100 / 68000 \text{ g/ml} = 0.00147 \text{ g/ml}$. $0.00147 \text{ g/ml} = 0.00147 / 100 \text{ g\%} = 0.147 \text{ g\%} = 0.147 / 1000 \text{ mg\%} = 147 \text{ mg\%}$
- $C = A / PR = 100g / 1,00,000g \times 0.68 = 100 / 68000 \text{ g/ml} = 0.00147 \text{ g/ml}$.
- $0.00147 \text{ g/ml} = 0.00147 / 100 \text{ g\%} = 0.147 \text{ g\%} = 0.147 / 1000 \text{ mg\%} = 147 \text{ mg\%}$
- Widmark's Formula is widely used in Blood alcohol analysis for multiple purposes using the type of alcohol, Sex of the person and body weight.
- The widmark formula for Urine = $\frac{3}{4}$ PCR

- Blood is the commonly used means to calculate the level of alcohol consumption and intoxication, other than Breath, Urine, and Saliva.
- Preservatives used while collecting blood are Sodium fluoride Potassium oxalate
- Sodium fluoride
- Potassium oxalate
- Other ways to identify blood alcohol levels are
- $C = A / PR = 100g / 1,00,000g \times 0.68 = 100 / 68000 \text{ g/ml} = 0.00147 \text{ g/ml}$.
- $0.00147 \text{ g/ml} = 0.00147 / 100 \text{ g\%} = 0.147 \text{ g\%} = 0.147 / 1000 \text{ mg\%} = 147 \text{ mg\%}$
- Sodium fluoride
- Potassium oxalate

Kozelka Test	Qualitative tests (Macro method)
Hena Test	
Cavett Test	Qualitative test (Micro method)
Gas Chromatography	Used in Medicolegal cases Qualitative + Quantitative test.
Breathalyzer Test	It follows Henry's law and uses infrared rays to detect alcohol vapour in exhaled breath, converting this measurement into an estimated blood alcohol concentration. The current legal level of alcohol for driving is 30 mg% . (Motor Vehicle Act)

Solution for Question 2:

Correct Answer: C) Run amok

Explanation:

- Run Amok can occur after consuming a high dose of cannabis and is marked by a sudden psychological disturbance.
- It starts with a period of depression, followed by violent outbursts and attempts to harm others.
- The episode typically ends with the individual either surrendering or committing suicide.

Effects of cannabis use	
-------------------------	--

Acute	Chronic
Psychiatric symptoms Euphoria (increased subjective experiences and disorientation) Disturbed thought pattern Altered concepts of time and space, lapse of attention Impaired immediate memory function Well-being and self-confidence Rapid and impaired speech Increased awareness of stimuli Fear of dying Physical symptoms Increased appetite and thirst Nausea and dizziness Ataxia or tremors Conjunctival injection Somnolence	Hashish psychosis (Option B) Auditory and visual hallucination Delusion of persecution Run amok begins with a period of depression, followed by violent outbursts aimed at harming others, and ends with either surrender or suicide. Amotivational syndrome (Option A) Loss of interest in day-to-day activity Lethargy Loss of age-appropriate behaviour Other effects: Decreased testosterone levels leading to gynecomastia, low sperm count

Korsakoff psychosis (Option D) : It is associated with thiamine deficiency and is usually seen in chronic alcoholics. It is characterized by confabulation, anterograde, and retrograde amnesia.

Solution for Question 3:

Correct Answer: D) Phencyclidine

Explanation:

The scenario suggests features consistent with the abuse of "Love Boat," which refers to Cannabis dipped in Phencyclidine (PCP) .

Source	Synthesised from piperazine, cyclohexanone, and potassium cyanide
Street names	Angel dust.
Mode of intake	Smoking; Insufflation; Ingestion; Parenteral (rare)
Combination	Mixed with parsley, mint, oregano, or marijuana. Tragic magico: Crack (cocaine) is dipped in PCP and smoked Love boat: Cannabis is dipped in PCP and smoked.
Mechanism	Blocking NMDA receptor-mediated neurotransmission and presynaptic monoamine reuptake, increasing dopamine and noradrenaline synaptic levels.

Clinical features	Acute psychosis and agitation (violent behaviour). PCP: Symptoms like confusion, disorientation, amnesia, catatonia, dystonic movements, mydriasis, tachycardia, hallucinations (auditory and visual), blank stare, nystagmus, blurred vision, etc. Cannabis: Symptoms like euphoria, disorientation, hallucinations (visual and auditory) , sedation, dysphoria, tachycardia, palpitations, hypotension, appetite stimulation, etc.
Management	Supportive Care and Monitoring Sedation with Benzodiazepines : Administer benzodiazepines (e.g., lorazepam or diazepam) to control agitation, anxiety, and seizures. Avoid antipsychotics as they may worsen agitation. Hydration and Renal Support : Provide intravenous fluids to maintain hydration and support renal function, especially in cases of myoglobinuria or acute renal failure.

Hemlock (*Conium maculatum*) (Option A) is a potent nerve poison that affects motor nerves. All parts of the plant are toxic. Specific manifestations include muscle weakness, paralysis, respiratory failure, confusion, miosis, and bradycardia. Historically, it was used to execute Socrates.



Chloral hydrate (Option B) , also known as a dry wine, Mickey Finn, or knockout drops , is a colourless crystalline substance with a distinct acrid odour. It is often administered in combination with alcohol and can cause sedation, dizziness, and respiratory depression. Overdose may lead to confusion, hypotension, and coma.



Cocaine (Option C) is a potent central nervous system stimulant derived from the leaves of the coca plant (*Erythroxylum coca*), commonly abused for its euphoric effects. Specific manifestations include increased heart rate, hypertension, dilated pupils, anxiety, and severe complications such as arrhythmias, seizures, and respiratory failure.



Amyl nitrite can be used to control heart rate and blood pressure.

Solution for Question 4:

Correct Answer: D) Heroin + Cocaine

Explanation:

The scenario describes the effects of Speedball addiction, which is a combination of Heroin and Cocaine.

Composition	Heroin + Cocaine
Route	Injected, Smoked (Moon Rock)
Effects	Produce a double effect of the initial high "kick" of cocaine, followed by the subsequent euphoric "rush" of heroin.
Clinical features	The presence of needle marks along with the features of: Cocaine: Tachypnea, tachycardia, restlessness, anxiety, loss of appetite, insomnia, hypertension, tremors, fever, mydriasis, and hallucinations. Heroin: It is an opioid & causes reduced respiration, hypotension, bradycardia, delirium, hallucinations, slurred speech, sedation, coma, ataxia, muscle weakness, hypothermia, miosis, and needle marks.
Fatal dose	50 mg
Fatal period	12 hours

LSD is a synthetic derivative of ergot alkaloids from the *Claviceps purpurea* fungus, is ingested and often leads to symptoms such as mydriasis, vertigo, tachycardia, hypertension, sweating, piloerection, hyperthermia, tachypnea, euphoria, and vivid hallucinations

Morphine derived from opium, is a natural alkaloid known for its effects, including orthostatic hypotension, respiratory depression, and pain relief for cancer patients. It may also cause bronchospasm, hypothermia, increased intracranial pressure, constipation, euphoria, and sedation. Naloxone and naltrexone are used as antidotes.

Solution for Question 5:

Correct Answer: D) Dimercaprol

Explanation:

The case is diagnosed as severe acute lead poisoning with encephalopathy, where BAL (Dimercaprol) is administered immediately alongside supportive measures.

Management of Lead Poisoning	
Severe acute poisoning with encephalopathy	BAL or Dimercaprol (administered immediately) (Option D) CaNa ₂ EDTA (calcium disodium edetate) (contraindicated in renal failure) For Cerebral edema: Mannitol Furosemide Corticosteroids (Dexamethasone) Continue until the patient is asymptomatic, then start oral chelation (D-penicillamine or DMSA).
Severe acute poisoning without encephalopathy (Blood Lead level >70 mcg/100 ml)	BAL EDTA When BL <40 mcg/100 ml Discontinue BAL Continue EDTA Change to oral chelation (D-penicillamine or DMSA).
Moderate poisoning (Blood Lead level: 45-70 mcg/100 ml)	EDTA When BL <40 mcg/100ml: Oral chelation (D-penicillamine or DMSA)
Mild poisoning (Blood Lead level: 20-35 mcg/100 ml)	D-penicillamine (Option C) or DMSA (dimercaptosuccinic acid) (Option B)
Supportive measures	Thiamine Calcium gluconate (Option A) Magnesium sulphate DMSA (succimer)

- BAL (dimercaprol) and EDTA are used together because EDTA can redistribute lead to the brain, while BAL helps to prevent this redistribution.

Solution for Question 6:

Correct Answer: C) Ingestion of lead dust or paint chips

Explanation:

The clinical features, including circumoral pallor, irritability, abdominal pain, and fatigue, suggest a diagnosis of lead poisoning. In children, the most common route of exposure is ingestion of lead dust or paint.

Routes of Lead Poisoning

Route	Example
Inhalational (Option A)	Plumbing factories (most common) Pipe factories Paint factories
Oral (ingestion)	Paint (most common route in children) (Option B) Water (contaminated) (Option C)
Through skin (Option D)	Absorption through the scalp (sindoor) In chemical factory workers.

Types of Lead Compounds

Compounds	Uses
Lead acetate (sugar of lead)	Therapeutics: used as an astringent and topical sedative for sprains previously. (most common)
Lead Sulfide	Collyrium (Surma) (least common)
Lead tetroxide	Vermillion (sindoor)
Tetraethyl lead	Antiknock agent (mixed with petrol to prevent detonation in internal combustion engines) Causes encephalopathy
Lead carbonate (white lead)	Paints
Lead oxide (litharge)	Glazing of pottery and enamelware

Solution for Question 7:

Correct Answer: C) Atomic absorption spectroscopy and BAL

Explanation:

- The scenario presents a case of Arsenic Poisoning , which can be detected using atomic absorption spectrometry and neutron activation analysis.
- Management involves the use of BAL (dimercaprol) , while Naloxone serves as the antidote for opioid poisoning.

Investigations of Arsenic Poisoning

Tests	The following tests estimate the concentration of arsenic in hair, bone, and nails: Neutron activation analysis (NAA) (Option A) Atomic absorption spectroscopy (AAS) (Option C) The following tests are obsolete now: Marsh test (Option B) Reinsch test Gutzeit test (Option D)
Urine levels	>50 microns in 24 hours indicates arsenic poisoning (Best)
Blood level	Serum arsenic > 0.9 µg/dl
Hair level	Best sample: Pubic hair (used for long-term exposure detection).
Radiography	Abdominal X-ray : May show arsenic presence in the gastrointestinal tract in cases of acute poisoning.
Others	CBC , serum electrolytes, liver and renal function tests. ECG and continuous cardiac monitoring in symptomatic patients. Chest X-ray : For severe poisoning or pulmonary involvement. Monitoring and medical surveillance are required for employees exposed to arsenic.

Management of Arsenic Poisoning

Supportive treatment	Gastric lavage, airway, breathing, and circulation management (intravenous fluids, cardiac monitoring, etc.).
Medical treatment	Freshly prepared hydrated ferric oxide (chemical antidote)
Chelating agents	BAL (Dimercaprol) (Option C) DMSA (Dimercapto succinic acid) or DMPS (Dimercapto propane sulfonic acid) Penicillamine
Other measures	Purgatives like Castor oil or Magnesium sulfate (remove unabsorbed poison from the intestine). Glucose-saline with sodium bicarbonate. Hemodialysis or exchange transfusion.

Solution for Question 8:

Correct Answer: D) It inhibits alcohol dehydrogenase

Explanation:

- Fomepizole, also known as 4-methylpyrazole, is a competitive inhibitor of alcohol dehydrogenase .
- It works by preventing the formation of formaldehyde and formic acid, which are toxic metabolites of methanol.

Methanol Poisoning	
Other names for methanol	Wood Alcohol/ Illicit Alcohol
Absorption and Metabolism	Absorbed through the stomach, intestine, also lungs and skin Alcohol dehydrogenase helps in forming Formaldehyde from Methanol Formic acid from Formaldehyde
Action	It is a CNS depressant and causes high serum osmolality Formic acid is responsible for Metabolic acidosis and Retinal Toxicity
Fatal Dose and Fatal Period	60- 200 ml and 24 - 36 hours
Symptoms	Initially similar to ethanol poisoning Severe Abdominal cramps (early symptom) Neck Stiffness Vertigo Dilated and fixed pupils Muscle Weakness Snowfield Vision and Optic neuritis Cardiac and CNS depression Coma Severe Metabolic Acidosis (Common cause of death)
Treatment	Gastric Lavage with 5% bicarbonate solution Activated Charcoal Antidote : Ethanol - Competitive inhibitor to Methanol Fomepizole - Competitive inhibitor of alcohol dehydrogenase Folinic or Folic acid : Increases excretion of Formic acid Sodium Bicarbonate: To manage acidosis Hemodialysis: Treatment of choice in severe poisoning
Postmortem Findings	Marked Cyanosis Absent postmortem clotting of blood Hemorrhagic necrosis of putamen
Hooch Tragedy is mass poisonings and deaths in India from consuming toxic illicit alcohol in the 2000s.	

Solution for Question 9:

Correct Answer: D) Wernicke's Encephalopathy

Explanation:

Wernicke's encephalopathy is characterised by a triad of Disorientation, ophthalmoplegia, and polyneuropathy (ataxia) due to thiamine (vitamin B1) deficiency, commonly seen in chronic alcoholics.

Chronic Alcoholism Manifestations	
Wernicke's Encephalopathy	Caused by Thiamine Deficiency Symptoms Global confusion, ataxia and Apathy (early symptoms) Polyneuropathy Ophthalmoplegia Disorientation, recent memory loss and loss of attention span Rx - IV Thiamine followed by oral thiamine.
Korsakoff Psychosis (Option C)	Caused by untreated Wernicke's Encephalopathy Symptoms Confabulation Retrograde and Anterograde Amnesia Difficulty in learning new things Severe Disorientation. Rx - IV Thiamine followed by oral thiamine
MarChiafava's Syndrome (Option B)	Caused by degeneration of Corpus callosum Symptoms Intellectual Disorientation Epilepsy, Ataxia Dysarthria Hallucinations Spastic paralysis of the limb Dementia Rx: Thiamine supplementation (especially in alcohol-related cases), antiepileptic drugs for seizures, and supportive therapies like physical and speech therapy for motor and speech deficits. Psychiatric care may be needed for hallucinations and cognitive impairment.
Alcohol Paranoia	Paranoid delusions are present with no hallucinations
Mallory Weiss Syndrome (Option A)	It involves tears in the esophageal mucosa caused by severe vomiting or retching, which can lead to gastrointestinal bleeding and mediastinitis.

Solution for Question 10:

Correct Answer: D) Disorientation

Explanation:

The above-mentioned symptoms and timeline align with Delirium Tremens, which is common 2 to 4 days after complete cessation of alcohol in chronic alcoholics.

Alcohol Withdrawal Symptoms		
Symptom	Time since the last drink	Explanation

Tremors	6 - 8 hours	First symptoms to occur Tremors affecting hands, legs and trunk
Hallucination	< 48 hours	Visual and Auditory Hallucinations Rx: Chlorpromazine
Seizures	> 1 day	Tonic-Clonic seizures Rum - fits Sometimes, it can happen when the person is drinking also
Delirium Tremens	2 - 4 days	It can happen due to sudden excessive consumption, stoppage, sudden injury, or acute infection in chronic alcoholics. It is a severe condition Disorientation to time, place, and person Insomnia (Option C) Restlessness, agitation, confusion, and temporary memory loss Coarse tremors (face, hand, tongue) Tachycardia, Tachypnea, Hypertension (Options A and B) (Autonomic Hyperactivity) Hallucinations are visual, auditory, and tactile. Rarely causes death due to shock, cardiac failure, or cerebral oedema IPC 84 - Crime is not punishable under this condition Rx: Diazepam (Agitation) Chlordiazepoxide (Withdrawal) Chlorpromazine (Sedation) Symptomatic management.

Solution for Question 11:

Correct Answer: D) 150 mg%

Explanation:

In the second phase of intoxication, i.e. Stage of Incoordination, the BAC is usually between 150 mg% to 250 mg%.

Stages of intoxication		
Stages	BAC in mg%	Symptoms

Excitement	30 - 150	There is slight impairment in the cognitive, sensory and motor functions at 30 mg% Impaired reaction time at 50 mg% Alcohol gaze nystagmus, low pain sensitivity, and poor concentration at 80 - 100mg% Nausea and slight movement incoordination is seen from 100 - 150 mg%
Incoordination	150 - 250	Nausea and Vomiting Sensory, Fine, and Skilled motor impairment Elevated heart rate Flushed face
Coma	> 300	Markedly affected speech (Thick and slurred speech), as well as sensory and motor functions. Extreme dizziness and a sense of fall are present. Positive McEwan's sign : The patient's pupils are contracted, which dilate on a stimulus like a pinch or a slap and later return to the contracted phase. Usually recovers in 8 to 10 hours with a hangover.

Pharmacokinetics:

- Alcohol absorption starts in the mouth, followed by the oesophagus, which is minimal, primarily occurring in the stomach (20%) and mostly in the small intestine (80%)
- Factors affecting the absorption of alcohol:

Factors increasing Absorption	Factors decreasing Absorption
Mixing of drinks Drinking with Carbonated drinks (due to the increased surface area carrying alcohol) Warm drink Gastrectomy patients Drinking on an Empty stomach Chronic alcoholism	Drinking after a mixed meal Cool Drink Achlorhydria Chronic gastritis In Pyloric spasm With Drugs reducing gastric motility (Atropine and Benzene)

- The rate of alcohol clearance from blood is 15 mg%.

Solution for Question 12:

Correct Answer: C) LSD

Explanation:

The symptoms described, including hallucinations, paranoia, and severe agitation, are commonly associated with a "bad trip" on LSD (lysergic acid diethylamide) .

Lysergic acid diethylamide	
Source	Claviceps purpurea or rye fungus (ergot alkaloid).
Sold as	Acid blotter Paper stamp Tiny tablets (microdots) Gelatin squares (window panes) Liquid or powder
Mode	Oral intake; Smoke (rare); Parenteral (rare)
Mechanism	5HT-1 receptor agonist (increases serotonin level)
Fatal dose	14 mg
Clinical manifestations	Good trip : Psychotropic effect consisting of euphoria including visual hallucinations, illusions, sensations of synesthesia (seeing smell or hearing colours) Bad trip : Psychotropic effect consisting of an acute panic reaction as paranoid delusion, violent behaviour, anxiety, automatism, or depersonalisation. Symptoms like sweating, dry mouth, mydriasis, hypertension, tachycardia, mood change, etc. Bizarre perceptual changes: Faces and body parts appear distorted, objects undulate, sounds may be amplified and distorted, and colours seem brighter with halos around them. Flashback phenomenon: The individual experiences a recurrence of the emotional and psychological aspects from the previous LSD trip.
Management	Anti-anxiety drugs (Low dose) Benzodiazepines (Diazepam) Prolonged talking involves verbal reassurance to keep the person grounded and reduce paranoia. Psychotherapy

- MDMA (Methylene-dioxy-methamphetamine) (Option A) , commonly known as Ecstasy, Molly, or the party/club drug, is a stimulant with hallucinogenic properties that induce euphoria. Specific manifestations include increased energy, emotional warmth, heightened sensory perception, and distorted time perception. It is often used in social settings like raves or clubs for its mood-enhancing effects.
- Amphetamine (Option B) is a CNS and cardiac stimulant that increases dopamine and norepinephrine levels, leading to euphoria. Clinical features include tachycardia, hypertension, seizures, hyperpyrexia, tremors, and pupillary dilatation. Psychiatric effects include anxiety, irritability, panic, and insomnia.
- Heroin (Diacetylmorphine) (Option D) , also known as Smack or Dope, causes euphoria, respiratory depression, hypotension, bradycardia , sedation, miosis , and coma . Other key effects include delirium, hallucinations , and muscle weakness .

Solution for Question 13:

Correct Answer: C) Blood lead level >40 micrograms/100 ml

Explanation:

- Blood lead level >40 micrograms/100 ml is the definitive diagnostic criterion for lead poisoning, indicating significant exposure requiring intervention.
- Microcytic-hypochromic anaemia with basophilic stippling (Option A), Increased urinary Coproporphyrin (Option B), and Free Erythrocyte Protoporphyrin (FEP) (Option C) are indicators of lead exposure or impaired heme synthesis but are not definitive diagnostic markers.

Laboratory findings of Lead poisoning:

Blood analysis	
CBC and Peripheral Blood Smear	Microcytic-hypochromic anemia Anisocytosis Poikilocytosis and Polycythemia Basophilic stippling (Option B) and Cabot rings in RBCs
Blood lead levels (BL)	>5 microgram/ 100 ml (lead exposure) >40 micrograms (appearance of signs and symptoms)
ZnP (Zinc Protoporphyrin) levels	35-50 microgram/ 100 ml.
FEP (Free Erythrocyte Protoporphyrin)	Elevated
Urine analysis	
ALA (Amino Levulinic Acid) levels	>5 micrograms/L
Urinary coproporphyrin level	>150 mcg /litre
Urine lead level	0.25 milligram/ litre.
Calcium disodium EDTA mobilisation test	This test is primarily conducted in children to determine whether they will respond to chelation therapy with a significant lead diuresis.

- Radiography - Lead osteopathy or Bone-line : In children, calcium deposition occurs at the growth plates of bones (specifically the lower end of the femur and the upper end of the

tibia), appearing as a radiopaque band over four weeks due to the inhibition of osteoclastic activity.



Solution for Question 14:

Correct Answer: C) Phossy jaw

Explanation:

The described scenario, with symptoms and an image showing circumoral pallor, indicates lead poisoning. In this condition, Burton's lines, basophilic stippling, and calcium deposition at bone ends are observed, while phossy jaw (seen in phosphorus poisoning) is absent.

Acute lead poisoning

- It's rare, and there may be exacerbations of chronic lead poisoning.
- Symptoms: Astringent or metallic taste, dry throat, thirst, burning abdominal pain, nausea, vomiting, diarrhoea, peripheral circulatory collapse, paraesthesia, depression, coma, and death.

Chronic lead poisoning (also known as Plumbism or Saturnism)	
Clinical manifestations	Description

Facial pallor	It is caused by vasospasm and specifically occurs circumorally, which is the earliest and most specific feature.
Anemia	Type: Microcytic-hypochromic. Enzymes inhibited: Ferrochelatase ALA-dehydratase Coproporphyrinogen oxidase Blood picture: Cabot rings in RBCs Microcytic-hypochromic Sideroblasts (nucleated red cells) Anisocytosis Poikilocytosis Polycythemia
Basophilic stippling or Punctate basophilia (Option B)	Multiple blue-coloured pinhead dots in RBCs due to the toxic action of lead on porphyrin metabolism (inhibition of enzyme 5- Pyrimidine nucleotidase) (PNN). Accumulation of ribosomes in RBC
Lead line or Burtonian lines (Option A)	The stippled blue line observed on the gums, resulting from a subepithelial deposit of granules that develops within one week, is seen in 50-70% of cases and is more common in individuals with carious teeth (bacteria form lead sulfide).
Lead osteopathy/Lead line (Bone-line) (Option D)	In children, calcium deposition occurs at the growth plates of bones (specifically the lower end of the femur and the upper end of the tibia), which appears as a radiopaque band and occurs over 4 weeks due to the inhibition of osteoclastic activity.
Colic and constipation (Painter's Colic or Devonshire colic)	Late presentation occurs in 85% of cases, typically at night, and is characterised by spasmodic pain relieved by pressure.
Lead palsy	Predominantly motor features include: Dry belly ache Wrist drop (Radial nerve) Foot drop (Common peroneal nerve)
Encephalopathy	This condition is seen in nearly all cases, is associated with tetraethyl lead exposure, primarily affects children, results in permanent brain damage in 85% of cases, and accounts for 25% of fatalities.
Reproductive system	Amenorrhea Dysmenorrhea Abortion Sterility derangements (both sexes)
Cardiovascular system and Kidneys	Hypertension (vascular constriction) Permanent arteriolar degeneration Nephritis
Gout	Saturnine gout (due to decreased excretion of uric acid)
Others	These include generalised weakness, anorexia, dyspepsia, emaciation, exhaustion, irritability, foul breath, headache, vertigo, alopecia, and dizziness.

Phossy jaw (Option C) , also known as glass jaw or Lucifer's jaw, refers to osteonecrosis of the jawbone . This condition is associated with phosphorus poisoning.

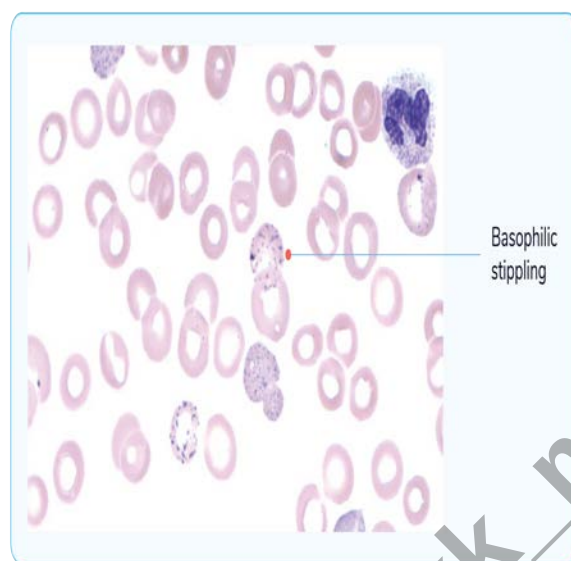


Solution for Question 15:

Correct Answer: B) Basophilic stippling

Explanation:

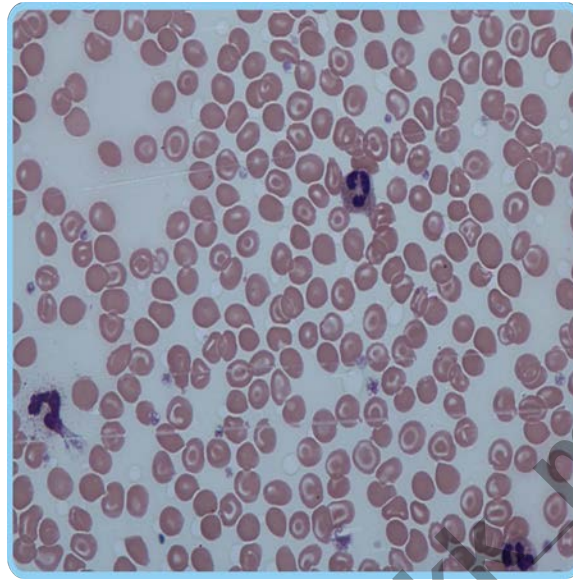
The scenario, featuring symptoms such as irritability, stomach pain, and the presence of blue gum lines (Burtonian lines), indicates chronic lead poisoning. The image shows basophilic stippling , a characteristic haematological finding in lead toxicity.



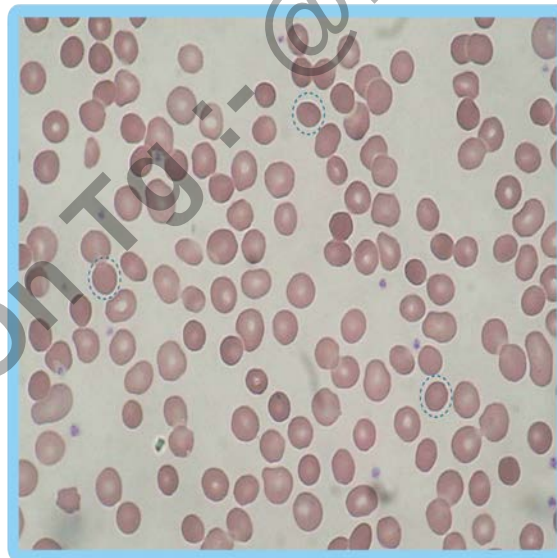
Hematological manifestations of Lead poisoning:

Hematological Manifestation	Description	Mechanism
Microcytic-Hypochromic Anemia	Small, pale RBCs (microcytes)	Inhibition of heme synthesis occurs through the inactivation of enzymes such as ALA dehydratase, ALA synthetase, decarboxylase, and ferrochelatase, due to lead interference.
Basophilic Stippling	Blue granules in RBCs on smear	Accumulation of ribosomes due to RNA degradation inhibition (due to the inhibition of the enzyme pyrimidine 5-nucleotidase).
Anisocytosis	Variation in RBC sizes	Ineffective erythropoiesis
Poikilocytosis	Variation in RBC shapes	Disruption of normal maturation processes
Elevated Free Erythrocyte Protoporphyrin (FEP)	Increased protoporphyrin levels	Impaired conversion of protoporphyrin to heme
Increased Zinc Protoporphyrin (ZnP)	Elevated levels in blood	Accumulation due to disruption of normal heme production

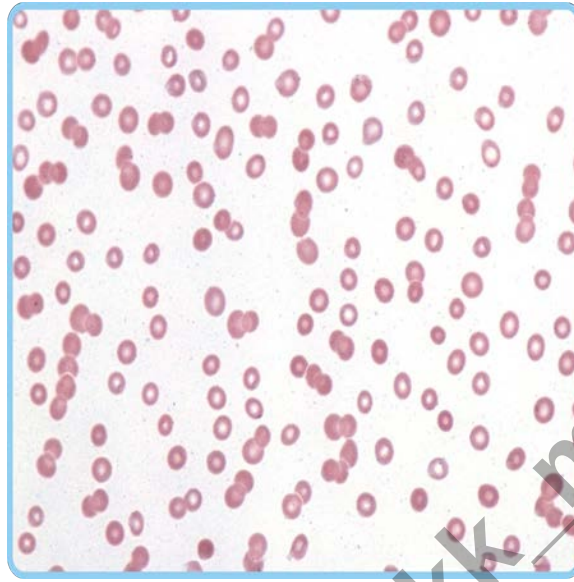
Target cells (Option A) are characterised by a central area of haemoglobin encircled by a pale ring, creating a target-like appearance. They are seen in liver disease, thalassemia, haemoglobin C disease, and post-splenectomy.



Spherocytes (Option C) are dense cells lacking central pallor, seen in conditions like autoimmune hemolytic anaemia, hereditary spherocytosis, and post-splenectomy.



Rouleaux formation (Option D) is the sticking together of red blood cells in linear arrays, commonly caused by high blood M protein levels. It is seen in various conditions, such as multiple myeloma, systemic lupus erythematosus (SLE), malaria, and HIV.



Solution for Question 16:

Correct Answer: B) Blocks RNA polymerase, preventing protein synthesis.

Explanation:

- The clinical scenario and given image suggest ingestion of Ricinus seeds.
- Ricin, a highly toxic protein derived from the seeds of *Ricinus communis*, blocks RNA Polymerase and inhibits protein synthesis.



Ricinus Communis (Castor):

- Seeds are smooth, flattened-oval, and mottled with light and dark brown markings. They have a bright, polished finish.
- Toxicity: The entire plant is toxic due to the water-soluble glycoprotein toxalbumin Ricin, a potent allergen with the highest concentration found in the seeds. The seeds are not toxic if intact but it is toxic if crushed or chewed.
- The seeds are not toxic if intact but it is toxic if crushed or chewed.
- Mechanism of action: Ricin inhibits protein synthesis by blocking RNA polymerase.
- Clinical Features: Symptoms may appear within 10 hours after ingestion. Gastrointestinal: Burning throat pain, abdominal cramps, nausea, thirst, vomiting, and often bloody diarrhea. Central Nervous System: Vertigo, drowsiness, delirium, convulsions, and coma. Other Symptoms include uremia, jaundice, rapid weak pulse, cold, clammy skin, and dehydration. In some cases, consciousness may be retained until death.
- Symptoms may appear within 10 hours after ingestion.
- Gastrointestinal: Burning throat pain, abdominal cramps, nausea, thirst, vomiting, and often bloody diarrhea.
- Central Nervous System: Vertigo, drowsiness, delirium, convulsions, and coma.
- Other Symptoms include uremia, jaundice, rapid weak pulse, cold, clammy skin, and dehydration.
- In some cases, consciousness may be retained until death.
- Fatal Dose: 1–10 µg/kg body weight (inhalation or injection); approximately 2 mg/kg (10–20 seeds) through ingestion.
- Fatal Period: 3–5 days.

- Ingestion of castor oil during pregnancy has been associated with teratogenic effects, including growth retardation, craniofacial dysmorphism, limb deformities, vertebral segmentation defects, and seizures in some cases.
- Treatment: No specific antidotes exist for toxalbumen exposures. Decontamination : Stomach wash, activated charcoal, and catharsis.
- No specific antidotes exist for toxalbumen exposures.
- Decontamination : Stomach wash, activated charcoal, and catharsis.
- The seeds are not toxic if intact but it is toxic if crushed or chewed.
- Symptoms may appear within 10 hours after ingestion.
- Gastrointestinal: Burning throat pain, abdominal cramps, nausea, thirst, vomiting, and often bloody diarrhea.
- Central Nervous System: Vertigo, drowsiness, delirium, convulsions, and coma.
- Other Symptoms include uremia, jaundice, rapid weak pulse, cold, clammy skin, and dehydration.
- In some cases, consciousness may be retained until death.
- No specific antidotes exist for toxalbumen exposures.
- Decontamination : Stomach wash, activated charcoal, and catharsis.

Supportive Care : IV fluids and monitoring for hypoglycemia and complications of hypovolemia. Urine alkalinisation may help prevent the crystallisation of haemoglobin in severe cases

Solution for Question 17:

Correct Answer: C

Explanation:

The scenario indicates chronic Arsenic Poisoning , characterized by anemia, polyneuritis, and hyperpigmentation in a raindrop pattern , along with radiopaque lines on the nails known as Aldrich-Mees lines, as shown in the image.

Clinical manifestations of Arsenic Poisoning:

	Acute Arsenic Poisoning	Chronic Arsenic Poisoning
--	-------------------------	---------------------------

Dermal	Hair loss Mees lines (Aldrich-Mees line or Reynold's line): Transverse bands of opacity in nails. (Option C)	Melanoses (neck, eyelids, nipples) Bowen's disease Facial oedema Hyperkeratosis Black foot disease (Peripheral gangrene) Hyperpigmentation (raindrop pattern) Skin cancer (commonly squamous cell carcinoma).
Ocular	Conjunctivitis Lacrimation	Diminution of Vision
Gastrointestinal	Abdominal pain Metallic taste Garlicky odour Dysphagia Vomiting Diarrhoea (bloody or rice water)	Anorexia Nausea Vomiting Diarrhea Weight loss
Airway	Upper airway irritation	Nasal septum perforation Laryngitis Bronchitis
Liver	Fatty degeneration	Hepatomegaly Jaundice Cirrhosis
Kidney	Oliguria Uraemia	Nephritic changes
Neurological	Hyperpyrexia Convulsions Coma	Encephalopathy Polyneuritis (glove & stocking type) Tremor Ataxia Limb tenderness Difficulty in walking
Haematological	–	Anaemia Leucopenia Thrombocytopenia Basophilic stippling (Rare in arsenic poisoning and common in lead poisoning) Karyorrhexis Pancytopenia
Cardiac	Tachycardia Hypotension Cardiac arrhythmias	Hypertension Myocarditis

Burtonian lines (Option A) are stippled blue lines seen on the gums, resulting from subepithelial deposits of granules. They are particularly indicative of lead poisoning but may also occur in cases of silver, bismuth, copper, and iron poisoning.



Kayser-Fleischer ring (Option B) is a greenish-brown deposit of copper that forms around the cornea of the eye, and it is observed in Copper poisoning and Wilson's disease .



Circumoral facial pallor (Option D) resulting from vasospasm associated with lead poisoning.



Solution for Question 18:

Correct Answer: A) Rigor mortis sets in quickly

Explanation:

The scenario, featuring throat pain, vomiting, and purging, along with autopsy findings of keratosis and white streaks on the nails, suggests Arsenic Poisoning, where rigor mortis sets in late and lasts for an extended period.

Postmortem findings of Arsenic Poisoning

External findings	Internal findings
Emaciation Raindrop pigmentation Keratosis Alopecia White streaks on nails (Aldrich Mees' lines) Jaundice and muscle wasting, Nasal septum ulceration. Sunken eyeballs Cyanosed and shrunken body. Rigor mortis (sets in late and persists for a prolonged duration) (Option A)	Reddening of stomach mucosa (red velvet or strawberry) due to gastrointestinal congestion. (Option B) Flea-bitten appearance of the stomach (due to focal haemorrhages). (Option D) The inflammation and presence of "rice water" contents in the intestines. Subendocardial haemorrhages in the heart. (Option C) Fatty degeneration of the heart, liver, and kidneys. Along with routine viscera and body fluids, the items preserved for chemical analysis are: A piece of long bone (preferably femur) A bunch of pulled scalp hair A wedge of muscle A small portion of skin (from the back of the trunk)

Solution for Question 19:

Correct Answer: B) Cardiac arrhythmia

Explanation:

Cardiac arrhythmia is a fatal complication of cocaine use.

Cocaine	
Origin	Leaves of <i>Erythroxylum coca</i>
Local names	Coke, snow, Cadillac, White lady
Physical properties	Colourless, odourless, crystalline, bitter taste
Mechanism of action	It is a potent central nervous system (CNS) stimulant that produces initial euphoria and increased energy, followed by depression as its effects wear off. It enhances alertness and activity while stimulating the autonomic nervous system, leading to increased heart rate and elevated blood pressure.
Route of administration	Snorting, Smoking, IV injection
Signs of abuse	Septal perforation Crack hands (multiple blackened, hyperkeratotic lesions on the fingers and palms). Black staining of tongue and teeth
Effects	Stages Stage of excitement Feeling of well-being Loss of depression and fatigue Excited and restless Tingling sensation in hands and feet Increased heart rate Increase in temperature Rapid and deep respiration Stage of depression Profuse perspiration Convulsion Reduced respiratory rate
Complications	Cardiac arrhythmia (Option B) Cardiac failure Cardiovascular collapse Cerebrovascular accident Aortic dissection Myocardial infarction Skin necrosis Intracerebral hemorrhage Formication (Cocaine Bugs)

Management of fatal dose	Maintain airway, breathing and circulation If ingested: Gastric lavage with potassium permanganate If snorted: Wash the nasal mucous membrane with water If injected: Ligature over the injection site Amyl nitrite via inhalation route Thiamine 100 mg IV injection Naloxone hydrochloride 2mg IV injection Symptomatic management for convulsions using barbiturates
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Respiratory depression (Option A) is a complication that is often associated with the use of opium, alcohol, barbiturates, and benzodiazepine. Cocaine is a stimulant, and it does not cause respiratory depression.

Urinary retention (Option C) is a complication that can occur in cases of datura/atropine poisoning. The patient has a dry mouth, dysphagia, delirium, a dilated pupil, urinary retention, respiratory paralysis, and dry skin.

Optic neuritis and atrophy (Option D) occur in methyl alcohol poisoning. Methyl alcohol is converted into formic acid, which causes optic neuritis and atrophy of the optic nerve, which can result in irreversible blindness.

Solution for Question 20:

Correct Answer: B) 1-d, 2-c, 3-a, 4-b, 5-e

Explanation:

Drugs	Local names
Cocaine	Coke, snow, Cadillac, white lady, crack (cocaine + baking soda + water)
MDMA	Molly, love drug, ecstasy
Heroin	Smack, brown sugar, Black tar, speedball (mixed with cocaine)
Cannabis	Pot, grass, dope, weed, hash, Mary Jane, M.J
LSD	Acid, Alice, Angel in the Sky, Lucy
Ketamine	K, Special K, Kit-Kat, K Hole

Methamphetamine	Ice, Glass, Crystal meth, Meth, Speed
-----------------	---------------------------------------

Solution for Question 21:

Correct Answer: C) Hashish oil > Charas > Ganja > Bhang

Explanation:

The list of different forms of cannabis is given below in order of increasing potency,

Form of cannabis	Percentage of active principle
Bhang	15%
Ganja	15-25%
Charas	25-40%
Hashish Oil	>50%

Charas:



Ganja:



Hashish oil:



- The active principle in cannabis is Tetrahydrocannabinol (THC), which is a CNS stimulant. Other constituents are Cannabinol and Cannabidiol.
- THC is metabolized in the liver and is lipid-soluble, allowing it to easily cross the blood-brain barrier.
- Different forms of cannabis: Bhang - Decoction made of dried cannabis leaves and fruit shoots. Majoon - sweet prepared from bhang Ganja - Flower tops of cannabis, which is smoked in a pipe or hukkah. Charas - Resin from the leaves and flower tops . Hashish oil - Concentrated dark viscous liquid extracted from the cannabis plant.
- Bhang - Decoction made of dried cannabis leaves and fruit shoots.

- Majoon - sweet prepared from bhang
- Ganja - Flower tops of cannabis, which is smoked in a pipe or hukkah.
- Charas - Resin from the leaves and flower tops .
- Hashish oil - Concentrated dark viscous liquid extracted from the cannabis plant.
- Fatal dose Charas - 2 grams Ganja - 8 grams Bhang - 10 grams/kg body weight THC - 30 mg/kg body weight
- Charas - 2 grams
- Ganja - 8 grams
- Bhang - 10 grams/kg body weight
- THC - 30 mg/kg body weight
- Bhang - Decoction made of dried cannabis leaves and fruit shoots.
- Majoon - sweet prepared from bhang
- Ganja - Flower tops of cannabis, which is smoked in a pipe or hukkah.
- Charas - Resin from the leaves and flower tops .
- Hashish oil - Concentrated dark viscous liquid extracted from the cannabis plant.
- Charas - 2 grams
- Ganja - 8 grams
- Bhang - 10 grams/kg body weight
- THC - 30 mg/kg body weight

Solution for Question 22:

Correct Answer: C) Datura seed has a smooth surface, while capsicum seeds have numerous depressions

Explanation:

Datura seed has numerous small depressions, while capsicum seed has a smooth surface.

Features	Datura seeds	Capsicum seeds
Size (Option A)	Large and thick	Small and thin
Shape (Option B)	Kidney shaped	Rounded

Color (Option C)	Dark or yellowish-brown	Pale - yellow
Surface	Numerous small depressions	Smooth
Margin	Laterally compressed and double edge at the convex border	Convex border is simple and sharp
Smell	Odourless	Pungent
Taste	Bitter	Pungent
Cut surface (longitudinal)	The embryo is curved outward at the hilum	The embryo is curved inward like the figure '6.'
Active principle	hyoscine (scopolamine), hyoscyamine and traces of atropine	Capsaicin and capsin
Effects	Antimuscarinic effects on the CNS	It acts as an irritant poison, causing pain and irritation.



Datura seeds



Capsicum seeds



Solution for Question 23:

Correct Answer: B) Physostigmine

Explanation:

- The patient's symptoms suggest anticholinergic toxicity , likely due to the ingestion of Datura seeds.
- Physostigmine 1mg IV (hourly) is used as an antidote for datura poisoning.
- Datura contains hyoscine (scopolamine), hyoscyamine and traces of atropine
- Mechanism of Action: Initially, it stimulates the higher centers of the brain and then activates the motor centers. Later, it leads to depression of the vital centers in the medulla, resulting in paralysis.



- Clinical features: Dryness of mouth and throat Dysphagia Difficulty in talking Hoarseness of voice Flushed face Pupil dilatation Photophobia + Diplopia Agitation and restlessness Delirium Urinary retention Dry and hot skin Exfoliation of skin may occur
- Dryness of mouth and throat
- Dysphagia
- Difficulty in talking
- Hoarseness of voice
- Flushed face
- Pupil dilatation
- Photophobia + Diplopia
- Agitation and restlessness
- Delirium
- Urinary retention
- Dry and hot skin

- Exfoliation of skin may occur
- Patients can end up in a coma, or death can occur due to respiratory paralysis.
- Treatment Gastrointestinal decontamination: Administer activated charcoal to absorb toxins. Use a cathartic if indicated. Antidote: Administer Physostigmine 1-2 mg IV slowly, repeating as necessary based on symptom response. Symptomatic treatment: Consider Pilocarpine 5 mg subcutaneously for severe anticholinergic effects. Supportive care: Monitor vital signs and address any respiratory or cardiovascular issues.
- Gastrointestinal decontamination: Administer activated charcoal to absorb toxins. Use a cathartic if indicated.
- Antidote: Administer Physostigmine 1-2 mg IV slowly, repeating as necessary based on symptom response.
- Symptomatic treatment: Consider Pilocarpine 5 mg subcutaneously for severe anticholinergic effects.
- Supportive care: Monitor vital signs and address any respiratory or cardiovascular issues.
- Dryness of mouth and throat
- Dysphagia
- Difficulty in talking
- Hoarseness of voice
- Flushed face
- Pupil dilatation
- Photophobia + Diplopia
- Agitation and restlessness
- Delirium
- Urinary retention
- Dry and hot skin
- Exfoliation of skin may occur
- Gastrointestinal decontamination: Administer activated charcoal to absorb toxins. Use a cathartic if indicated.
- Antidote: Administer Physostigmine 1-2 mg IV slowly, repeating as necessary based on symptom response.
- Symptomatic treatment: Consider Pilocarpine 5 mg subcutaneously for severe anticholinergic effects.
- Supportive care: Monitor vital signs and address any respiratory or cardiovascular issues.

Toxins and Antidotes:

Toxicity	Antidote	Mechanism of action
Datura/anticholinergics toxicity	Physostigmine (Option B)	Inhibits acetylcholinesterase enzyme
Opium	Naloxone (Option A)	Opioid receptor antagonist
Organophosphates	Atropine Pralidoxime	Muscarinic antagonist Cholinesterase reactivation
Acetaminophen	N-acetyl-cysteine	Replenishes Glutathione
Methanol	Fomepizole (Option D)	Inhibitor of alcohol dehydrogenase
Benzodiazepine	Flumazenil	Inhibits GABA receptors
Cyanide	Eli Lilly antidote Kit (3-step approach)	Amyl nitrate: Oxidises haemoglobin to methemoglobin, which scavenges cyanide.
Sodium nitrate: causes cyanide to detach from the heme group of cytochrome oxidase. Amyl nitrate is used as a temporary measure until IV Sodium nitrate can be given.		
Sodium thiosulphate: It activates an enzyme that converts cyanide to a less toxic agent, thiocyanate.		

Pyridostigmine (Option C): While it also inhibits acetylcholinesterase, it does not penetrate the blood-brain barrier, making it less suitable for addressing the central nervous system effects of anticholinergic toxicity.

Solution for Question 24:

Correct Answer: C) Commonly used for arrow and cattle poisoning.

Explanation:

- The scenario of blistering and severe itching on her arms after applying the juice from the seed, as shown in the image, suggests Semecarpus Anacardium.
- It is used to create artificial bruises (malingering) but is not commonly used for Arrow and cattle poisoning.

- *Abrus precatorius* is used as an arrow poison and ideal cattle poison.

Semecarpus Anacardium:

- Commonly known as the marking nut or bhilawanol (washermen use its juice to inscribe identification numbers on clothes).
- Seeds: Black, cone/heart-shaped with a rough projection at the base. They have a thick pericarp that contains an acrid, brownish, oily juice that turns black upon exposure to air.
- Black, cone/heart-shaped with a rough projection at the base.
- They have a thick pericarp that contains an acrid, brownish, oily juice that turns black upon exposure to air.
- Black, cone/heart-shaped with a rough projection at the base.
- They have a thick pericarp that contains an acrid, brownish, oily juice that turns black upon exposure to air.



- Active principles: Semecarpol and Bhilawanol (Option B)
- Signs and Symptoms: Skin exposure: Applying the juice can cause irritation, itching, and painful blisters filled with acrid serum. Eczematous eruptions resembling bruises may also occur, potentially evolving into ulcers with sloughing. Constitutional symptoms include fever and painful micturition, often accompanied by brown-colored urine. Oral ingestion: A large dose can cause blisters in the mouth and throat, severe gastrointestinal irritation, dyspnea, tachycardia, hypotension, cyanosis, loss of reflexes, delirium, coma, and potentially death. (Option D) Fatal dose: Approximately 5–10 g (5–8 seeds). Fatal period: 12–24 hours.
- Skin exposure: Applying the juice can cause irritation, itching, and painful blisters filled with acrid serum. Eczematous eruptions resembling bruises may also occur, potentially evolving into ulcers with sloughing.

- Constitutional symptoms include fever and painful micturition, often accompanied by brown-colored urine.
- Oral ingestion: A large dose can cause blisters in the mouth and throat, severe gastrointestinal irritation, dyspnea, tachycardia, hypotension, cyanosis, loss of reflexes, delirium, coma, and potentially death. (Option D)
- Fatal dose: Approximately 5–10 g (5–8 seeds).
- Fatal period: 12–24 hours.
- Treatment: Rinse the affected skin with soap and water. Apply bland liniments. Provide demulcent drinks and symptomatic treatment.
- Rinse the affected skin with soap and water.
- Apply bland liniments.
- Provide demulcent drinks and symptomatic treatment.
- Medico-legal Aspects: Criminal Use: The juice can be introduced vaginally as a form of punishment for infidelity or applied to the cervical os for illegal abortions. Malingering: The juice can simulate conjunctivitis or falsely substantiate assault claims, as the lesions resemble bruises (Option A)
- Criminal Use: The juice can be introduced vaginally as a form of punishment for infidelity or applied to the cervical os for illegal abortions.
- Malingering: The juice can simulate conjunctivitis or falsely substantiate assault claims, as the lesions resemble bruises (Option A)
- Skin exposure: Applying the juice can cause irritation, itching, and painful blisters filled with acrid serum. Eczematous eruptions resembling bruises may also occur, potentially evolving into ulcers with sloughing.
- Constitutional symptoms include fever and painful micturition, often accompanied by brown-colored urine.
- Oral ingestion: A large dose can cause blisters in the mouth and throat, severe gastrointestinal irritation, dyspnea, tachycardia, hypotension, cyanosis, loss of reflexes, delirium, coma, and potentially death. (Option D)
- Fatal dose: Approximately 5–10 g (5–8 seeds).
- Fatal period: 12–24 hours.
- Rinse the affected skin with soap and water.
- Apply bland liniments.
- Provide demulcent drinks and symptomatic treatment.
- Criminal Use: The juice can be introduced vaginally as a form of punishment for infidelity or applied to the cervical os for illegal abortions.

- Malingering: The juice can simulate conjunctivitis or falsely substantiate assault claims, as the lesions resemble bruises (Option A)

Solution for Question 25:

Correct Answer: A) Abrin

Explanation:

- The given scenario and image suggest the ingestion of seeds of *Abrus precatorius*, and Abrin is the primary active principle found in the seeds.
- It is a highly toxic protein belonging to the class of toxins known as toxalbumins.



Abrus precatorius (Rati, Gunchi, Jequirity):

- It belongs to the Leguminosae family and is found throughout India.
- Ideal Arrow poison/Cattle poison,
- Needles are used in conjunction with the seeds to create injuries (resembles snake bites)
- Identification of Seeds: Seeds are egg-shaped and scarlet in colour with a black spot at one end. Seeds are tasteless and odourless.
- Seeds are egg-shaped and scarlet in colour with a black spot at one end
- Seeds are tasteless and odourless.

- Active Principles: Abrin: A thermolabile toxalbumin. Abrine: An amino acid. Hemagglutinin: A lipolytic enzyme. Abralin: A glycoside.
- Abrin: A thermolabile toxalbumin.
- Abrine: An amino acid.
- Hemagglutinin: A lipolytic enzyme.
- Abralin: A glycoside.
- Signs and Symptoms: Abdominal pain, nausea, vomiting, and bloody diarrhoea. Weakness, cold sweat, hand tremors, weak pulse, rectal bleeding, tachycardia, headache, dilated pupils, hallucinations, drowsiness, tetany, and circulatory collapse. Symptoms may appear within 6 hours but can be delayed for 1–3 days.
- Abdominal pain, nausea, vomiting, and bloody diarrhoea.
- Weakness, cold sweat, hand tremors, weak pulse, rectal bleeding, tachycardia, headache, dilated pupils, hallucinations, drowsiness, tetany, and circulatory collapse.
- Symptoms may appear within 6 hours but can be delayed for 1–3 days.
- If the extract is injected S/c, symptoms resemble those of a viperine snakebite.
- Fatal Dose: 90–120 mg of abrin via injection or ingestion of 1–2 crushed seeds.
- Fatal Period: 3–5 days.
- Treatment: Injection of anti-abrin. Dissection of the needle used for injection. Oral administration of a hydrochloric acid-pepsin mixture. Maintaining urine alkalinity.
- Injection of anti-abrin.
- Dissection of the needle used for injection.
- Oral administration of a hydrochloric acid-pepsin mixture.
- Maintaining urine alkalinity.
- Seeds are egg-shaped and scarlet in colour with a black spot at one end
- Seeds are tasteless and odourless.
- Abrin: A thermolabile toxalbumin.
- Abrine: An amino acid.
- Hemagglutinin: A lipolytic enzyme.
- Abralin: A glycoside.
- Abdominal pain, nausea, vomiting, and bloody diarrhoea.
- Weakness, cold sweat, hand tremors, weak pulse, rectal bleeding, tachycardia, headache, dilated pupils, hallucinations, drowsiness, tetany, and circulatory collapse.
- Symptoms may appear within 6 hours but can be delayed for 1–3 days.
- Injection of anti-abrin.
- Dissection of the needle used for injection.

- Oral administration of a hydrochloric acid-pepsin mixture.
 - Maintaining urine alkalinity.
-

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Previous Year Questions

1. A patient came to the hospital with muscle weakness, diarrhea, and bradycardia. He was diagnosed with organophosphate poisoning for which atropine was administered. After atropine administration, which of the following changes are not seen?

- A. Bradycardia
- B. Muscle weakness
- C. Diaphoresis
- D. Nasal discharge

2. A patient presented with altered sensorium, hypotension and constricted pinpoint pupil. Which of the following is responsible for poisoning

- A. Opioids
- B. Datura
- C. Cocaine
- D. Amphetamine

3. Match the poisons and site of action A. Datura 1. CVS B. Nicotine 2. CNS C. Nitric acid 3. SKIN D. Nux vomica (strychnine) 4. SPINAL CORD

A. Datura	1. CVS
B. Nicotine	2. CNS
C. Nitric acid	3. SKIN

D. Nux vomica (strychnine)	4. SPINAL CORD
----------------------------	----------------

- A. A-2,B-1,C-3,D-4
 - B. A-1,B-2,C-3,D-4
 - C. A-1,B-2,C-4,D-3
 - D. A-4,B-3,C-1,D-2
-

4. A 28-year-old male patient was found unconscious in his backyard, the leaves of a plant similar to the one shown below were found scattered around him. He is now presenting with symptoms of nausea, vomiting, abdominal pain, diarrhea, and irregular heart rate. Identify the plant given below likely responsible for these symptoms?



- A. Yellow oleander
- B. Ricinus communis
- C. Datura
- D. Papaver somniferum

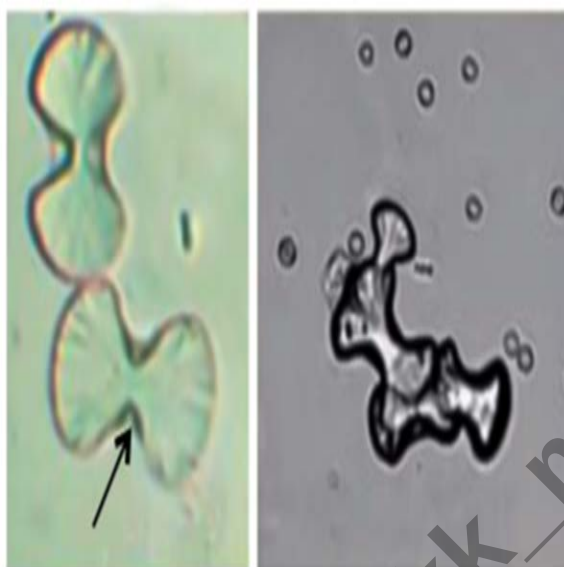
5. A 50-year-old man with a history of chronic alcoholism goes to a liquor store and purchases two bottles of liquor. Approximately one hour later, after consuming alcohol, he presents to the emergency department with complaints of abdominal discomfort, acute confusion, vision loss, and an unsteady gait. What is the most appropriate treatment for his condition?

- A. Naloxone
 - B. Ethyl alcohol
 - C. Flumazenil
 - D. Atropine
-

6. Which of the following substances are antidotes for cyanide poisoning? Amyl nitrate Sodium nitrate Sodium thiosulphate Benzyl thiocyanate

- A. 1, 2, and 3 only
 - B. 1, 2, 3, and 4
 - C. 1, and 2 only
 - D. 1, 3, and 4 only
-

7. A patient presents with paraesthesia, tetany, muscle twitching, and muscle cramps, along with urinary crystals as shown in the image. These symptoms are typical of poisoning with what substance?



- A. Cystine
- B. Xanthine
- C. Ammonium urate
- D. Calcium oxalate

8. A 40-year-old female consumed a white-colored rodenticide powder and presents with a potassium level of 2.5 mEq/L. On examination, she has an irregular pulse and muscle paralysis. ECG shows multiple ventricular ectopics. What is the suspected poison?

- A. Aluminum phosphide
- B. Zinc phosphide
- C. Warfarin toxicity
- D. Barium carbonate

9. A chronic alcoholic came with complaints of agitation, global confusion, disorientation, hallucination, fever, hypertension, diaphoresis, and autonomic hyperactivity after cessation of alcohol for 48 hours. Which of the following is the diagnosis?

- A. Delirium tremens
- B. Wernicke Encephalopathy
- C. Korsakoff Psychosis
- D. Magnan Phenomena

10. A patient was admitted with complaints of tachycardia and arrhythmias. On examination, multiple scratch marks were observed on the body. The probable drug of abuse in this patient is:

- A. Cocaine
- B. Morphine
- C. Datura
- D. Cannabis

11. Which of the following is used for the treatment of Datura poisoning?

- A. Physostigmine
- B. Atropine
- C. Sodium nitrite
- D. Neostigmine

12. In cases of alcohol abuse, in what substance is the blood typically preserved during an autopsy?

- A. Sodium fluoride
 - B. Formalin
 - C. Rectified spirit
 - D. Sodium chloride
-

13. Gastric lavage is contraindicated in which of the following poisonings?

- A. Strychnine
 - B. Paraldehyde
 - C. Hydrocarbons
 - D. Iron tablets
-

14. Police brought a person from the railway track with features of dry dilated pupil, dry skin, slurring of speech, altered sensorium. The poisoning is?

- A. Morphine
 - B. Cannabis
 - C. Datura
 - D. Alcohol
-

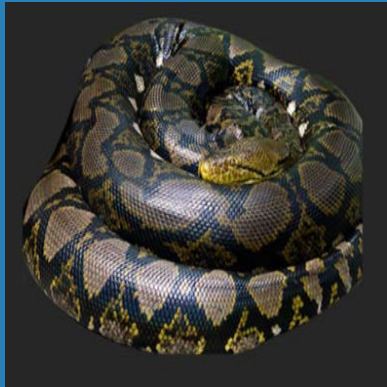
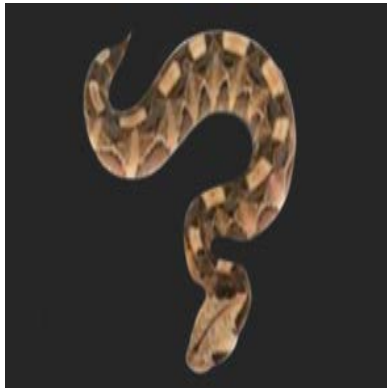
15. Aldrich mees lines and raindrop pigmentation in hand is seen in which poisoning?

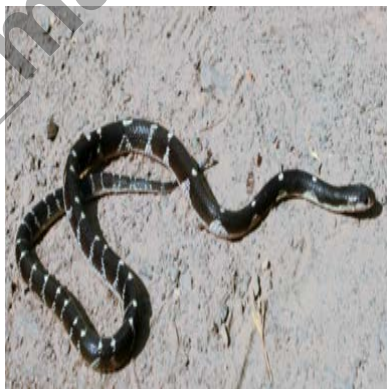
- A. Arsenic
- B. Thallium
- C. Cadmium
- D. Lead

16. Saturnine gout is seen in:

- A. Lead poisoning
- B. Cadmium poisoning
- C. Beryllium poisoning
- D. Mercury poisoning

17. Match the following : 1. Krait A. 2. Cobra B. 3. Python C. 4. Viper D.

1. Krait	A.	
2. Cobra	B.	

3. Python	C.	
4. Viper	D.	

A. 1-D, 2-C, 3-A, 4-B

B. 1-A, 2-B, 3-C, 4-D

C. 1-D, 2-C, 3-B, 4-A

D. 1-D, 2-A, 3-B, 4-C

18. Identify the plant which produces a deliriant toxin.



- A. Erythroxylum coca
- B. Datura
- C. Hyoscine
- D. Digitalis purpurea

19. A 30 year old apparently healthy man who was carrying laxatives and enema apparatus developed abdominal pain at the airport and an x-ray was done which appears as shown below. Which of the following is the likely diagnosis?



- A. Bezoar syndrome
- B. Pica due to anaemia
- C. Body packer syndrome
- D. Constipation due to fecalith

20. Assertion: Most common hematological finding in case of lead poisoning is eosinophilia Reason: Lead inhibits ALA dehydratase

- A. Both assertion and reason are true and the reason is the correct explanation for the assertion
 - B. Both assertion and reason are true, but the reason is not the correct explanation for the assertion
 - C. Assertion is false, but the reason is true
 - D. Both assertion and reason are false
 - E. The assertion is false but reason is true
-

21. Identify the active toxin extracted from the poisonous plant in the image below?



- A. Calotropis gigantea
- B. Digitalis
- C. Ricin
- D. 1-hyoscyamine

22. Identify the seeds shown in the image below:



- A. Tropicana canna
- B. Nigella sativa
- C. Datura seeds
- D. Opium seeds

23. What is the cause of death in smoke inhalation ?

- A. Histotoxic hypoxia
- B. Ischemic hypoxia
- C. Anemic hypoxia
- D. All of the above

24. Match the following: i. Heroin ii. MDMA iii. Joint iv. LSD Ganja Acid Brown sugar Amphetamine

i. Heroin ii. MDMA iii. Joint iv. LSD

Ganja Acid Brown sugar Amphetamine

- A. i -2, ii-1, iii-4, iv-3
 - B. i -3, ii-2, iii-1, iv-4
 - C. i -4, ii-2, iii-1, iv-3
 - D. i -3, ii-4, iii-1, iv-2
-

25. Which of the following is the incorrect statement regarding the image shown below?



- A. Only the roots of this plant are poisonous
 - B. Causes AV block
 - C. Atropine is not specific for aconite poisoning
 - D. Grow in sub Himalayan range
-

26. Match the following poisons with their probable mechanism of Action? A. Zinc chloride B. Chloral hydrate C. Quinine D. Potassium carbonate 1. Abortifacient 2.

Irritant 3. Corrosive 4. Stupefying

A. Zinc chloride B. Chloral hydrate C. Quinine D. Potassium permanganate

- A. A-2, B-4, C-1, D-3
- B. A-4, B-3, C-2, D-1
- C. A-4, B-2, C-1, D-3
- D. A-2, B-3, C-4, D-1

27. Which of the following statements is not true regarding the management of snakebite in India?

- A. Atropine and neostigmine are very useful in the management of krait bite
- B. Neostigmine and ventilatory support are used in treatment along with ASV
- C. ASV is the mainstay of treatment
- D. Current ASV in India is not effective against humped pit viper

28. Which metallic poisoning Mimics cholera?

- A. Arsenic
- B. Chromium
- C. Mercury
- D. Lead

29. Which of the following poisons present with bluish discoloration of the stomach mucosa on post mortem examination?

- A. Oxalic acid
- B. Sodium amytal
- C. Soneryl
- D. Arsenic

30. Which of the following types of toxins has the maximum damage potential?

- A. Irritant poison
- B. Corrosive poison
- C. Alcohol
- D. Opioid

31. Match the following poisons and test: Column A Column B
1. Marsh test 2. Silver nitrate paper test 3. Cavett test 4. Marquis test
A. Opium B. Aluminium phosphide C. Cyanide D. Alcohol E. Arsenic

Column A	Column B
1. Marsh test 2. Silver nitrate paper test	3. Cavett test 4. Marquis test

- A. 1-E, 2-B, 3-A, 4-D
- B. 1-E, 2-C, 3-B, 4-A
- C. 1-E, 2-B, 3-D, 4-A
- D. 1-E, 2-C, 3-B, 4-D

32. A 28 year old man presented with dry mouth, urinary retention, and constipation after consuming an unknown substance. On examination, his PR 110

bpm, and his pupils were dilated. What did he most likely consume?

- A. Heroin
 - B. Morphine
 - C. Malathion
 - D. Belladonna
-

33. Which of the following are toxins? Atropine Colchicine Curare Arsenic

- A. All are correct
 - B. 1, 2, 3 are correct
 - C. 1, 3 are correct
 - D. 3, 4 are correct
-

34. Which of the following statements about digoxin toxicity are true: A. Pain in abdomen, burning sensation, diarrhea seen in digoxin toxicity B. Vision changes like Transient ambylopia, photophobia, may be noted in digoxin toxicity C. Specific antidote for cardiac arrhythmias is lignocaine 100 mg IV or novocaine or propranolol. D. Death occurs from Respiratory failure.

- A. A, B, C
 - B. D, A, C
 - C. A, B, D
 - D. D, B
-

35. What is the poisoning associated with increased gastrointestinal secretions and diarrhea?

- A. Cocaine
 - B. Crack
 - C. Cannabis
 - D. Ketamine
-

36. A farmer presents with a painful snakebite wound on his right foot. There are two fang marks with swelling and bleeding at the bite site. What is the first step in management?

- A. Polyvalent ASV
 - B. Sling or splint leading to immobilization
 - C. Venom suction
 - D. See the snake brought by a relative
-

37. Blood during autopsy is preserved in?

- A. NaF
 - B. NaCl
 - C. Formalin
 - D. Rectified spirit
-

38. Unknown substance consumption leading to tachycardia and red-colored halo is seen with?

- A. Cannabis
- B. Crack/cocaine
- C. LSD

D. Heroin

39. Acute liver failure is a clinical feature of which of the following?

- A. OP poisoning
 - B. Amanita Phalloides poisoning
 - C. Belladonna poisoning
 - D. Morphine poisoning
-

40. In case of corrosive acid ingestion, stomach is opened along the?

- A. Lesser curvature
 - B. Greater curvature
 - C. Vertical
 - D. Pylorus
-

41. Identify the method of substance abuse as shown in the image below.



- A. Bagging
 - B. Snorting
 - C. Sniffing
 - D. Huffing
-

42. Inhaling the cloth soaked in drug is:

- A. Bagging
 - B. Spraying
 - C. Huffing
 - D. Sniffing
-

43. Cause of Opisthotonus is?

- A. Cocaine
- B. Codeine

- C. Strychnine
 - D. Curare
-

44. What is the preferred treatment for acute arsenic poisoning?

- A. Ipecac
 - B. Penicillamine
 - C. Activated charcoal
 - D. Dimercaprol
-

45. What is the active principle of the poison shown in the image?



- A. Ricin
- B. Bhilawanol
- C. Abrin
- D. Calotropin

46. A young teenage boy with a history of drug abuse had been instilled with a shot of intravenous cocaine at a party. He is rushed to the emergency room by his friends. Some of the symptoms were diaphoresis, headache, and acute coronary spasm. Which of the following symptoms noted would go against the likely diagnosis of cocaine poisoning?

- A. Myocardial ischemia
- B. Bradycardia
- C. Hyperthermia
- D. Agitation

47. A little boy was playing in a garden. Suddenly, he collapsed and was rushed to the hospital. His friend who played with him in the garden informed the doctors that he ate one of the fruits in the garden post which he seemed to have developed these symptoms. He also had irritability, restlessness, dry hot skin, and was unable to pass urine and stools. Identify the poison and its appropriate antidote.

- A. Datura, Pralidoxime
- B. Datura, Physostigmine
- C. Yellow oleander, Digoxin
- D. Yellow oleander, Physostigmine

48. Black foot disease with peripheral neuropathy is seen in which metal poisoning?

- A. Arsenic
- B. Lead

- C. Mercury
 - D. Cadmium
-

49. A female was given morphine sulfate during labor for pain following which she developed respiratory distress. Which of the following will be the correct antidote?

- A. Naloxone
 - B. Atropine
 - C. Pralidoxime
 - D. Epinephrine
-

50. Which of the following is correct regarding sample collection in as case of alcohol intoxication?

- A. Sample area cleaned with spirit
 - B. Sample area cleaned with soap and water
 - C. Sample should be preserved without preservative
 - D. Only one urine sample is required
-

51. Which of the following are seen in patients with chronic arsenic poisoning? 1. Raindrop pigmentation 2. Peripheral neuritis 3. Mees line 4. Luminous vomit

- A. 1, 2, 3 are correct
 - B. 2, 3 are correct
 - C. 1, 4 are correct
 - D. Only 1 is correct
-

52. A male patient was presented with a history of consumption of an unknown substance. He had increased sweating, lacrimation, diarrhea and bradycardia. What's the management?

- A. Atropine
 - B. Calcium gluconate
 - C. Pralidoxime
 - D. Obidoxime
-

53. Hooch tragedy is related to:

- A. Phosphorus
 - B. Mercury
 - C. Ethanol
 - D. Methanol
-

54. Which of these is not a cardiac poison?

- A. Aconite
 - B. Cerbera thevetia
 - C. Nicotiana tabacum
 - D. Atropa belladonna
-

55. A 72-year-old farmer was presented in the hospital with pinpoint pupil & increased secretions. The diagnosis is

- A. Alcohol poisoning
 - B. Opioid poisoning
 - C. Organophosphorus poisoning
 - D. Datura poisoning
-

56. Identify the given plant.:



- A. Datura Alba
 - B. Argemone Mexicana
 - C. Nerium Odorum
 - D. Papaver Somniferum
-

57. A farmer was sleeping in the field, and he felt a sting on his leg. He saw something moving away quickly. He then got drowsy and was taken to the hospital. He developed pain around the site and continued to bleed profusely from the wound site. The wound became red with blisters. Which of the following is the

most likely cause?

- A. Viper
 - B. Cobra
 - C. Wasp bite
 - D. Scorpion
-

58. What could be the possible reason for a patient experiencing seizures, with multiple episodes and taking herbal medications, who demonstrates arching of the back during these episodes, and remains conscious in between them?

- A. Strychnine
 - B. Ricinus
 - C. Nerium odorum
 - D. Datura
-

59. A child before playing consumed fruit from the garden. After some time he developed a high fever, confusion, photophobia, and unable to urinate. What are the likely causative agent and the appropriate antidote used in this case?

- A. Datura, Pralidoxime
 - B. Datura, Physostigmine
 - C. Yellow oleander, Pralidoxime
 - D. Yellow oleander, Physostigmine
-

60. A 50-year-old factory worker was brought to the emergency room with complaints of headache, vomiting and blurring of vision after he consumed local spirit. Which of the following is used for the treatment of his condition?

- A. Fomepizole
 - B. Flumazenil
 - C. N-acetylcysteine
 - D. Naloxone
-

61. Which of the following is not a symptom associated with chronic cocaine abuse?

- A. Nasal/palatal perforation.
 - B. Dilated pupils, rapid pulse and sweating.
 - C. Black tongue and teeth.
 - D. Hallucination of insects crawling on the skin (formication).
-

62. In which poisoning is cherry red hypostasis seen on post-mortem examination?

- A. CO Poisoning
 - B. Hydrogen sulphide
 - C. Yellow phosphorus
 - D. Hydrogen cyanide
-

63. A 28-year-old male is brought to the emergency room. On examination, his PR is 50 bpm, BP is 80/60 mmHg, RR is 7 breaths/min and his temperature is 95° F. He has a pinpoint pupil and is disoriented. What is the most probable diagnosis?

- A. Benzodiazepine overdose
- B. Cocaine intoxication
- C. Opioid withdrawal

D. Opioid intoxication

64. A 23-year-old farm worker from a remote village was brought to the hospital 8 hours later, following a snake bite at 5 in the morning. The patient complained of headache, blurred vision, tingling sensation around the mouth, dizziness and vertigo and had multiple episodes of vomiting since the bite. He has signs of respiratory depression and bradycardia (HR<50). The local wound showed mild swelling and blistering. There are no apparent bleeding manifestations. He was administered 10 vials of antivenom. Which of the following is the subsequent step of management?

- A. Check the blood coagulation profile
 - B. Neostigmine with atropine
 - C. Epinephrine injection
 - D. I.V hydrocortisone
-

65. In this case, which poisoning do you suspect if a person employed in a dye factory complains of symptoms such as nausea, vomiting, dark bloody stools, conjunctivitis, and a burning sensation in the throat and stomach?

- A. Potassium permanganate
 - B. Arsenic
 - C. Thallium
 - D. Lead
-

66. Luminescent stools are associated with which of the following?

- A. Mercury

- B. Cadmium
 - C. Phosphorus
 - D. Oxalic acid
-

67. Which of the following acts as a physiological antidote to scorpion venom?

- A. Physostigmine
 - B. Norepinephrine
 - C. Prazosin
 - D. Atropine
-

68. Gastric lavage can be performed in case of which of the following poisoning?

- A. Sulphuric acid
 - B. Strychnine
 - C. Carbolic acid
 - D. Kerosene oil
-

69. A 40 year old patient was brought by the police in an unconscious state to the casualty with a history of unknown poisoning. On examination the patient had needle track marks over the upper limbs. Patient had increased blood pressure, increased heart rate and hyperthermia. Which poisoning should be suspected here?

- A. Heroine
- B. Cocaine
- C. Cannabis

D. LSD

70. A patient was brought to the casualty with greenish color skin and a rotten egg odour. The most probable poisoning could be.

- A. Carbon monoxide
 - B. Nitrite
 - C. Cyanide
 - D. Hydrogen sulfide
-

71. A child accidentally took a bottle full of iron tablets. Which drug is used as an antidote for the poisoning?

- A. Luspatercept
 - B. Deferoxamine
 - C. Deferiprone
 - D. DMSA
-

72. A man was found to be lying dead on the ground. His friend saw a snake passing in the area and took a picture of it and showed it to the doctor (image below). What is the most probable cause of death of this man?



- A. Shock
- B. Respiratory failure
- C. Coagulopathy
- D. Renal failure

73. Cherry red post mortem lividity is typically seen in:

- A. Opioid poisoning
- B. Carbon monoxide poisoning
- C. Hydrogen sulphide poisoning
- D. Phosphorus poisoning

74. A child who consumed an unidentified fruit is brought to your emergency department presenting with the following indications

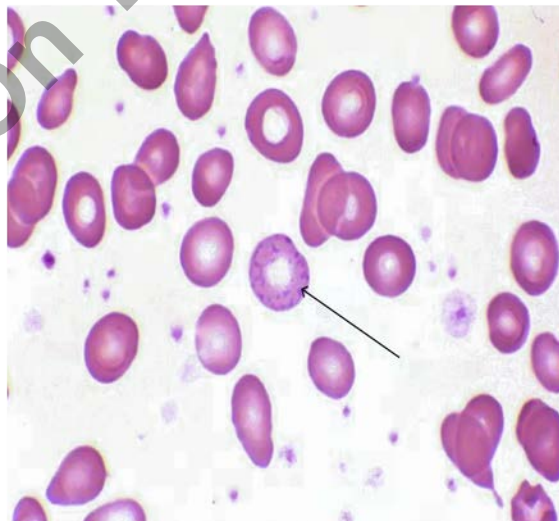
- A. Poppy seeds poisoning

- B. Datura poisoning
 - C. Mushroom poisoning
 - D. Cannabis poisoning
-

75. After 72 hours of Cessation of alcohol, which of the following withdrawal symptoms can be seen in the alcoholic patients?

- A. Seizures
 - B. Delirium tremens
 - C. Tremors
 - D. Hallucinations
-

76. Which of the following diagnoses give the hematological picture as given below?



- A. Minamata disease

- B. Saturnism
 - C. Chronic iron toxicity
 - D. Arsenic poisoning
-

77. Please rearrange the following options to reflect the correct sequence of symptoms observed in cases of iron toxicity. Healing with scarring Latent phase Gastrointestinal Irritation Liver failure and hypoglycemia

- A. 2, 3, 1, 4
 - B. 2, 3, 4, 1
 - C. 3, 2, 4, 1
 - D. 3, 4, 1, 2
-

78. A young teenage boy presents to the emergency with a history of drug abuse at a party a couple of hours ago. Some of the symptoms were diaphoresis, headache, anxiety and hyperactivity. On examination, there was tachycardia and hypertension and his pupils were dilated and fixed. Which of the following substances might he have taken?

- A. Alcohol
 - B. Cocaine
 - C. Opioid
 - D. Cannabis
-

79. In America, Asia, Africa and Australia, a plant is considered a noxious weed. This weed is thought to be a contributor to allergic respiratory issues, contact dermatitis, and cattle and human mutagenicity. Its allelopathy significantly reduces

crop output. Additionally, this weed's aggressive dominance endangers biodiversity, what is this plant?

- A. *Parthenium hysterophorus*
 - B. *Datura stramonium*
 - C. *Digitalis purpurea*
 - D. *Calotropis gigantea*
-

80. Which of the following choices best characterizes the typical symptoms observed in a patient who has been bitten by the snake depicted in the accompanying image?



- A. Non-poisonous, Dry bite
 - B. Severe edema, Necrosis and Bleeding
 - C. Severe muscle necrosis, Rhabdomyolysis
 - D. Ptosis, Paralysis and Amputation
-

81. Which of the following poisons can be found even in skeletonized remains of the body?

- A. Arsenic
 - B. Carbon monoxide
 - C. Hydrogen sulphide
 - D. Phosphorus
-

82. What is the drug that commonly leads to physical dependence?

- A. Ketamine
 - B. Heroin
 - C. LSD
 - D. Phencyclidine
-

83. In an autopsy of suspected poisoning, which organ, along with the stomach, part of the intestine, and liver, is preserved?

- A. Kidney
 - B. Lungs
 - C. Brain
 - D. Heart
-

84. Which poisoning is caused by a bullet retained inside the body?

- A. Iron
- B. Phosphorus

C. Nitrocellulose

D. Lead

85. A patient with Pink color skin & mucosa, a bitter almond smell from breath, and frothy discharge. The cause of death is:

A. H₂S poisoning

B. CO poisoning

C. HCN poisoning

D. OPC poisoning

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	1
Question 5	2
Question 6	1
Question 7	4
Question 8	4
Question 9	1
Question 10	1
Question 11	1
Question 12	1
Question 13	3
Question 14	3
Question 15	1
Question 16	1
Question 17	1
Question 18	1

Question 19	3
Question 20	3,5
Question 21	4
Question 22	2
Question 23	3
Question 24	4
Question 25	1
Question 26	1
Question 27	1
Question 28	1
Question 29	2
Question 30	2
Question 31	3
Question 32	4
Question 33	2
Question 34	1
Question 35	4
Question 36	2
Question 37	1
Question 38	3
Question 39	2

Question 40	2
Question 41	1
Question 42	3
Question 43	3
Question 44	4
Question 45	2
Question 46	2
Question 47	2
Question 48	1
Question 49	1
Question 50	2
Question 51	1
Question 52	1
Question 53	4
Question 54	4
Question 55	3
Question 56	2
Question 57	1
Question 58	1
Question 59	2
Question 60	1

Question 61	2
Question 62	1
Question 63	4
Question 64	2
Question 65	2
Question 66	3
Question 67	3
Question 68	3
Question 69	2
Question 70	4
Question 71	2
Question 72	2
Question 73	2
Question 74	2
Question 75	2
Question 76	2
Question 77	3
Question 78	2
Question 79	1
Question 80	4
Question 81	1

Question 82	2
Question 83	1
Question 84	4
Question 85	3

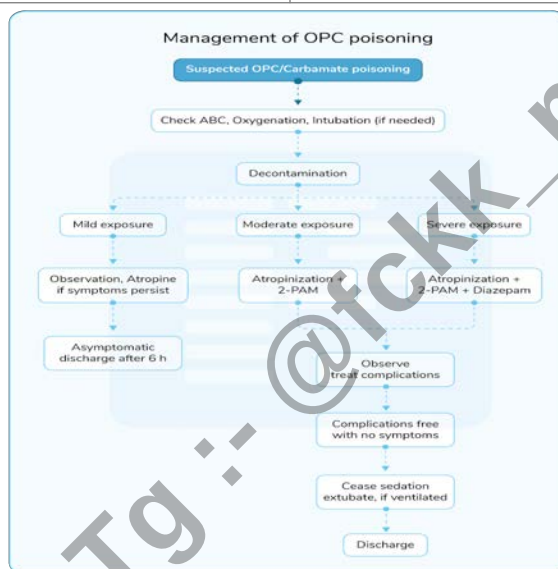
Solution for Question 1:

- Organophosphate (OP) poisoning leads to excessive cholinergic activity due to inhibition of acetylcholinesterase (AChE), resulting in accumulation of acetylcholine (ACh) at synapses. This leads to overstimulation of muscarinic and nicotinic receptors.
- Atropine is a muscarinic receptor antagonist, which reverses muscarinic effects but does not act on nicotinic receptors.
- Muscle weakness is a nicotinic effect of organophosphate poisoning, and atropine does not counteract nicotinic receptors.

OPC poisoning:

Symptoms	Muscarinic: Lethargy, Salivation, Diarrhea, Lacrimation, Urination, Miosis, Muscle weakness, Bronchorrhea, Bronchospasm, Bradycardia, Emesis (Mnemonic: DUMB BELLS)
Nicotinic: Muscle weakness, Fasciculation, Areflexia, Muscle paralysis	
CNS: Headache, Tremor, Confusion, Slurred speech, Coma, Convulsion	
Laboratory Diagnosis	RBC/True Cholinesterase (specific) Plasma/Pseudocholinesterase (Most common) Falls rapidly with organophosphorus poisoning Inexpensive, sensitive) Atropine test : If 2 mg of Atropine relieves symptoms, it confirms OP poisoning P-nitro phenol test for confirmation.

Management	Resuscitative measures (Airway-Breathing-Circulation) Decontamination Gastric lavage with 1:5000 KMnO ₄ Activated charcoal Atropine sulfate (2-4 mg IV every 10-15 minutes until atropinization) Signs of atropinization: Heart rate>80 beats/min, pupils no longer pinpoint, clear chest on auscultation with no wheeze, systolic BP>80mmHg, Dry axillae Oximes (Pralidoxime) to decrease muscarinic and CNS symptoms removes phosphate group reduces atropine requirement
Cause of Death	Respiratory Muscle Paralysis (RMP)



Bradycardia (Option A): Atropine blocks muscarinic receptors in the heart, leading to an increase in heart rate (tachycardia), so bradycardia would not persist.

Diaphoresis (Option C): Atropine reduces sweating (diaphoresis) by blocking muscarinic receptors in sweat glands.

Nasal discharge (Option D): Atropine dries up secretions, including nasal discharge, by blocking muscarinic receptors.

Solution for Question 2:

Correct Option A - Opioid poisoning

- Mechanism: Opioids, such as heroin or morphine, exert their effects by binding to opioid receptors in the central nervous system. This binding results in central nervous system depression, leading to altered sensorium, respiratory depression, and pinpoint pupils.
- Clinical Features: Altered mental status Respiratory depression Constricted pupils (miosis) Hypotension
- Altered mental status
- Respiratory depression
- Constricted pupils (miosis)
- Hypotension
- Treatment: Opioid toxicity is managed with the administration of naloxone, an opioid receptor antagonist, which can reverse the effects of opioid overdose.
- Altered mental status
- Respiratory depression
- Constricted pupils (miosis)
- Hypotension

Incorrect Options:

Options B - Datura:

- Datura contains anticholinergic alkaloids, leading to anticholinergic toxicity. The characteristic features include altered mental status, dilated pupils (mydriasis), dry skin, and hallucinations. However, the described symptoms of constricted pinpoint pupils are not typical of Datura poisoning.
- Treatment: Supportive care, including symptomatic treatment and administration of physostigmine (an acetylcholinesterase inhibitor).

Option C - Cocaine:

- Cocaine is a stimulant that increases sympathetic activity. Clinical features of cocaine toxicity include agitation, hypertension, dilated pupils, and hyperthermia. Constricted pinpoint pupils are not consistent with cocaine
- Treatment : Benzodiazepines for sedation, antihypertensive agents, and cooling measures in cases of hyperthermia.

Option D - Amphetamine:

- Similar to cocaine, amphetamines stimulate the release of neurotransmitters, leading to increased sympathetic activity. The clinical presentation includes agitation, hypertension, dilated pupils, and hyperthermia. Constricted pinpoint pupils are not characteristic of amphetamine toxicity.

- Treatment: Supportive care, including sedation with benzodiazepines and management of complications.

Solution for Question 3:

Correct Option A - A-2, B-1, C-3, D-4:

- Datura : Datura, commonly known as Jimson Weed, induces poisoning through its tropane alkaloids, primarily impacting the Central Nervous System (CNS). Symptoms include dryness of skin and mouth, difficulty in swallowing, dilated pupils as an early sign, a staggering gait resembling drunkenness, and slurred speech. Muttering delirium and drowsiness may also manifest, reflecting the profound CNS effects.

Datura Stramonium



Sparks

- Nicotine : Nicotine stimulates the cardiovascular system, causing an increase in heart rate and blood pressure. This effect is attributed to the release of adrenaline and noradrenaline, contributing to its addictive nature and potential cardiovascular risks associated with tobacco use.
- Nitric acid: Nitric acid is a corrosive substance that can cause severe skin damage upon contact, leading to burns and tissue destruction. The acid's caustic nature results in chemical burns, inflammation, and potential scarring, emphasizing the importance of prompt and thorough first aid measures, such as washing the affected area with water.

- Nux Vomica (Strychnine): Strychnine, found in Nux Vomica, acts on the spinal cord. It blocks inhibitory neurotransmitters, leading to uncontrolled muscle contractions and spasms, primarily affecting the spinal cord.

Strychnine Seeds aka Dog Buttons



Solution for Question 4:

Correct Option A - Yellow Oleander (*Thevetia peruviana*)

- Yellow oleander contains toxic cardiac glycosides, which can affect the heart's rhythm and function. Ingestion of any part of the plant can lead to symptoms like nausea, vomiting, abdominal pain, diarrhea, irregular heart rate, and even death. The symptoms described in the case (nausea, vomiting, abdominal pain, diarrhea, irregular heart rate) match those associated with yellow oleander poisoning.

Incorrect Options:

Option B - *Ricinus communis* (Castor Bean Plant): *Ricinus communis* contains the deadly toxin ricin. While its seeds are highly toxic, causing symptoms such as abdominal pain, vomiting, diarrhea, and potentially fatal organ damage, the symptoms described in the case (irregular heart rate) don't typically align with ricin poisoning. Ricin primarily affects the gastrointestinal system and other organs.

Option C - Datura: Datura contains tropane alkaloids that can lead to symptoms like hallucinations, delirium, dry mouth, blurred vision, and an increased heart rate. While some symptoms could match (such as irregular heart rate), the prominent gastrointestinal symptoms described in the case (nausea, vomiting, abdominal pain, diarrhea) are not characteristic of Datura poisoning.

Option D - Papaver somniferum (Opium Poppy): While opium poppy can produce opiates like morphine and codeine, the symptoms associated with its ingestion include respiratory depression, sedation, pinpoint pupils, and potential overdose symptoms. The symptoms described in the case (nausea, vomiting, abdominal pain, diarrhea, irregular heart rate) do not align well with opium poppy poisoning.

Solution for Question 5:

Correct Option B - Ethyl alcohol

- The patient's presentation with abdominal discomfort, acute confusion, vision loss, and an unsteady gait after consuming alcohol are suggestive of methanol poisoning.
- Ethyl alcohol (ethanol) is used as an antidote for methanol poisoning because it competes with methanol for the same enzyme, alcohol dehydrogenase, thereby preventing the formation of toxic metabolites (formaldehyde and formic acid) from methanol.
- This treatment helps to reduce the toxic effects and allows time for methanol to be excreted unchanged by the kidneys.

Incorrect Options: Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 6:

Correct Option A - 1, 2, and 3 only:

- The antidote used in the cyanide poisoning includes Amyl nitrate, Sodium nitrate, and Sodium thiosulphate.
- This is called the Lily cyanide antidote kit

Incorrect Options:

Option B, C, and D: These are incorrect options.

Solution for Question 7:

Correct Option D - Calcium oxalate:

- The above case scenario is diagnostic of Oxalic acid poisoning.
- Signs and symptoms Fulminant poisoning- Burning, sour or bitter taste in the mouth with a sense of constriction around the throat and burning pain from the mouth to stomach, radiating all over the abdomen. Nausea and eructation are immediately followed by vomiting which contains altered blood, mucus and Acute poisoning- Tingling and numbness of fingers and limbs, weakness, paresthesia, carpopedal spasm*, hyperirritability of peripheral nerves (Chvostek/Weiss sign), tetany, convulsions, coma and death. Delayed poisoning- Characterized by nephritis—uremia, scanty urine, hematuria, albuminuria, oxaluria (envelope-shaped calcium oxalate crystals in the urine are seen under microscope).
- Fulminant poisoning- Burning, sour or bitter taste in the mouth with a sense of constriction around the throat and burning pain from the mouth to stomach, radiating all over the abdomen. Nausea and eructation are immediately followed by vomiting which contains altered blood, mucus and
- Acute poisoning- Tingling and numbness of fingers and limbs, weakness, paresthesia, carpopedal spasm*, hyperirritability of peripheral nerves (Chvostek/Weiss sign), tetany, convulsions, coma and death.
- Delayed poisoning- Characterized by nephritis—uremia, scanty urine, hematuria, albuminuria, oxaluria (envelope-shaped calcium oxalate crystals in the urine are seen under microscope).
- Fulminant poisoning- Burning, sour or bitter taste in the mouth with a sense of constriction around the throat and burning pain from the mouth to stomach, radiating all over the abdomen. Nausea and eructation are immediately followed by vomiting which contains altered blood, mucus and

- Acute poisoning- Tingling and numbness of fingers and limbs, weakness, paresthesia, carpopedal spasm*, hyperirritability of peripheral nerves (Chvostek/Weiss sign), tetany, convulsions, coma and death.
- Delayed poisoning- Characterized by nephritis—uremia, scanty urine, hematuria, albuminuria, oxaluria (envelope-shaped calcium oxalate crystals in the urine are seen under microscope).

Incorrect Options:

Option A - Cystine:

- Cystine stones are hexagonal or benzene shaped.

Option B - Xanthine:

- Xanthine stones are very uncommon in urine.

Option C - Ammonium urate:

- Uric acid crystals are rosette shaped.

Solution for Question 8:

Correct Option D - Barium carbonate:

- Barium carbonate is a rodenticide which presents with classical features of areflexia and paralysis.
- Other symptoms include- Tingling sensation, tremors, cramps, stiffness of the muscles, paralysis of the tongue and larynx, myoclonus, myalgia, flaccid quadriplegia, hypertension, arrhythmia, ectopic beats, ventricular fibrillation, irregular pulse, shock, cardiac arrest, pulmonary edema, mydriasis, vertigo, headache, confusion, convulsions.
- Postmortem findings: Non-specific. Submucosal hemorrhages may be seen in the GIT.

Incorrect Options :

Option A - Aluminum phosphide:

- Aluminum phosphide presents with hypotension, arrhythmias, acute CHF, shock, dyspnea, cyanosis, Oliguria, renal failure. altered mental state, restlessness, convulsions, acute hypoxic encephalopathy, coma. muscle wasting, tenderness, bleeding diathesis.

Option B - Zinc phosphide:

- Zinc phosphide presents with dyspnea, lethargy, hypotension, cardiac arrhythmias, pulmonary edema, metabolic acidosis, convulsions, circulatory collapse, coma and death

Option C - Warfarin toxicity:

- Warfarin toxicity, the patient presents with abnormal bleeding, dark urine, dark stools.

Solution for Question 9:

Correct Option A - Delirium tremens:

- Delirium tremens is a severe form of alcohol withdrawal that typically occurs 48 to 96 hours after the last drink. It is characterized by: Confusion and disorientation Hallucinations (often visual) Autonomic hyperactivity Coarse tremors
- Confusion and disorientation
- Hallucinations (often visual)
- Autonomic hyperactivity
- Coarse tremors
- Confusion and disorientation
- Hallucinations (often visual)
- Autonomic hyperactivity
- Coarse tremors

Incorrect Options:

Option B - Wernicke Encephalopathy: Wernicke encephalopathy is a neurological disorder caused by thiamine (vitamin B1) deficiency commonly associated with chronic alcoholism. It is characterized by a classic triad of symptoms:

- G - Global confusion
- O - Ophthalmoplegia
- A - Ataxia
- P - Polyneuropathy

Option C - Korsakoff Psychosis: Korsakoff psychosis is a chronic condition that often follows Wernicke encephalopathy, characterized by:

- C - Confabulation (filling the memory of the brain with stories)

- A - Anterograde amnesia
- R - Retrograde amnesia (rare)

Option D - Magnan Phenomena: Magnan phenomena refer to tactile hallucinations, such as the sensation of insects crawling on or under the skin, associated with cocaine use rather than alcohol withdrawal.

Solution for Question 10:

Correct Option A - Cocaine:

- Cocaine is a powerful central nervous system (CNS) stimulant known for causing tachycardia and arrhythmias, which are common symptoms associated with its abuse. Additionally, a distinctive symptom of chronic cocaine abuse is formication—a tactile hallucination where individuals feel like insects are crawling on their skin, leading to scratching and multiple scratch marks observed on the body. This symptom is also referred to as "cocaine bugs" or Magnan syndrome.

Incorrect Options:

- Options B, C, and D are not associated with Magnan syndrome.

Solution for Question 11:

Correct Option A - Physostigmine:

- Physostigmine is the antidote of choice for Datura poisoning because it is a reversible acetylcholinesterase inhibitor. It helps counteract the anticholinergic effects caused by the tropane alkaloids (such as atropine, hyoscine, and hyoscyamine) present in Datura.

Incorrect Options:

- Options B, C, and D are not the mainstay management of datura poisoning.

Solution for Question 12:

Correct Option A - Sodium fluoride:

- For toxic analysis, like in alcohol abuse, the ideal preservative for blood is sodium fluoride.

Incorrect Options:

- Options B, C, and D are not the preferred substances to preserve blood.
-

Solution for Question 13:

Correct Option C - Hydrocarbons:

- Hydrocarbons (such as gasoline, kerosene, or other petroleum products) are highly dangerous if aspirated into the lungs. They can cause severe chemical pneumonitis and aspiration pneumonia, so gastric lavage is contraindicated in hydrocarbon poisoning.

Incorrect Options:

- Options A, B, and D are not contraindications of gastric lavage.
-

Solution for Question 14:

Correct Option C - Datura:

Datura is a plant known for its anticholinergic properties. It can cause symptoms of anticholinergic toxicity, which include dry skin, dilated pupils (mydriasis), slurred speech, and altered sensorium. This is consistent with the clinical presentation described.

Incorrect Options

Option A - Morphine: Morphine is an opioid and typically causes miosis (constricted pupils), not mydriasis. It can lead to respiratory depression and altered sensorium, but the dry skin and dilated pupils are not characteristic of morphine poisoning.

Option B - Cannabis: Cannabis can cause altered sensorium and slurred speech but is not associated with dry skin or dilated pupils. Cannabis effects are typically less severe and do not match the full clinical picture described.

Option D - Alcohol: Alcohol can cause altered sensorium and slurred speech, but it is not associated with dry skin or dilated pupils. Alcohol typically causes miosis and does not have the anticholinergic effects seen with Datura.

Solution for Question 15:

Correct Option A - Arsenic:

Aldrich-Mees lines (transverse white lines across the nails) and raindrop pigmentation (brownish discoloration) on the hands are characteristic signs of arsenic poisoning. Arsenic is known to cause these distinctive dermatological and nail changes. Arsenic interferes with the keratinization process in the nails, leading to the formation of Mees lines, and can cause pigmentary changes in the skin.

Incorrect Options:

Option B - Thallium:

Thallium poisoning can cause symptoms like hair loss, peripheral neuropathy, and gastrointestinal issues, but it is not typically associated with Aldrich-Mees lines or raindrop pigmentation. Thallium poisoning does have its own set of diagnostic signs but these specific ones are not characteristic.

Option C - Cadmium:

Cadmium poisoning is associated with kidney damage and bone disease, not with Aldrich-Mees lines or raindrop pigmentation. Cadmium does not produce the same dermatological findings as arsenic.

Option D - Lead:

Lead poisoning is associated with basophilic stippling of red blood cells, but it does not typically present with Aldrich-Mees lines or raindrop pigmentation. Lead affects different body systems

and has different clinical manifestations.

Solution for Question 16:

Correct Option A - Lead poisoning:

- Saturnine gout refers to gout-like symptoms caused by chronic lead poisoning.
- Lead exposure can interfere with the kidneys' ability to excrete uric acid, leading to hyperuricemia (elevated levels of uric acid in the blood).
- This can result in the deposition of urate crystals in the joints, causing gout-like symptoms.
- This condition was historically seen in people exposed to lead, such as those who drank wine sweetened with lead or worked in industries with high lead exposure.

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This condition was historically seen in people exposed to lead, such as those who drank wine sweetened with lead or worked in industries with high lead exposure.

Incorrect Options:

Option B - Cadmium poisoning: Cadmium poisoning primarily affects the kidneys and can cause kidney damage, but it is not specifically associated with Saturnine gout.

Option C - Beryllium poisoning: Beryllium poisoning is associated with lung disease (berylliosis) and is not directly linked to Saturnine gout.

Option D - Mercury poisoning: Mercury poisoning primarily affects the nervous system and can lead to symptoms such as tremors, cognitive impairments, and sensory disturbances. It is not specifically associated with Saturnine gout.

Solution for Question 17:

Correct Option A - 1-D; 2-C; 3-A; 4-B:

- This is the correct matching. The images are correctly matched with their corresponding snake species.

Incorrect Options:

Option B - 1-A, 2-B, 3-C, 4-D: This is not the correct matching. The images are mismatched with their corresponding snake species.

Option C - 1-D, 2-C, 3-B, 4-A : This is not the correct matching. The images are not matched correctly with their corresponding snake species.

Option D - 1-D, 2-A, 3-B, 4-C : This is not the correct matching. The images are not matched correctly with their corresponding snake species.

Solution for Question 18:

Correct Option A - Erythroxylum coca:

- Erythroxylum coca, commonly known as the coca plant, contains the alkaloid cocaine.
- While cocaine is primarily known as a powerful stimulant, it can induce delirium and hallucinations at high doses or in certain circumstances.
- Thus, Erythroxylum coca can produce a deliriant effect when consumed in large quantities.

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Incorrect Options:

Option B - Datura: Datura plants, such as Datura stramonium, are known for producing deliriant toxins. They contain tropane alkaloids like scopolamine, hyoscyamine, and atropine, which are well-known for their deliriant properties. However, Datura is not the correct option in this context.



Option C - Hyoscine : Hyoscine, also known as scopolamine, is an alkaloid found in various plants, including Datura. It is known for its deliriant effects. However, in this question, the correct option is not Hyoscine itself but rather the plant that produces the deliriant toxin.

Option D - Digitalis purpurea : Digitalis purpurea, commonly known as foxglove, is not associated with producing deliriant toxins. It contains cardiac glycosides that primarily affect the heart.



Solution for Question 19:

Correct Option C - Body packer syndrome:

- Body packer's syndrome, also called the surgical mules syndrome is a planned activity to smuggle drugs.
- The drug to be smuggled is usually wrapped in polythene or condom, and swallowed/planted inside vagina or rectum. Keywords such as Laxatives, enema apparatus and airport points towards it.
- The x-ray shows cylindrical objects which may be containers for the drugs

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The x-ray shows cylindrical objects which may be containers for the drugs

Incorrect Options:

Option A - Bezoar Syndrome: It is a syndrome in which there is a tightly packed collection of something inside the stomach. Usually, bezoar occurs for a collection of hair inside the stomach, but it can be any indigestible natural/unnatural object.

Option B - Pica due to anemia: It does not present as shown in the case history.

Option D - Constipation due to fecalith: It mostly shows excessive bloating, not shown in the x ray

Solution for Question 20:

Correct

Option C. The assertion is false but reason is true:

- This option suggests that the assertion (eosinophilia as the most common hematological finding in lead poisoning) is false, but the reason (lead inhibits ALA dehydratase) is true. Therefore, option C is correct.
- Eosinophilia is not the most common hematological finding in lead poisoning, Most common haematological finding in lead poisoning is microcytic hypochromic anemia with ringed sideroblasts and basophilic stippling.
- The primary problem in lead poisoning is blockage of enzymes of heme synthesis , which are ALA and Ferrochelatase.

Incorrect Option

Option A . Both assertion and reason are true, and the reason is the correct explanation for the assertion:

- This option suggests that eosinophilia is the most common hematological finding in cases of lead poisoning because lead inhibits ALA dehydratase. Therefore, option A is incorrect.

Answer: B

- Both assertion and reason are true, but the reason is not the correct explanation for the assertion: This option is incorrect
- This option is incorrect
- This option is incorrect

Option D. Both assertion and reason are false:

- This option states that both the assertion and reason are false. Therefore, option D is incorrect.

Solution for Question 21:

Correct Option D - 1-hyoscyamine:

- The above image shows Atropa Belladonna , also called the deadly nightshade. It produces an active poisonous component called hyoscyamine.
- The toxin may also be found in datura Stramonium (Jimson weed)
- Hyoscyamine is an isomer of Atropine and has Anticholinergic and Antimuscarinic activity.
- It is used as an antiemetic, anti-secretory and antispasmodic agent in the treatment of nausea, motion sickness, allergic rhinitis etc.
- Side effects include dryness of the mouth and eyes, tachycardia, palpitations, decreased sweating, and can also produce anxiety and restlessness

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Incorrect Options:

Option A - Calotropis Gigantea: It is a purple colored flower which has an active poisonous principle called Calotropin, Calotoxin etc.

Calotropis



Option B - Digitalis: The image below shows the plant digitalis purpurea. It is also called Purple Fox Glove

Digitalis Purpurea



Option C: Ricin - It is produced by a castor plant, also called Arandi.

Solution for Question 22:

Correct Option B - Nigella sativa:

- The seeds shown in the image resemble the seeds of Nigella sativa, also known as black cumin or black seed. Nigella sativa is a flowering plant that is native to Southwest Asia and has been used for centuries in traditional medicine and culinary practices.
- The seeds of Nigella sativa are small, black, and triangular in shape. They have a distinct aroma and taste. These seeds are known for their potential health benefits and are used in various cuisines and traditional remedies.
- In traditional medicine, Nigella sativa seeds have been attributed with several medicinal properties, including anti-inflammatory, antioxidant, and antimicrobial effects. They are often used for their potential benefits in improving digestion, boosting the immune system, and relieving respiratory conditions.
- Culturally, Nigella sativa seeds are used in cooking and baking as a spice or flavoring agent. They add a nutty and slightly peppery taste to dishes and are commonly used in bread, curries, and pickles.

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Culturally, Nigella sativa seeds are used in cooking and baking as a spice or flavoring agent. They add a nutty and slightly peppery taste to dishes and are commonly used in bread, curries, and pickles.

Incorrect Options:

Option A - Tropicana canna : Tropicana canna refers to a specific variety of Canna plant, which is known for its colorful flowers. The seeds of Tropicana canna are typically not used or recognized for their medicinal or culinary purposes. Therefore, this option is incorrect.

Option C - Datura seeds : Datura is a genus of flowering plants that contains several species, including Datura stramonium. While Datura seeds do exist, they have a distinct appearance and are not similar to the seeds shown in the image. Datura seeds are larger, rounder, and typically have a spiky or prickly texture. Therefore, this option is incorrect.

Option D - Opium seeds : Opium seeds come from the opium poppy plant (*Papaver somniferum*). These seeds are small and kidney-shaped, quite different from the triangular seeds shown in the image. Opium seeds are mainly recognized for their role in opium production and are not typically used in culinary or medicinal applications. Therefore, this option is incorrect.

Solution for Question 23:

Correct Option C - Anemic hypoxia:

- Smoke inhalation can lead to anemic hypoxia as the cause of death. Anemic hypoxia occurs when there is a reduced oxygen-carrying capacity of the blood. In the case of smoke inhalation, the inhaled smoke can contain toxic substances such as carbon monoxide (CO) and hydrogen cyanide (HCN). These substances can interfere with the normal binding of oxygen to hemoglobin in red blood cells, resulting in reduced oxygen transport throughout the body.
- Carbon monoxide (CO) is particularly problematic as it has a strong affinity for hemoglobin and forms carboxyhemoglobin, which prevents the binding of oxygen. Hydrogen cyanide (HCN) interferes with cellular respiration and inhibits the utilization of oxygen by cells.

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Incorrect Options:

Option A - Histotoxic hypoxia : Histotoxic hypoxia refers to the inability of the body tissues to effectively use oxygen, even when an adequate amount of oxygen is available. It typically occurs due to the impairment of cellular metabolism or function, such as in cases of certain drug overdoses. Histotoxic hypoxia is not the primary cause of death in smoke inhalation.

Option B - Ischemic hypoxia : Ischemic hypoxia is caused by a reduced blood supply to tissues, resulting in insufficient oxygen delivery. It can occur due to conditions like cardiac arrest or

severe blood loss. While smoke inhalation can lead to tissue damage and potential ischemia in specific cases, it is not the primary cause of death in smoke inhalation.

Option D - All of the above: This option is incorrect because all the given options are not correct.

Solution for Question 24:

Correct Option D - i -3, ii-4, iii-1, iv-2:

- Heroin: Heroin is a highly addictive opioid drug derived from morphine. It is commonly abused for its euphoric effects. Heroin is correctly paired with option 3 .
- MDMA: "Eve" is a term commonly used to refer to the drug MDMA, also known as Ecstasy. MDMA is a synthetic psychoactive drug with stimulant and hallucinogenic effects. Therefore, " Eve" is correctly paired with option 4 .
- Joint: A joint refers to a rolled cigarette containing cannabis (marijuana). Cannabis is commonly consumed for its psychoactive effects. Therefore, "Joint" is correctly paired with option 1 .
- LSD: LSD (lysergic acid diethylamide) is a potent hallucinogenic drug known for its psychedelic effects. It can induce visual and auditory hallucinations. Therefore, LSD is correctly paired with option 2 .
- i. Heroin ii. MDMA iii. Joint iv. LSD 3.Brown sugar 4.Amphetamine 1.Ganja 2.Acid

Heroin: Heroin is a highly addictive opioid drug derived from morphine. It is commonly abused for its euphoric effects. Heroin is correctly paired with option 3 .

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Joint: A joint refers to a rolled cigarette containing cannabis (marijuana). Cannabis is commonly consumed for its psychoactive effects. Therefore, "Joint" is correctly paired with option 1 .

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i. Heroin ii. MDMA iii. Joint iv. LSD	3.Brown sugar 4.Amphetamine 1.Ganja 2.Acid
---------------------------------------	--

Incorrect Options:

Option A - i-2, ii-1, iii-4, iv-3: This choice incorrectly pairs Heroin with option 2, Eve with option 1, Joint with option 4, and LSD with option 3. The correct pairings have been explained earlier.

Option B - i-3, ii-2, iii-1, iv-4 : This choice incorrectly pairs Heroin with option 3, Eve with option 2, Joint with option 1, and LSD with option 4. Again, the correct pairings have been provided above.

Option C - i-4, ii-2, iii-1, iv-3 : This choice incorrectly pairs Heroin with option 4, Eve with option 2, Joint with option 1, and LSD with option 3. The correct pairings have been explained earlier.

Solution for Question 25:

Correct Option A - Only the roots of this plant are poisonous:

- Aconite, also known as monkshood or wolfsbane, is a highly toxic plant. It contains various alkaloids, including aconitine, which is responsible for its poisonous properties.
- The incorrect statement in this case is that only the roots of the plant are poisonous.
- In reality, all parts of the aconite plant are considered poisonous, including the roots, stems, leaves, and flowers.
- The plant contains toxic alkaloids throughout its structure, not just in the roots. Therefore, it is important to exercise caution and avoid direct contact or ingestion of any part of the aconite plant.

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The plant contains toxic alkaloids throughout its structure, not just in the roots. Therefore, it is important to exercise caution and avoid direct contact or ingestion of any part of the aconite plant.

Incorrect Options:

Option B - Causes AV block: Aconite is known to have potent cardiac effects. It can cause significant cardiac toxicity, including cardiac arrhythmias and disturbances in conduction, such as AV block. These effects are attributed to the toxic alkaloids present in the plant, particularly aconitine. Therefore, it is correct to state that aconite can cause AV block.

Option C - Atropine is not specific for aconite poisoning: In cases of aconite poisoning, the primary treatment is supportive care and immediate medical attention. Activated charcoal may be administered to help prevent absorption of the toxin. Additionally, supportive measures such as cardiac monitoring, intravenous fluids, and management of arrhythmias may be necessary. Atropine is not specific for aconite poisoning and may not effectively counteract the toxic effects of aconite alkaloids.

Option D - Grow in sub Himalayan range: This statement is correct it grows in Himalayan range and used in various kind of herbal medicine.

Solution for Question 26:

Correct Option A - A-2, B-4, C-1, D-3:

- (A) Zinc chloride: (2) Irritant
- (B) Chloral hydrate: (4) Stupefying
- (C) Quinine: (1) Abortifacient
- (D) Potassium carbonate: (3) Corrosive

(A) Zinc chloride: (2) Irritant

(B) Chloral hydrate: (4) Stupefying

(C) Quinine: (1) Abortifacient

(D) Potassium carbonate: (3) Corrosive

Explanation:

- Zinc chloride is an irritant and can cause irritation and damage to the gastrointestinal tract.
- Chloral hydrate is a sedative-hypnotic drug that can induce a state of stupor or unconsciousness.
- Quinine has been historically used as an abortifacient, which means it can induce abortion or termination of pregnancy.
- Potassium carbonate is corrosive and can cause damage to tissues upon contact.

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Chloral hydrate is a sedative-hypnotic drug that can induce a state of stupor or unconsciousness.

Quinine has been historically used as an abortifacient, which means it can induce abortion or termination of pregnancy.

Potassium carbonate is corrosive and can cause damage to tissues upon contact.

Incorrect Options:

Option B, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 27:

Correct Option A - Atropine and neostigmine are very useful in the management of krait bite:

- The false statement regarding the management of snakebite in India is Atropine and neostigmine are very useful in the management of krait bite.
- Atropine and neostigmine are not typically used in the management of snake bites.
- Kraits are venomous snakes found in India, and their bites primarily result in neurotoxic effects.
- The mainstay of treatment for snakebites, including krait bites, is administration of anti-snake venom (ASV) to neutralize the venom's effects.
- Supportive measures such as ventilatory support may be required in severe cases.

Incorrect Options:

Option B - Neostigmine and ventilatory support are used in treatment along with ASV:

- In certain cases of snakebite, but this is not specific to krait bites.

Option C - ASV is the mainstay of treatment:

- ASV is indeed the mainstay of treatment for snakebites, including krait bites.

Option D - Current ASV in India is not effective against humped pit viper:

- The statement regarding the current ASV in India not being effective against humped pit viper is not mentioned in the options, so we cannot evaluate its accuracy based on the given choices.

Solution for Question 28:

Correct Option A - Arsenic:

Features of Arsenic Poisoning :

- A - Aldrich-Mees line/Reynold's line (present over nails), Aphrodisiac (increase libido)
- R - Red velvety mucosa, Raindrop pigmentation, Rashes (fading measles rash)
- S - Subendocardial hemorrhage/Sheehan hemorrhage
- E - Excess hyperpigmentation of palms and soles
- N - Neuritis (sensory > motor)
- I - Iron oxide antidote (hydrated ferric oxide)
- C - Mimics cholera

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R - Red velvety mucosa, Raindrop pigmentation, Rashes (fading measles rash)

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E - Excess hyperpigmentation of palms and soles

N - Neuritis (sensory > motor)

I - Iron oxide antidote (hydrated ferric oxide)

C - Mimics cholera

Incorrect Options:

Option B, C & D do not mimic cholera.

Solution for Question 29:

Correct Option B - Sodium amytal:

- Sodium amytal is known to cause bluish discoloration of the stomach mucosa on post-mortem examination

Incorrect Options:

Option A - Oxalic acid: Oxalic acid is not typically associated with bluish discoloration of the stomach mucosa on post-mortem examination.

Option C - Soneryl: Soneryl is the trade name for a medication called amitriptyline, which is a tricyclic antidepressant. It is not known to cause bluish discoloration of the stomach mucosa.

Option D - Arsenic: It is known to cause red velvety discoloration of the stomach mucosa.

Solution for Question 30:

Correct Option B - Corrosive poison:

- Corrosive poisons have the maximum damage potential as they can cause severe chemical burns and tissue damage.
- Corrosive poisons, such as strong acids and alkalis, can cause extensive damage when they come into contact with the body.
- They can lead to severe chemical burns, tissue destruction, and long-term complications.
- Irritant poisons, alcohol, and opioids can also cause harm, but they generally do not have the same level of damage potential as corrosive poisons.

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They can lead to severe chemical burns, tissue destruction, and long-term complications.

Irritant poisons, alcohol, and opioids can also cause harm, but they generally do not have the same level of damage potential as corrosive poisons.

Incorrect Options:

Option A - Irritant poison: While irritant poisons can cause harm, they generally have a lower damage potential compared to corrosive poisons.

Option C - Alcohol: Alcohol can be toxic in excessive amounts or when combined with certain substances, but it typically does not have the same level of damage potential as corrosive poisons.

Option D -Opioid: Opioids are potent drugs that can have serious health effects, including overdose and addiction. However, in terms of damage potential, corrosive poisons generally pose a higher risk.

Solution for Question 31:

Correct Option C - 1-E, 2-B, 3-D, 4-A:

Column A lists the poisons , and Column B lists the corresponding tests .

- Marsh test : Marsh test is used to detect the presence of arsenic . It involves generating arsine gas from the sample and using it to produce a characteristic brown or black deposit on a cold surface.
- Silver nitrate paper test : The silver nitrate paper test is used to detect the presence of cyanide . When exposed to cyanide, the silver nitrate paper turns yellow or brown.
- Cavett test : The Cavett test is used to detect the presence of alcohol . It involves using potassium dichromate and sulfuric acid to oxidize the alcohol, producing a colour change.
- Marquis test : Marquis test is used to detect the presence of opium alkaloids . It involves the addition of Marqui's reagent to the sample, resulting in a characteristic color change. It can be used to test cocaine, opiates and phenethylamines.
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Incorrect Options:

Option A, B, & D are incorrect.

Solution for Question 32:

Correct Option D - Belladonna:

- Belladonna, also known as deadly nightshade, contains tropane alkaloids such as atropine and scopolamine. These substances are known to cause anticholinergic effects, which include dry mouth, urinary retention, constipation, increased heart rate (tachycardia), and dilated pupils (mydriasis).

Incorrect Options:

Option A - Heroin : Heroin is an opioid drug that acts on opioid receptors in the central nervous system. While opioids like heroin can cause respiratory depression, they typically do not produce significant anticholinergic effects. Common effects of heroin use include euphoria, drowsiness, pinpoint pupils (miosis), respiratory depression, and constipation. Dry mouth, urinary retention, and dilated pupils are not typical symptoms associated with heroin use.

Option B - Morphine: Morphine is another opioid drug with similar effects to heroin. Like heroin, morphine primarily acts on opioid receptors and can cause respiratory depression and constipation. However, it does not typically produce pronounced anticholinergic effects such as dry mouth, urinary retention, and dilated pupils.

Option C - Malathion : Malathion is an organophosphate insecticide. Ingestion of malathion can lead to symptoms such as excessive salivation, sweating, and constricted pupils (miosis), rather than the anticholinergic effects described in the case.

Solution for Question 33:

Correct Option B - 1, 2, 3 are correct:

- 1, 2 and 3 are correct. Atropine, colchicine, and curare are all toxins.
- Atropine, colchicine, and curare are substances that possess toxic properties.

- Atropine is a naturally occurring substance found in certain plants, colchicine is a toxic alkaloid found in the autumn crocus plant, and curare refers to a group of natural toxins obtained from South American plants. Arsenic, on the other hand, is not classified as a toxin, but rather as a toxic element.

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Incorrect Options:

Option A - All are correct: This option is incorrect because arsenic is not considered a toxin.

Option C - 1, 3 are correct: This option is incorrect because it fails to include colchicine (option 2), which is indeed a toxin.

Option D - 3, 4 are correct: This option is incorrect because it includes arsenic (option 4), which is not a toxin

Solution for Question 34:

Correct Option A - A, B, C:

Statements A: Pain in abdomen, burning sensation, diarrhea seen in digoxin toxicity .This statement is true. Gastrointestinal symptoms, like Nausea, vomiting, pain in abdomen, burning sensation, diarrhea.

Statements B : Vision changes like Transient ambylopia, photophobia, may be noted in digoxin toxicity: This statement is true. Digoxin toxicity can cause Transient ambylopia, blurring, photophobia, scotoma, diplopia, color aberration

Statements C: Specific antidote for cardiac arrhythmias is lignocaine 100 mg IV or novocaine or propranolol. This statement is true.

Manifestations of digoxin toxicity:

- Digoxin toxicity can present with various signs and symptoms. The earliest manifestations often involve the gastrointestinal system, Nausea, vomiting, pain in abdomen, burning sensation, diarrhea.
- Cardiovascular effects : Bradycardia, extra systoles, ventricular tachycardia and fibrillation, atrial fibrillation, faintness, precordial oppression, heart block.
- Neurological effects : Headache, fatigue, confusion, anxiety, depression, disorientation, drowsiness, hallucinations, delirium.
- Respiratory system: Labored and sighing respiration
- The patient becomes drowsy and the condition may deepen into coma. Convulsions may precede death.
- Death occurs from cardiovascular collapse.

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Death occurs from cardiovascular collapse.

Treatment

- ECG monitoring is necessary as a guide to treatment. Gastric lavage is done with a solution of tannic acid. Activated charcoal is given. Purgatives may be given. Atropine is given in a dose of 0.6 mg IV to treat bradycardia. Potassium chloride may be given to reduce extra systoles. Specific antidote for cardiac arrhythmias is lignocaine 100 mg IV or novocaine or propranolol. Trisodium EDTA may help to lower serum calcium. Symptomatic treatment
- Gastric lavage is done with a solution of tannic acid.
- Activated charcoal is given.
- Purgatives may be given.
- Atropine is given in a dose of 0.6 mg IV to treat bradycardia.
- Potassium chloride may be given to reduce extra systoles.
- Specific antidote for cardiac arrhythmias is lignocaine 100 mg IV or novocaine or propranolol.

- Trisodium EDTA may help to lower serum calcium.
- Symptomatic treatment
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Gastric lavage is done with a solution of tannic acid.

Activated charcoal is given.

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Atropine is given in a dose of 0.6 mg IV to treat bradycardia.

Potassium chloride may be given to reduce extra systoles.

Specific antidote for cardiac arrhythmias is lignocaine 100 mg IV or novocaine or propranolol.

Trisodium EDTA may help to lower serum calcium.

Symptomatic treatment

Incorrect Options:

Option B, C & D are incorrect.

Statements D : Death occurs from Respiratory failure. This is a false statement.

- Death occurs from cardiovascular collapse.

Solution for Question 35:

Correct Option D - Ketamine:

- Ketamine acts primarily by blocking N-methyl-D-aspartate (NMDA) receptors in the brain, which can lead to altered perception, dissociation, and hallucinations.
- Additionally, ketamine can also affect other systems in the body, including the gastrointestinal system.
- Increased secretion and diarrhea are common gastrointestinal side effects of ketamine intoxication.
- Ketamine can stimulate gastrointestinal secretions and motility, leading to excess fluid secretion into the intestines and diarrhea.

Incorrect Options:

Options A, B, C: Cocaine, Crack, and Cannabis are not typically associated with increased secretion and diarrhea.

Solution for Question 36:

Correct Option B - Sling or splint leading to immobilization:

Initial management of snake bite involves the "RIGHT strategy."

- R: Reassure
- I: Immobilize
- G H: Go to the Hospital
- T: Tell the appropriate history and symptoms to the attending doctor.

Therefore, the best option for this question is "Sling or splint leading to immobilization."

Incorrect Options:

Option A - Polyvalent ASV: Polyvalent antivenom (Polyvalent ASV) might be required later in the treatment process, depending on the severity of the bite and the type of snake involved. However, the initial priority is to immobilize the affected limb to prevent the spread of venom.

Option C - Venom suction: Venom suction is to be avoided in case of a snake bite.

Option D - See the snake brought by a relative: This is not necessary as the ASV available is effective against all the common venomous snakes in India. Hence, ASV is administered regardless of the snake that caused the bite.

Solution for Question 37:

Correct Option A - NaF:

Preservative used for blood in autopsy:

- Sodium fluoride: Inhibits glycolysis.
- Potassium oxalate: Anticoagulant.

Incorrect Options:

Options B and D: Rectified spirit and NaCl are used to preserve organs and are preferred in the respective order.

Option C - Formalin: Formalin, also known as formaldehyde solution, is a commonly used fixative in histopathology.

Solution for Question 38:

Correct Options C - LSD:

- Tachycardia and visual hallucinations of red-colored halo are indicative of consumption of LSD.
- LSD - Lysergic acid Diethylamide: Acid blotter or paper stamp or taken as fumes.
- Obtained from rye fungus (ergot).
- Increases serotonin level in the brain (only psychological dependence).
- Associated with the trip (most common)
- Good trip Visual hallucinations and illusions
- Visual hallucinations and illusions
- Bad trip Anxiety Panic attacks
- Anxiety
- Panic attacks

- Flashback phenomenon is seen (re-experiencing the past experience).
- Visual hallucinations and illusions
- Anxiety
- Panic attacks

Incorrect Options:

Option A - Cannabis: Cannabis (marijuana) use typically does not cause significant tachycardia. While cannabis can induce visual hallucinations in some individuals, they are more commonly associated with LSD and other hallucinogens.

Option B - Crack/cocaine: Crack/cocaine use is known to cause tachycardia due to its stimulant effects on the central nervous system. Cocaine can sometimes induce tactile hallucinations of bugs crawling on the skin (formication).

Option D - Heroin: Heroin use is more commonly associated with symptoms such as sedation, respiratory depression, and pinpoint pupils (miosis).

Solution for Question 39:

Correct Option B - Amanita Phalloides poisoning:

- Amanita poisoning causes acute liver failure. It is a wild mushroom known as the death cap mushroom. Amanita poisoning presents with nausea, vomiting, diarrhea, enlarged and tender liver, jaundice, oliguria, mental confusion, convulsions and coma

Incorrect Options:

Option A - OP poisoning: OP poisoning presents with pinpoint pupils and diaphoresis, liver failure is not seen

Option C - Belladonna poisoning: Belladonna poisoning presents with tachycardia, hallucinations not liver failure

Option D - Morphine poisoning: Morphine poisoning presents with respiratory depression and pinpoint pupils, liver failure is not seen

Solution for Question 40:

Correct Option B - Greater curvature:

- In cases of corrosive acid ingestion, if surgical intervention is necessary to address severe damage, the stomach is typically surgically opened along the greater curvature.
- Here's an explanation of the correct option and the other options: Lesser Curvature: The maximum damage is on the lesser curvature Greater Curvature: Because, in cases like acid or corrosive poisoning, the maximum damage is on the lesser curvature (Magenstrasse). To examine the lesser curvature, it opened from greater curvature by the double ligature method. Vertical: The term "vertical" is not typically used to describe the surgical opening of the stomach. Instead, surgical approaches are often described in relation to specific anatomical features like the curvatures. Pylorus: The pylorus is the lower part of the stomach that connects to the small intestine. In cases of corrosive acid ingestion, the surgical focus is often on the affected areas of the stomach, and the pylorus may or may not be involved depending on the extent of the damage.
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Incorrect Options:

- Option A, C and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 41:

Correct Option A - Bagging:

- Bagging refers to a method of inhaling substances by placing them inside a bag and then inhaling or breathing in the vapors. The bag is typically held over the nose and mouth, allowing the person to inhale the substance contained within the bag. This method is often associated with the abuse of volatile substances or inhalants, such as solvents, aerosols, or gases.

Bagging refers to a method of inhaling substances by placing them inside a bag and then inhaling or breathing in the vapors. The bag is typically held over the nose and mouth, allowing the person to inhale the substance contained within the bag. This method is often associated with the abuse of volatile substances or inhalants, such as solvents, aerosols, or gases.

Incorrect Options:

Option B - Snorting: Snorting involves the inhalation of powdered substances, typically drugs, through the nose. The substance is crushed into a fine powder and then snorted through the nostrils using a straw, rolled-up paper, or other devices. This method allows the drug to be absorbed through the nasal mucosa and quickly enter the bloodstream.

Option C - Sniffing: Sniffing refers to the act of inhaling or smelling substances, often volatile or aromatic substances, to experience their scent or aroma. It is commonly associated with the evaluation of odors, such as in perfumery or wine tasting. In a forensic context, sniffing may be relevant in detecting or identifying certain substances or odors related to a crime scene.

Option A - Huffing : Huffing involves the deliberate inhalation of volatile substances or inhalants directly from their container or source. The person typically places their mouth or nose directly on the container and inhales deeply, allowing the vapors or fumes to enter their respiratory system. This method is associated with the misuse of chemicals, solvents, or other volatile substances for recreational purposes.

Solution for Question 42:

Correct Option C - Huffing:

- Huffing refers to the act of inhaling volatile substances or drugs by saturating a cloth or other material and then inhaling the fumes.

- Huffing is a form of inhalant abuse, and it can have harmful effects on the body, including intoxication and potential damage to the respiratory system.

Huffing refers to the act of inhaling volatile substances or drugs by saturating a cloth or other material and then inhaling the fumes.

Huffing is a form of inhalant abuse, and it can have harmful effects on the body, including intoxication and potential damage to the respiratory system.

Incorrect Options:

Option A - Bagging: Bagging refers to a method of inhaling substances by placing them inside a bag and then inhaling or breathing in the vapors. The bag is typically held over the nose and mouth, allowing the person to inhale the substance contained within the bag

Option B - Spraying : Spraying typically involves the dispersal of liquid substances in a fine mist or spray form.

Option D - Sniffing : Sniffing refers to the act of inhaling substances directly through the nose.

Solution for Question 43:

Correct Option C - Strychnine:

- Alkaloid from seeds of strychnos nux-vomica causes opisthotonus.
- The convulsions are most marked in antigravity muscles, so that the body typically arches in hyperextension.

Incorrect Options:

- Option A, B and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 44:

Correct Option D - Dimercaprol:

- In treating acute arsenic poisoning, the primary choice is D. Dimercaprol.
- Dimercaprol, also known as British Anti-Lewisite (BAL) , is an agent used in chelation therapy - a medical procedure that involves the administration of chelating agents to remove heavy metals from the body. Dimercaprol forms stable, non-toxic complexes with arsenic which are then excreted from the body making this the preferred treatment for acute arsenic poisoning.

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Treatment:

- First and appropriate step is to removal of exposure
- Chelation
- BAL (British anti lewisite/ Dimercaprol) Given as deep IM injection It is oil-based solution It is very painful.
- Given as deep IM injection
- It is oil-based solution
- It is very painful.
- DMSA (succimer): Oral formulation.
- Supportive measures

First and appropriate step is to removal of exposure

Chelation

BAL (British anti lewisite/ Dimercaprol)

- Given as deep IM injection
- It is oil-based solution
- It is very painful.

Given as deep IM injection

It is oil-based solution

It is very painful.

DMSA (succimer): Oral formulation.

Supportive measures

Incorrect Options:

Option A - Ipecac: This is a substance used to induce vomiting, mainly in cases of certain types of poisoning. However, it's not typically employed in arsenic poisoning due to the risk of causing further damage to the gastrointestinal tract.

Option B - Penicillamine: This is a chelating agent used primarily in the treatment of Wilson's disease (a rare genetic disorder of copper metabolism) and rheumatoid arthritis. It's not the first-line treatment for arsenic poisoning.

Option C - Activated Charcoal: This is commonly used in the emergency treatment of certain kinds of poisoning to prevent the poison from being absorbed from the stomach into the body. However, it's not effective in arsenic poisoning as arsenic is rapidly absorbed and not effectively adsorbed by activated charcoal.

Solution for Question 45:

Correct Option B - Bhilawanol:

- The correct answer for the active principle of the poison shown in the image of the marking nut is Bhilawanol . Bhilawanol is a toxic phenolic constituent found in the seeds or nuts of the plant *Semecarpus anacardium* . It is responsible for the toxic effects associated with the consumption or contact with the marking nut.

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Incorrect Options:

Option A - Ricin: Ricin is a toxic protein found in the seeds of the castor oil plant (*Ricinus communis*). It is not relevant to the marking nut (*Semecarpus anacardium*).

Option C - Abrin: Abrin is a toxic protein found in the seeds of the rosary pea (*Abrus precatorius*). It is not associated with the marking nut.

Option D - Calotropin: Calotropin is a cardiac glycoside found in plants of the *Calotropis* genus, such as *Calotropis gigantea* and *Calotropis procera*. It is not present in the marking nut.

Solution for Question 46:

Correct Option B - Bradycardia:

- In the context of cocaine poisoning, the symptoms typically include diaphoresis (excessive sweating), headache, acute coronary spasm leading to myocardial ischemia (reduced blood flow to the heart), and agitation. However, bradycardia (slow heart rate) is not commonly associated with cocaine poisoning. Instead, cocaine use usually leads to tachycardia (rapid heart rate) as one of its effects on the cardiovascular system. Therefore, the presence of bradycardia would go against the likely diagnosis of cocaine poisoning.

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Incorrect Options:

Option A - Myocardial ischemia: Myocardial ischemia refers to inadequate blood supply to the heart muscle, often leading to chest pain or discomfort. In the context of cocaine poisoning, one of the significant adverse effects is acute coronary spasm, which can result in myocardial ischemia. Therefore, the presence of myocardial ischemia would actually support the diagnosis of cocaine poisoning, making this option incorrect.

Option C - Hyperthermia: Hyperthermia, or abnormally high body temperature, is a common manifestation of cocaine poisoning. Cocaine stimulates the sympathetic nervous system, leading to increased body temperature, diaphoresis (excessive sweating), and hyperthermia. Thus, the presence of hyperthermia would be consistent with cocaine poisoning, making this option incorrect.

Option D - Agitation: Agitation is a characteristic symptom of cocaine intoxication. Cocaine acts as a central nervous system stimulant, leading to heightened arousal, restlessness, and agitation. Therefore, the presence of agitation would support the diagnosis of cocaine poisoning, making this option incorrect.

Solution for Question 47:

Correct Option B - Datura, Physostigmine:

- Datura is a plant that contains alkaloids, such as scopolamine and atropine , which have potent anticholinergic effects.
- Ingestion of Datura can lead to the symptoms described in the scenario, including irritability, restlessness, dry hot skin, and urinary and bowel retention.
- Physostigmine is a reversible acetylcholinesterase inhibitor that acts by increasing the levels of acetylcholine in the body.
- It is an appropriate antidote for anticholinergic poisoning, including Datura poisoning. By inhibiting acetylcholinesterase, physostigmine can counteract the effects of excessive anticholinergic activity and alleviate the symptoms.
- Therefore, in the given scenario, the correct option is "Datura, Physostigmine" because physostigmine is the appropriate antidote for reversing the anticholinergic effects of Datura poisoning.

Datura is a plant that contains alkaloids, such as scopolamine and atropine , which have potent anticholinergic effects.

Ingestion of Datura can lead to the symptoms described in the scenario, including irritability, restlessness, dry hot skin, and urinary and bowel retention.

Physostigmine is a reversible acetylcholinesterase inhibitor that acts by increasing the levels of acetylcholine in the body.

It is an appropriate antidote for anticholinergic poisoning, including Datura poisoning. By inhibiting acetylcholinesterase, physostigmine can counteract the effects of excessive anticholinergic activity and alleviate the symptoms.

Therefore, in the given scenario, the correct option is "Datura, Physostigmine" because physostigmine is the appropriate antidote for reversing the anticholinergic effects of Datura poisoning.

Incorrect Options:

Option A. - Datura , pralidoxime: Pralidoxime, is an antidote used for organophosphate poisoning, not specifically for Datura poisoning. Organophosphates are a different class of toxic compounds that inhibit acetylcholinesterase, but they have different mechanisms of action compared to Datura alkaloids.

Option C and D - Yellow oleander, Digoxin and Yellow oleander, Physostigmine:

- Yellow oleander (*Thevetia peruviana*) is a plant that contains cardiac glycosides, particularly Thevetin A and Thevetin B. Ingestion of yellow oleander can lead to cardiac toxicity, including arrhythmias, conduction disturbances, and cardiac arrest. However, the symptoms described in the scenario, such as irritability, restlessness, dry hot skin, and urinary and bowel retention, are not typical of yellow oleander poisoning.
- Digoxin, on the other hand, is a cardiac glycoside commonly used in the treatment of heart failure and certain cardiac arrhythmias.
- It has a narrow therapeutic index, and overdose or toxicity can result in various cardiac manifestations, including arrhythmias.
- However, it does not cause the specific symptoms mentioned in the scenario, such as irritability, restlessness, and dry hot skin.
- Therefore, the option "Yellow oleander, Digoxin" is incorrect because it does not match the symptoms described in the scenario and does not involve the appropriate poison-antidote relationship.

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Digoxin, on the other hand, is a cardiac glycoside commonly used in the treatment of heart failure and certain cardiac arrhythmias.

It has a narrow therapeutic index, and overdose or toxicity can result in various cardiac manifestations, including arrhythmias.

However, it does not cause the specific symptoms mentioned in the scenario, such as irritability, restlessness, and dry hot skin.

Therefore, the option "Yellow oleander, Digoxin" is incorrect because it does not match the symptoms described in the scenario and does not involve the appropriate poison-antidote relationship.

Solution for Question 48:

Correct Option A - Arsenic:

- Black foot disease , also known as Blackfoot or arsenicosis, is a condition associated with chronic arsenic poisoning .
- It is a severe form of peripheral vascular disease in which blood vessels of the lower limb are severely damaged eventually resulting in gangrene .
- Prolonged exposure to high levels of arsenic, typically through contaminated drinking water or food, can lead to peripheral neuropathy and other health issues.

Black foot disease , also known as Blackfoot or arsenicosis, is a condition associated with chronic arsenic poisoning .

It is a severe form of peripheral vascular disease in which blood vessels of the lower limb are severely damaged eventually resulting in gangrene .

Prolonged exposure to high levels of arsenic, typically through contaminated drinking water or food, can lead to peripheral neuropathy and other health issues.

Incorrect Options:

Option B - Lead:

- Lead poisoning can lead to developmental delays, learning difficulties, behavior problems, and impaired growth. In adults, it can cause neurological symptoms, such as headaches, memory loss, and concentration problems, as well as abdominal pain and reproductive issues. It is not associated with black foot disease.

Option C - Mercury:

- The symptoms and effects of mercury poisoning depend on the form of mercury involved: Elemental mercury vapor exposure primarily affects the respiratory system and can cause lung damage, respiratory distress, and neurological symptoms such as tremors and cognitive impairments. Methylmercury, a form of organic mercury found in contaminated fish and seafood, primarily affects the nervous system and can cause developmental issues in children, as well as neurological and cardiovascular problems in adults. It is not associated with black foot disease.
- Elemental mercury vapor exposure primarily affects the respiratory system and can cause lung damage, respiratory distress, and neurological symptoms such as tremors and cognitive impairments.
- Methylmercury, a form of organic mercury found in contaminated fish and seafood, primarily affects the nervous system and can cause developmental issues in children, as well as neurological and cardiovascular problems in adults.
- It is not associated with black foot disease.

- Elemental mercury vapor exposure primarily affects the respiratory system and can cause lung damage, respiratory distress, and neurological symptoms such as tremors and cognitive impairments.
- Methylmercury, a form of organic mercury found in contaminated fish and seafood, primarily affects the nervous system and can cause developmental issues in children, as well as neurological and cardiovascular problems in adults.
- It is not associated with black foot disease.

Elemental mercury vapor exposure primarily affects the respiratory system and can cause lung damage, respiratory distress, and neurological symptoms such as tremors and cognitive impairments.

Methylmercury, a form of organic mercury found in contaminated fish and seafood, primarily affects the nervous system and can cause developmental issues in children, as well as neurological and cardiovascular problems in adults.

It is not associated with black foot disease.

Option D - Cadmium:

- Cadmium primarily affects the kidneys and can cause kidney damage or failure over time. It can also have adverse effects on the bones, respiratory system, and cardiovascular system. Symptoms of cadmium poisoning may include fatigue, weakness, cough, chest pain, and gastrointestinal disturbances. Prolonged exposure to high levels of cadmium can increase the risk of lung cancer and other cancers. It is not associated with black foot disease.

Solution for Question 49:

Correct Option A - Naloxone:

- Naloxone is an opioid receptor antagonist that rapidly reverses the effects of opioids, including respiratory depression. It competes with opioids for binding to opioid receptors, displacing them and reversing the respiratory depression .
- Morphine sulfate is an opioid analgesic that can cause respiratory depression as a side effect. If a patient develops respiratory distress or respiratory depression due to opioid overdose, naloxone is the appropriate antidote

Naloxone is an opioid receptor antagonist that rapidly reverses the effects of opioids, including respiratory depression. It competes with opioids for binding to opioid receptors, displacing them

and reversing the respiratory depression .

Morphine sulfate is an opioid analgesic that can cause respiratory depression as a side effect. If a patient develops respiratory distress or respiratory depression due to opioid overdose, naloxone is the appropriate antidote

Incorrect Options:

Option B - Atropine: Atropine is not the correct antidote for respiratory distress caused by morphine sulfate. Atropine is an anticholinergic medication used to treat muscarinic effects of organophosphorus compounds (OPC) . It does not reverse the effects of opioids.

Option C - Pralidoxime: Pralidoxime is an antidote used to treat poisoning by certain types of organophosphorus nerve agents or organophosphate insecticides by reactivation of acetylcholine esterase. It is not effective for opioid overdose.

Option D - Epinephrine: Epinephrine is primarily used for the treatment of severe allergic reactions (anaphylaxis), cardiac arrest and certain other emergency situations. It is not the appropriate antidote for respiratory distress caused by opioid overdose.

Solution for Question 50:

Correct Option B - Sample area cleaned with soap and water:

- The Sample area should be cleaned with soap and water instead of spirits or any other substance that might contain alcohol.
- Also 1:1000 mercuric chloride can be used to clean the area from where sample is to be collected.

The Sample area should be cleaned with soap and water instead of spirits or any other substance that might contain alcohol.

Also 1:1000 mercuric chloride can be used to clean the area from where sample is to be collected.

Blood sample collection in Alcohol intoxication

- The Sample area should be cleaned with soap and water also 1:1000 mercuric chloride can be used to clean the area from where sample is to be collected.
- Spirits or any other substance that might contain alcohol should be avoided as it might affect lab values.

- 100 mg NaF and 30 mg potassium oxalate for 10 ml of blood followed by thorough shaking is used for sample preservation.
- Test name : Kozelka and Hine's test
- Principle: Alcohol containing test substance reduces potassium dichromate solution

The Sample area should be cleaned with soap and water also 1:1000 mercuric chloride can be used to clean the area from where sample is to be collected.

Spirits or any other substance that might contain alcohol should be avoided as it might affect lab values.

100 mg NaF and 30 mg potassium oxalate for 10 ml of blood followed by thorough shaking is used for sample preservation.

Test name : Kozelka and Hine's test

Principle: Alcohol containing test substance reduces potassium dichromate solution

Incorrect Options:

Option A - Sample area cleaned with spirit: This option is incorrect as the presence of alcohol in spirit might effect the sample alcohol levels.

Option C - Sample should be preserved without preservative : This option is incorrect as 100 mg NaF and 30 mg potassium oxalate for 10 ml of blood followed by thorough shaking is used for sample preservation.

Option D - Only one urine sample is required - This option is incorrect as 2 samples are taken one at the time of arrival and the other one 25 to 30 min after 1 st sample .

For reliable UAC results:

- Collect two urine samples, spaced about half to 1 hour apart.
- Ensure the bladder is fully emptied before the first void.
- The second urine sample reflects the alcohol concentration formed between the first and second voids.
- Differences in UAC values indicate the trend of the UAC curve (rising or falling).
- The alcohol content of the second void represents the average alcohol concentration of the urine between voids.
- Typically, a 1.33:1 ratio of urine alcohol to blood alcohol is utilized.

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Ensure the bladder is fully emptied before the first void.

The second urine sample reflects the alcohol concentration formed between the first and second voids.

Differences in UAC values indicate the trend of the UAC curve (rising or falling).

The alcohol content of the second void represents the average alcohol concentration of the urine between voids.

Typically, a 1.33:1 ratio of urine alcohol to blood alcohol is utilized.

Solution for Question 51:

Correct Option A - 1, 2, 3 are correct:

The correct answer for the manifestations of chronic arsenic poisoning is A. 1, 2 and 3 are correct.

Raindrop pigmentation: This is a sign of chronic arsenic poisoning. It refers to the appearance of dark, 'raindrop-like' spots on the skin particularly on the torso and limbs. It's a form of hyperpigmentation caused by the buildup of arsenic in the body over time.

Peripheral neuritis: Chronic arsenic poisoning can also lead to peripheral neuritis, which is inflammation of the peripheral nerves. This can result in a variety of symptoms including weakness, numbness and pain, especially in the hands and feet.

Mees line: This refers to horizontal, white lines that can appear across the nails. These lines can be seen in a variety of conditions associated with systemic stress or illness, including arsenic poisoning.

- Treatment of arsenic poisoning : Remove the patient from the source of poison BAL (British anti lewisite) / Dimercaprol
- Remove the patient from the source of poison
- BAL (British anti lewisite) / Dimercaprol

Treatment of arsenic poisoning :

- Remove the patient from the source of poison
- BAL (British anti lewisite) / Dimercaprol

Remove the patient from the source of poison

BAL (British anti lewisite) / Dimercaprol

Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 52:

Correct Option A - Atropine:

- The correct management in this case would be A. Atropine .
- From the symptoms that the patient is displaying - excessive sweating, tearing, diarrhea, muscle weakness, and a rapid heart rate - we can infer that he may have ingested an organophosphorus compound. These compounds are found in some pesticides. They work by disrupting the normal function of the nervous system leading to an overabundance of the neurotransmitter acetylcholine.
- Atropine is normally used as the initial treatment in these types of cases. It's a n anticholinergic , which means it works by blocking the action of acetylcholine which helps to lessen the symptoms caused by this neurotransmitter's excess. It helps to restore the balance that the organophosphorus compound has upset.

The correct management in this case would be A. Atropine .

From the symptoms that the patient is displaying - excessive sweating, tearing, diarrhea, muscle weakness, and a rapid heart rate - we can infer that he may have ingested an organophosphorus compound. These compounds are found in some pesticides. They work by disrupting the normal function of the nervous system leading to an overabundance of the neurotransmitter acetylcholine.

Atropine is normally used as the initial treatment in these types of cases. It's a n anticholinergic , which means it works by blocking the action of acetylcholine which helps to lessen the symptoms caused by this neurotransmitter's excess. It helps to restore the balance that the organophosphorus compound has upset.

Antidotes for organophosphate poisoning:

Atropine sulphate

- It is a specific antidote (counteracts the muscarinic effects of organophosphate)

It is a specific antidote (counteracts the muscarinic effects of organophosphate)

Oximes

- Oximes reactivates acetylcholinesterase by removing the phosphoryl group which has been inactivated by phosphorylation due to an organophosphate.

Oximes reactivates acetylcholinesterase by removing the phosphoryl group which has been inactivated by phosphorylation due to an organophosphate.

Incorrect Options:

Option B, C and D are incorrect. Atropine is the initial treatment of choice in patients with organophosphate poisoning.

Solution for Question 53:

Correct Option D - Methanol:

- The term 'Hooch Tragedy' is generally used to refer to mass poisonings that occur due to the consumption of illicitly distilled or home-made alcohol, colloquially referred to as 'hooch.' These tragedies typically occur when methanol, a highly toxic substance, is used in the production of this illicit alcohol. Methanol poisoning can lead to a range of serious health problems, including blindness and death.

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Clinical features

- Abdominal pain/ vomiting [MC presenting complaint]
- Eye disturbances d/t formaldehyde & formic acid Retinal edema Blurring of vision is seen k/a Snow field vision Formaldehyde accumulates in the optic nerve & causes Optic neuritis, optic atrophy & eventually blindness
- Retinal edema
- Blurring of vision is seen k/a Snow field vision
- Formaldehyde accumulates in the optic nerve & causes Optic neuritis, optic atrophy & eventually blindness
- Metabolic acidosis d/t excess accumulation of Formic acid

Abdominal pain/ vomiting [MC presenting complaint]

Eye disturbances d/t formaldehyde & formic acid

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Retinal edema

Blurring of vision is seen k/a Snow field vision

Formaldehyde accumulates in the optic nerve & causes Optic neuritis, optic atrophy & eventually blindness

Metabolic acidosis d/t excess accumulation of Formic acid

Incorrect Options:

Option A - Phosphorus : While phosphorus is indeed a hazardous substance, it is not typically associated with alcohol production or consumption, and therefore would not be related to a ' hooch tragedy.'

Option B - Mercury: Mercury is a heavy metal that is dangerous if ingested, but it is not used in the production of alcohol and is not associated with hooch tragedies.

Option C - Ethanol : While ethanol is the type of alcohol that is safe for human consumption and is found in legally produced alcoholic beverages, hooch tragedies are specifically associated with the consumption of methanol, not ethanol.

Solution for Question 54:

Correct Option D - Atropa belladonna:

- The deadly nightshade plant, Atropa belladonna , is poisonous. Atropine, hyocyamine, and scopolamine are alkaloids found in their roots, leaves, and fruits. The alkaloids present are called tropane alkaloids, often known as belladonna alkaloids. The anticholinergic syndrome may result from Atropa belladonna poisoning. High doses of the plant may result in drowsiness and unconsciousness. Belladonna contains compounds that can affect the nervous system's operation as well as the production of saliva, sweat, changes in pupil size,

urine, and other bodily functions. Additionally, belladonna might raise blood pressure and pulse rate but is not considered a cardiac poison that can affect cardiac function.

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Signs and Symptoms:

- Dryness in skin and mouth
- Dysphagia: Difficulty in swallowing
- Dilated pupil (very earliest symptom)
- Drunken Gait
- Dysarthria: Slurred speech
- Muttering Delirium
- Drowsiness
- Death
- Carphologia: It feels like it is a kind of hallucination, imaginary threads, and pill-rolling movement.

Dryness in skin and mouth

Dysphagia: Difficulty in swallowing

Dilated pupil (very earliest symptom)

Drunken Gait

Dysarthria: Slurred speech

Muttering Delirium

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Death

Carphologia: It feels like it is a kind of hallucination, imaginary threads, and pill-rolling movement.



Cardiac Poisons

Mnemonic: Queen DONAC

- Caused by Q uinine D igitalis O leander (yellow, white/pink) N icotina tabacum A conite C erebra odellum
- Q uinine
- D igitalis
- O leander (yellow, white/pink)
- N icotina tabacum
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Caused by

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Q uinine

D igitalis

O leander (yellow, white/pink)

Nicotiana tabacum

Aconite

Cerbera odellum

Incorrect Options:

Option A - Aconite: The sodium channels are persistently activated by Aconitine's high affinity binding to their open state in the voltage-sensitive sodium channel at site 2, which results in arrhythmias from triggered activity.

Option B - Cerbera thevetia: This plant's seeds contain extremely toxic cardiac glycosides that can lead to electrolyte imbalances, vomiting, dizziness, and cardiac dysrhythmias such as conduction block that affects the sinus and AV nodes.

Option C - Nicotiana tabacum: Chronic nicotine use can increase the risk of heart attack and stroke as well as cause cardiovascular disease.

Solution for Question 55:

Correct Option C - Organophosphorus poisoning:

Organophosphorus poisoning generally occurs after respiratory, oral, or dermal exposure to nerve agents or organophosphorus pesticides. The onset symptoms of this specific disease are excessive lacrimation and saliva production, tiny pupils (miosis), vomiting, diarrhea, excessive sweating, confusion, etc. So, as farmers are experiencing pinpoint pupils and increased secretion, the diagnosis is organophosphorus poisoning.

- The primary treatment of Organophosphorus poisoning in emergency cases is stabilization. The experts will stabilize the breathing, decontaminate the body to stop further exposure and use intravenous fluids to clear the system from the toxins.
- But, in non-emergency cases, the healthcare provider will give supportive therapy and stabilize the breathing using atropine (drug).

Incorrect Options:

Option A - Alcohol poisoning: Alcohol poisoning is a severe and deadly condition. It quickly affects heart rate, breathing, gag reflex, and body temperature. So, it often leads to coma and even death.

Option B - Opioid poisoning: Opioid poisoning occurs when there is excessive stimulation on the pathway of opiate. It eventually leads to lower respiratory effort and death. It can also cause pinpoint pupils but there is no increased body secretions.

Option D: Datura poisoning: Datura poisoning after 60 minutes of ingestion. The symptoms are dry skin and mouth, respiratory arrest, urinary retention, delirium, and tachycardia.

Solution for Question 56:

Correct Option B - Argemone Mexicana:

- They're large and showy, with six bright yellow petals arranged around a central capsule . Keep in mind though, the petals sometimes have a slightly crumpled look, which gives the flowers a bit of an interesting flair. Then there's the seed pod, which is another key identifier. Once the flowers have bloomed and fallen off, you'll find seed capsules, which are prickly, oblong, and start off green but turn a dark brown or black as they mature. The foliage of the Argemone mexicana is blue-green in color, lobed, spiky, and has a somewhat thistle-like appearance. It has a prickly feel because of the presence of spines or bristles, which is a distinct characteristic of the plant. The plant as a whole has a very upright growth habit.

Incorrect Options:

Option A - Datura Alba : Large trumpet-shaped flowers are a distinguishing feature of the Datura species, which are either spreading annuals or short-lived perennials. The fruit, a spiny capsule with many seeds, is produced by the fragrant blooms, which can be white, yellow, pink, or purple.



Option C - *Nerium Odorum* : A strongly 5-lobed fringed corolla encircles the central corolla tube, and the blooms are white, pink to red, growing in clusters at the ends of each branch and about 2.5–5 cm (1-2 in) in diameter.



Option D - *Papaver Somniferum* : The flowers can be up to 3-10 cm (1-4 in) in diameter and typically have four white, mauve, or crimson petals. The base of the petals occasionally has dark patterns.



Solution for Question 57:

Correct Option A - Viper:

- The scenario described in the question aligns most closely with the typical symptoms of a viper bite.
- Vipers are known for their potent hemotoxic venom , which affects the body's blood clotting ability and can lead to significant local tissue damage.
- Symptoms of a viper bite can include severe local pain, bleeding, swelling and blistering at the wound site, all of which have been mentioned in the scenario. Additionally, systemic symptoms such as drowsiness could also occur, especially in severe cases.
- Bleeding from bite site is the most important feature, as it indicates viper's bite & is important sign to start ASV immediately

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Symptoms of Viper Snakebite

- Local symptoms: Pain Swelling Blisters Bleeding from the bite site is the most important feature, as it indicates a viper's bite & is an important sign to start ASV immediately Necrosis Regional lymphadenopathy: It is also an important sign for starting ASV These features are commonly referred as painful progressive swelling. It is commonly a/w Viper bites

- Pain

- Swelling

- Blisters

- Bleeding from the bite site is the most important feature, as it indicates a viper's bite & is an important sign to start ASV immediately

- Necrosis

- Regional lymphadenopathy: It is also an important sign for starting ASV

- These features are commonly referred as painful progressive swelling. It is commonly a/w Viper bites

- Vasculotoxic/ Hemotoxic features: Commonly seen with viper bites DIC [Disseminated Intravascular Coagulation] is the most important complication of viper bites. Spontaneous bleeding is seen everywhere I.e epistaxis, bleeding gums, hematemesis, melena, hemoptysis, hematochezia, and bleeding from the injection site are seen Local bleeding/ systemic bleeding is an indication for viper bite D/t DIC, a person can land-up in renal failure Rx: ASV, FFP [Fresh Frozen Plasma], Local surgical debridement for local necrosis

- Commonly seen with viper bites

- DIC [Disseminated Intravascular Coagulation] is the most important complication of viper bites.

- Spontaneous bleeding is seen everywhere

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- Local bleeding/ systemic bleeding is an indication for viper bite

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- Rx: ASV, FFP [Fresh Frozen Plasma], Local surgical debridement for local necrosis

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- Pain

- Swelling

- Blisters
- Bleeding from the bite site is the most important feature, as it indicates a viper's bite & is an important sign to start ASV immediately
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Swelling

Blisters

Bleeding from the bite site is the most important feature, as it indicates a viper's bite & is an important sign to start ASV immediately

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Regional lymphadenopathy: It is also an important sign for starting ASV

These features are commonly referred as painful progressive swelling. It is commonly a/w Viper bites

Vasculotoxic/ Hemotoxic features:

- Commonly seen with viper bites
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D/t DIC, a person can land-up in renal failure

Rx: ASV, FFP [Fresh Frozen Plasma], Local surgical debridement for local necrosis

Incorrect Options:

Option B - Cobra: Cobra venom is primarily neurotoxic, affecting the nervous system and leading to symptoms such as difficulty breathing, paralysis, and potentially death due to respiratory failure. While local symptoms like pain and swelling can occur, bleeding and blistering are not as common as in viper bites.

Option C - Wasp bite: Wasps can cause painful stings, and in some individuals, a severe allergic reaction can occur. However, profuse bleeding and blistering at the sting site are not typical of a wasp sting.

Option D - Scorpion: Most scorpion stings cause symptoms such as pain, numbness, or tingling around the sting area. Some scorpions can cause more severe symptoms, but these are typically systemic (such as difficulty breathing, muscle twitching, or irregular heartbeat) rather than severe local reactions like profuse bleeding or blistering.

Solution for Question 58:

Correct Option A - Strychnine:

- The symptoms described in the question - multiple episodes of seizures with consciousness between episodes and arching of the back (a condition known as opisthotonus) suggest strychnine poisoning . Strychnine, a potent neurotoxin, is commonly found in certain types of seeds and has historically been used in rat poison. Its toxicity primarily affects the nervous system, causing severe muscular convulsions . These symptoms are congruent with the scenario provided.

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Incorrect Options:

Option B - Ricinus: Ricinus is the plant from which castor oil is derived. The seeds contain the toxic compound ricin, which is one of the most poisonous naturally occurring substances. However, ricin poisoning typically causes severe abdominal pain, vomiting, and diarrhea. It does not usually result in seizures or opisthotonus, which were the primary symptoms described in the question.

Option C - Nerium odorum: This plant, also known as oleander, contains toxins that affect the heart. Oleander poisoning can cause a variety of symptoms, including an irregular or slow heartbeat, stomach pain, nausea, and vomiting, but it is not typically associated with seizures or arching of the back.

Option D - Datura: Datura is a plant that contains anticholinergic alkaloids. Anticholinergic poisoning can cause symptoms such as fever, confusion, photophobia, and difficulty urinating. While it may cause delirium and hallucinations, seizures are not a common symptom.

Solution for Question 59:

Correct Option B - Datura, Physostigmine:

- The symptoms exhibited by the child, namely high fever, confusion, photophobia (sensitivity to light), and inability to urinate, indicate anticholinergic toxicity. The Datura plant, also known as jimsonweed or devil's trumpet, produces toxins with strong anticholinergic properties. These substances include scopolamine and atropine. If ingested, they can cause severe symptoms, much like the ones the child is experiencing.
- Physostigmine is the appropriate antidote in this case because it is a reversible cholinesterase inhibitor that increases the levels of acetylcholine in the body. This action effectively counteracts the anticholinergic effects of the Datura toxins. Physostigmine can cross the blood-brain barrier, so it can help alleviate central nervous system symptoms like confusion and photophobia.

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the anticholinergic effects of the Datura toxins. Physostigmine can cross the blood-brain barrier, so it can help alleviate central nervous system symptoms like confusion and photophobia.

Treatment

- Gastric lavage is done to decontaminate the pt
- Physostigmine, which is a cholinergic drug is given (Antidote) OPC isn't given to these pts as they are poisonous Neostigmine doesn't cross the blood brain barrier. Thus, physostigmine is preferred Pilocarpine doesn't have any practical importance
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- Neostigmine doesn't cross the blood brain barrier. Thus, physostigmine is preferred
- Pilocarpine doesn't have any practical importance
- Supportive measures Treat hyperthermia, delirium etc.
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Supportive measures

- Treat hyperthermia, delirium etc.

Treat hyperthermia, delirium etc.

Incorrect Options:

Option A - Datura, Pralidoxime: The causative agent is correct, but Pralidoxime is not the appropriate antidote for Datura poisoning. Pralidoxime is used for poisoning by organophosphates and nerve gases, which act by inhibiting acetylcholinesterase. While this seems similar to the action of Physostigmine, these substances have different mechanisms and pralidoxime is not effective against anticholinergic poisoning caused by plants like Datura.

Option C - Yellow oleander, Pralidoxime : Yellow oleander is a toxic plant, but it does not cause symptoms similar to the ones experienced by the child. The ingestion of yellow oleander leads to cardiac toxicity, not anticholinergic toxicity. Moreover, Pralidoxime is not the antidote for yellow oleander poisoning. The correct treatment for yellow oleander poisoning usually involves the use

of Digoxin-specific antibody fragments.

Option D - Yellow oleander, Physostigmine : Yellow oleander plant does not lead to the symptoms described. Although Physostigmine is a good antidote for certain poisonings, it would not be effective for yellow oleander poisoning.

Solution for Question 60:

Correct Option A - Fomepizole:

- Fomepizole is used as an antidote in confirmed or suspected methanol poisoning. Fomepizole is a competitive inhibitor of alcohol dehydrogenase, the enzyme that catalyses the initial steps in the metabolism of ethylene glycol and methanol to their toxic metabolites.

Incorrect Options:

Option B - Flumazenil: Flumazenil is a benzodiazepine antagonist that is used for the complete or partial reversal of the sedative effects caused by benzodiazepines in various clinical settings, such as induced general anaesthesia for diagnostic and therapeutic procedures. The patient is suspected of methanol poisoning and hence, usage of flumazenil is not recommended.

Option C - N-acetylcysteine: N-acetylcysteine (NAC) is an acetylated form of the amino acid L-cysteine. It has antioxidant, anti-inflammatory, and mucolytic properties. NAC is converted to L-cysteine after ingestion, which, in turn, is converted into glutathione, a powerful antioxidant. It is used in paracetamol poisoning, and not in methanol poisoning.

Option D - Naloxone: Naloxone belongs to a class of drugs known as opioid antagonists. It works by blocking the effects of the opioid in the brain. Since the patient has a methanol poisoning, he doesn't require naloxone.

Solution for Question 61:

Correct Option B - Dilated pupils, rapid pulse and sweating :

Seen in acute cocaine poisoning

- Mechanism of action of cocaine: Inhibits the synaptic reuptake of Epinephrine, Norepinephrine, Dopamine, Serotonin.
- All these substances will increase synaptic cleft.
- This will cause Sympathomimetic Toxidrome

Chronic Cocaine Abuse

- AKA Cocainism, Cocainomania or Cocainophagia
- Tolerance for high doses
- Rush (When excited) and crash (when depressed)
- Triad of cocaine abuse Nasal / Palatal perforation Black tongue and teeth Gangrene of periphery (Nose tip, ear lobule)
- Nasal / Palatal perforation
- Black tongue and teeth
- Gangrene of periphery (Nose tip, ear lobule)
- Cocaine bugs is aka Magnan symptom or formication. This is tactile hallucination. It seems like insects, mice or rats are crawling on skin.
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- Treatment Antidote : Amyl nitrite to control heart rate and BP Supportive and Symptomatic treatment Fatal dose : Around 1g
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- Fatal dose : Around 1g

Antidote : Amyl nitrite to control heart rate and BP

Supportive and Symptomatic treatment

Fatal dose : Around 1g

Sympathomimetic Toxidrome Complications		
	Small Dose	High Dose
Heart Rate	Increase	Cardiac arrhythmia
Blood vessels	Constrictive	Myocardial Infarction
Blood Pressure	Increase	Intracerebral brain stroke
Respiratory rate	Increase	Increase
Temperature	Increase	Crack fever
Sweating	Increase	Increase
Pupil	Dilated	Fixed
CNS Effects	Excitation Euphoria	Delirium Excitation

Incorrect Options:

Option A - Nasal/palatal perforation: Chronic cocaine abuse can lead to damage and perforation of the nasal passages and palate. Prolonged use of cocaine can cause the erosion of the delicate tissues in these areas, resulting in perforation.

Option C - Black tongue and teeth : Chronic cocaine abuse can cause discoloration of the tongue and teeth. The blackening of the tongue, known as "black tongue," can occur due to the drying effect of cocaine on the mouth. Additionally, prolonged cocaine use can lead to tooth decay and staining, resulting in blackened or discolored teeth.

Option D - Hallucination of insects crawling on the skin: "Formication" refers to a tactile hallucination commonly associated with cocaine abuse. Individuals experiencing formication may feel sensations similar to insects crawling on or under their skin. This hallucination is a result of the stimulant effects of cocaine on the central nervous system.

Solution for Question 62:

Correct Option A - CO Poisoning:

- Carbon monoxide poisoning causes cherry red discoloration of the skin, mucus membranes, conjunctiva, nail beds, areas of hypostasis, blood tissues and internal organs. This coloration changes to dark green and then to brown as decomposition starts to occur.

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Incorrect Options:

Option B - Hydrogen sulphide : Hydrogen Sulphide poisoning causes a rotten egg smell in the victim. The general signs of asphyxia are present along with signs of changing of the blood and viscera colour to greenish-purple and not cherry-red.

Option C - Yellow Phosphorus: Yellow Phosphorus is a potent hepatotoxin. Post-mortem findings include garlicky odour along with icterus of the body due to liver failure as a part of multiple organ failure which causes the death of the victim.

Option D - Hydrogen cyanide: Hydrogen cyanide is an extremely lethal poison. Post-mortem findings of hydrogen cyanide poisoning include glistening and prominent dilated pupils, frothy mouth, brick-red blood and cheeks due to impaired oxygenation of the tissues. The internal organs may be oedematous due to congestion.

Solution for Question 63:

Correct Option D - Opioid intoxication:

- Opioid intoxication is characterized by respiratory depression, bradycardia, and pinpoint pupils.

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Incorrect Options:

Option A - Benzodiazepine overdose: Benzodiazepines are central nervous system depressants that can cause respiratory depression, hypotension, and decreased level of consciousness. However, benzodiazepine overdose is less likely to cause bradycardia and pinpoint pupils. The patient's presentation is more consistent with opioid intoxication.

Option B - Cocaine intoxication: Cocaine is a stimulant drug that can cause tachycardia, hypertension, hyperthermia, dilated pupils, and agitation. The patient's presentation is inconsistent with cocaine intoxication. Cocaine use is more likely to cause hyperthermia, tachycardia, and dilated pupils, rather than hypothermia, bradycardia, and pinpoint pupils.

Option C - Opioid withdrawal: Opioid withdrawal is characterized by symptoms such as sweating, nausea, vomiting, diarrhea, dilated pupils, and agitation. The patient's presentation is inconsistent with opioid withdrawal. Opioid withdrawal is more likely to cause tachycardia, rather than bradycardia, and elevated respiratory rate, rather than decreased respiratory rate.

Solution for Question 64:

Correct Option B - Neostigmine with atropine:

- The subsequent step of management for this patient with snakebite envenomation would be Neostigmine with Atropine.
- The symptoms described suggest that the patient may be experiencing neurotoxic effects of the snake venom, which can cause respiratory depression, bradycardia, and other neurological symptoms.
- Antivenom is an important component of snakebite treatment, but it may not reverse the neurological effects of the venom, which can persist even after neutralization of the venom's systemic effects.
- Neostigmine with Atropine is a treatment option for such cases of neurotoxicity, as it can help to reverse neuromuscular paralysis caused by snake venom.

Incorrect Options:

Option A - Check the blood coagulation profile: Checking the blood coagulation profile may be necessary in cases of snakebite envenomation caused by venom that can cause coagulopathy. However, in this case, there are no apparent bleeding manifestations described

Option C - Epinephrine injection: Epinephrine injection is not indicated for this patient because the symptoms described are not consistent with anaphylaxis, which is the primary indication for epinephrine administration.

Option D - I.V hydrocortisone: I.V hydrocortisone is not indicated for this patient because it is not effective in treating the neurotoxic effects of snake venom.

Solution for Question 65:

Correct Option B - Arsenic:

- Arsenic poisoning can occur through exposure to arsenic compounds.
- It can lead to various symptoms, including gastrointestinal issues such as nausea, vomiting, and bloody stools.
- Additionally, arsenic exposure can cause conjunctivitis (inflammation of the conjunctiva) and a burning sensation in the throat and stomach.
- Therefore, considering the presented symptoms, arsenic poisoning is a plausible explanation.

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It can lead to various symptoms, including gastrointestinal issues such as nausea, vomiting, and bloody stools.

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Therefore, considering the presented symptoms, arsenic poisoning is a plausible explanation.

Chronic Arsenic Poisoning

- Aka Arsenicosis [i.e toxicity with arsenic]
- Sources of chronic arsenic poisoning are Occupational exposure: Commonly seen in people working in smelting industries, insecticide industries, rat killer industries etc. Exposure

through food ingestion, for homicidal purpose Water contaminated with arsenic

- Occupational exposure: Commonly seen in people working in smelting industries, insecticide industries, rat killer industries etc.
- Exposure through food ingestion, for homicidal purpose
- Water contaminated with arsenic
- MC seen in West bengal, Bangladesh etc
- It leads to Hydroarsenicism

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Exposure through food ingestion, for homicidal purpose

Water contaminated with arsenic

MC seen in West bengal, Bangladesh etc

It leads to Hydroarsenicism

Incorrect Options:

Option A - Potassium permanganate : Potassium permanganate is a chemical compound commonly used as a disinfectant, oxidizing agent, and in water treatment. However, poisoning due to potassium permanganate is rare, especially in a dye factory setting. Symptoms of potassium permanganate poisoning typically include irritation and burns on the skin, but the other symptoms mentioned in the question, such as bloody stools and conjunctivitis, are not commonly associated with potassium permanganate poisoning. Therefore, this option is unlikely to be the correct answer.

Option C - Thallium : Thallium is a highly toxic heavy metal that can cause severe poisoning when ingested, inhaled, or absorbed through the skin. However, the symptoms described in the question, such as dark bloody stools and conjunctivitis, are not typically associated with thallium poisoning. Thallium poisoning often presents with neurological symptoms, such as peripheral neuropathy and hair loss. Therefore, thallium poisoning is unlikely based on the given symptoms.

Option D - Lead : Lead poisoning can occur through the ingestion or inhalation of lead-containing substances. While lead exposure can lead to various health issues, the symptoms mentioned in

the question, such as dark bloody stools, conjunctivitis, and a burning sensation in the throat and stomach, are not typical of lead poisoning. Lead poisoning is more commonly associated with neurological symptoms, developmental delays in children, and other systemic effects. Therefore, lead poisoning is unlikely based on the provided symptoms.

Solution for Question 66:

Correct Option C - Phosphorus:

- Luminescent stools are associated with phosphorus.
- High levels of phosphorus ingestion or phosphorus poisoning can sometimes cause stools to exhibit a luminescent or glowing appearance.

Incorrect Options:

Option A - Mercury: Mercury is a toxic heavy metal that can cause various health issues when ingested. However, it is not typically associated with luminescent stools.

Option B - Cadmium: Cadmium is another toxic heavy metal that can be harmful to human health. Like mercury, it is not commonly associated with luminescent stools.

Option D - Oxalic acid: Oxalic acid is a naturally occurring compound found in various foods. Ingesting oxalic acid in normal amounts does not typically result in luminescent stools.

Solution for Question 67:

Correct Option C - Prazosin:

- Prazosin is a medication classified as an alpha-1 adrenergic antagonist. It works by blocking the action of certain receptors in the body, specifically the alpha-1 adrenergic receptors.
- In the case of scorpion envenomation, scorpion venom typically contains compounds that can stimulate the release of catecholamines, such as norepinephrine, leading to sympathetic

overactivity. This can result in symptoms such as hypertension and increased heart rate.

- By blocking the alpha-1 adrenergic receptors, prazosin helps to counteract the effects of the excessive sympathetic stimulation caused by scorpion venom. It helps to relax smooth muscles in blood vessels, reducing blood pressure and improving symptoms associated with scorpion envenomation.

Prazosin is a medication classified as an alpha-1 adrenergic antagonist. It works by blocking the action of certain receptors in the body, specifically the alpha-1 adrenergic receptors.

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By blocking the alpha-1 adrenergic receptors, prazosin helps to counteract the effects of the excessive sympathetic stimulation caused by scorpion venom. It helps to relax smooth muscles in blood vessels, reducing blood pressure and improving symptoms associated with scorpion envenomation.

Incorrect Options:

Option A - Physostigmine: Physostigmine is a medication that acts as an acetylcholinesterase inhibitor, increasing the availability of acetylcholine in the body. While it can be used as an antidote for certain types of poisoning, such as anticholinergic toxicity, it is not typically used as a specific antidote for scorpion venom.

Option B - Norepinephrine: Norepinephrine is a naturally occurring neurotransmitter and hormone that is involved in the sympathetic nervous system response. While norepinephrine might be elevated in scorpion envenomation, it is not considered a specific physiological antidote to scorpion venom.

Option D - Atropine: Atropine is an anticholinergic medication that blocks the action of acetylcholine in the body. It is commonly used to counteract the effects of excessive acetylcholine stimulation. However, in the case of scorpion envenomation, the primary issue is excessive sympathetic activity rather than excessive cholinergic activity. Therefore, atropine is not the preferred physiological antidote for scorpion venom.

Solution for Question 68:

Correct Option C - Carbolic acid:

- Carbollic acid, also known as phenol, is a toxic substance that can cause severe poisoning if ingested.
- Gastric lavage, or stomach pumping, can be performed in cases of carbollic acid poisoning.
- The procedure involves introducing a tube into the stomach to wash out the toxic substance and reduce its absorption into the body.
- Gastric lavage is considered beneficial in removing any remaining carbollic acid from the stomach and can be an important part of the treatment protocol for carbollic acid poisoning.

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Gastric lavage is considered beneficial in removing any remaining carbollic acid from the stomach and can be an important part of the treatment protocol for carbollic acid poisoning.

Incorrect Options:

Option A - Sulphuric acid: Sulphuric acid is a highly corrosive and dangerous substance. In cases of sulphuric acid poisoning, gastric lavage is generally not recommended as it may cause further damage to the gastrointestinal tract. Immediate medical attention and appropriate management are crucial in cases of sulphuric acid poisoning to minimize harm and provide necessary treatment.

Option B - Strychnine: Strychnine is a highly toxic alkaloid that affects the nervous system. Gastric lavage is typically not performed in cases of strychnine poisoning. The primary focus of treatment for strychnine poisoning involves managing symptoms, such as muscle spasms and seizures, providing supportive care, and administering specific antidotes as appropriate.

Option D - Kerosene oil: Kerosene oil ingestion can lead to chemical pneumonia or aspiration pneumonia if it enters the lungs. Gastric lavage is not routinely recommended for kerosene oil poisoning. Treatment primarily involves supportive care, monitoring for any respiratory complications, and providing appropriate respiratory support if needed.

Solution for Question 69:

Correct Option B - Cocaine:

Needle track marks over upper limbs, increased blood pressure, increased heart rate, and increased temperature (hyperthermia) – all of these signs symptoms indicate the poisoning of a deliriant poison Cocaine.

- Mechanism of action of cocaine : Inhibits the synaptic reuptake of Epinephrine, Norepinephrine, Dopamine, Serotonin.
- All these substances will increase synaptic cleft. This will cause Sympathomimetic Toxidrome.

Sympathomimetic Toxidrome Complications		
	Small Dose	High Dose
Heart Rate	Increase	Cardiac arrhythmia
Blood vessels	Constrictive	Myocardial function
Blood Pressure	Increase	Intracerebral brain stroke
Respiratory rate	Increase	Increase
Temperature	Increase	Crack fever
Sweating	Increase	Increase
Pupil	Dilated	Fixed
CNS Effects	Excitement Euphoria	Delirium Excitation

Incorrect Options:

Option A - Heroin: Excessive heroin ingestion leads to hypothermia, slowed breathing, blue lips and fingernails, cold and clammy skin, convulsions and coma.

Option C - Cannabis: Cannabis poisoning leads to these symptoms: Small doses: Causes euphoria and disorientation; Large doses: Psychiatric problems- Euphoria, hallucination, increase self-confidence.

Option D - LSD: LSD (Lysergic acid diethylamide – synthetic chemical made from ergot) overdose leads to hyperthermia, blurred vision, rapid heartbeat, insomnia, dry mouth, etc. Adverse outcomes include features of psychosis, extreme anxiety, aggression towards others.

Solution for Question 70:

Correct Option D - Hydrogen sulfide:

- Based on the given symptoms of greenish skin color and a rotten egg odor, the most probable poisoning, in this case, would be hydrogen sulfide (H₂S) poisoning.
- Hydrogen sulfide is a toxic gas that is commonly found in various industrial settings, such as petroleum refining, sewer systems, and certain chemical processes. It has a distinct odor resembling rotten eggs.
- Exposure to high levels of hydrogen sulfide can be extremely dangerous and potentially fatal.
- The characteristic greenish discoloration of the skin is known as sulfhemoglobinemia, which occurs as a result of hydrogen sulfide poisoning. Hydrogen sulfide binds to hemoglobin in the blood, forming sulfhemoglobin, which gives the skin a greenish appearance.
- Therefore, based on the symptoms described, the most probable poisoning in this case is hydrogen sulfide (H₂S) poisoning.

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Therefore, based on the symptoms described, the most probable poisoning in this case is hydrogen sulfide (H₂S) poisoning.

Incorrect Options:

Option A - Carbon monoxide: Carbon monoxide (CO) poisoning, on the other hand, typically causes a cherry red coloration of the skin rather than a greenish color.

Option B - Nitrite: Nitrite poisoning is associated with symptoms such as headache, dizziness, and shortness of breath, but it does not cause greenish skin discoloration.

Option C - Cyanide: Cyanide poisoning may cause a bluish discoloration of the skin, but it does not produce a rotten egg odor.

Solution for Question 71:

Correct Option B - Deferoxamine:

- Deferoxamine also known as Desferrioxamine is used as an antidote for iron.
- Fatal dose of iron is 20 to 30 grams.
- Treatment: Stomach wash with 5% sodium bicarbonate solution. Instil 5 to 10 gm of Deferoxamine at the end of lavage. Give plenty of egg and milk to form iron-protein complexes. (3) Magnesium hydroxide 1% solution orally. Deferoxamine 1g i.m. followed by 500 mg 4th hourly for 2 doses and finally 500 mg 4 to 12 hourly up to a maximum of 6 g in 24 hours. It can also be given in infusion 15 mg/kg/hr in normal saline. Haemodialysis or exchange transfusion in severe cases.
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Haemodialysis or exchange transfusion in severe cases.

Incorrect Options:

Option A - Luspatercept: Luspatercept is used to treat anemia in beta-thalassemia, myelodysplastic syndromes, and primary myelofibrosis.

Option C - Deferiprone: Deferiprone chelates iron but is used in the treatment of iron overload in thalassemia (For children 3 years & above and adults).

Option D - DMSA: DMSA (Dimercapto succinic acid) is a radioisotope. It is used for metal chelation, and is also used in imaging and therapy

Solution for Question 72:

Correct Option B - Respiratory failure:

- The most probable cause of death in this scenario would be respiratory failure.
- The common krait (*Bungarus caeruleus*) is a highly venomous snake found in South Asia, including India. Its venom contains neurotoxins that affect the nervous system, particularly the respiratory muscles and the diaphragm.
- When a person is bitten by a common krait, the venom can cause paralysis of the respiratory muscles, leading to respiratory failure. This means that the person is unable to breathe adequately on their own, resulting in a lack of oxygen supply to the body and vital organs.
- While shock, coagulopathy (blood clotting disorder), and renal failure are potential complications associated with snakebites, the rapid onset of respiratory failure and subsequent hypoxia (low oxygen levels) are the most immediate and life-threatening consequences of envenomation by the common krait.

- It is important to seek immediate medical attention and provide supportive care, including artificial ventilation, in cases of suspected snakebite to prevent or manage respiratory failure and other complications. Antivenom administration may also be necessary to neutralize the snake venom and improve the patient's outcome.

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It is important to seek immediate medical attention and provide supportive care, including artificial ventilation, in cases of suspected snakebite to prevent or manage respiratory failure and other complications. Antivenom administration may also be necessary to neutralize the snake venom and improve the patient's outcome.

Option A, C, D :

The venom of common krait contains neurotoxins that affect the nervous system, particularly the respiratory muscles and the diaphragm, leading to respiratory failure.

Shock , coagulopathy and renal failure are highly unlikely to be caused by this toxin.

Solution for Question 73:

Correct Option B - Carbon monoxide poisoning:

Cherry red postmortem staining is associated with carbon monoxide poisoning.

- Mechanism of action of CO : It has 200-300 times more affinity to Hb than O₂ . CO combines with hb to form carboxyhemoglobin. Carboxy Hb reduces O₂ in blood, tissues.

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- Signs and symptoms : Depends on percent of CO present in blood
- <10% - No symptoms
- 10-20% - Breathless on moderate exertion, headache, weakness.
- 20-30% - Severe headache, irritability, emotional instability, defective memory.
- 30-40% - Cherry red discolouration skin, mucous membrane, vision decreased, nausea and vomiting.
- 40-50% - Resemble acute alcoholic intoxication
- 50-60% - Syncope and coma
- >80% - Death
- Treatment: Antidote 100% O₂ at atmospheric pressure given by a tight fitting mask.

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>80% - Death

Treatment: Antidote 100% O₂ at atmospheric pressure given by a tight fitting mask.

Postmortem findings:

- Cherry-red post-mortem staining.

Cherry-red post-mortem staining.

- Blisters in dependent or frictional areas— Axilla, buttocks, inner thigh, calves, wrists, and knees. Also seen in barbiturates poisoning.
- In delayed deaths Bilateral and symmetrical necrosis and cavitation seen in globus pallidus and putamen i.e., basal ganglia change.

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In delayed deaths Bilateral and symmetrical necrosis and cavitation seen in globus pallidus and putamen i.e., basal ganglia change.

Incorrect Options:

Option A - Opioid poisoning: Black postmortem staining is seen.

Option C - Hydrogen sulphide poisoning: It produces greenish- blue/ bluish-green discolouration.

Option D - Phosphorus poisoning: Dark brown or yellowish discolouration is seen.

Solution for Question 74:

Correct Option B - Datura poisoning:

- Datura/ devil's trumpet is a deliriant poison (other than Cannabis, Cocaine). The symptoms given are due to Datura poisoning.
- If it is a white flower plant, it is Datura alba. If it is a purple dark color flower it is Datura niger. Datura stramonium AKA Jimson's weed is also known as thorn apple.
- Entire plant is poisonous. Active principle: Hyoscine (Scopolamine)
- 1 Fruit almost contains 100s of seeds.
- Fatal dose is 100- 125 seeds.
- Used as a roadside poison or railway poison.
- There is no criminal responsibility for a person who has been given Datura because his actions are involuntary intoxication. 328 IPC - Hurt with poison with criminal intention.
- Shows anticholinergic effects.

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DATURA STRAMONIUM



Signs and Symptoms of datura poisoning.

- Dryness in skin and mouth
- Dysphagia: Difficulty in swallowing
- Dilated pupil (very earliest symptom)
- Drunken Gait
- Dysarthria: Slurred speech
- Muttering Delirium
- Drowsiness
- Death
- Carpalgia: It feels like it is a kind of hallucination, imaginary threads, and pill-rolling movement.

Dryness in skin and mouth

Dysphagia: Difficulty in swallowing

Dilated pupil (very earliest symptom)

Drunken Gait

Dysarthria: Slurred speech

Muttering Delirium

Drowsiness

Death

Carpologia: It feels like it is a kind of hallucination, imaginary threads, and pill-rolling movement.

Treatment

- ■: Stomach wash and activated charcoal.
- Physostigmine is the antidote choice for datura used to counteract the anticholinergic

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Physostigmine is the antidote choice for datura used to counteract the anticholinergic

Incorrect Options:

Option A - Poppy seeds poisoning: 50 grams (5-6 tablespoons) of Poppy seeds might be sufficient to cause poisoning. Symptoms include severe muscle spasms, severe muscle pain, seizures, cramps, and stiffness or abnormal movements.

Option C - Mushroom poisoning: Mushroom poisoning might cause severe gastrointestinal upsets, nausea, vomiting and diarrhea to devastating manifestations which might be neurologic complications, kidney and liver failure.

Option D - Cannabis poisoning:

- Cannabis poisoning leads to these symptoms: Small doses : Causes euphoria and disorientation Large doses Psychiatric problems : Euphoria, hallucination, increase self-confidence.
- Small doses : Causes euphoria and disorientation
- Large doses Psychiatric problems : Euphoria, hallucination, increase self-confidence.
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Small doses : Causes euphoria and disorientation

Large doses

- Psychiatric problems : Euphoria, hallucination, increase self-confidence.

Psychiatric problems : Euphoria, hallucination, increase self-confidence.

Solution for Question 75:

Correct Option B - Delirium tremens:

- Delirium tremens is a severe and potentially life-threatening withdrawal symptom that can occur in alcoholic patients after 72 hours of cessation of alcohol.
- It typically manifests with symptoms such as severe confusion, disorientation, hallucinations, agitation, rapid heartbeat, high blood pressure, and fever.
- Delirium tremens requires immediate medical attention.

Delirium tremens is a severe and potentially life-threatening withdrawal symptom that can occur in alcoholic patients after 72 hours of cessation of alcohol.

It typically manifests with symptoms such as severe confusion, disorientation, hallucinations, agitation, rapid heartbeat, high blood pressure, and fever.

Delirium tremens requires immediate medical attention.

Incorrect Options:

Option A - Seizures:

- Seizures can indeed be a withdrawal symptom in alcoholic patients, but they typically occur within the first 48 hours of alcohol cessation rather than after 72 hours.

Option C - Tremors :

- Tremors are one of the most common early withdrawal symptoms seen within the first few hours to a couple of days after alcohol cessation.
- They are characterized by involuntary shaking of the hands, arms, or other body parts. While tremors are a common symptom, they are not specific to the 72-hour mark.

Option D - Hallucinations :

- Hallucinations can be part of the withdrawal syndrome in alcoholic patients, but they often occur within the first 48 hours rather than after 72 hours. They can be visual, auditory, or tactile in nature.

Solution for Question 76:

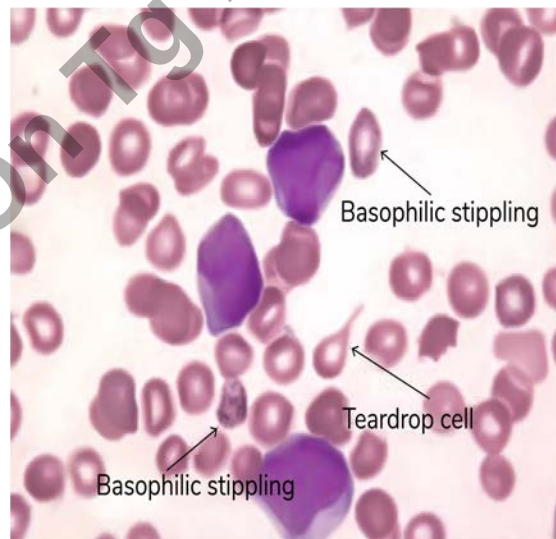
Correct Option B - Saturnism:

- Saturnism, also known as chronic lead poisoning, can cause a variety of hematological abnormalities, including anemia, basophilic stippling, and the presence of nucleated red blood cells (RBCs) in the peripheral blood.
- In severe cases, lead poisoning can also lead to bone marrow suppression, which can result in pancytopenia.
- However, it is important to note that the hematological picture of lead poisoning can vary depending on the level and duration of exposure, as well as individual factors such as age and nutritional status.

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However, it is important to note that the hematological picture of lead poisoning can vary depending on the level and duration of exposure, as well as individual factors such as age and nutritional status.



Incorrect Options:

Option A - Minamata disease : While it is possible that mercury toxicity may lead to some degree of hematological abnormalities, such as anemia, the typical hematological picture of Minamata disease does not include specific features that would clearly distinguish it from other causes of

anemia.

Option C - Chronic iron toxicity: Chronic iron toxicity can lead to a hematological picture of microcytic hypochromic anemia, which is characterized by small red blood cells that contain insufficient hemoglobin. While this is a specific hematological feature that can be seen in chronic iron toxicity, it is not unique to this condition and can also occur in other forms of anemia.

Option D - Arsenic poisoning: Arsenic poisoning can lead to a variety of hematological abnormalities, including anemia, leukopenia, and thrombocytopenia. The specific hematological picture may vary depending on the severity and duration of exposure. However, the presence of anemia alone is not specific enough to diagnose arsenic poisoning, as anemia can also occur in many other conditions.

Solution for Question 77:

Correct Option C - 3, 2, 4, 1:

- (3) Gastrointestinal Irritation : As iron toxicity progresses, it can cause damage to the lining of the gastrointestinal tract and irritation. This can lead to gastrointestinal bleeding, which is characterized by the presence of blood in the vomit or stool. Gastrointestinal bleeding is a serious complication of iron toxicity and requires immediate medical attention.
- (2) Latent phase : The initial phase of iron toxicity after GI irritation may often be asymptomatic, meaning that there are no noticeable symptoms. During this phase, the individual may not experience any immediate effects or manifestations of iron toxicity. However, it is important to note that this phase can be transient, and symptoms can develop as the toxicity progresses.
- (4) Liver failure and hypoglycemia : In severe cases of iron toxicity, especially when a large amount of iron has been ingested, it can lead to liver failure and hypoglycemia. Iron overload can cause significant damage to the liver, impairing its normal function. This can result in liver failure, which is a life-threatening condition. Additionally, iron toxicity can also cause hypoglycemia, a condition characterized by low blood sugar levels.
- (1) Healing with scarring : In the context of iron toxicity, pyloric stricture occurs as a late-stage complication. It is typically seen after significant ingestion of iron, usually in the form of iron tablets or supplements. Pyloric stricture can cause symptoms such as vomiting, abdominal pain, and difficulty in food passage.

- Therefore , the correct order of symptomatology seen in iron toxicity is 3 (Gastrointestinal Irritation), 2 (Latent phase), 4 (Liver failure and hypoglycemia), and 1 (Healing with scarring).

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Incorrect Options:

- Option A, B and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 78:

Correct Option B - Cocaine:

- Cocaine is a strongly addictive stimulant that increases the levels of vigilance, focus, and vigour.
- Agitation, paranoia, hallucinations, delusions, violence, and thoughts of suicide and murder are some of the symptoms.
- Routes of Cocaine abuse Inhalation/ Snorting is MC route Onset of action occurs immediately with inhalation Smoking IV Injection Oral ingestion
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- Onset of action occurs immediately with inhalation
- Smoking
- IV Injection
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Routes of Cocaine abuse

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Inhalation/ Snorting is MC route

- Onset of action occurs immediately with inhalation

Onset of action occurs immediately with inhalation

Smoking

IV Injection

Oral ingestion

Effects of Cocaine intake

- Cocaine is sympathomimetic toxidrome [i.e it increases sympathetic outflow]
- It inhibits the reuptake of Noradrenaline/ Dopamine in the presynaptic membrane
- Thus, lots of Noradrenaline/ Dopamine molecules accumulate in the synaptic cleft resulting in excess sympathetic effect

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It inhibits the reuptake of Noradrenaline/ Dopamine in the presynaptic membrane

Thus, lots of Noradrenaline/ Dopamine molecules accumulate in the synaptic cleft resulting in excess sympathetic effect

Incorrect Options:

Option A - Alcohol: Common signs and symptoms include weakness, fatigue, thirst, headache, muscle aches, nausea, stomach discomfort, vertigo, sensitivity to light and sound, anxiety,

irritability, perspiration, and elevated blood pressure.

Option C - Opioid: While certain opioid side effects, such as tiredness and constipation, may be modest, more severe side symptoms, including as shallow breathing, a slowed heartbeat, and loss of consciousness, may be significant and indicate an overdose.

Option D - Cannabis: Disorientation's negative repercussions include diminished perception, a potential rise in periodontal disease and caries risk, enhanced risk of oral mucositis-related inflammation and mouth drool.

Solution for Question 79:

Correct Option A - Parthenium hysterophorus:

- In America, Asia, Africa, and Australia, parthenium hysterophorus is considered a noxious weed.
- Parthenium hysterophorus is an aggressive annual herbaceous weed that is found everywhere. It is now one of the seven most dangerous and destructive weeds in the world.
- P. hysterophorus affects food and fodder crops because its pollen and dust cause allergic contact dermatitis in people.
- The condition known as dermatitis, which is a T cell-mediated immune damage, causes itchy, erythematous papules and papulovesicular lesions on exposed body parts.
- People who are exposed to this plant for an extended length of time experience symptoms such as skin rashes, dermatitis, asthma, allergic rhinitis, hay fever, black spots, burning, and eye blisters.
- Additionally, parthenium hysterophorus produces choking, severe papular erythematous rashes, diarrhoea, and dyspnea.

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Additionally, parthenium hysterophorus produces choking, severe papular erythematous rashes, diarrhoea, and dyspnea.

Incorrect Options:

Option B - Datura stramonium: A common annual plant called Datura stramonium (DS) contains the drugs atropine, hyoscyamine, and scopolamine, which can cause severe anticholinergic syndrome and lead to poisoning.

Option C - Digitalis purpurea: The hazardous kind of flowering plant known as the foxglove, or Digitalis purpurea, belongs to the Plantaginaceae family of plantains. The same cardiac glycoside, digitoxin, which is extracted from the leaves, is used as a treatment for heart failure.

Option D - Calotropis gigantea: Calotropis gigantea is a huge shrub or small tree that is native to India, southern China, Malaysia, and Indonesia but has been widely cultivated in tropical regions all over the world. It is also known as a crown flower or giant milkweed. It is distinguished by its upright growth, milky sap, crown-like flowers that are pale purple or white, and white woolly stems and leaves that eventually become subglabrous. No significant issues with insects or diseases. Severe irritation might result from sap in the eyes. Mucous membranes are irritated by sap as well.

Solution for Question 80:

Correct Option D - Bleeding, Paralysis and Amputation:

- The snake in the image is a highly venomous species of snake called Bungarus caeruleus, also referred to as the common krait. It causes several neurotoxic symptoms and muscle paralysis.

The snake in the image is a highly venomous species of snake called Bungarus caeruleus, also referred to as the common krait. It causes several neurotoxic symptoms and muscle paralysis.

Neurotoxic symptoms are:

- Ptosis: Most Important and Common Symptom - Dropping of Eyelid
- Diplopia : Paralysis of extra ocular Muscles.

- Dysphagia : Paralysis of pharyngeal muscles.
- Dysphonia : Paralysis of vocal codes.
- Dysarthria : Paralysis of tongue muscles.
- Dyspnea : Paralysis of respiratory muscles.
- Stridor : Respiratory difficulty.
- Descending Paralysis
- Death due to respiratory muscles paralysis leading to respiratory failure.

Ptosis: Most Important and Common Symptom -Dropping of Eyelid

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Dyspnea : Paralysis of respiratory muscles.

Stridor : Respiratory difficulty.

Descending Paralysis

Death due to respiratory muscles paralysis leading to respiratory failure.

Incorrect Options:

Option A - Non-poisonous, Dry bite: The absence of venom being injected into the victim during a snakebite incidence is known as a "dry bite" from a snake

Option B - Severe edema, Necrosis and Bleeding: This is brought on by viper bites, which cause more severe local reactions than those from other snakes. Within 15 minutes, swelling may be noticeable, and it might become huge within two to three days. Bite marks from some rattlesnakes and Asian pit vipers indicate necrosis.

Option C - Severe muscle necrosis, Rhabdomyolysis: These symptoms are due to bites by Russell's vipers, which increase renal ischemia and cause acute kidney injury by haemorrhage, hypotension, disseminated intravascular coagulation (DIC), intravascular hemolysis, and rhabdomyolysis.

Solution for Question 81:

Correct Option A - Arsenic:

- Arsenic is a poison that can be found even in skeletonized remains of the body. It has the ability to accumulate in bones over time, making it detectable even after the decomposition of soft tissues.
- This is because arsenic is a heavy metal, it replaces phosphorus and gets absorbed into the bones, where it may remain for years.

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This is because arsenic is a heavy metal, it replaces phosphorus and gets absorbed into the bones, where it may remain for years.

Incorrect Options:

Option B - Carbon monoxide: Carbon monoxide is a gas that is produced during the incomplete combustion of carbon-containing materials. It is unlikely to be found in skeletonized remains since it is a volatile gas that dissipates quickly and does not accumulate in the body or bones.

Option C - Hydrogen sulphide: Hydrogen sulphide is a toxic gas that has a strong odor of rotten eggs. Similar to carbon monoxide, it is unlikely to be found in skeletonized remains as it is a volatile gas that dissipates rapidly and does not accumulate in the body or bones.

Option D - Phosphorus: Phosphorus is a chemical element that can be toxic in certain forms, such as white phosphorus. However, it is not typically found in skeletonized remains as it is highly reactive and tends to undergo chemical reactions or degradation over time.

Solution for Question 82:

Correct Option B - Heroin:

- Heroin, also known as diamorphine, is an opioid drug that is derived from morphine. It is highly addictive and is known to cause physical dependence.
- When individuals use heroin regularly, their body adapts to the presence of the drug, and they develop a tolerance to its effects. This means that over time, they require higher doses of heroin to achieve the desired effect.

- If a person becomes physically dependent on heroin and abruptly stops using it or significantly reduces its dosage, they may experience withdrawal symptoms.
- These symptoms can be severe and include cravings, restlessness, muscle and bone pain, insomnia, gastrointestinal distress, and flu-like symptoms.

Incorrect Options:

Option A - Ketamine: Ketamine is a dissociative anesthetic that produces hallucinogenic effects. While it can lead to psychological dependence, it is not typically associated with significant physical dependence.

Option C - LSD: LSD (Lysergic acid diethylamide) is a hallucinogenic drug that does not typically produce physical dependence. However, it can lead to psychological dependence or addiction in some individuals.

Option D - Phencyclidine: Phencyclidine PCP is a dissociative drug that can cause hallucinations, distorted perceptions, and other psychological effects. It is not commonly associated with physical dependence, but it can lead to psychological dependence in some cases.

Solution for Question 83:

Correct Option A - Kidney:

- In an autopsy of suspected poisoning, the kidney is preserved because it is the primary organ responsible for filtering and excreting toxins from the body. The kidney can provide crucial evidence of poisoning.

Incorrect Options:

Option B - Lungs: While the lungs may show signs of poisoning in some instances, they are not typically preserved as a routine procedure in a poisoning autopsy.

Option C - Brain : The brain is not directly involved in the filtration or detoxifying of toxins, so it is not commonly preserved in a poisoning autopsy.

Option D - Heart: The heart's primary function is to pump blood and distribute oxygen, so it is not explicitly preserved in a poisoning autopsy unless there is suspicion of cardiac involvement.

Solution for Question 84:

Correct Option D - Lead:

- Lead poisoning can occur due to a bullet in an individual's body.
- The poisoning generally occurs when the bullet lead reaches in the synovial cavity of the chest, hip, and pleural space.
- The clinical features of lead poisoning are : A - Anemia B - Basophilic stippling, Burtonian line. C - Colic and constipation. D - Drops (wrist drop). E - Encephalopathy, common in toddlers. F - Facial pallor (earliest and most consistent sign). G - Gout (decreases uric acid excretion).
- A - Anemia
- B - Basophilic stippling, Burtonian line.
- C - Colic and constipation.
- D - Drops (wrist drop).
- E - Encephalopathy, common in toddlers.
- F - Facial pallor (earliest and most consistent sign).
- G - Gout (decreases uric acid excretion).
- Lab Investigation: Search for polycythemia, anemia, anisocytosis, eosinophilia, poikilocytosis, and punctuate basophilia in the peripheral smear. In the lab investigation, the blood sample shows increased levels of lead and erythrocyte porphyrin. The urine samples show increased coproporphyrinogen, ALA, and erythrocyte porphyrin. On the radiology tests, the expert finds bone lines, the presence of lead chips, total quantity of lead present in the body.
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Incorrect Options:

Option A - Iron:

- Iron poisoning mostly happens due to consuming iron supplements. Therefore, it is best to take advice from a healthcare professional regarding dosage.

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Option B - Phosphorus:

- Phosphorus is very dangerous for our body, but the retained bullet cannot be the cause of phosphorus poisoning.
- The primary reasons for Phosphorus poisoning are tumor lysis syndrome, rhabdomyolysis, and impaired renal function.

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Option C - Nitrocellulose:

- When an individual inhales Nitrocellulose, it can cause Nitro cellulose poisoning. So, it is one of the incorrect choices.

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Solution for Question 85:

Correct Option C - HCN poisoning:

- HCN or Hydrogen Cyanide is a pale-blue or almost colorless gas or liquid with a bitter almond taste.
- When it interferes with using oxygen in the human body, it directly affects our blood vessels, heart, brain, and lungs.
- Additionally, the pink color of mucosa and skin appears because of the excessive infectious hemoglobin saturation.
- Symptoms of HCN poisoning: Difficulty breathing or shortness of breath Rapid and shallow breathing Rapid heart rate (tachycardia) Chest pain or tightness Dizziness or lightheadedness Headache Confusion or disorientation Seizures or convulsions Loss of consciousness Nausea and vomiting Abdominal pain or cramps Skin discoloration (bluish appearance) Eye irritation or redness (if exposed to HCN gas) Cyanide-like odor (bitter

almond odor) may be present in some cases

- Difficulty breathing or shortness of breath
- Rapid and shallow breathing
- Rapid heart rate (tachycardia)
- Chest pain or tightness
- Dizziness or lightheadedness
- Headache
- Confusion or disorientation
- Seizures or convulsions
- Loss of consciousness
- Nausea and vomiting
- Abdominal pain or cramps
- Skin discoloration (bluish appearance)
- Eye irritation or redness (if exposed to HCN gas)
- Cyanide-like odor (bitter almond odor) may be present in some cases
- Treatment: Activated charcoal to absorb the toxin. In severe cases, Cyanokit, hydroxocobalamin, and Eli Lilly's antidote
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HCN or Hydrogen Cyanide is a pale-blue or almost colorless gas or liquid with a bitter almond taste.

When it interferes with using oxygen in the human body, it directly affects our blood vessels, heart, brain, and lungs.

Additionally, the pink color of mucosa and skin appears because of the excessive infectious hemoglobin saturation.

Symptoms of HCN poisoning:

- Difficulty breathing or shortness of breath
- Rapid and shallow breathing
- Rapid heart rate (tachycardia)
- Chest pain or tightness
- Dizziness or lightheadedness
- Headache
- Confusion or disorientation

- Seizures or convulsions
- Loss of consciousness
- Nausea and vomiting
- Abdominal pain or cramps
- Skin discoloration (bluish appearance)
- Eye irritation or redness (if exposed to HCN gas)
- Cyanide-like odor (bitter almond odor) may be present in some cases

Difficulty breathing or shortness of breath

Rapid and shallow breathing

Rapid heart rate (tachycardia)

Chest pain or tightness

Dizziness or lightheadedness

Headache

Confusion or disorientation

Seizures or convulsions

Loss of consciousness

Nausea and vomiting

Abdominal pain or cramps

Skin discoloration (bluish appearance)

Eye irritation or redness (if exposed to HCN gas)

Cyanide-like odor (bitter almond odor) may be present in some cases

Treatment:

- Activated charcoal to absorb the toxin.
- In severe cases, Cyanokit, hydroxocobalamin, and Eli Lilly's antidote

Activated charcoal to absorb the toxin.

In severe cases, Cyanokit, hydroxocobalamin, and Eli Lilly's antidote

Incorrect Options:

Option A - H₂S poisoning: H₂S poisoning is also a severe and fatal condition. The primary symptoms of this condition are headaches, tremors, nausea, delirium, convulsions, disturbed equilibrium, eye and skin irritation, rapid unconsciousness, and eventually death.

Option B - CO poisoning: CO poisoning can also be deadly. The most common signs of CO poisoning are vomiting, upset stomach, dizziness, headache, chest pain, and confusion. If anyone inhales excessive CO, it has the potential to kill that individual.

Option D - OPC poisoning: OPC poisoning is a known class of insecticides that occurs when a person is exposed to organophosphates. The condition's primary symptoms are a runny nose, glassy eyes, blurry vision, agitation, muscle weakness, narrowed pupil, etc. In severe cases, it leads to coma or collapse.

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Thanatology

1. A 50-year-old female is found deceased in her home, and upon examination, it is noted that her stomach contains undigested food from a heavy meal. Based on the examination findings, what is the most likely time frame for the patient's death after her last meal?

- A. 1 to 2 hours
- B. 3 to 4 hours
- C. 5 to 8 hours
- D. 10 to 12 hours

2. A forensic expert is examining a dead body that was found in a hot and moist place. There is adipocere formation all over the body including the internal organs. What does this finding signify? It gives a rough idea of the time since death. It is formed due to postmortem hydrolysis and hydrogenation of saturated fats. Organisms like aerobic bacteria help this process. It is usually found in sites with fat deposits.

- A. 1, 2, and 4
- B. 1, 2, and 3
- C. 1 and 4
- D. All are correct

3. A forensic investigator examines a body in a room at 22°C. The normal body temperature is 37°C, and the rectal temperature is 31°C. The rate of temperature fall is approximately 0.6°C per hour. What is the estimated time since death?

- A. 8 hours

- B. 9 hours
 - C. 10 hours
 - D. 12 hours
-

4. What is the correct order of presenting sequence in rigor mortis after death?

- A. Fingers and toes → Upper limbs → Lower limbs → Abdomen
 - B. Upper limbs → Abdomen → Lower limbs → Fingers and toes
 - C. Lower limbs → Abdomen → Upper limbs → Fingers and toes
 - D. Fingers and toes → Upper limbs → Abdomen → Lower limbs
-

5. A dead body is being examined by the forensic surgeon and maggots are found. Which of the following life stage of maggots is correctly matched with their time?

- A. Larvae formation: 3-4 days
 - B. Pupae formation: 3-6 days
 - C. Adult fly formation from pupae: 1-2 days
 - D. Entire cycle: 8-20 days summer
-

6. A 50-year-old male passed away due to a myocardial infarction. The family requested embalming for preservation of the body before the funeral. Which of the following statements about embalming is incorrect?

- A. Produces permanent chemical rigidity that mimics rigor mortis.
- B. Sodium citrate is used in embalming fluid as an anticoagulant.
- C. Embalming makes it easier to interpret postmortem injuries.
- D. Femoral artery is commonly used for injecting embalming fluid.

7. Which of the following factors does not favour the process in the given image?



- A. Hot environment
- B. Free air movement
- C. Contact of the body with water
- D. Chronic antimony poisoning

8. According to Casper's dictum, the rate of decomposition varies depending on the environment. Which of the following correctly represents this relationship?

- A. Decomposition in deeply buried bodies is faster than in water.
- B. Decomposition in water occurs at the same rate as in air.
- C. Decomposition in air is faster than in soil.
- D. Decomposition in air occurs eight times faster than in water.

9. A decomposing body is found in the woods. Upon examination, it is noted that the skull sutures have separated, and the skin on the hands and feet has sloughed off in a 'glove and stocking' manner. Liquefied brain matter is observed. What is the most likely time frame of death?

- A. 12–24 hours
- B. 24–48 hours
- C. 3–5 days
- D. 5–10 days

10. During an autopsy performed on a body 20 hours after death, the pathologist notices small blisters filled with gas on the lower abdomen and thighs. What is the likely cause of these blisters?



- A. Antemortem burn injury
- B. Decomposition due to gas formation
- C. Friction-induced blisters

D. Chemical exposure

11. A 35-year-old male is found deceased in an apartment, and the body is examined 48 hours post-mortem. A distinct branching pattern is observed along the superficial blood vessels after 24 hours, as shown below. What is the most likely cause of the skin pattern observed in the image?



- A. Marbling
 - B. Adipocere
 - C. Filigree burns
 - D. Postmortem lividity
-

12. A body arrived at the forensic department for autopsy, presenting with distinct features, including a greenish discoloration of the abdomen, a strong odor, and gas formation within the tissues (as shown in the image provided below). What enzyme is primarily responsible for these characteristics?



- A. Hyaluronidase
- B. Collagenase
- C. Lecithinase
- D. Protease

13. A body was discovered in a lake near a village, and it needs to be determined whether the individual was alive at the time of drowning or if the body was dumped postmortem. Which of the following postmortem signs would most likely help in distinguishing between the two situations?

- A. Rigor mortis
 - B. Cold stiffening
 - C. Cadaveric spasm
 - D. Pugilistic attitude
-

14. Which of the following conditions can lead to early onset rigor mortis? Cholera Strychnine poisoning Tetanus Asphyxia Pneumonia

- A. 1 and 4
 - B. 1, 2, and 3
 - C. 2, 3, and 5
 - D. All are correct
-

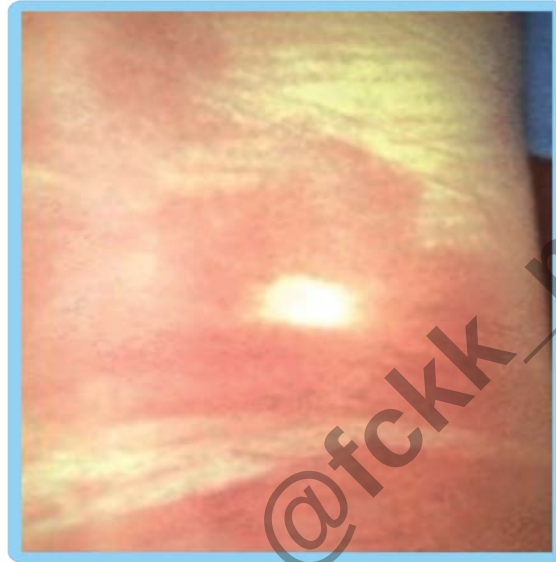
15. According to the Bishop's tripod of life, which of the following is not a criterion for establishing death?

- A. Irreversible cessation of circulation
 - B. Irreversible cessation of respiration
 - C. Total absence of reflexes
 - D. Irreversible cessation of brain activity
-

16. A 32-year-old male with a known history of chronic alcohol abuse was found unconscious at home. He was noted to have respiratory depression, pinpoint pupils, and altered mental status upon initial examination. Despite prompt resuscitation efforts, he succumbed to his condition. A postmortem examination revealed that the body had turned a dark black color. What is the most likely cause of death?

- A. Hydrogen sulfide toxicity
 - B. Carbon monoxide poisoning
 - C. Methanol adulteration
 - D. Opioid overdose
-

17. A body has been presented to the forensic department for autopsy, exhibiting typically visible post-mortem changes given below as an image, including blanching and a shift of pattern occurring upon handling and moving the body. Based on these observations, what is the approximate time of death?



- A. Immediately after death
- B. Within 4 hours
- C. After 12 hours
- D. After 24 hours

18. A 35-year-old male is discovered deceased in his apartment. During examination, livor mortis is observed primarily on the front part of his body, as illustrated below. What was the most likely position of the body at the time of death based on this distribution of livor mortis?



- A. Prone
- B. Supine
- C. Head downwards
- D. Vertical suspension

19. A forensic pathologist is examining the body of a 50-year-old male found deceased under suspicious circumstances, with elevated core temperature within two hours after death. Which of the following conditions is not typically associated with this postmortem finding?

- A. Burns
- B. Tetanus
- C. Septicemia
- D. Pontine hemorrhage

20. A forensic investigator is examining a 45-year-old male found deceased in his apartment. To determine the time of death using algor mortis, which of the following sites is ideal for measuring core body temperature?

- A. Nostrils
 - B. Rectum
 - C. Subhepatic region
 - D. External auditory meatus
-

21. A 45-year-old male is found dead in his apartment with no signs of external trauma. Based on the provided image of the eye showing hazy corneas and sunken eyeballs, what is the most likely time since death?



- A. Less than 2 hour
 - B. 3 - 4 hours
 - C. 6 - 7 hours
 - D. 10 - 12 hours
-

22. The appearance of facial pallor after death is an example of which category of postmortem changes?

- A. Immediate changes
 - B. Early changes
 - C. Late changes
 - D. Suspended animation
-

23. A 45-year-old male is brought dead to the hospital. The doctor ties a ligature around the base of the fingers and notices no bluish discoloration or edema. Which test is the doctor using to check for the absence of circulation?

- A. Icard's test
 - B. Magnus test
 - C. Diaphanous test
 - D. Pressure test
-

24. A 45-year-old woman is declared dead after a cardiac arrest. 2 hours later, a forensic expert strikes the quadriceps with a reflex hammer and observes contraction. What does this indicate about the state of the patient's tissues?

- A. Molecular death has occurred
 - B. Muscle tissues have lost all responsiveness
 - C. Zsako's phenomenon is present
 - D. Muscles are in rigor mortis
-

25. Suspended animation is most likely to occur after which type of poisoning?

- A. Cyanide poisoning
 - B. Strychnine poisoning
 - C. Phosphorus poisoning
 - D. Barbiturate poisoning
-

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Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	3
Question 4	2
Question 5	2
Question 6	3
Question 7	3
Question 8	3
Question 9	3
Question 10	2
Question 11	1
Question 12	3
Question 13	3
Question 14	2,3
Question 15	3
Question 16	4
Question 17	2
Question 18	1

Question 19	1
Question 20	2
Question 21	2
Question 22	2
Question 23	2
Question 24	3
Question 25	4

Solution for Question 1:

Correct Answer: C) 5 to 8 hours

Explanation: Heavy meal typically takes 5 to 8 hours to empty from the stomach, so the presence of undigested food suggests the patient died within that time frame after eating.

Gastric Emptying:

- The stomach usually starts emptying 10 minutes after eating.
- Light meal: Empties in 1 to 2 hours. (Option A)
- Medium-sized meal: 3 to 4 hours. (Option B)
- Heavy meal: 5 to 8 hours. (Option C)
- Order of stomach emptying is as follows: Carbohydrate > Protein > Fat
- Fluids and semi-fluids leave the stomach within 2 hours.
- Water ingested with solid meals empties rapidly and separately.
- Solid foods identifiable by the naked eye are usually ingested within 2 hours of death.

Variability in gastric emptying:

- Head injury, shock, or stress : Inhibit gastric juice secretion, stomach motility, and pyloric opening, leading to undigested food remaining for over 24 hours.
- Illness or emotional stress: Prolong gastric emptying for many hours.
- Sleep and coma: Digestion is delayed.

- Strong alcohol: Irritates the mucosa, delaying emptying.

Estimating time of death:

- Stomach contents provide clues about the last meal and time of death if the meal type is known.
- Full stomach with undigested food suggests death within 2 to 4 hours after the last meal.
- Presence of urine in the bladder and feces in the large intestine may suggest death occurred late at night.

10 to 12 hours (Option D): This option is incorrect because it suggests a longer time frame than typical for heavy meal digestion, as undigested food in the stomach indicates a more recent time of death, usually within 5 to 8 hours.

Solution for Question 2:

Correct Answer: A) 1, 2, and 4

Explanation:

Adipocere/Saponification	
This is a change of decomposition where there is a development of an offensive, sweet rancid-smelling, soft, whitish, or grayish-white, crumbly, waxy, and greasy material (similar to soap) that occurs in fatty tissues of a dead body.	
Time required for formation	Hot and humid environment: by the end of 1 week. Temperate countries: starts after 3 weeks and ends by about 3 months.
Mechanism of formation	The formation of saturated fatty acids due to postmortem hydrolysis and hydrogenation of body fats leads to adipocere.
Sites	Usual sites: Face, buttocks, breasts, and abdomen Internal sites: Small muscles turn gray and are dehydrated and thin Large muscles are red/pink Fat is completely converted to adipocere Intestines and lungs are parchment-like The liver is prominent and retains its shape

Factors influencing adipocere formation	Temperature: Heat speeds up, and cold retards adipocere formation Moisture: Water helps with adipocere formation Organisms: Bacteria like <i>Clostridium perfringens</i> and other anaerobes assist in the reaction by producing lecithinase Built: Faster in mature newborns and obese people Age: Fetuses <7 months do not show adipocere Air current: Retards adipocere formation Running water: Retarded due to wash out of electrolytes
Medicolegal importance	It is the surest sign of death. Gives a rough estimate of the time of death. Features of the face stay preserved and help in identification. Cause of death may be found as injuries might be preserved. Some idea about the place of disposal can be made.



Solution for Question 3:

Correct Answer: C) 10 hours

Explanation:

Formula for Estimation of Time Since Death (TSD):

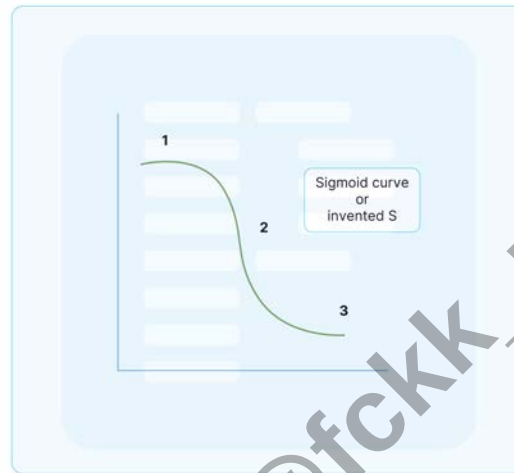
$$\bullet \text{ TSD (in hours)} = \frac{\text{Normal body temperature} - \text{Rectal temperature}}{\text{Rate of temperature falls per hour}}$$

Rate of temperature falls per hour

In this scenario: Estimated TSD = (37°C - 31°C) = 10 hours

0.6°C

Phases in Reduction of Core Body Temperature (CBT):



- The cooling pattern of a body after death follows a sigmoid, biexponential, or inverted S-shape.
- 3 Phases of Fall of CBT: Isothermic Phase: No significant loss of heat or fall in core (rectal) temperature. Intermediate Phase: CBT starts to fall in a regular, linear, and constant pattern. Terminal Phase: Rate of cooling gradually declines as the body temperature approaches the environmental temperature.
- Isothermic Phase: No significant loss of heat or fall in core (rectal) temperature.
- Intermediate Phase: CBT starts to fall in a regular, linear, and constant pattern.
- Terminal Phase: Rate of cooling gradually declines as the body temperature approaches the environmental temperature.
- Isothermic Phase: No significant loss of heat or fall in core (rectal) temperature.
- Intermediate Phase: CBT starts to fall in a regular, linear, and constant pattern.
- Terminal Phase: Rate of cooling gradually declines as the body temperature approaches the environmental temperature.

Rate of Fall in CBT: After the isothermic phase, the body cools at a linear rate of 0.4 to 0.6°C per hour for the next 12 to 16 hours.

Factors Affecting Rate of Cooling:

- Temperature Difference: Greater temperature difference between the body and surroundings leads to faster cooling.

- Build of the Cadaver: Smaller body surface area results in faster cooling; children and the elderly cool more quickly than adults.
- Physique: Lean bodies cool faster than fat bodies.
- Environment: Ventilated spaces promote faster cooling, while humid air or water cools quicker than dry air.
- Coverings: Clothing or coverings slow cooling.

Medicolegal Importance:

- Temperature-based time of death estimation is more useful in cold and temperate climates where more people die indoors.
- In tropical climates, temperature falls may be minimal, reducing its reliability.

Solution for Question 4:

Correct Answer: B) Upper limbs → Abdomen → Lower limbs → Fingers and toes

Explanation:

Nysten's Law states that rigor mortis occurs in a specific sequence, starting with larger muscle groups and progressing to smaller ones.

The sequence in which external muscles are affected:

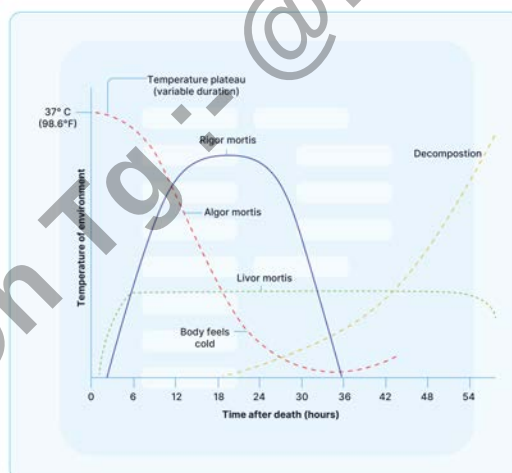
Myocardium → Eyelid muscles → Neck → Lower jaw → Muscles of face → Muscles of chest → Upper limbs → Abdomen → Lower limbs → Fingers and toes

Rigor mortis

Rigor mortis (RM), or cadaveric rigidity, refers to the stiffening of muscles after death, resulting from biochemical changes within muscle cells due to ATP depletion.



Progression of Rigor Mortis:



The progression of rigor mortis can be divided into three main phases.

Phase	Description	Timeframe
Primary Relaxation	ATP levels are initially 100%, allowing muscle relaxation. Sphincters relax, leading to the passage of feces and urine.	0 to 60 minutes after death

Rigor Mortis Onset	Onset occurs around 60 minutes as ATP depletes to ~85%. Muscles stiffen as actin and myosin filaments bind. Maximum rigidity is reached approximately 12 hours postmortem, with significant stiffness noticeable, especially in larger muscles.	60 minutes to 12 hours after death
Resolution	Following peak rigidity, a secondary relaxation phase begins as proteins decompose (proteolysis of actin-myosin). Rigor mortis resolves completely after about 12 hours, leading to further decomposition.	12 to 24 hours after death

Rules of Progression

1. Nysten's Rule

Rigor mortis progresses in a specific sequence, starting from the myocardium to the eyelids, neck, upper limbs, abdomen, and lower limbs.

- Order of Appearance:

Myocardium → Eyelid muscles → Neck → Lower jaw → Muscles of face → Muscles of chest → Upper limbs → Abdomen → Lower limbs → Fingers and toes

- Order of Disappearance:

Fingers and toes → Lower limbs → Abdomen → Upper limbs → Muscles of chest → Muscles of face → Lower jaw → Neck → Eyelid muscles → Myocardium

2. Shapiro's Rule

Rigor mortis is a biochemical process that occurs simultaneously across all muscles, with visibility depending on muscle bulk.

3. Rule of 12

Onset occurs at 60 minutes, peaks at 12 hours, remains for another 12 hours, and resolves completely thereafter.

Solution for Question 5:

Correct Answer: B) Pupae formation: 3-6 days

Explanation:

The larvae of maggots form reddish or brown pupae within 3 to 6 days.

Forensic entomology:

- Forensic entomology is the branch of science that deals with the study of insects and other arthropods found in dead bodies.
- These can help identify the time since death, the length of the body's exposure, and whether the body was moved.

Techniques used: Forensic entomologists employ various methods to estimate the time of death, including:

- Species succession
- Larval weight and length
- Accumulated degree hour technique (requires specific data for accuracy)

Egg Laying Behavior:

- Flies such as the common house fly (*Musca domestica*), green bottle fly (*Lucilia sericata*), and blue bottle fly (*Calliphora vomitoria*) lay eggs in moist areas of the body (e.g., around the nose, mouth, eyes, and armpits).
- Egg laying can occur as soon as 8–9 hours post-mortem .

Maggots, the larvae of flies, aid in early decomposition.

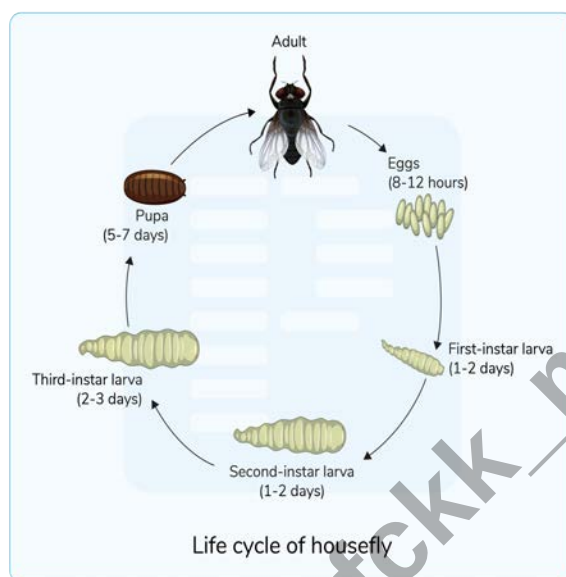
Insects involved include flesh flies (*Sarcophagidae*) and blowflies (*Calliphoridae*).

Development Stages of Maggots:

- Eggs hatch and produce larvae in 8-24 hours in summer. (Option A)
- Larvae are pale white , 3–9 mm long, with a thinner mouth end and no legs.
- The larvae undergo changes: First instar to second instar in 1–2 days Second instar to third instar in another 1–2 days
- First instar to second instar in 1–2 days
- Second instar to third instar in another 1–2 days
- Pupae are reddish or brown and are formed in 3 to 6 days .
- Pupae become adult flies in 3 to 6 days. (Option C)
- The complete lifecycle may take 5 to 10 days in summer and 8 to 20 days in winter . (Option D)
- First instar to second instar in 1–2 days
- Second instar to third instar in another 1–2 days

Medicolegal importance:

Determining the time of death: done by identifying the species of maggot and the developmental stage present in the body.



Solution for Question 6:

Correct Answer: C) Embalming makes it easier to interpret postmortem injuries.

Explanation: Embalming can complicate the interpretation of injuries and the detection of poisons, as it alters body conditions.

Embalming	Aka thanatopraxia Embalming is treatment of the dead body with antiseptics and preservatives to prevent putrefaction and preserve the body.
Effect	Produces chemical stiffening similar to rigor mortis, and normal rigor does not develop. Embalming rigidity is permanent. (Option A)
Best Time	< 6 hours of death.
Best injection technique	Discontinuous injection and drainage.
Best vessel for injection	Femoral artery (Option C)
Best embalming technique	High pressure and low flow rate.

Skin color in Embalming	Jaundiced Bodies: Greenish discoloration Normal Individual: Gray color
Methods of Embalming	Arterial embalming Cavity embalming Hypodermic embalming Surface embalming
Disadvantages	Difficult to interpret any injury or disease. (Option C) Determination of poisons becomes challenging. Blood grouping may not be possible. Thrombi and emboli may be dislocated/dislodged.
Medicolegal Importance	Embalming before autopsy in a medicolegal case invites liability under Section 201 IPC, causing the disappearance of evidence or giving false information to screen the offender. Disrespecting the corpse can lead to applicability of Section 297 IPC.
Image	

Composition of a Typical Embalming fluid:

Preservative	Formalin Methanol
Germicide	Phenol
Fungicide	Thymol
Buffer	Sodium borate
Anticoagulant	Sodium citrate (Option B)
Wetting agent	Glycerin
Controls pH	Sodium chloride
Cosmetic	Eosin
Perfume	Soluble wintergreen
Vehicle	Water

Solution for Question 7:

Correct Answer: C) Contact of the body with water

Explanation:

The given image is of the phenomenon of mummification which is supported by a hot environment with free air movement as these factors help in the drying of the body by loss of water content by evaporation but contact with water will hinder this process.

Mummification	
It is the rapid dehydration/desiccation and shriveling of the dead body from evaporation of water, with the preservation of natural appearances and features of the body. It is a modification of putrefaction.	
Features	Begins in the exposed parts of the body like the face, hands, and feet. Then extends to the entire body and internal organs. The skin may be translucent, shrunken, brittle, leathery, dry, and rusty-brown color.
Time for mummification	3-12 months or longer.
Factors favoring mummification	Hot environment. (Option A) Dry atmosphere. Free air movement. (Option B) Contact of the body with absorbing media. Poisoning: chronic arsenic or antimony poisoning favors the process. (Option D)
Medicolegal importance	Climate and time of death. Determine the time of death. Identification is easy.



Solution for Question 8:

Correct Answer: C) Decomposition in air is faster than in soil.

Explanation:

According to Casper's dictum, the rate of decomposition in air is indeed faster than in soil due to better oxygen availability and temperature regulation.

- Putrefaction begins above 10°C & the optimal range is 21–38°C .
- It is arrested below 0°C & above 45°C
- Optimal temperature aids in tissue breakdown and microbial growth.
- First amino acids to disappear from bone during decomposition are proline and hydroxyproline.
- Bones that are 100 years old will have 7 amino acids.

Poisons delaying putrefaction

These substances can either preserve tissues or inhibit microorganisms , leading to a slower rate of putrefaction.

- Carbolic acid
- Zinc chloride
- Strychnine
- Antimony
- Arsenic
- Nux vomica

Casper's dictum

- The rate of decomposition in the air is twice as rapid as in water and eight times as rapid in deeply buried bodies.
- (e.g., 1 week of decomposition in air = 2 weeks in water = 8 weeks in soil at similar temperatures)
- Air > Water > Earth
- Decomposition in water is slow due to: Lack of air Protection provided by clothing Rapid cooling of the body
- Lack of air
- Protection provided by clothing
- Rapid cooling of the body

- Lack of air
- Protection provided by clothing
- Rapid cooling of the body

Solution for Question 9:

Correct Answer: C) 3–5 days

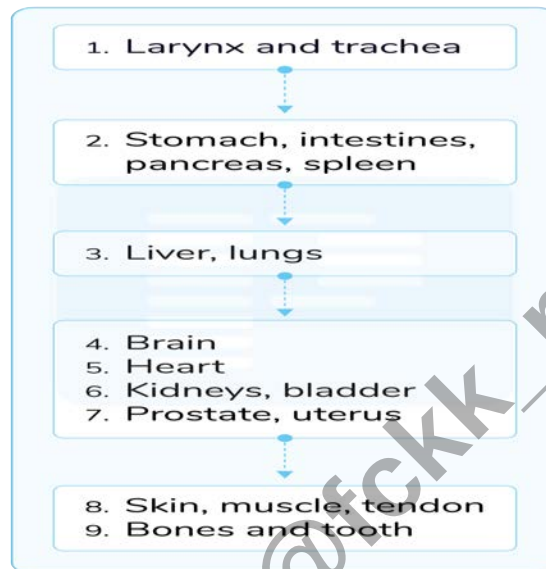
Explanation:

At 3–5 days postmortem, significant putrefactive changes include skull suture separation, skin sloughing off in a "glove and stocking" manner, and liquefaction of brain matter. These advanced decomposition signs indicate this specific time frame.

Duration	Changes
12–24 hours (Option A)	Gas accumulates in the abdominal cavity, causing tension. Blood-tinged froth may emerge from the nostrils and mouth (postmortem purge) . Eyes become soft and collapsed; the cornea appears white and flattened.
24–48 hours (Option B)	Subcutaneous tissue becomes emphysematous. Swelling occurs in breasts (females) and in the scrotum and penis (males). The tongue protrudes, and blisters form on the lower trunk and thighs.
48–72 hours	Prolapse of the uterus and anus may occur. Possible postmortem delivery of a fetus. Postmortem staining shifts from original areas, and the face becomes swollen and dark green to black, complicating identification. Hair and nails loosen .
3–5 days (Option C)	Anterior and premolar teeth become loose . Skull sutures may separate, allowing liquefied brain matter to escape, particularly in children. The skin on hands and feet may slough off in a 'glove and stocking' manner. Heavy maggot infestation can occur, resulting in skin destruction.
5–10 days (Option D)	Colliquative putrefaction (liquefaction) occurs during this period. The abdomen may burst open, and puffiness decreases as gas escapes from damaged areas. Soft and firm tissues transform into a thick, semisolid black mass, and cartilage and ligaments become softened.

Putrefaction sequence

Organs decompose at different rates based on their structure, vascularity, and exposure to air and bacteria. The general order of putrefaction is as follows:



- First organs to decompose are the larynx and trachea due to their direct contact to air
- The virgin uterus is the last organ to undergo putrefaction, while the gravid uterus, or uterus shortly after delivery, decomposes much more quickly.
- Prostate is the last organ to decompose in males.
- The overall last organ to decompose is bone/tooth.

Solution for Question 10:

Correct Answer: B) Decomposition due to gas formation

Explanation:

Postmortem gas production occurs as a result of bacterial activity, leading to gas accumulation between the epidermis and dermis, which manifests as blisters. This process typically occurs within 18 to 24 hours after death.

- H₂S
- Phosphorated hydrogen

- Ammonia
- CO₂,
- Mercaptans and methane .

Post-mortem blister	
	Gas formation in the blood vessels can push blood stained fluid, air, or liquid fat between the epidermis and dermis, leading to the formation of small blisters. Occurs within 18 to 24 hours. Blisters typically appear on the lower surfaces of the trunk and thighs , areas where hypostatic edema causes tissue fluid buildup.
Postmortem gas rigidity	
	It is also known as gas stiffening , this is when the body becomes rigid due to the accumulation of putrefactive gases.
Post mortem purge	
	Gas accumulation in the abdomen forces the diaphragm upwards compressing the lungs and heart, and the bloodstained froth exude from the mouth and nostrils. This can be misinterpreted as bleeding from antemortem injury.

Ante-mortem blister vs Post-mortem blister

Features	Ante-mortem blister	Post-mortem blister
Cause	Burns (Option A)	Decomposition (Option B)
Content	Inflammatory fluid	Gas
Base	Hyperemia	Pale
Vital reaction	Present	Absent

Solution for Question 11:

Correct Answer: A) Marbling

Explanation:

The branching pattern along the blood vessels is typical of marbling , which occurs due to bacterial decomposition of hemoglobin into sulfmethemoglobin. This phenomenon usually appears 24 hours after death.

Marbling	
Definition	Refers to a characteristic pattern in the skin caused by decomposition.
Mechanism	After death, bacteria spread through blood vessels, decomposing hemoglobin into sulfmethemoglobin, which causes greenish or reddish-brown staining of the vessel walls.
Pattern	Marbled pattern follows a linear branching pattern, resembling tree branches, which initially appears red and later transitions to a greenish hue.
Onset and Timing	Begins to develop 24 hours after death. Becomes prominent by 36-48 hours.
Color Changes Over Time	Skin's reddish-green marbling may turn dark green or nearly black within 3 to 4 days.
Areas of Marbling	Shoulders, roots of the limbs, thighs, sides of the abdomen, chest, and neck.
Medicolegal Importance	Helps in determining the time since death
Image	

Adipocere (Option B) usually forms several weeks after death and involves specific changes in fatty tissues. In contrast, marbling exhibits a distinct branching discoloration that occurs within 24 to 48 hours due to bacterial action on hemoglobin, making marbling the more relevant phenomenon in this scenario.



Filigree burns (Option C) are characterized by superficial, irregular markings resulting from electrical discharge that develop within minutes to an hour. In contrast, marbling involves the bacterial decomposition of hemoglobin, creating a distinct branching pattern in the skin, which does not occur immediately and is not indicative of electrical injury.



Postmortem lividity (Option D) typically appears as a uniform discoloration rather than a distinct branching pattern as seen in marbling, making it less relevant in this scenario.



Solution for Question 12:

Correct Answer: C) Lecithinase

Explanation:

Lecithinase, primarily produced by the anaerobic bacterium *Clostridium perfringens*, breaks down lecithin, a phospholipid found in cell membranes. The activity of lecithinase disrupts cell membrane integrity, leading to cellular breakdown, gas formation, and the characteristic greenish discoloration of the skin due to the release of various gases and pigments during decomposition. These changes, along with the strong odor, are indicative of advanced decomposition, commonly associated with putrefaction processes.

Decomposition

Decomposition is the biological process by which organic substances break down into simpler matter, critical for recycling nutrients in ecosystems.

It occurs through two main mechanisms: Autolysis and Putrefaction.

Autolysis	Putrefaction
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Autolysis is the self-digestion of tissues caused by the body's own enzymes after death. Characteristics of Autolysis: Initiation: Begins immediately after death when metabolic processes cease. Mechanism: Enzymatic action leads to cell lysis, altering tissue appearance and texture. Examples of Autolytic Changes: Clouding of the Cornea: The first observable external change, losing transparency due to cellular breakdown. Changes in Brain Glandular Tissue: The first internal change, where neuronal and glial cells undergo lysis. Note: Aseptic Autolysis: Occurs in intrauterine fetal demise, leading to mummification without bacterial involvement.

Putrefaction is driven by bacterial activity, resulting in tissue breakdown and gas production. Characteristics of Putrefaction: Bacterial Involvement: Primarily caused by *Clostridium perfringens*. Lytic Enzyme: The main enzyme involved is lecithinase, which breaks down cell membranes. Stages of Putrefaction: Color Change: The first internal change is reddish-brown discoloration of the aortic intima. The first external change is green discoloration in the right iliac fossa due to the breakdown of hemoglobin. Mechanism of Color Change: Hydrogen Sulfide Production: Bacteria produce hydrogen sulfide, which forms a green compound with hemoglobin. The cecum is the primary site of discoloration due to high bacterial activity. 2. Gas Formation: Gas production leads to bloating, with gases like carbon dioxide, methane, and hydrogen sulfide. 3. Liquefaction of Tissue: Tissues become liquefied, losing structural integrity, contributing to the characteristic odor of decomposition. Timing of Discoloration: Discoloration begins within 12-18 hours in summer and 1-2 days in winter, influenced by temperature and bacterial activity.

Solution for Question 13:

Correct Answer: C) Cadaveric spasm

Explanation:

Cadaveric spasm is an instantaneous stiffening of muscles occurring at the time of death, often in cases of violent death or extreme stress. If found in a body recovered from a lake, it suggests the person was alive and experiencing extreme stress (such as drowning) at the time of death, supporting the conclusion of antemortem drowning.

Differential Diagnosis of Rigor Mortis

1. Heat stiffening / Pugilistic attitude (Option D)

Definition	Also known as the boxing attitude, pugilistic attitude, or fencing attitude, occurs when the body is exposed to extreme temperatures ($\geq 65^{\circ}\text{C}$).
Mechanism	Coagulation of muscle proteins, similar to how proteins coagulate when cooked.

Features	The body adopts a flexed attitude with stiff joints. Hyperextended neck. Claw-like fingers. No normal rigor mortis is seen because the heat coagulates the proteins before rigor can occur.
Example	Observed in cases of individuals exposed to high temperatures, such as in a fire.

2. Cold stiffening (Option B)

Definition	Cold stiffening occurs when the body is exposed to extremely low temperatures ($\leq -5^{\circ}\text{C}$).
Mechanism	Involves the freezing of body fluids and the hardening of subcutaneous tissues, creating stiffness in the body.
Features	Joints and tissues become stiff as the fluids in the body freeze. Cold stiffening disappears when the body returns to normal temperatures, allowing normal rigor mortis to become apparent.
Example	Observed in cases of hypothermia or individuals exposed to extremely cold environments.

3. Gas Stiffening

Definition	Gas stiffening occurs during decomposition as gases build up within the body.
Mechanism	Gas accumulation from bacterial activity in the intestines and other tissues.
Features	Stiffness in the body caused by bloating due to gas formation. Protrusion of the tongue and eyeballs may occur due to internal pressure from gas.
Example	Often observed in bodies that have been deceased for a period, especially in warm environments that promote decomposition.

4. Cadaveric Spasm

Definition	Cadaveric spasm, also known as cataleptic rigidity or instantaneous rigor, is an ante-mortem phenomenon occurring immediately at the time of death.
Mechanism	ATP depletion in a specific group of muscles that were contracted just before death.

Features	Muscles that were in use at the time of death remain stiff and rigid immediately after death. No primary relaxation phase, but secondary relaxation occurs during decomposition. Cannot be artificially induced, making it an exclusively ante-mortem finding.
Medico-Legal Importance	Cadaveric spasm helps in determining whether drowning was ante-mortem or post-mortem. In cases of antemortem drowning, the spasm can cause the victim to tightly grip objects like weeds or mud, indicating they were alive and struggling before death. This provides evidence of active efforts during the drowning process. Conversely, if no cadaveric spasm is present, it may suggest post-mortem immersion, indicating that the body was placed in water after death.

Solution for Question 14:

Correct Answer: B) 1, 2, and 3

Explanation:

- Cholera primarily causes rapid dehydration and electrolyte imbalance, which can lead to death but typically does not cause early rigor mortis.
- Strychnine is a potent neurotoxin that causes severe muscle spasms and convulsions. The increased muscular activity and sustained contraction can lead to early onset rigor mortis.
- Tetanus, caused by the bacterium *Clostridium tetani*, results in sustained muscle contractions (spasms). Similar to strychnine poisoning, these contractions can result in early rigor mortis.

Factors Influencing Rigor Mortis

- Age: Rigor mortis is usually absent in fetuses younger than 7 months of intrauterine life (IUL) due to underdeveloped actin and myosin filaments.
- Muscle Bulk: Individuals with greater muscle mass may experience a delayed onset of rigor mortis due to larger energy reserves, while those with lower muscle bulk may show an earlier onset due to quicker ATP depletion.

Onset of rigor mortis under different conditions

Condition	Mechanism
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Early Onset	
Violent Death	Rapid muscle activity and ATP depletion due to trauma (e.g., firearms, cut throat).
Chronic Wasting Diseases	Muscle wasting leads to lower ATP reserves (e.g., tuberculosis, cholera).
Poisoning	Toxins (e.g., strychnine, cyanide) and conditions (e.g., tetanus) increase muscle contractions, depleting ATP.
Heat Stroke	Elevated body temperature accelerates biochemical reactions, promoting ATP depletion.
Late Onset	
Asphyxia	Insufficient oxygen during death slows ATP production, delaying rigor mortis.
Pneumonia and Hemorrhage	Impaired oxygenation and circulation reduce energy production at death.
No Rigor Mortis	
Extreme Cold	Hypothermia significantly reduces metabolic processes, potentially delaying rigor mortis.
Decomposing Bodies	Rapid decomposition hinders the development of rigor mortis.
Rigor Mortis in Fetus	Fetuses younger than 7 months lack developed actin and myosin filaments, preventing rigor mortis.

Solution for Question 15:

Correct Answer: C) Total absence of reflexes

Explanation:

Total absence of reflexes is not a part of the Bishop's tripod of life.

- Thanatology (Greek thanatos: death) is the scientific study of death in all its aspects, including its cause and phenomena. It also includes bodily changes that accompany death (postmortem changes) and their medico-legal significance.

- S.46, I.P.C / 2(6) BNS : Death denotes the death of a human being unless the contrary appears from the context.
- Taphonomy is often used to refer to the study of postmortem processes that affect the preservation and recovery of human remains. This helps reconstruct the circumstances surrounding the death event.
- Bishop's tripod of life

Irreversible stoppage of the:

- Circulation - Heart (Option A)
- Respiration - Lungs (Option B)
- Brain functions (Option D)

Solution for Question 16:

Correct Answer: D) Opioid overdose

Explanation:

Opioid overdose presents with classic symptoms, including respiratory depression, altered mental status, and pinpoint pupils. The dark black coloration observed in the body can be due to the hypoxic state and the subsequent effects on blood and tissue, potentially leading to black discoloration

Given the patient's history of chronic alcohol use, which may have contributed to opioid dependence or overdose.

Colours observed in postmortem staining (livor mortis) under various conditions

Condition	Color	Explanation
Normal (Natural Death)	Bluish-Purple	Deoxygenated blood pooling in the lower body parts.
Opium Poisoning	Black	Respiratory depression leads to severe hypoxia, causing a black discoloration.
Carbon Monoxide (CO) Poisoning (Option B)	Cherry Red	Formation of carboxyhemoglobin, which gives a bright red hue to the blood.

Hydrogen Sulfide (HS) Poisoning (Option A)	Bluish-Green	Sulfhemoglobin is formed due to HS exposure, causing a blue-green staining.
Cyanide Poisoning	Bright Red / Brick Red	Cyanide prevents cells from using oxygen, keeping blood highly oxygenated, giving a bright red color.
Phosphorus, Aniline, Nitrite, Bromide Poisoning	Dark Brown	These agents cause methemoglobinemia, where hemoglobin becomes oxidized, resulting in a dark brown color.
Methanol Poisoning (Option C)	Purple	Acidosis from methanol metabolism leads to a purple discoloration.
Hypothermia	Pink	Cold slows metabolism, preserving oxygen in the blood, which leads to a pinkish color.
Clostridium welchii Infection (Septic Abortion)	Bronze	Bacterial toxins cause hemolysis, leading to bronze-colored staining.
Phosphorus and Chloroform Poisoning	Yellow	Severe liver damage causes jaundice, leading to a yellow tint in postmortem staining.
Strychnine Poisoning	Intense Purple	Muscle rigidity and asphyxia result in extreme oxygen deprivation, causing a deep purple discoloration.

Solution for Question 17:

Correct Answer: B) Within 4 hours

Explanation:

Post-mortem lividity generally begins to appear within 1-4 hours after death, as blood settles in dependent parts of the body, creating reddish-purple discoloration.

The presence of blanching indicates that the blood remains fluid and has not yet clotted.

Additionally, the ability to shift the lividity pattern upon handling suggests that it is not fixed, which typically occurs after 6 to 12 hours. Therefore, these observations indicate that the time of death is approximately within 4 hours.

Appearance of Hypostasis

Onset	Begins approximately 30 minutes after death. Initially unfixed and can shift with body position changes.
Visibility	Becomes visibly apparent within 4 hours and reaches maximum intensity between 6-12 hours.
Fixation	Typically it becomes fixed by around 8 hours (between 6-12 hours). Staining won't move after fixation.
Confirmation Test	Apply pressure to the area: Blanching indicates blood is still within vessels, staining is not yet fixed. No blanching means staining is fixed, indicating breakdown of vessels and leakage into tissues.

Solution for Question 18:

Correct Answer: A) Prone

Explanation:

Lividity appearing on the front part of the body suggests the individual was lying face down, which corresponds to the prone position at the time of death.

Livor Mortis

Other Names	Postmortem staining, subcutaneous hypostasis, cadaveric lividity, suggillations, cogitation, vibices, and darkening of death.
Definition	Bluish or purplish-red discoloration caused by the gravitational settling of blood in the toneless capillaries and venules of the dependent parts of the dead body.
Site	Superficial layers of the dermis
Mechanism	Blood in capillaries and venules → Stagnates and sinks in dependent parts due to gravity → Bluish purple discoloration due to deoxy hemoglobin.
Begins	0.5 - 1 hour
Fully Developed	Within 4 hours

Fixation of Livor Mortis	Typically, within 6-12 hours
Confirmation of Fixation	Apply thumb pressure against the skin for a few seconds → Blanching → Livor mortis not fixed.
Primary Lividity	Occurs as lividity reaches maximum intensity between 6 to 12 hours after death.
Secondary Lividity	Shifting of lividity can occur if the body is moved within a few hours after death.
Duration of Persistence	Staining persists until putrefaction begins.
Distribution	Depends on position of the body:
Supine (Option B)	Appears on the back of neck, entire back, flanks, except areas under pressure (contact pallor).
Prone (Option A)	Appears on the front part of the body.
Side	Appears on the dependent side.
Head downwards (Option C)	Appears as a blackened face and neck.
Vertical suspension (Option D)	Appears on legs and hands (glove and stocking appearance) and external genitalia.
Drowning	Appears on the face, upper chest, hands, lower arms, feet, and calves.
Livor Mortis Absent	Excessive blood loss Severe anemia Death from wasting diseases Lobar pneumonia
Medicolegal Importance	Helps estimate time of death. Indicates body posture at death. Indicates if the body was moved post-death. Distribution may indicate the cause of death.
Image	

Solution for Question 19:

Correct Answer: A) Burns

Explanation:

Elevated body temperature within the first two hours after death points toward a diagnosis of Postmortem caloricity. Burns do not typically cause postmortem caloricity.

Postmortem Caloricity:

- Refers to the condition where the body temperature remains elevated for the first two hours after death.
- Causes of Postmortem Caloricity:

Disturbed heat regulation	Increased heat production (excessive muscle contraction)	Excessive bacterial activity
Sunstroke Pontine hemorrhage (Option D)	Tetanus (Option B) Strychnine poisoning	Septicaemia (Option C) Cholera

Solution for Question 20:

Correct Answer: B) Rectum

Explanation:

Rectum is the ideal site for core temperature measurement in algor mortis, except in cases of sodomy.

Algor mortis	Refers to the cooling of the body after death, where the core body temperature equilibrates with the surrounding environment. First detectable sign of death, followed by rigor mortis.
Onset	About 30 minutes to 1 hour after death.
Mechanism	Death → Heat loss (conduction, convection, radiation, evaporation) → ↓Body core temperature
Sites for Core Temperature Measurement	Rectum (8-10 cm above the anus): Ideal site for measurement except in cases of sodomy. (Option B) External auditory meatus (Option D) Subhepatic region (Option C) Nostrils (up to the cribriform plate) (Option A) Intracerebral (through the orbit)
Instruments for Core Temperature Measurement	Thanatometer/Chemical thermometer: 25 cm long thermometer with a range of 0 to 50°C. Electronic thermocouple with a digital readout

Timing of Reading	Body temperature readings and environmental temperature should be recorded at 1 to 2-hour intervals to obtain the rate of fall of temperature.
Effect of Illness on Cooling	In serious illness, circulation begins to fail before death, causing hands and feet to cool before the rest of the body.
Medicolegal Importance	Inner core temperature measurements are important for estimating the time since death and are more reliable than surface temperature measurements.

Solution for Question 21:

Correct Answer: B) 3 - 4 hours

Explanation:

The image shows tache noire that forms after 3-4 hours, indicating that this case fits within that time frame. Corneal haziness usually develops within 2 hours, and sunken eyeballs develop within minutes. These findings support an estimated time of death around 3-4 hours ago.

Postmortem Changes in Eye:

- **Medicolegal Importance:** Changes in the retina and vitreous humor are more significant for estimating the time since death.

Part of Eye	Features	Time Since Death (TSD)
Retina	Kevorkian Sign: Fragmentation or segmentation of blood column in retinal vessels (cattle trucking or shunting) Occurs within minutes and lasts about an hour. Seen using an ophthalmoscope.	Few minutes
Vitreous Humor	Vitreous humor's accessibility and resistance to putrefaction make it ideal for forensic analysis. Potassium along with hypoxanthine increases due to autolysis of choroid and retinal cells. Potassium levels provide accurate TSD estimates, unaffected by external factors. The Sturmer and Gantner formula is used to estimate TSD from potassium levels.	-

Sclera	Tache Noire : Appears as two yellow triangles on the sclera (base on limbus, apex at medial or lateral canthus), which darken to brown and then black. Caused by desiccation and deposition of cellular debris, mucus, and dust on exposed conjunctiva and the sclera underneath. Forms if eyelids remain open for 3-4 hours postmortem.	3 - 4 hours
Cornea	Cornea becomes hazy within 2 hours of death due to drying and dust deposition. Delayed for about 2 hours if lids are closed after death. Transient haziness can be removed by applying water, but permanent opacity occurs after 10-12 hours due to decomposition.	Within 2 hours
Eyeball	Eyes appear sunken and become softer within minutes due to reduced intraocular tension. Intraocular pressure drops from 10-22 mmHg during life to zero within 2 hours.	Within 2 hours (IOP becomes zero)

Solution for Question 22:

Correct Answer: B) Early changes

Explanation:

The appearance of facial pallor after death is an example of Early postmortem change.

The changes that occur after death, which assist in estimating the time of death and distinguishing between death and suspended animation, can be categorized as follows:

Changes After Death		
Immediate Changes (Option A)	Early Changes (Option B)	Late Changes (Option C)

Irreversible cessation of: Brain function Circulation Respiration	Loss of elasticity of the skin Facial pallor Primary muscle relaxation Contact pallor and flattening Changes in the eyes Algor mortis Livor mortis Rigor mortis	Putrefaction Adipocere formation Mummification
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Solution for Question 23:

Correct Answer: B) Magnus test

Explanation:

The Magnus test (ligature test) involves applying a ligature at the base of the fingers. If no bluish discoloration or edema is seen, it indicates a lack of circulation.

Tests to Detect Stoppage of Respiration
Winslow's Test: A mirror or a surface of water placed on the chest shows no movement or reflection of light. Feather Test: A feather or fine cotton fibers held in front of the nostrils show no movement. Mirror Test: A mirror held near the mouth and nostrils shows no haziness, indicating no exhalation.
Tests to Detect Stoppage of Circulation
Magnus Test (Ligature Test): No bluish discoloration or edema appears on the fingers when a ligature is applied at their base. (Option B) Diaphanous Test (Transillumination Test): The web space between the fingers fails to show redness when transilluminated from behind. (Option C) Icard's Test: Injected fluorescein dye does not produce yellowish-green discoloration, unlike in a living person. (Option A) Pressure Test: Fingernails remain pale and do not turn reddish after applying and releasing firm pressure. (Option D)

Solution for Question 24:

Correct Answer: C) Zsako's phenomenon is present

Explanation:

The observed muscle response in the quadriceps upon stimulation is characteristic of Zsako's phenomenon, which occurs during the supravital period shortly after death. This phenomenon

indicates that the muscle fibers are still reactive and viable for a short time post-mortem.

Supravital period

- The "supravital period" or "intermediary life" refers to the time during which certain tissues remain viable even after irreversible cardiac arrest.
- Organ harvesting for transplantation is done during this period.

Tendon reaction (Zsako's phenomenon)

- It occurs after somatic death when molecular death has not occurred.
- Occurs during the supravital period .
- It is a stage of Primary relaxation .
- Striking the lower part of the quadriceps femoris, about 10 cm above the kneecap, with a reflex hammer, causes the patella to move upward due to full muscle contraction.
- Seen within 2–3 hrs after death .
- This response is caused by propagated excitation of muscle fibers, after death.

Molecular death (Option A) indicates complete cessation of cellular functions, which is not the case here, as the muscle shows a response.

Muscle tissues are completely non-responsive (Option B) , which implies that the muscles are entirely inactive, which contradicts the observed muscle response.

If rigor mortis (Option D) were fully established, there would be no muscle response.

Solution for Question 25:

Correct Answer: D) Barbiturate poisoning

Explanation:

Barbiturates are CNS depressants that, in large doses, cause deep sedation and respiratory depression , potentially leading to suspended animation where vital signs like heart rate, respiration, and reflexes may become undetectable, mimicking death, but the condition is reversible with timely intervention.

Types of death

Feature	Somatic Death	Molecular Death
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Definition	Complete and irreversible cessation of brain function, circulation, and respiration	Progressive disintegration of body tissues, leading to the death of individual cells.
Onset	Occurs before molecular death.	Follows somatic death, occurring 1-2 hours after vital functions cease.
Tissue Viability	Tissues and cells remain alive and functioning.	Tissues and cells are dead and non-functioning, with no metabolic activity.
Response to Stimuli	Muscles can respond to thermal, electrical, or chemical stimuli.	No response to external stimuli.
Confirmation	Identified by flat ECG and EEG readings, and absence of breath sounds	Confirmed through rigor mortis, algor mortis, postmortem staining, and putrefaction.
Resemblance	May resemble conditions like suspended animation, coma, or hypothermia.	Does not resemble any other condition.

The "supravital period" or "intermediary life" refers to the time during which certain tissues remain viable even after irreversible cardiac arrest.

Suspended Animation (Apparent Death)
Suspended animation refers to a condition where vital signs, such as heartbeat and respiration, are undetectable using standard clinical methods due to temporary interruption or reduction of these functions to minimal levels. Mechanism The metabolic rate significantly decreases, allowing cells to meet their oxygen needs through dissolved oxygen in body fluids. Types Voluntary: Occurs in individuals practicing yoga or in trance states. Involuntary: Occurs in conditions like hypothermia, barbiturate (Option D) or opiate poisoning, newborns, drowning, electrocution, heatstroke, cholera, post-anesthesia, shock, cerebral concussion, or insanity. Management Resuscitation can be achieved using cardiac massage, electrical stimulation, and artificial respiration. An ECG or EEG is required before issuing a death certificate to confirm death.

Cyanide poisoning (Option A) causes rapid tissue asphyxia and death due to cellular hypoxia, but it does not lead to suspended animation, as there is no temporary or reversible cessation of vital signs.

Strychnine poisoning (Option B) leads to severe muscle convulsions and death from respiratory paralysis, without a state of suspended animation.

Phosphorus poisoning (Option C) results in progressive organ failure, especially liver damage, but does not cause suspended animation.

Previous Year Questions

1. A body brought for autopsy is cold and has developed rigor mortis in all 4 limbs and livor mortis at the back/fixed but no signs of putrefaction. What will be the estimated time since death?

- A. 12-24 hrs
- B. 6-12 hrs
- C. 24-48 hrs
- D. 36-48 hrs

2. What is the approximate age of a person in whom the medial end of the clavicle is fused and also the sternum is completely fused:

- A. <18 years
- B. 18-22 years
- C. 22-25 years
- D. >25 years

3. Cadaveric spasm seen in all except ?

- A. Suicidal gun-shot wound
- B. Electrocution
- C. Drowning
- D. Traumatic asphyxia

4. All are used in identification of dead body except

- A. Teeth
 - B. Blood
 - C. Nail
 - D. Muscle
-

5. Among the listed options, which substance is not utilized as a chelating agent in cases of heavy metal poisoning?

- A. Deferiprone
 - B. EDTA
 - C. Penicillamine
 - D. Oximes
-

6. The dead body shows the following finding as in image . What is this case?



- A. Hanging
- B. Throttling
- C. Choking
- D. Smothering

7. In medicolegal examination accused 18 year old male claims he is 16 year old. Which joint x-ray should be done?

- A. Head & shoulder
- B. Elbow and ankle
- C. Knee and wrist
- D. Elbow & hip

8. Which of the following is true regarding adipocere formation?

- A. Preservation by saponification

- B. High temperature needed
- C. Cool and dry climate needed
- D. Occurs within minutes to hours

9. Which of the following eye signs can be appreciated in the image?



- A. Tache noire
- B. Kevorkian sign
- C. Corneal changes
- D. Vitreous humour changes

10. The most reliable criteria among Gustafson's criteria for age estimation is?

- A. Root resorption
- B. Paradentosis
- C. Translucency of root

D. Attrition

11. You are posted as an intern in casualty when an MLC case is brought by the police with the following appearance. She was declared brought dead. What would you mark as the likely cause of death before sending the patient for postmortem ?



- A. Drowning
- B. Hanging
- C. Strangulation
- D. Traumatic asphyxia

12. A patient who was involved in a road traffic accident 20 minutes ago has been brought in for autopsy. The post-mortem examination indicates rigidity in the neck, upper limbs, and lower limbs, with no signs of initial relaxation. What does this indicate?

- A. Rigor mortis

- B. Heat stiffness
- C. Cadaveric spasm
- D. Cold stiffness

13. Identify the correct pair: i. Mummification 1) ii. Adipocere 2) iii. Marbling 3) iv. Lividity of death 4)

i. Mummification	1)	
ii. Adipocere	2)	

iii. Marbling	3)	MUMMIFICATION 
iv. Lividity of death	4)	

A. i -4, ii-3, iii-2, iv-1

B. i -3, ii-4, iii-2, iv-1

C. i -3, ii-1, iii-4, iv-2

D. i -3, ii-4, iii-1, iv-2

14. What is the first postmortem change that occurs in a dead body?

A. Maggot formation

B. Putrefaction

C. Greenish discoloration of right iliac fossa

D. Mummification

15. Paradoxical undressing is seen in :

- A. Heart stress
 - B. Voyeurism
 - C. Hypothermia
 - D. LSD overdose
-

16. Which of the following is commonly used for preservation of viscera in toxicology autopsy?

- A. Glycerin
 - B. Rectified spirit
 - C. Formalin
 - D. Saturated salt solution
-

17. Identify the post-mortem change described below.:



- A. Tattooing
- B. Suggilation
- C. Putrefaction
- D. Decomposition

18. Which option is correct regarding techniques of organ removal during autopsy :
A. Rockitansky – In situ B. Virchow – en masse C. Ghon’s – en block D. Letulle – One by one

- A. A and C are correct
- B. A, B and C are correct
- C. B and D, A are correct
- D. A, B, C and D are correct

19. At what age, basiocciput fuses with basisphenoid?

- A. 18-22 years
 - B. 22-25 years
 - C. 14-16 years
 - D. 12-14 years
-

20. Which of the following trait is feature of the female skull ?

- A. Larger teeth
 - B. Mastoid medium-large
 - C. Mandible Rounded
 - D. Zygomatic is more pronounced
-

21. A girl was allegedly kidnapped by a man. In court, she says that she is a major and is married to the man, X ray images of her elbow, wrist and pelvis are given. What is the approximate age of the girl?





- A. 14 to 15 years
- B. 16 to 17 years
- C. 18 to 19 years
- D. 21 to 22 years

22. What is the estimated gestational age of the fetus discovered in the water canal, considering its crown heel length measured at 25 cm?

- A. 6 months
 - B. 5 months
 - C. 4 months
 - D. 3 months
-

23. Which of the following bones is most accurate for sex determination?

- A. Skull
 - B. Pelvis
 - C. Long bones
 - D. Sternum
-

24. A man was found dead due to starvation. Which of the following findings will be seen on autopsy?

- A. Distended gallbladder
 - B. Enlarged liver
 - C. Expanded lungs
 - D. Hypertrophied heart
-

25. Rearrange the following to correspond with alterations in fingerprints.: 1. Incomplete atrophy of ridges 2. Less of pattern with ridge atrophy 3. Altered ridges 4. Permanent loss of fingerprints 5. Distance between ridges changed but pattern retained A. Radiation B. Scleroderma C. Celiac disease D. Acromegaly E. Dermatitis

- A. 1-E, 2-C, 3-B, 4-A, 5-D
- B. 1-D, 2-B, 3-C, 4-E, 5-A
- C. 1-C, 2-A, 3-D, 4-B, 5-E
- D. 1-D, 2-C, 3-A, 4-E, 5-B

26. Which is the first organ to putrefy?

- A. Brain
- B. Heart
- C. Kidney
- D. Prostate

27. From the dense vegetation near the highway, the police have found the remains of a burnt body. The location was discovered with the help of a sniffer dog, as they were tracing a 25-year-old IT worker who had gone missing the night before. To determine the identity of the victim, the team needs to gather a suitable sample. Which type of tissue from a charred body is most effective for identification purposes?

- A. Hair
- B. Bone
- C. Teeth
- D. Blood

28. Identical twins have similarities in all of the following except?

- A. Blood group
 - B. DNA fingerprinting
 - C. Fingerprint pattern
 - D. Iris color
-

29. Which option provides the most accurate description of a posthumous child?

- A. Delivery of a macerated fetus
 - B. Child is delivered after death of biological father
 - C. Child is born to an unmarried couple
 - D. Child is abandoned by the parents
-

30. In a dead body, bloating of face, tongue protrusions were noted along with the findings in the image. Identify the phenomenon:



- A. Electrocution
 - B. Crocodile burns
 - C. Scalds
 - D. Putrefaction
-

31. What does the observation of contracted hands with no primary relaxation during a post-mortem examination indicate?

- A. Rigor Mortis
 - B. Livor Mortis
 - C. Cadaveric spasm
 - D. Algor Mortis
-

32. What is the term used to describe the comprehensive examination of death in all its aspects?

- A. Thanatology
 - B. Forensic anthropology
 - C. Toxicology
 - D. Molecular death
-

33. Following the recovery of a beheaded skull bone by the police, the relatives of a missing person want to confirm his identity. Which of the following techniques can be used to figure out a person's identity using the beheaded skull bone and photograph of the person?

- A. Superimposition

- B. Anthropometry
 - C. DNA fingerprinting
 - D. Cheiloscopy
-

34. The method of autopsy carried out en- masse to remove organs from tongue to prostate is:

- A. Virchow technique
 - B. Rokitansky technique
 - C. Ghon technique
 - D. Letulle technique
-

35. A 42-year-old HIV-positive patient died, and his body was brought for post-mortem. The method of autopsy to be done is___:

- A. Ghon technique
 - B. Letulle technique
 - C. Virchow technique
 - D. Rokitansky technique
-

36. What is the likely cause of a deceased person having a body temperature of 39 degrees Celsius?

- A. Cyanide poisoning
- B. Septicemia
- C. Corrosive poisoning
- D. Intra-abdominal hemorrhage

37. Post-mortem examination of a patient reveals the following. What is this finding known as?



- A. Hesitation cuts
- B. Grazed abrasion
- C. Laceration
- D. Patterned bruise

38. Identify the postmortem change given in the following image.:



- A. Marbling
 - B. Tattooing
 - C. Lividity
 - D. Filigree burns
-

39. A man was working in a field and he suddenly collapsed on a hot summer afternoon. On examination, there were no physical signs of dehydration. His serum electrolyte level was normal. The doctor found that the body temperature was 106 degrees Fahrenheit. Which of the following symptoms is least likely to be seen in this patient?

- A. Hot skin
 - B. Hypotension
 - C. Sweating
 - D. Disorientation
-

40. The phenomenon shown below is typically seen in the case of ?



- A. Case of torture phalanges.
- B. Case of suspension or hanging more than 72 hours
- C. Case of immersion in water for 36 hours
- D. Case of colliquative liquefaction due to immersion in hot water

41. Which of the following is true regarding the post-mortem change shown in the image given below?



- A. Seen within first 12 hours of death
- B. It occurs due to the accumulation of deoxygenated blood
- C. It results due to bacterial activity
- D. It is seen in case of lightning

42. A woman gave birth to twins. The father who believed that they did not belong to him requested DNA testing. After testing, it was found that one child did not belong to him. This case would be best described as?

- A. Superfetation
- B. Superfecundation
- C. Suppositious child
- D. Atavism

43. A child who is a victim under the POCSO Act is brought to the Department of forensic medicine for age estimation. The X-ray image of the hand is shown below.

What is the likely age of the child?



- A. 4 years
- B. 7 years
- C. 10 years
- D. 13 years

44. Identify the following post mortem change:



- A. Tattooing
- B. Suggillation
- C. Putrefaction
- D. Decomposition

45. In the case of drowning, the following finding is seen during autopsy. What is it known as?



- A. Cadaveric spasm
- B. Cutis anserina
- C. Rigor mortis
- D. Washerwoman's hands

46. Identify the odor produced by the change seen in the image below.:



- A. Odorless
- B. Pungent
- C. Putrid
- D. Offensive

47. Match the following with the respective ages of fusion :

1. Medial end of clavicle
2. Sacrum as a single bone **3. Crista scapulae** **4. Lambdoid suture**
A. 21-25 years **B. 2-50 years** **C. 50-70 years** **D. 20-25 years**

1. Medial end of clavicle 2. Sacrum as a single bone 3. Crista scapulae 4. Lambdoid suture

- A. 1-A, 2-B, 3-C, 4-D
- B. 1-A, 2-D, 3-B, 4-C
- C. 4-A, 3-B, 1-C, 2-D
- D. 1-A, 4-B, 3-C, 2-D

48. A pelvis is brought for forensic examination. Which of the following characteristics will identify the given pelvis as a male? Triangular obturator foramen U-shaped pubic angle Everted ischial tuberosity Large greater sciatic notch Large acetabulum Deep preauricular sulcus

- A. 2, 3, 4, 5
- B. 4, 5, 6
- C. Only 5
- D. 1 and 3

49. Please match the following findings observed during postmortem examination with their corresponding time of appearance: A. Adipocere formation B. Relaxation of urethral sphincter C. Marbling of veins D. Liquefaction of tissues 24 hrs 2-3 days 4-5 days 8-10 days

A. Adipocere formation B. Relaxation of urethral sphincter C. Marbling of veins D. Liquefaction of tissues

- A. A-4, B-1, C-2, D-3
- B. A-3, B-1, C-2, D-4
- C. A-4, B-2, C-1, D-3
- D. A-1, B-2, C-3, D-4

50. A skull was recovered from a forest. According to inquest papers, a girl had gone missing 15 days back. The skull was sent to the forensic research lab. Which of the following would identify it as a female skull? Large frontal and parietal eminence Heavy cheek bones Less pronounced glabella Square orbits Small mastoid

- A. 1 and 4 only
- B. 2, 3 and 4

- C. 3 and 5
 - D. 3, 4 and 5
-

51. The dead body of a 20 year old man found in the sea was brought in for postmortem examination. Which of the following findings would you see in seawater drowning? Hypernatremia Hyponatremia Hyperkalemia Myocardial anoxia Hemodilution

- A. 1 and 3
 - B. 1 and 4
 - C. 2, 3 and 5
 - D. 2, 4 and 5
-

52. Identify the fingerprint pattern shown in the image:



- A. Whorl
- B. Composite

C. Loop

D. Arch

53. Rigor mortis occurs first in:

A. Eyelids

B. Intestines

C. Myocardium

D. Neck

54. Which of the following joint is the best predictor of age of 16 to 17 years by X-ray?

A. Knee

B. Elbow

C. Hip

D. Ankle

55. What is the odor of the mummified body?

A. Odorless

B. Pungent

C. Putrid

D. Offensive

56. Which of the following is seen in freshwater drowning? A. Haemodilution B. Hypokalemia C. Hyponatremia D. Arrhythmia

- A. A, B, C & D
 - B. A, B & C
 - C. B, C & D
 - D. A, C & D
-

57. All of the following changes occur in chronic starvation except?

- A. Hyperthermia
 - B. Exhaustion
 - C. Heart atrophy
 - D. Hypotension
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	4
Question 4	3
Question 5	4
Question 6	4
Question 7	3
Question 8	1
Question 9	1
Question 10	3
Question 11	1
Question 12	3
Question 13	4
Question 14	3
Question 15	3
Question 16	4
Question 17	2
Question 18	1

Question 19	1
Question 20	3
Question 21	4
Question 22	2
Question 23	2
Question 24	1
Question 25	1
Question 26	1
Question 27	3
Question 28	3
Question 29	2
Question 30	4
Question 31	3
Question 32	1
Question 33	1
Question 34	4
Question 35	4
Question 36	2
Question 37	1
Question 38	1
Question 39	3

Question 40	3
Question 41	3
Question 42	2
Question 43	4
Question 44	2
Question 45	1
Question 46	1
Question 47	2
Question 48	3
Question 49	1
Question 50	3
Question 51	2
Question 52	4
Question 53	3
Question 54	4
Question 55	1
Question 56	4
Question 57	1

Solution for Question 1:

- rigor mortis appears in all muscles within 12 hours. If rigor mortis is fully developed in all four limbs, it suggests that up to 12 hours have passed since death.
- livor mortis becomes fixed in 7-8 hours. Fixed livor mortis indicates that 7-8 hours have passed since death.
- The absence of skin green discoloration (a sign of putrefaction) confirms that death occurred less than 24-48 hours ago.

Given that rigor mortis is fully developed and livor mortis is fixed, but the body has not yet begun to show signs of putrefaction, the most precise estimate for the time since death would be 6-12 hours. This aligns with the fixed livor mortis (7-8 hours) and the development of rigor mortis (up to 12 hours).

Livor Mortis

Other Names	Postmortem staining, subcutaneous hypostasis, cadaveric lividity, suggillations, cogitation, vibices, and darkening of death.
Definition	Bluish or purplish-red discoloration caused by the gravitational settling of blood in the toneless capillaries and venules of the dependent parts of the dead body.
Site	Superficial layers of the dermis
Mechanism	Blood in capillaries and venules → Stagnates and sinks in dependent parts due to gravity → Bluish purple discoloration due to deoxy hemoglobin.
Begins	0.5 - 1 hour
Fully Developed	Within 4 hours
Fixation of Livor Mortis	Typically, within 6-12 hours
Confirmation of Fixation	Apply thumb pressure against the skin for a few seconds → Blanching → Livor mortis not fixed.
Primary Lividity	Occurs as lividity reaches maximum intensity between 6 to 12 hours after death.
Secondary Lividity	Shifting of lividity can occur if the body is moved within a few hours after death.
Duration of Persistence	Staining persists until putrefaction begins.
Distribution	Depends on position of the body:
Supine	Appears on the back of neck, entire back, flanks, except areas under pressure (contact pallor).

Prone	Appears on the front part of the body.
Side	Appears on the dependent side.
Head downwards	Appears as a blackened face and neck.
Vertical suspension	Appears on legs and hands (glove and stocking appearance) and external genitalia.
Drowning	Appears on the face, upper chest, hands, lower arms, feet, and calves.
Livor Mortis Absent	Excessive blood loss Severe anemia Death from wasting diseases Lobar pneumonia
Medicolegal Importance	Helps estimate time of death. Indicates body posture at death. Indicates if the body was moved post-death. Distribution may indicate the cause of death.
Image	

The progression of rigor mortis can be divided into three main phases.

Phase	Description	Timeframe
Primary Relaxation	ATP levels are initially 100%, allowing muscle relaxation. Sphincters relax, leading to the passage of feces and urine.	0 to 60 minutes after death
Rigor Mortis Onset	Onset occurs around 60 minutes as ATP depletes to ~85%. Muscles stiffen as actin and myosin filaments bind. Maximum rigidity is reached approximately 12 hours postmortem, with significant stiffness noticeable, especially in larger muscles.	60 minutes to 12 hours after death
Resolution	Following peak rigidity, a secondary relaxation phase begins as proteins decompose (proteolysis of actin-myosin). Rigor mortis resolves completely after about 12 hours, leading to further decomposition.	12 to 24 hours after death

Solution for Question 2:

Correct Option D - >25 years:

- The approximate age of a person in whom the medial end of the clavicle is fused and the sternum is completely fused is typically greater than 25 years . The fusion of the clavicle and sternum occurs as part of the natural aging process and is considered a sign of skeletal maturity . Therefore, individuals who exhibit complete fusion of the clavicle and sternum are typically adults over the age of 25.

Incorrect Options:

- Option B, C & D are incorrect. Refer to the explanation of the correct answer.
-

Solution for Question 3:

Correct Option D - Traumatic asphyxia:

- Cadaveric spasm, also known as instantaneous rigor or cataleptic rigidity, is a phenomenon where a group of muscles, typically those that were in a state of intense contraction immediately before death, remain contracted after death. This condition is relatively rare and is different from rigor mortis, which is the gradual stiffening of muscles after death.
- Traumatic asphyxia, also known as crush asphyxia, occurs when a heavy weight or force compresses the chest, leading to rapid death. This condition does not typically cause cadaveric spasm because the death is usually instantaneous and involves pressure-induced cessation of the heart and lungs without significant pre-mortem muscle activity.

Incorrect Options:

Option A - Suicidal gun-shot wound:

- The presence of cadaveric spasm, characterized by the instantaneous rigor of muscles after death, is notably linked to cases involving intense emotional or physical trauma. This phenomenon is often observed in scenarios such as suicidal gunshot wounds, where the individual experiences a violent and traumatic event leading to their demise.

Option B - Electrocution:

- In cases of electrocution, where there is a sudden and severe electrical shock, the body may exhibit cadaveric spasm due to the abrupt nature of the event.

Option C - Drowning:

- Drowning involves a struggle for survival, leading to increased stress and physical exertion, which may trigger cadaveric spasms.
-

Solution for Question 4:

Correct Option C - Nail:

- Limited use due to being a dead tissue.
- While nails may have limited forensic use, they are not as commonly used as teeth or blood for positive identification. Nails are composed of dead tissue, which limits their utility in comparison to other more viable tissues.

Incorrect Options:

Option A - Teeth:

- Dental records and comparison of dental features.
- Dental characteristics are unique to individuals, and dental records can be used for positive identification. Comparisons of dental charts, X-rays, or other dental features aid in confirming a person's identity.

Option B - Blood:

- DNA analysis and blood typing.
- Blood can be used for DNA analysis, which is a highly reliable method of identification. Blood typing can also provide information about the deceased person, especially in cases where other methods may not be feasible.

Option D - Muscle:

- Limited use due to complexity and degradation.
 - Muscle tissue does contain DNA, and in certain situations, it can be used for DNA analysis. However, DNA extraction from muscle is not as common or straightforward as from blood.
-

Solution for Question 5:

Correct Option D - Oximes:

- Oximes are not typically considered chelating agents in the same way that certain compounds, such as EDTA (ethylenediaminetetraacetic acid) or ligands with multiple donor atoms, are. Chelating agents are molecules that can form multiple bonds to a metal ion, creating a ring-like structure known as a chelate.

Incorrect Options:

Option A – Deferiprone: It is a chelating agent in iron poisoning.

Option B - EDTA (Ethylenediaminetetraacetic acid): EDTA is a synthetic amino acid that forms stable complexes with a variety of metal ions. It is commonly used as a chelating agent in cases of lead poisoning. The EDTA binds to the lead ions in the bloodstream, forming a complex that can be excreted by the kidneys.

Option C – Penicillamine: Penicillamine is a chelating agent used in the treatment of Wilson's disease. It is also used to reduce cystine excretion in cystinuria and to treat patients with severe, active rheumatoid arthritis unresponsive to conventional therapy. Penicillamine is used as a form of immunosuppression to treat rheumatoid arthritis.

Solution for Question 6:

Correct Option D - Smothering:

Smothering is characterized by the presence of bruises around the perioral region and nail abrasions around the lips and nose. Petechial hemorrhages are also commonly observed in smothering cases. This form of asphyxia occurs when an object or material covers the nose and mouth, obstructing the airways and preventing breathing.

Incorrect Options:

Option A - Hanging:

In hanging cases, there is typically a ligature mark around the neck, and the pattern of petechiae is usually more prominent in the eyes and face rather than the perioral region. There are often no specific bruises around the lips or nail abrasions.

Option B - Throttling:

Throttling involves a deep bruise or contusion around the neck, commonly seen as a result of manual compression. This would not typically present with bruises specifically around the perioral region or nail abrasions.

Option C - Choking:

Choking usually does not result in significant external trauma or bruising around the perioral area. Instead, it is characterized by signs of airway obstruction without specific bruises around the lips or petechial hemorrhages.

Solution for Question 7:

Correct Option C - Knee and wrist:

Knee and wrist X-rays are commonly used to assess skeletal maturity and bone age in medico-legal examinations. These joints are particularly useful for evaluating the maturity of the bones and growth plates, which helps in estimating the chronological age of an individual. The epiphyseal ossification patterns in the knee and wrist are well-documented and reliable indicators of skeletal age.

Incorrect Options:

Option A - Head & shoulder:

X-rays of the head and shoulder are less useful for determining bone age because the skeletal maturity indicators are less distinct in these regions compared to the knee and wrist.

Option B - Elbow and ankle:

While the elbow and ankle can provide some information on bone age, they are not as commonly used or as reliable as the knee and wrist for this purpose. The elbow and ankle are not as definitive in assessing the maturity of growth plates.

Option D - Elbow & hip:

The elbow and hip can be used to assess bone age, but they are not as commonly employed as the knee and wrist. The hip is less frequently used for this purpose because of variability in ossification patterns.

Solution for Question 8:

Correct Option A - Preservation by saponification:

Adipocere formation, also known as "grave wax," is a process where body fat turns into a waxy substance due to saponification. This process involves the hydrolysis of fats in the body by bacteria, leading to the formation of a soapy, wax-like substance that helps in the preservation of the body.

Incorrect Options:

Option B - High temperature needed:

Adipocere formation generally occurs in a cool, moist environment. High temperatures are not conducive to this process; rather, the process is enhanced in cooler conditions which slow down decomposition and support the saponification process.

Option C - Cool and dry climate needed:

While a cool climate can favor adipocere formation, a dry climate is not ideal. Adipocere formation requires moisture to facilitate the saponification process. Therefore, a cool and moist environment is more conducive to adipocere formation.

Option D - Occurs within minutes to hours:

Adipocere formation is a relatively slow process that can take several weeks to months to develop fully, not minutes to hours.

Solution for Question 9:

Correct Option A - Tache Noire:

- Tâche noire sclérotique is seen in the given image. It appears as a triangular area of cell debris and mucus, with color changes from yellow to brown or black after 3 to 6 hours of death.

Incorrect Options:

Option B - Kevorkian sign: Kevorkian Sign or railroading phenomenon or cattle trucking, is characterized by the segmentation or fragmentation of retinal blood vessels.

Option C - Corneal changes: Corneal Changes include haziness of the cornea occurring within 1 hour and opacity developing after 6 hours post-mortem.

Option D - Vitreous humour changes: The vitreous humor resists putrefaction and exhibits changes like linear increases in potassium (K) and hypoxanthine, as well as a decrease in glucose.

Solution for Question 10:

Correct Option C - Translucency of root:

- Teeth play a crucial role in the identification of individuals, especially in forensic odontology. Gustafson's technique, developed by Gustafson in 1950, is a widely used method for age estimation in adults based on dental characteristics.
- Among Gustafson's criteria, the most reliable criterion for age estimation is the translucency of the root (Option C).
- The technique involves attributing scores for various age-related changes observed in teeth, such as attrition, periodontal ligament retractions, secondary dentin formations, root translucency, and root resorption.
- The translucency of the root is considered one of the most reliable indicators of age, as it gradually increases with age due to the deposition of secondary dentin and changes in the root structure.

Teeth play a crucial role in the identification of individuals, especially in forensic odontology. Gustafson's technique, developed by Gustafson in 1950, is a widely used method for age estimation in adults based on dental characteristics.

Among Gustafson's criteria, the most reliable criterion for age estimation is the translucency of the root (Option C).

The technique involves attributing scores for various age-related changes observed in teeth, such as attrition, periodontal ligament retractions, secondary dentin formations, root translucency, and root resorption.

The translucency of the root is considered one of the most reliable indicators of age, as it gradually increases with age due to the deposition of secondary dentin and changes in the root

structure .

Incorrect Options:

Option A, B & D - Root resorption, Parodontosis & Attrition:

- While these criteria are also considered in Gustafson's technique, the translucency of the root is considered the most reliable indicator of age estimation among Gustafson's criteria.

While these criteria are also considered in Gustafson's technique, the translucency of the root is considered the most reliable indicator of age estimation among Gustafson's criteria.

Solution for Question 11:

Correct Option A - Drowning :

- Drowning occurs when a person's airway is submerged in water or another liquid, leading to asphyxia. The presence of frothing from the nostrils can be indicative of water entering the airway during the drowning process. This frothy material consists of a mixture of water, air and mucus .

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Incorrect Options :

Option B - Hanging: Hanging is a form of asphyxia caused by constriction of the neck due to a ligature or suspension. Frothing from the nostrils is not typically associated with hanging.

Option C - Strangulation: Strangulation involves the external compression of the neck, leading to asphyxia. Frothing from the nostrils is not commonly observed in cases of strangulation.

Option D - Traumatic asphyxia: Traumatic asphyxia is a condition that occurs when there is a sudden and severe compression of the chest, leading to a lack of oxygen supply to the body. Frothing from the nostrils is not a characteristic feature of traumatic asphyxia.

Solution for Question 12:

Correct Option C - Cadaveric spasm:

- Cadaveric spasm occurs due to ATP depletion, and is usually seen in accidental cases, in which the person performs sudden physical movements in distress just before death.
- It affects voluntary group of muscles (legs, neck, eyes) and short group of muscles(hands).
- Aka instantaneous rigor i.e. Occurs immediately after death.
- It is spasm of group of voluntary muscles which are used at the time of death.
- It is purely Antemortem Origin
- It cannot be produced artificially
- An important component of cadaveric spasm is the absence of primary relaxation .

Aka instantaneous rigor i.e. Occurs immediately after death.

It is spasm of group of voluntary muscles which are used at the time of death.

It is purely Antemortem Origin

It cannot be produced artificially

Incorrect Options:

Option A - Rigor mortis: Rigor mortis is a postmortem change resulting in the stiffening of the body muscles due to chemical changes in their myofibrils . Rigor mortis helps in estimating the time since death as well to ascertain if the body had been moved after death.

Option B - Heat stiffness: The muscle proteins undergo coagulation and as a result the body posture stiffen up. It usually occurs in temperatures above 65 °C

Option D - Cold stiffness: This occurs due to freezing of the body fluids, due to sub zero temperatures of around <-5■

Solution for Question 13:

Correct Option D - i -3, ii-4, iii-1, iv-2:

- Mummification : Mummification is the preservation of a dead body, typically achieved through the removal of moisture and the use of chemicals, natural processes, or

environmental conditions. It involves drying out the body to prevent decomposition. Therefore, it is correctly paired with option 3.

- Adipocere : Adipocere, also known as "grave wax," is a waxy substance that forms on a corpse under specific conditions, such as when the body is buried in a moist environment or submerged in water. It is a result of the hydrolysis of fats in the body. Thus, it is correctly paired with option 4.

- Marbling : Marbling, in the context of death, refers to the discoloration or mottling of the skin after death. It occurs due to the settling of blood and the pooling of blood in the lower parts of the body. Therefore, it is correctly paired with option 1.

- Lividity of death : Lividity of death, also known as livor mortis , is the process by which blood settles in the lower parts of the body after death, resulting in a purple or reddish discoloration of the skin. It occurs due to the gravitational pull on the blood. Thus, it is correctly paired with option 2.

- Therefore, the correct pair is : i - Mummification (3) ii - Adipocere (4) iii - Marbling (1) iv - Lividity of death (2)

- i - Mummification (3)

- ii - Adipocere (4)

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- i - Mummification (3)

- ii - Adipocere (4)

- iii - Marbling (1)

- iv - Lividity of death (2)

Incorrect Options:

- Option A, B and C are incorrect as they do not match the correct pairings

Solution for Question 14:

Correct Options C -Greenish discoloration of right iliac fossa:

- Greenish discoloration of the right iliac fossa, known as the Greenish Line of Hochenegg, is one of the early signs of decomposition and is observed before any other changes such as maggot formation, putrefaction, or mummification .

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Incorrect Options:

Option A - Maggot formation: Maggot formation occurs during the later stages of decomposition, after the initial discoloration of the right iliac fossa.

Option B - Putrefaction: Putrefaction, the breakdown of body tissues, occurs after the initial discoloration.

Option D - Mummification: Mummification is a specific form of postmortem preservation and does not occur as the first change in a dead body.

Solution for Question 15:

Correct Ans: C

- Paradoxical undressing is a phenomenon that can occur in cases of severe hypothermia.
- When a person is experiencing extreme cold and their core body temperature drops significantly, paradoxical undressing refers to the behaviour of the individual inexplicably

removing their clothing, often resulting in being found partially or completely naked.

- Peripheral vasoconstriction caused by cold temperatures may also contribute to discomfort, leading to clothing removal.

Incorrect option:

Option A . Heart stress: Heart stress, also known as cardiac stress or cardiac strain, refers to excessive strain or demand placed on the heart. It is not associated with paradoxical undressing.

Option B. Voyeurism: Voyeurism is a paraphilic disorder characterized by deriving sexual pleasure from observing others without their consent. It is unrelated to paradoxical undressing.

Option D. LSD overdose: LSD (Lysergic acid diethylamide) is a hallucinogenic drug, and an overdose of LSD can lead to various physiological and psychological effects. However, paradoxical undressing is not a recognized symptom or consequence of LSD overdose.

Solution for Question 16:

Correct Option D - Saturated salt solution:

- Saturated salt solution is commonly used for the preservation of viscera in toxicology autopsy. It is a hypertonic solution that helps to prevent decomposition and maintain the integrity of the tissues for further toxicological analysis. The high salt concentration creates an osmotic environment that inhibits bacterial growth and enzymatic degradation.
- Various methods of preserving viscera: Saturated salt solution Formalin fixation Freezing Alcohol preservation Formaldehyde solution Formalin-alcohol solution Frozen section Liquid nitrogen
- Saturated salt solution
- Formalin fixation
- Freezing
- Alcohol preservation
- Formaldehyde solution
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- Liquid nitrogen

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Various methods of preserving viscera:

Saturated salt solution

Formalin fixation

Freezing

Alcohol preservation

Formaldehyde solution

Formalin-alcohol solution

Frozen section

Liquid nitrogen

Incorrect Options:

Option A - Glycerin: Glycerin is not commonly used for the preservation of viscera in toxicology autopsy. While it has some preservative properties, it is not as effective as saturated salt solution for maintaining tissue integrity and preventing decomposition.

Option B - Rectified spirit: Rectified spirit, which is a type of alcohol, is not typically used for preservation in toxicology autopsy. It may have some disinfectant properties but is not suitable for long-term tissue preservation.

Option C - Formalin: Formalin is commonly used for general tissue preservation and fixation in anatomical pathology, but it is not specifically preferred for toxicology autopsy. It may affect the results of toxicological analyses and can interfere with certain tests.

Solution for Question 17:

Correct Option B - Suggillation:

- Suggillation, also known as livor mortis or lividity, refers to the discoloration of the skin that occurs after death due to the settling of blood in the dependent areas of the body.

- This happens because, after death, blood circulation ceases, and gravity causes blood to pool in the lowest parts of the body .
- This results in a purplish-red or bluish discoloration of the skin in those areas.

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This results in a purplish-red or bluish discoloration of the skin in those areas.

Incorrect Options:

Option A - Tattooing : Tattooing refers to the permanent insertion of ink or pigment into the dermis layer of the skin. It is not a post-mortem change but rather a deliberate form of body modification performed during a person's lifetime.

Option C - Putrefaction : Putrefaction is the process of decomposition of organic matter after death. It involves the breakdown of tissues by bacteria and other microorganisms, leading to the release of foul-smelling gases and the decay of body tissues. While putrefaction is a post-mortem change, it is not the correct option in this case.

Option D - Decomposition : Decomposition refers to the process by which organic substances break down into simpler forms. It includes various chemical and physical changes that occur after death, such as the breakdown of proteins, fats, and carbohydrates. Putrefaction is one of the stages of decomposition. While decomposition is a general term for post-mortem changes, it is not the specific term used to describe the discoloration of the skin seen in suggillation.

Solution for Question 18:

Correct Option A - A and C:

- Rockitansky – In situ: This technique refers to the removal of organs one by one while keeping them in their original position within the body. It allows for a more detailed examination and dissection of each organ individually. eg. in cases of HIV, Hep B etc.
- Ghon's – en block : Ghon's technique involves the removal of multiple organs as a single unit or block.

- This technique is commonly used in organ transplantation procedures to preserve the integrity of the organs and minimize ischemic time.
- Organs removed as organ blocks and subsequently dissected

Incorrect Option:

Option B - A, B, and C are correct :

- Rockitansky – In situ: Explained above.
- B is incorrect – Virchows – One by One: Technique, also known as the "one by one" technique, is a method used in autopsies. hence does not match as written above.
- Ghon's – en block: Explained above.

Option C - B and D, A are correct:

- Virchows – One by one : Technique, also known as the "one by one" technique, is a method used in autopsies. It involves the sequential removal of organs for examination and analysis, allowing for a systematic and thorough assessment of each organ individually.
- Letulle method , also known as " en masse ". In this method, the entire organ block is removed as one . hence does not match as written above.

Option D - A, B, C and D are correct:

- B and D both are incorrect

Solution for Question 19:

Correct Option A - 18-22 years:

- The fusion of the basiocciput (the base of the occipital bone) with the basisphenoid (part of the sphenoid bone) occurs ib 18-22 years

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Skull Sutures

- 1st Suture: Metopic at 9 months
- Last Suture: Squamous temporal ≥ 80 yr
- Sagittal (best) Posterior ■ 20-40 yrs Anterior ■ 40-50 yrs Middle ■ >50 yrs
- Posterior ■ 20-40 yrs

- Anterior ■ 40-50 yrs
- Middle ■ >50 yrs
- Coronal Upper half 50-60yrs Lower half 40-50 yrs
- Upper half 50-60yrs
- 50-60yrs
- Lower half 40-50 yrs
- 40-50 yrs
- Lambdoid Upper half 50-60 yrs Lower half 60-70 yrs
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- 50-60 yrs
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- Basiocciput with Basisphenoid 18-21 years
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Lower half

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Lambdoid

- Upper half 50-60 yrs
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- Lower half 60-70 yrs
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- 60-70 yrs

60-70 yrs

Basiocciput with Basisphenoid

- 18-21 years

18-21 years

Incorrect Options:

Option B - 22-25 years: By the age of 22-25 years, the fusion of the basiocciput with the basisphenoid has usually already taken place.

Option C - 14-16 years: The fusion of the basiocciput with the basisphenoid occurs later than 14-16 years of age. At this stage, the cranial bones are still undergoing development and growth.

Option D - 12-14 years: The fusion of the basiocciput with the basisphenoid occurs later than 12-14 years of age. During early adolescence, the cranial bones are still in the process of development and have not yet fused completely.

Solution for Question 20:

Correct Option C - Mandible Rounded:

- Mandible Rounded in female and Mandible is squared in male

Incorrect Options:

Option A - Larger teeth : The size of teeth is not a distinguishing feature between male and female skulls. Tooth size can vary among individuals, but it is not directly related to sexual dimorphism.

Option B - Mastoid medium-large : Mastoid medium-large in male, in females it is small.

Option D - Zygomatic is more pronounced : Zygomatic is more pronounced in male and it is less pronounced in female.

Solution for Question 21:

Correct Option D - 21 to 22 years:

- To determine the approximate age of an individual based on X-ray images of their elbow, wrist, and pelvis, a method called skeletal age assessment can be used. This method relies on the assessment of ossification centers and fusion of growth plates in these areas, which occur at predictable ages during development. The X-ray findings are consistent with the girl's age as 21-22 years.
- Findings: Elbow : In a 21-22 year old female, the epiphyseal plates in the elbow region should be closed, indicating the completion of growth and fusion. The bones of the elbow joint, including the humerus, radius, and ulna, should appear fully developed and fused. The joint space should be minimal, and the bony contours should be well-defined. Wrist : At this age, the growth plates in the wrist region should also be closed, indicating the completion of growth. The carpal bones of the wrist, including the scaphoid, lunate, triquetrum, pisiform, trapezium, trapezoid, capitate, and hamate, should be well-formed and have fused together. The bony architecture of the wrist should be well-defined. Pelvis : In a 21-22 year old female, the pelvis should have completed its growth and reached its adult form. The ilium, ischium, and pubis bones of the pelvis should be fully developed and fused. The sacroiliac joints should be stable, and the symphysis pubis should be fused with minimal or no joint space.
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Incorrect Options :

- Option A, B and C are incorrect as they are inconsistent with the above X-ray findings
- Option A, B and C are incorrect as they are inconsistent with the above X-ray findings

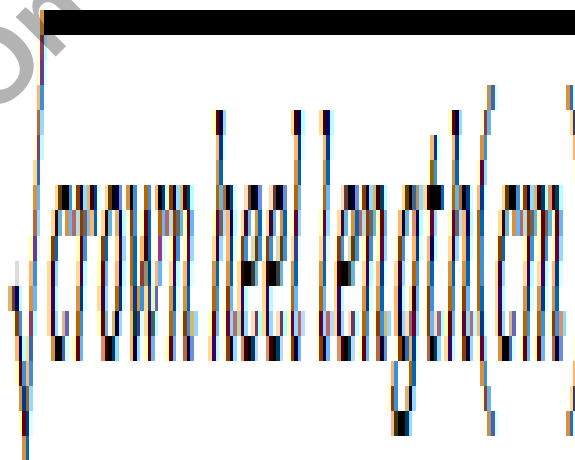
Solution for Question 22:

Correct Option B - 5 months:

The calculation of age of fetus can be done by two methods

- Rule of Hasse (for upto 5 months of age) - calculates Age(months) = crown heel length(cm)
 - Morrison Rule (for more than 5 months of age)- Age(months) = Crown heel length (cm) / 5
- Since 3 options are of 5 months or less, Rule of hasse can be used first to see if we come to an answer. Square root of 25 = 5 cm which is the answer
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Square root of 25 = 5 cm which is the answer

Incorrect Options:

Option A - 6 months: 6 months would've been the answer if crown heel length was 30 cm.

Option C - 4 months: 4 months would've been the answer if crown heel length was 16 cm

Option D - 3 months: 3 months would've been the answer if crown heel length was 9 cm.

Solution for Question 23:

Correct Option B - Pelvis:

- Single best bone for sex identification is the pelvis.
- It provides 95% accurate sex differentiation.

Single best bone for sex identification is the pelvis.

It provides 95% accurate sex differentiation.

Incorrect Options:

Option A - Skull:

- Skull bone provides only 90-92% accuracy, whereas pelvis provides 95% and thus is not a correct option

Option C - Long bones:

- Long bones provide only 80-85 % accuracy and are best used in combination with other bones.

Option D - Sternum:

- Ashley's rule is used to differentiate sex through the estimation of length of sternal bone. However, individually it does not provide accurate assessment of sex differentiation.

- Ashley's rule: Sternal bone in males is usually >149mm, whereas in females it is <149.

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Ashley's rule: Sternal bone in males is usually >149mm, whereas in females it is <149.

Solution for Question 24:

Correct Option A - Distended Gallbladder:

- Starvation may occur from the actual withholding of food or from the administration of unsuitable food.
- In the case of starvation, all organs will shrink (se in size) except gallbladder, because there is no food. Bile gets accumulated. The only organ which maintains the size is the brain .
- In the case of starvation (no food and water), the person dies in 10-12 days.
- If the person is provided only with water, the person can survive 6-8 weeks.
- Cause of Death: Death occurs from exhaustion, circulatory failure due to brown atrophy of the heart, or intercurrent infection. Dehydration and hypothermia contribute to death.
- Postmortem findings: Extreme emaciation and the general reduction in size and weight of all organs, except the brain.
- Gall bladder is distended with bile.
- Brown atrophy of the heart.
- Rigor mortis sets in and disappears early.
- Face is pale with inelastic and pigmented skin.
- Ankle edema.
- Fat is completely absent in subcutaneous tissue sparing the female breast and orbit.
- Stress fracture due to demineralization and osteomalacia.
- Wall of the intestine appears like tissue paper with atrophy of mucosa and all layers are also bile stained.

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Ankle edema.

Fat is completely absent in subcutaneous tissue sparing the female breast and orbit.

Stress fracture due to demineralization and osteomalacia.

Wall of the intestine appears like tissue paper with atrophy of mucosa and all layers are also bile stained.

Incorrect Options:

Option B - Enlarged liver: In starvation all organs shrink, so does the liver.

Option C - Expanded lungs: Only organ which distends is the gallbladder, except that all organs shrink. Hence, the lungs don't expand.

Option D - Hypertrophied heart: Only organ which maintains the size is the brain, the only organ which does not shrink is the gallbladder. Heart may shrink to 5 to 6 ounces from 11 ounces

Solution for Question 25:

Correct Option A - 1-E, 2-C, 3-B, 4-A, 5-D:

- (1) Incomplete atrophy of ridges: This characteristic is matched with dermatitis (option e) . Dermatitis refers to inflammation of the skin, which can cause changes in the fingerprint patterns . In cases of dermatitis, there may be incomplete atrophy or flattening of the ridges, leading to altered fingerprint patterns.
- Dermatitis refers to inflammation of the skin, which can cause changes in the fingerprint patterns . In cases of dermatitis, there may be incomplete atrophy or flattening of the ridges, leading to altered fingerprint patterns.
- (2) Less pattern with ridge atrophy: This characteristic is matched with celiac disease (option c) . Celiac disease is an autoimmune disorder that affects the small intestine when gluten is ingested. In individuals with celiac disease, ridge atrophy can occur, leading to a decrease in the pattern visibility or clarity of fingerprints.
- Celiac disease is an autoimmune disorder that affects the small intestine when gluten is ingested.
- In individuals with celiac disease, ridge atrophy can occur, leading to a decrease in the pattern visibility or clarity of fingerprints.
- (3) Altered ridges: This characteristic is matched with scleroderma (option b) . Scleroderma is a chronic autoimmune disease that affects the connective tissues, including the skin. In cases of scleroderma, the skin can become thickened and hardened, leading to alterations in the ridge patterns of fingerprints.
- Scleroderma is a chronic autoimmune disease that affects the connective tissues, including the skin. In cases of scleroderma, the skin can become thickened and hardened, leading to alterations in the ridge patterns of fingerprints.
- (4) Permanent loss of fingerprints: This characteristic is matched with radiation (option a). Radiation therapy, particularly in high doses, can cause damage to the skin and underlying tissues. In some cases, this can lead to permanent loss or disappearance of the fingerprint patterns.
- Radiation therapy, particularly in high doses, can cause damage to the skin and underlying tissues. In some cases, this can lead to permanent loss or disappearance of the fingerprint patterns.
- (5) Distance between ridges changed but pattern retained: This characteristic is matched with acromegaly (option d). Acromegaly is a hormonal disorder that occurs due to the excessive production of growth hormone in adulthood. In individuals with acromegaly, the bones and tissues can enlarge, including the hands and fingers. This growth can lead to changes in the distance between ridges, but the overall pattern of the fingerprints remains intact.
- This characteristic is matched with acromegaly (option d).
- Acromegaly is a hormonal disorder that occurs due to the excessive production of growth hormone in adulthood.

- In individuals with acromegaly, the bones and tissues can enlarge, including the hands and fingers. This growth can lead to changes in the distance between ridges, but the overall pattern of the fingerprints remains intact.

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In individuals with acromegaly, the bones and tissues can enlarge, including the hands and fingers. This growth can lead to changes in the distance between ridges, but the overall pattern of the fingerprints remains intact.

Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 26:

Correct Option A - Brain:

- When it comes to decomposition or putrefaction of the body after death, the brain is generally the first organ to undergo significant decomposition.

When it comes to decomposition or putrefaction of the body after death, the brain is generally the first organ to undergo significant decomposition.

Incorrect Options:

Option B - Heart: Heart may undergo changes after death, such as loss of contractility and blood circulation, it does not typically decompose as rapidly as the brain.

Option C - Kidney: Kidneys can be affected by postmortem changes, they generally do not decompose as quickly as the brain.

Option D - Prostate: Prostate may undergo changes after death, but it does not decompose as rapidly as the brain.

Solution for Question 27:

Correct Option C - Teeth:

- The best tissue for identification purposes is the teeth (Option C)
- Teeth are highly durable and resistant to heat, making them more likely to survive the burning process compared to other tissues.
- Dental records, including dental charts, X-rays, and dental impressions, can be used to compare with the teeth recovered from the burnt body to establish a positive identification.

The best tissue for identification purposes is the teeth (Option C)

Teeth are highly durable and resistant to heat, making them more likely to survive the burning process compared to other tissues.

Dental records, including dental charts, X-rays, and dental impressions, can be used to compare with the teeth recovered from the burnt body to establish a positive identification.

Incorrect Options:

Option A - Hair: While hair can provide some information for identification, it is more susceptible to damage by fire and may not be as reliable as teeth.

Option B - Bone: Bone can also provide information for identification, but it may be severely damaged or fragmented in cases of extensive burning.

Option D - Blood: Blood is not a suitable tissue for identification purposes in a burnt body, as it would likely be destroyed during the burning process.

Solution for Question 28:

Correct Option C - Fingerprint pattern:

- Fingerprint patterns are not solely determined by genetic factors but also influenced by other developmental factors during the formation of the ridges and patterns on the fingers. As a result, even identical twins will have distinct fingerprint patterns.

Incorrect Options:

Option A - Blood group: Identical twins typically have the same blood group.

Option B - DNA fingerprinting: Identical twins will have virtually identical DNA profiles as they share the same genetic information

Option D - Iris color: Identical twins often have similar iris colors as it is influenced by genetic factors. While there can be slight variations in shade or intensity, the overall color of the iris is usually similar between identical twins.

Solution for Question 29:

Correct Option B - Child is delivered after death of biological father:

- Child is delivered after the death of the biological father: This is the correct option.
- A posthumous child refers to a child who is born after the death of their biological father.

Child is delivered after the death of the biological father: This is the correct option.

A posthumous child refers to a child who is born after the death of their biological father.

Incorrect Options:

Option A - Delivery of a macerated fetus: Maceration, refers to the softening and disintegration of fetal tissues due to prolonged retention in the uterus after fetal death.

Option C - Child is born to an unmarried couple: It is unrelated to the concept.

Option D - Child is abandoned by the parents: This is not the correct description of a posthumous child.

Solution for Question 30:

Correct Option D - Putrefaction:

- Putrefaction refers to the decomposition or decay of a body after death.

- It is a natural process that occurs as a result of the breakdown of tissues by bacteria and enzymes in the body.
- Various signs and changes are observed during putrefaction, including bloating of the face and other body parts, as well as tongue protrusion.

Incorrect Options:

Option A & B - Electrocution & Crocodile burns: Electrocution refers to the injury or death caused by an electric shock. While electrocution can cause various physical signs and symptoms, such as burns and internal organ damage, it does not specifically result in bloating of the face, tongue protrusion, or putrefaction. Electrocution-related findings would typically involve specific burn patterns or injury marks associated with the electrical source. Crocodile burns are seen in high voltage electric burns.

Option C - Scalds: Scalds are burns caused by contact with hot liquids or steam. While scalds can result in various skin-related changes, such as blisters and skin peeling, they do not lead to the specific findings of bloating of the face, tongue protrusion, or putrefaction. Scalds are acute injuries and do not involve postmortem processes.

Solution for Question 31:

Correct Option C - Cadaveric spasm:

- Cadaveric spasm, or cataleptic rigidity, occurs when the muscles that are contracted during life, become stiff and rigid immediately, after death, without passing into the stage of primary relaxation.
- The spasm originates by normal nervous stimulation of the muscles and hence, is usually limited to a single group of involuntary muscles and frequently involves the hands.
- It is caused due to ATP depletion.

Incorrect Options:

Option A - Rigor Mortis: Rigor Mortis, also called death stiffening or cadaveric rigidity, is a state of stiffening of muscles, sometimes with slight shortening of the fibres. There is individual cell death that occurs when rigor mortis sets in. It affects all muscles in the body including voluntary and non-voluntary, contracted or uncontracted, unlike cadaveric spasms.

Option B - Livor Mortis: Livor Mortis, or post-mortem hypostasis, is the purplish discoloration of skin in the dependent parts of the body caused by the collection of blood in those parts due to

gravitational pull. The purplish colour is due to the deoxyhemoglobin.

Option D - Algor Mortis: Algor Mortis is the cooling of the body post death. The body does not follow Newton's law when losing temperature. The fall in temperature is because of the cessation of heat production by the somatic cells of the body.

Solution for Question 32:

Correct Option A - Thanatology:

- Thanatology deals with death in all aspects including somatic, systemic and clinical, and molecular or cellular death.

Incorrect Options:

Option B - Forensic anthropology : Forensic anthropology involves the study of human remains. It involves applying the skeletal analysis, descriptive data, body marks, and photographs of the individual together to get to forensic inferences. It only deals with one aspect of death and not all of them.

Option C - Toxicology: Toxicology is the study of analysing suspect samples for the presence of toxins. It only deals with toxins that can be a cause of death- one aspect of death and not all of them.

Option D - Molecular death: Molecular death is the death of cells and tissues individually. It occurs one to two hours after the stoppage of the vital functions. It only deals with one aspect of death and not all of them.

Solution for Question 33:

Correct Option A - Superimposition:

- The superimposition technique is a method used in forensic anthropology to assist in the analysis of an unknown skull.

- The technique involves using the ante mortem image of the victim superimposed by the image of the recovered skull.
- The principle behind the study is that the shape of the face is related to the bony contours of the facial part of the skin.

The superimposition technique is a method used in forensic anthropology to assist in the analysis of an unknown skull.

The technique involves using the ante mortem image of the victim superimposed by the image of the recovered skull.

The principle behind the study is that the shape of the face is related to the bony contours of the facial part of the skin.

Incorrect Options:

Option B - Anthropometry: Anthropometry or Bertillon system involves analysis of the descriptive data of individuals, body marks on individuals, the individual's body measurements and the photographs of particular aspects of the individuals. It does not require an ante mortem photograph of the face of the victim.

Option C - DNA fingerprinting: DNA Fingerprinting is a method of forensic analysis that uses the study of analysing DNA found on articles of the crime scene and victim. The DNA has 95% of silent genes which are very specific to different individuals. The number of repeats in a repetitive DNA can be used to match with suspect DNAs to come to a conclusion. This technique does not require an ante mortem photograph of the face of the victim.

Option D - Cheiloscopy: Cheiloscopy is a technique of forensic analysis that involves human identification based on lips. It uses a method similar to that of fingerprint analysis. It does not require an ante mortem photograph of the face of the victim and is mainly a study to detect the accused from a set of suspects.

Solution for Question 34:

Correct Option D - Letulle technique:

Letulle technique is named after Maurice Letulle which involves the removal of organs together in one mass from tongue to prostate.

- This method of autopsy is thus also called “en masse”.

- The anatomy of the organs and organ systems can be preserved in the greatest possible way due to this.
- Forensic pathologists can get a better overall picture due to the preservation of anatomical linkages.

This method of autopsy is thus also called "en masse" .

The anatomy of the organs and organ systems can be preserved in the greatest possible way due to this.

Forensic pathologists can get a better overall picture due to the preservation of anatomical linkages.

Incorrect Options:

Option A - Virchow technique : This approach is named after Rudolf Virchow. Instead of removing all organs at once, this method involves the sequential removal and examination of individual organs . It's a more time-consuming process but provides the opportunity to inspect each organ in detail, in its anatomic position. The Virchow technique does not fit the description in the question because it does not involve the "en masse" removal of organs.

Option B - Rokitansky technique : In situ dissection (inside the body itself) is done in Rokitansky technique.

Option C - Ghon's technique : Anatomical relationships are maintained by using the Ghon's approach, often known as "en bloc" . Instead of one enormous block, the thoracic, abdominal, and pelvic organs are removed using this procedure in three separate blocks. The dissection is made much more manageable by dividing the huge block into these three portions while still keeping the anatomy between the organs.

Solution for Question 35:

Correct Option D - Rokitansky procedure:

- The Rokitansky procedure is a local (or in situ) dissection that begins at the neck and proceeds downward while also removing the organ in one piece.
- The larynx is punctured first to separate the esophagus and pharynx, followed by the larynx and trachea, and finally, the chest organs are severed to expose the abdominal organs.

- As the patient was HIV positive, the use of this technique prevents the spread of the disease.

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The larynx is punctured first to separate the esophagus and pharynx, followed by the larynx and trachea, and finally, the chest organs are severed to expose the abdominal organs.

As the patient was HIV positive, the use of this technique prevents the spread of the disease.

Incorrect Options:

Option A - Ghon technique : The thoracic, cervical, and abdominal organs are removed using the "en bloc" procedure in the Ghon technique, which is comparable to the Rokitansky but does not use an in situ dissection hence is not a useful form of autopsy for an HIV-positive individual.

Option B - Letulle technique : The cervical, thoracic, abdominal, and pelvic organs are removed as a single mass using the Letulle technique, sometimes known as the en masse technique, and are then subdivided into organ blocks. The linkages between the organs and the circulatory supply can both be preserved using this procedure but it cannot prevent the spread of infections such as HIV.

Option C - Virchow technique : Instead of removing all organs at once, this method involves the sequential removal and examination of individual organs. It's a more time-consuming process but provides the opportunity to inspect each organ in detail, in its anatomic position. This method is not useful for the autopsy of an HIV-positive individual as it can cause the spread of disease.

Solution for Question 36:

Correct Option B - Septicemia:

- Septicemia, or sepsis, refers to a severe infection that spreads through the bloodstream.
- This condition causes systemic inflammation and could lead to multiple organ dysfunction or failure.
- A significant symptom of sepsis is high body temperature or fever, which aligns with the scenario described in the question. A body temperature of 39 degrees Celsius (102.2 degrees Fahrenheit) suggests an active systemic infection, making septicemia the most probable cause.

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Causes of Post-mortem calorificity:

- Causes include: Septicemia (cholera, typhoid). Sun stroke (pontine hemorrhage). Strychnine poisoning (tetanus, exercise).
- Septicemia (cholera, typhoid).
- Sun stroke (pontine hemorrhage).
- Strychnine poisoning (tetanus, exercise).
- In septicemia, there is increased bacterial activity.
- In sunstroke, heat regulation interferes.
- In strychnine poisoning, there is increased muscle activity.

Causes include:

- Septicemia (cholera, typhoid).
- Sun stroke (pontine hemorrhage).
- Strychnine poisoning (tetanus, exercise).

Septicemia (cholera, typhoid).

Sun stroke (pontine hemorrhage).

Strychnine poisoning (tetanus, exercise).

In septicemia, there is increased bacterial activity.

In sunstroke, heat regulation interferes.

In strychnine poisoning, there is increased muscle activity.

Incorrect Options:

Option A - Cyanide poisoning: Cyanide is a potent and rapidly acting poison. Symptoms of cyanide poisoning include headache, confusion, nausea, and can result in cardiac arrest. Cyanide poisoning may lead to death swiftly but it typically does not result in a sustained elevated body temperature, especially not after death.

Option C - Corrosive poisoning: Corrosive substances can cause severe damage to the tissues they come into contact with. However, these substances do not typically cause fever. In cases of

corrosive poisoning, the primary symptoms are related to the areas of contact, which could include the mouth, throat, and stomach.

Option D - Intra-abdominal hemorrhage: This condition refers to bleeding within the abdomen, which could be due to several causes such as trauma or a ruptured blood vessel. While intra-abdominal hemorrhage is a serious condition, it does not generally cause a high body temperature.

Solution for Question 37:

Correct Option A - Hesitation cuts:

- Hesitation cuts refer to superficial wounds, often found on the wrists or forearms, which are thought to be self-inflicted as part of a suicide attempt. These wounds are usually not immediately life-threatening being relatively shallow in nature. This contrasts with more decisive, deeper cuts that may pose a greater immediate danger due to more severe blood loss.
- Features of hesitational cuts: Multiple cuts Superficial, linear/ parallel cuts Seen in accessible parts of the body [Forearm, neck, thigh, abdomen etc] Suggestive of suicidal tendency
- Multiple cuts
- Superficial, linear/ parallel cuts
- Seen in accessible parts of the body [Forearm, neck, thigh, abdomen etc]
- Suggestive of suicidal tendency

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Features of hesitational cuts:

- Multiple cuts
- Superficial, linear/ parallel cuts
- Seen in accessible parts of the body [Forearm, neck, thigh, abdomen etc]
- Suggestive of suicidal tendency

Multiple cuts

Superficial, linear/ parallel cuts

Seen in accessible parts of the body [Forearm, neck, thigh, abdomen etc]

Suggestive of suicidal tendency

Incorrect Options:

Option B - Grazed abrasion: Grazed abrasions are brought on by the body being dragged across a rough surface, which causes the epidermis to scrape. The depth of grazed injuries varies according to the body's speed and the surface's roughness. The given image does not indicate a grazed abrasion.

Option C - Laceration: A laceration is a wound that results from soft bodily tissue being torn. This kind of wound frequently has uneven and jagged edges. The image is not indicative of laceration.

Option D - Patterned bruise : A patterned injury is one that has a clear pattern that may mimic the features of the object that caused it. The pattern may be brought on by the body coming into contact with a patterned surface or being struck by a weapon or other instrument. The image doesn't show a patterned bruise.

Solution for Question 38:

Correct Option A - Marbling:

- Marbling is a postmortem change that occurs in the body after death. It refers to the development of a network of dark, irregularly shaped lines or patches on the skin. These lines or patches seen in the image resemble the patterns found in marble, hence the term marbling.

Incorrect Options:

Option B - Tattooing: The given image is not of Tattooing.

Option C - Lividity: The given image is not of Lividity

Option D - Filigree burns: The given image is not of Filigree burns

Solution for Question 39:

Correct Option C - Sweating:

- In the given scenario, with a body temperature of 106 degrees Fahrenheit, the symptom that is least likely to be seen in this patient is sweating.
- Heatstroke, also known as sunstroke, is a severe form of heat-related illness characterized by elevated body temperature.
- In heatstroke, the body's thermoregulatory mechanisms fail, leading to a rapid rise in body temperature.
- Sweating is the body's natural mechanism to cool down, but in cases of severe heatstroke, the body's ability to sweat may be impaired

In the given scenario, with a body temperature of 106 degrees Fahrenheit, the symptom that is least likely to be seen in this patient is sweating.

Heatstroke, also known as sunstroke, is a severe form of heat-related illness characterized by elevated body temperature.

In heatstroke, the body's thermoregulatory mechanisms fail, leading to a rapid rise in body temperature.

Sweating is the body's natural mechanism to cool down, but in cases of severe heatstroke, the body's ability to sweat may be impaired

Incorrect Options:

Option A - Hot skin: This option is commonly seen in cases of heatstroke. When the body's thermoregulatory mechanisms fail, the body temperature rises, and the skin can become hot to the touch. This symptom is often associated with heatstroke and is not least likely to be seen in the given scenario.

Option B - Hypotension: Heatstroke can lead to dehydration and a decrease in blood volume, which can result in hypotension (low blood pressure). Hypotension is a common finding in severe heatstroke and is not least likely to be seen in the given scenario.

Option D - Disorientation: Heatstroke can cause neurological symptoms, including disorientation and altered mental status. The rise in body temperature can affect brain function and lead to confusion, disorientation, and even loss of consciousness. Disorientation is commonly observed in severe heatstroke and is not least likely to be seen in the given scenario

Solution for Question 40:

Correct Option C- Case of immersion in water for 36 hours:

- Immersion in water for an extended period can lead to specific postmortem changes.
- These changes include maceration of the skin, separation of the epidermis, and swelling due to water absorption.
- Additionally, there may be a softening and detachment of the skin, known as a "washerwoman's hands" or "glove-like" appearance.
- Other features that can be observed include skin slippage, wrinkling, and a pale appearance.

Incorrect Options:

Option A- Case of torture phalanges: Torture can cause various injuries and damage to the phalanges, such as fractures, dislocations, or soft tissue injuries. Torture-related injuries would have different characteristics and patterns than those resulting from water immersion.

Option B- Case of suspension or hanging more than 72 hours: Prolonged suspension or hanging can lead to characteristic postmortem changes, such as livor mortis (postmortem lividity), rigor mortis (stiffening of the body), and changes in the neck due to ligature marks or fractures.

Option D- Case of colliquative liquefaction due to immersion in hot water: Colliquative liquefaction refers to the process of tissue breakdown and liquefaction that occurs in certain conditions, such as prolonged immersion in warm or hot water.

Solution for Question 41:

Correct Option C - It results from bacterial activity:

- The image shows post-mortem marbling.
- Post-mortem marbling appears as discoloration of prominent superficial veins to a green-brown color.
- Clostridium welchii bacteria form Hydrogen Sulphide gas in the superficial veins, which combine with the Hb to form Sulfhemoglobin (SulfHb), which leads to these changes.

The image shows post-mortem marbling.

Post-mortem marbling appears as discoloration of prominent superficial veins to a green-brown color.

Clostridium welchii bacteria form Hydrogen Sulphide gas in the superficial veins, which combine with the Hb to form Sulfhemoglobin (SulfHb), which leads to these changes.

Incorrect Options:

Option A - Seen within first 12 hours of death: Post-mortem marbling can begin to appear after 24 hrs

Option B - It occurs due to the accumulation of deoxygenated blood: Post-mortem marbling appears due to the formation of sulfhemoglobin.

Option D - It is seen in the case of lightning: Lightning injuries do not correspond to the vascular channels or vessels and is associated with pink color changes.

Solution for Question 42:

Correct Option B - Superfecundation:

- Superfecundation refers to the fertilization of two or more ova from different acts of sexual intercourse, in the same cycle, resulting in the conception of multiple embryos with different genetic fathers. In the given scenario, the woman gave birth to twins, but DNA testing revealed that one child did not belong to the father.
- This indicates that two separate acts of sexual intercourse with different partners occurred, leading to the fertilization of two different ova by two different spermatozoa. As a result, the twins have different genetic fathers, confirming the occurrence of superfecundation.

Superfecundation refers to the fertilization of two or more ova from different acts of sexual intercourse, in the same cycle, resulting in the conception of multiple embryos with different genetic fathers. In the given scenario, the woman gave birth to twins, but DNA testing revealed that one child did not belong to the father.

This indicates that two separate acts of sexual intercourse with different partners occurred, leading to the fertilization of two different ova by two different spermatozoa. As a result, the twins have different genetic fathers, confirming the occurrence of superfecundation.

Incorrect Options:

Option A - Superfetation: Superfetation refers to the occurrence of a second pregnancy during an ongoing pregnancy, where a woman conceives another fetus while already pregnant. This is not applicable in the given scenario as there is no indication of a second pregnancy occurring during an ongoing pregnancy.

Option C - Suppositious child: A suppositious child refers to a child who is falsely or fraudulently claimed or substituted as one's own. In the given scenario, there is no indication of false claims or substitution of the child, as the DNA testing revealed that one child did not belong to the father. Therefore, suppositious child is not the appropriate term for this case.

Option D - Atavism: Atavism refers to the reappearance of a characteristic or trait in an organism that is not present in its immediate ancestors but was present in previous generations. It is not applicable in the given scenario, as the issue at hand is related to the genetic paternity of the twins, rather than the reappearance of ancestral traits.

Solution for Question 43:

Correct Option D - 13 years:

- The X-ray image of the hand shows that all carpal bones are visible and the lower epiphysis of the first metacarpal has not yet united with shaft .
- These findings indicate that the child's skeletal development is consistent with an age of approximately 13 years.

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These findings indicate that the child's skeletal development is consistent with an age of approximately 13 years.

Incorrect Option:

Option A - 4 years: By the age of 4 years, the carpal bones in the hand would not be fully developed, and the lower epiphysis of the first metacarpal would not be fused.

Option B - 7 years: By the age of 7 years, most of the carpal bones would be visible on X-ray, but the lower epiphysis of the first metacarpal would usually not be fused.

Option C - 10 years: By the age of 10 years, the carpal bones would be well-developed and most of them would be fused. The lower epiphysis of the first metacarpal would not be fused by this

age.

Solution for Question 44:

Correct Option B - Suggillation:

- Suggillation , also known as livor mortis or postmortem lividity , refers to the discoloration of the skin that occurs after death. It happens as a result of the settling of blood in the dependent parts of the body, due to the gravitational pull . This pooling of blood gives a reddish or purplish discoloration to the skin, which can be observed in the areas closest to the ground or pressure points as seen in the given image.

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Incorrect Options:

Option A - Tattooing: Tattooing is not a postmortem change but rather a deliberate process of injecting ink into the skin during a person's lifetime to create permanent designs or patterns.

Option C - Putrefaction: Putrefaction refers to the process of decomposition of body tissues after death. It involves the breakdown of proteins by bacteria and enzymes, leading to the release of foul-smelling gases, bloating of the body, and eventual disintegration. While putrefaction is a postmortem change, it is not the correct answer in this context.

Option D - Decomposition: Decomposition refers to the process by which organic matter breaks down into simpler substances after death. It involves various physical, chemical, and biological changes, including putrefaction, autolysis (self-digestion of cells), and skeletonization. While decomposition is a general term for the overall process of breakdown, it is not the specific answer related to the given context.

Solution for Question 45:

Correct Option A - Cadaveric spasm:

- A condition known as cadaveric spasm occurs when a group of muscles that were actively utilised soon before death suddenly become stiff and inflexible.
- The majority of cases of this "instantaneous rigour" affect the hands, while extremely occasionally cadaveric spasms might affect the entire body.
- It can happen in attacks that end in a struggle, in suicides, in drowning instances , etc. In such an instance, the victim's hand appears stiff and clinched, gripping/grasping the attacker's clothing, buttons, hair, etc., possibly holding the suicide weapon, or holding weeds, gravel, mud, etc. from the water bed.
- It is a specific antemortem finding and is useful for commenting on the method of death , whereas rigour mortis provides information regarding the amount of time since death.

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It is a specific antemortem finding and is useful for commenting on the method of death , whereas rigour mortis provides information regarding the amount of time since death.

Incorrect Options:

Option B - Cutis anserina: Commonly referred to as goosebumps, this short local change in the skin occurs when tiny muscles contract in response to cold, fright, or excitement. This skin alteration is the result of a series of events that begin with a stimulus like cold or fright.

Option C - Rigor mortis: Rigor mortis is the stiffening or rigidity of the body after death. It happens when adenosine triphosphate (ATP) levels fall below a certain point. Due to the muscle cells' lack of integrity, calcium ions flood muscle fibres after death. These ions cause the actin and myosin filaments to bind, which results in contraction. The ATP-driven pumping of calcium ions back into the muscle cells' sarcoplasmic reticulum causes the muscles to relax. The absence of ATP prevents the muscles from relaxing and keeps the actinomyosin complex formed during the contraction, which makes the muscles rigid and hard.

Option D - Washerwoman's hands: Skin maceration, which starts happening in warm water within minutes and takes longer to become noticeable in cold water immersion, is the main indicator of immersion. The skin starts to seem saggy, pallid, and wrinkled, similar to a "

washerwoman's skin."

Solution for Question 46:

Correct Option A - Odorless:

- The change seen in the image is the mummification of body.
- The bodies, if protected, can be preserved for years, and they are practically odorless.
- Marked dehydration before death favours mummification.
- It takes 3 months to 1 year for the entire body to be mummified.
- The absence of moisture and the continuous action of dry warm air are necessary for this process.

The change seen in the image is the mummification of body.

The bodies, if protected, can be preserved for years, and they are practically odorless.

Marked dehydration before death favours mummification.

It takes 3 months to 1 year for the entire body to be mummified.

The absence of moisture and the continuous action of dry warm air are necessary for this process.

Incorrect Options:

Option B - Pungent: Decomposition is the main reason for the pungent strong odour that surrounds a dead body. The tissues of a decomposing body are broken down by bacteria and enzymes, generating numerous gases and chemicals that add to the odour.

Option C - Putrid: The advanced phases of decomposition are what cause the putrid odour that is associated with a dead body. The body's tissues disintegrate and produce a variety of gases and chemicals throughout this process, creating a vile and repulsive odour.

Option D - Offensive: The decomposition process is what gives a dead corpse its offensive odour. As the corpse decomposes, bacteria and enzymes disintegrate the tissues, generating gases and substances that give off a very foul odour.

Solution for Question 47:

Correct Option B - 1-A, 2-D, 3-B, 4-C:

- (1) Medial end of clavicle: The correct match is option A - 21-25 years .
- The medial end of the clavicle refers to the innermost portion of the collarbone. The fusion of the medial end of the clavicle with the shaft of the clavicle typically occurs around the ages of 21-25 years. This fusion marks the completion of growth and development of the clavicle.
- (2) Sacrum as a single bone: The correct match is option D - 20-25 years .
- The sacrum is a large triangular bone at the base of the spine, composed of fused vertebrae. The fusion of the sacrum, where individual segments of the sacral vertebrae become a single bone, generally occurs between the ages of 20-25 years. This fusion contributes to the stability of the pelvic region.
- (3) Crista scapulae: The correct match is option B - 50 years .
- The crista scapulae refers to a bony ridge present on the scapula, also known as the shoulder blade. The fusion of the crista scapulae typically occurs around the age of 50 years. This fusion is a normal age-related change and does not have any significant functional implications.
- (4) Lambdoid suture: The correct match is option C - 50 - 70 years.
- The lambdoid suture is a fibrous joint that connects the parietal bones and the occipital bone in the skull. The upper part of lambdoid suture fuses at 50-60 years of age and the lower part fuses at 60-70 years of age.

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(3) Crista scapulae: The correct match is option B - 50 years .

The crista scapulae refers to a bony ridge present on the scapula, also known as the shoulder blade. The fusion of the crista scapulae typically occurs around the age of 50 years. This fusion is a normal age-related change and does not have any significant functional implications.

(4) Lambdoid suture: The correct match is option C - 50 - 70 years.

The lambdoid suture is a fibrous joint that connects the parietal bones and the occipital bone in the skull. The upper part of lambdoid suture fuses at 50-60 years of age and the lower part fuses at 60-70 years of age.

Incorrect Options:

- Option A, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 48:

Correct Option - C: Only 5

Large acetabulum

- The acetabulum is the socket in the pelvis that forms the hip joint. In males, the acetabulum tends to be larger and deeper compared to females.
- A large acetabulum is an indicator of a male pelvis.

The acetabulum is the socket in the pelvis that forms the hip joint. In males, the acetabulum tends to be larger and deeper compared to females.

A large acetabulum is an indicator of a male pelvis.

Incorrect Options:

Options A, B and D:

- (1) Triangular obturator foramen: In males, the obturator foramen is typically more oval or rounded in shape, while in females, it tends to be more triangular.
- (2) U-shaped pubic angle : In males, the pubic angle tends to be more narrow and V-shaped, while in females, it is wider and U-shaped. However, a U-shaped pubic angle is not exclusive to males and can also be found in some females.
- (3) Everted ischial tuberosity: In males, the ischial tuberosity tends to be more projected or pronounced compared to females. However, the presence of an everted (pronounced) ischial tuberosity is not a definitive indicator of a male pelvis.
- (4) Large greater sciatic notch: In males, the greater sciatic notch is generally narrower compared to females, where it tends to be wider. A large greater sciatic notch is more commonly seen in female pelvises, so it does not indicate a male pelvis.

- (6) Deep preauricular sulcus: This characteristic is not directly related to determining the sex of the pelvis and is not exclusive to either males or females.

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(6) Deep preauricular sulcus: This characteristic is not directly related to determining the sex of the pelvis and is not exclusive to either males or females.

Solution for Question 49:

Correct Option A - A-4, B-1, C-2, D-3:

- (A) Adipocere formation: The correct match is option (4): 8-10 days.
- Adipocere formation, also known as grave wax, is a postmortem change that occurs when body fat undergoes hydrolysis and saponification. This process is influenced by environmental factors, such as moisture and temperature. Adipocere formation typically takes place in the later stages of decomposition, usually around 8-10 days after death.
- (B) Relaxation of urethral sphincter: The correct match is option (1): 24 hours.
- After death, the muscles of the body start to relax due to the absence of vital functions. The relaxation of the urethral sphincter, which controls the release of urine, is one of the early postmortem changes. It occurs relatively soon after death, typically within 24 hours.
- (C) Marbling of veins: The correct match is option (2): 2-3 days.
- The marbling of veins, also known as hypostasis or livor mortis, is the pooling and settling of blood in the dependent parts of the body after death. This phenomenon occurs due to gravity and the cessation of blood circulation. The marbling of veins usually becomes visible

around 2-3 days after death.

- (D) Liquefaction of tissues: The correct match is option (3) 4-5 days.

- Liquefaction of tissues, also referred to as autolysis, is a postmortem change that involves the breakdown of tissues by the action of enzymes released from cells. This process leads to the softening and liquefaction of body tissues. The liquefaction of tissues generally occurs around 4-5 days after death.

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(B) Relaxation of urethral sphincter: The correct match is option (1): 24 hours.

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Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 50:

Correct Option C - 3 and 5:

- Statement (3) Less pronounced glabella: The glabella refers to the smooth area between the eyebrows and above the nasal bridge. In general, females tend to have a smoother and less pronounced glabella compared to males. A smoother glabella can be an indication of a female skull.
- Statement (5) Small mastoid: The mastoid process is a bony prominence located behind the ear. In general, males tend to have a more robust and prominent mastoid process compared to females. A narrower mastoid can be indicative of a female skull.

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Incorrect Options:

Option A, B & D:

- Statement (2) Heavy cheekbones: This characteristic is not included in the correct answer. Heavy cheekbones, also known as prominent zygomatic arches, can occur in both males and females. It is not a reliable indicator of gender as the size and prominence of the cheekbones can vary among individuals regardless of their sex.
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- Statement (4) Square orbits: This characteristic is not included in the correct answer. The shape of the eye sockets or orbits does not reliably indicate the gender of the skull. The shape of the orbits can vary among individuals and is not specifically associated with either males or females.
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- The shape of the eye sockets or orbits does not reliably indicate the gender of the skull. The shape of the orbits can vary among individuals and is not specifically associated with either males or females.
- Statement (1) Large frontal and parietal eminence: In general, females tend to have less prominent frontal and parietal eminences compared to males. Eminences refer to the prominent areas or bulges on the skull. Therefore, a large frontal and parietal eminence can be an indication of a male skull rather than a female skull.

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Solution for Question 51:

Correct Option B - 1 and 4:

- (1) Hypernatremia: This characteristic is included in the correct answer (1 and 4).
- Seawater contains a higher concentration of salt (sodium chloride) compared to the body's internal fluids . When a person drowns in seawater, they inhale and ingest seawater, leading to an increase in sodium levels in the body. Hypernatremia, which refers to an elevated sodium concentration, can be observed in cases of seawater drowning.
- (4) Myocardial anoxia : This characteristic is included in the correct answer (1 and 4).
- Myocardial anoxia, which refers to a lack of oxygen supply to the heart muscle, can be observed in cases of seawater drowning . When a person drowns in seawater, the inhaled water can enter the lungs and impair the exchange of oxygen and carbon dioxide. This can lead to hypoxia (lack of oxygen) and subsequent myocardial anoxia.

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Incorrect Options:

Options A, C & D :

- (2) Hyponatremia : This characteristic is not included in the correct answer.
- Hyponatremia is a condition characterized by low levels of sodium in the blood. In cases of seawater drowning, hyponatremia is unlikely to occur because seawater has a higher salt concentration than the body's fluids. Hyponatremia is more commonly associated with freshwater drowning, where water with a lower salt concentration is ingested or aspirated.
- (3) Hyperkalemia : This characteristic is not included in the correct answer.
- Hyperkalemia refers to an elevated potassium level in the blood.
- While hyperkalemia can occur in certain drowning scenarios, such as freshwater drowning, it is not typically associated with seawater drowning. Seawater contains relatively low levels of potassium, so it is less likely to cause hyperkalemia.
- (5) Hemodilution: This characteristic is not included in the correct answer.\
- Hemodilution refers to a decrease in the concentration of red blood cells or other blood components due to an increase in the fluid volume. While seawater drowning can cause fluid

shifts and changes in blood composition, hemodilution is not a specific finding associated with seawater drowning.

(2) Hyponatremia : This characteristic is not included in the correct answer.

Hyponatremia is a condition characterized by low levels of sodium in the blood. In cases of seawater drowning, hyponatremia is unlikely to occur because seawater has a higher salt concentration than the body's fluids. Hyponatremia is more commonly associated with freshwater drowning, where water with a lower salt concentration is ingested or aspirated.

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Hyperkalemia refers to an elevated potassium level in the blood.

While hyperkalemia can occur in certain drowning scenarios, such as freshwater drowning, it is not typically associated with seawater drowning. Seawater contains relatively low levels of potassium, so it is less likely to cause hyperkalemia.

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Hemodilution refers to a decrease in the concentration of red blood cells or other blood components due to an increase in the fluid volume. While seawater drowning can cause fluid shifts and changes in blood composition, hemodilution is not a specific finding associated with seawater drowning.

Solution for Question 52:

Correct Option D - Arch:

- An arch fingerprint pattern is the simplest and least common type.
- It appears as a continuous wave-like pattern without any deltas or cores. In an arch pattern, the ridges enter from one side of the fingerprint and exit from the opposite side, forming a slight upward or downward curve.

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It appears as a continuous wave-like pattern without any deltas or cores. In an arch pattern, the ridges enter from one side of the fingerprint and exit from the opposite side, forming a slight upward or downward curve.

Incorrect Options:

Option A - Whorl : A whorl fingerprint pattern consists of one or more concentric circles or spiral-like patterns. It appears as a circular or oval shape with ridges that form a spiral or circular pattern. Whorls can be subdivided into several categories, such as plain whorl, central pocket loop whorl, double loop whorl, and accidental whorl.

Option B - Composite : A composite fingerprint pattern is a combination of two or more different fingerprint patterns. It typically consists of a combination of loops, whorls, or arches in the same print. For example, a composite pattern may have a loop on one side and a whorl on the other.

Option C - Loop : A loop fingerprint pattern is characterized by ridges that enter from one side of the fingerprint, form a curve or loop, and exit from the same side. Loops can be further categorized as radial loops or ulnar loops based on the direction the loop opens towards (towards the thumb or little finger, respectively). The majority of fingerprint patterns fall into the loop category.

Solution for Question 53:

Correct Option C - Myocardium:

- Rigor mortis refers to the stiffening of the body's muscles after death due to biochemical changes in muscle tissue.
- It typically begins within a few hours after death and progresses gradually over the next 6 to 12 hours.
- Rigor mortis first appears in the heart muscle within an hour after death.

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Rigor mortis first appears in the heart muscle within an hour after death.

Incorrect Options:

Option A, B & D - Eyelids, Intestines & Neck:

- Among the voluntary muscles, rigor mortis usually develops sequentially and follows a descending pattern, the so-called Nysten's law: It first appears in the muscles of the eye lids (orbicularis oculi) (3–5 h), then in jaw, facial muscles (4–5 h), neck, thorax (5–7 h) upper limb

(from shoulder to the hand) (7–9 h), abdomen, lower limb (from the hip to the foot) (9–11 h), and lastly in the small muscles of fingers and toes (11–12 h).

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Solution for Question 54:

Correct Option D - Ankle:

- For estimation of age by X-ray - At 16 to 17 years - Ankle joint

Incorrect Options:

Option A - Knee: At 18 to 19 years - knee, shoulder and wrist joint

Option B - Elbow : At age of 16 years elbow joint

Option C - Hip : At age of 17 to 18 years

Solution for Question 55:

Correct Option A: Odorless

- Mummification is a process characterized by the rapid dehydration and desiccation of a dead body, resulting in the preservation of its natural appearances and features through evaporation of water. Here are the key points about mummification: Definition: Rapid dehydration and desiccation of the body, preserving its natural appearance. Modification of putrefaction: Mummification is a form of dry decomposition, distinct from the wet decomposition seen in putrefaction. Physical changes: The entire body loses weight, becomes thin, stiff, and brittle. It also becomes odorless. Prevention of decomposition: The process of normal decomposition, which involves the growth of microorganisms, is prevented or significantly retarded in mummification. This is due to the rapid dehydration and lack of moisture, which inhibits microbial growth.
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Prevention of decomposition: The process of normal decomposition, which involves the growth of microorganisms, is prevented or significantly retarded in mummification. This is due to the rapid dehydration and lack of moisture, which inhibits microbial growth.

Incorrect Options:

Option B - Pungent: This option is not correct. A pungent odor refers to a strong, sharp, and often unpleasant smell. It is not typically associated with a mummified body, as the process of mummification involves the preservation of the body and the prevention of decomposition.

Option C - Putrid: This option is not correct. The term "putrid" refers to a foul or rotten smell associated with decomposition. A mummified body does not have a putrid odor because the mummification process hinders decomposition.

Option D - Offensive: This option is not correct. An offensive odor refers to a strong, unpleasant, and offensive smell. Similar to the previous options, a mummified body does not have an offensive odor due to the preservation process that prevents decomposition.

Solution for Question 56:

Correct Option D - A, C & D:

In freshwater drowning, the following are seen:

- (A) Haemodilution - Freshwater ingestion during drowning can cause hemodilution, leading to a decrease in the concentration of blood components.
- (C) Hyponatremia : Freshwater ingestion can result in dilution of sodium levels in the blood, leading to hyponatremia.
- (D) Arrhythmia : Imbalances in electrolytes, such as hyponatremia and other factors related to drowning, can contribute to cardiac arrhythmias.
- (B) Hypokalemia is not typically associated with freshwater drowning. Hypokalemia is more commonly seen in saltwater drowning due to the loss of potassium-rich fluids from the body.

(A) Haemodilution - Freshwater ingestion during drowning can cause hemodilution, leading to a decrease in the concentration of blood components.

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(B) Hypokalemia is not typically associated with freshwater drowning. Hypokalemia is more commonly seen in saltwater drowning due to the loss of potassium-rich fluids from the body.

Incorrect Options:

- Option A, B and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 57:

Correct Option A - Hyperthermia:

- Hyperthermia refers to elevated body temperature. In chronic starvation, the body's metabolic rate decreases, and thermoregulation mechanisms work to conserve heat rather than generate excess heat. Therefore, hyperthermia is not a characteristic change seen in chronic starvation.

Incorrect Options:

Option B - Exhaustion: Chronic starvation leads to the depletion of energy reserves in the body, resulting in fatigue, weakness, and exhaustion.

Option C - Heart atrophy: In chronic starvation, the body conserves energy by reducing its metabolic rate. This can result in muscle wasting, including the heart muscle, leading to heart atrophy.

Option D - Hypotension: Hypotension refers to low blood pressure. In chronic starvation, the body tries to conserve fluids and maintain blood volume, which can result in decreased blood pressure.

Forensic Psychiatry

1. Which of the following best describes the core principle of the M'Naughten Rule in determining legal insanity?

- A. The accused was suffering from a delusion at the time of the offence.
- B. The accused could not control their actions due to a mental disorder.
- C. The accused could not understand the nature or wrongness of their actions due to a mental defect.
- D. The accused had a history of psychiatric illness before the offence.

2. Which of the following is NOT a rule for criminally insane individuals?

- A. Curren's Rule
- B. Durham's Rule
- C. American Law Institute Test
- D. Morrison's Rule

3. Identify the FALSE statement on civil responsibilities of the insane from the following options:

- A. A contract is void if one party lacks mental capacity at the time of the agreement.
- B. Mentally ill individuals can serve as guardians for their own property.
- C. The Hindu Marriage Act permits annulment if one party is mentally unfit at the time of the ceremony.
- D. A holographic will must be entirely in the testator's handwriting and meet specific validity criteria.

4. Which of the following best differentiates hallucination from illusion in perception disorders?

- A. Hallucination occurs with external stimuli, while illusion occurs without any stimuli.
 - B. Hallucination is a false perception without an external stimulus, while illusion is a misinterpretation of a real external stimulus.
 - C. Hallucination and illusion are both voluntary perceptions of external objects.
 - D. Hallucination is always visual, while illusion is always auditory.
-

5. Match the following types of delusions with their appropriate descriptions. 1. Othello Syndrome a) Belief that one is dead or putrefying. 2. Cotard Syndrome b) Belief that a spouse is unfaithful. 3. de Clérambault's Syndrome c) Belief of being infested with parasites. 4. Ekbom Syndrome d) Belief that a higher-status person loves the individual.

1. Othello Syndrome	a) Belief that one is dead or putrefying.
2. Cotard Syndrome	b) Belief that a spouse is unfaithful.
3. de Clérambault's Syndrome	c) Belief of being infested with parasites.
4. Ekbom Syndrome	d) Belief that a higher-status person loves the individual.

- A. 1-a, 2-c, 3-b, 4-a
 - B. 1-d, 2-c, 3-b, 4-a
 - C. 1-c, 2-a, 3-d, 4-b
 - D. 1-b, 2-a, 3-d, 4-c
-

6. A 45 y/o man is brought to the psychiatric clinic by his wife. He claims that his wife has been replaced by an impostor who looks exactly like her but is not his real wife. He reports feeling anxious and confused about why others cannot see the difference. His medical history includes a previous diagnosis of paranoid schizophrenia. What is the most likely diagnosis?

- A. Delusion of doubles
- B. Capgras syndrome
- C. Fregoli's phenomenon
- D. Schizoaffective disorder

7. A 25 y/o woman presents to the dermatologist with intense itching and anxiety, convinced that tiny insects are crawling on her skin. She frequently inspects her body and applies various lotions to prevent what she believes are infestations, although no evidence of insects or bites is found on examination. She is aware that her fear may be exaggerated, but she cannot stop feeling anxious about it. What is the most likely diagnosis?

- A. Delusional parasitosis
- B. Acarophobia
- C. Acrophobia
- D. Agoraphobia

8. Which of the following statements about the Mental Healthcare Act (MHCA) 2017 is NOT true?

- A. The Act aims to align mental healthcare services with the United Nations Convention on the Rights of Persons with Disabilities.
- B. It allows for both voluntary and involuntary admissions, with specific protocols for each.
- C. The MHCA 2017 came into effect on July 7, 2020, replacing the previous Mental Health Act of 1987.

D. It introduces advance directives that enable individuals to express their treatment preferences in advance.

9. A 34-year-old man is brought to the psychiatric unit after assaulting a bystander without provocation. He reports months of auditory hallucinations with threatening voices, disorganized thoughts, and paranoid delusions. He is distressed, has poor hygiene, has a flat affect, and shows no signs of manipulation or substance abuse. What is the most likely explanation for his symptoms?

- A. True insanity
 - B. Feigned insanity
 - C. Delusional disorder
 - D. Personality disorder
-

10. A 25 y/o man is brought to the ED after being found wandering outside his home late at night, appearing confused and disoriented. He has no recollection of the events that occurred while he was outside. This behaviour is consistent with which of the following?

- A. Delirium
 - B. Somnambulism
 - C. Acute Psychosis
 - D. Hypnosis
-

11. Which of the following statements is true about Psychosis?

- A. In psychosis, contact with reality is often preserved.
- B. Individuals with psychosis typically exhibit empathy

- C. Insight is always present
 - D. An organic causative factor is present.
-

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Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	2
Question 4	2
Question 5	4
Question 6	2
Question 7	2
Question 8	3
Question 9	1
Question 10	2
Question 11	4

Solution for Question 1:

Correct Answer: C) The accused could not understand the nature or wrongness of their actions due to a mental defect.

Explanation:

The M'Naughten Rule specifies that to establish legal insanity, it must be proven that the accused had a mental defect preventing them from understanding either the nature of their act or that it was wrong at the time of the offence.

Insanity or unsoundness of mind refers to a mental disorder that impacts an individual's personality, mental state, critical thinking abilities, emotional processes, and interactions with others in society.

Lucid Interval:

- Definition : A period during insanity when symptoms completely disappear.
- Judgment : Individual can soundly judge their actions and are legally liable.
- Variability : Duration varies between individuals and over time for the same individual.
- Legal Implications : Difficulty in determining mental state at the time of an offence complicates full responsibility.
- Common in : Conditions like mania and melancholia.
- During a lucid interval, an individual is capable of sound judgment and understanding their actions, making them legally responsible for their conduct. If a crime is committed during this period, according to the McNaghten Rule, the individual could be held accountable for their actions.

Sec. 84 IPC deals with the criminal responsibility of insane persons.

It states that: "Nothing is an offence which is done by a person who, at the time of doing it, because of unsoundness of mind, is incapable of knowing the nature of the act, or what he is doing is either wrong or contrary to the law".

BNS Section 22: Nothing is an offence done by a person who, at the time of doing it, because of unsoundness of mind, is incapable of knowing the nature of the act, or that he is doing what is either wrong or contrary to law.

McNaughten Rule:

- The key principle states that every person is presumed sane until proven otherwise.
- To establish a defence of insanity, it must be proven that the accused, at the time of the act, had a mental disease that prevented them from understanding the nature of the act or recognising it as wrong.
- Legal Test of Insanity (Right or Wrong Test): A person is not criminally responsible if they did not know the nature of their act or that it was wrong at the time of the crime.

Requirements include:

- Evidence of a mental disease.
- The mental disease must be present at the time of the crime.
- The condition must be severe enough to prevent understanding that the act is wrong or illegal.

The accused was suffering from a delusion at the time of the offence. (Option A): This may support an insanity defence but doesn't fully capture the McNaughten Rule's focus on

understanding.

The accused could not control their actions due to a mental disorder. (Option B): This suggests a lack of control, which is not the core principle of the M'Naughten Rule, which emphasises understanding over control.

The accused had a history of psychiatric illness before the offence (Option D): While relevant, prior psychiatric history alone doesn't establish insanity; the focus is on the mental state at the time of the offence.

Solution for Question 2:

Correct Answer: D) Morrison's Rule

Explanation:

Morrison's Rule is not a recognised rule related to the criminal responsibility of insane individuals.

Morrison's Rule states that after five months of fetal development, the crown-heel length in centimetres can be divided by five to estimate the fetal age in months. This provides a simple method for assessing gestational age based on fetal size.

Criminal responsibility of insane persons rules:

- Durham's Rule (1954) : An accused is not criminally responsible if their unlawful act results from a mental disease or defect. (Option B ruled out)
- Curren's Rule (1964) : An accused is not criminally responsible if, due to a mental disease or defect, they were unable to regulate their conduct according to the law at the time of the act. (Option A ruled out)
- American Law Institute Test (1970) : A person is not responsible for criminal actions if, due to mental disease or defect, they lack adequate capacity to understand the wrongfulness of their actions or to conform to legal requirements. (Option C ruled out)
- Brawner Rule (1972) : Insanity determinations should be made by a jury, which has the discretion to decide the "insanity question."
- The Irresistible Impulse : This rule posits that an individual may recognize an act as illegal but, due to mental impairment, cannot control their actions. For example, Lorena Bobbitt was acquitted in 1994 by arguing that an irresistible impulse led her to commit her crime.

Solution for Question 3:

Correct Answer: B) Mentally ill individuals can serve as guardians for their own property.

Explanation:

Courts generally appoint guardians to manage the property of mentally ill individuals, as those individuals are not deemed competent to act as guardians for their own interests.

- Children under 12 and individuals deemed insane cannot validly consent to harm from actions taken in good faith for their benefit; parental or guardian consent is required (Sec. 89 IPC). In emergencies, loco parentis allows responsible adults, like teachers or school principals, to consent to treatment for a child when parents are unavailable.

Civil responsibility involves an individual's legal liability for actions that violate the law, with the presumption of mental competence unless proven otherwise.

- The Hindu Marriage Act allows annulment if one party is mentally unfit at the ceremony. (Option C ruled out)
- Under the Muslim Marriages Act, women can divorce for a husband's insanity within two years, while men can divorce freely.
- Holographic Will: This is a will written entirely in the testator's handwriting. It must still meet the specific validity criteria, including being signed in front of witnesses. (Option D ruled out)
- Wills may be valid if made by deaf, dumb, or blind persons, during lucid intervals, or even if made shortly before the testator's suicide. They are invalid if made by those deemed imbecilic, intoxicated, or under delusion.

Solution for Question 4:

Correct Answer: B) Hallucination is a false perception without an external stimulus, while illusion is a misinterpretation of a real external stimulus.

Explanation:

Hallucinations involve perceptions without any actual external stimuli, while illusions are distortions or misinterpretations of real stimuli. For example, seeing objects that do not exist (hallucination) versus mistaking a rope for a snake (illusion).

Hallucination:

- Hallucination is a false perception experienced by the senses without any external stimulus. It can occur in conditions like insanity, high fever, drug intoxication, and withdrawal. (Option A ruled out)
- Types of Hallucinations: (Option D ruled out) Visual : Seeing non-existent sights, common in disorders like schizophrenia and conversion disorder. Auditory (most common type) : Hearing sounds or voices without a source, prevalent in schizophrenia. Olfactory : False perception of smells, often associated with schizophrenia and temporal lobe epilepsy. Gustatory : Experiencing tastes without food, noted in certain medical conditions. Tactile : Feeling sensations like crawling insects on the skin, seen in cocaine use and schizophrenia. Psychomotor : Feeling movement in body parts without actual movement. Lilliputian : Perception of objects being smaller than they are, more like an illusion.
- Visual : Seeing non-existent sights, common in disorders like schizophrenia and conversion disorder.
- Auditory (most common type) : Hearing sounds or voices without a source, prevalent in schizophrenia.
- Olfactory : False perception of smells, often associated with schizophrenia and temporal lobe epilepsy.
- Gustatory : Experiencing tastes without food, noted in certain medical conditions.
- Tactile : Feeling sensations like crawling insects on the skin, seen in cocaine use and schizophrenia.
- Psychomotor : Feeling movement in body parts without actual movement.
- Lilliputian : Perception of objects being smaller than they are, more like an illusion.
- Auditory hallucinations are the most common , followed by visual ones. Hallucinations are involuntary and can lead to dangerous behaviours. Some auditory hallucinations occurring while falling asleep (hypnagogic) or waking up (hypnopompic) are considered normal experiences.
- Visual : Seeing non-existent sights, common in disorders like schizophrenia and conversion disorder.
- Auditory (most common type) : Hearing sounds or voices without a source, prevalent in schizophrenia.
- Olfactory : False perception of smells, often associated with schizophrenia and temporal lobe epilepsy.
- Gustatory : Experiencing tastes without food, noted in certain medical conditions.

- Tactile : Feeling sensations like crawling insects on the skin, seen in cocaine use and schizophrenia.
- Psychomotor : Feeling movement in body parts without actual movement.
- Lilliputian : Perception of objects being smaller than they are, more like an illusion.

Illusion:

- Illusion is a false interpretation of an external object or stimulus that has real existence. (Option A ruled out) It can be classified as: Universal Illusions: Common to all individuals and unchanging, such as the appearance of converging railway tracks. Personal Illusions: Vary between individuals, such as misidentifying a doctor as a family member or mistaking a dog for a lion.
- Universal Illusions: Common to all individuals and unchanging, such as the appearance of converging railway tracks.
- Personal Illusions: Vary between individuals, such as misidentifying a doctor as a family member or mistaking a dog for a lion.
- A sane person can correct these false impressions, while an insane person persists in believing them despite clear evidence.
- Illusions are often associated with psychoses, particularly of the organic type.
- Universal Illusions: Common to all individuals and unchanging, such as the appearance of converging railway tracks.
- Personal Illusions: Vary between individuals, such as misidentifying a doctor as a family member or mistaking a dog for a lion.

Hallucination and illusion are both voluntary perceptions of external objects. (Option C): Both hallucinations and illusions are involuntary; individuals cannot control these experiences, especially hallucinations.

Solution for Question 5:

Correct Answer: D) 1-b, 2-a, 3-d, 4-c

Explanation:

Delusion:

Delusion is a firmly held false belief based on incorrect inferences about external reality, despite evidence to the contrary. It is a symptom of schizophrenia and is not typically seen in neurotic disorders.

Medico-legal Importance: Diminished responsibility can apply to individuals committing unlawful acts due to delusions.

Types of Delusions:

- Delusion of Grandeur : Belief of being very important or wealthy, often seen in mania.
- Delusion of Poverty : Conviction of impending loss of all material possessions.
- Delusion of Infidelity/Jealousy (Othello Syndrome) : Belief that a spouse is unfaithful, leading to extreme actions. Males are more commonly affected.
- Delusion of Reference : Belief that media and others refer to the individual, often negatively.
- Delusion of Persecution : Belief that one is being harmed or poisoned by others, leading to potential violence.
- Delusion of Influence/Control : Belief that one's thoughts and actions are controlled by external forces.
- Hypochondriacal Delusion : Persistent fear of having a serious illness based on unrealistic interpretations.
- Delusion of Self-Reproach/Criticism : Self-blame for imagined offences, potentially leading to suicide.
- Nihilistic Delusion : Belief that one does not exist or that the world does not exist, often seen in depression.
- Delusion of Doubles (Doppelgänger) : Patients believe that another person has been physically transformed into themselves.
- Erotomania (de Clérambault's Syndrome) : Belief that someone of higher social status is in love with the individual.
- Capgras Syndrome : Belief that a close person has been replaced by an exact double.
- Fregoli's Phenomenon : Strangers are identified as familiar people.
- Ekbom Syndrome (Delusions of Parasitosis) : Belief of being infested with parasites.
- Folie à Deux : Shared delusions between two people.
- Cotard Delusion/Syndrome : Belief that one is dead, does not exist, is putrefying, or has lost blood or internal organs.

Solution for Question 6:

Correct Answer: B) Capgras syndrome

Explanation:

The patient believes that his wife has been replaced by an impostor, which is characteristic of Capgras syndrome.

- Delusion of Doubles (Doppelgänger) : Patients believe that another person has been physically transformed into themselves. (Option A ruled out)
- Capgras Syndrome : A delusion where a patient believes a close person has been replaced by an exact double.
- Fregoli's Phenomenon : A delusion where strangers are misidentified as familiar people from the patient's life. (Option C ruled out)

Associated Mental Illnesses : Delusions are typically seen in schizophrenia, but not in neurotic illnesses.

Medico-Legal Relevance :

- Delusions can impact legal responsibility under the doctrine of diminished responsibility.
- A person may not be fully accountable for unlawful acts if the delusion impairs their reasoning.

Schizoaffective disorder (Option D) : While this disorder involves psychotic symptoms, the specific delusion of impostor replacement points to Capgras syndrome.

Solution for Question 7:

Correct Answer: B) Acarophobia

Explanation:

Acarophobia is the fear of mites or small insects crawling on the skin, which fits the patient's symptoms of irrational anxiety about infestations without objective evidence.

Phobias and Impulses:

- Phobias : Classified under Neurotic and Somatoform Disorders. Defined as a persistent, unrealistic, and intense fear of an object or situation, often recognised as irrational by the individual. Types of phobias: Agoraphobia : Fear of open spaces, crowds, or leaving home; commonly affects women and may involve panic attacks. (Option D) Social Phobia : Fear of social interactions or situations (e.g., public speaking). Specific Phobia : Fear of particular objects or situations, examples include:

- Classified under Neurotic and Somatoform Disorders.

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- Social Phobia : Fear of social interactions or situations (e.g., public speaking).

- Specific Phobia : Fear of particular objects or situations, examples include:

- Agoraphobia : Fear of open spaces, crowds, or leaving home; commonly affects women and may involve panic attacks. (Option D)

- Social Phobia : Fear of social interactions or situations (e.g., public speaking).

- Specific Phobia : Fear of particular objects or situations, examples include:

Acrophobia	Fear of heights (Option C)
Acarophobia	Fear of mites/insects (Option B)
Claustrophobia	Fear of confined spaces

Needle Phobia	Fear of injections
Algophobia	Fear of pain
Xenophobia	Abnormal fear of strangers
Gamophobia	Fear of marriage/relationships
Zoophobia	Fear of animals

- Impulses : Highlighted in forensic psychiatry due to potential links to criminal behaviour. Defined as sudden, irresistible forces driving a person to act without motive or forethought.
- Highlighted in forensic psychiatry due to potential links to criminal behaviour.
- Defined as sudden, irresistible forces driving a person to act without motive or forethought.
- Highlighted in forensic psychiatry due to potential links to criminal behaviour.
- Defined as sudden, irresistible forces driving a person to act without motive or forethought.

Types of impulses:

Kleptomania	Compulsive stealing
Dipsomania	Compulsive drinking of alcohol
Pyromania	Urge to set fires
Mutilomania	Desire to harm animals
Oniomania	Compulsive shopping
Trichotillomania	Compulsive hair-pulling
Suicidal Impulse	The urge to commit suicide, often triggered by intoxication
Homicidal Impulse	Urge to commit homicide, linked to chronic intoxication

Delusional parasitosis (Option A) : In delusional parasitosis, patients have a fixed false belief of being infested with parasites, typically without insight into the irrationality of their belief.

Causes of Delusional Parasitosis

Primary Psychiatric Disorders:

- Schizophrenia
- Bipolar disorder
- Depression
- Anxiety
- OCD

- Illness anxiety disorder

Medical Causes:

- Illnesses : Hyperthyroidism, B12/folate deficiencies, diabetes.
- Neurological : Dementia, stroke, multiple sclerosis.
- Substance Abuse : Methamphetamine, alcohol withdrawal, cocaine.
- Infections : HIV, tuberculosis, leprosy, syphilis.
- Medications : Topiramate, ciprofloxacin, steroids, ketoconazole.

Solution for Question 8:

Correct Answer: C) The MHCA 2017 came into effect on July 7, 2020, replacing the previous Mental Health Act of 1987

Explanation:

The MHCA 2017 came into effect on July 7, 2018 , not 2020.

Mental Healthcare Act (MHCA) 2017 in India:

- Overview of the Act : Purpose : Aims to provide comprehensive mental healthcare and services, aligning with the UN Convention on the Rights of Persons with Disabilities (2006). (Option A ruled out) Implementation Date : Enacted on July 7, 2018, replacing the Mental Health Act of 1987.
- Purpose : Aims to provide comprehensive mental healthcare and services, aligning with the UN Convention on the Rights of Persons with Disabilities (2006). (Option A ruled out)
- Implementation Date : Enacted on July 7, 2018, replacing the Mental Health Act of 1987.
- Key Provisions : Rights Protection : Focuses on protecting and promoting the rights of individuals during mental healthcare delivery; ensures treatment in the least restrictive environment. Admission Procedures : Specifies detailed protocols for voluntary and involuntary admissions; all admissions require notification to a Mental Health Review Board (MHRB) for stays exceeding 30 days. (Option B ruled out) Advance Directives : Introduces advance directives and nominated representatives, allowing individuals to express treatment preferences in advance . (Option D ruled out)
- Rights Protection : Focuses on protecting and promoting the rights of individuals during mental healthcare delivery; ensures treatment in the least restrictive environment.

- Admission Procedures : Specifies detailed protocols for voluntary and involuntary admissions; all admissions require notification to a Mental Health Review Board (MHRB) for stays exceeding 30 days. (Option B ruled out)
- Advance Directives : Introduces advance directives and nominated representatives, allowing individuals to express treatment preferences in advance . (Option D ruled out)
- Implementation Challenges : Resource Limitations : Acknowledges limited mental healthcare resources in India; provides a 10-year timeline for meeting international standards in human resource development for mental healthcare. Transition from Previous Legislation : Introduces stringent involuntary admission procedures compared to MHA 1987, requiring legal formalities that may delay treatment.
- Resource Limitations : Acknowledges limited mental healthcare resources in India; provides a 10-year timeline for meeting international standards in human resource development for mental healthcare.
- Transition from Previous Legislation : Introduces stringent involuntary admission procedures compared to MHA 1987, requiring legal formalities that may delay treatment.
- Conclusion : The MHCA 2017 is a progressive step towards improving mental health care in India. Success depends on effective implementation despite existing resource constraints.
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- Success depends on effective implementation despite existing resource constraints.
- Purpose : Aims to provide comprehensive mental healthcare and services, aligning with the UN Convention on the Rights of Persons with Disabilities (2006). (Option A ruled out)
- Implementation Date : Enacted on July 7, 2018, replacing the Mental Health Act of 1987.
- Rights Protection : Focuses on protecting and promoting the rights of individuals during mental healthcare delivery; ensures treatment in the least restrictive environment.
- Admission Procedures : Specifies detailed protocols for voluntary and involuntary admissions; all admissions require notification to a Mental Health Review Board (MHRB) for stays exceeding 30 days. (Option B ruled out)
- Advance Directives : Introduces advance directives and nominated representatives, allowing individuals to express treatment preferences in advance . (Option D ruled out)
- Resource Limitations : Acknowledges limited mental healthcare resources in India; provides a 10-year timeline for meeting international standards in human resource development for mental healthcare.
- Transition from Previous Legislation : Introduces stringent involuntary admission procedures compared to MHA 1987, requiring legal formalities that may delay treatment.
- The MHCA 2017 is a progressive step towards improving mental health care in India.
- Success depends on effective implementation despite existing resource constraints.

Solution for Question 9:

Correct Answer: A) True insanity

Explanation:

Given the presence of long-standing auditory hallucinations, paranoid delusions, disorganized thoughts, flat affect (a severe reduction in emotional expressiveness.), and poor hygiene, these are classic features of a psychotic disorder and suggest a genuine mental illness rather than feigned symptoms.

True vs. Feigned Insanity:

Individuals may feign insanity to avoid criminal responsibility.

Feature	True Insanity (Option A)	Feigned Insanity (Option B)
Onset	Gradual	Sudden
Motive	Absent	Present (e.g., after a crime)
Predisposing Factors	Usually present (e.g., family history)	Absent
Signs and Symptoms	Uniform, specific	Variable, exaggerated
Activity	Careless, persists even when unobserved	Only present when observed
Mood	Excited, depressed, fluctuating	Overacted to appear abnormal
Facial Expression	Peculiar, vacant, worried	Frequently changing, exaggerated
Insomnia	Present	Inconsistent, resolves in a day or two
Exertion	Can endure exertion, fatigue	Breaks down after short exertion
Habits	Dirty and filthy	Clean, not neglected
Dressing	Careless	Reasonably proper
Skin and Lips	Dry, harsh	Normal
Tongue	Coated	Clean

Repeated Examinations	Doesn't mind	Resists for fear of detection
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Delusional disorder (Option C) involves fixed, non-bizarre delusions without prominent hallucinations or disorganised thinking, whereas this patient has broader psychotic symptoms like hallucinations and thought disorganisation.

Personality disorder (Option D) does not typically present with acute psychotic symptoms such as auditory hallucinations or severe disorganisation.

Solution for Question 10:

Correct Answer: B) Somnambulism

Explanation:

Somnambulism, or sleepwalking, involves automatic motor activity during sleep with no recollection upon awakening, consistent with the patient's behaviour.

Criminal Responsibility in Specific Cases:

- Delirium: Type: Organic mental disorder. Definition: An acute reversible mental disorder characterised by confusion and impaired consciousness. Context: Occurs in individuals with high fever, overwork, mental stress, acute poisoning, or chronic alcohol/drug intoxication. (Option A) Criminal Responsibility : Individuals may not be responsible for criminal acts due to impaired consciousness and irrational behaviour.
- Type: Organic mental disorder.
- Definition: An acute reversible mental disorder characterised by confusion and impaired consciousness.
- Context: Occurs in individuals with high fever, overwork, mental stress, acute poisoning, or chronic alcohol/drug intoxication. (Option A)
- Criminal Responsibility : Individuals may not be responsible for criminal acts due to impaired consciousness and irrational behaviour.
- Run Amok : (Run Amok) Definition: Describes an individual engaging in frenzied violence without clear provocation. Cultural Context: Aligns with Section 84 of the IPC , stating that a person is not liable for acts committed under unsoundness of mind. Causes of run amok: Chronic poisoning of Cannabis sativa
- Definition: Describes an individual engaging in frenzied violence without clear provocation.

- Cultural Context: Aligns with Section 84 of the IPC , stating that a person is not liable for acts committed under unsoundness of mind.
- Causes of run amok: Chronic poisoning of Cannabis sativa
- Somnambulism (Sleepwalking) : Category: Parasomnia involving automatic motor activity during sleep with no recollection upon waking. (Option B) Criminal Responsibility : Individuals committing criminal acts during sleepwalking episodes are not held responsible.
- Category: Parasomnia involving automatic motor activity during sleep with no recollection upon waking. (Option B)
- Criminal Responsibility : Individuals committing criminal acts during sleepwalking episodes are not held responsible.
- Type: Organic mental disorder.
- Definition: An acute reversible mental disorder characterised by confusion and impaired consciousness.
- Context: Occurs in individuals with high fever, overwork, mental stress, acute poisoning, or chronic alcohol/drug intoxication. (Option A)
- Criminal Responsibility : Individuals may not be responsible for criminal acts due to impaired consciousness and irrational behaviour.
- Definition: Describes an individual engaging in frenzied violence without clear provocation.
- Cultural Context: Aligns with Section 84 of the IPC , stating that a person is not liable for acts committed under unsoundness of mind.
- Causes of run amok: Chronic poisoning of Cannabis sativa
- Category: Parasomnia involving automatic motor activity during sleep with no recollection upon waking. (Option B)
- Criminal Responsibility : Individuals committing criminal acts during sleepwalking episodes are not held responsible.

Sleep disorders in different phases of the cycle:

Sleep apnea	It occurs in the deeper sleep phases, primarily affecting stage N3 and REM sleep.
Narcolepsy	It involves skipping initial sleep phases and directly entering REM sleep, leading to persistent daytime sleepiness. This results in reduced N3 deep sleep and irregular sleep patterns.
REM sleep disorder	It involves acting out dreams with vocalizations and movements due to disrupted muscle atonia during REM sleep.

Somnambulism or sleepwalking	It occurs during non-REM sleep, typically in the N3 phase. This is when individuals can make purposeful movements without acting out their dreams, which happens in REM sleep.
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- Sleep Drunkenness : Definition: A form of hypersomnia where individuals experience confusion and disorientation during the transition from sleep to wakefulness. Criminal Responsibility : Individuals are not responsible for violent acts performed while in this state.
- Definition: A form of hypersomnia where individuals experience confusion and disorientation during the transition from sleep to wakefulness.
- Criminal Responsibility : Individuals are not responsible for violent acts performed while in this state.
- Kleptomania : Type : Impulse disorder characterised by a compulsion to steal items of little value. Criminal Responsibility : Courts may consider kleptomania as a mental disorder affecting impulse control (not guilty under the New Hampshire Doctrine), which could influence sentencing or lead to treatment rather than punishment. (MHCA 2017)
- Type : Impulse disorder characterised by a compulsion to steal items of little value.
- Criminal Responsibility : Courts may consider kleptomania as a mental disorder affecting impulse control (not guilty under the New Hampshire Doctrine), which could influence sentencing or lead to treatment rather than punishment. (MHCA 2017)
- Hypnosis : (Option D) Context: Not explicitly addressed but related to "delusion of influence/control". Criminal Responsibility : Complex determination involving the degree of hypnosis influence, nature of suggestions, and individual's mental state.
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- Criminal Responsibility : Complex determination involving the degree of hypnosis influence, nature of suggestions, and individual's mental state.
- Definition: A form of hypersomnia where individuals experience confusion and disorientation during the transition from sleep to wakefulness.
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- Criminal Responsibility : Courts may consider kleptomania as a mental disorder affecting impulse control (not guilty under the New Hampshire Doctrine), which could influence sentencing or lead to treatment rather than punishment. (MHCA 2017)
- Context: Not explicitly addressed but related to "delusion of influence/control".
- Criminal Responsibility : Complex determination involving the degree of hypnosis influence, nature of suggestions, and individual's mental state.

Acute Psychosis (Option C): This condition involves a loss of contact with reality, hallucinations, or delusions, which the patient does not exhibit in this case.

Solution for Question 11:

Correct Answer: D) An organic causative factor is present.

Explanation:

Psychosis vs Neurosis

Features	Psychosis	Neurosis
1. Contact with reality	Lost	Preserved (Option A ruled out)
2. Interpersonal behavior	Marked disturbance in personality and behaviour	Preserved
3. Empathy	Absent	Present (Option B ruled out)
4. Insight	Absence of understanding of current symptoms (insight absent)	Symptoms are recognized as undesirable (insight present) (Option C ruled out)
5. Organic causative factor	Present	Absent
6. Symptoms	Delusions, illusions and hallucinations	Usually physical or psychic symptoms
7. Dealing with reality	Capacity is grossly impaired	Preserved
8. Examples	Dementia, schizophrenia	Anxiety, phobia, depression, conversion disorder.